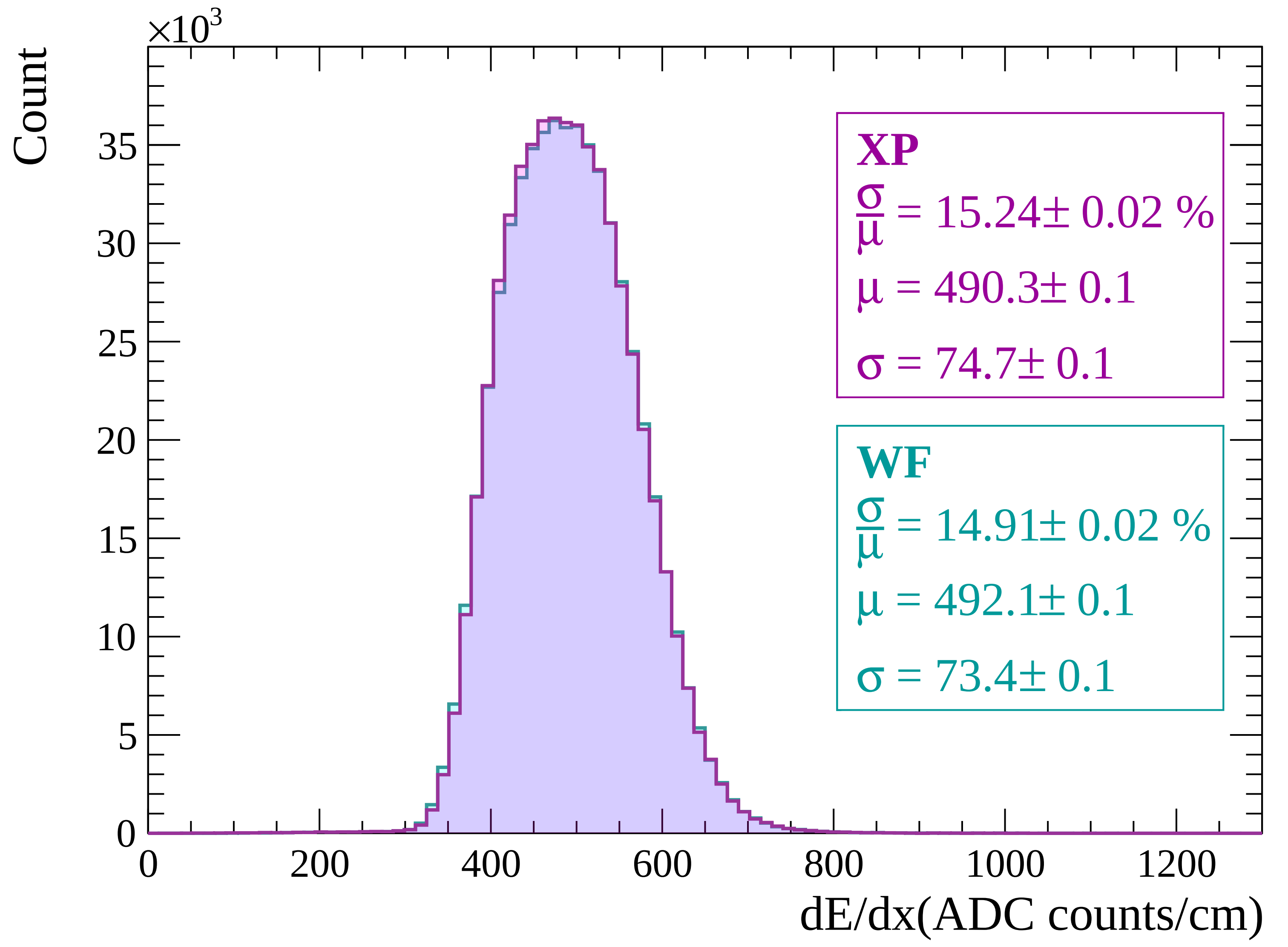
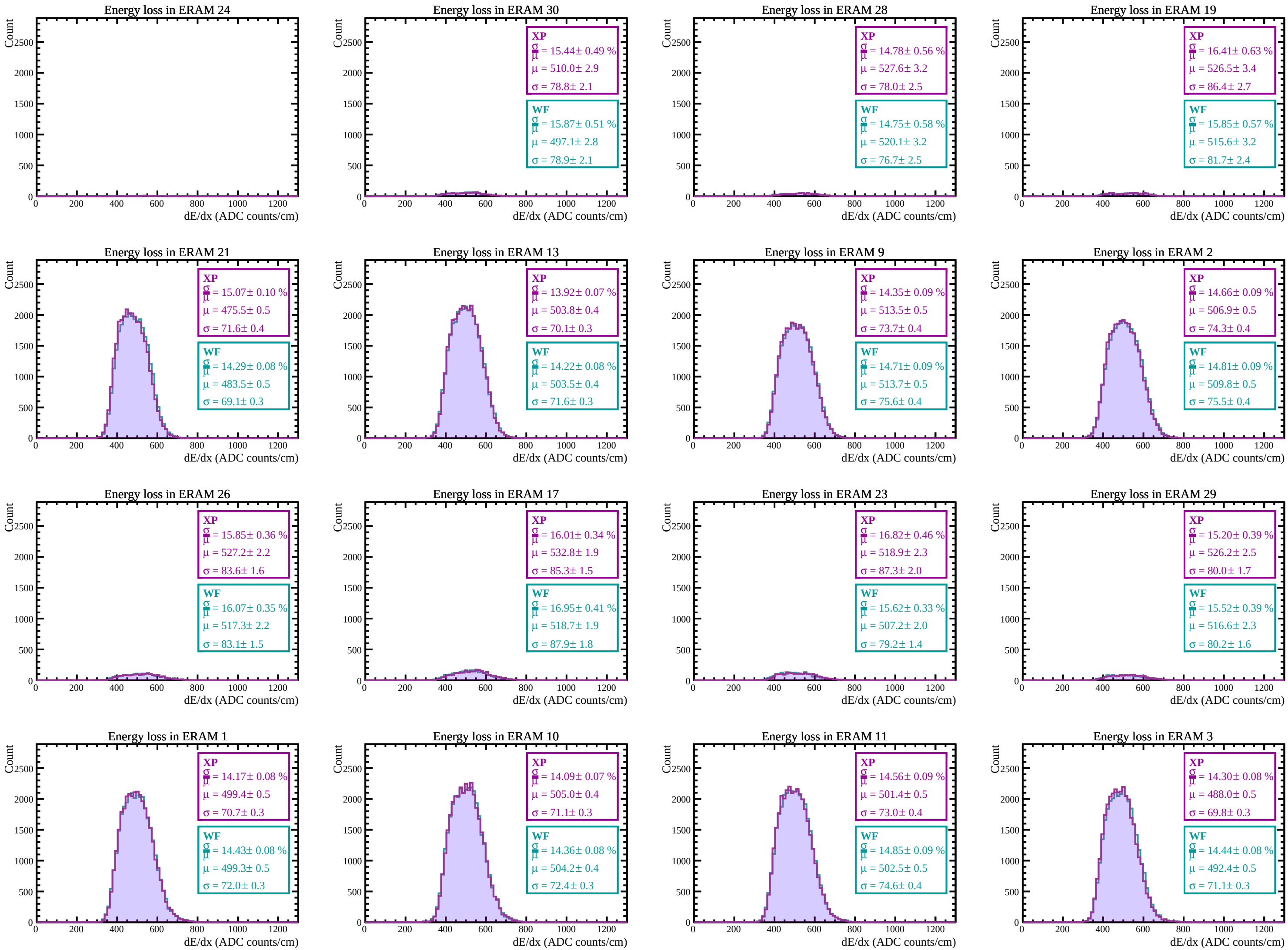
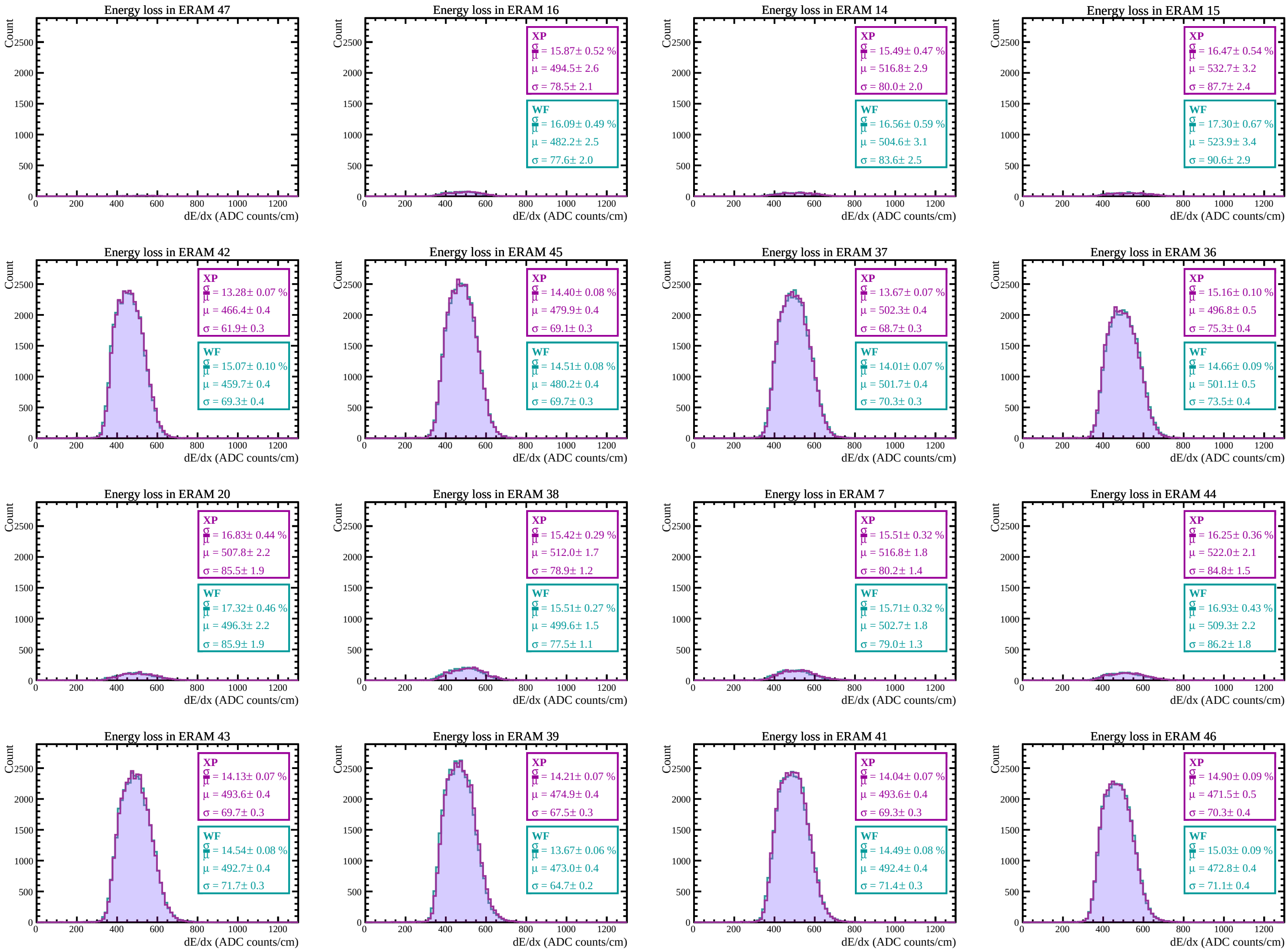
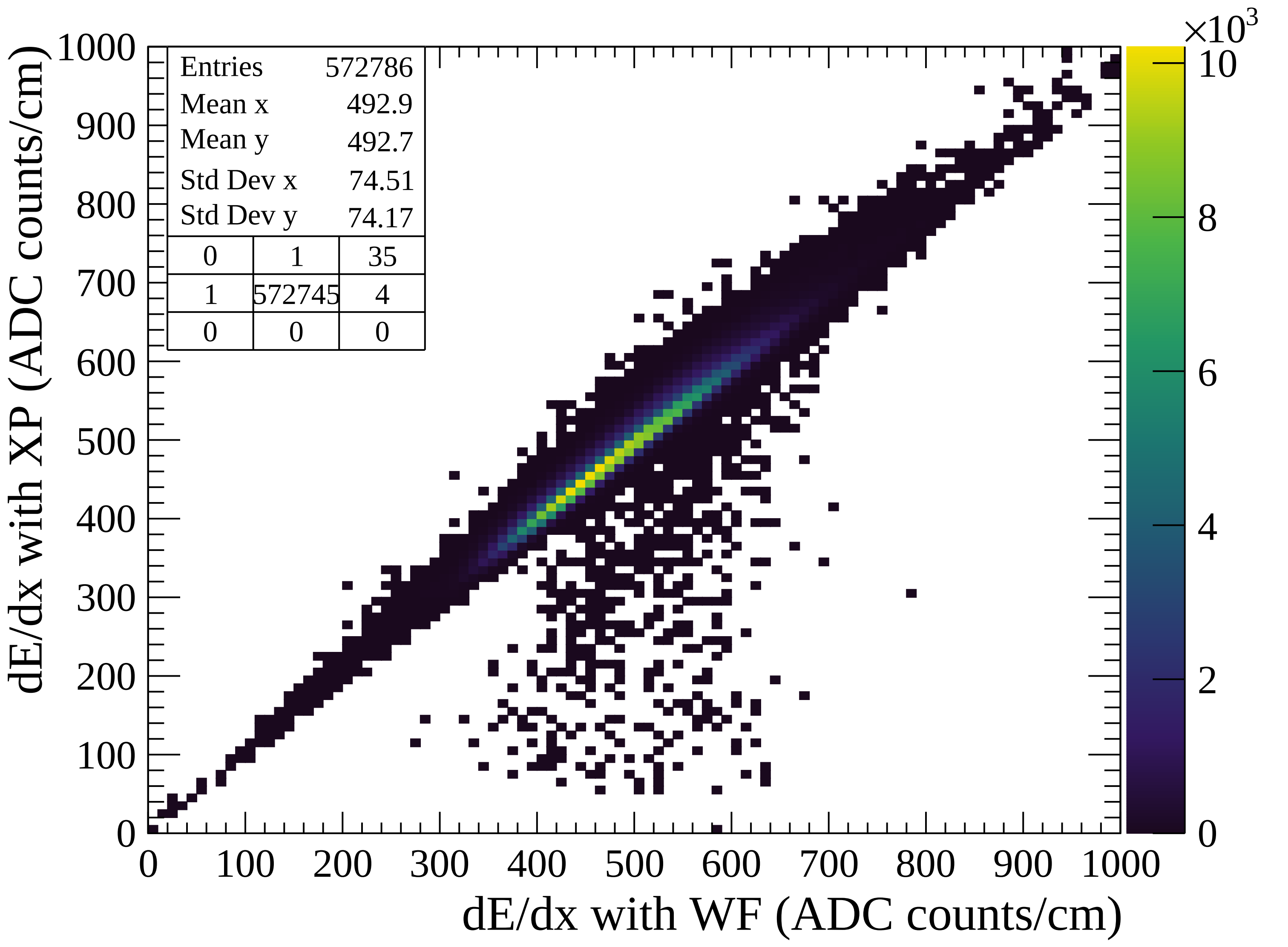


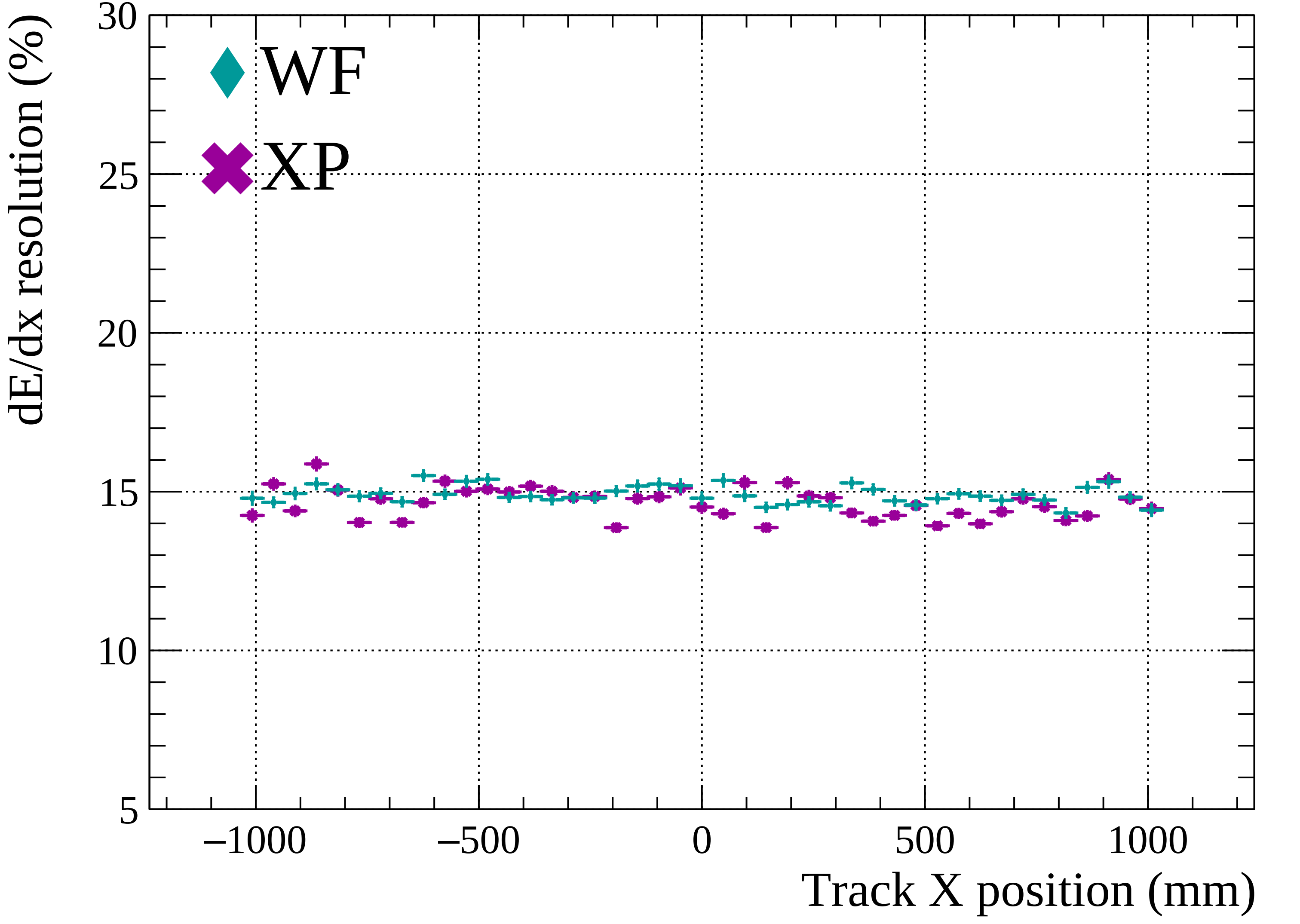
×10^3
Count
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
σ/μ = 15.24± 0.02 %
μ = 490.3± 0.1
σ = 74.7± 0.1
WF
σ/μ = 14.91± 0.02 %
μ = 492.1± 0.1
σ = 73.4± 0.1


Energy loss in ERAM 24
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 30
XP
σ/μ = 15.44± 0.49 %
μ = 510.0± 2.9
σ = 78.8± 2.1
WF
σ/μ = 15.87± 0.51 %
μ = 497.1± 2.8
σ = 78.9± 2.1
Energy loss in ERAM 28
XP
σ/μ = 14.78± 0.56 %
μ = 527.6± 3.2
σ = 78.0± 2.5
WF
σ/μ = 14.75± 0.58 %
μ = 520.1± 3.2
σ = 76.7± 2.5
Energy loss in ERAM 19
XP
σ/μ = 16.41± 0.63 %
μ = 526.5± 3.4
σ = 86.4± 2.7
WF
σ/μ = 15.85± 0.57 %
μ = 515.6± 3.2
σ = 81.7± 2.4
Energy loss in ERAM 21
XP
σ/μ = 15.07± 0.10 %
μ = 475.5± 0.5
σ = 71.6± 0.4
WF
σ/μ = 14.29± 0.08 %
μ = 483.5± 0.5
σ = 69.1± 0.3
Energy loss in ERAM 13
XP
σ/μ = 13.92± 0.07 %
μ = 503.8± 0.4
σ = 70.1± 0.3
WF
σ/μ = 14.22± 0.08 %
μ = 503.5± 0.4
σ = 71.6± 0.3
Energy loss in ERAM 9
XP
σ/μ = 14.35± 0.09 %
μ = 513.5± 0.5
σ = 73.7± 0.4
WF
σ/μ = 14.71± 0.09 %
μ = 513.7± 0.5
σ = 75.6± 0.4
Energy loss in ERAM 2
XP
σ/μ = 14.66± 0.09 %
μ = 506.9± 0.5
σ = 74.3± 0.4
WF
σ/μ = 14.81± 0.09 %
μ = 509.8± 0.5
σ = 75.5± 0.4
Energy loss in ERAM 26
XP
σ/μ = 15.85± 0.36 %
μ = 527.2± 2.2
σ = 83.6± 1.6
WF
σ/μ = 16.07± 0.35 %
μ = 517.3± 2.2
σ = 83.1± 1.5
Energy loss in ERAM 17
XP
σ/μ = 16.01± 0.34 %
μ = 532.8± 1.9
σ = 85.3± 1.5
WF
σ/μ = 16.95± 0.41 %
μ = 518.7± 1.9
σ = 87.9± 1.8
Energy loss in ERAM 23
XP
σ/μ = 16.82± 0.46 %
μ = 518.9± 2.3
σ = 87.3± 2.0
WF
σ/μ = 15.62± 0.33 %
μ = 507.2± 2.0
σ = 79.2± 1.4
Energy loss in ERAM 29
XP
σ/μ = 15.20± 0.39 %
μ = 526.2± 2.5
σ = 80.0± 1.7
WF
σ/μ = 15.52± 0.39 %
μ = 516.6± 2.3
σ = 80.2± 1.6
Energy loss in ERAM 1
XP
σ/μ = 14.17± 0.08 %
μ = 499.4± 0.5
σ = 70.7± 0.3
WF
σ/μ = 14.43± 0.08 %
μ = 499.3± 0.5
σ = 72.0± 0.3
Energy loss in ERAM 10
XP
σ/μ = 14.09± 0.07 %
μ = 505.0± 0.4
σ = 71.1± 0.3
WF
σ/μ = 14.36± 0.08 %
μ = 504.2± 0.4
σ = 72.4± 0.3
Energy loss in ERAM 11
XP
σ/μ = 14.56± 0.09 %
μ = 501.4± 0.5
σ = 73.0± 0.4
WF
σ/μ = 14.85± 0.09 %
μ = 502.5± 0.5
σ = 74.6± 0.4
Energy loss in ERAM 3
XP
σ/μ = 14.30± 0.08 %
μ = 488.0± 0.5
σ = 69.8± 0.3
WF
σ/μ = 14.44± 0.08 %
μ = 492.4± 0.5
σ = 71.1± 0.3

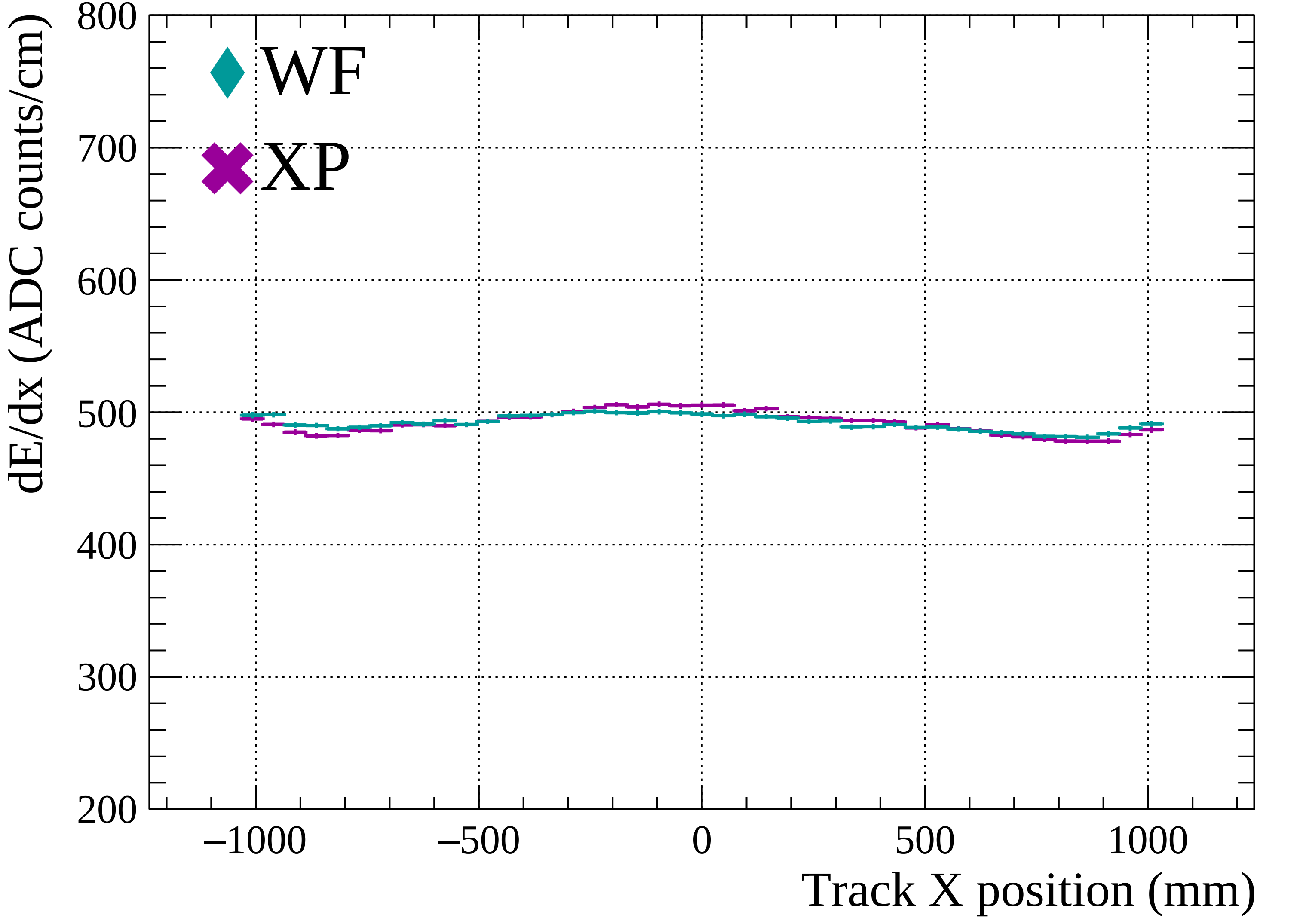


Energy loss in ERAM 47
Energy loss in ERAM 16
XP
σ/μ = 15.87± 0.52 %
μ = 494.5± 2.6
σ = 78.5± 2.1
WF
σ/μ = 16.09± 0.49 %
μ = 482.2± 2.5
σ = 77.6± 2.0
Energy loss in ERAM 14
XP
σ/μ = 15.49± 0.47 %
μ = 516.8± 2.9
σ = 80.0± 2.0
WF
σ/μ = 16.56± 0.59 %
μ = 504.6± 3.1
σ = 83.6± 2.5
Energy loss in ERAM 15
XP
σ/μ = 16.47± 0.54 %
μ = 532.7± 3.2
σ = 87.7± 2.4
WF
σ/μ = 17.30± 0.67 %
μ = 523.9± 3.4
σ = 90.6± 2.9
Energy loss in ERAM 42
XP
σ/μ = 13.28± 0.07 %
μ = 466.4± 0.4
σ = 61.9± 0.3
WF
σ/μ = 15.07± 0.10 %
μ = 459.7± 0.4
σ = 69.3± 0.4
Energy loss in ERAM 45
XP
σ/μ = 14.40± 0.08 %
μ = 479.9± 0.4
σ = 69.1± 0.3
WF
σ/μ = 14.51± 0.08 %
μ = 480.2± 0.4
σ = 69.7± 0.3
Energy loss in ERAM 37
XP
σ/μ = 13.67± 0.07 %
μ = 502.3± 0.4
σ = 68.7± 0.3
WF
σ/μ = 14.01± 0.07 %
μ = 501.7± 0.4
σ = 70.3± 0.3
Energy loss in ERAM 36
XP
σ/μ = 15.16± 0.10 %
μ = 496.8± 0.5
σ = 75.3± 0.4
WF
σ/μ = 14.66± 0.09 %
μ = 501.1± 0.5
σ = 73.5± 0.4
Energy loss in ERAM 20
XP
σ/μ = 16.83± 0.44 %
μ = 507.8± 2.2
σ = 85.5± 1.9
WF
σ/μ = 17.32± 0.46 %
μ = 496.3± 2.2
σ = 85.9± 1.9
Energy loss in ERAM 38
XP
σ/μ = 15.42± 0.29 %
μ = 512.0± 1.7
σ = 78.9± 1.2
WF
σ/μ = 15.51± 0.27 %
μ = 499.6± 1.5
σ = 77.5± 1.1
Energy loss in ERAM 7
XP
σ/μ = 15.51± 0.32 %
μ = 516.8± 1.8
σ = 80.2± 1.4
WF
σ/μ = 15.71± 0.32 %
μ = 502.7± 1.8
σ = 79.0± 1.3
Energy loss in ERAM 44
XP
σ/μ = 16.25± 0.36 %
μ = 522.0± 2.1
σ = 84.8± 1.5
WF
σ/μ = 16.93± 0.43 %
μ = 509.3± 2.2
σ = 86.2± 1.8
Energy loss in ERAM 43
XP
σ/μ = 14.13± 0.07 %
μ = 493.6± 0.4
σ = 69.7± 0.3
WF
σ/μ = 14.54± 0.08 %
μ = 492.7± 0.4
σ = 71.7± 0.3
Energy loss in ERAM 39
XP
σ/μ = 14.21± 0.07 %
μ = 474.9± 0.4
σ = 67.5± 0.3
WF
σ/μ = 13.67± 0.06 %
μ = 473.0± 0.4
σ = 64.7± 0.2
Energy loss in ERAM 41
XP
σ/μ = 14.04± 0.07 %
μ = 493.6± 0.4
σ = 69.3± 0.3
WF
σ/μ = 14.49± 0.08 %
μ = 492.4± 0.4
σ = 71.4± 0.3
Energy loss in ERAM 46
XP
σ/μ = 14.90± 0.09 %
μ = 471.5± 0.5
σ = 70.3± 0.4
WF
σ/μ = 15.03± 0.09 %
μ = 472.8± 0.4
σ = 71.1± 0.4
Count
dE/dx (ADC counts/cm)

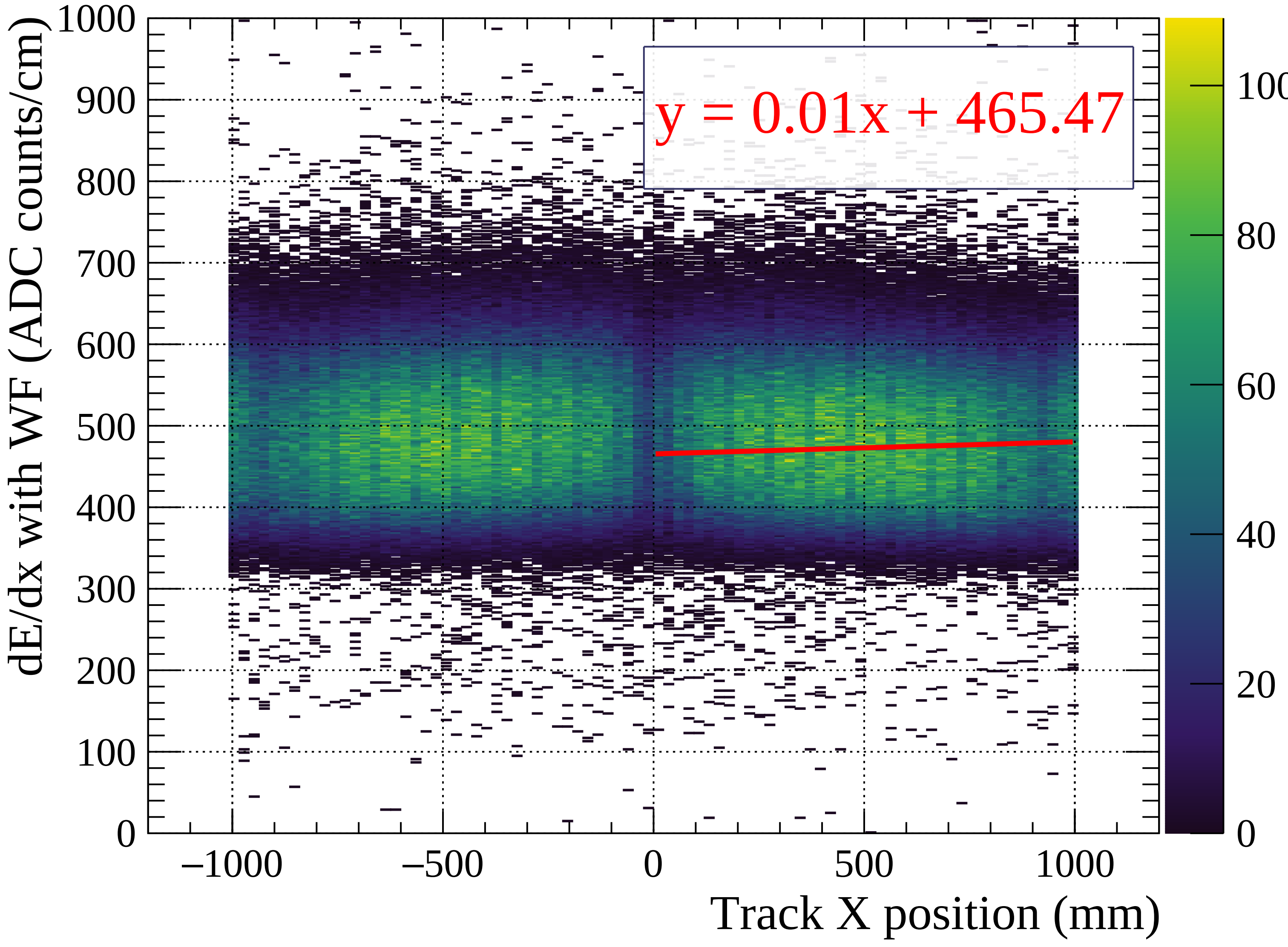

Entries 572786
Mean x 492.9
Mean y 492.7
Std Dev x 74.51
Std Dev y 74.17
0 1 35
1 572745 4
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0
100
200
300
400
500
600
700
800
900
1000
0
2
4
6
8
10



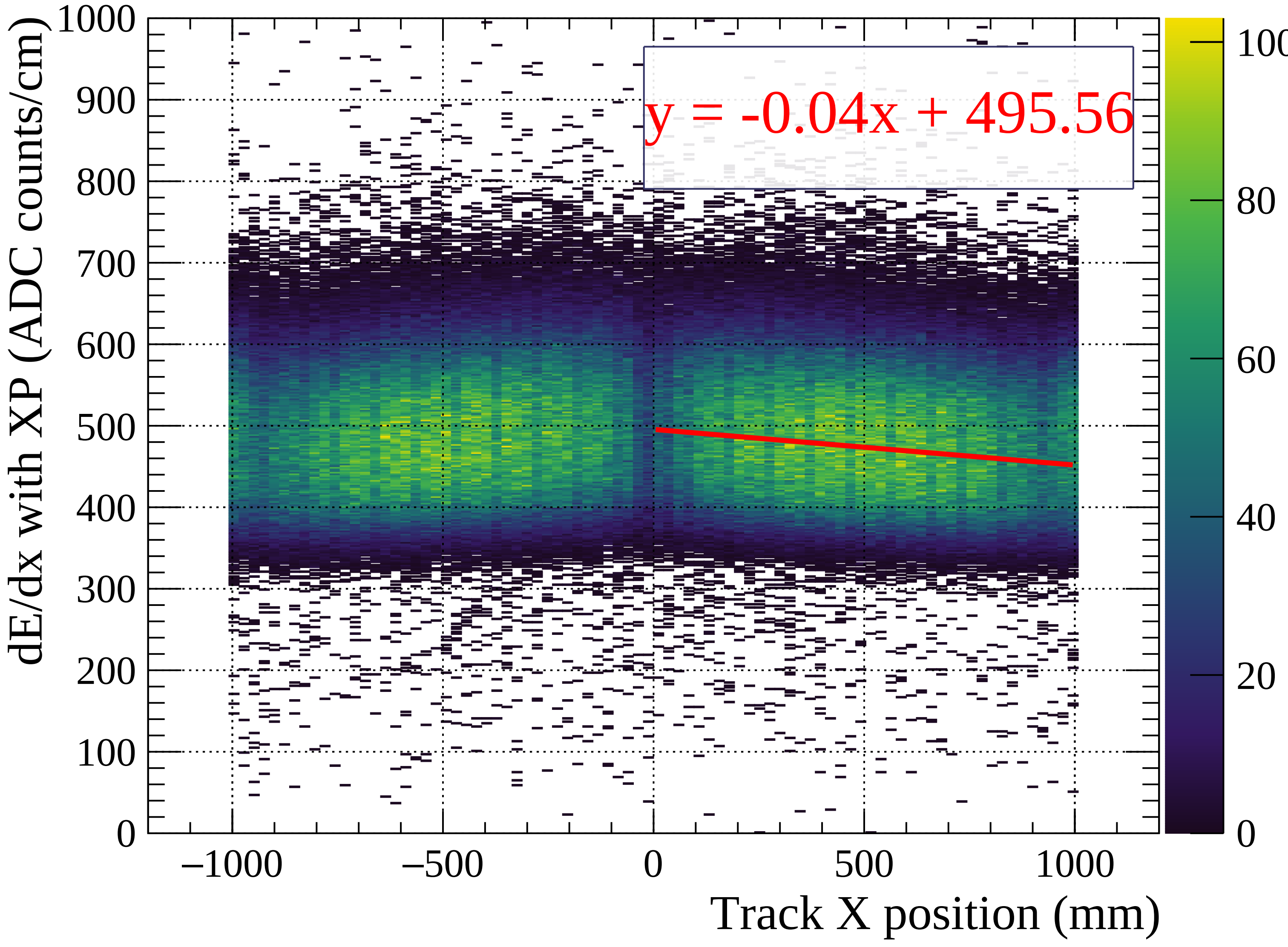
WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)



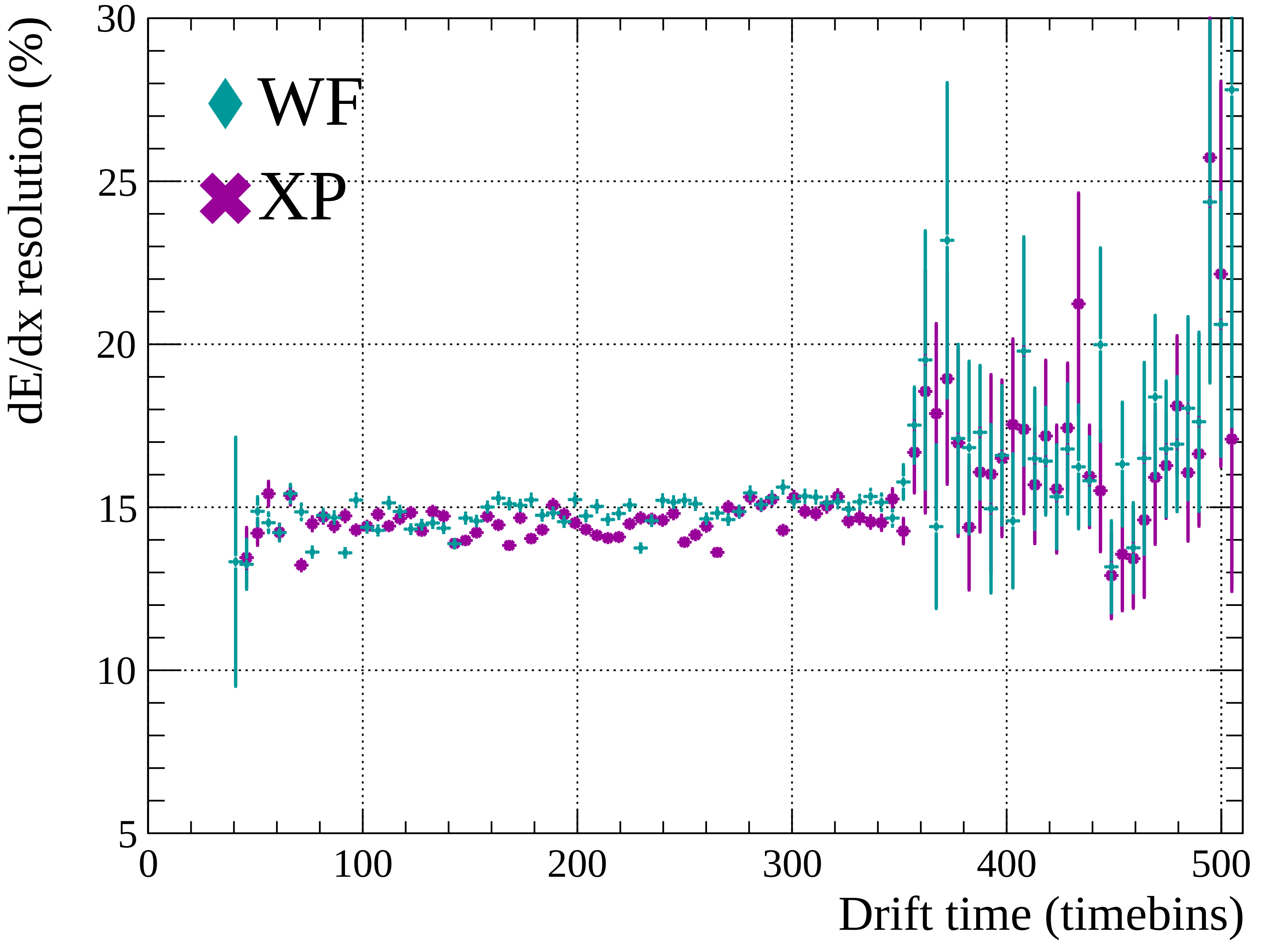
WF
XP
800
700
600
500
400
300
200
−1000
−500
0
500
1000
dE/dx (ADC counts/cm)
Track X position (mm)



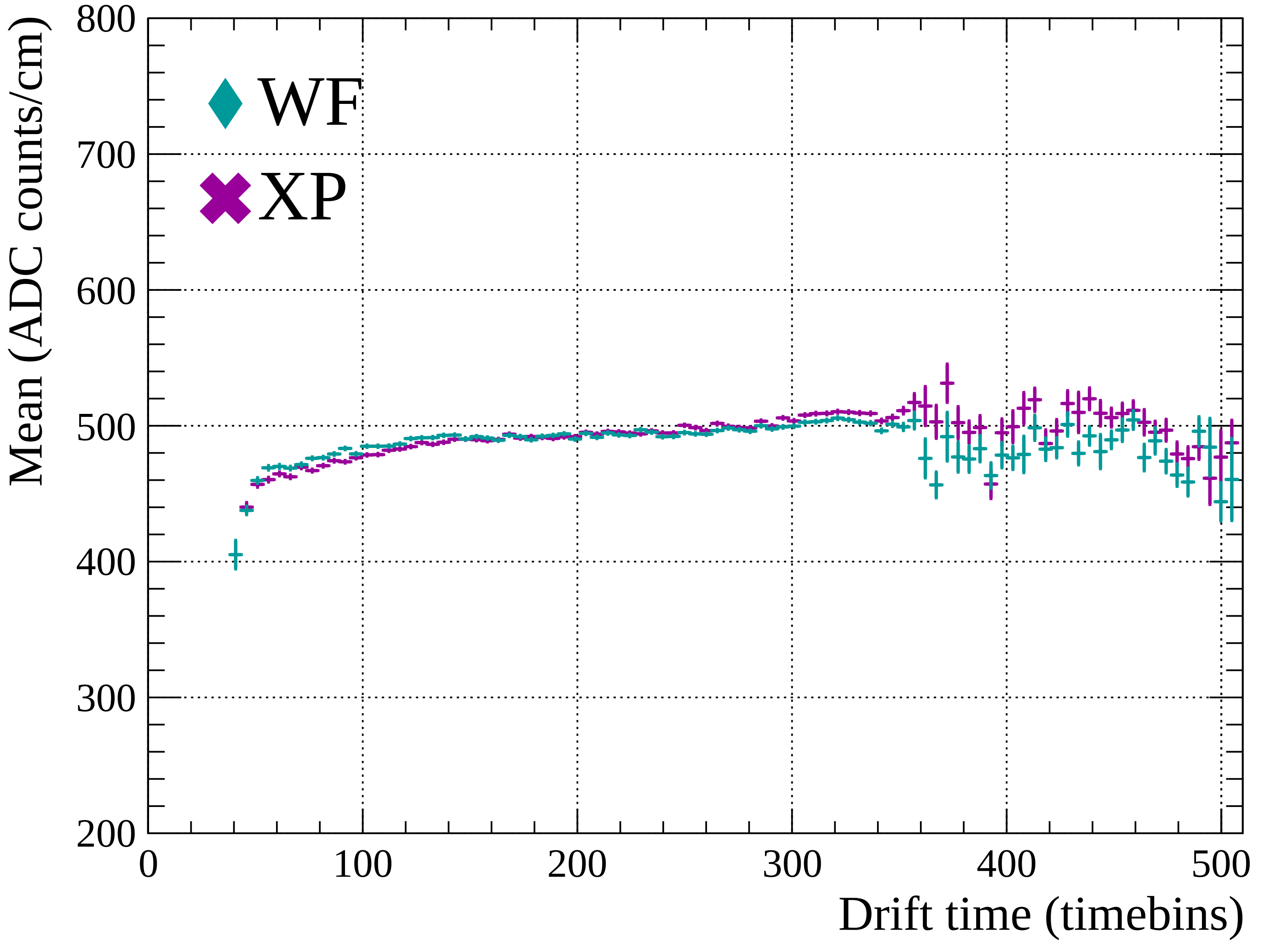
y = 0.01x + 465.47
dE/dx with WF (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
100
80
60
40
20
0



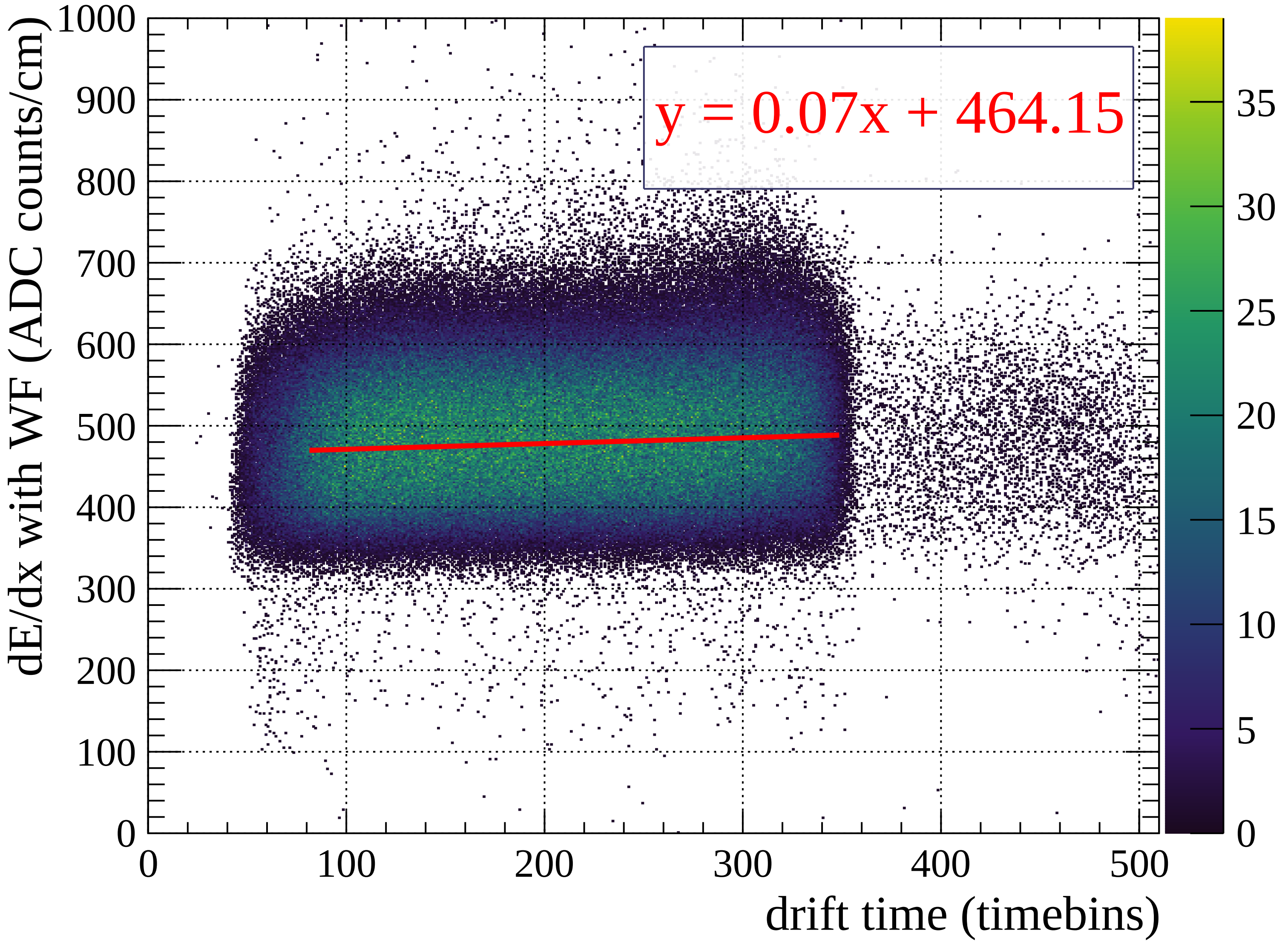
y = -0.04x + 495.56
dE/dx with XP (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
100
80
60
40
20
0



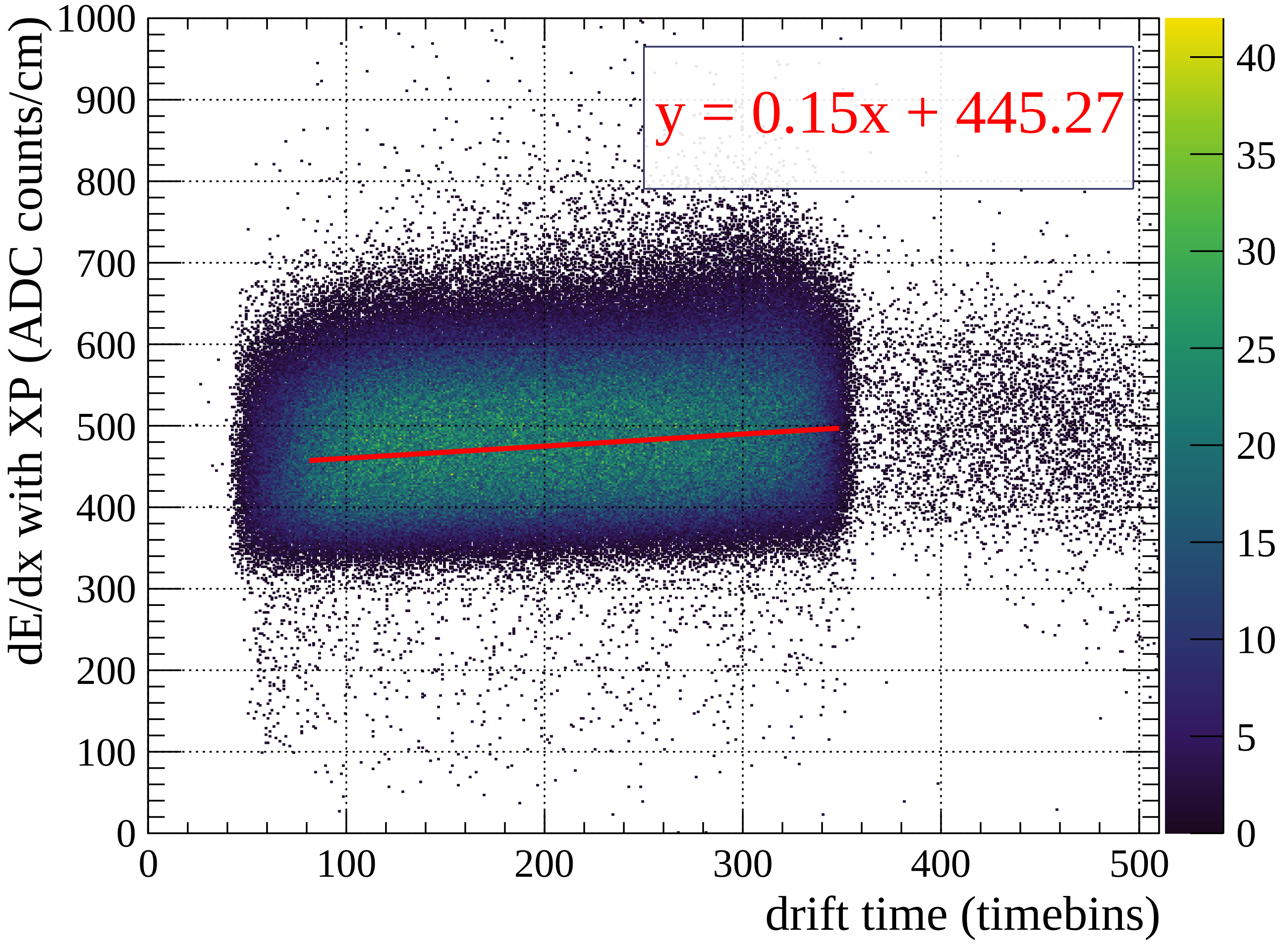
dE/dx resolution (%)
Drift time (timebins)
WF
XP
0
100
200
300
400
500
5
10
15
20
25
30

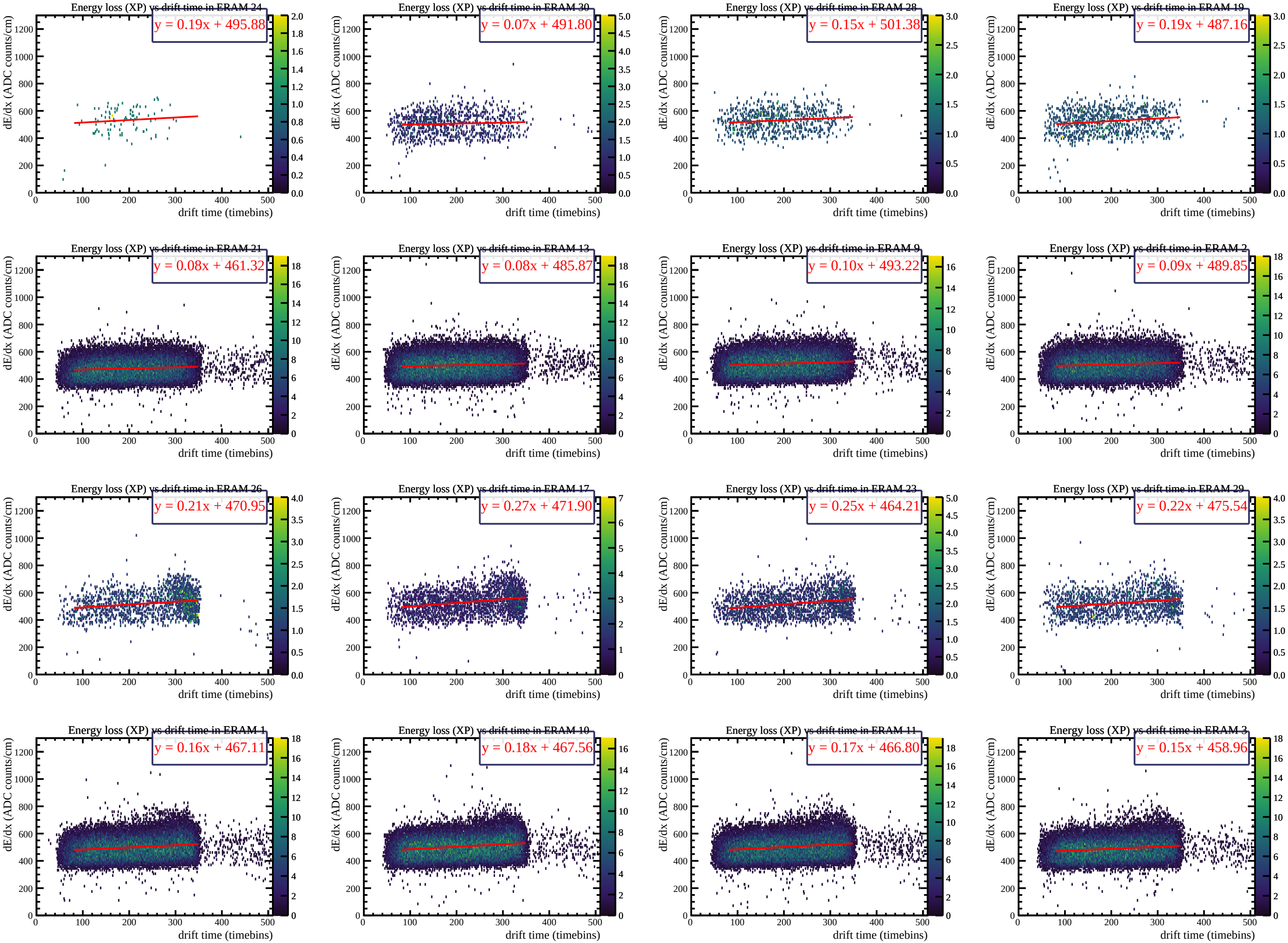


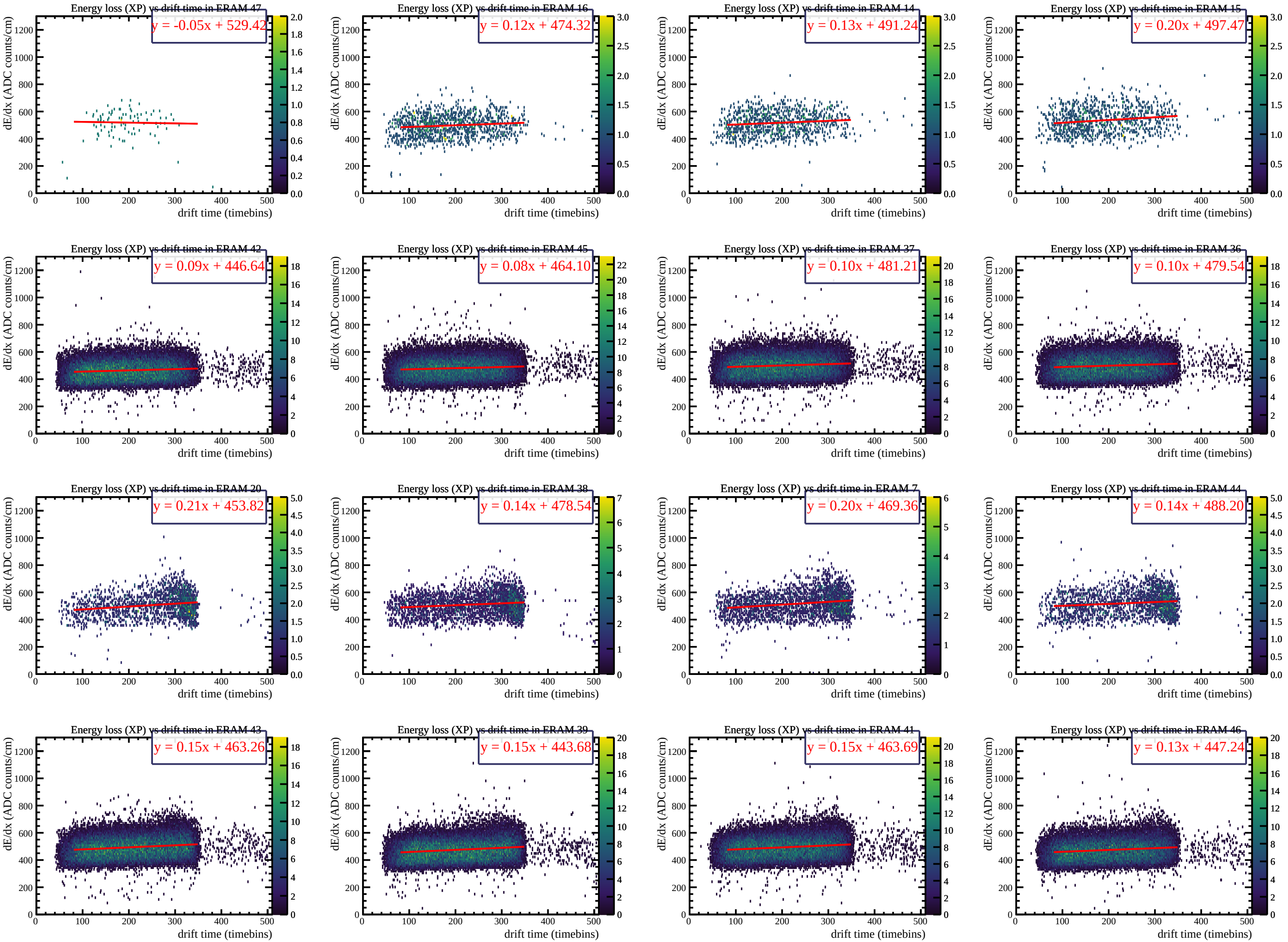
WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
200
300
400
500
600
700
800
0
100
200
300
400
500



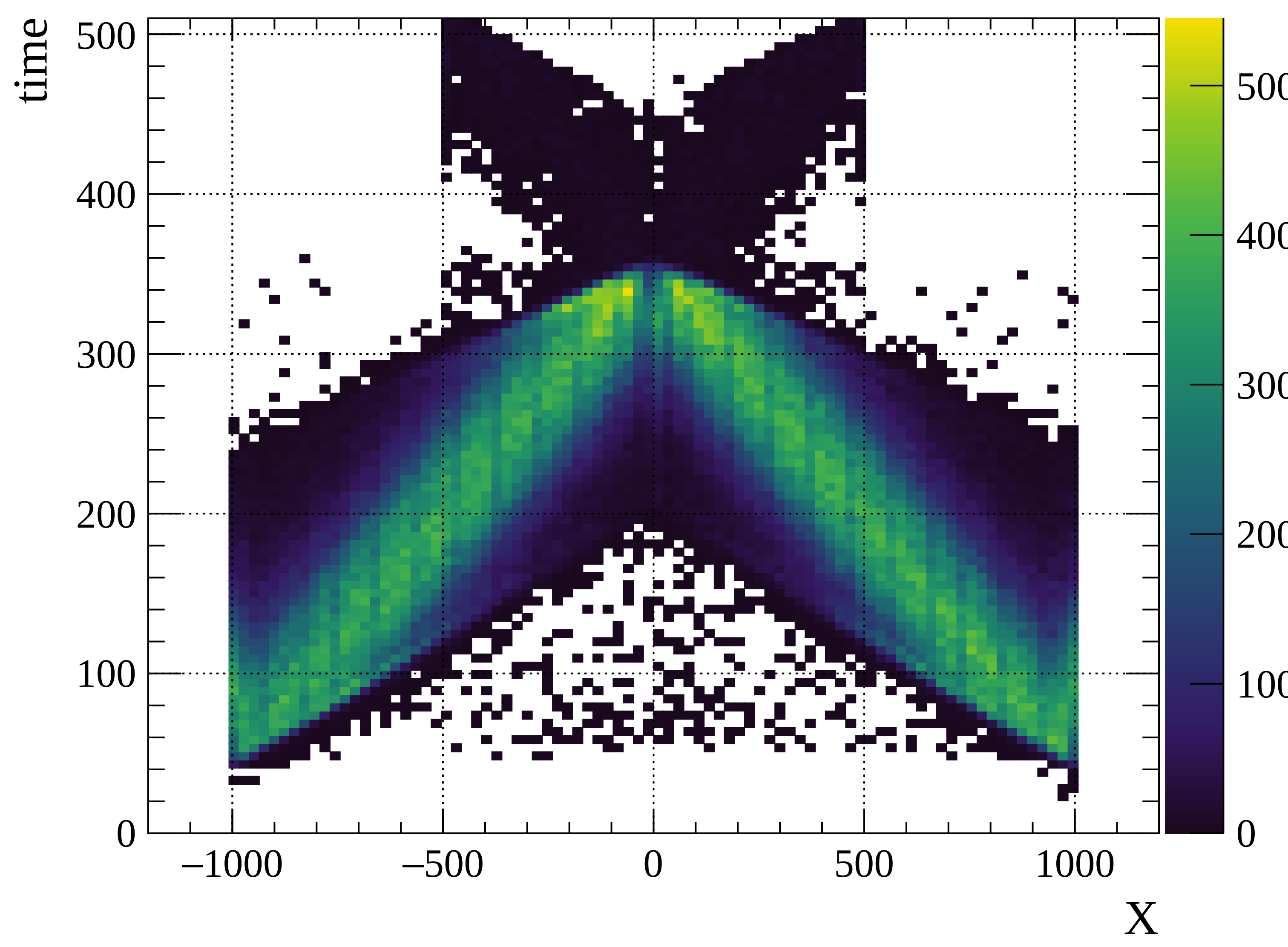
y = 0.07x + 464.15
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35

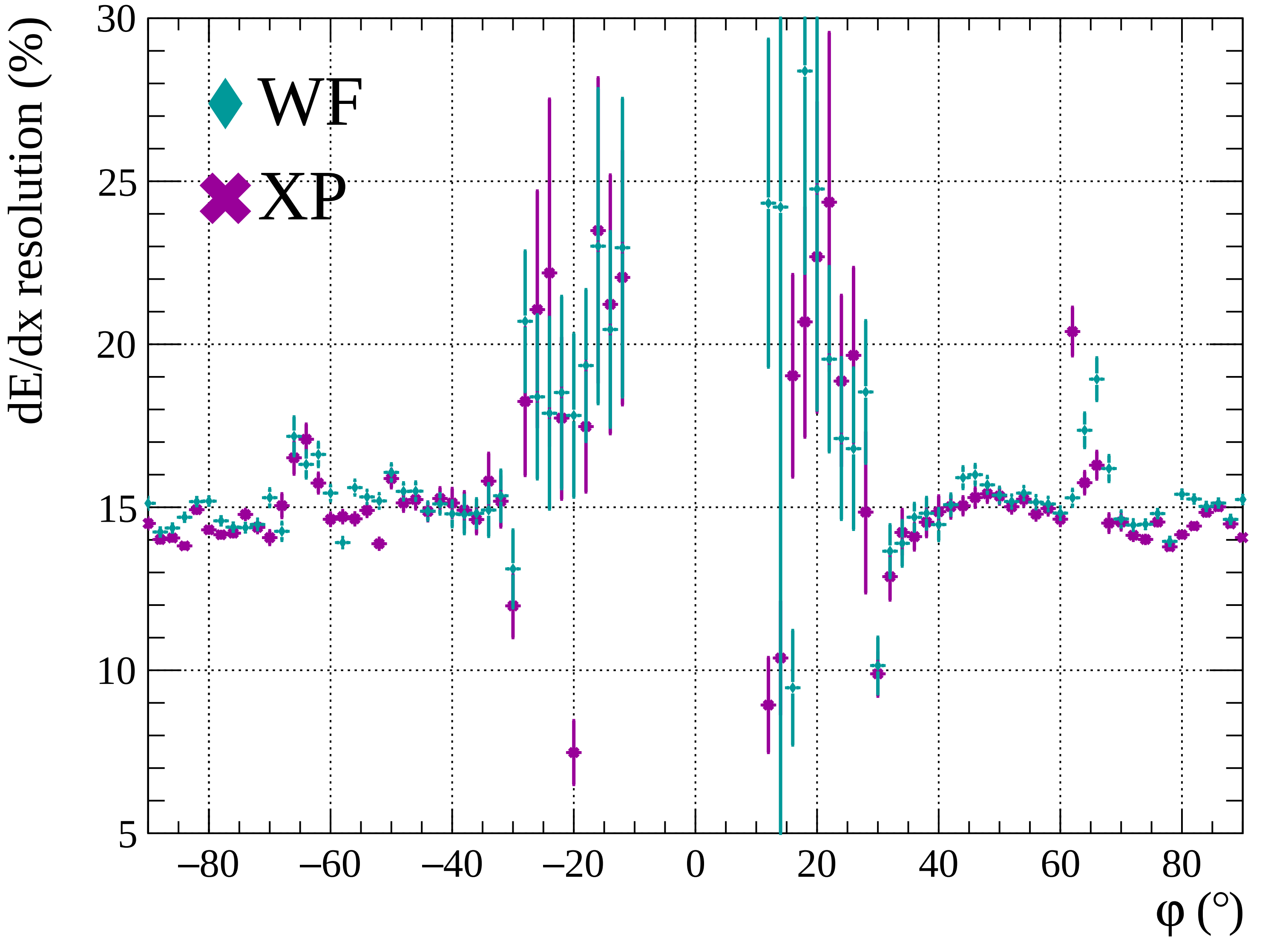

y = 0.15x + 445.27
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40


Energy loss (XP) vs drift time in ERAM 24
y = 0.19x + 495.88
Energy loss (XP) vs drift time in ERAM 30
y = 0.07x + 491.80
Energy loss (XP) vs drift time in ERAM 28
y = 0.15x + 501.38
Energy loss (XP) vs drift time in ERAM 19
y = 0.19x + 487.16
Energy loss (XP) vs drift time in ERAM 21
y = 0.08x + 461.32
Energy loss (XP) vs drift time in ERAM 13
y = 0.08x + 485.87
Energy loss (XP) vs drift time in ERAM 9
y = 0.10x + 493.22
Energy loss (XP) vs drift time in ERAM 2
y = 0.09x + 489.85
Energy loss (XP) vs drift time in ERAM 26
y = 0.21x + 470.95
Energy loss (XP) vs drift time in ERAM 17
y = 0.27x + 471.90
Energy loss (XP) vs drift time in ERAM 23
y = 0.25x + 464.21
Energy loss (XP) vs drift time in ERAM 29
y = 0.22x + 475.54
Energy loss (XP) vs drift time in ERAM 1
y = 0.16x + 467.11
Energy loss (XP) vs drift time in ERAM 10
y = 0.18x + 467.56
Energy loss (XP) vs drift time in ERAM 11
y = 0.17x + 466.80
Energy loss (XP) vs drift time in ERAM 3
y = 0.15x + 458.96
dE/dx (ADC counts/cm)
drift time (timebins)

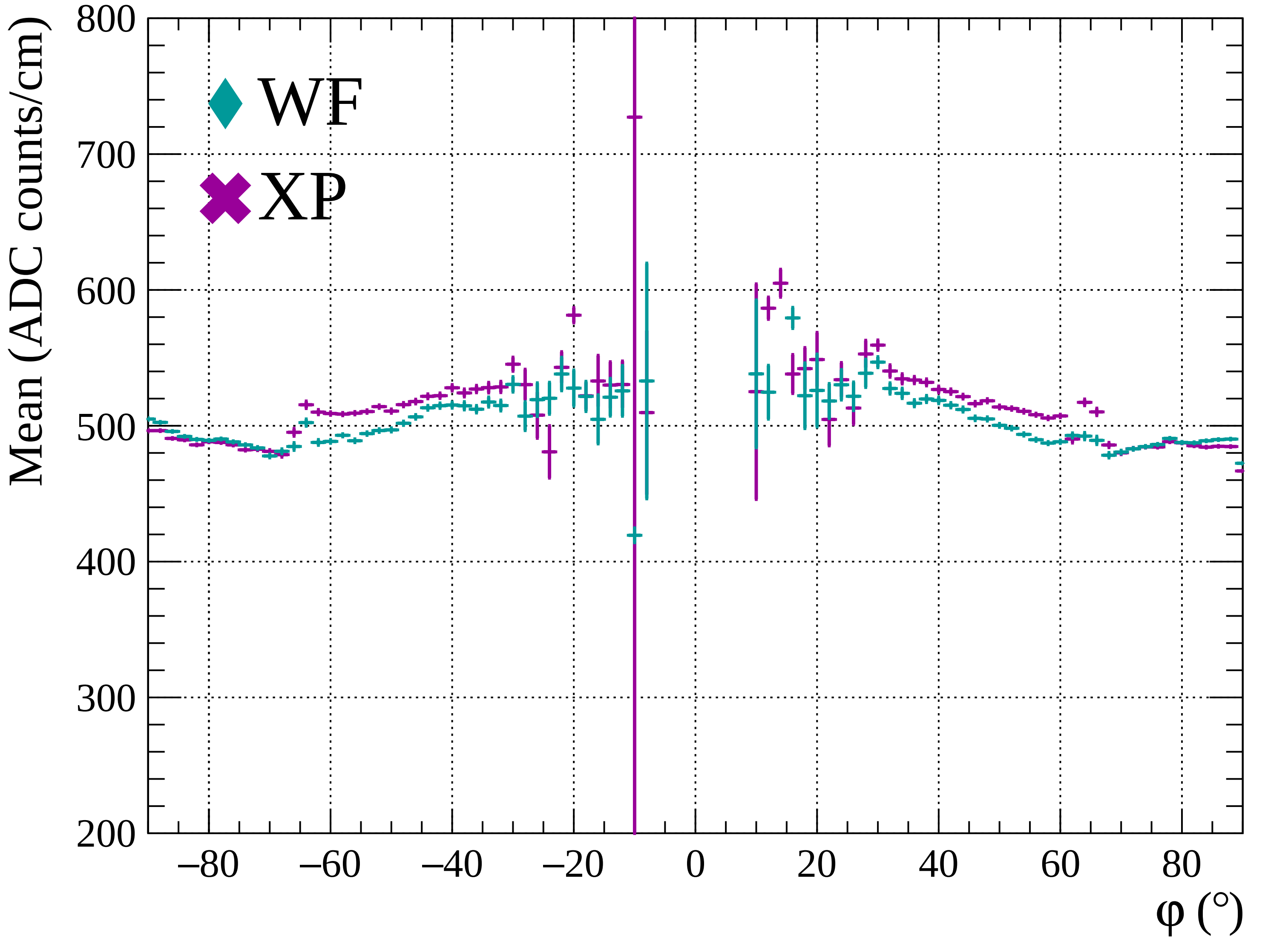


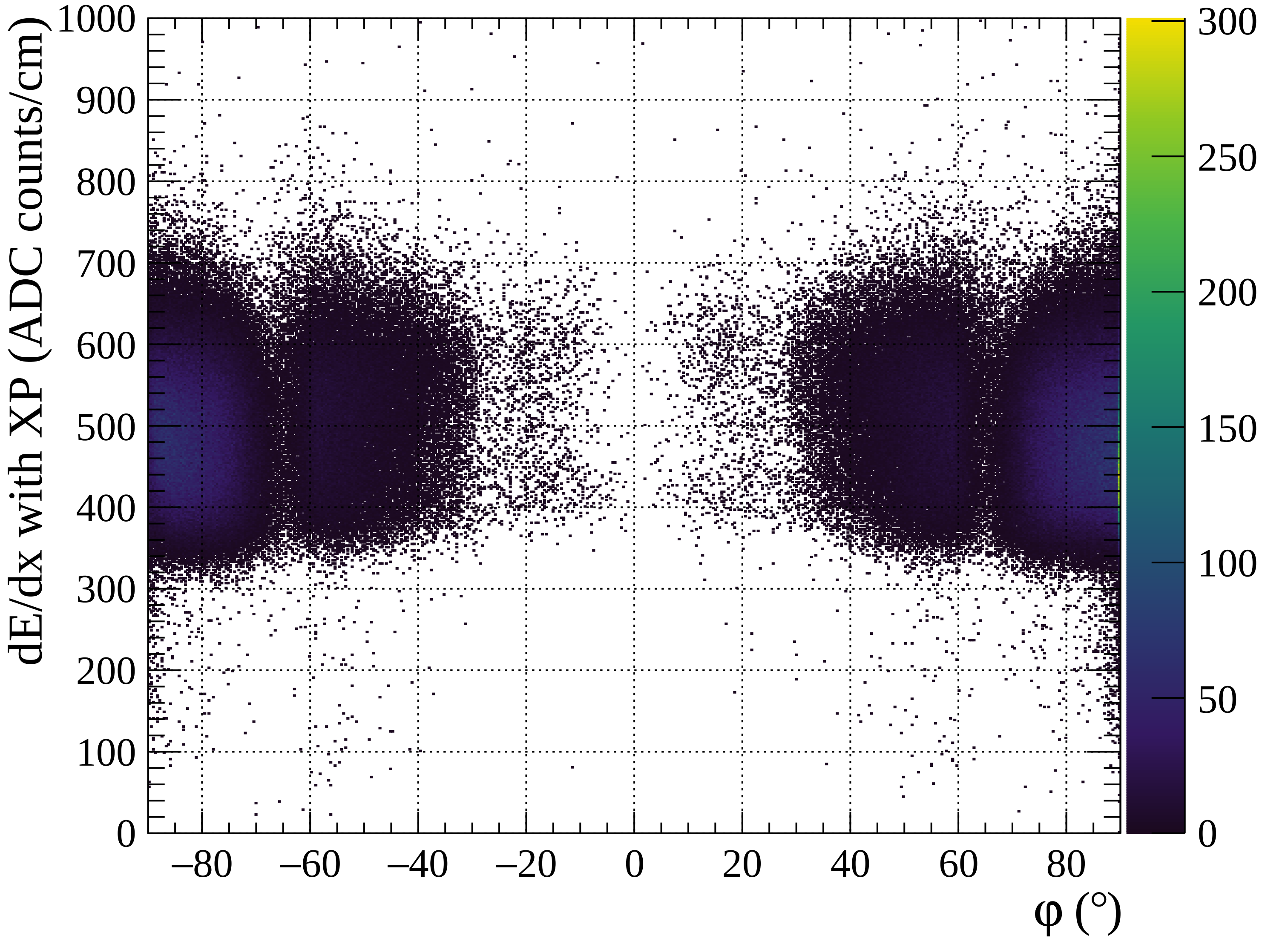
Energy loss (XP) vs drift time in ERAM 47
y = -0.05x + 529.42
Energy loss (XP) vs drift time in ERAM 16
y = 0.12x + 474.32
Energy loss (XP) vs drift time in ERAM 14
y = 0.13x + 491.24
Energy loss (XP) vs drift time in ERAM 15
y = 0.20x + 497.47
Energy loss (XP) vs drift time in ERAM 42
y = 0.09x + 446.64
Energy loss (XP) vs drift time in ERAM 45
y = 0.08x + 464.10
Energy loss (XP) vs drift time in ERAM 37
y = 0.10x + 481.21
Energy loss (XP) vs drift time in ERAM 36
y = 0.10x + 479.54
Energy loss (XP) vs drift time in ERAM 20
y = 0.21x + 453.82
Energy loss (XP) vs drift time in ERAM 38
y = 0.14x + 478.54
Energy loss (XP) vs drift time in ERAM 7
y = 0.20x + 469.36
Energy loss (XP) vs drift time in ERAM 44
y = 0.14x + 488.20
Energy loss (XP) vs drift time in ERAM 43
y = 0.15x + 463.26
Energy loss (XP) vs drift time in ERAM 39
y = 0.15x + 443.68
Energy loss (XP) vs drift time in ERAM 41
y = 0.15x + 463.69
Energy loss (XP) vs drift time in ERAM 46
y = 0.13x + 447.24
dE/dx (ADC counts/cm)
drift time (timebins)


time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
500
400
300
200
100
0

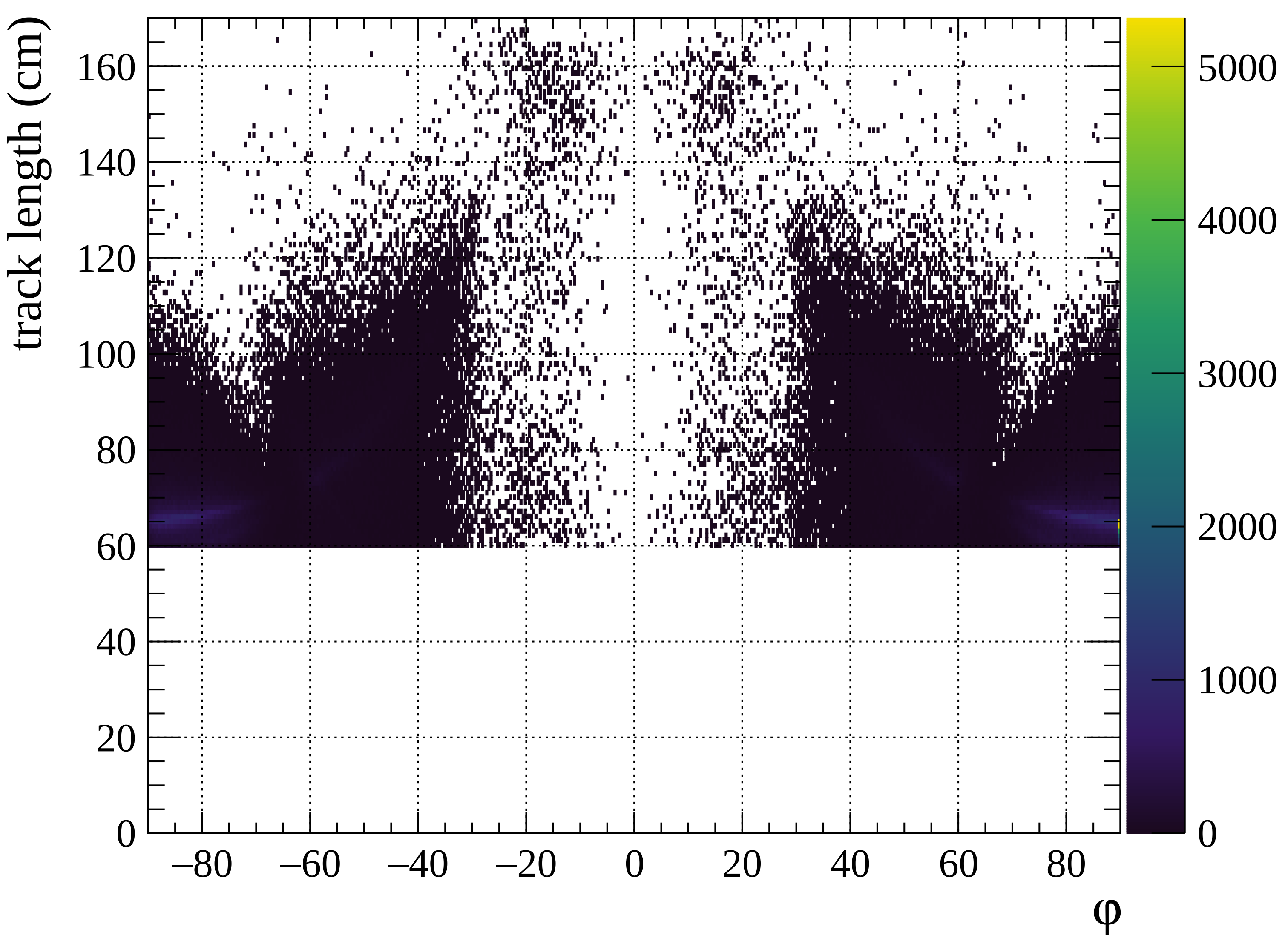


WF
XP
dE/dx resolution (%)
φ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

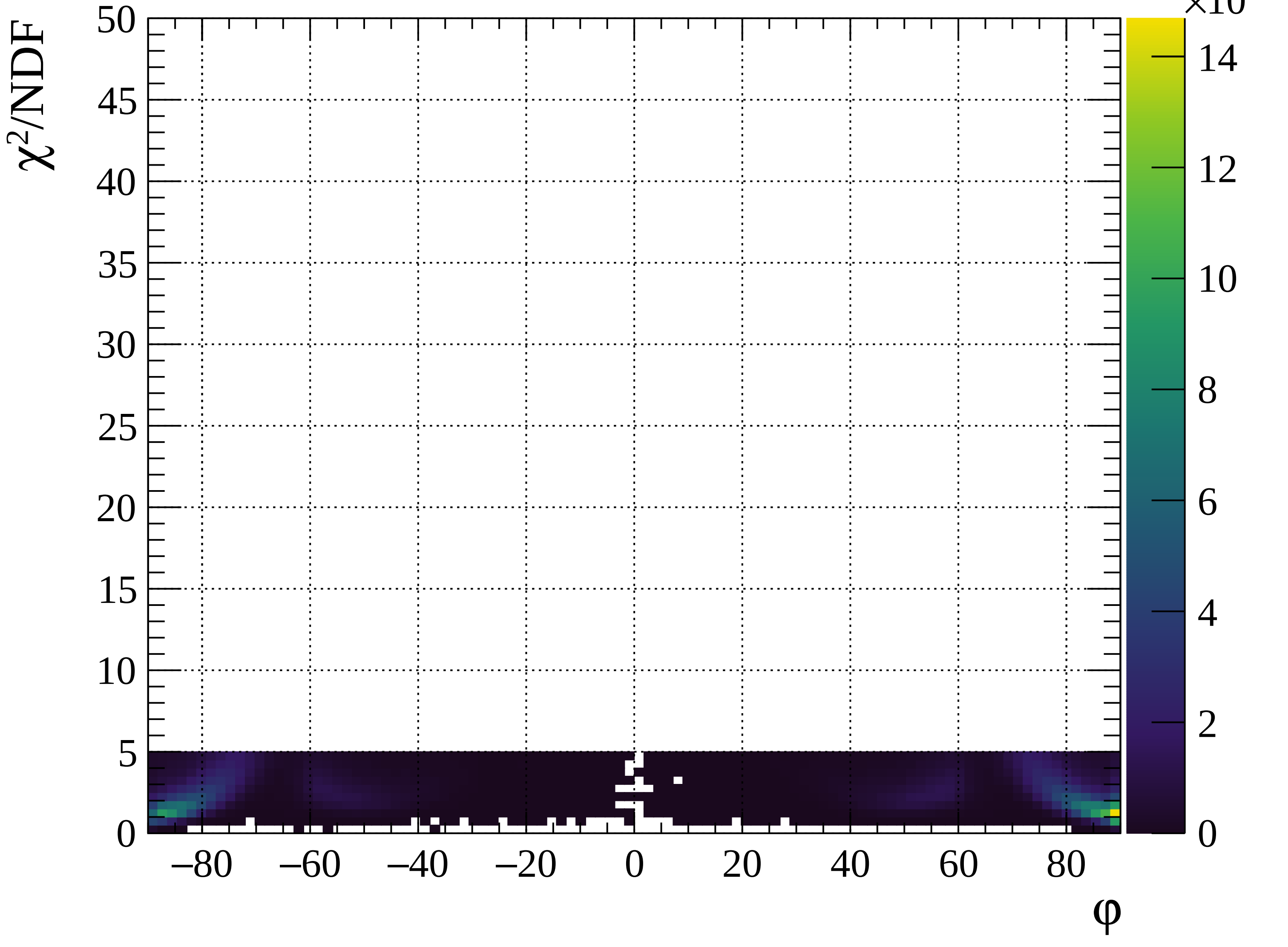

WF
XP
Mean (ADC counts/cm)
φ (°)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80



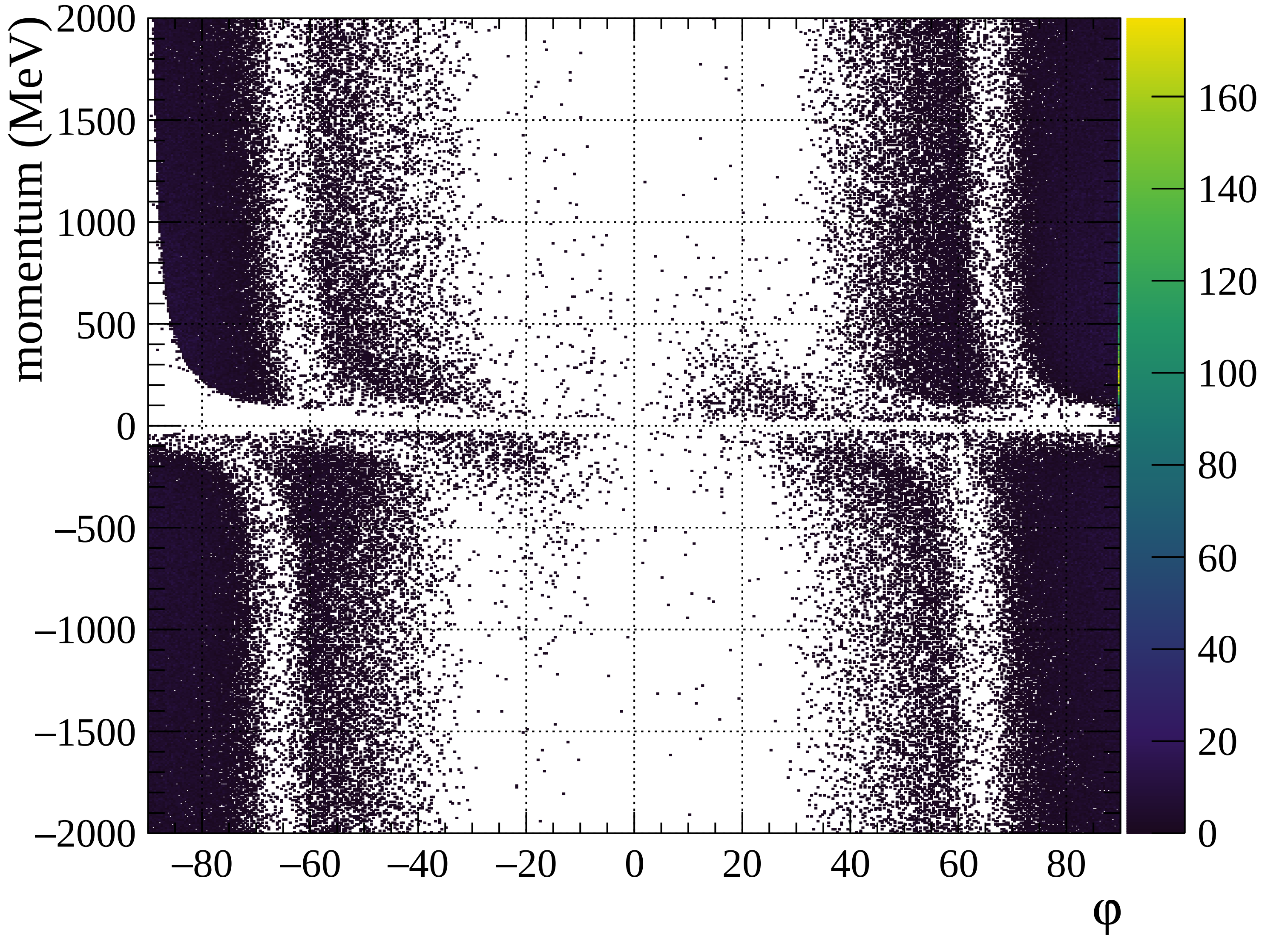
dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250
300



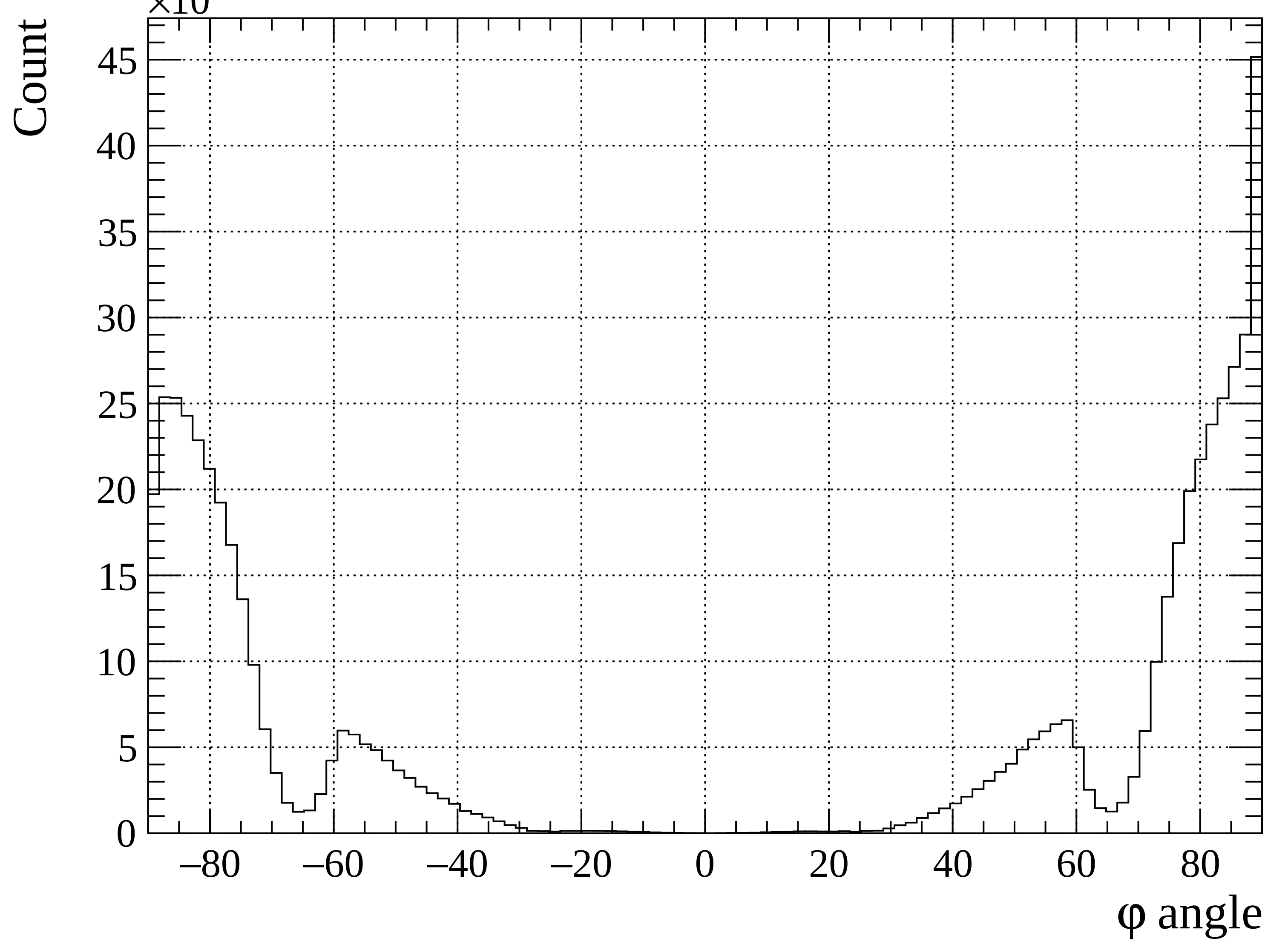
track length (cm)
φ



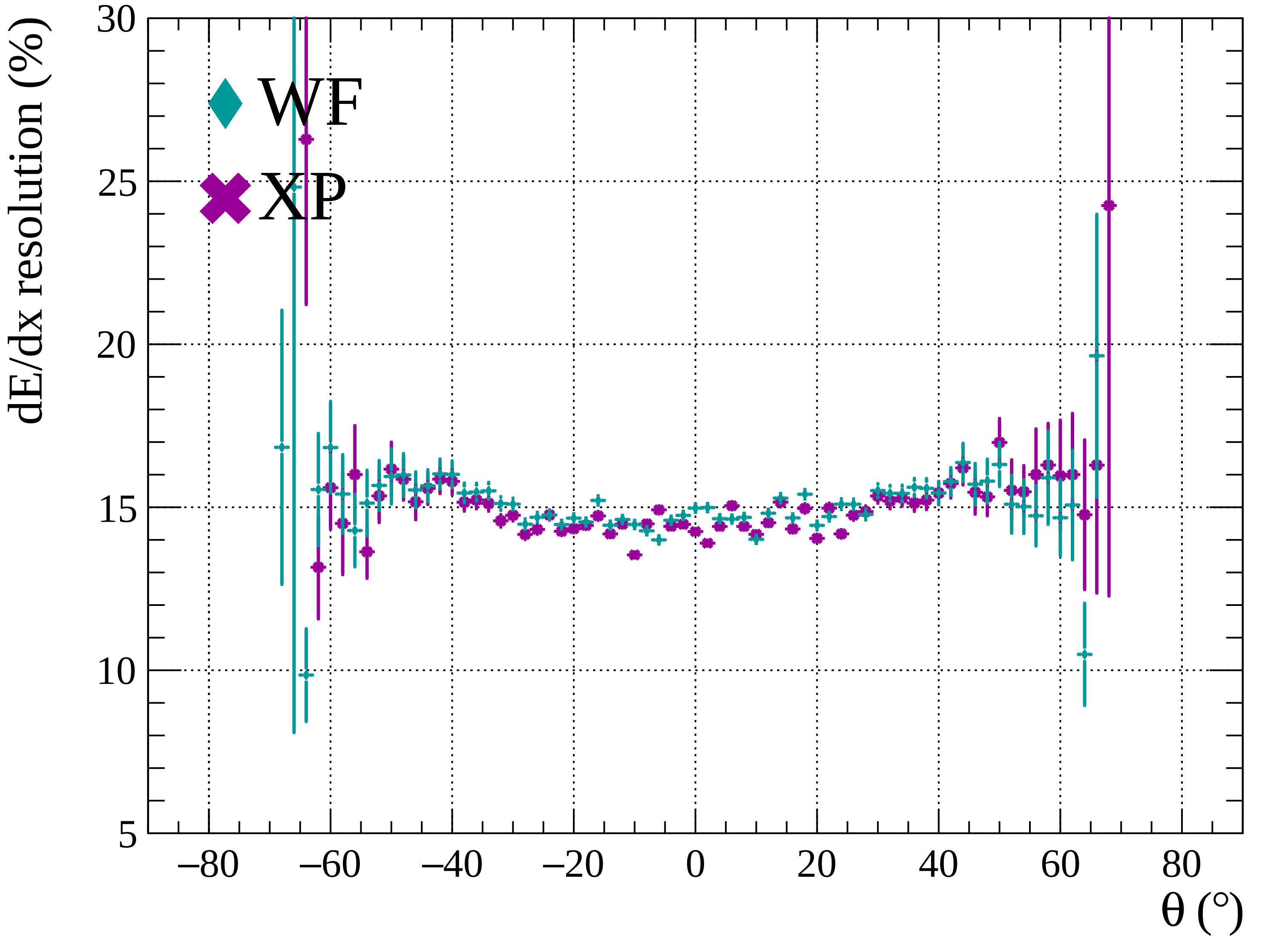
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
×10
14
12
10
8
6
4
2
0



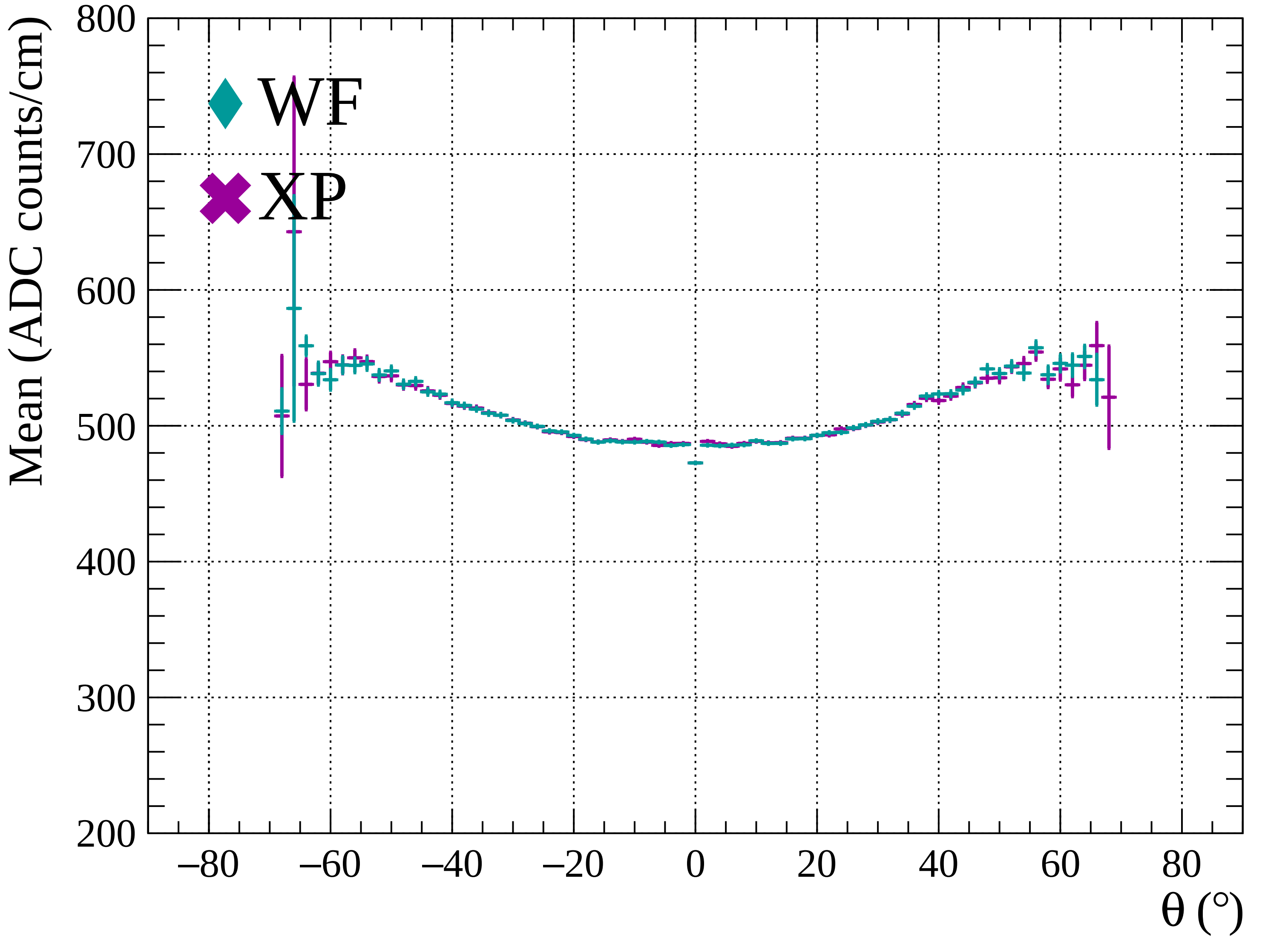
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
160
140
120
100
80
60
40
20
0



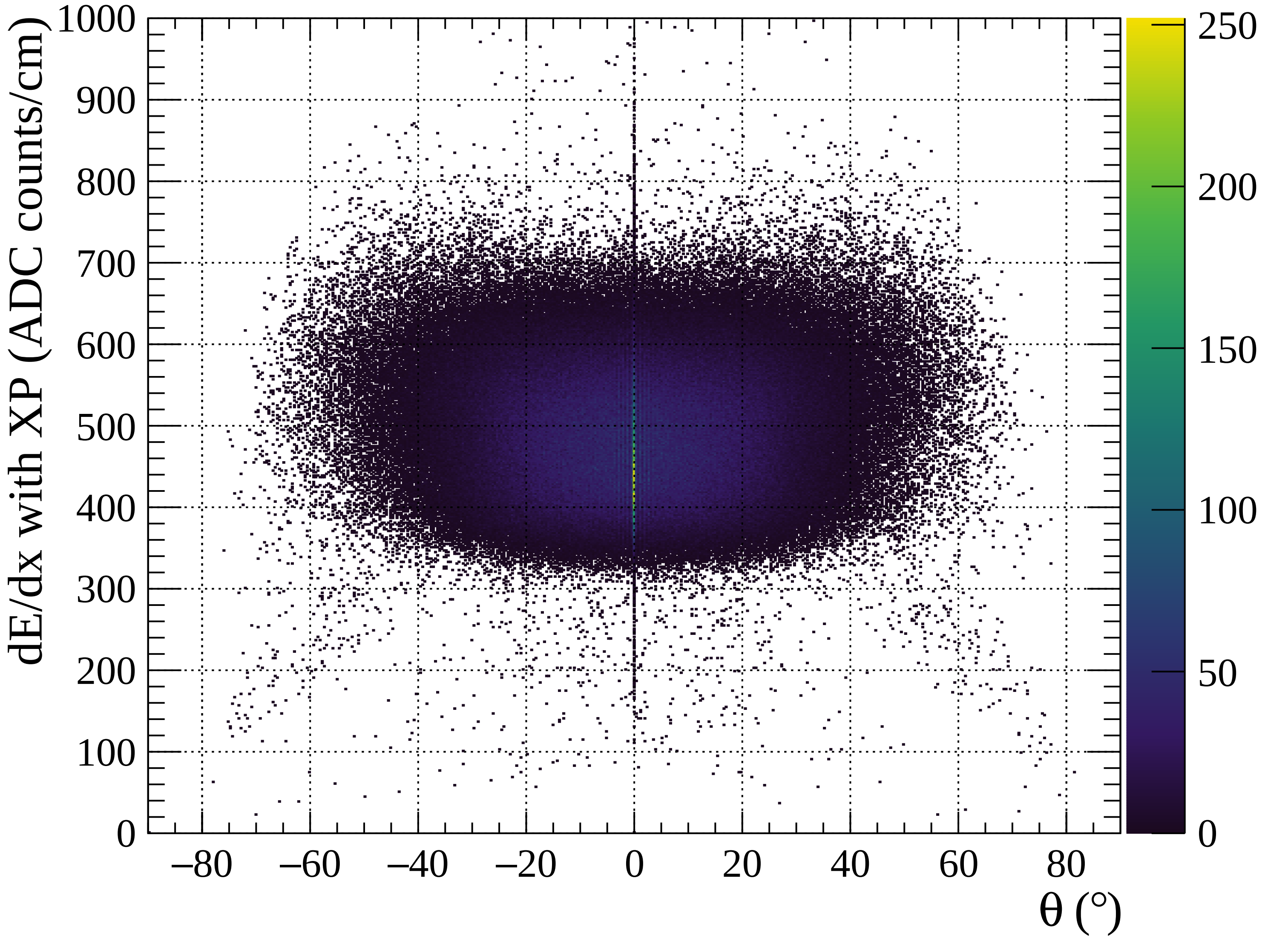
×10
Count
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

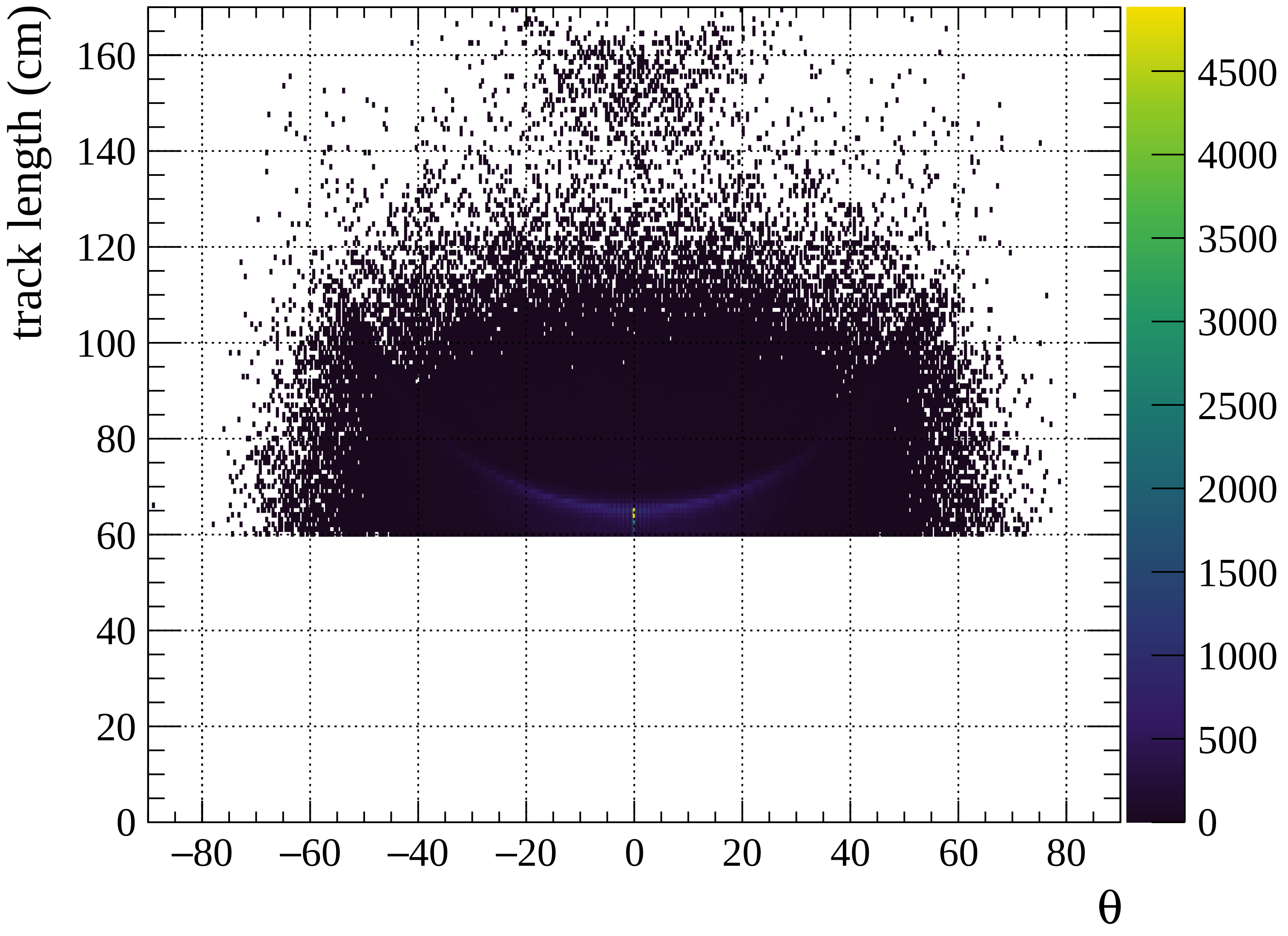


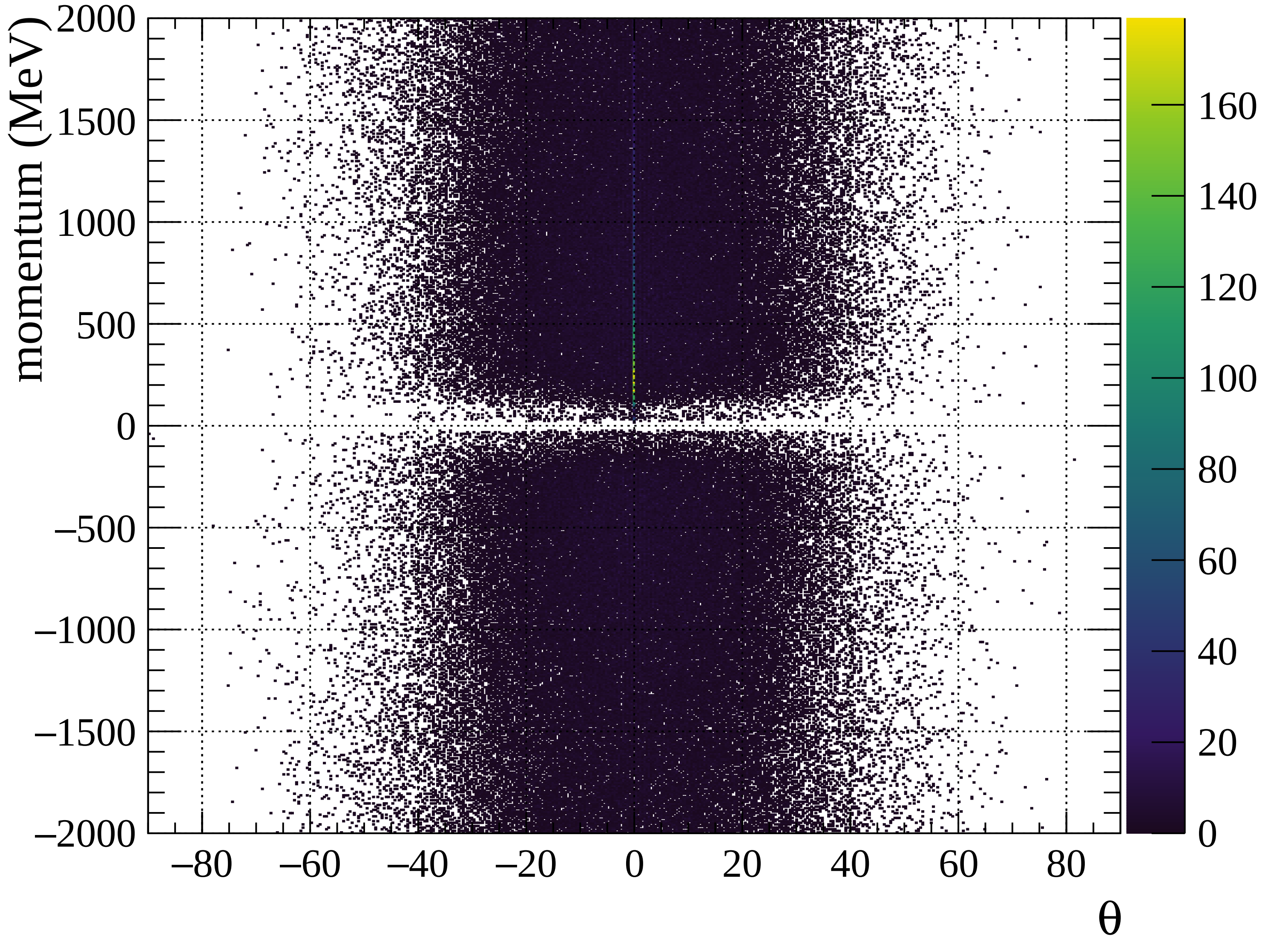
WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

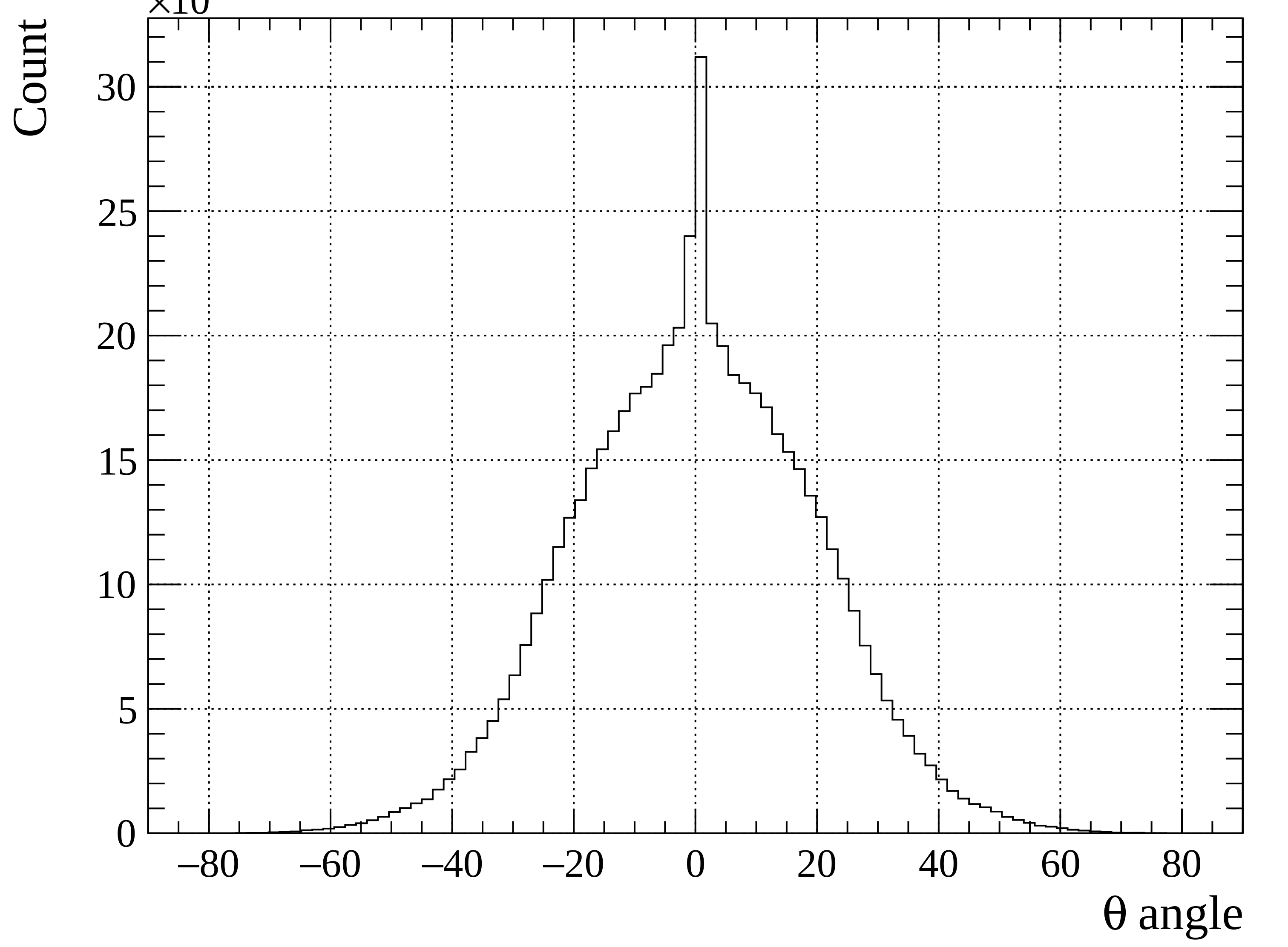


WF
XP
Mean (ADC counts/cm)
θ (°)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80

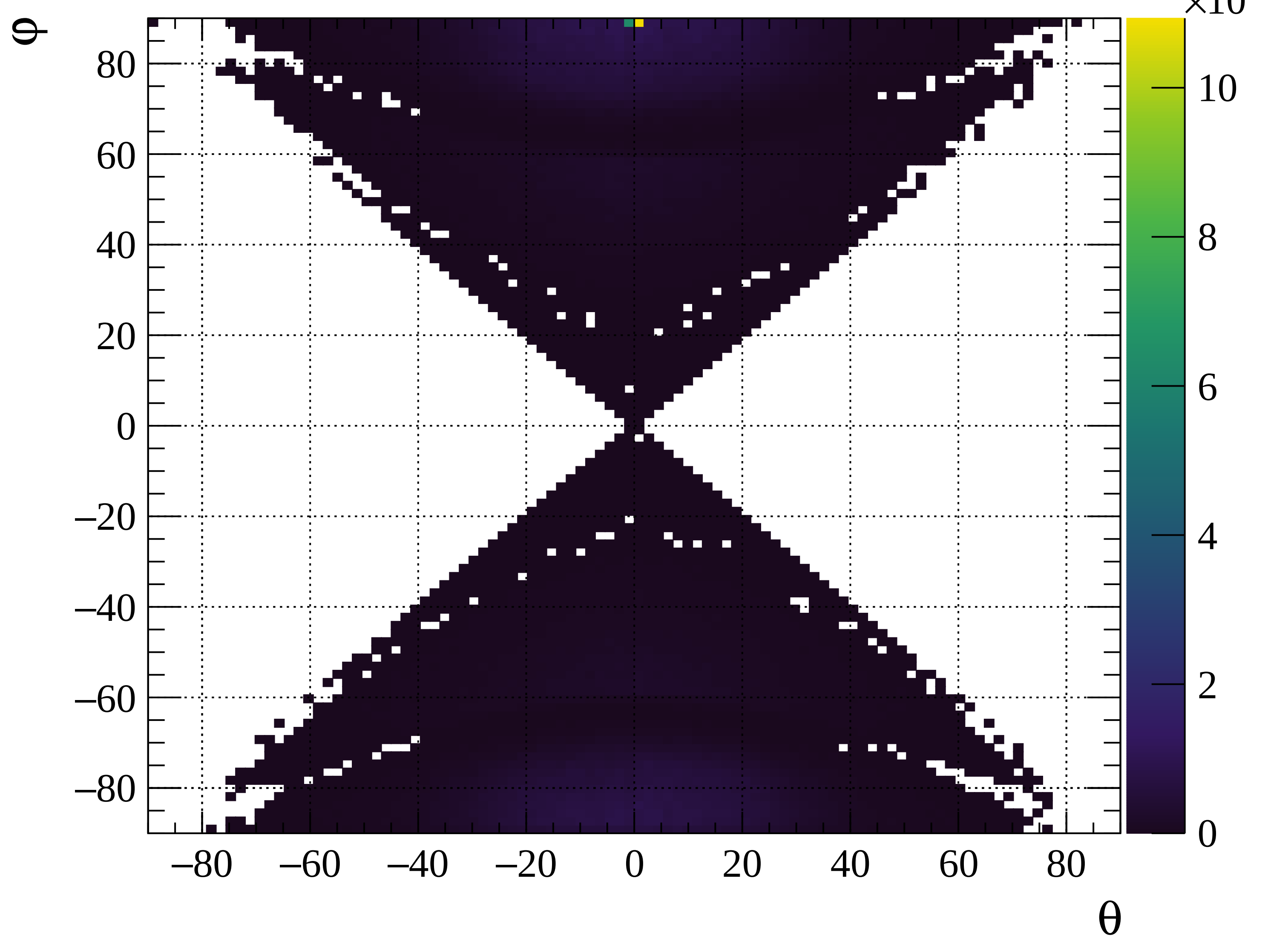

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
250
200
150
100
50
0

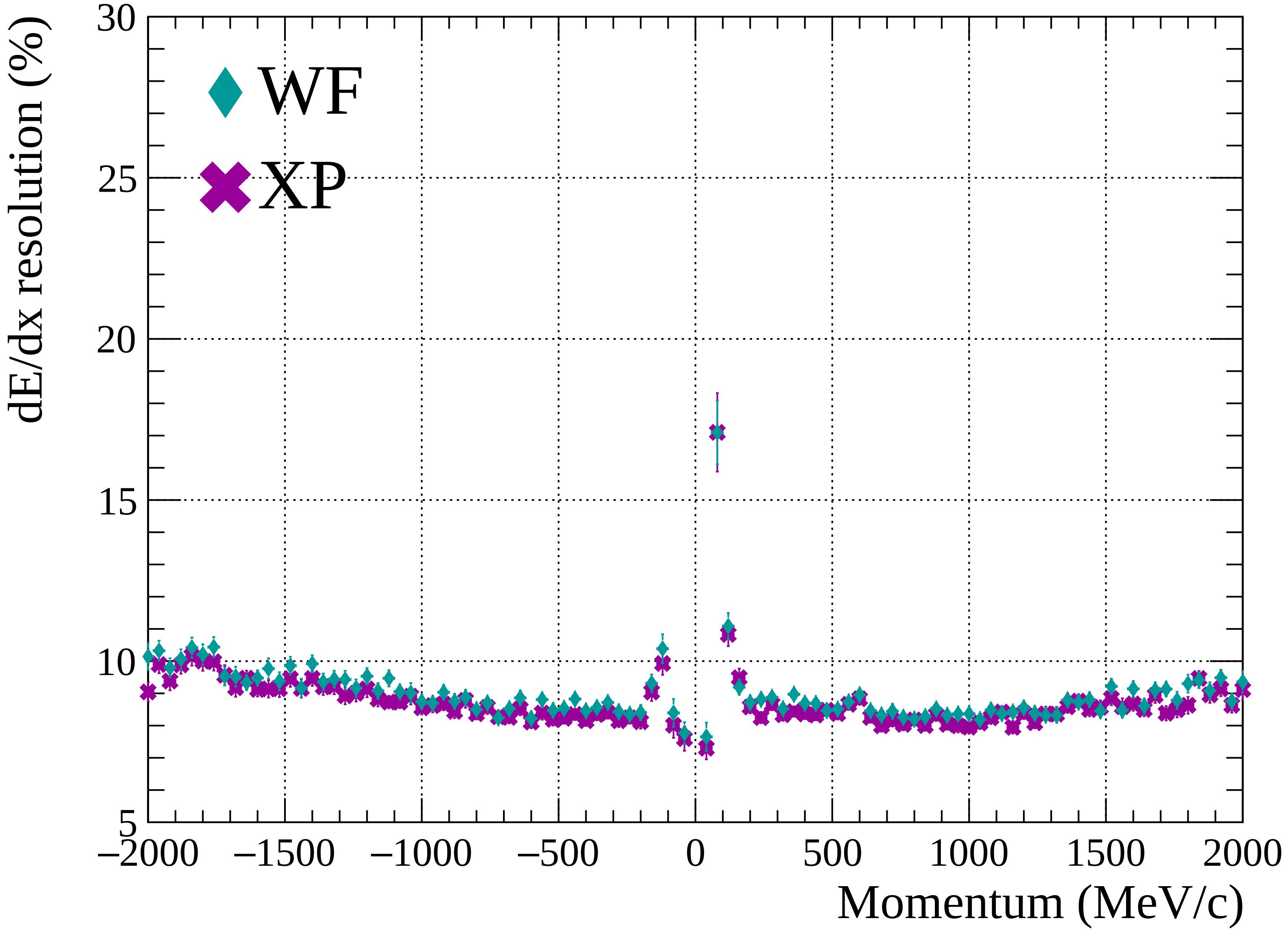

track length (cm)
θ
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
4500
4000
3500
3000
2500
2000
1500
1000
500
0

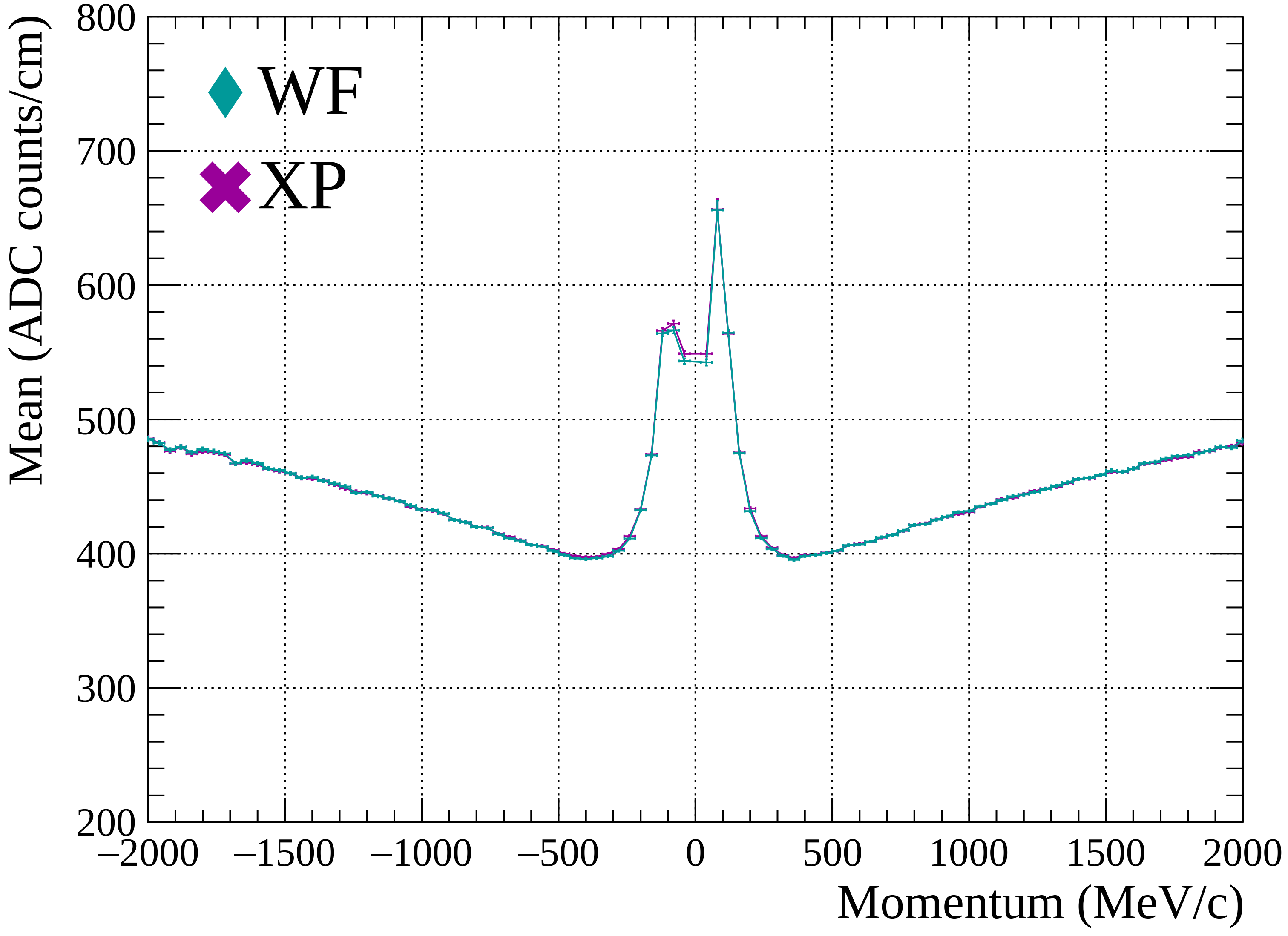

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
160
140
120
100
80
60
40
20
0

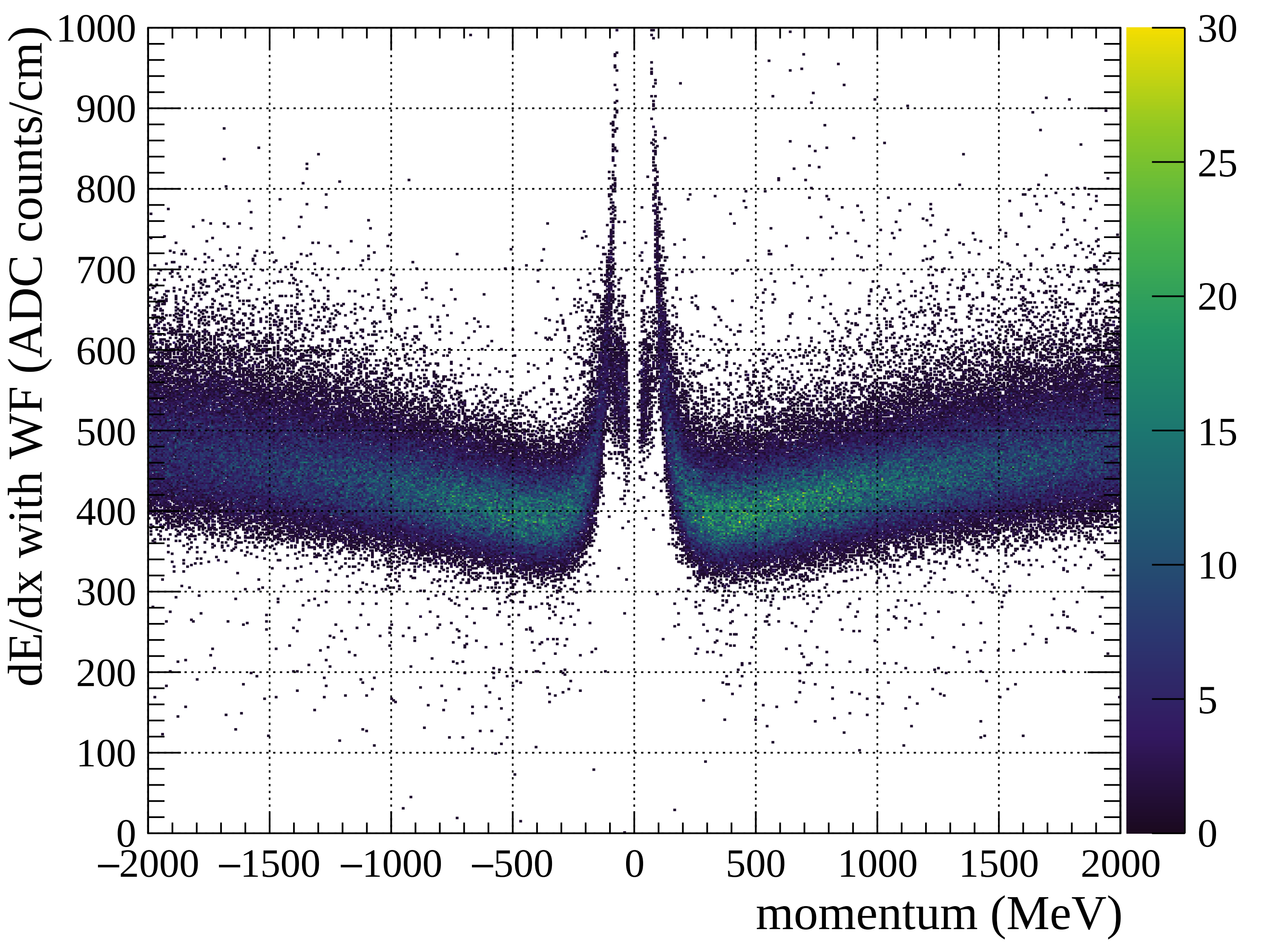


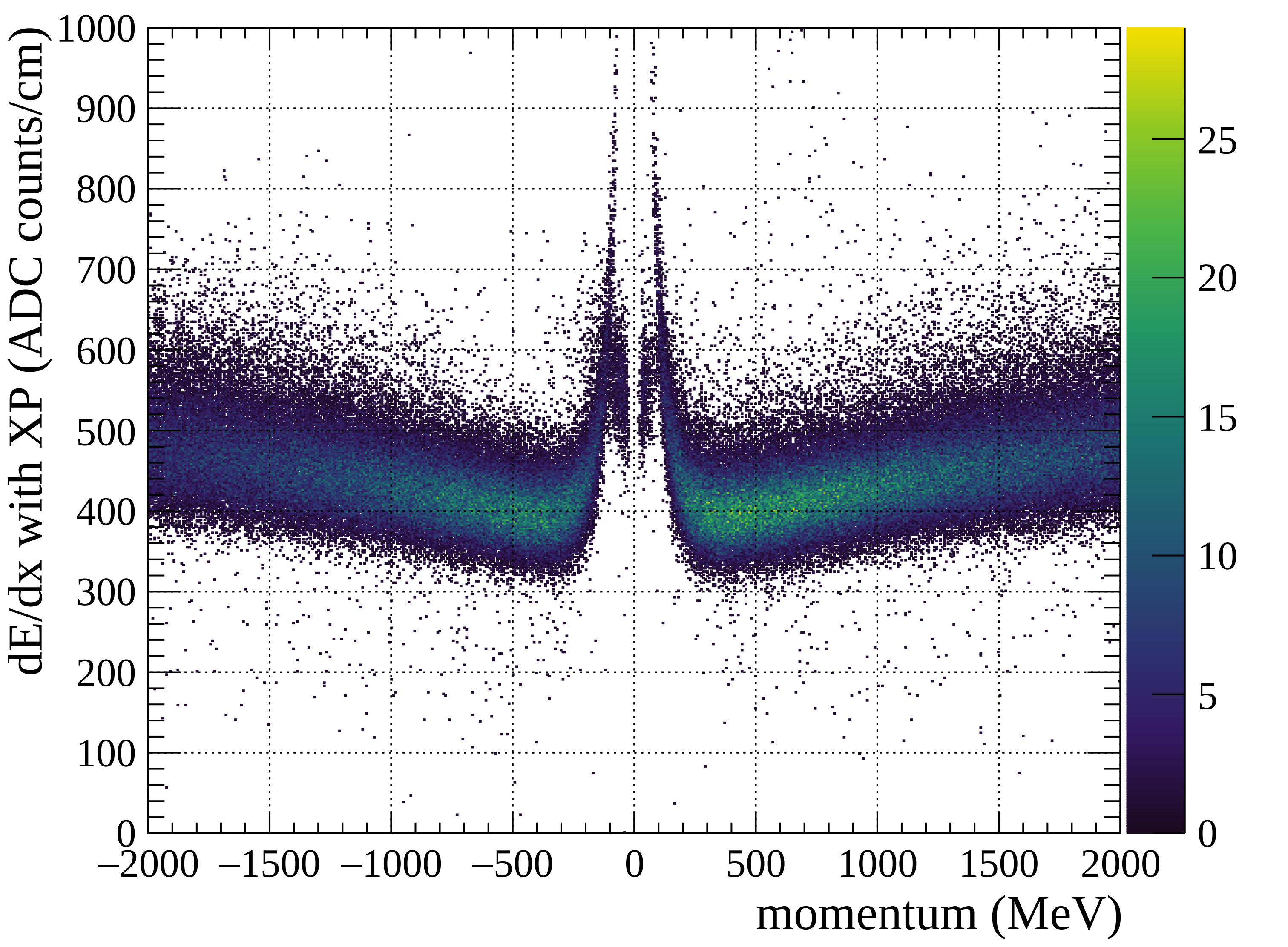
×10
Count
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ angle


φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
×10
10
8
6
4
2
0

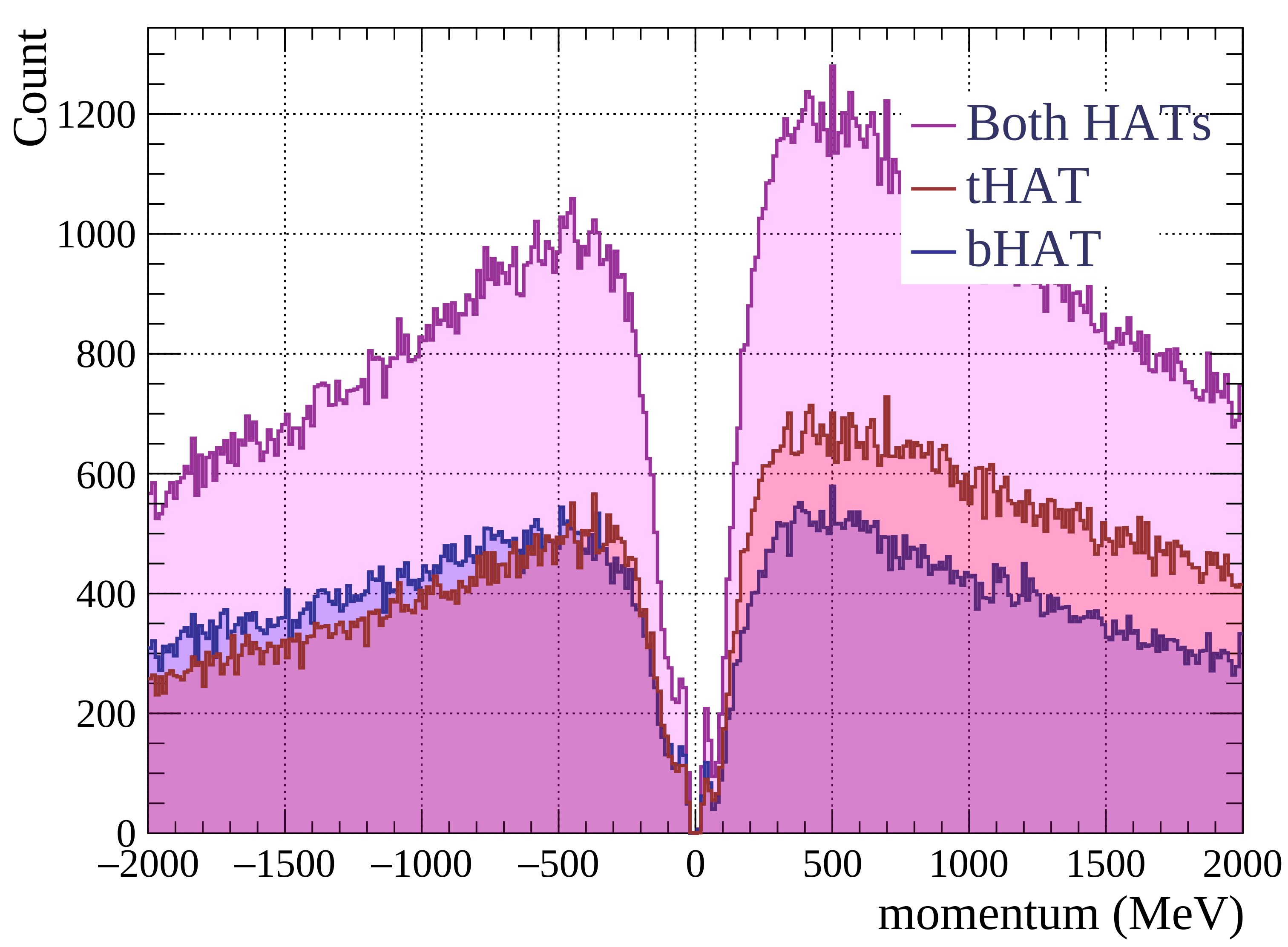

dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−2000
−1500
−1000
−500
0
500
1000
1500
2000

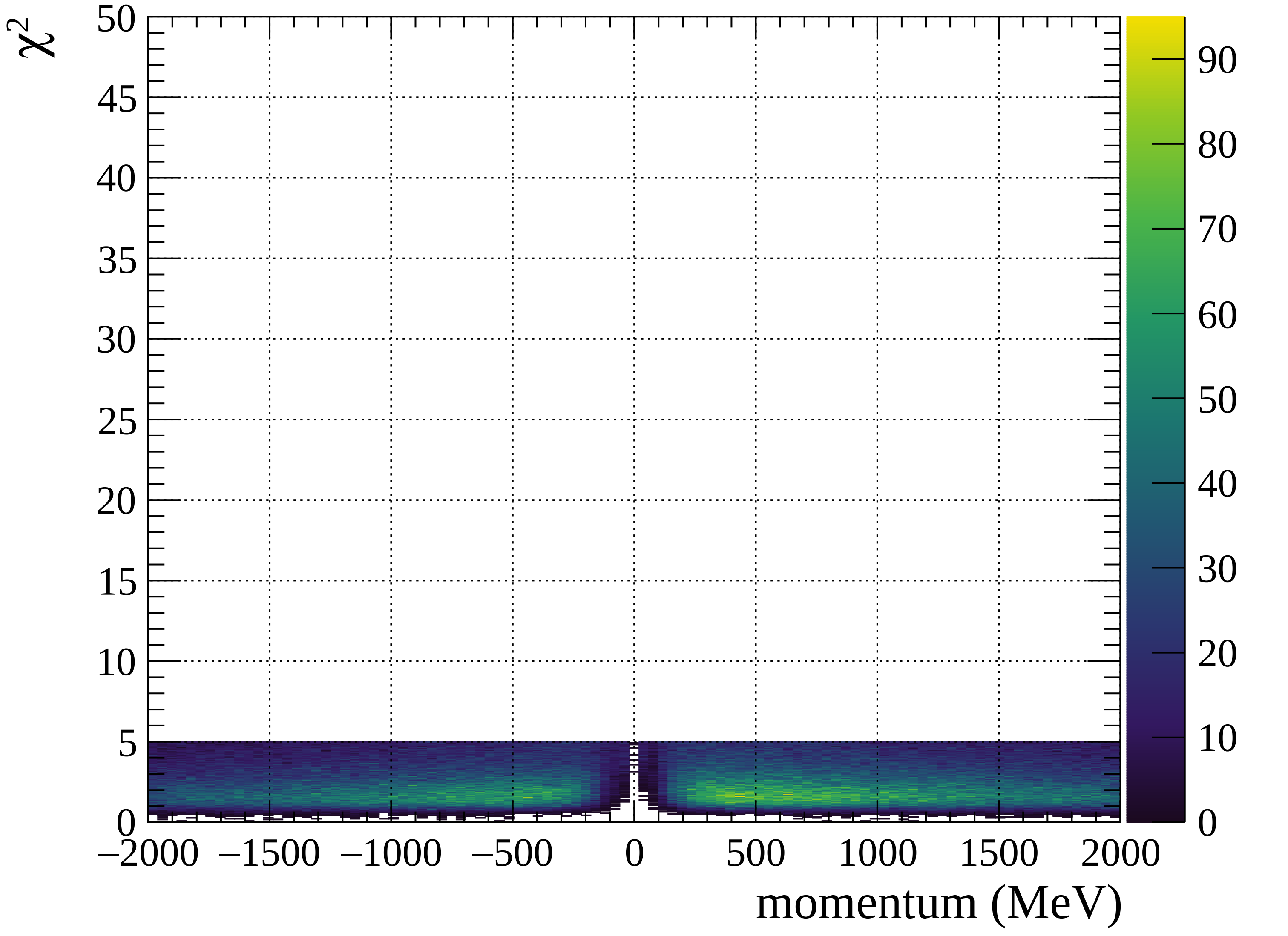

dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
5
10
15
20
25
30



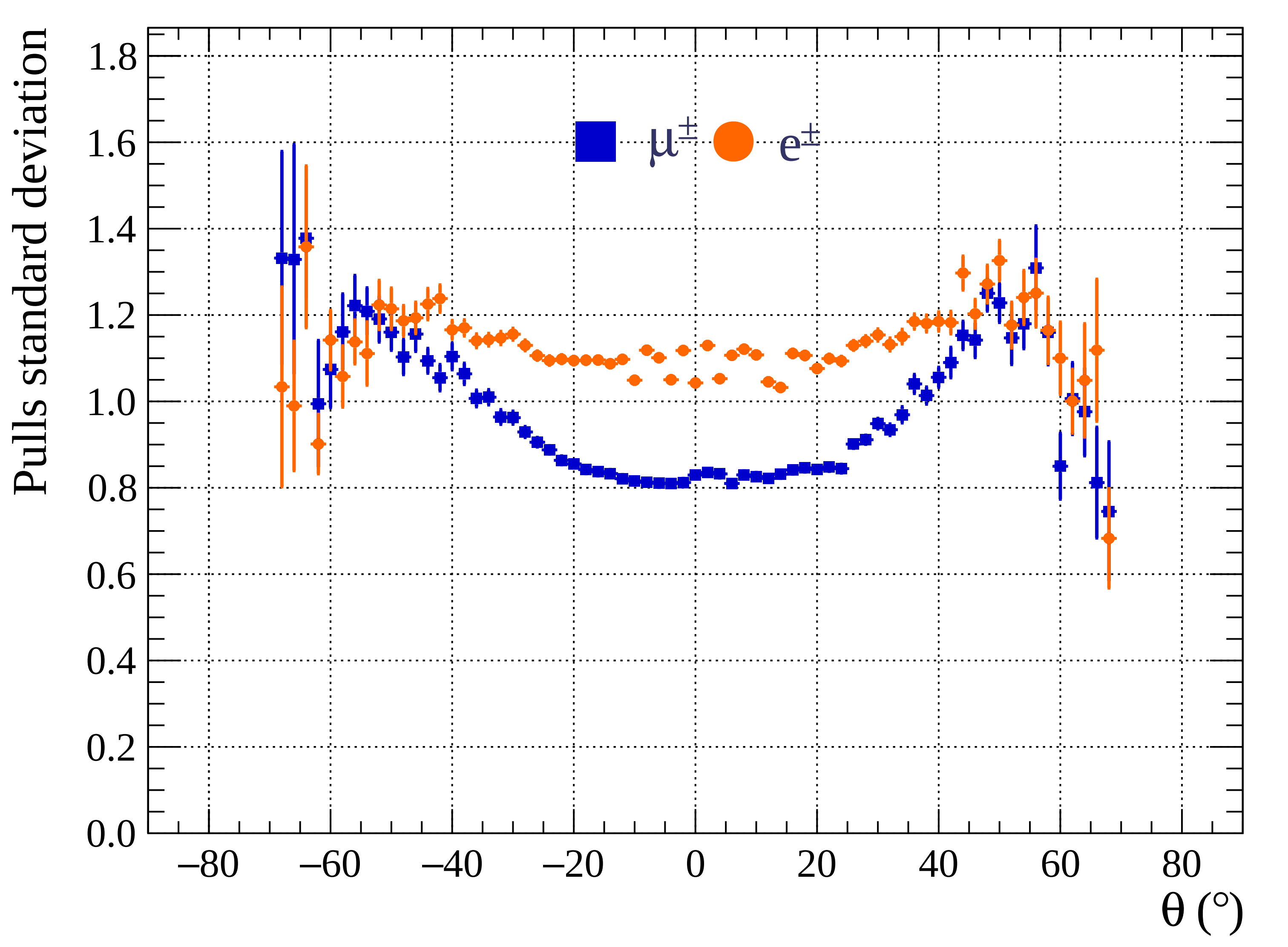
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
25
20
15
10
5
0



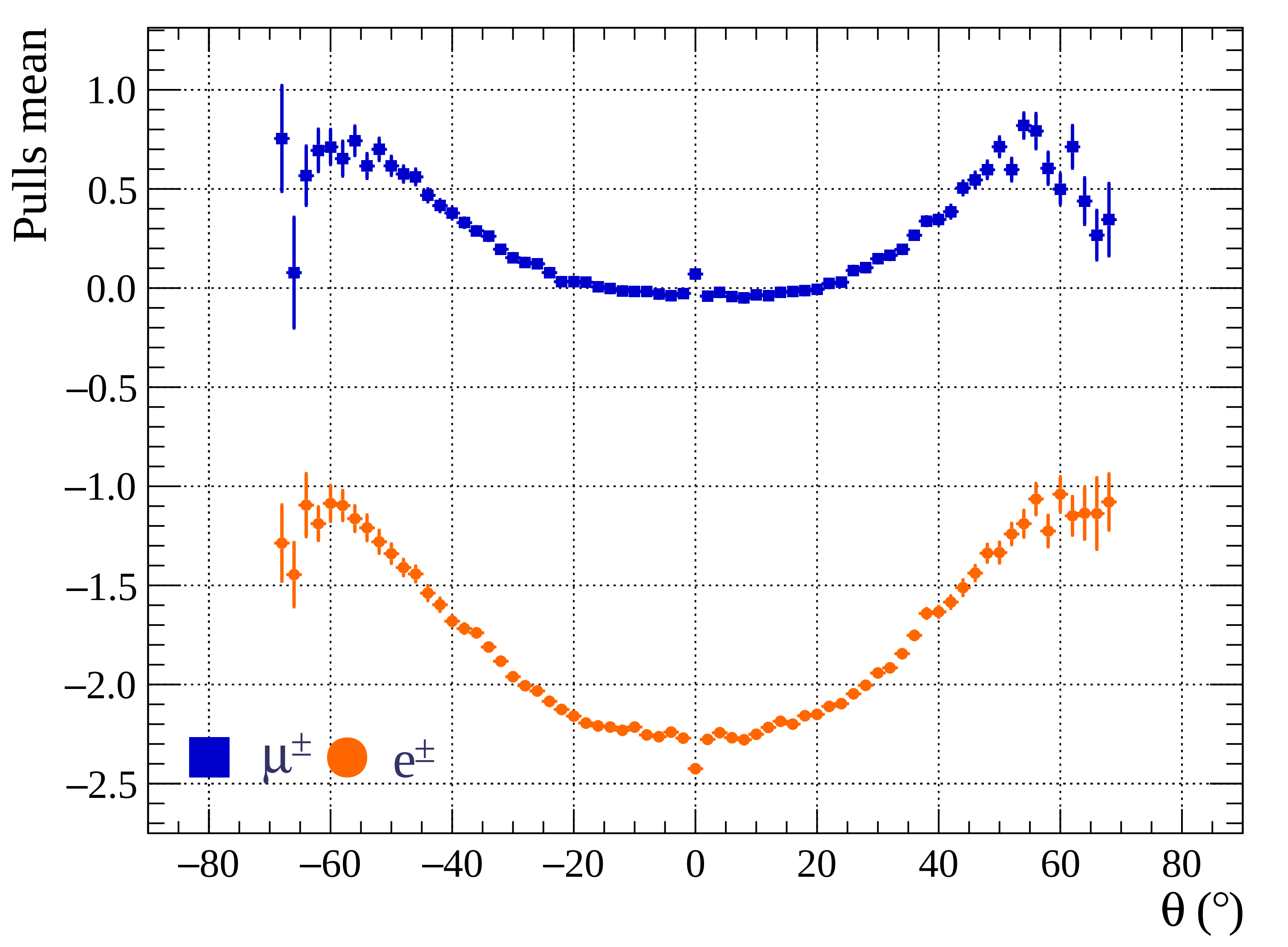
Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
200
400
600
800
1000
1200
−2000
−1500
−1000
−500
0
500
1000
1500
2000



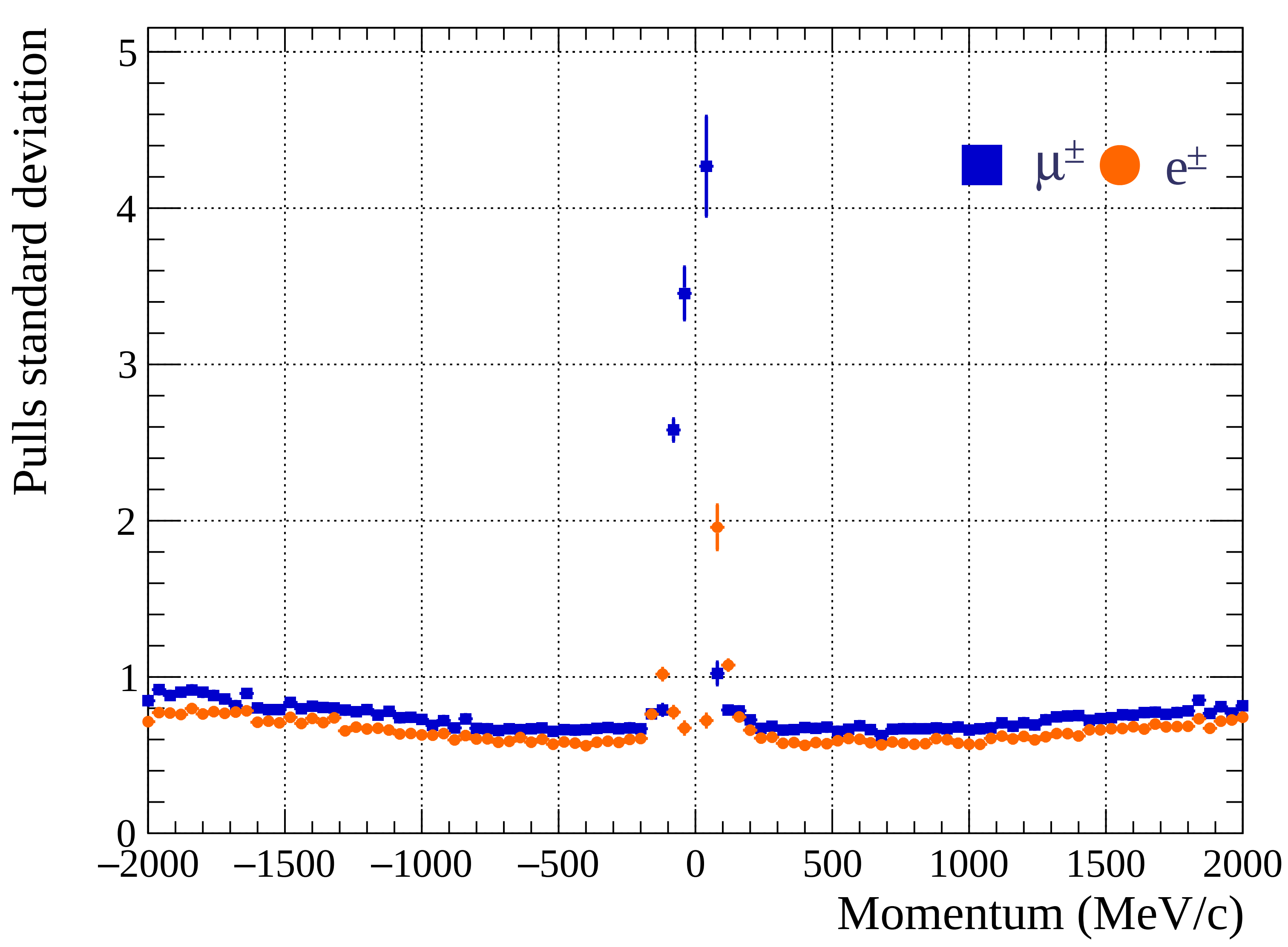
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
90
80
70
60
50
40
30
20
10
0



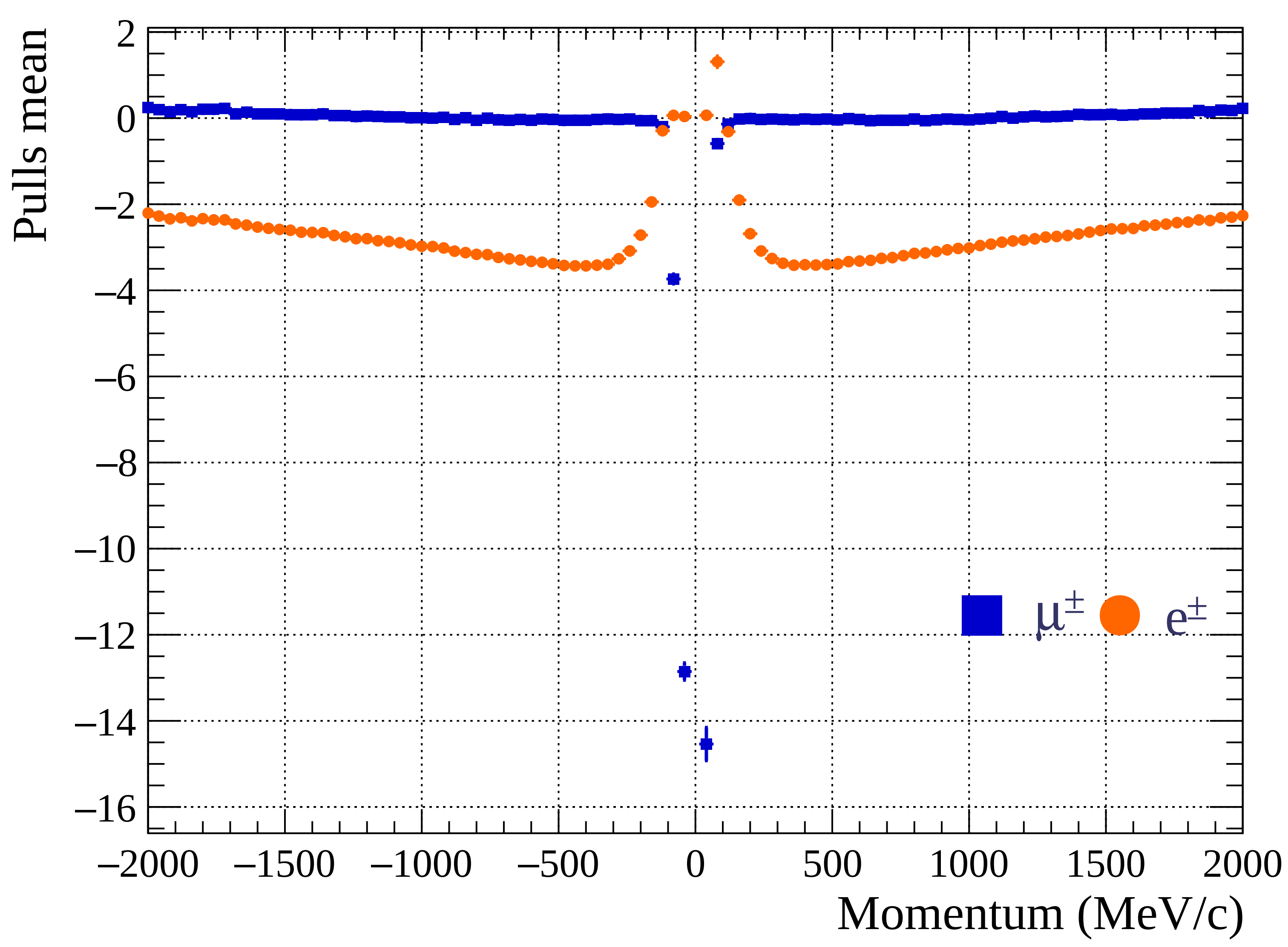
Pulls standard deviation
μ±
e±
θ (°)
-80
-60
-40
-20
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8

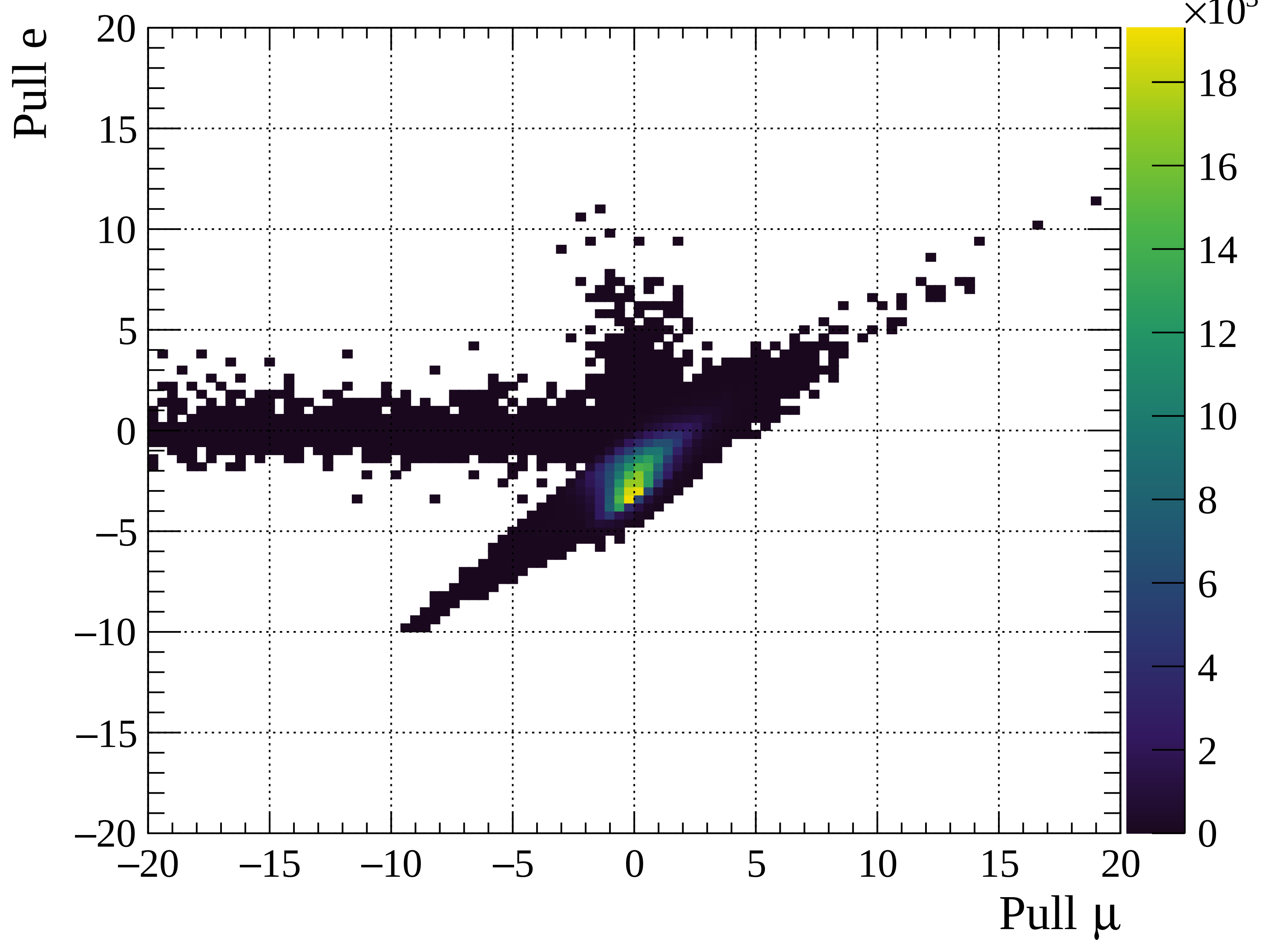


Pulls mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

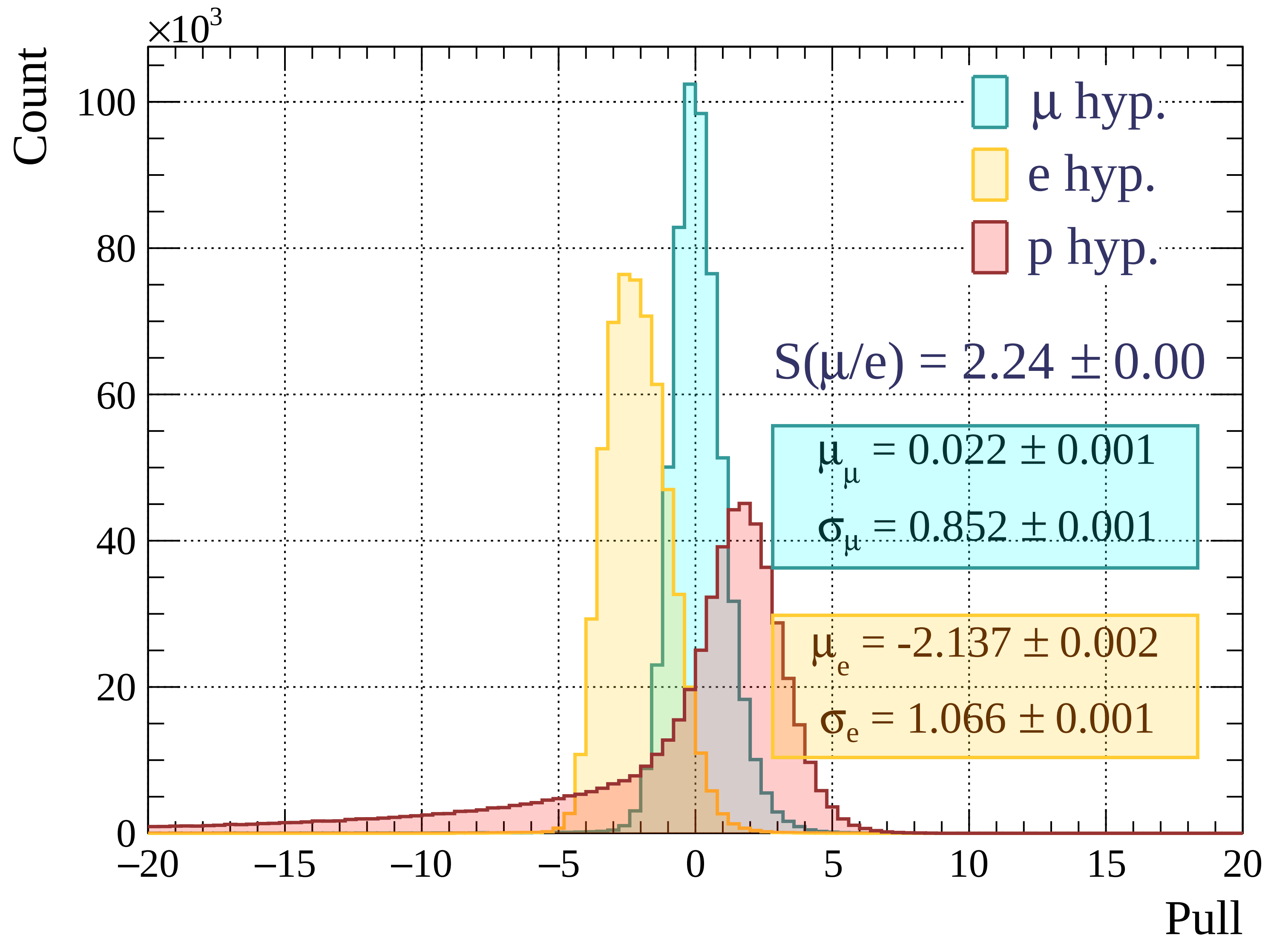


Pulls standard deviation
Momentum (MeV/c)
μ±
e±
0
1
2
3
4
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

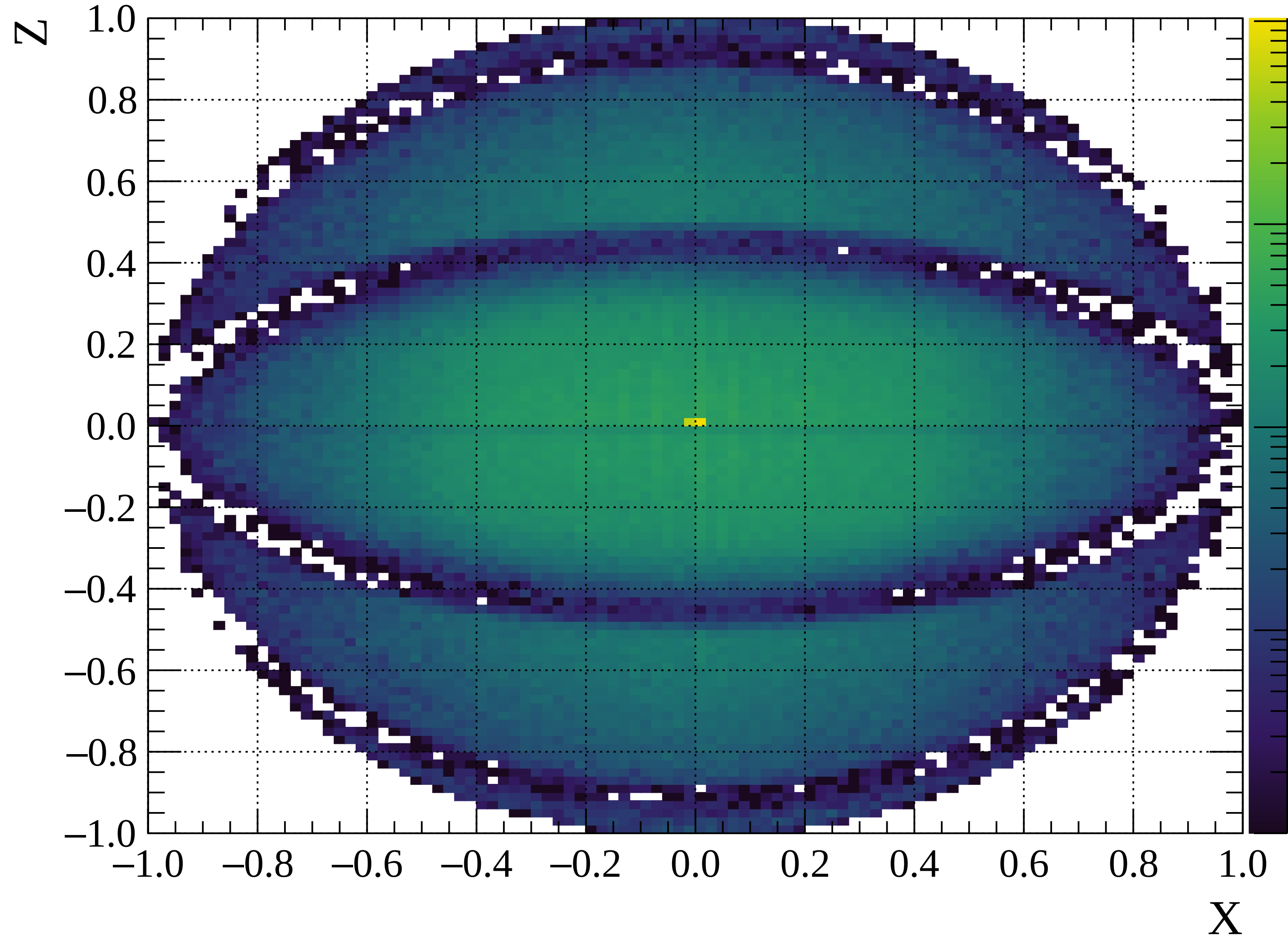

Pulls mean
2
0
−2
−4
−6
−8
−10
−12
−14
−16
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

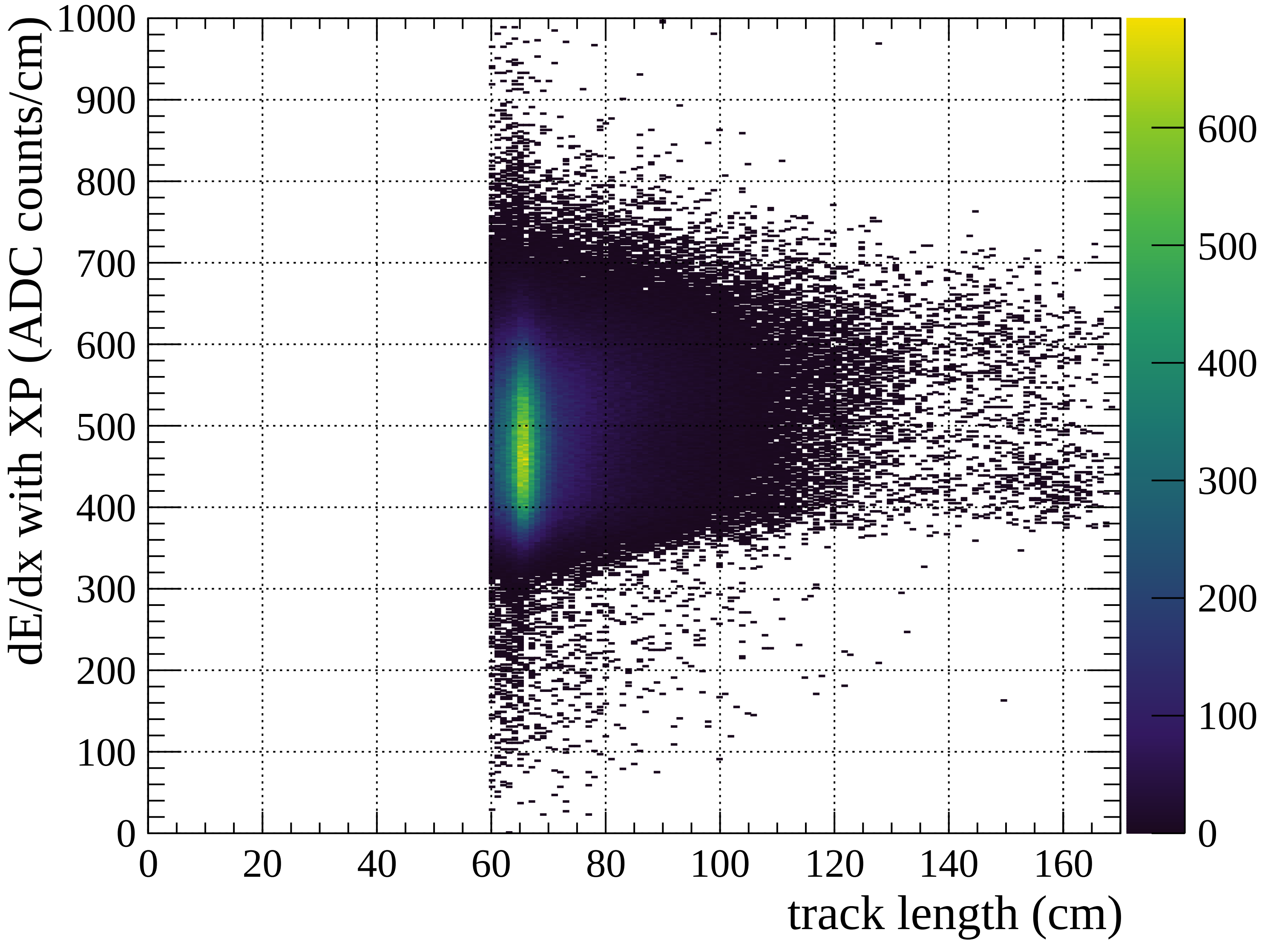


Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
18
16
14
12
10
8
6
4
2
0

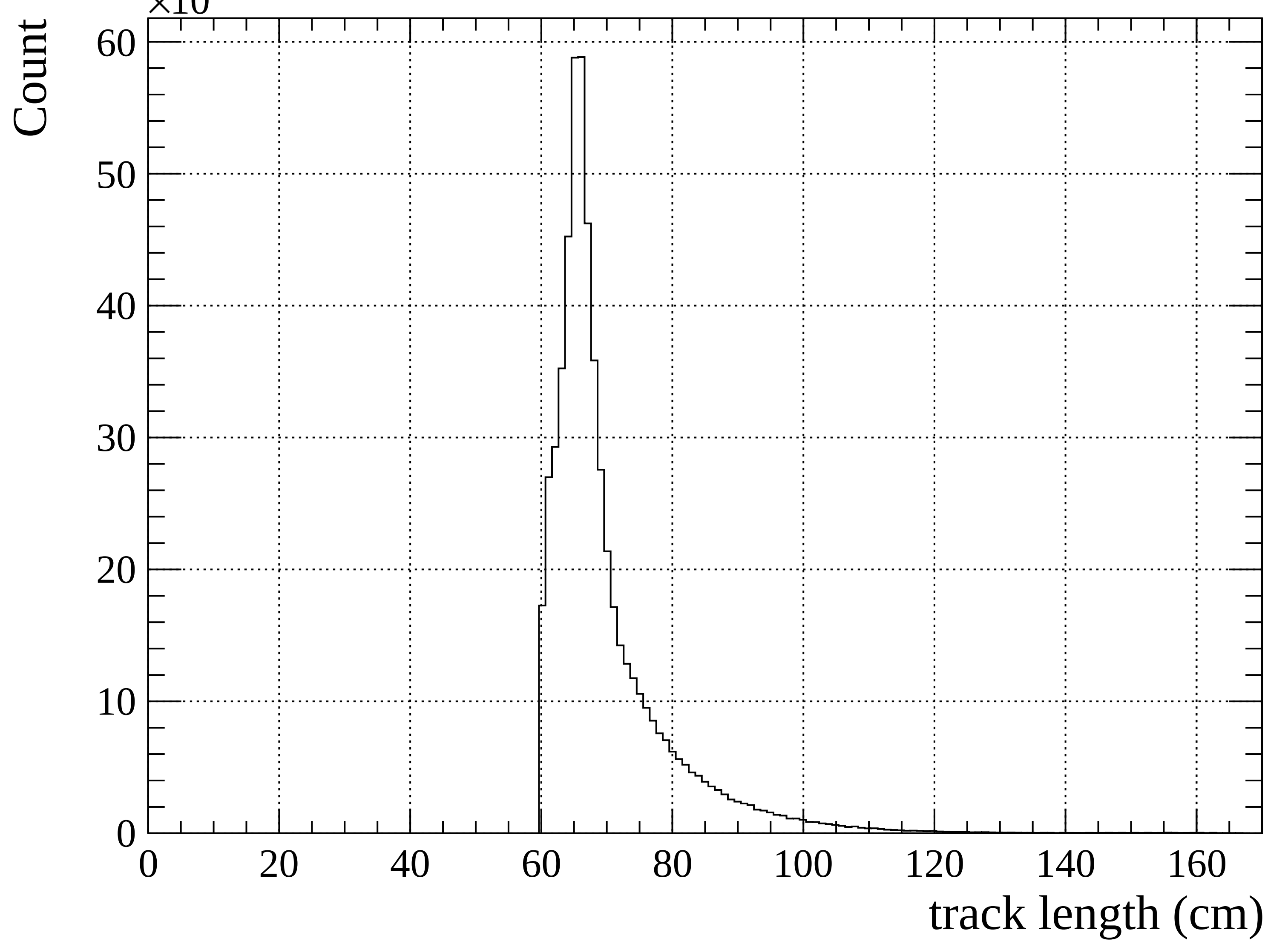


×10^3
Count
100
80
60
40
20
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.24 ± 0.00
μ_μ = 0.022 ± 0.001
σ_μ = 0.852 ± 0.001
μ_e = -2.137 ± 0.002
σ_e = 1.066 ± 0.001

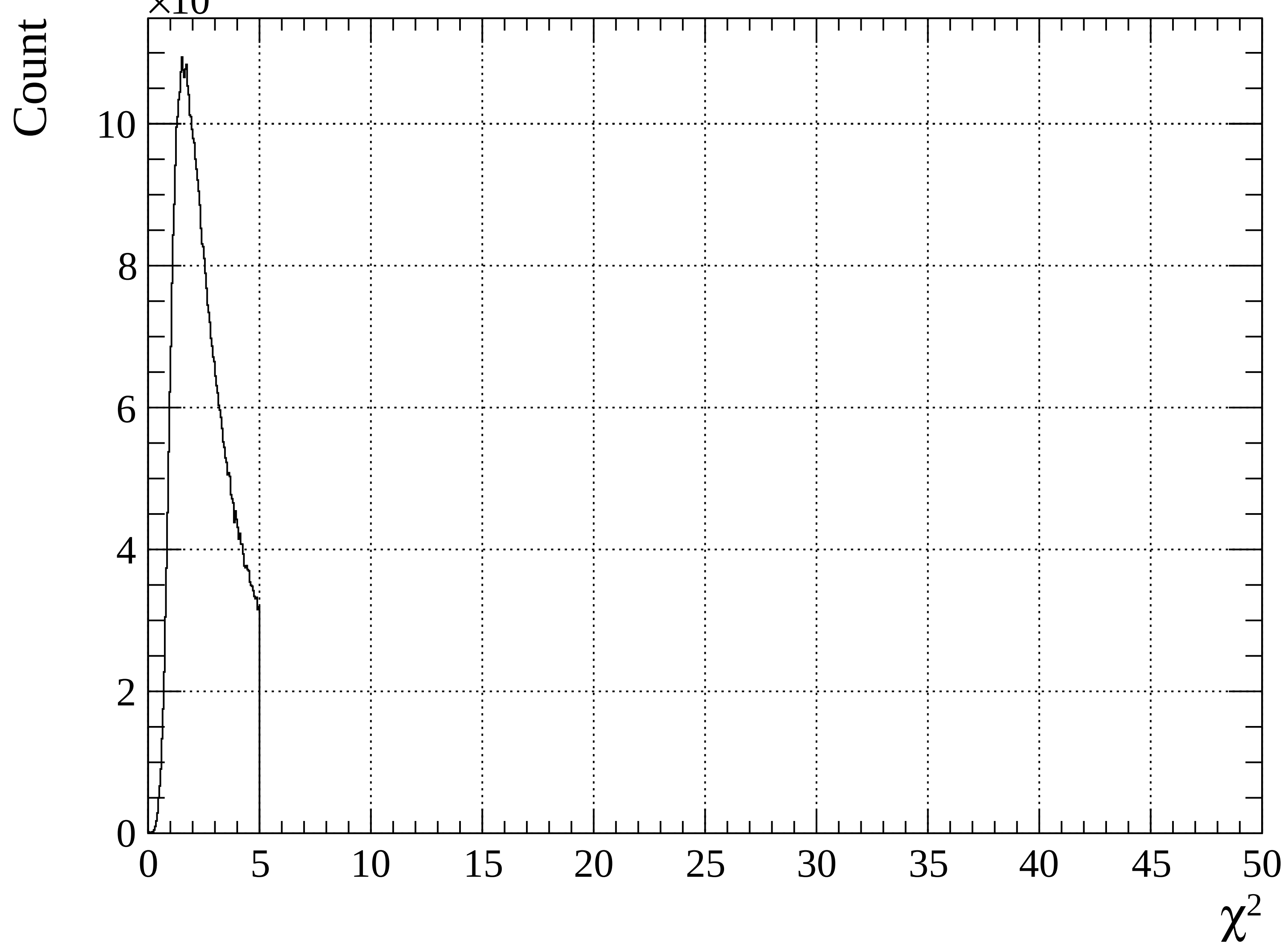

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



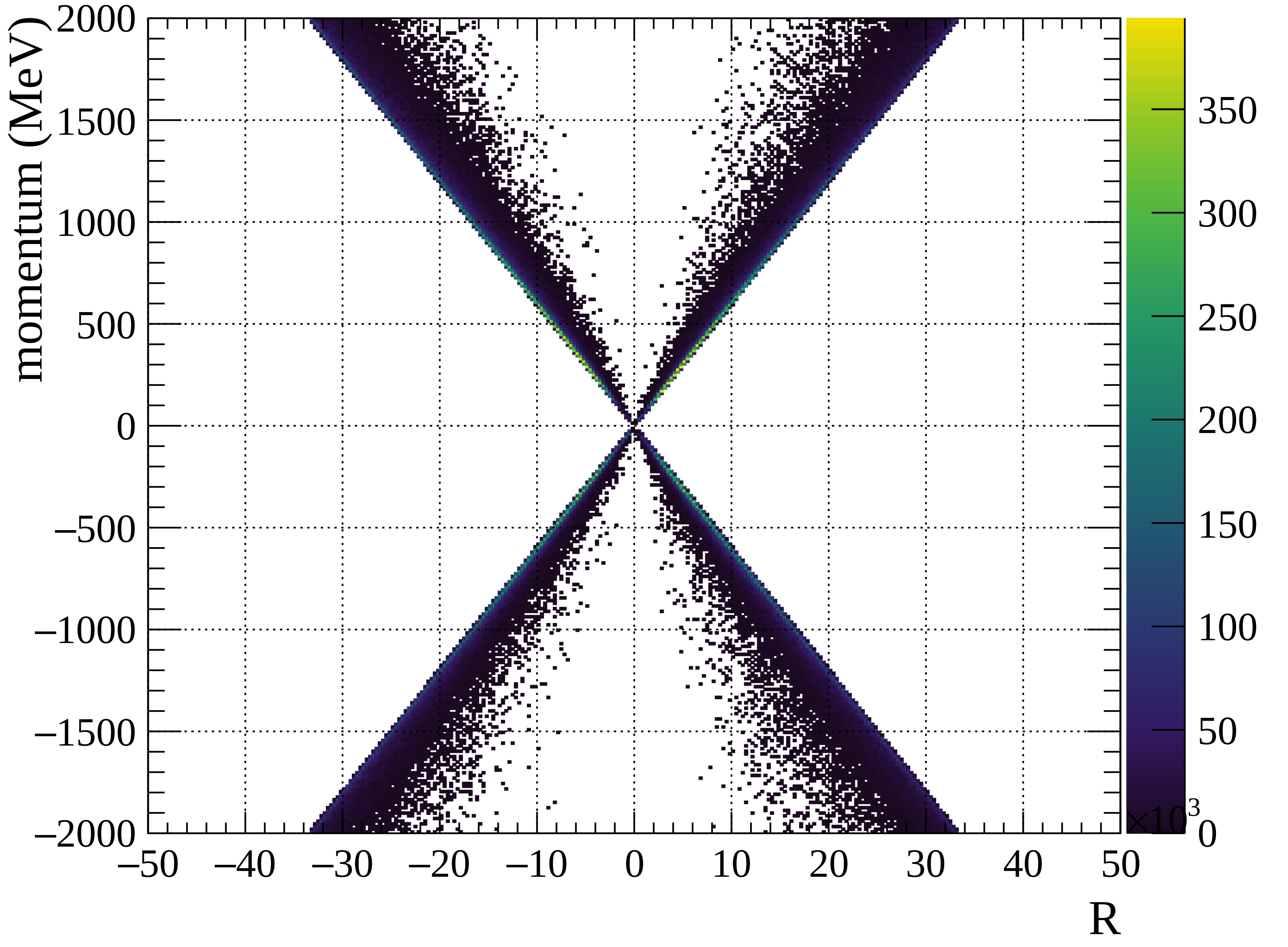
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
100
200
300
400
500
600
700
800
900
1000

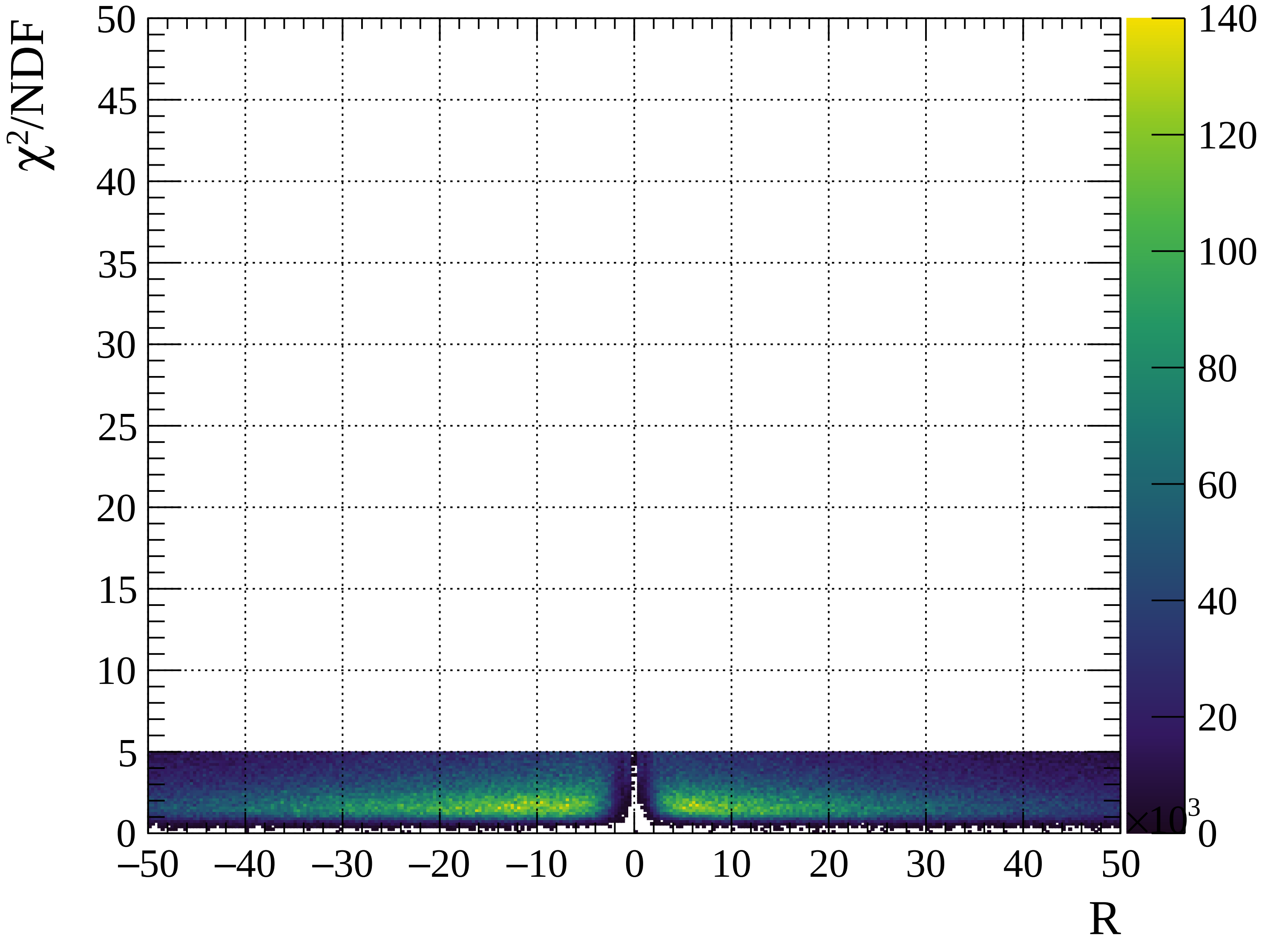


×10
Count
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
track length (cm)

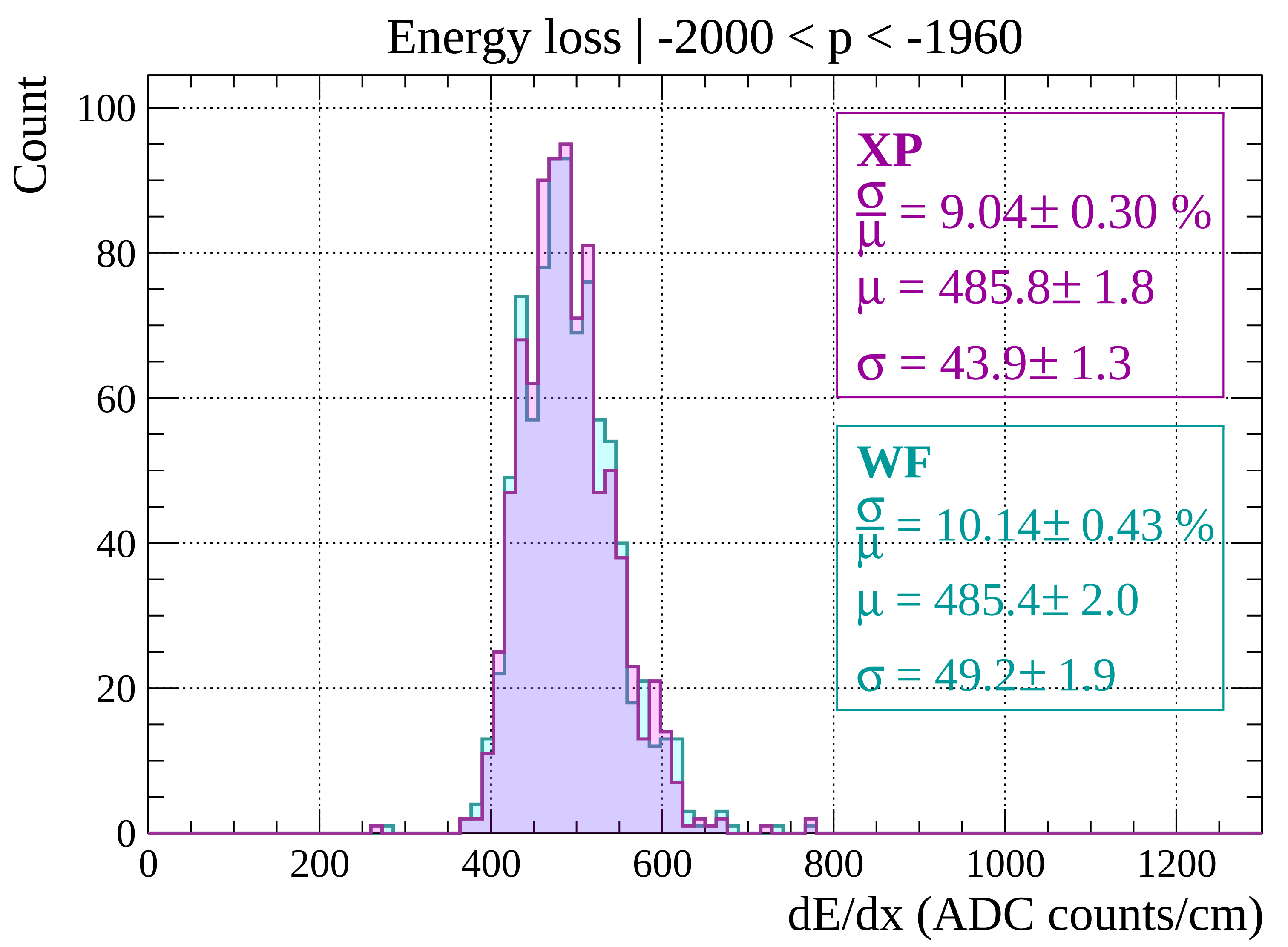


×10
Count
0
2
4
6
8
10
0
5
10
15
20
25
30
35
40
45
50
χ^2

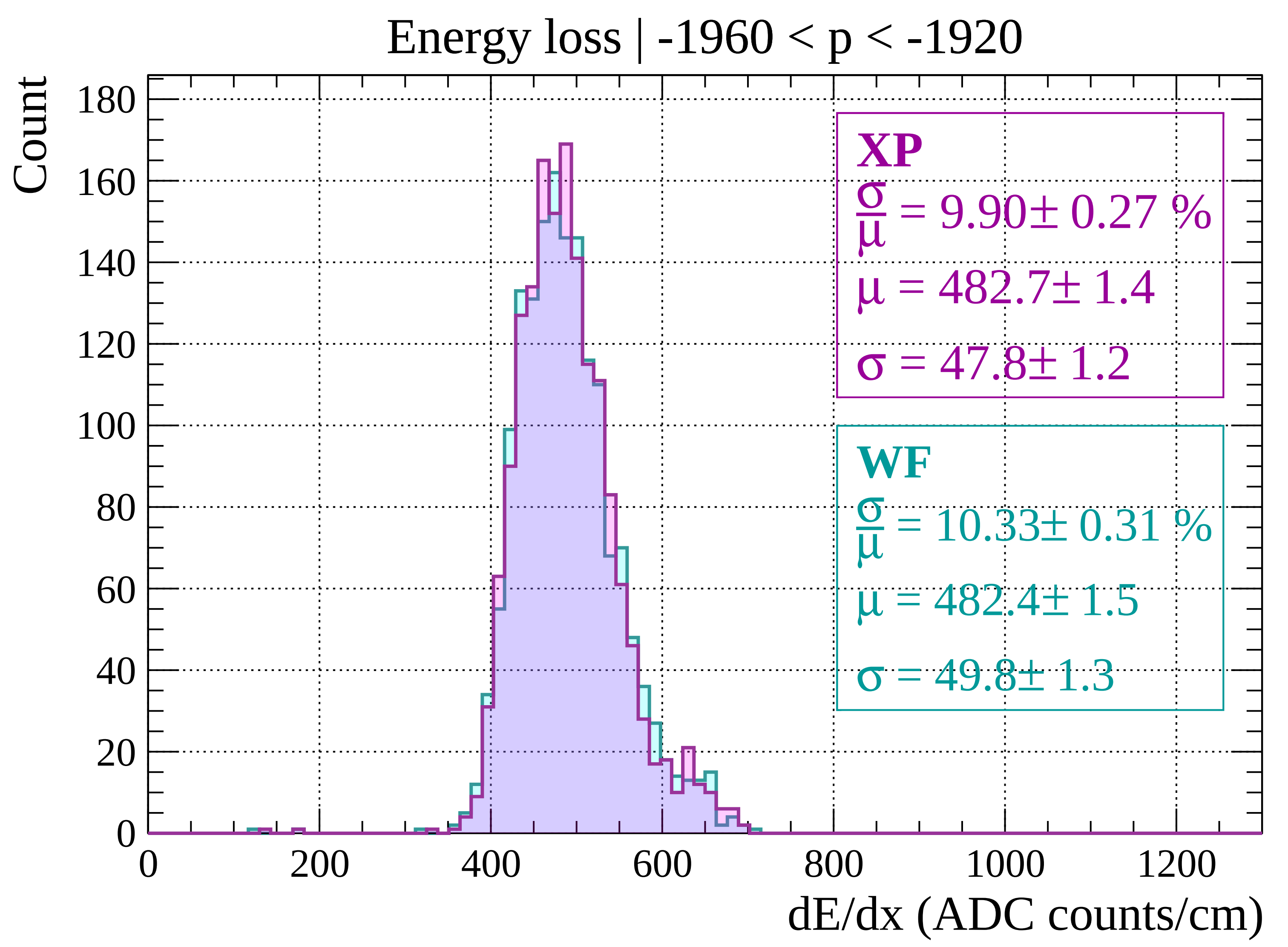

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
350
300
250
200
150
100
50
0
×10^3



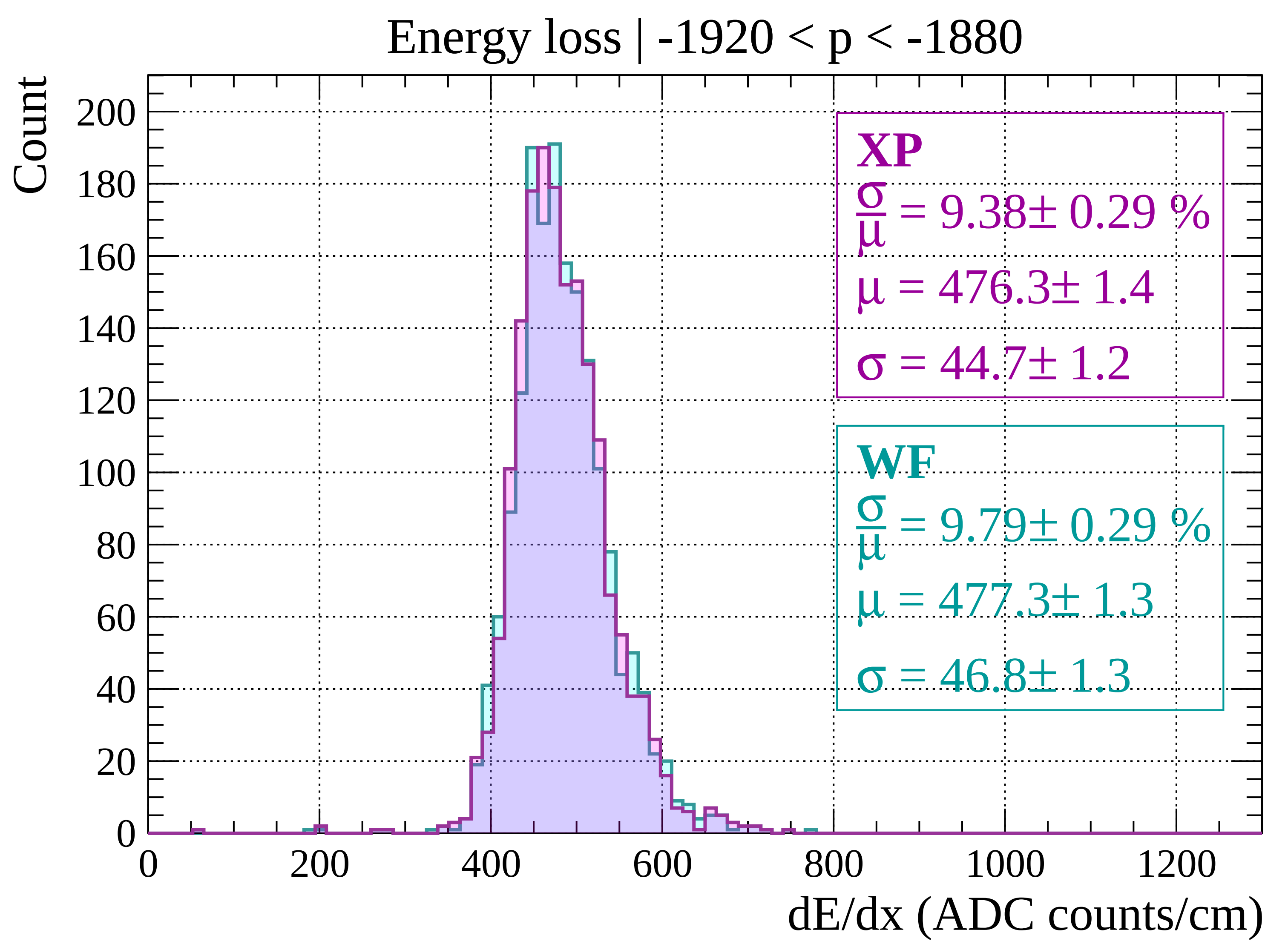
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
140
120
100
80
60
40
20
0
$\times 10^3$



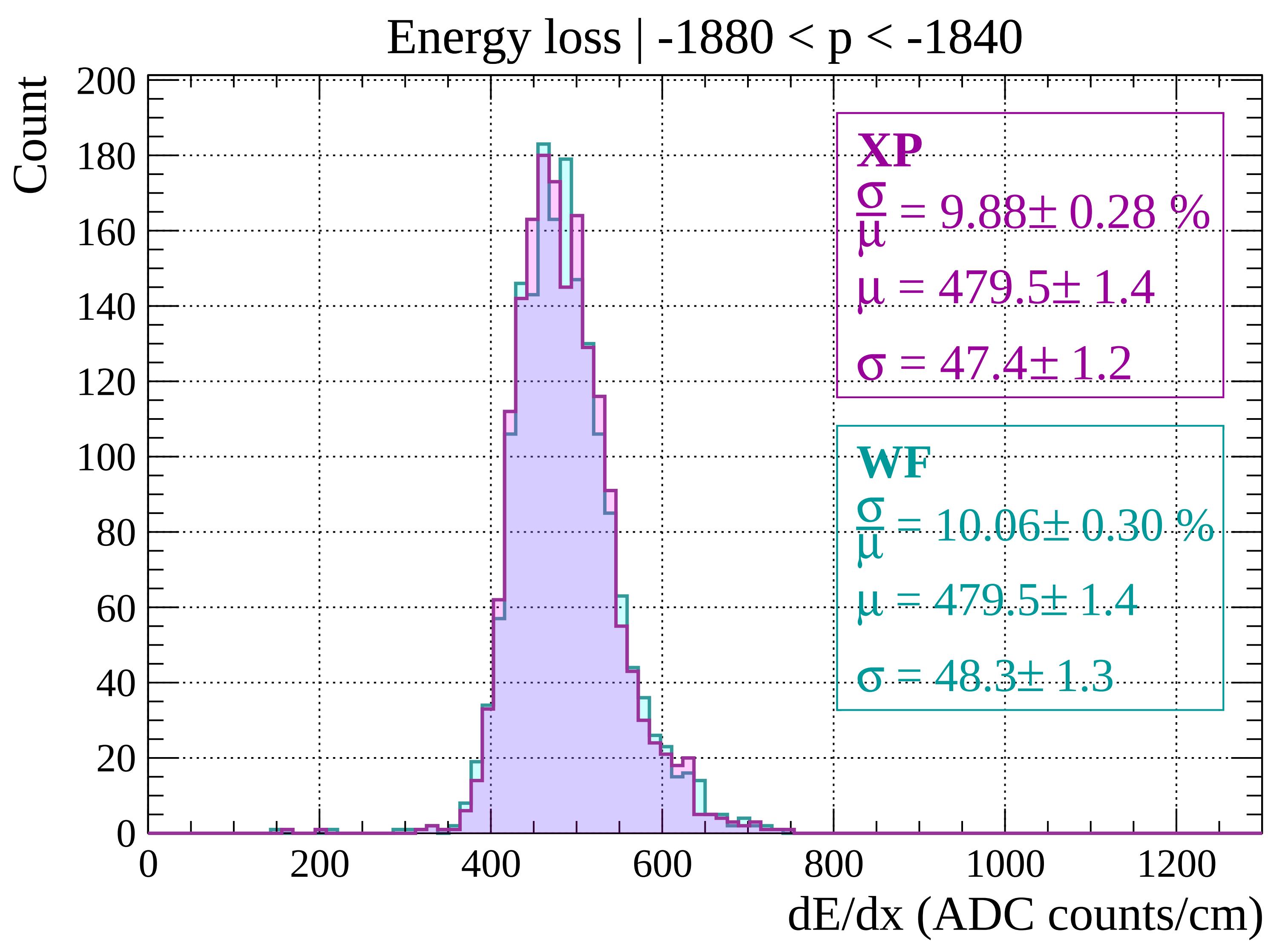
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 9.04± 0.30 %
μ = 485.8± 1.8
σ = 43.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 10.14± 0.43 %
μ = 485.4± 2.0
σ = 49.2± 1.9



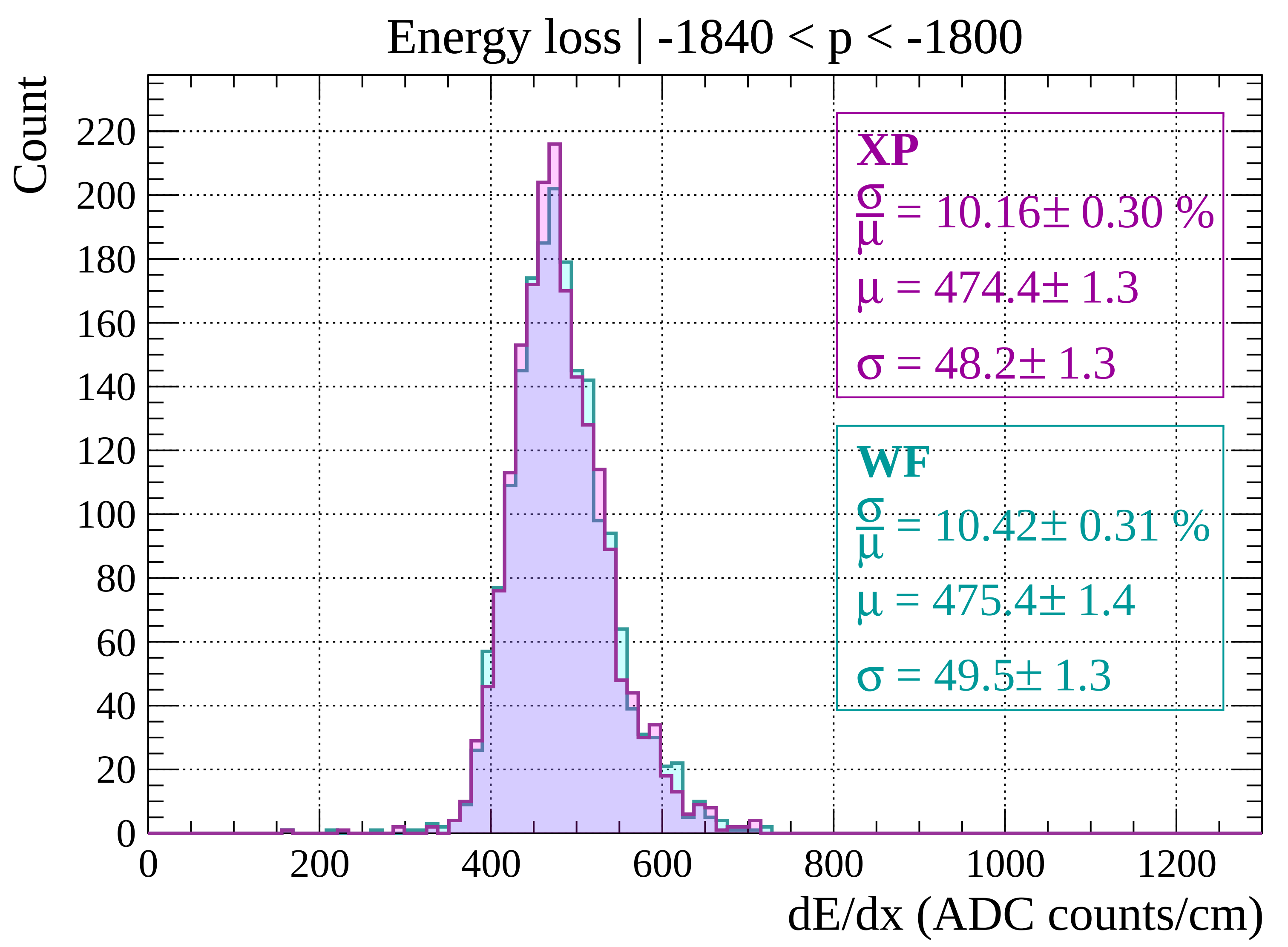
Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.90± 0.27 %
μ = 482.7± 1.4
σ = 47.8± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.33± 0.31 %
μ = 482.4± 1.5
σ = 49.8± 1.3



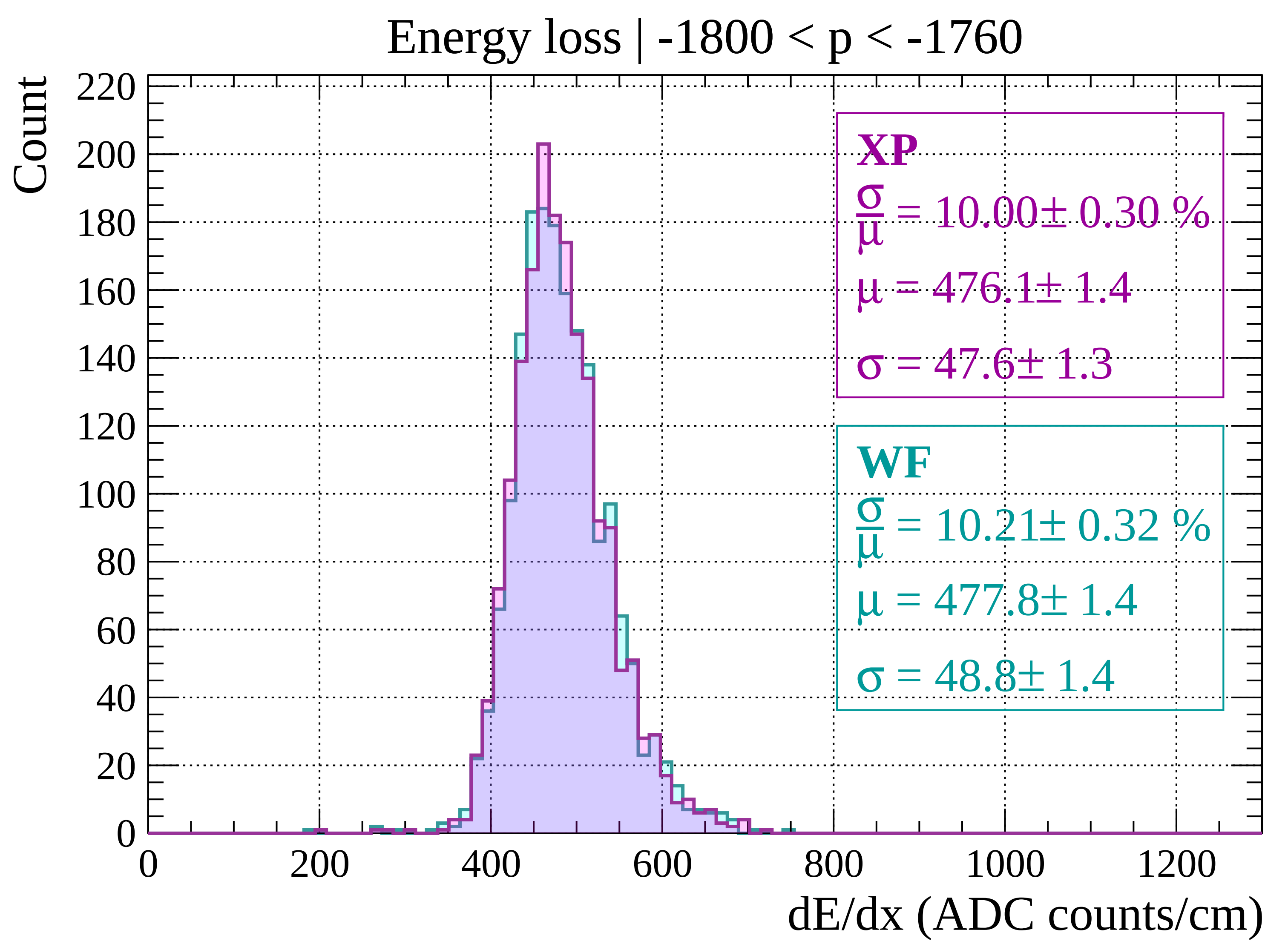
Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.38± 0.29 %
μ = 476.3± 1.4
σ = 44.7± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.79± 0.29 %
μ = 477.3± 1.3
σ = 46.8± 1.3
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200

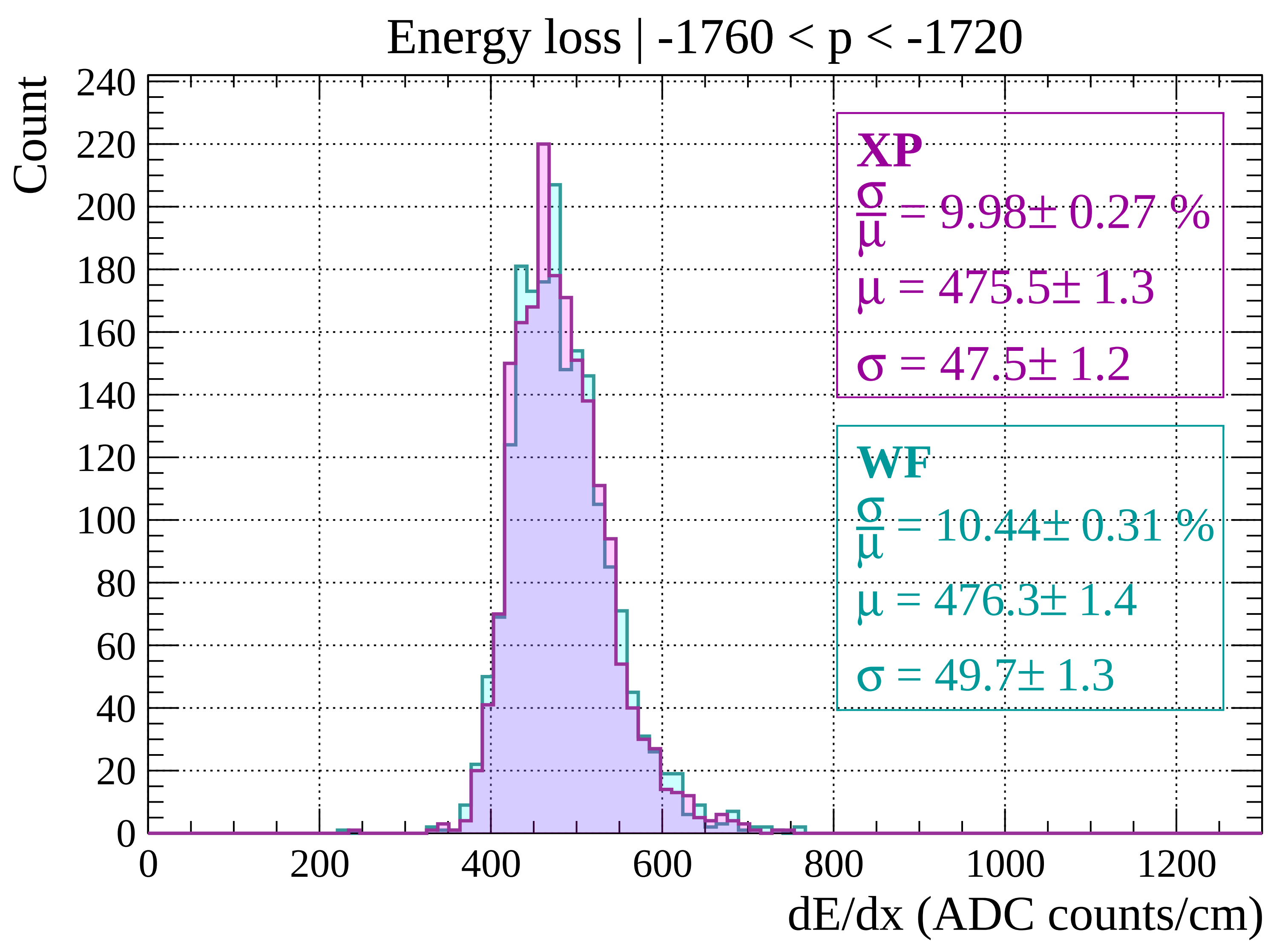


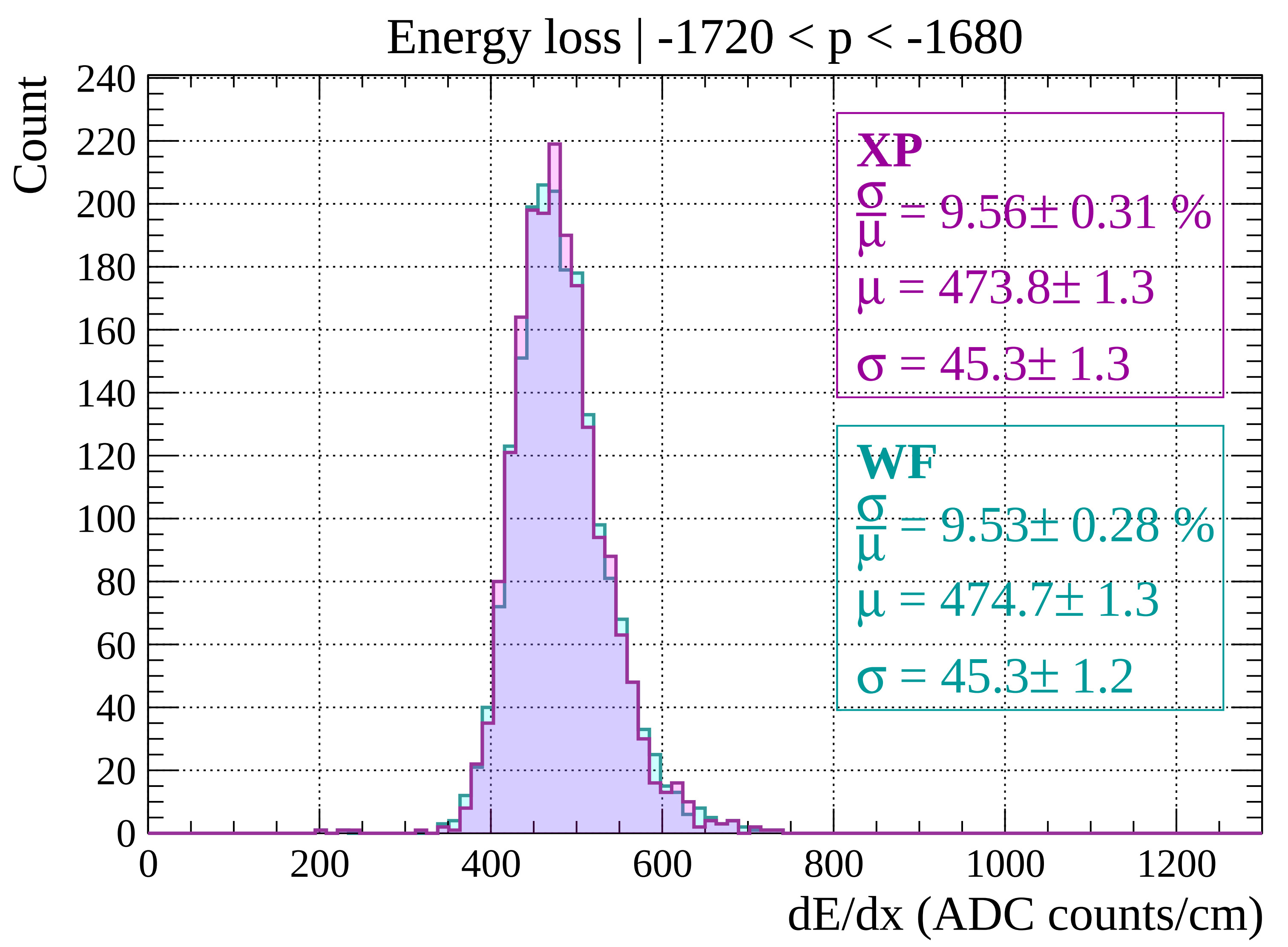
Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 9.88± 0.28 %
μ = 479.5± 1.4
σ = 47.4± 1.2
WF
σ/μ = 10.06± 0.30 %
μ = 479.5± 1.4
σ = 48.3± 1.3



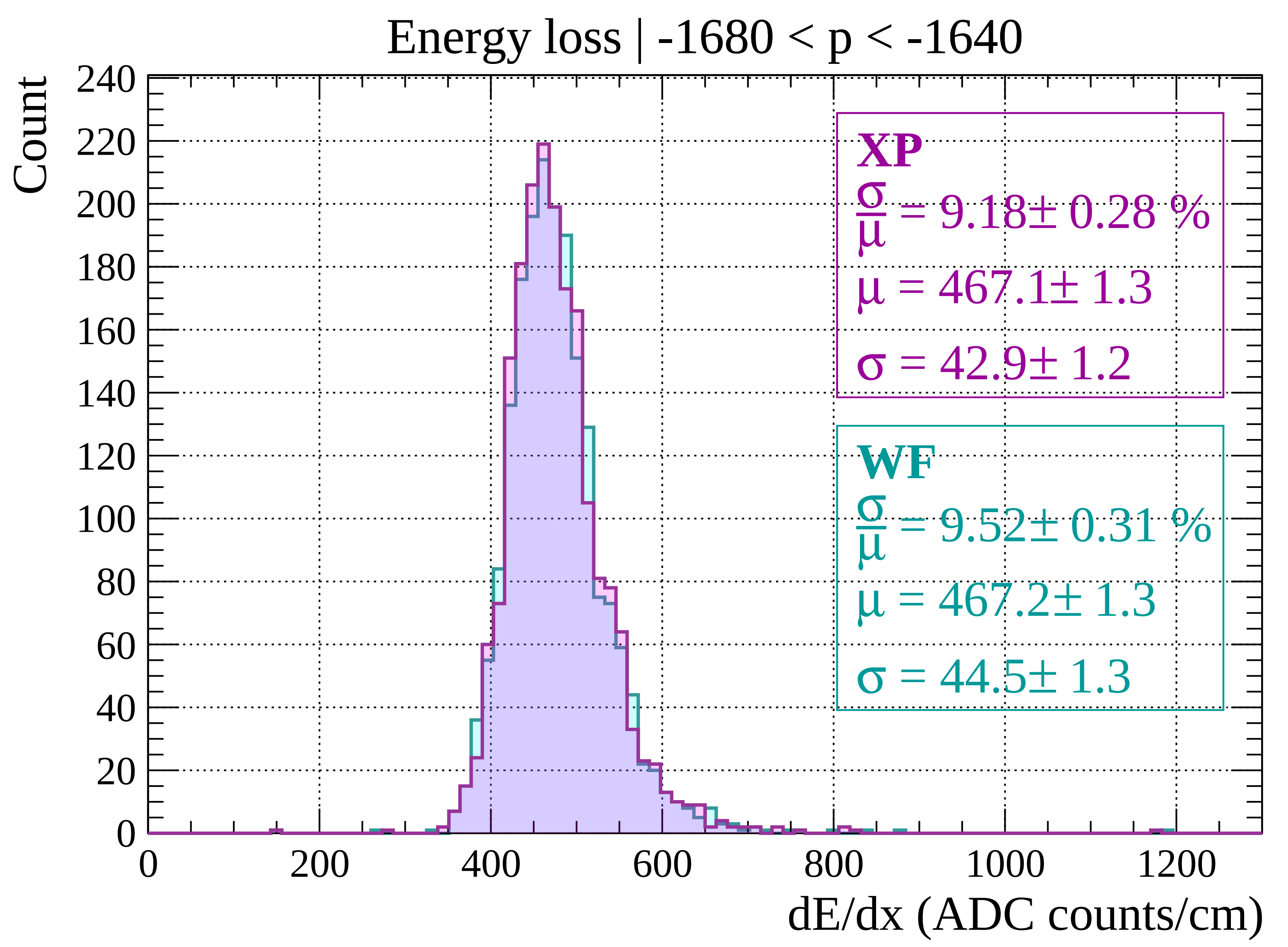
Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 10.16± 0.30 %
μ = 474.4± 1.3
σ = 48.2± 1.3
WF
σ/μ = 10.42± 0.31 %
μ = 475.4± 1.4
σ = 49.5± 1.3

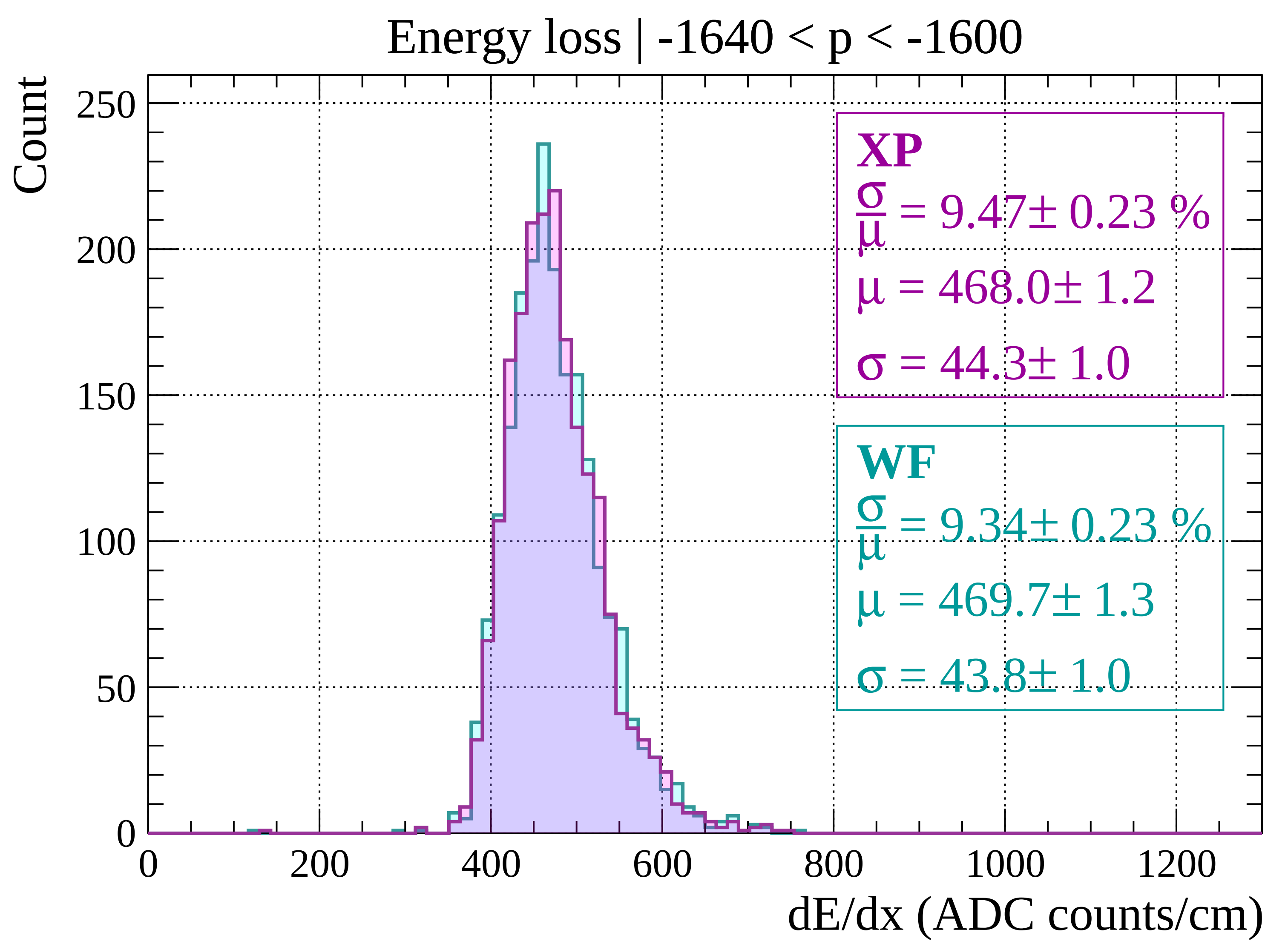

Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.00± 0.30 %
μ = 476.1± 1.4
σ = 47.6± 1.3
WF
$\frac{\sigma}{\mu}$ = 10.21± 0.32 %
μ = 477.8± 1.4
σ = 48.8± 1.4


Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.98± 0.27 %
μ = 475.5± 1.3
σ = 47.5± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.44± 0.31 %
μ = 476.3± 1.4
σ = 49.7± 1.3
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200

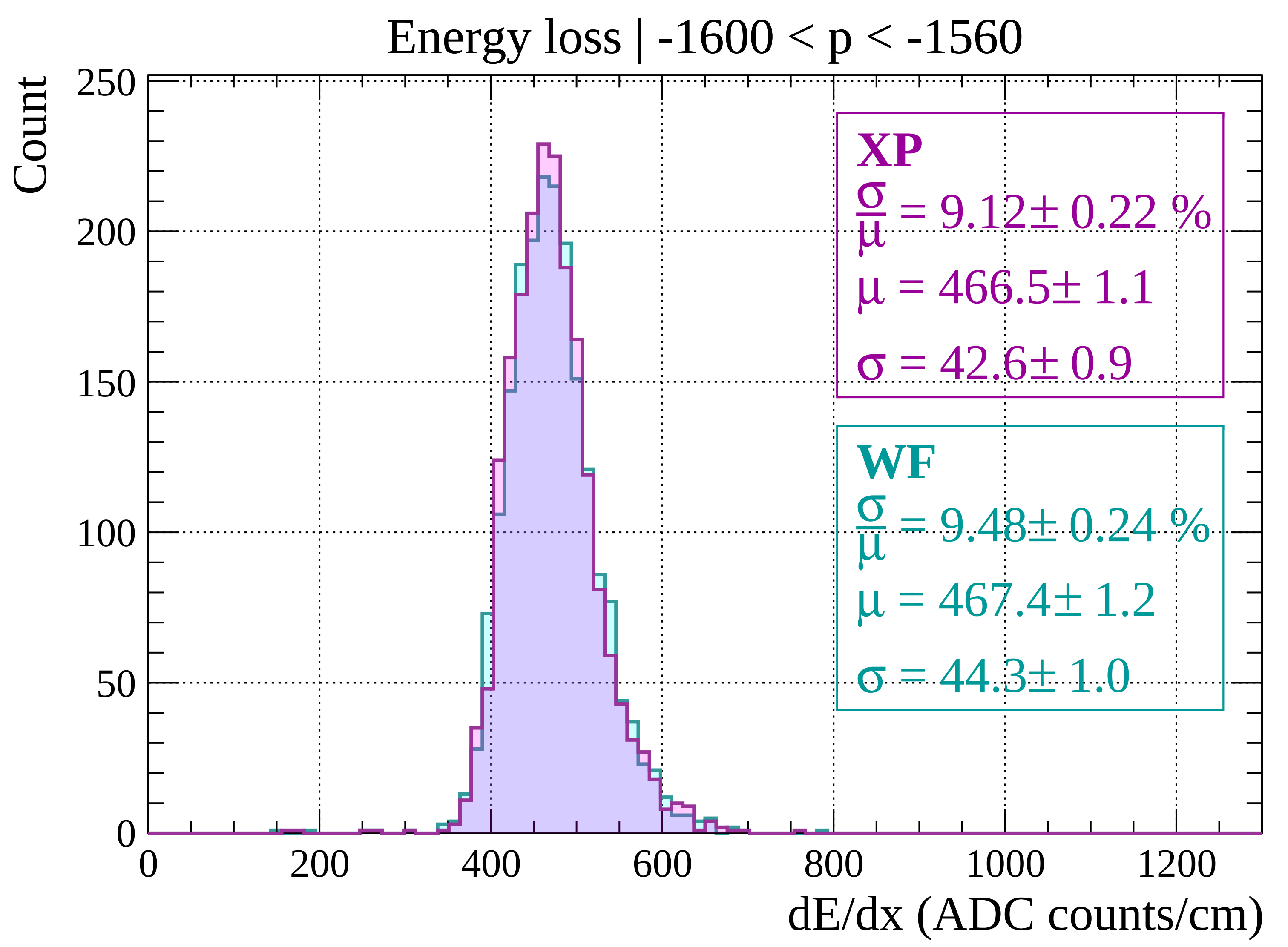


Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.56± 0.31 %
μ = 473.8± 1.3
σ = 45.3± 1.3
WF
σ/μ = 9.53± 0.28 %
μ = 474.7± 1.3
σ = 45.3± 1.2

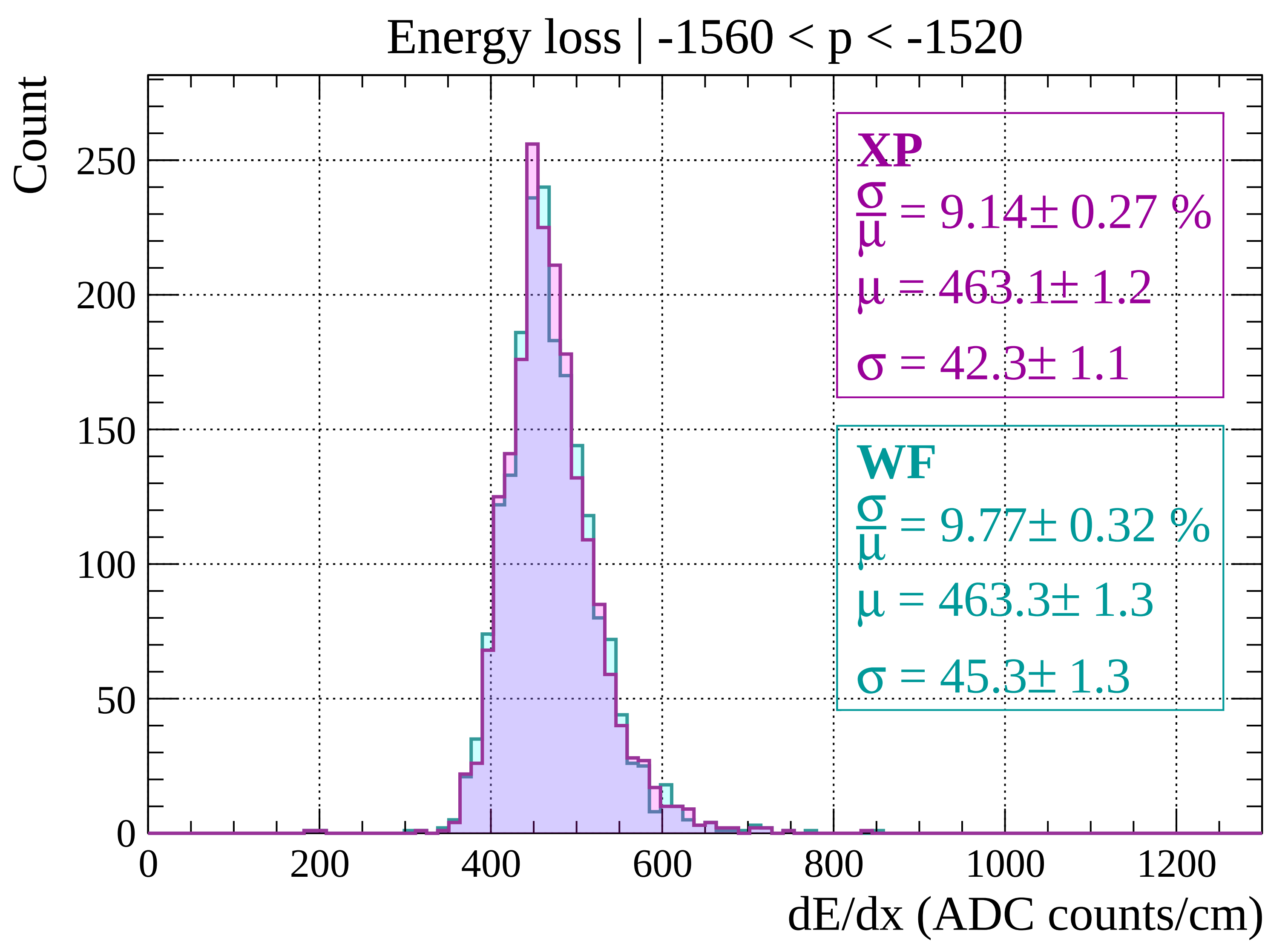

Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.18± 0.28 %
μ = 467.1± 1.3
σ = 42.9± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.52± 0.31 %
μ = 467.2± 1.3
σ = 44.5± 1.3



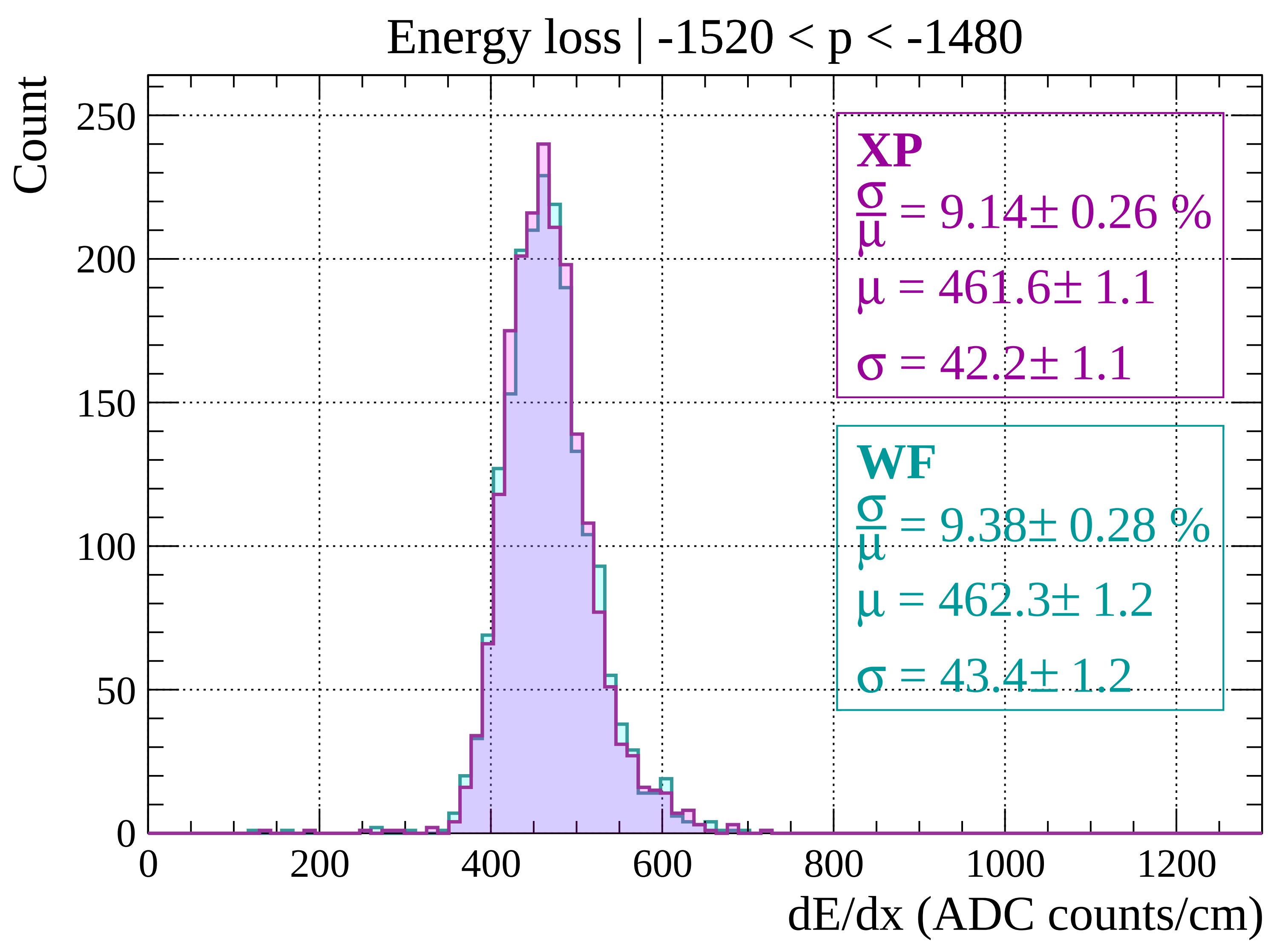
Energy loss | -1640 < p < -1600
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.47± 0.23 %
μ = 468.0± 1.2
σ = 44.3± 1.0
WF
σ/μ = 9.34± 0.23 %
μ = 469.7± 1.3
σ = 43.8± 1.0



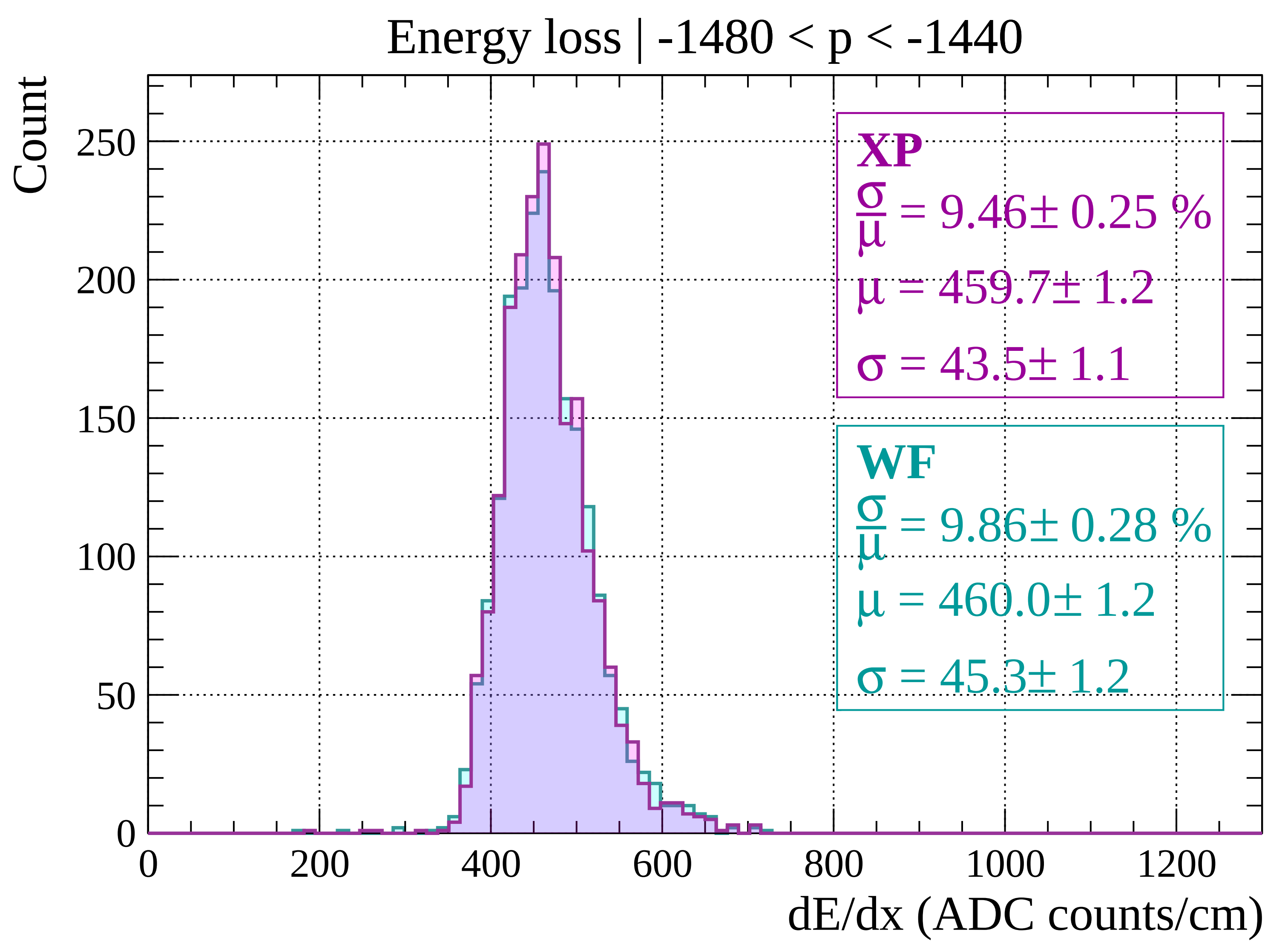
Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.12± 0.22 %
μ = 466.5± 1.1
σ = 42.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.48± 0.24 %
μ = 467.4± 1.2
σ = 44.3± 1.0

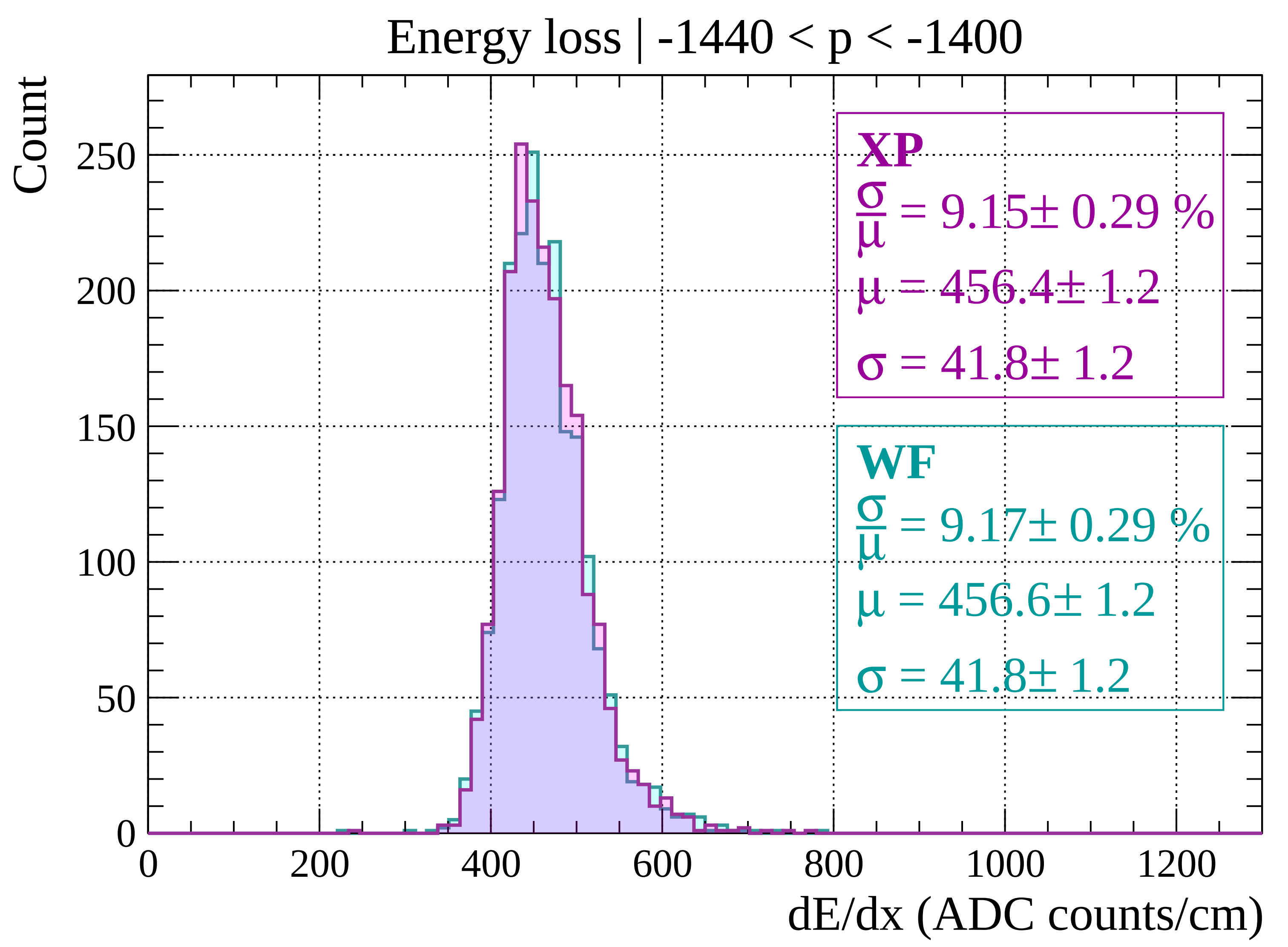


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 9.14± 0.27 %
μ = 463.1± 1.2
σ = 42.3± 1.1
WF
σ/μ = 9.77± 0.32 %
μ = 463.3± 1.3
σ = 45.3± 1.3

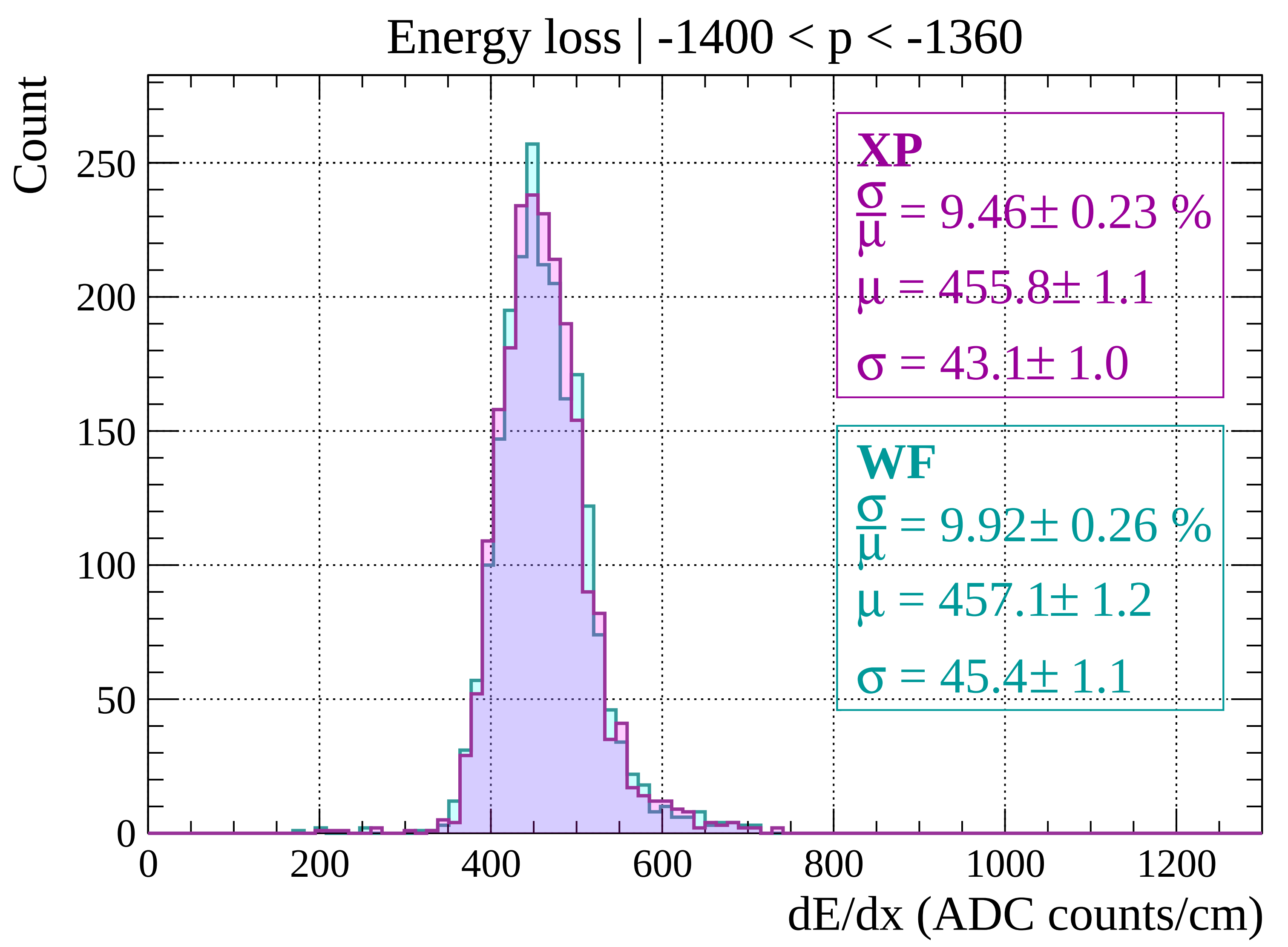


Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
$\frac{\sigma}{\mu} = 9.14 \pm 0.26$ %
$\mu = 461.6 \pm 1.1$
$\sigma = 42.2 \pm 1.1$
WF
$\frac{\sigma}{\mu} = 9.38 \pm 0.28$ %
$\mu = 462.3 \pm 1.2$
$\sigma = 43.4 \pm 1.2$

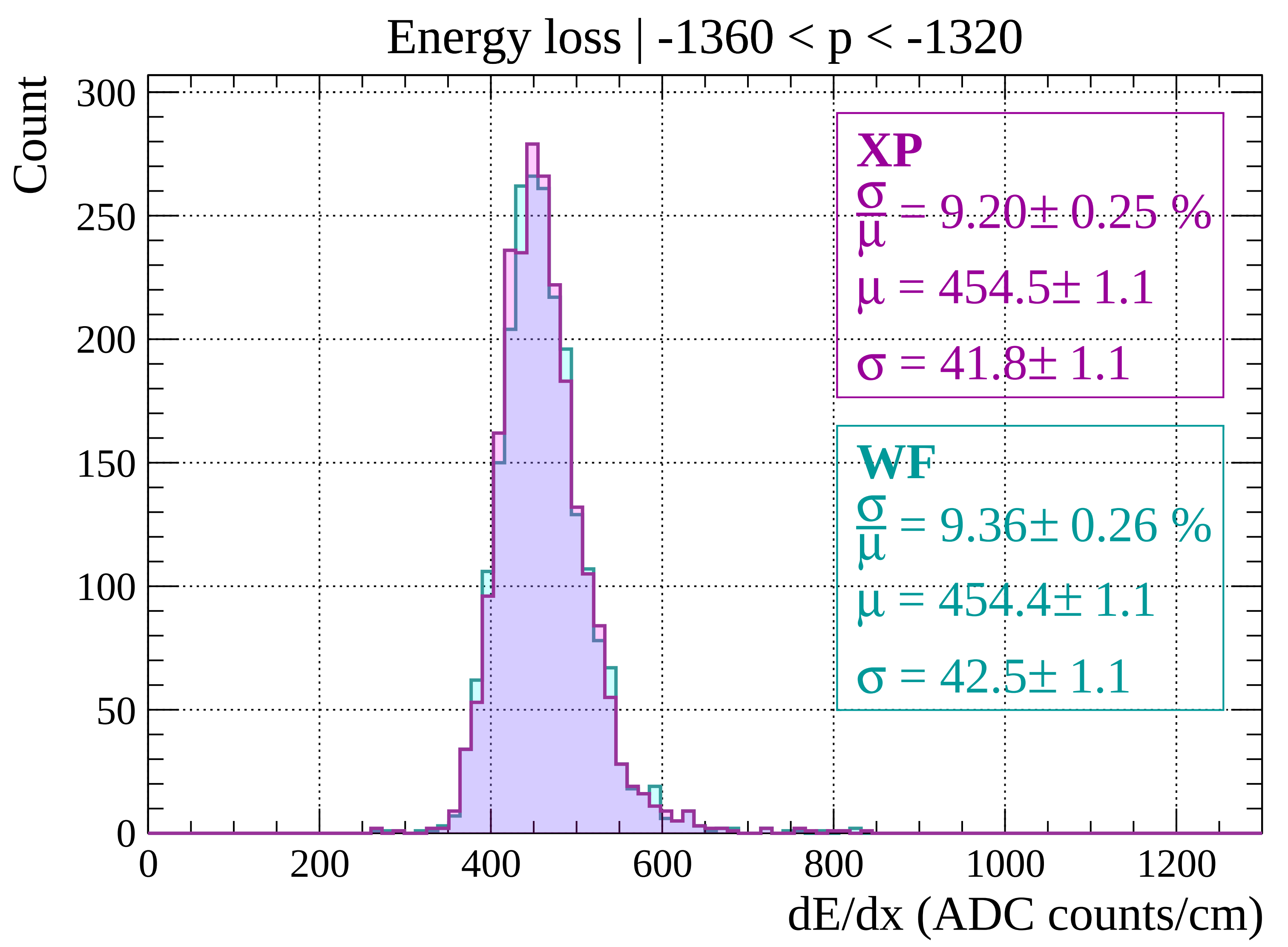

Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 9.46± 0.25 %
μ = 459.7± 1.2
σ = 43.5± 1.1
WF
σ/μ = 9.86± 0.28 %
μ = 460.0± 1.2
σ = 45.3± 1.2



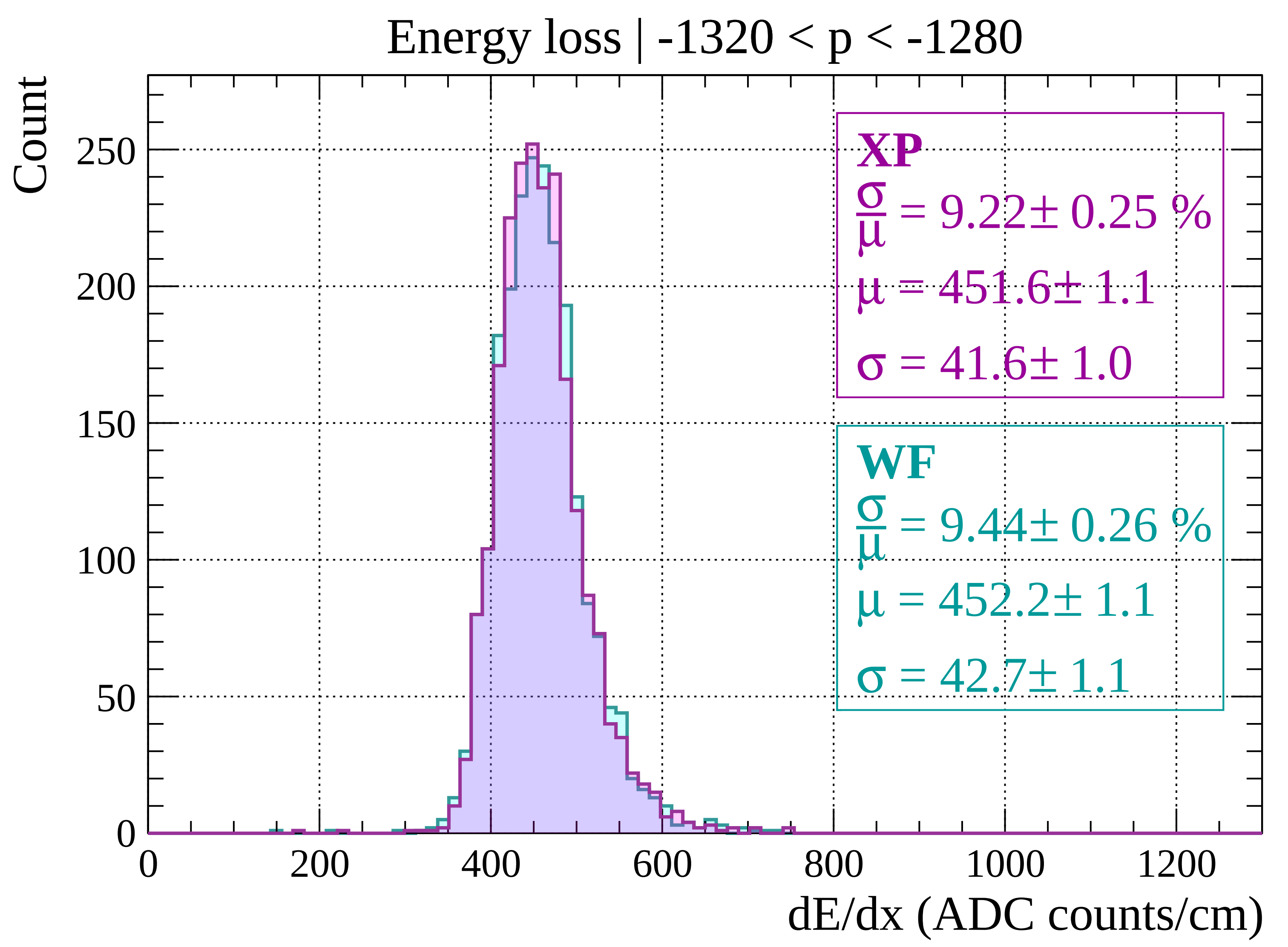
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 9.15± 0.29 %
μ = 456.4± 1.2
σ = 41.8± 1.2
WF
σ/μ = 9.17± 0.29 %
μ = 456.6± 1.2
σ = 41.8± 1.2



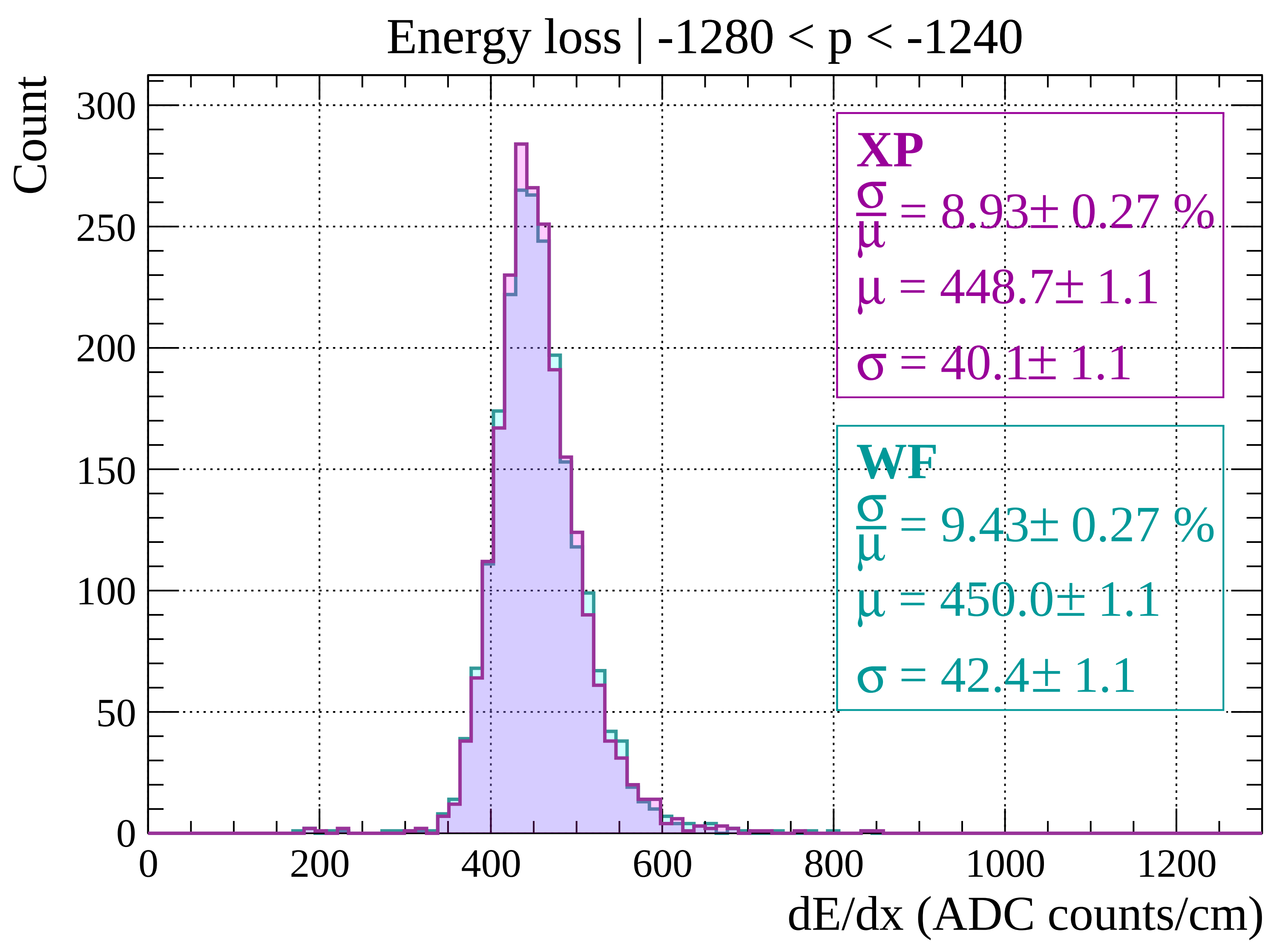
Energy loss | -1400 < p < -1360
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.46± 0.23 %
μ = 455.8± 1.1
σ = 43.1± 1.0
WF
σ/μ = 9.92± 0.26 %
μ = 457.1± 1.2
σ = 45.4± 1.1



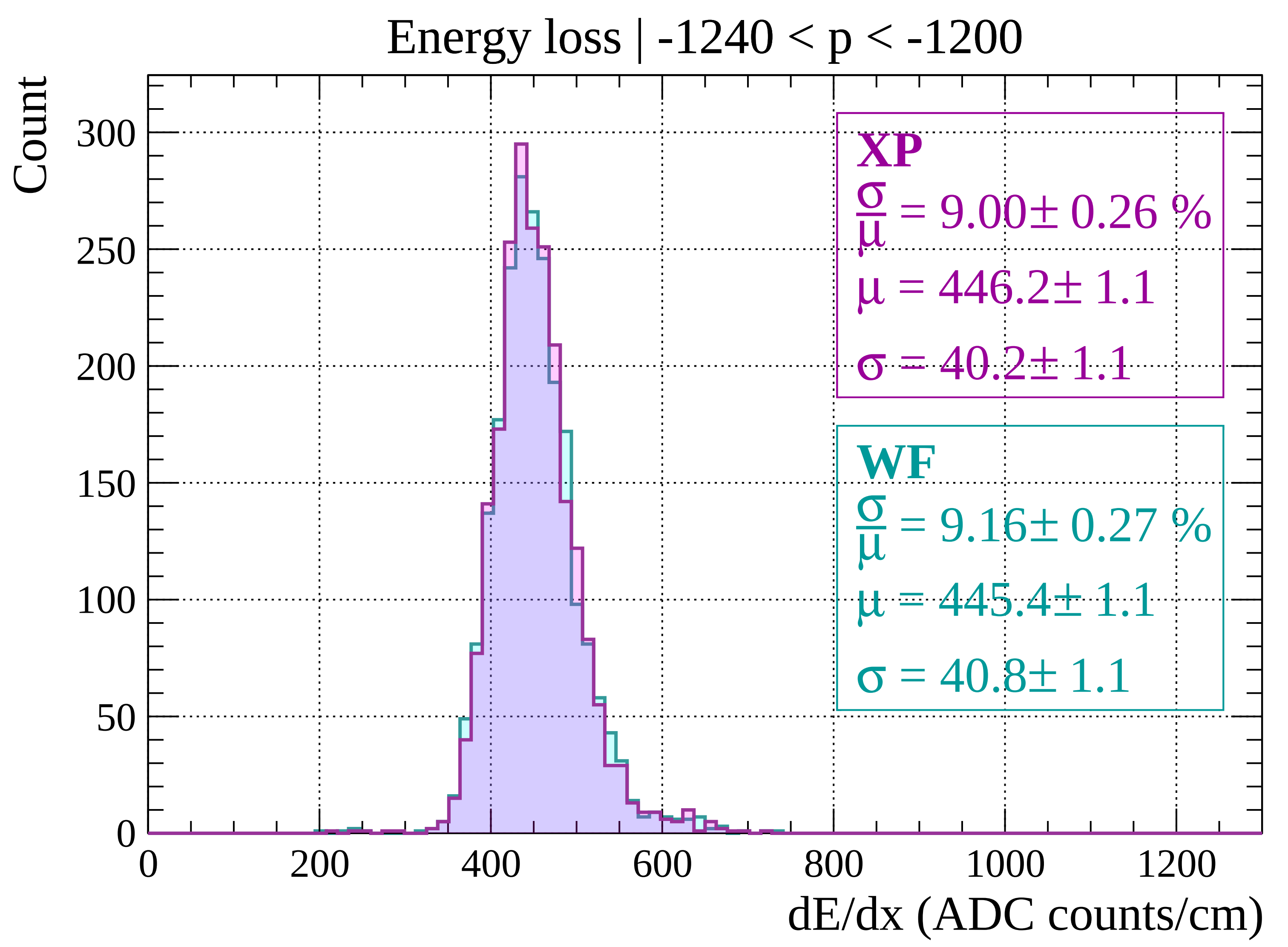
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.20± 0.25 %
μ = 454.5± 1.1
σ = 41.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.36± 0.26 %
μ = 454.4± 1.1
σ = 42.5± 1.1
0
200
400
600
800
1000
1200
50
100
150
250
300



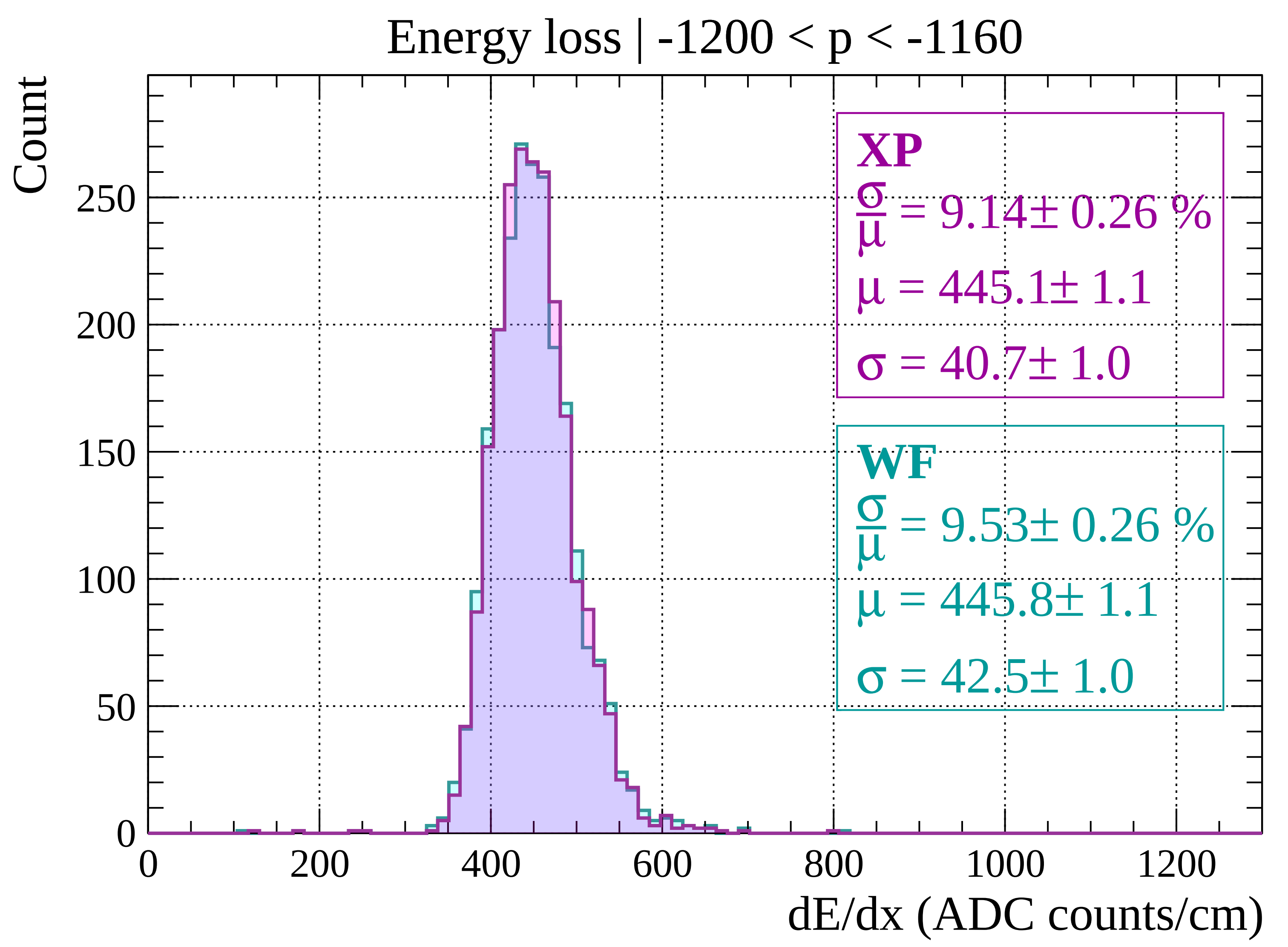
Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.22± 0.25 %
μ = 451.6± 1.1
σ = 41.6± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.44± 0.26 %
μ = 452.2± 1.1
σ = 42.7± 1.1
0
50
100
150
200
250
0
200
400
600
800
1000
1200



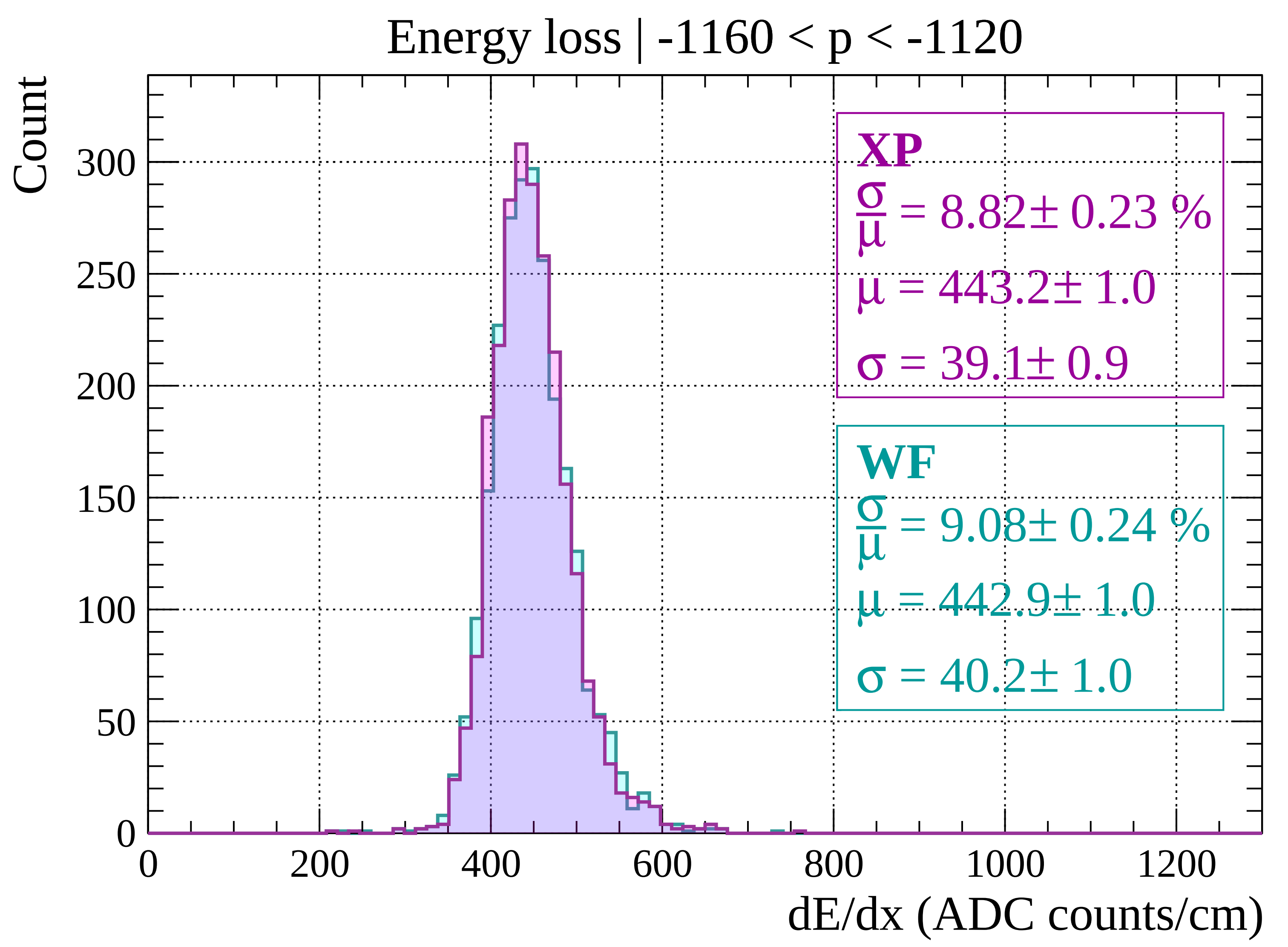
Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.93± 0.27 %
μ = 448.7± 1.1
σ = 40.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.43± 0.27 %
μ = 450.0± 1.1
σ = 42.4± 1.1



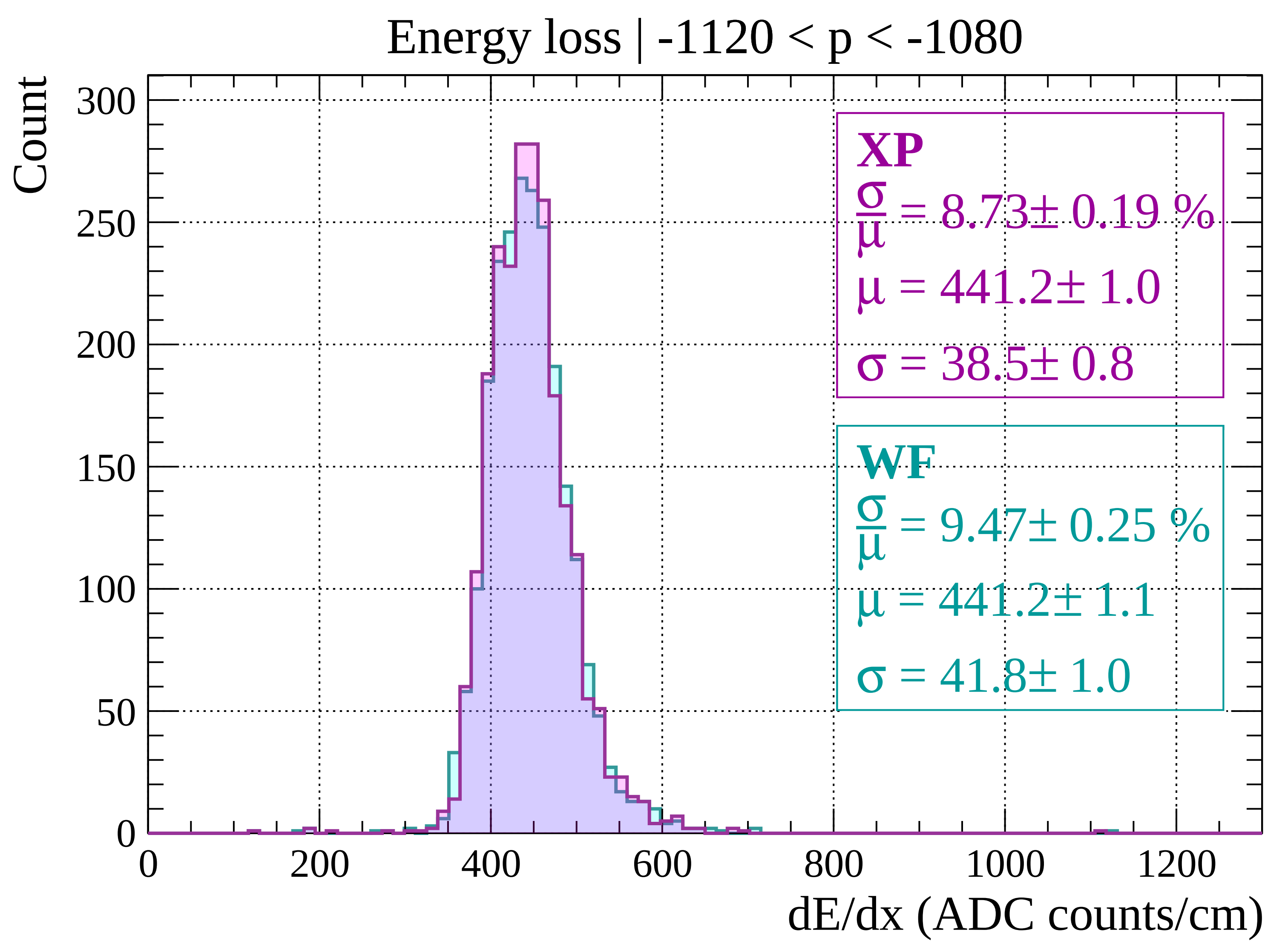
Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 9.00± 0.26 %
μ = 446.2± 1.1
σ = 40.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.16± 0.27 %
μ = 445.4± 1.1
σ = 40.8± 1.1

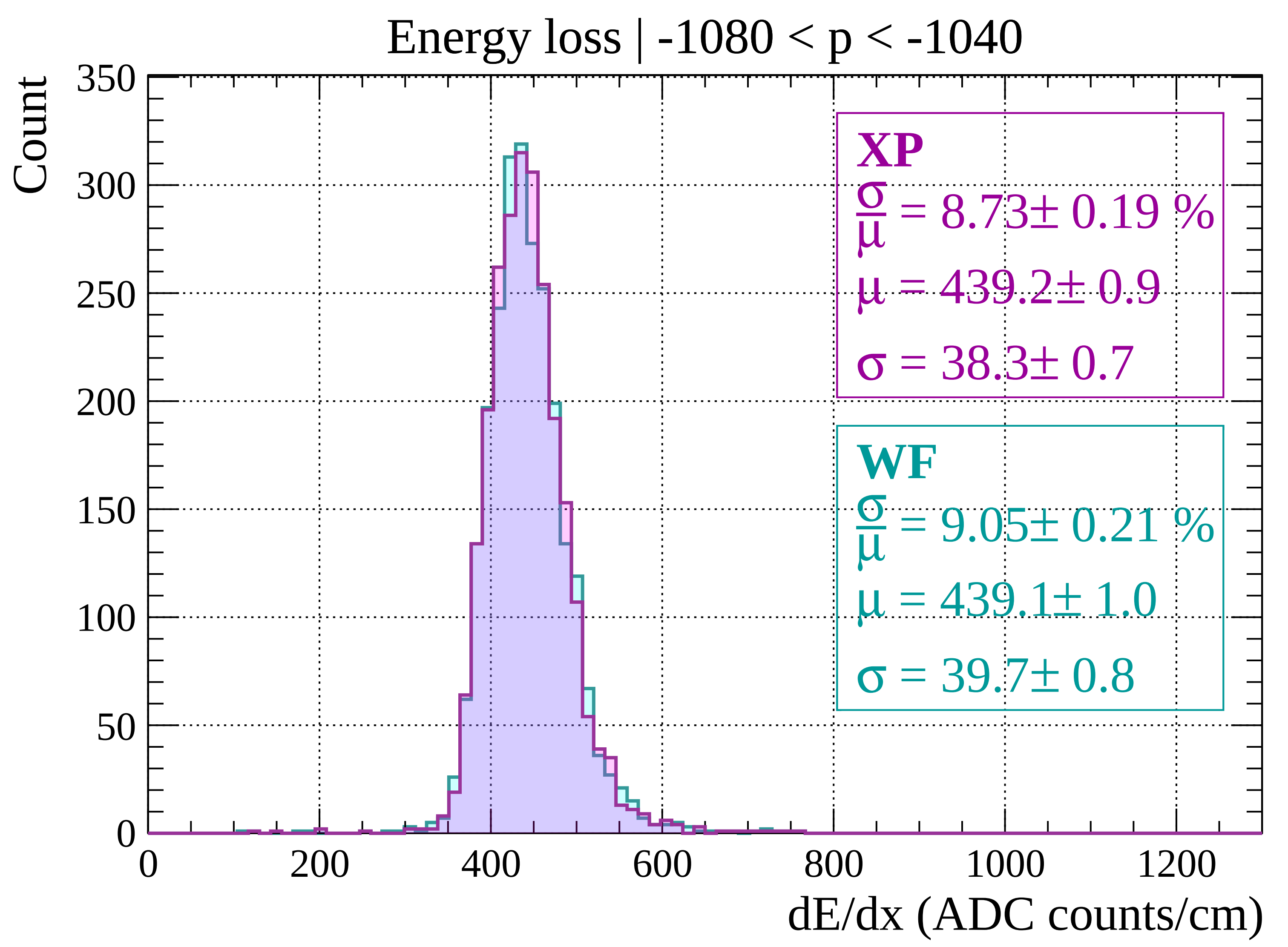


Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 9.14± 0.26 %
μ = 445.1± 1.1
σ = 40.7± 1.0
WF
σ/μ = 9.53± 0.26 %
μ = 445.8± 1.1
σ = 42.5± 1.0

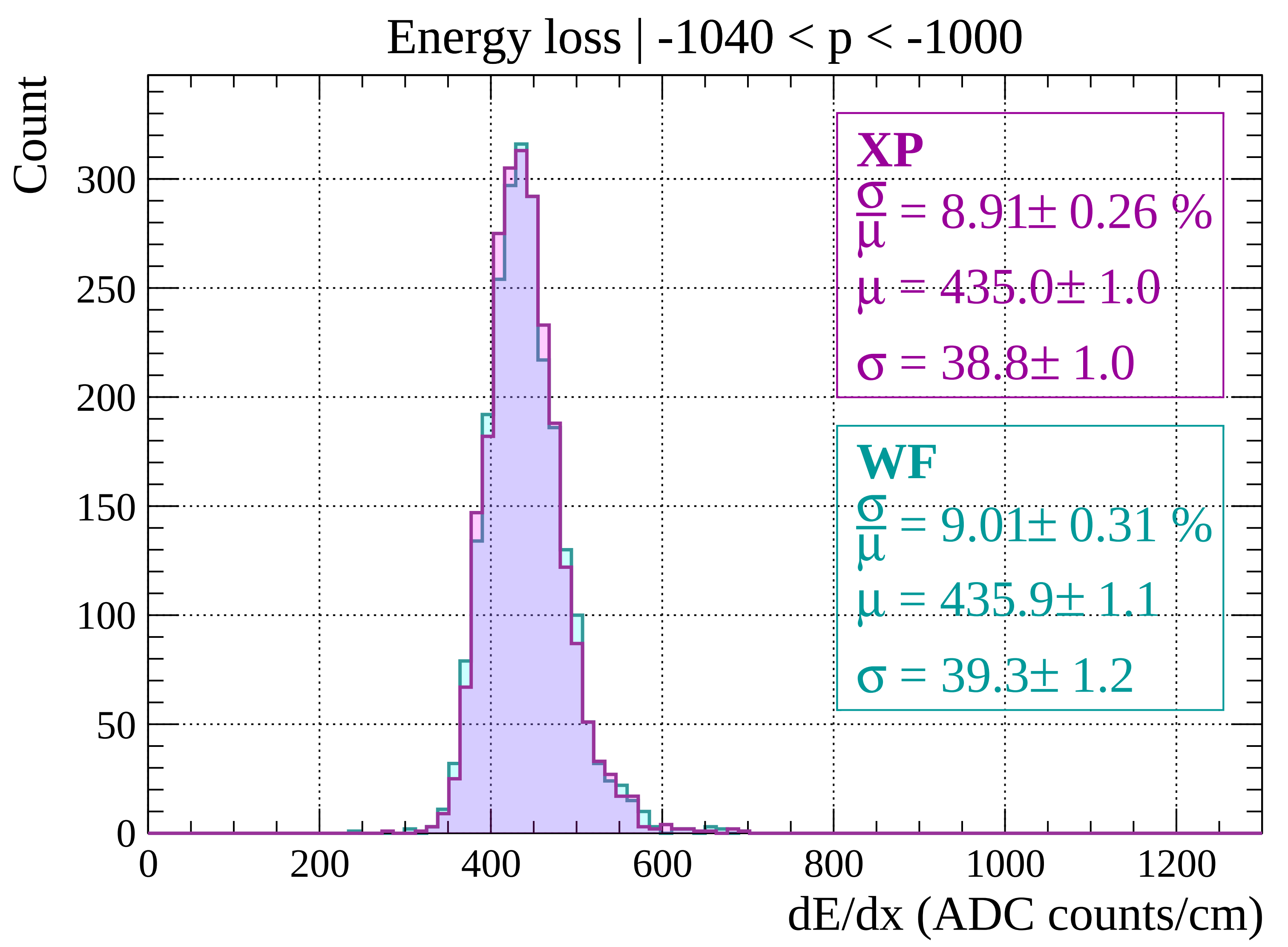


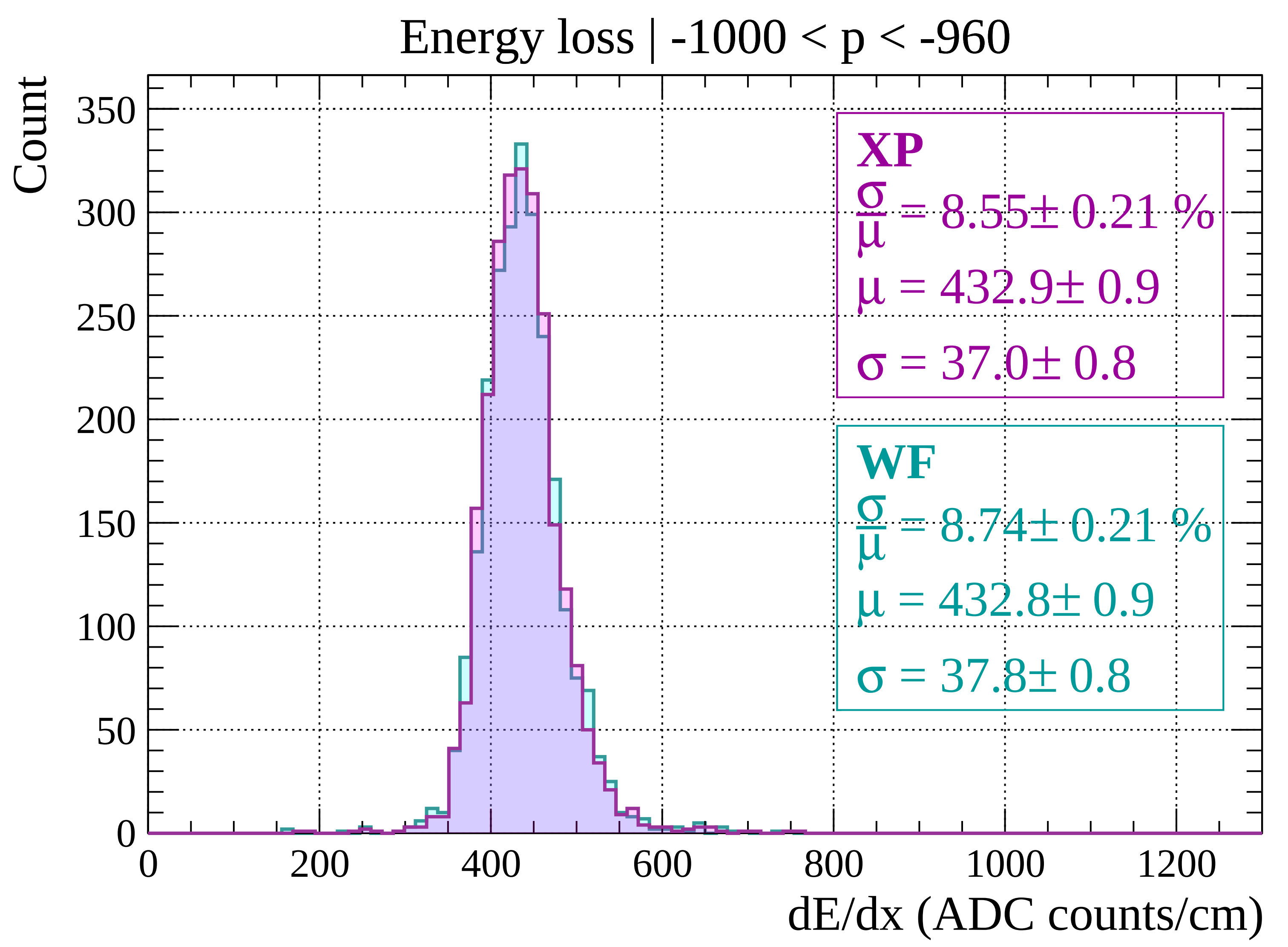
Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.82± 0.23 %
μ = 443.2± 1.0
σ = 39.1± 0.9
WF
σ/μ = 9.08± 0.24 %
μ = 442.9± 1.0
σ = 40.2± 1.0


Energy loss | -1120 < p < -1080
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.73± 0.19 %
μ = 441.2± 1.0
σ = 38.5± 0.8
WF
σ/μ = 9.47± 0.25 %
μ = 441.2± 1.1
σ = 41.8± 1.0

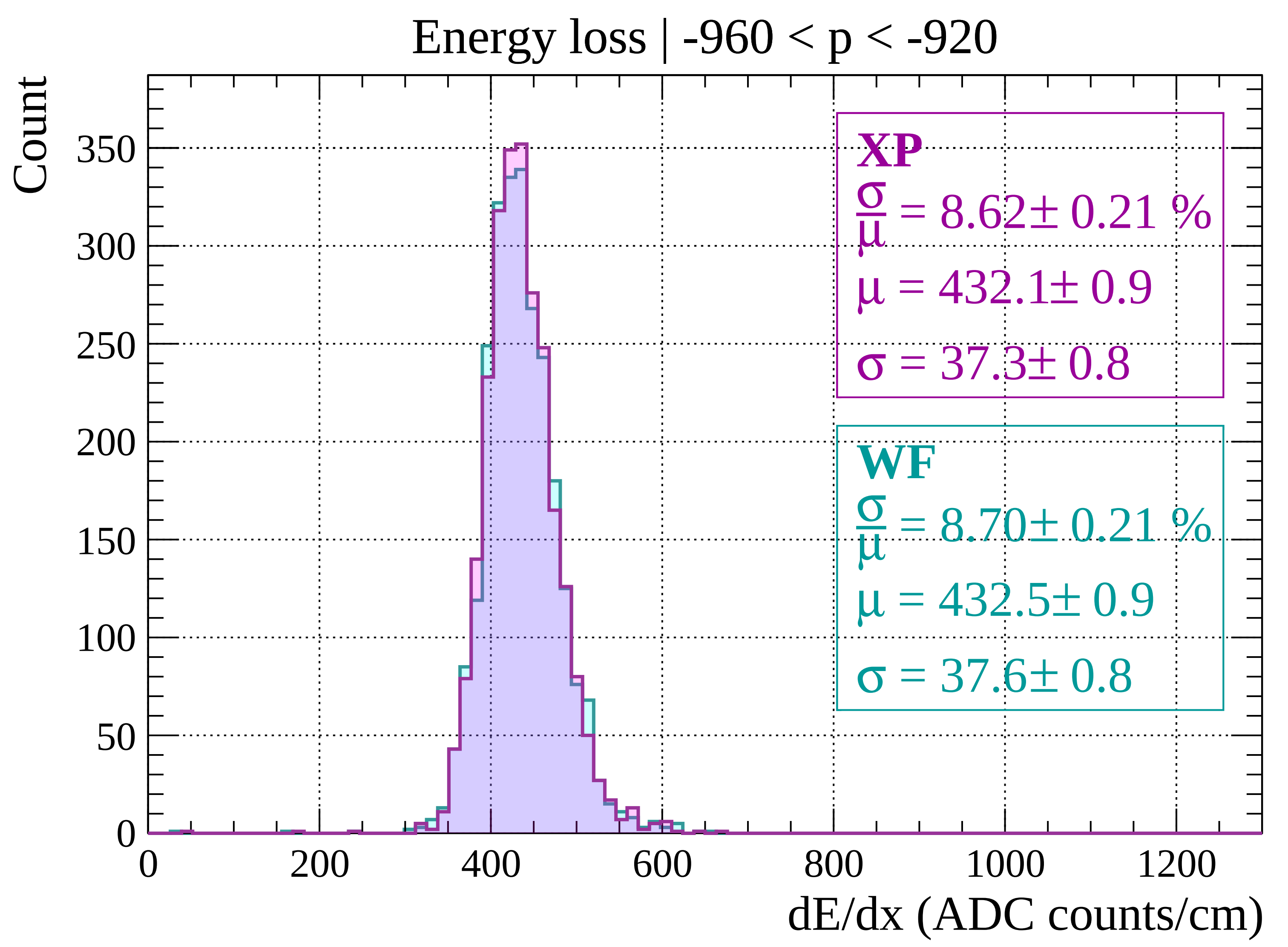


Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.73± 0.19 %
μ = 439.2± 0.9
σ = 38.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.05± 0.21 %
μ = 439.1± 1.0
σ = 39.7± 0.8

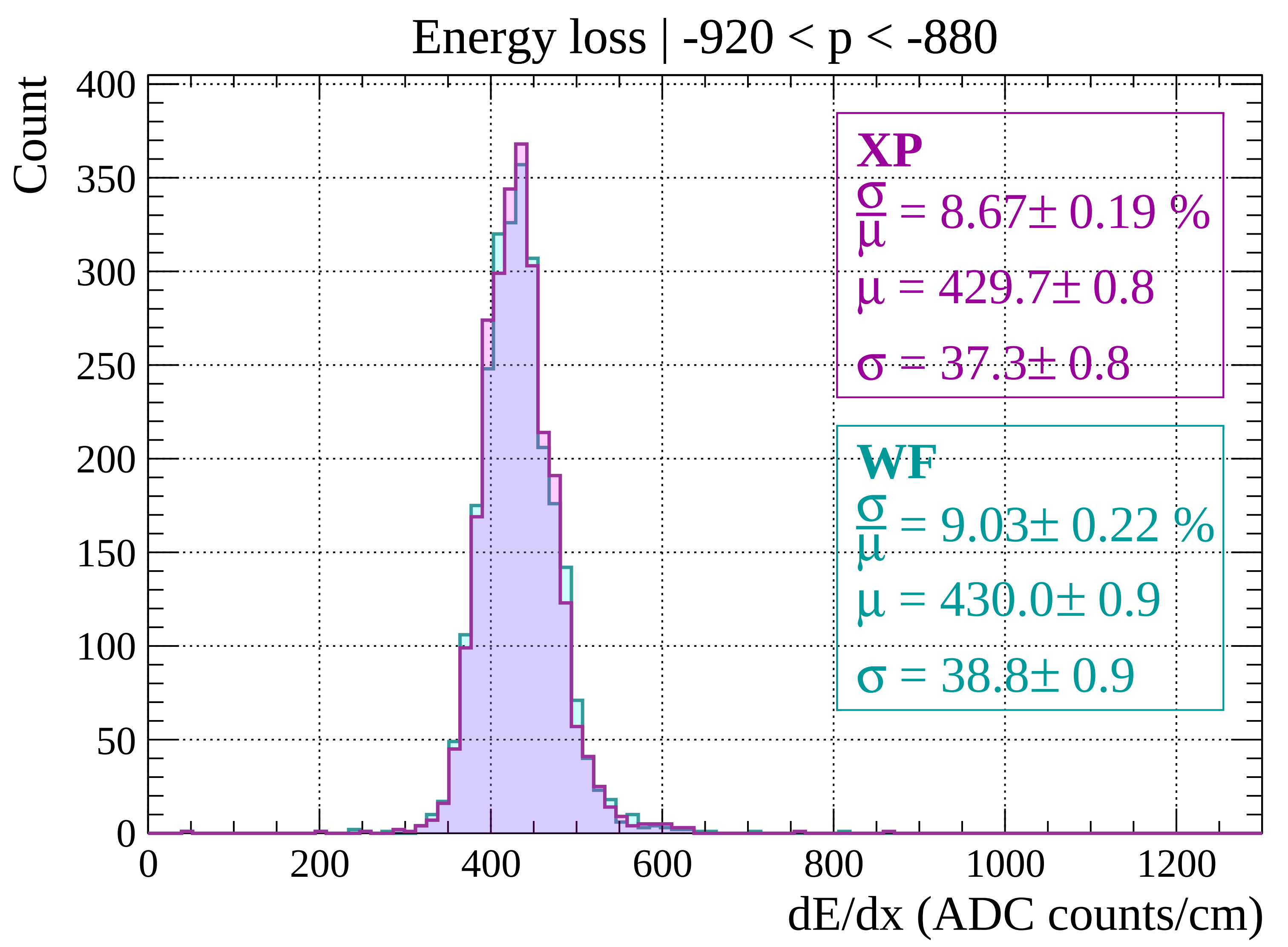

Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.91± 0.26 %
μ = 435.0± 1.0
σ = 38.8± 1.0
WF
σ/μ = 9.01± 0.31 %
μ = 435.9± 1.1
σ = 39.3± 1.2

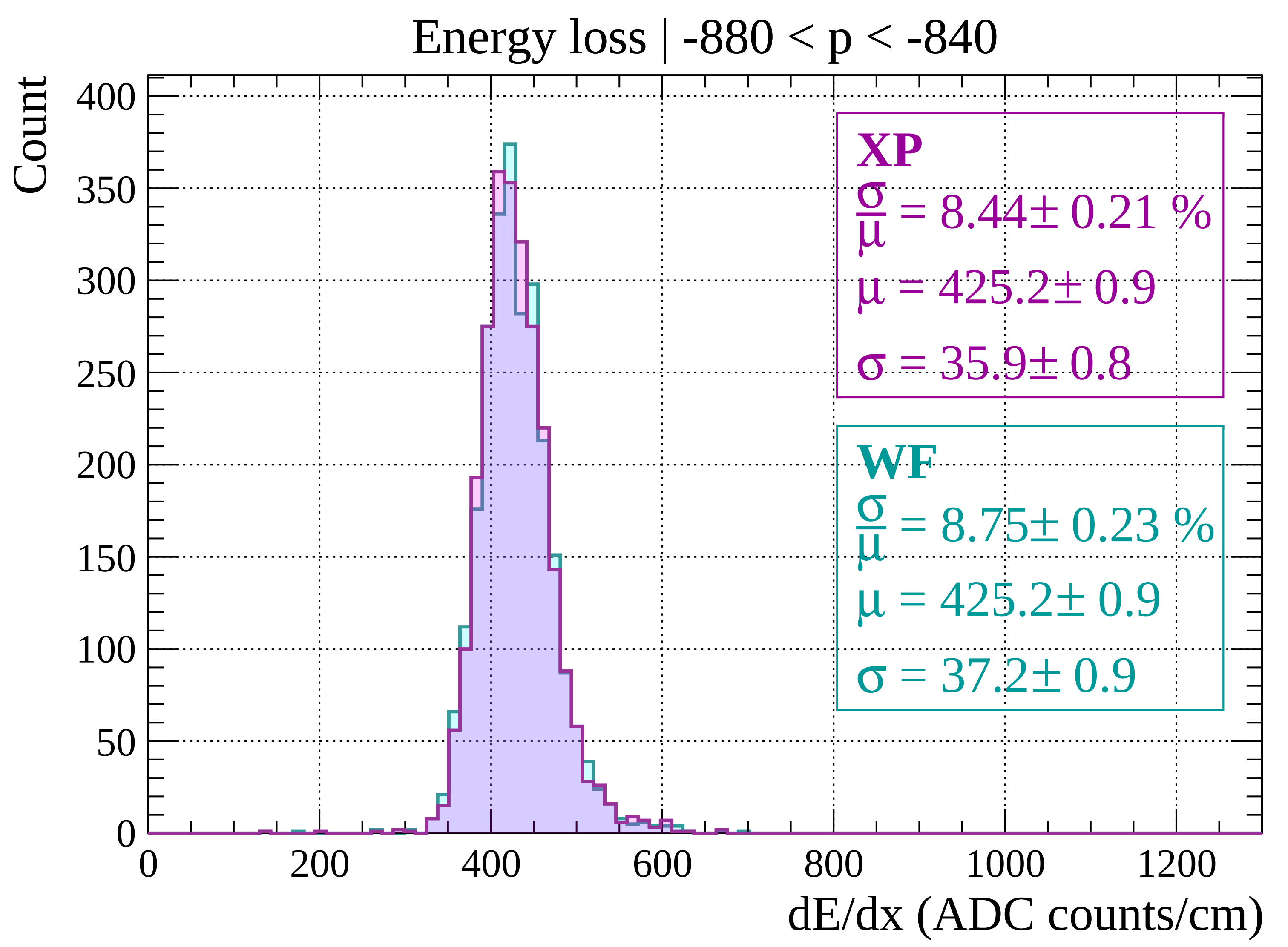


Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.55± 0.21 %
μ = 432.9± 0.9
σ = 37.0± 0.8
WF
σ/μ = 8.74± 0.21 %
μ = 432.8± 0.9
σ = 37.8± 0.8

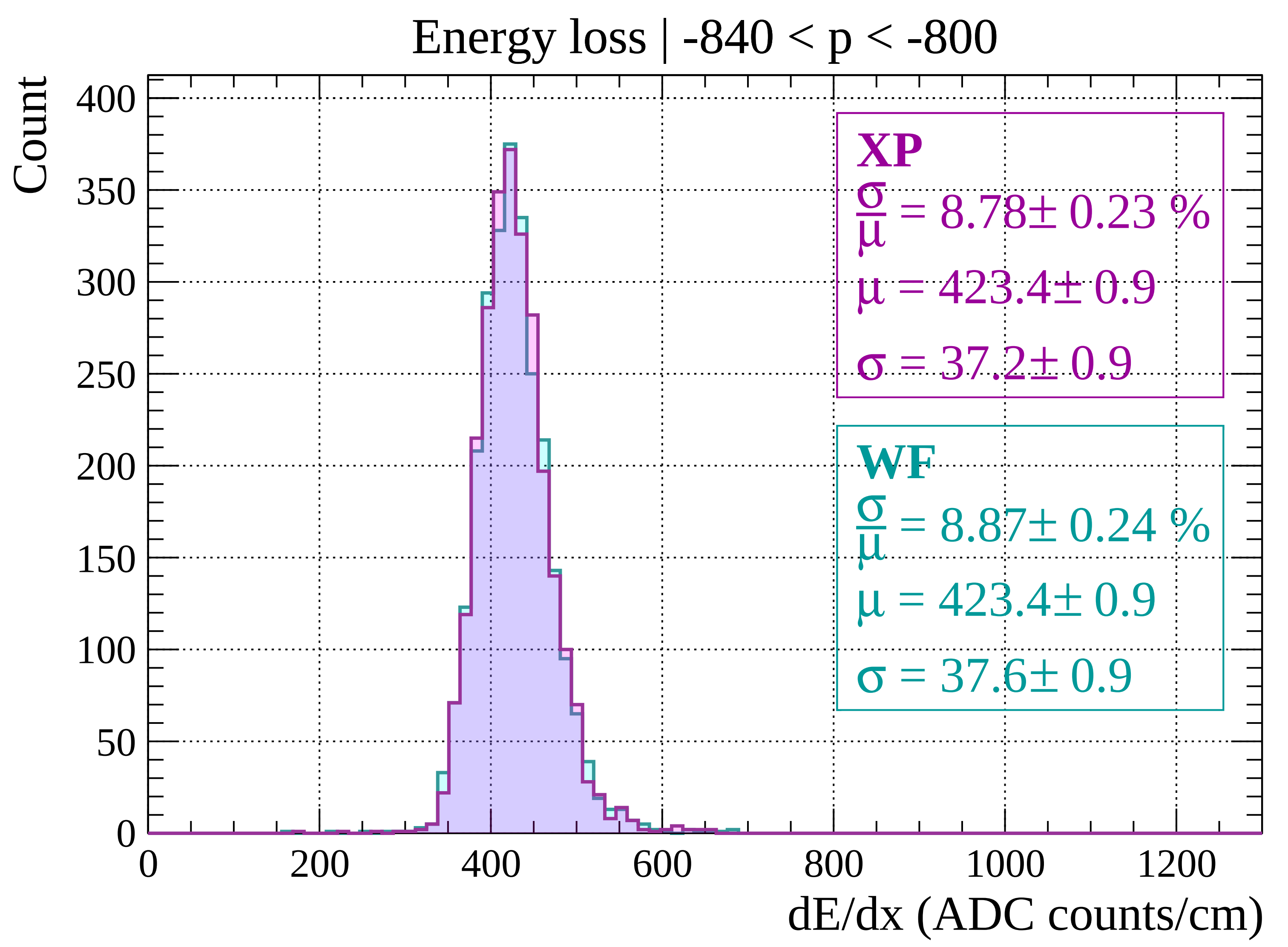


Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 8.62± 0.21 %
μ = 432.1± 0.9
σ = 37.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.70± 0.21 %
μ = 432.5± 0.9
σ = 37.6± 0.8

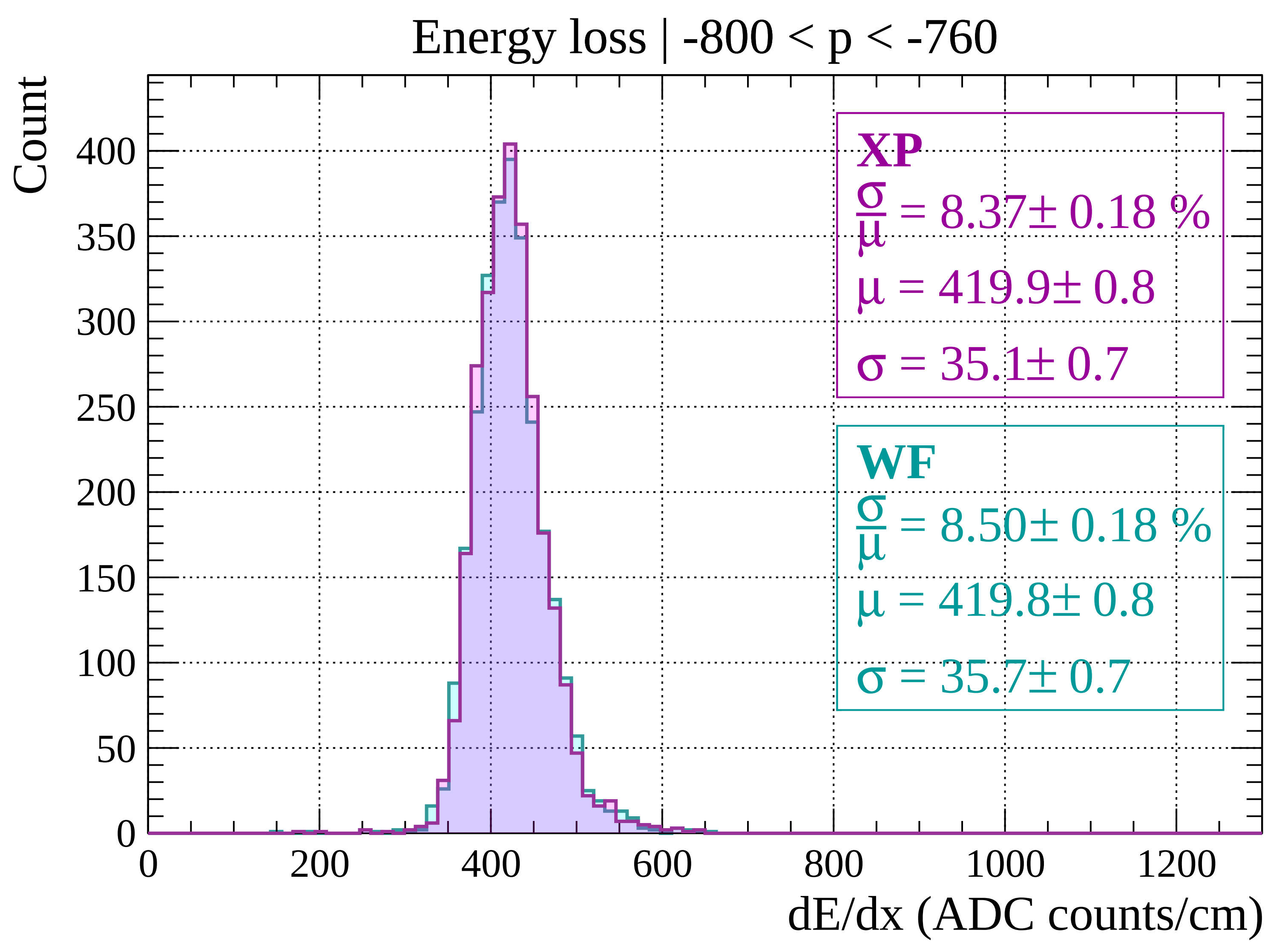

Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.67± 0.19 %
μ = 429.7± 0.8
σ = 37.3± 0.8
WF
σ/μ = 9.03± 0.22 %
μ = 430.0± 0.9
σ = 38.8± 0.9
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400



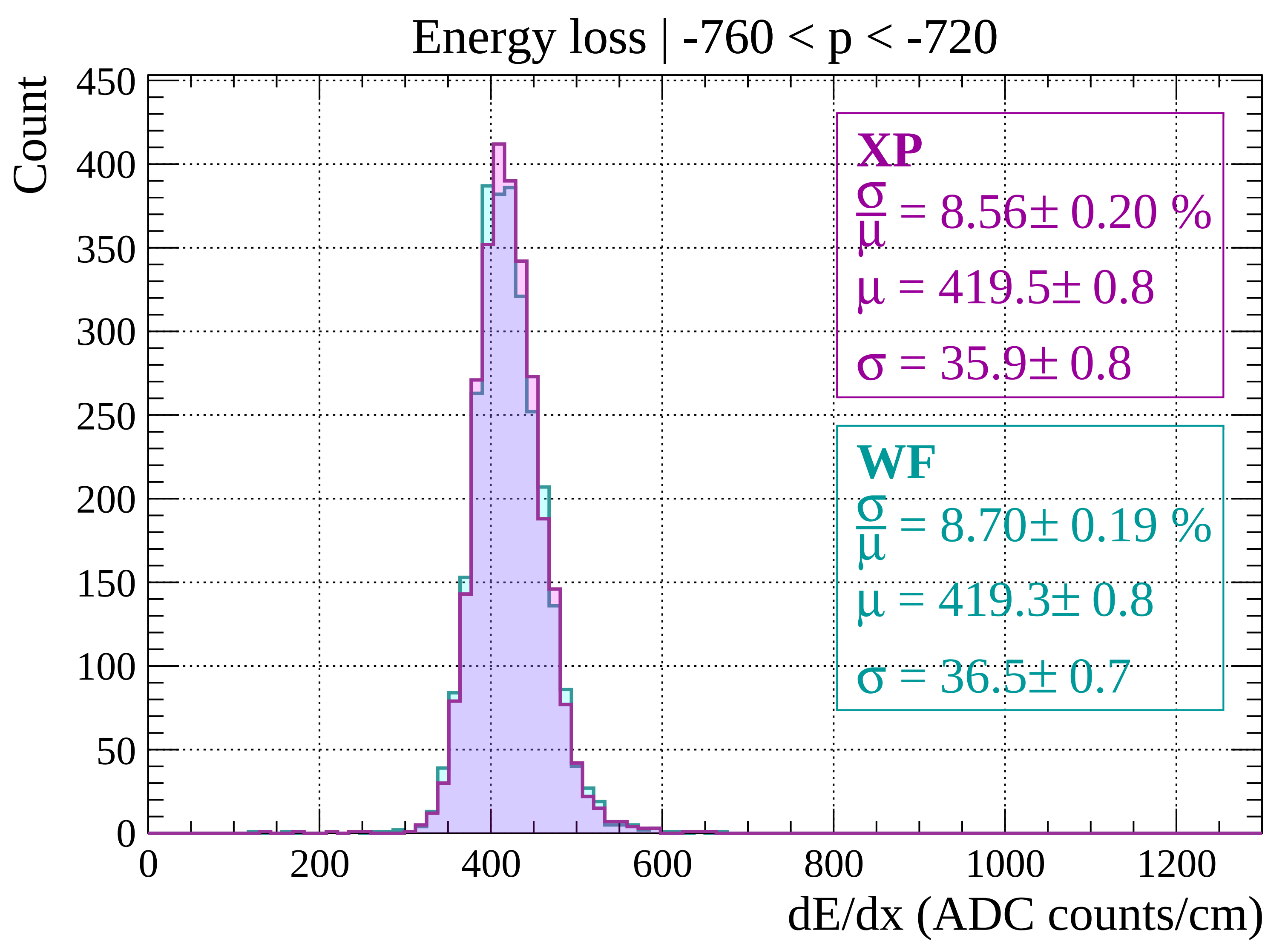
Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.44± 0.21 %
μ = 425.2± 0.9
σ = 35.9± 0.8
WF
σ/μ = 8.75± 0.23 %
μ = 425.2± 0.9
σ = 37.2± 0.9



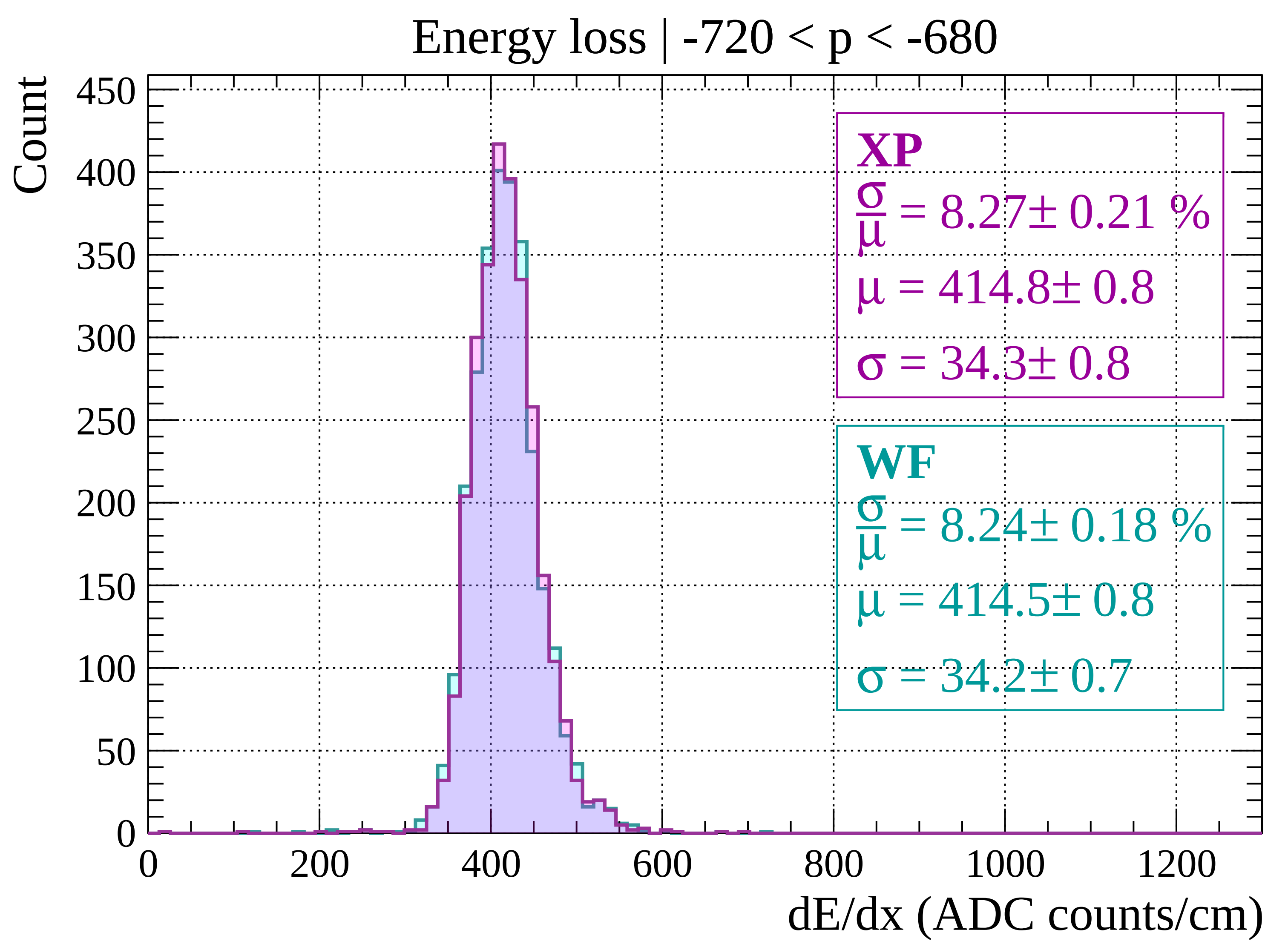
Energy loss | -840 < p < -800
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.78± 0.23 %
μ = 423.4± 0.9
σ = 37.2± 0.9
WF
σ/μ = 8.87± 0.24 %
μ = 423.4± 0.9
σ = 37.6± 0.9



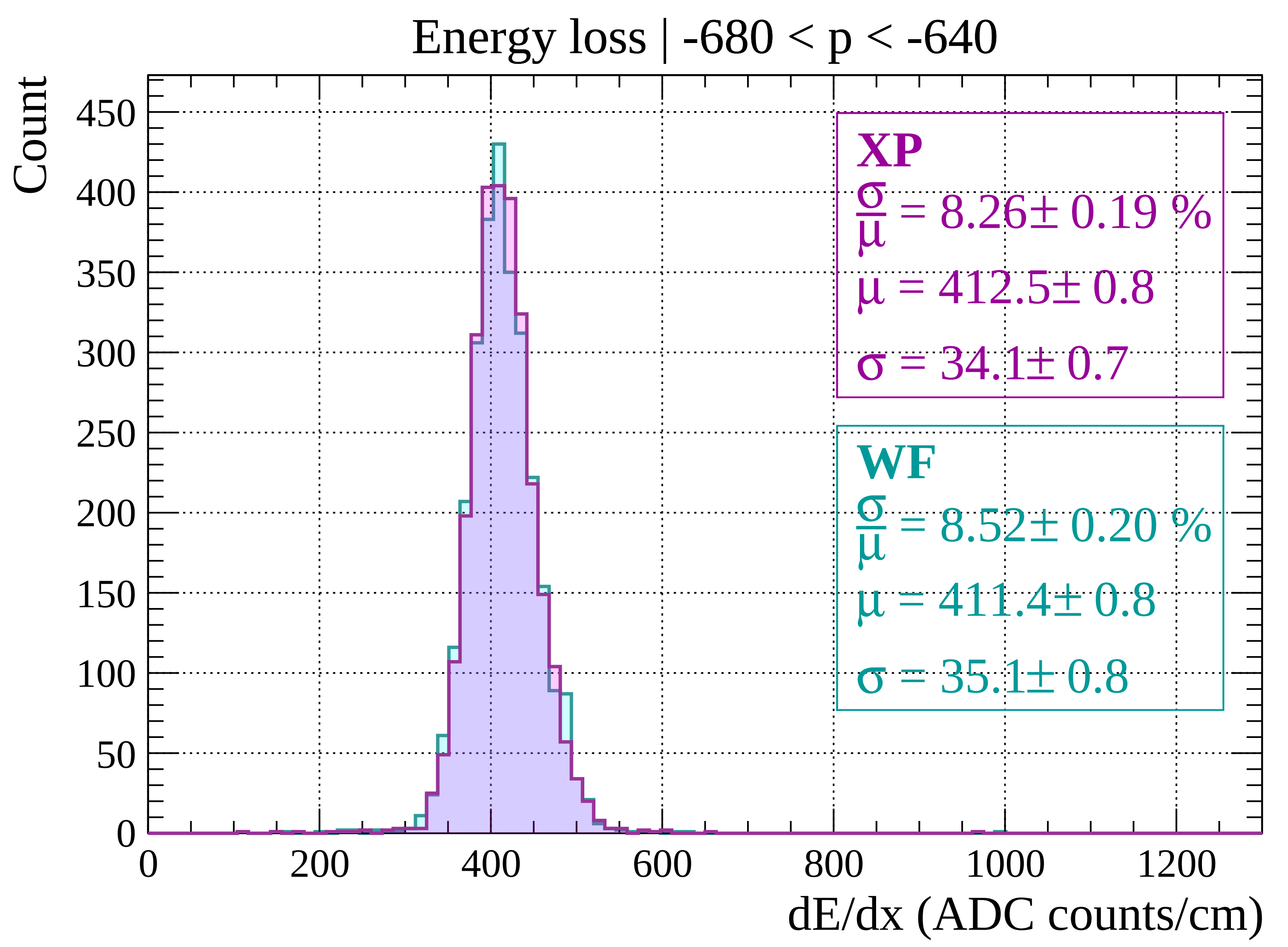
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 8.37± 0.18 %
μ = 419.9± 0.8
σ = 35.1± 0.7
WF
σ/μ = 8.50± 0.18 %
μ = 419.8± 0.8
σ = 35.7± 0.7

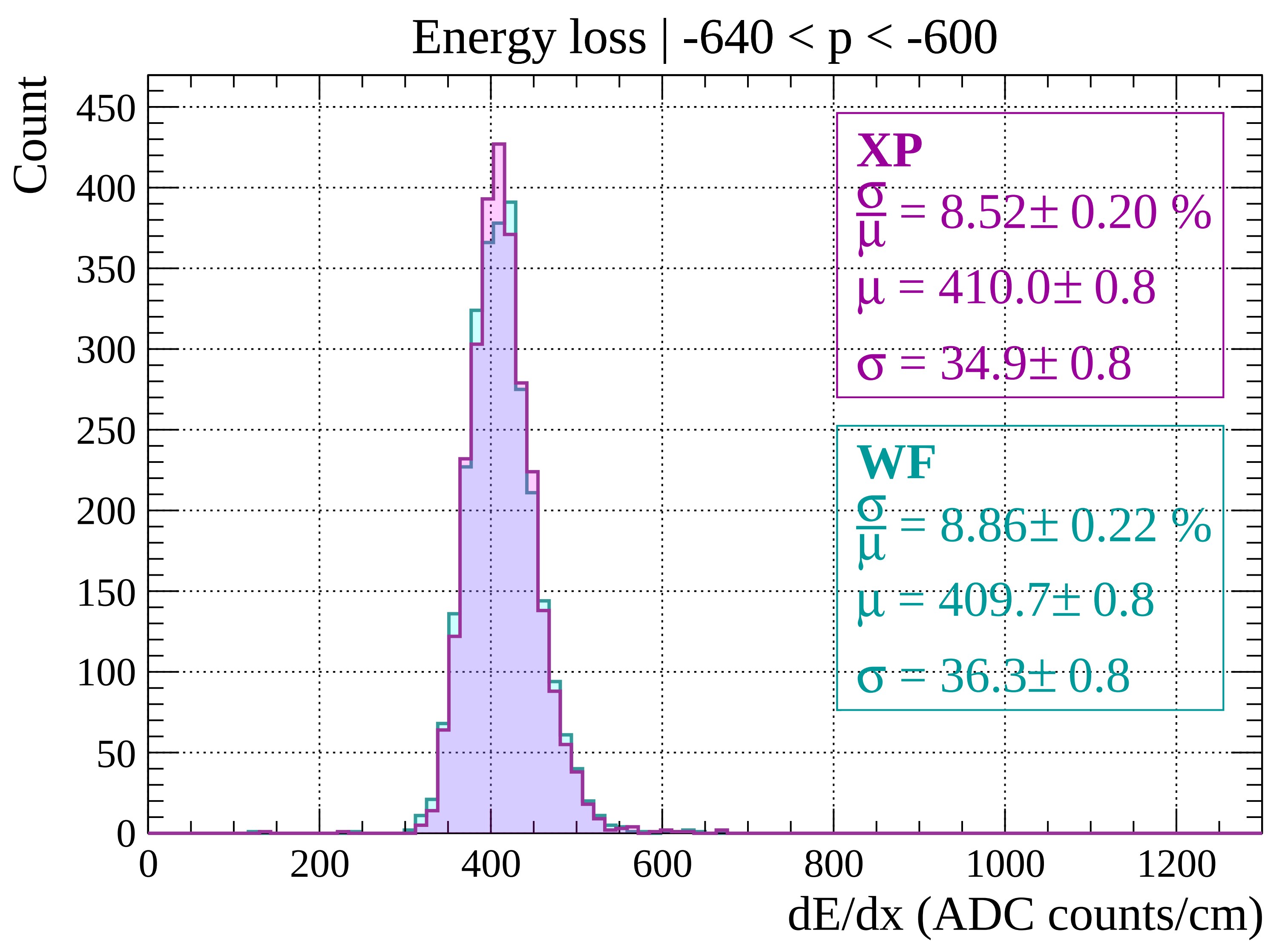


Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.56± 0.20 %
μ = 419.5± 0.8
σ = 35.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.70± 0.19 %
μ = 419.3± 0.8
σ = 36.5± 0.7

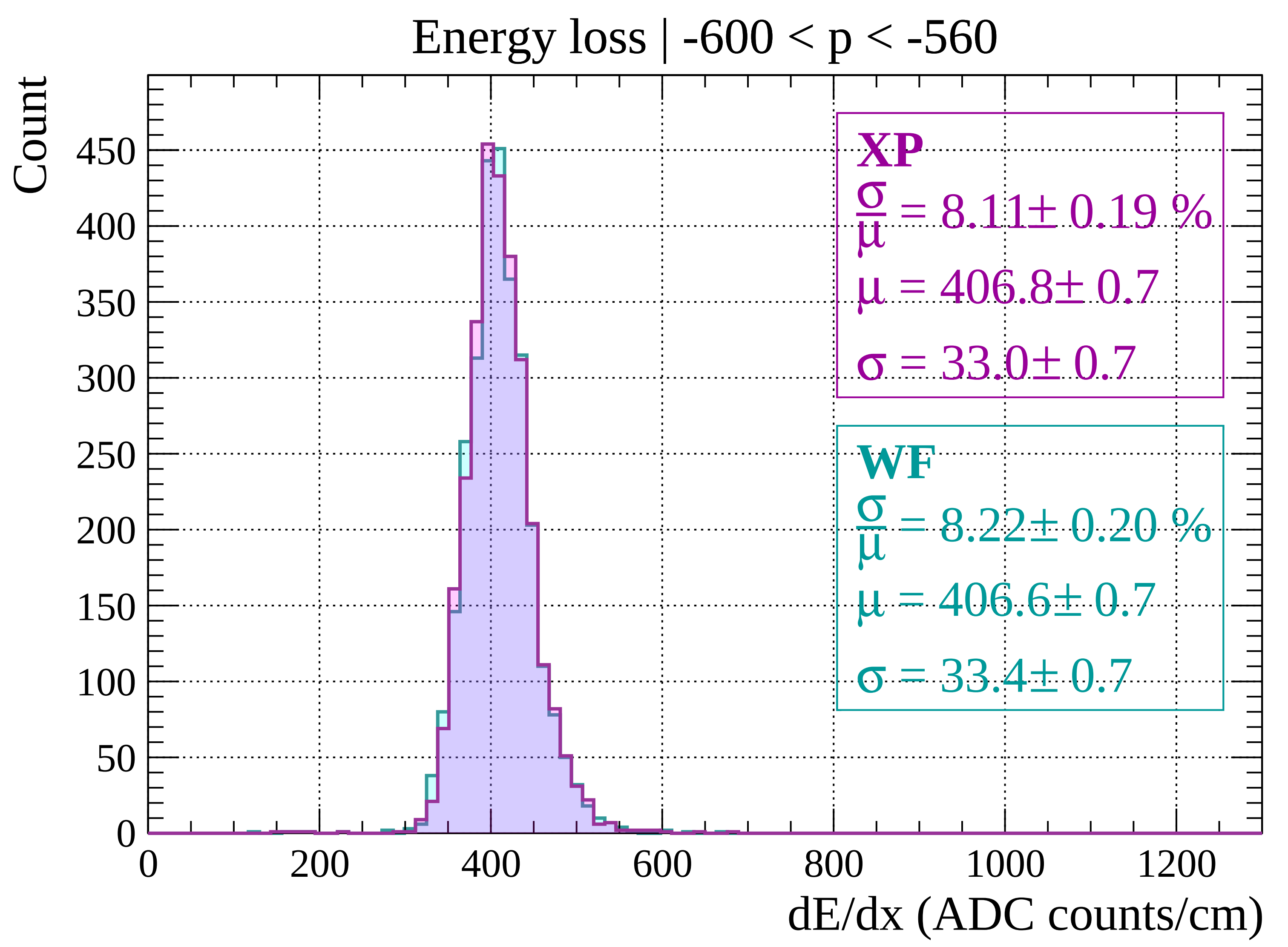


Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.27± 0.21 %
μ = 414.8± 0.8
σ = 34.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.24± 0.18 %
μ = 414.5± 0.8
σ = 34.2± 0.7
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

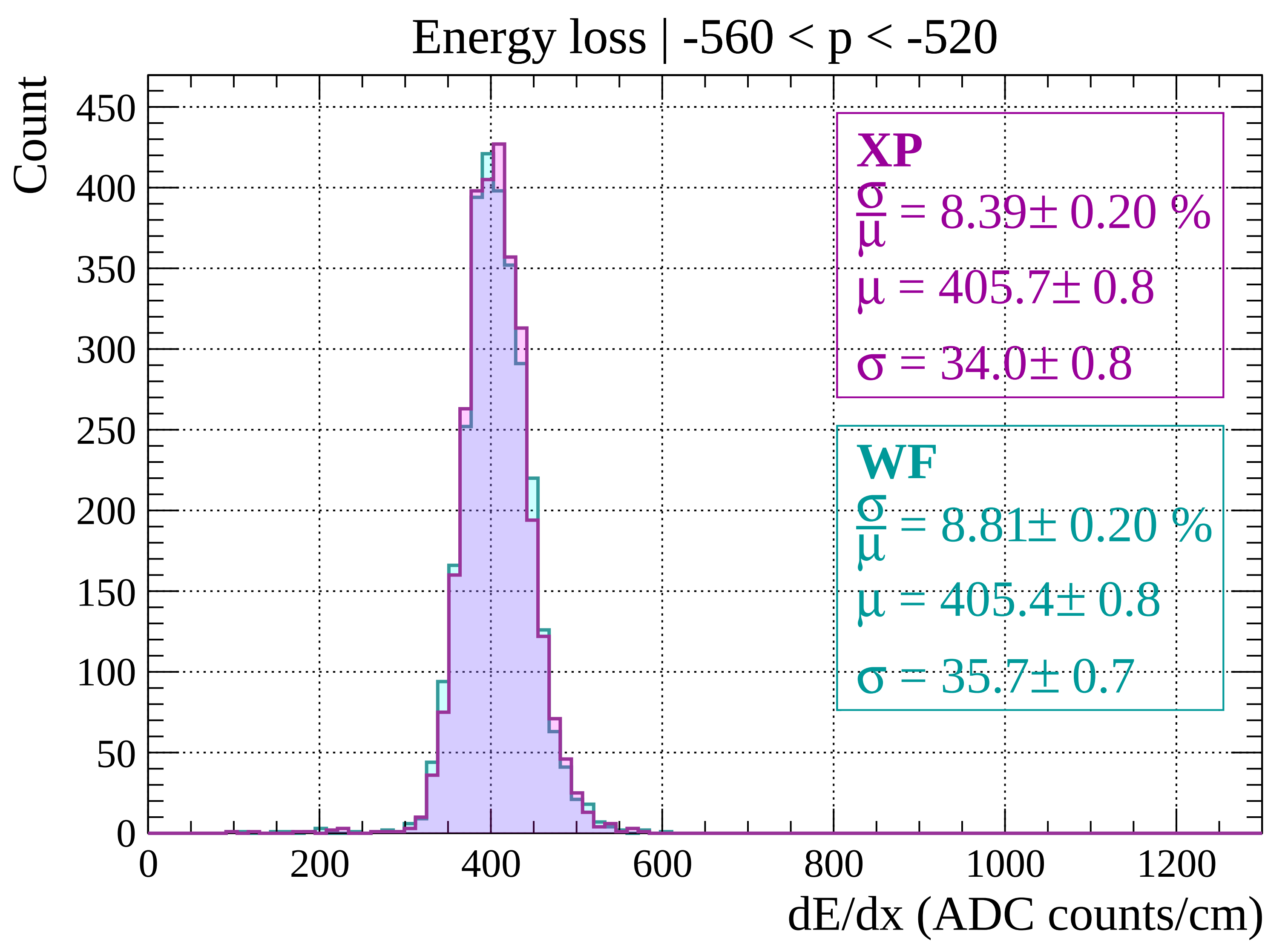

Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.26± 0.19 %
μ = 412.5± 0.8
σ = 34.1± 0.7
WF
σ/μ = 8.52± 0.20 %
μ = 411.4± 0.8
σ = 35.1± 0.8



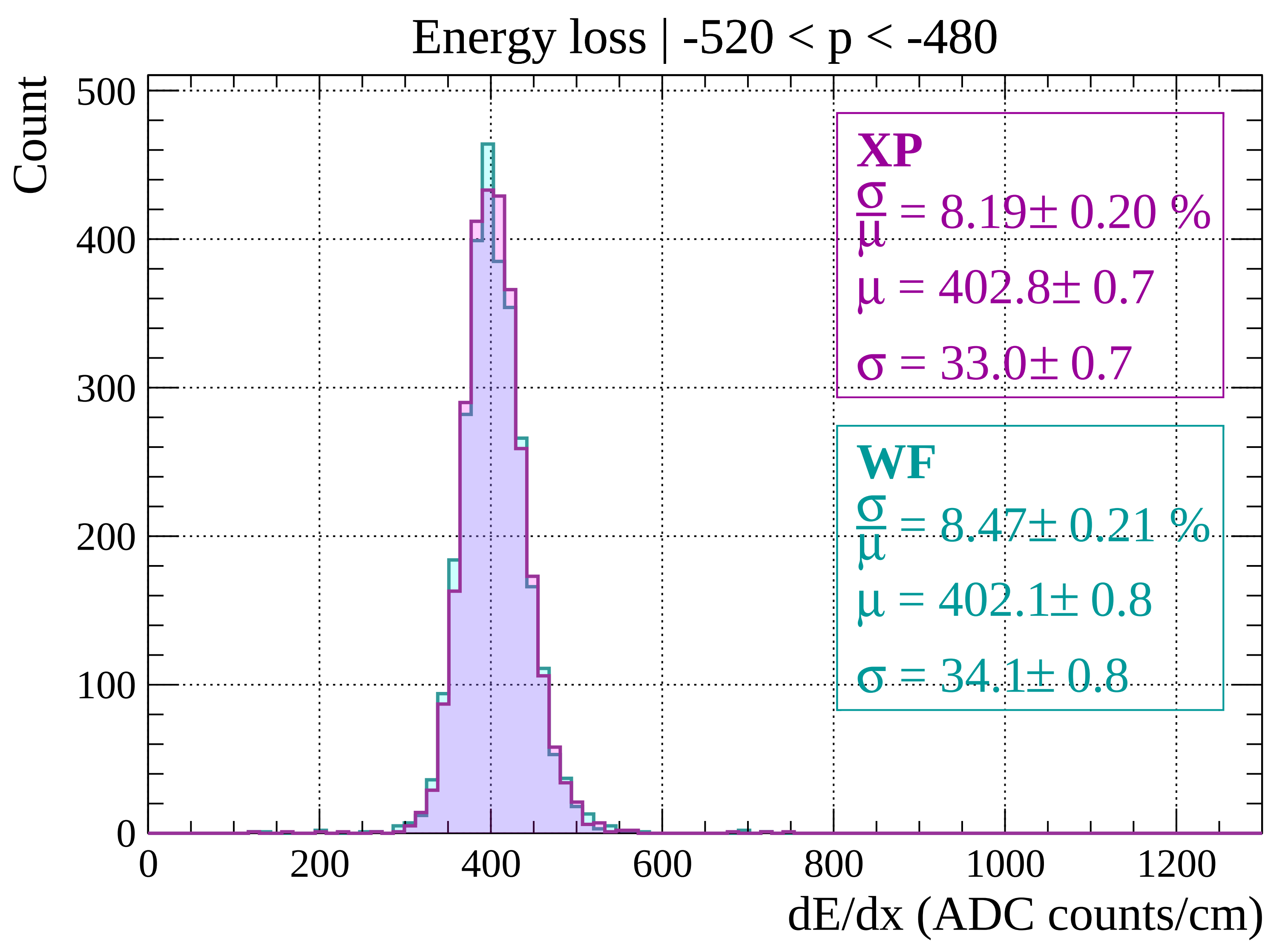
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.52 \pm 0.20$ %
$\mu = 410.0 \pm 0.8$
$\sigma = 34.9 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 8.86 \pm 0.22$ %
$\mu = 409.7 \pm 0.8$
$\sigma = 36.3 \pm 0.8$



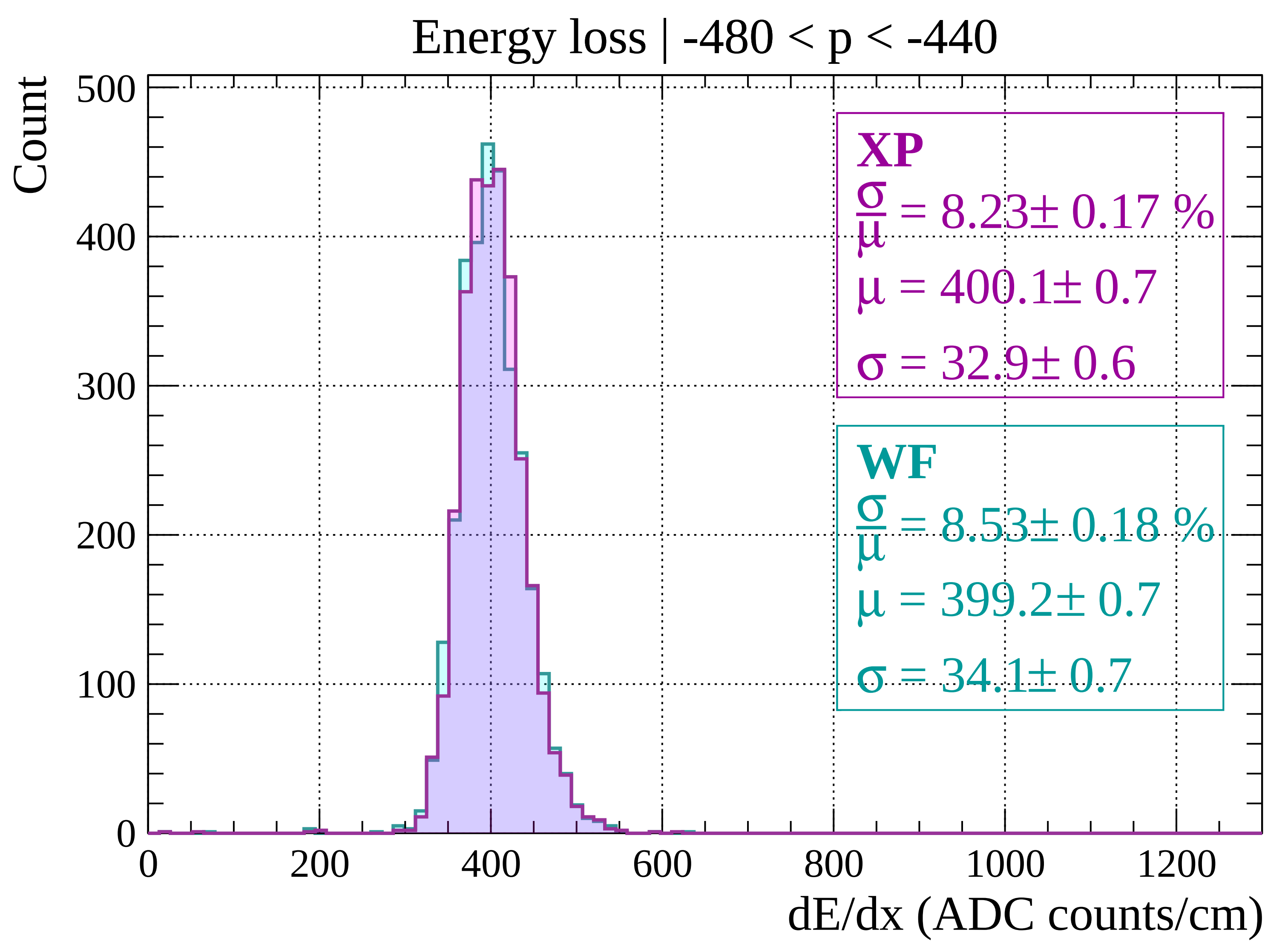
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.11± 0.19 %
μ = 406.8± 0.7
σ = 33.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.22± 0.20 %
μ = 406.6± 0.7
σ = 33.4± 0.7
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

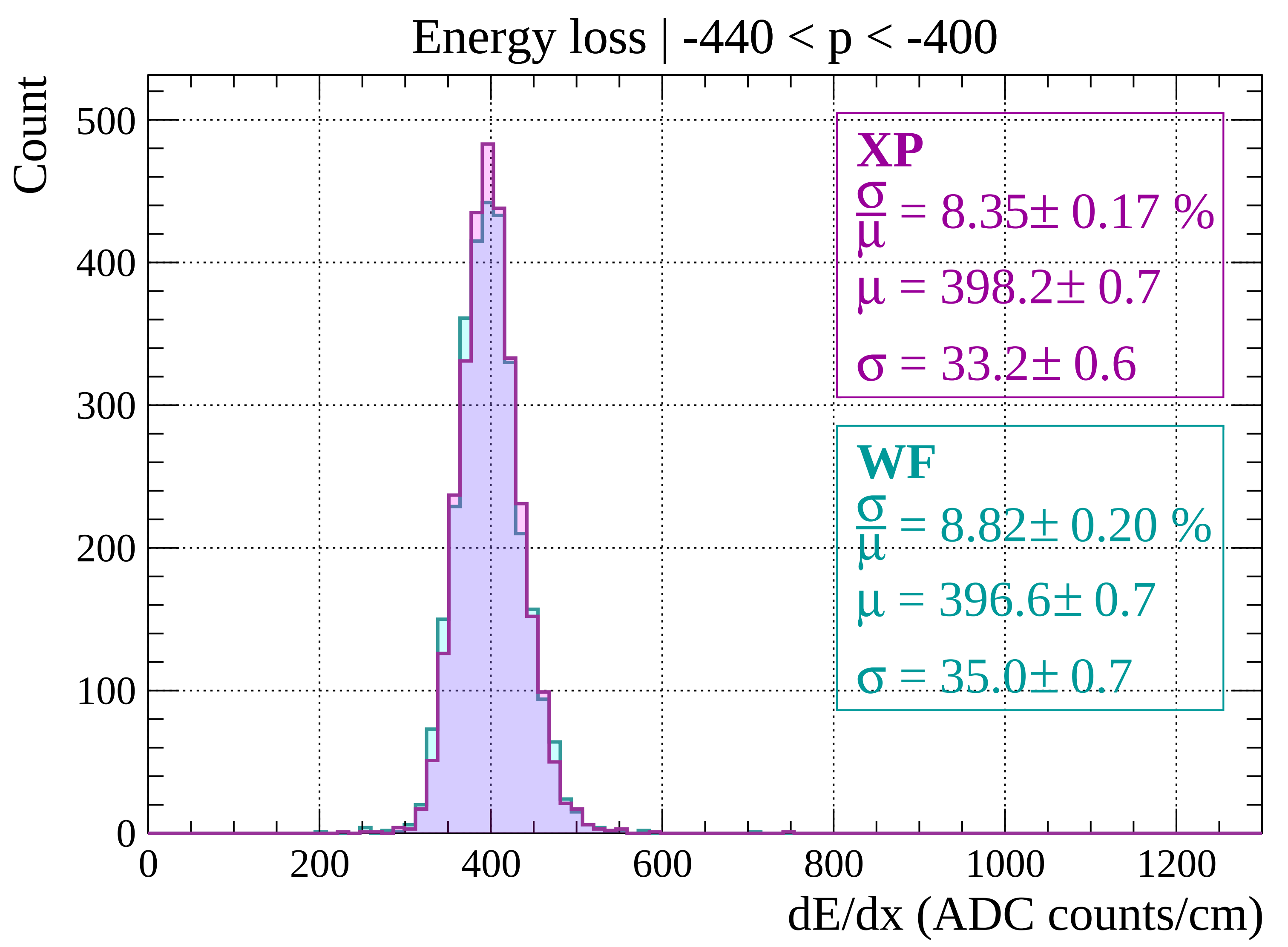


Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.39± 0.20 %
μ = 405.7± 0.8
σ = 34.0± 0.8
WF
σ/μ = 8.81± 0.20 %
μ = 405.4± 0.8
σ = 35.7± 0.7

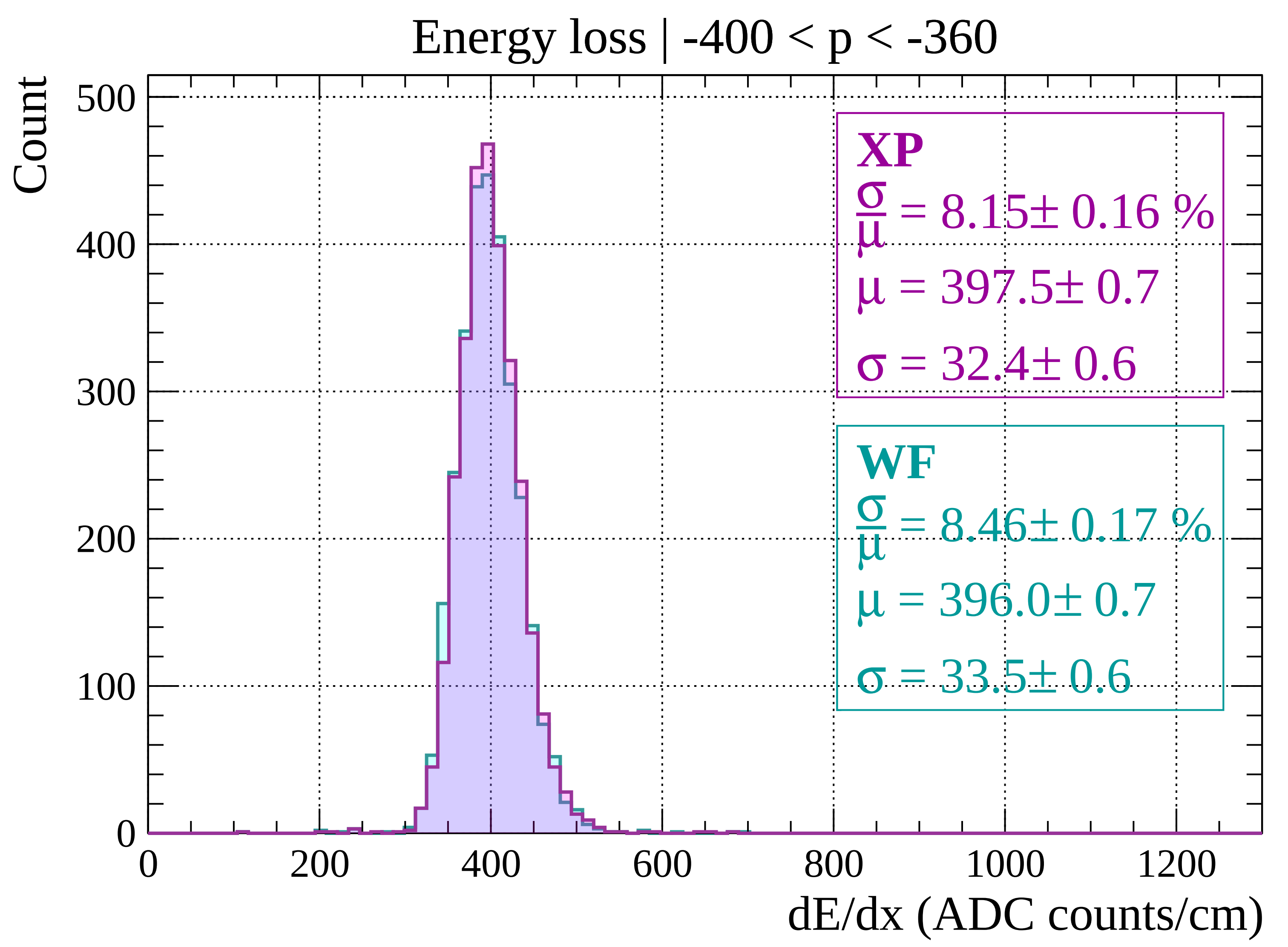


Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.19± 0.20 %
μ = 402.8± 0.7
σ = 33.0± 0.7
WF
σ/μ = 8.47± 0.21 %
μ = 402.1± 0.8
σ = 34.1± 0.8

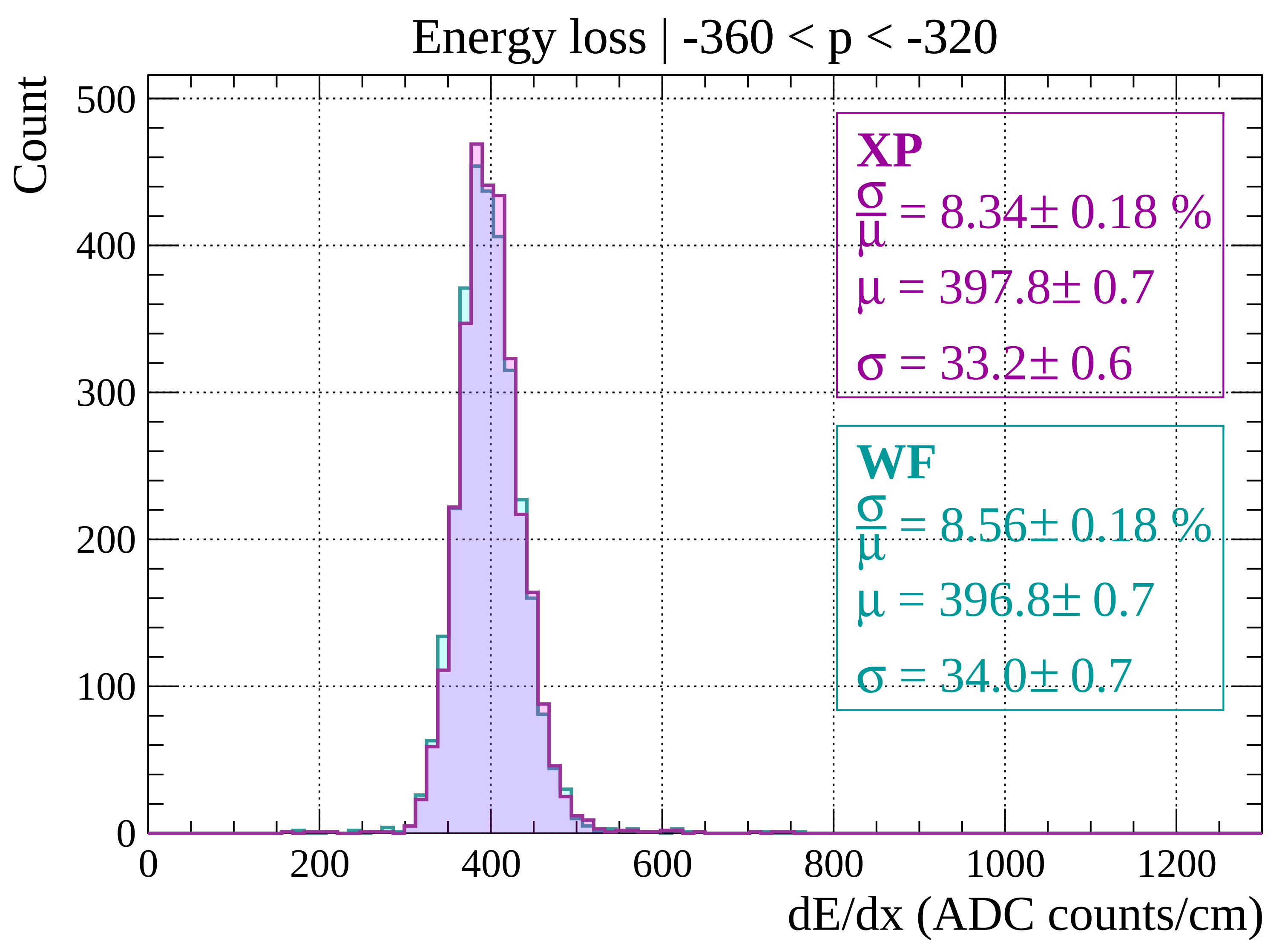

Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.23± 0.17 %
μ = 400.1± 0.7
σ = 32.9± 0.6
WF
σ/μ = 8.53± 0.18 %
μ = 399.2± 0.7
σ = 34.1± 0.7

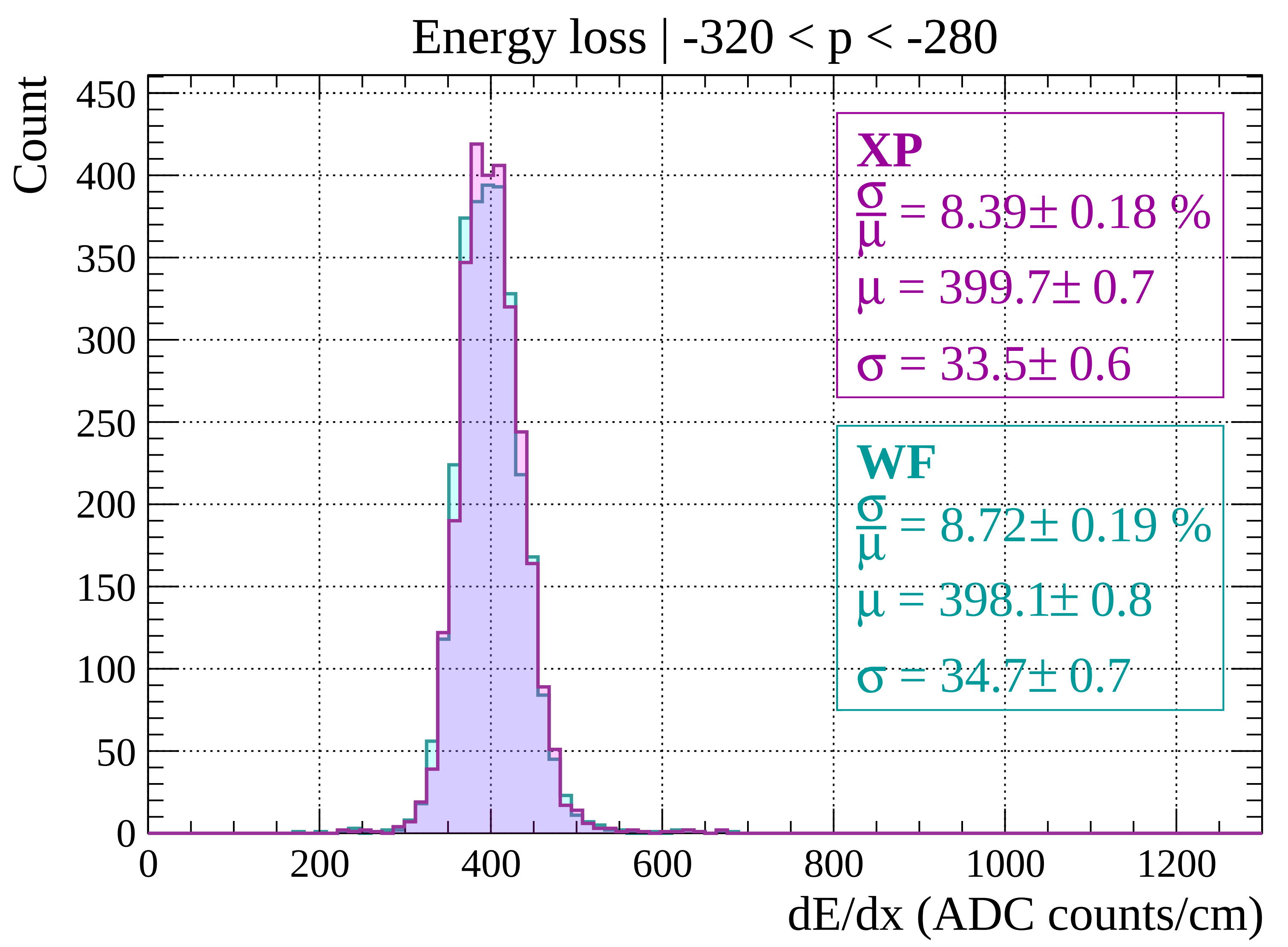


Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.35± 0.17 %
μ = 398.2± 0.7
σ = 33.2± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.82± 0.20 %
μ = 396.6± 0.7
σ = 35.0± 0.7

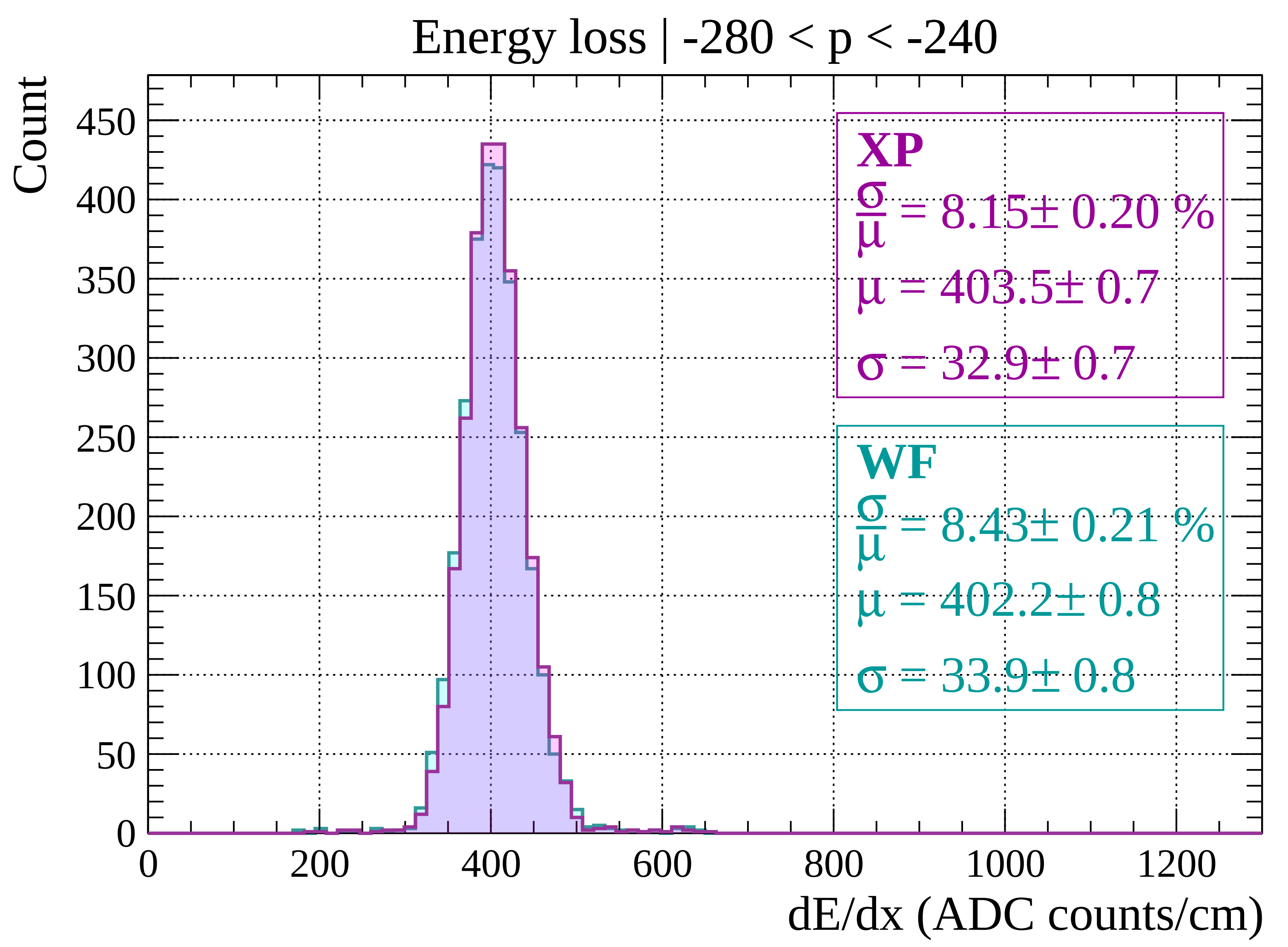


Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.15± 0.16 %
μ = 397.5± 0.7
σ = 32.4± 0.6
WF
σ/μ = 8.46± 0.17 %
μ = 396.0± 0.7
σ = 33.5± 0.6

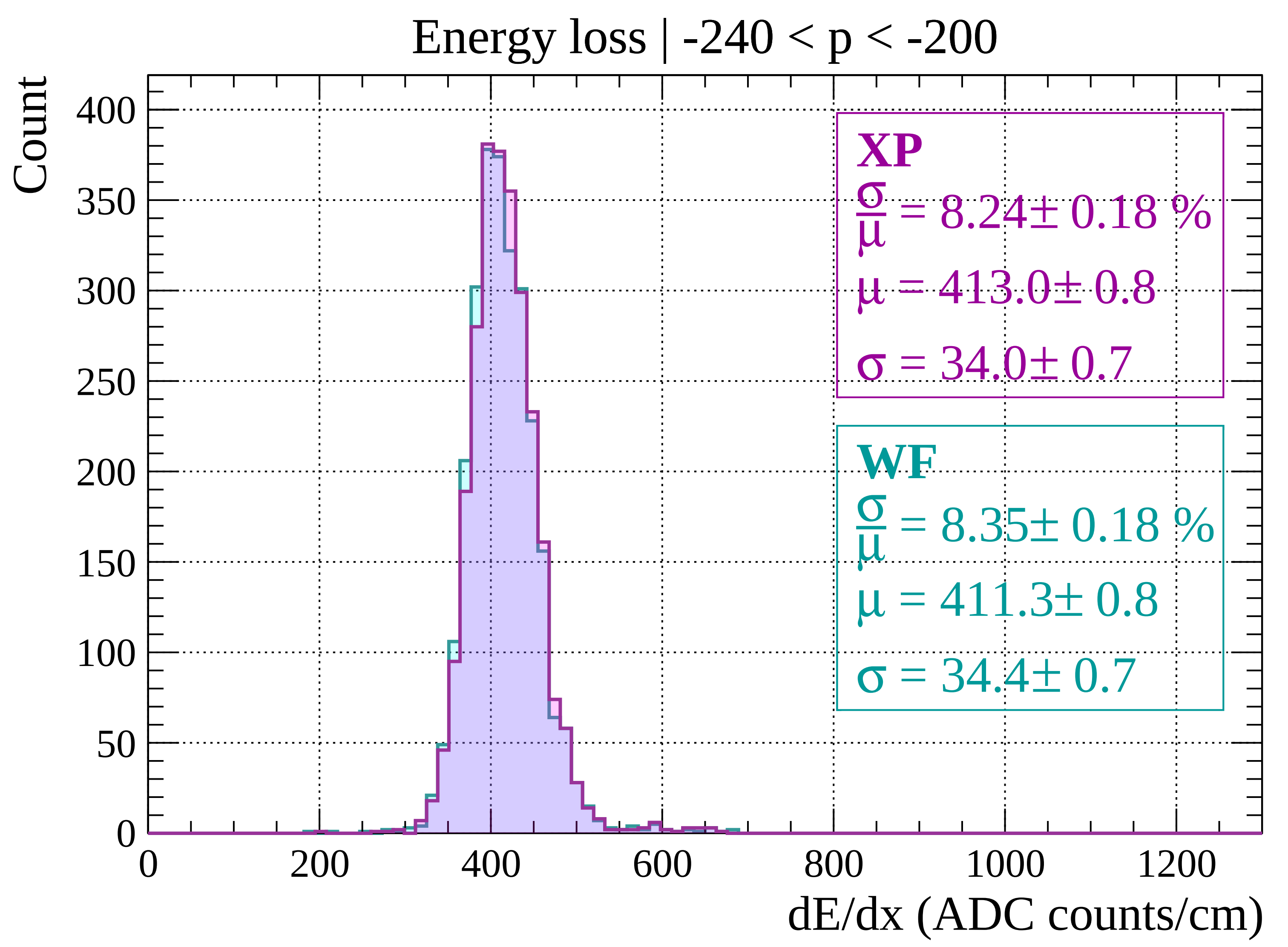

Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.34 \pm 0.18$ %
$\mu = 397.8 \pm 0.7$
$\sigma = 33.2 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 8.56 \pm 0.18$ %
$\mu = 396.8 \pm 0.7$
$\sigma = 34.0 \pm 0.7$



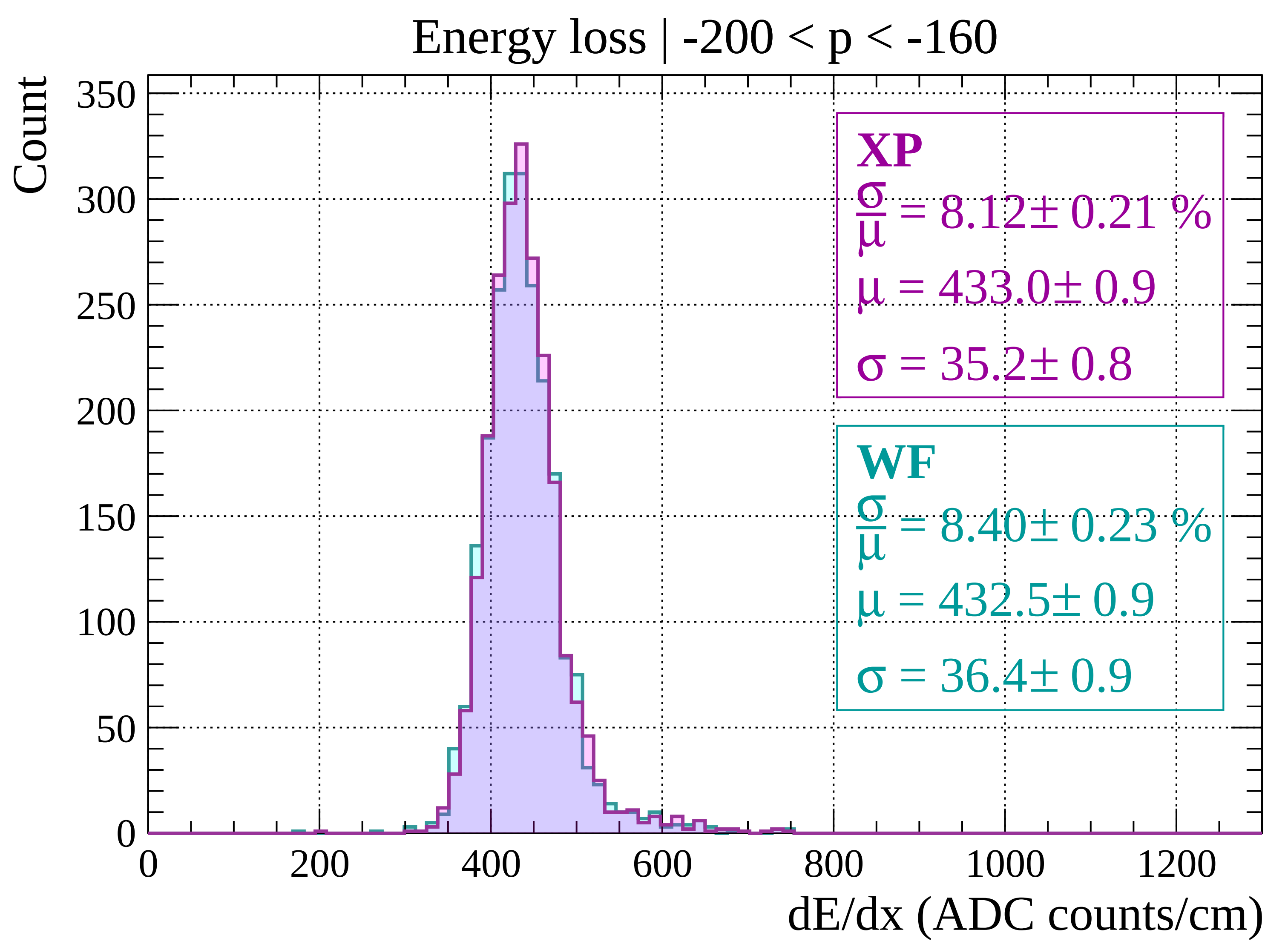
Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP
$\frac{\sigma}{\mu}$ = 8.39± 0.18 %
μ = 399.7± 0.7
σ = 33.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.72± 0.19 %
μ = 398.1± 0.8
σ = 34.7± 0.7



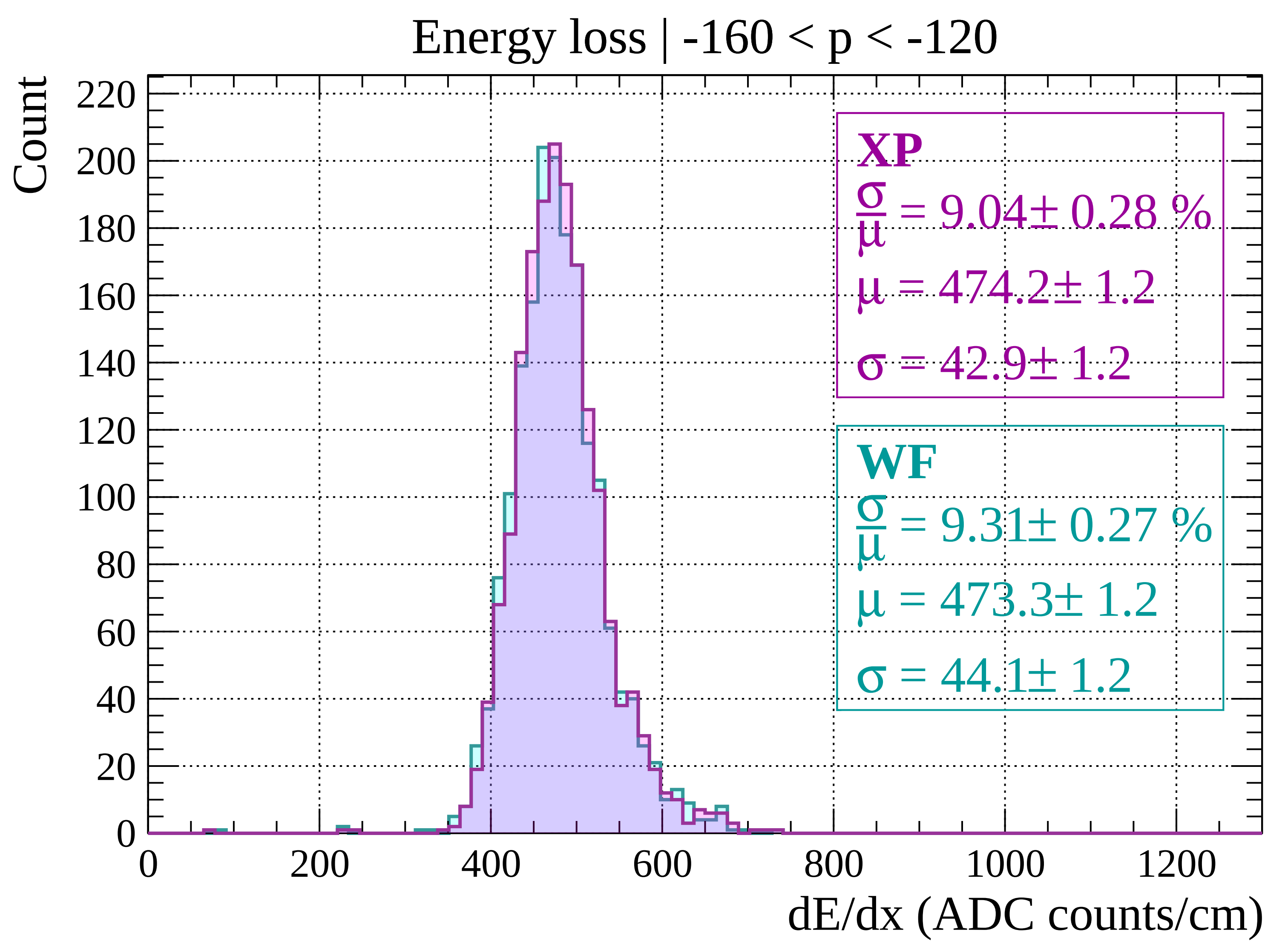
Energy loss | -280 < p < -240
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.15± 0.20 %
μ = 403.5± 0.7
σ = 32.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.43± 0.21 %
μ = 402.2± 0.8
σ = 33.9± 0.8



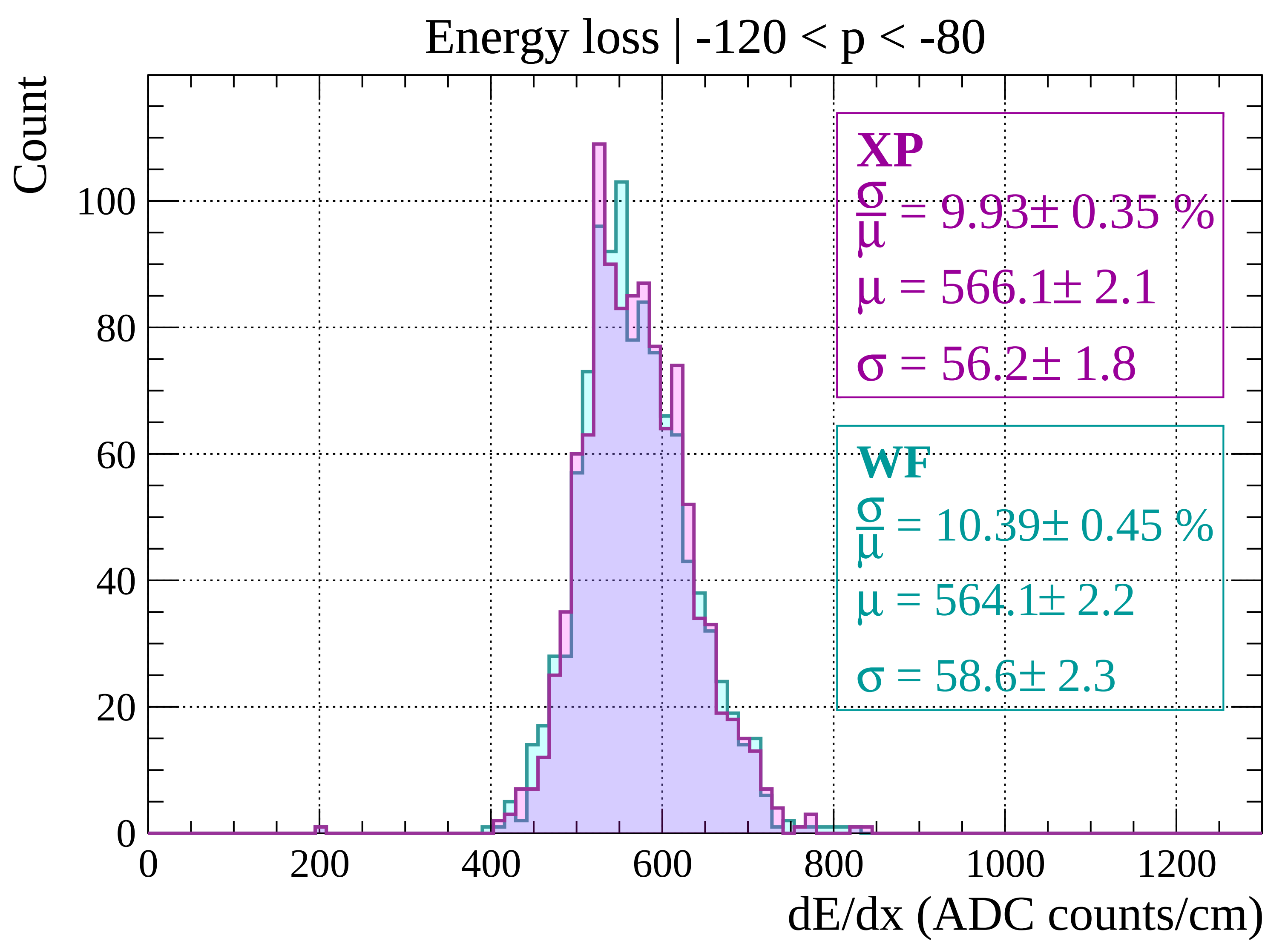
Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 8.24± 0.18 %
μ = 413.0± 0.8
σ = 34.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.35± 0.18 %
μ = 411.3± 0.8
σ = 34.4± 0.7



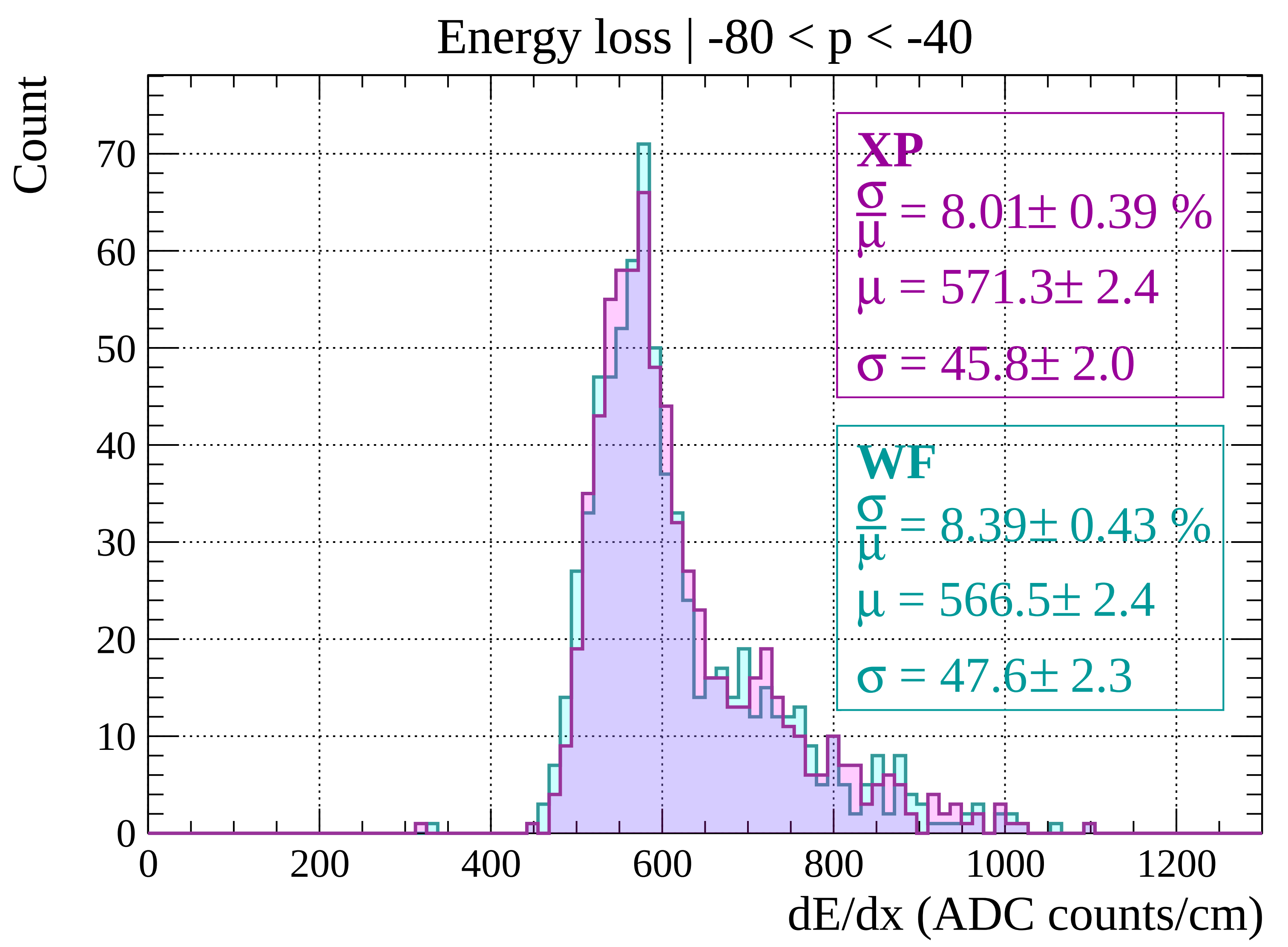
Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.12 ± 0.21 %
μ = 433.0 ± 0.9
σ = 35.2 ± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.40 ± 0.23 %
μ = 432.5 ± 0.9
σ = 36.4 ± 0.9

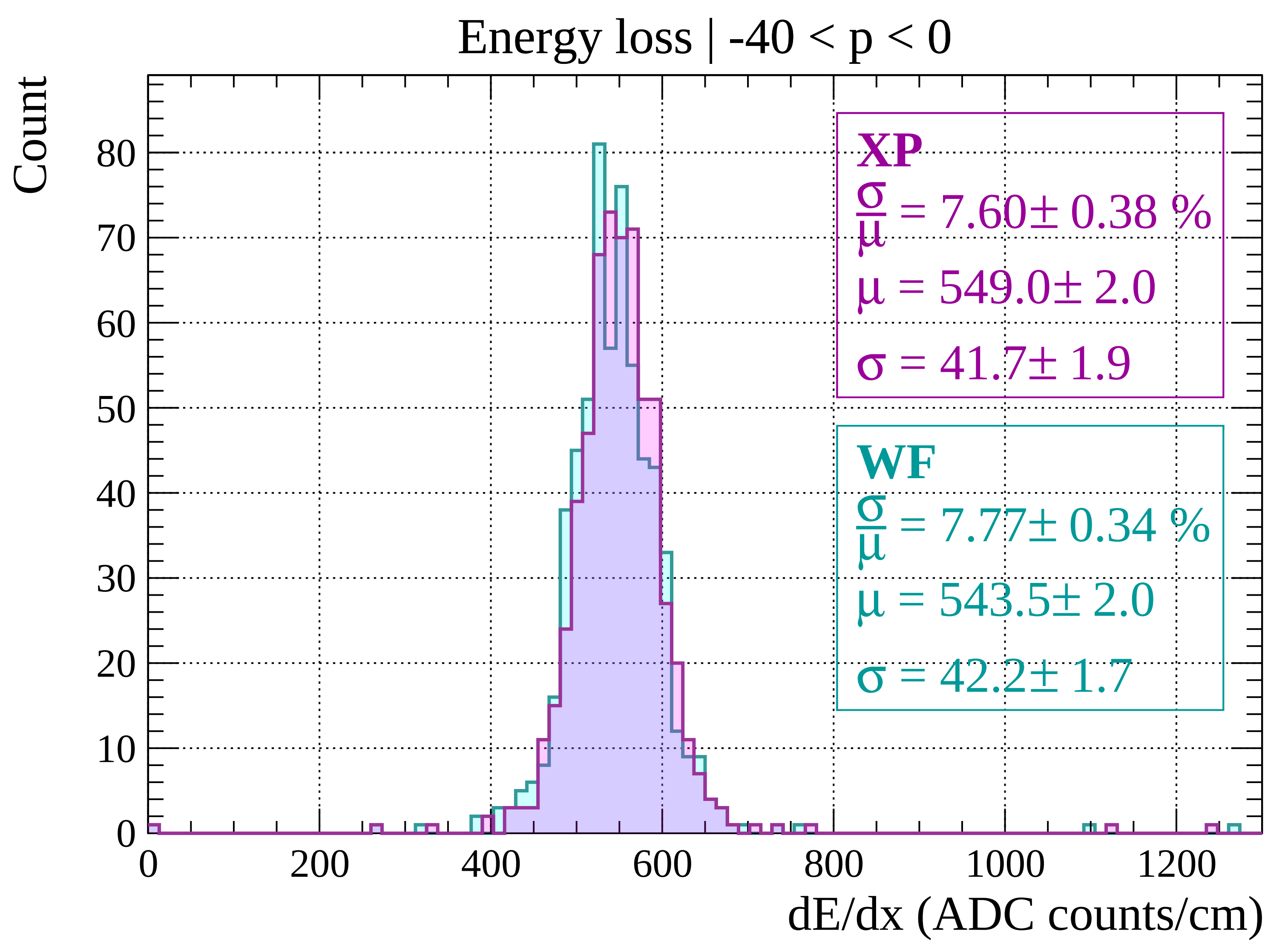


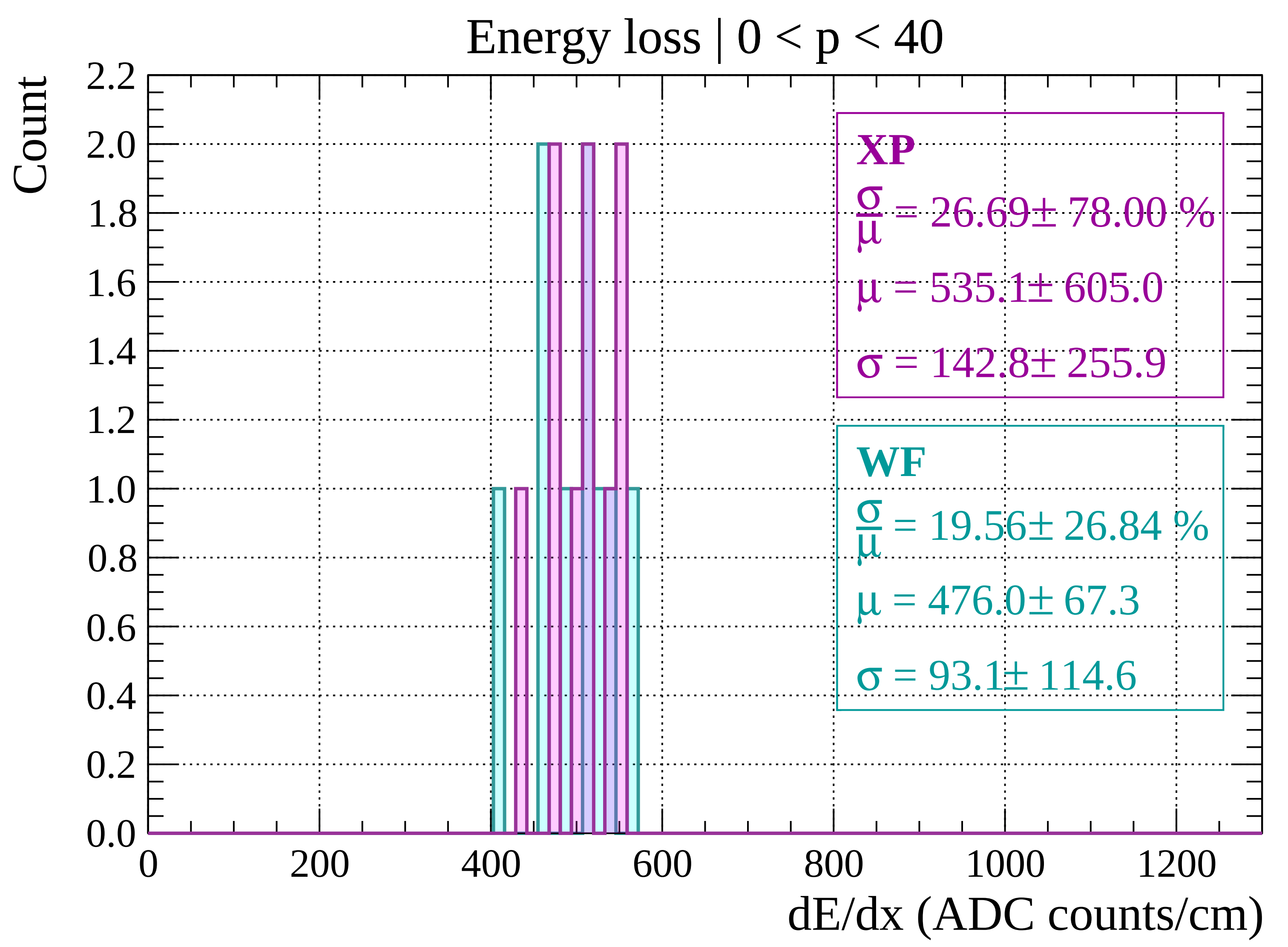
Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 9.04± 0.28 %
μ = 474.2± 1.2
σ = 42.9± 1.2
WF
σ/μ = 9.31± 0.27 %
μ = 473.3± 1.2
σ = 44.1± 1.2



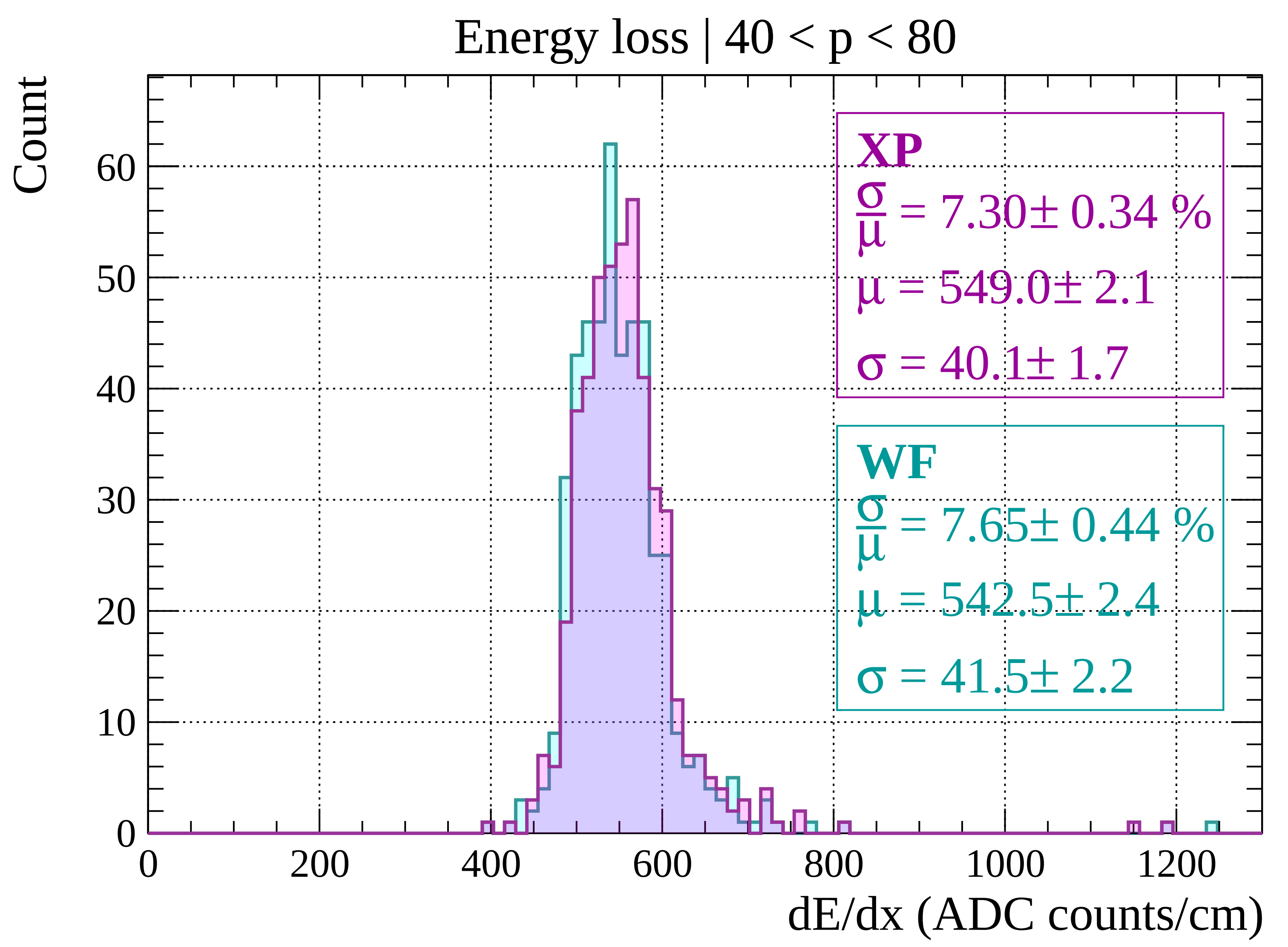
Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.93± 0.35 %
μ = 566.1± 2.1
σ = 56.2± 1.8
WF
$\frac{\sigma}{\mu}$ = 10.39± 0.45 %
μ = 564.1± 2.2
σ = 58.6± 2.3

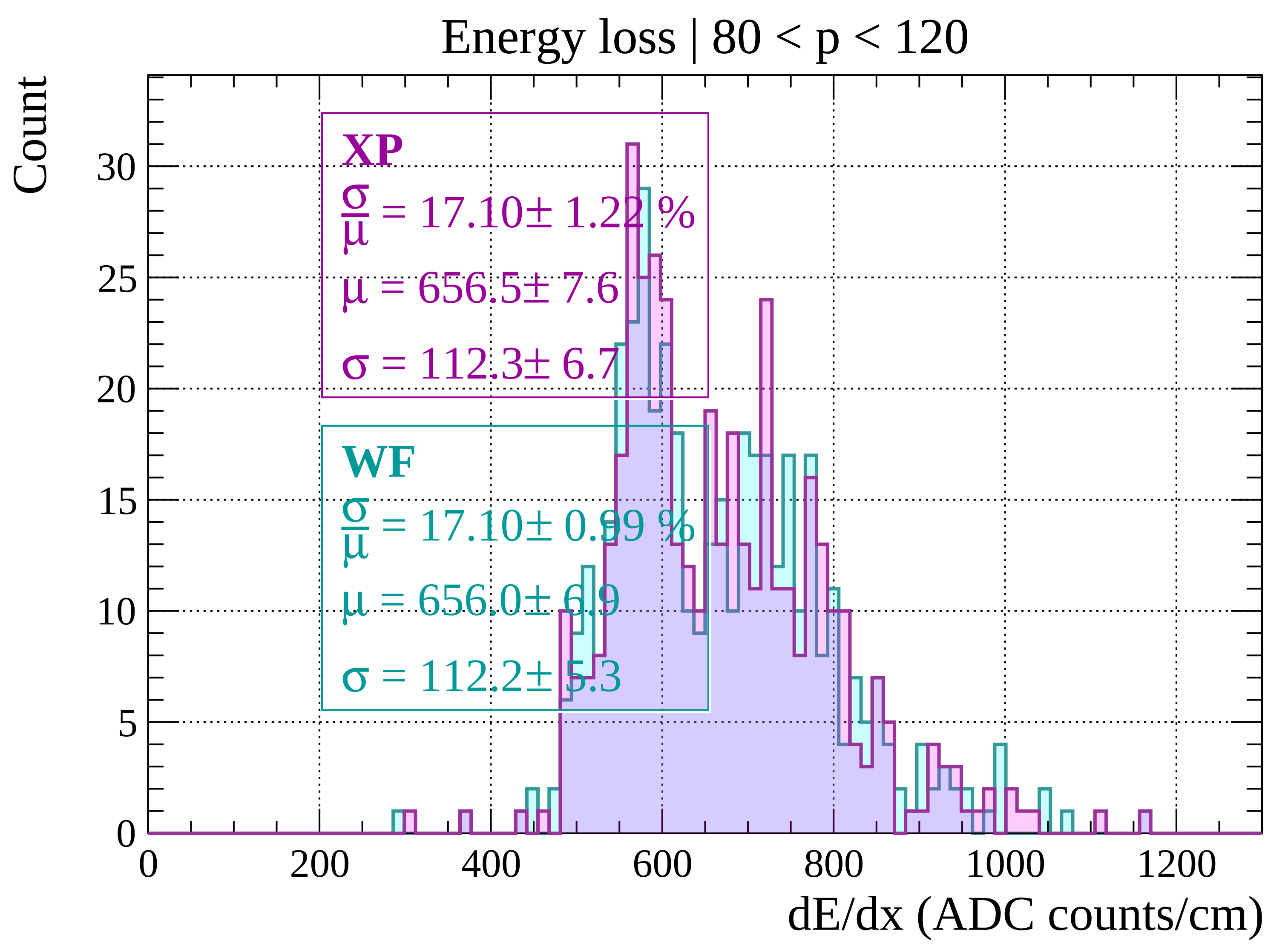

Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 8.01± 0.39 %
μ = 571.3± 2.4
σ = 45.8± 2.0
WF
σ/μ = 8.39± 0.43 %
μ = 566.5± 2.4
σ = 47.6± 2.3


Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.60 \pm 0.38$ %
$\mu = 549.0 \pm 2.0$
$\sigma = 41.7 \pm 1.9$
WF
$\frac{\sigma}{\mu} = 7.77 \pm 0.34$ %
$\mu = 543.5 \pm 2.0$
$\sigma = 42.2 \pm 1.7$

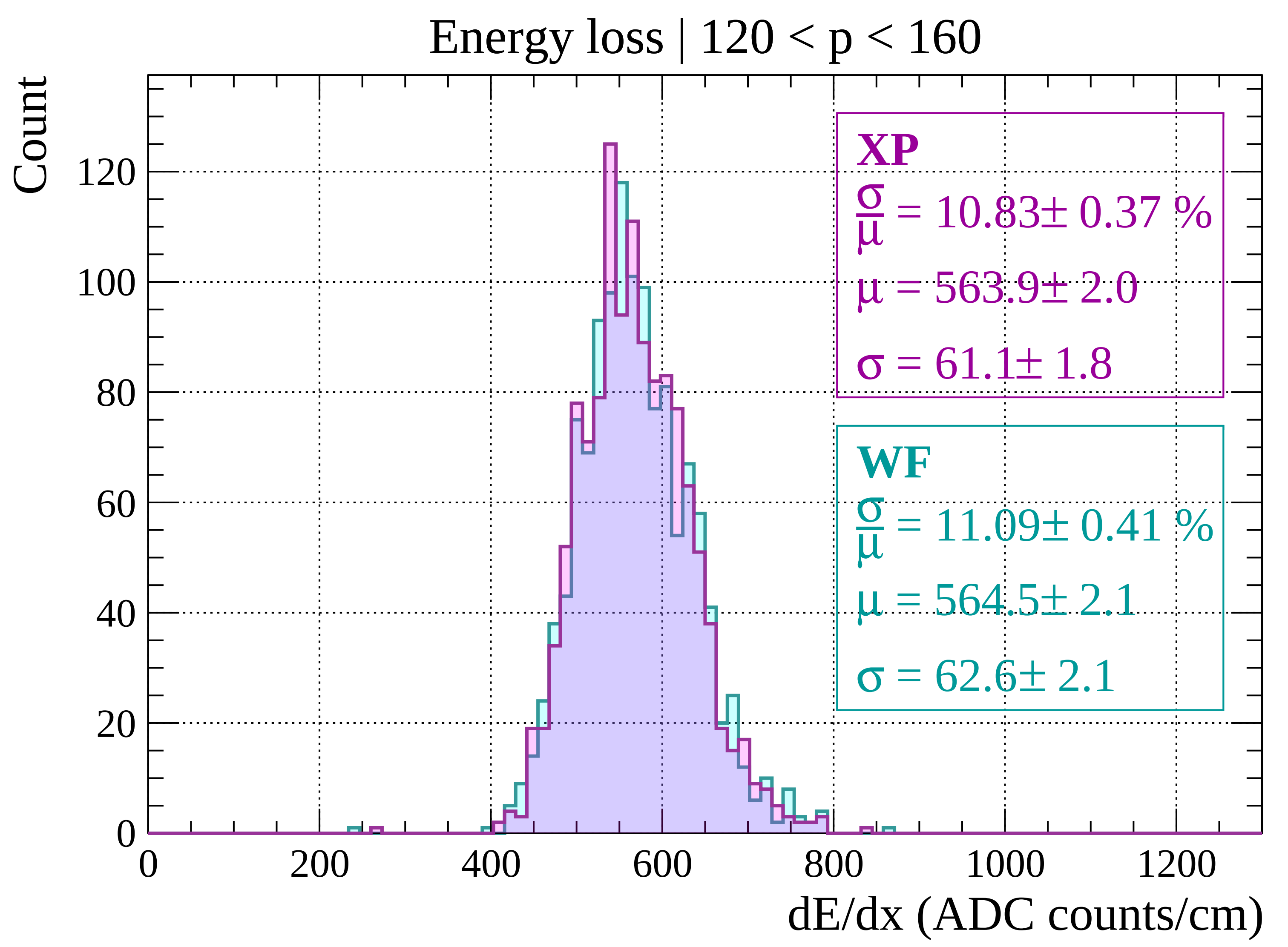


Energy loss | 0 < p < 40
Count
XP
$\frac{\sigma}{\mu}$ = 26.69± 78.00 %
μ = 535.1± 605.0
σ = 142.8± 255.9
WF
$\frac{\sigma}{\mu}$ = 19.56± 26.84 %
μ = 476.0± 67.3
σ = 93.1± 114.6
dE/dx (ADC counts/cm)

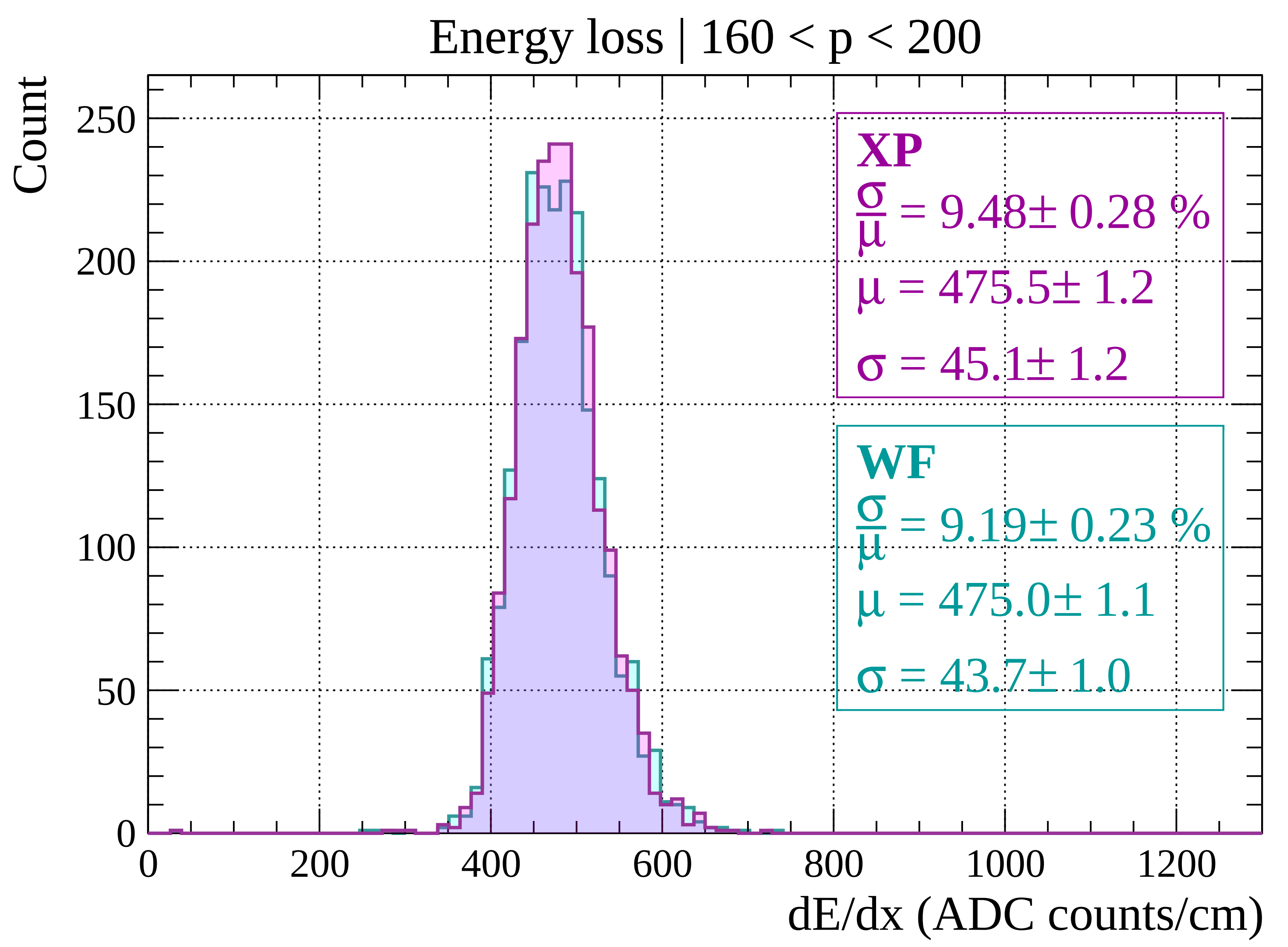

Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 7.30± 0.34 %
μ = 549.0± 2.1
σ = 40.1± 1.7
WF
σ/μ = 7.65± 0.44 %
μ = 542.5± 2.4
σ = 41.5± 2.2

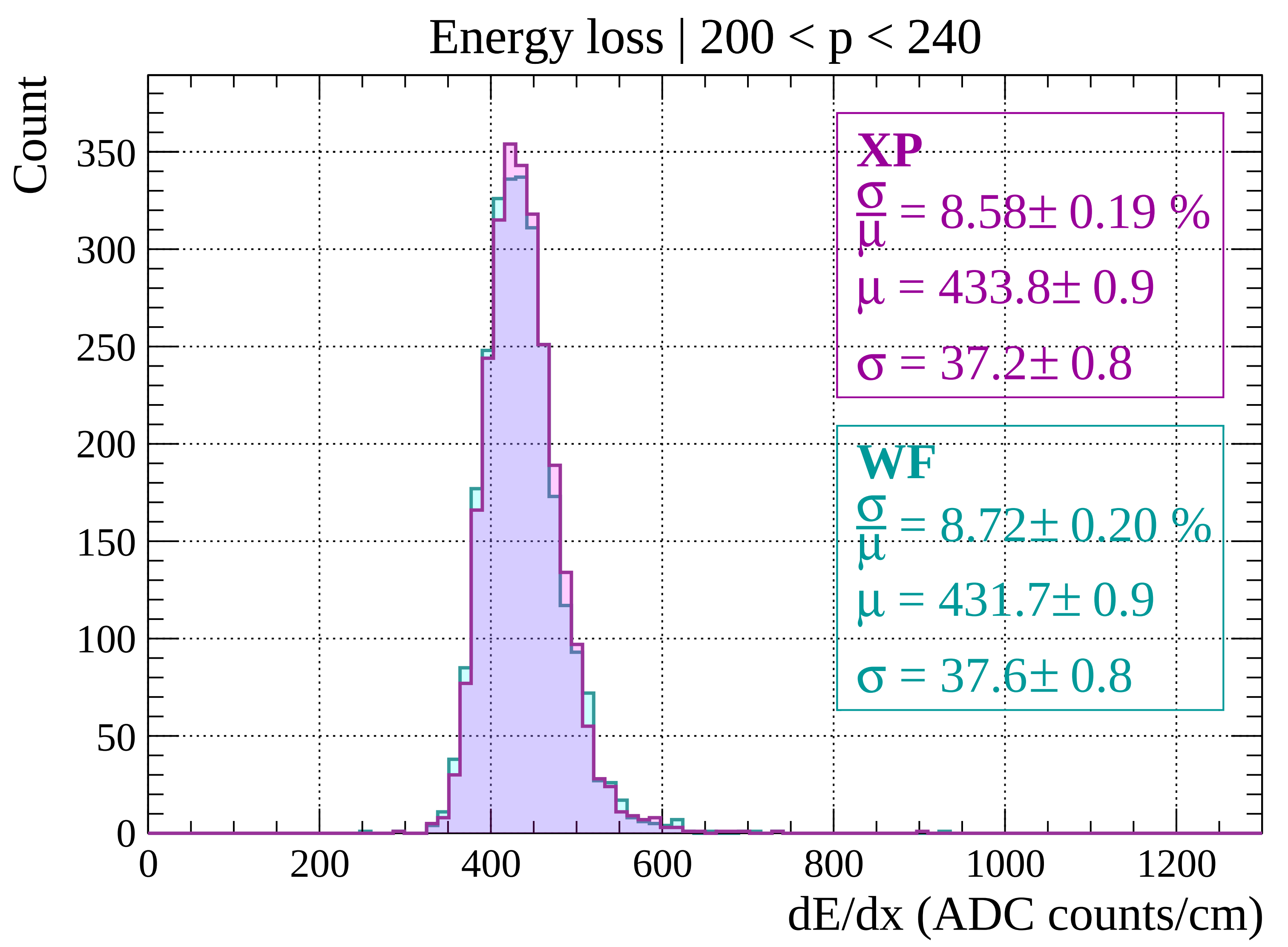


Energy loss | 80 < p < 120
Count
XP
σ/μ = 17.10± 1.22 %
μ = 656.5± 7.6
σ = 112.3± 6.7
WF
σ/μ = 17.10± 0.99 %
μ = 656.0± 6.9
σ = 112.2± 5.3
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

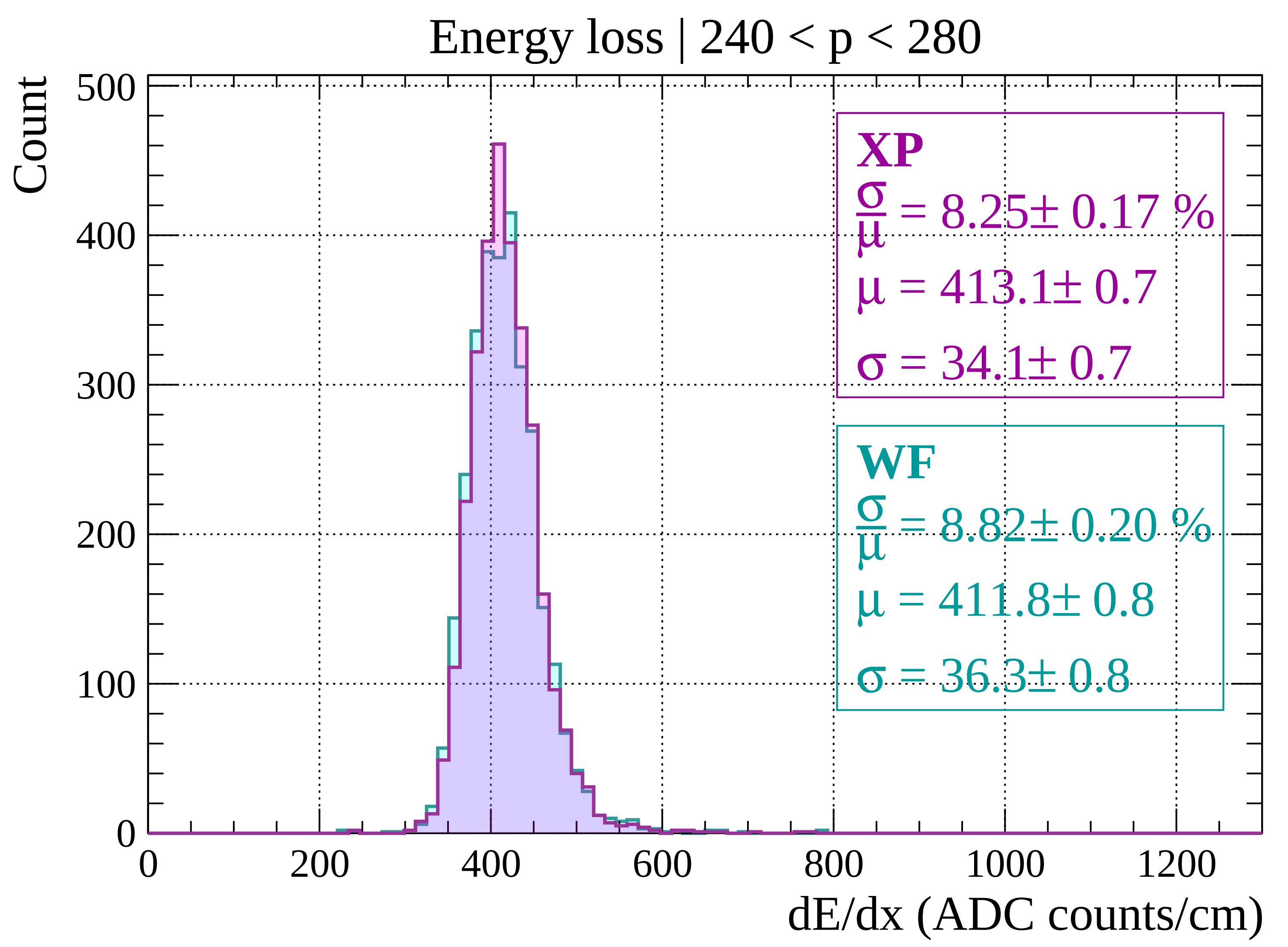


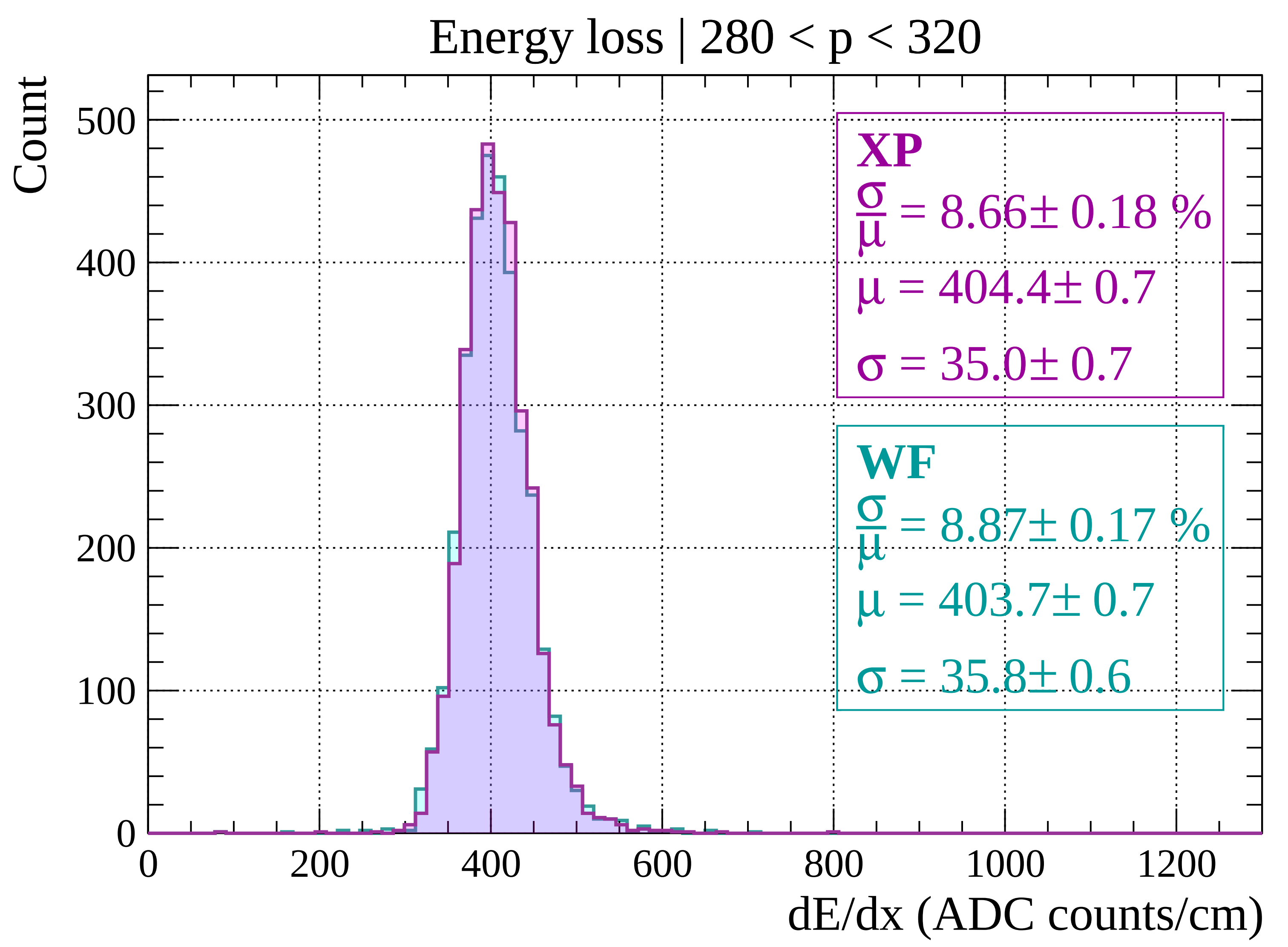
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 10.83± 0.37 %
μ = 563.9± 2.0
σ = 61.1± 1.8
WF
σ/μ = 11.09± 0.41 %
μ = 564.5± 2.1
σ = 62.6± 2.1


Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.48± 0.28 %
μ = 475.5± 1.2
σ = 45.1± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.19± 0.23 %
μ = 475.0± 1.1
σ = 43.7± 1.0

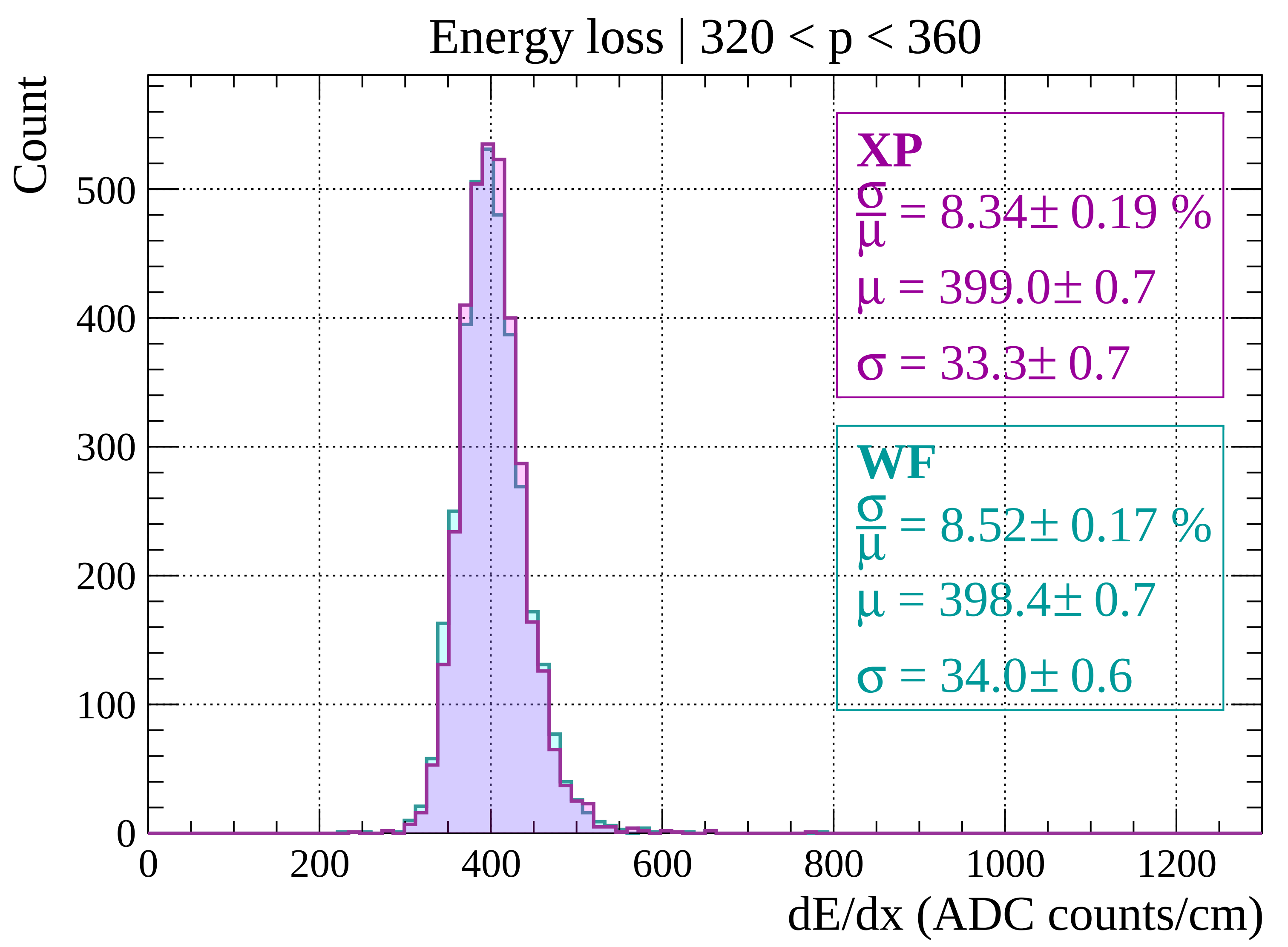


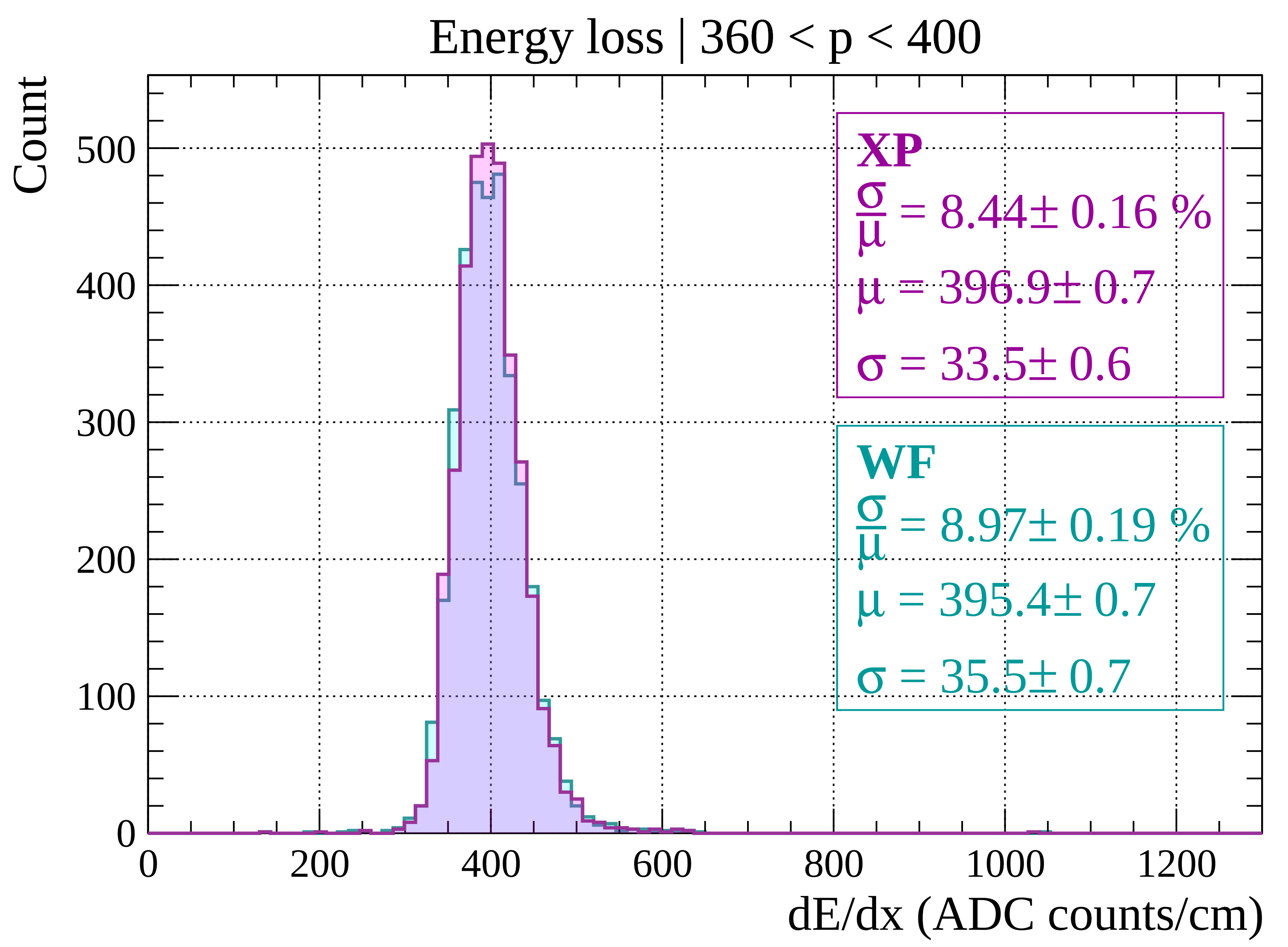
Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP
σ/μ = 8.58± 0.19 %
μ = 433.8± 0.9
σ = 37.2± 0.8
WF
σ/μ = 8.72± 0.20 %
μ = 431.7± 0.9
σ = 37.6± 0.8

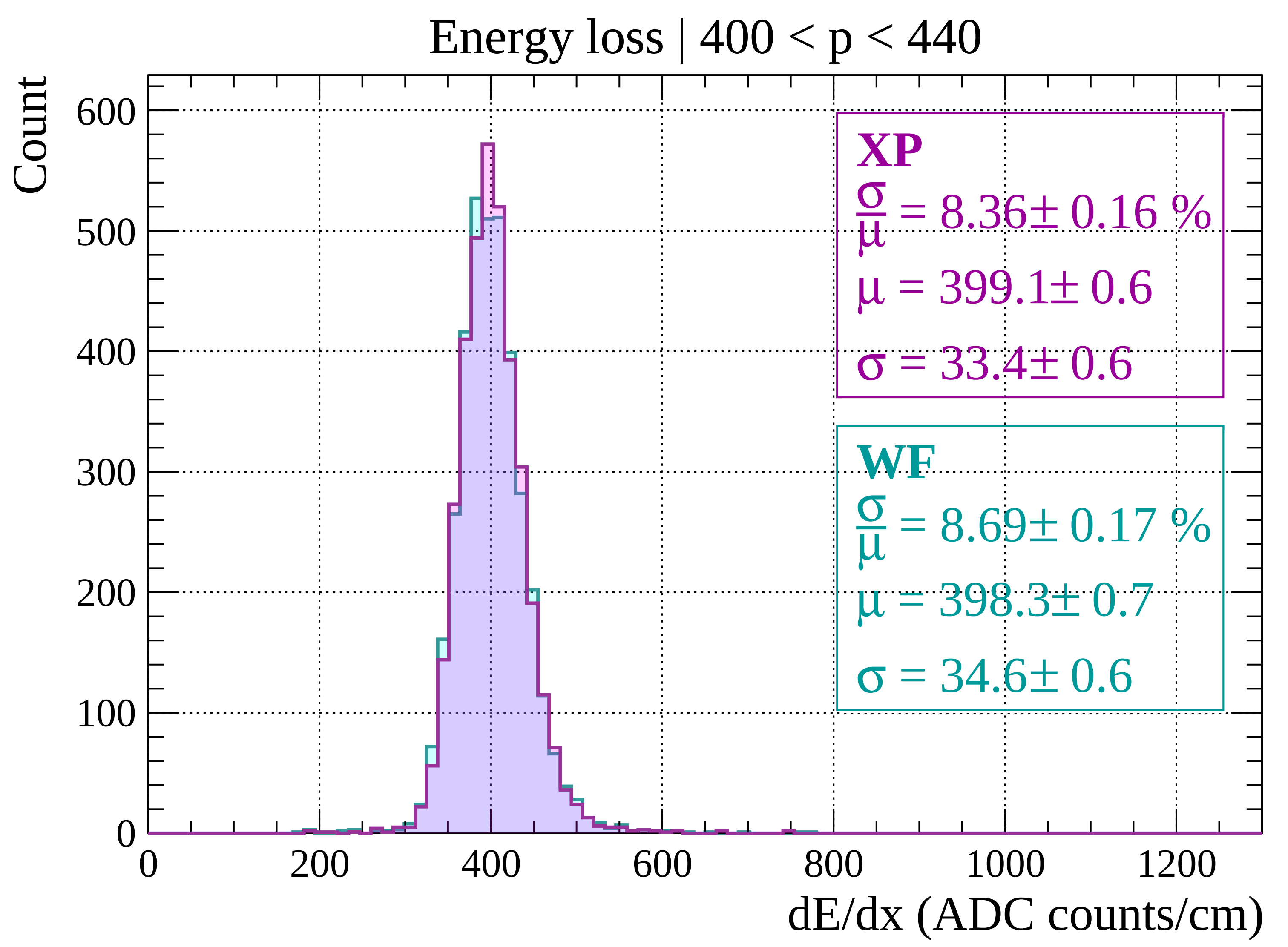

Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.25± 0.17 %
μ = 413.1± 0.7
σ = 34.1± 0.7
WF
σ/μ = 8.82± 0.20 %
μ = 411.8± 0.8
σ = 36.3± 0.8



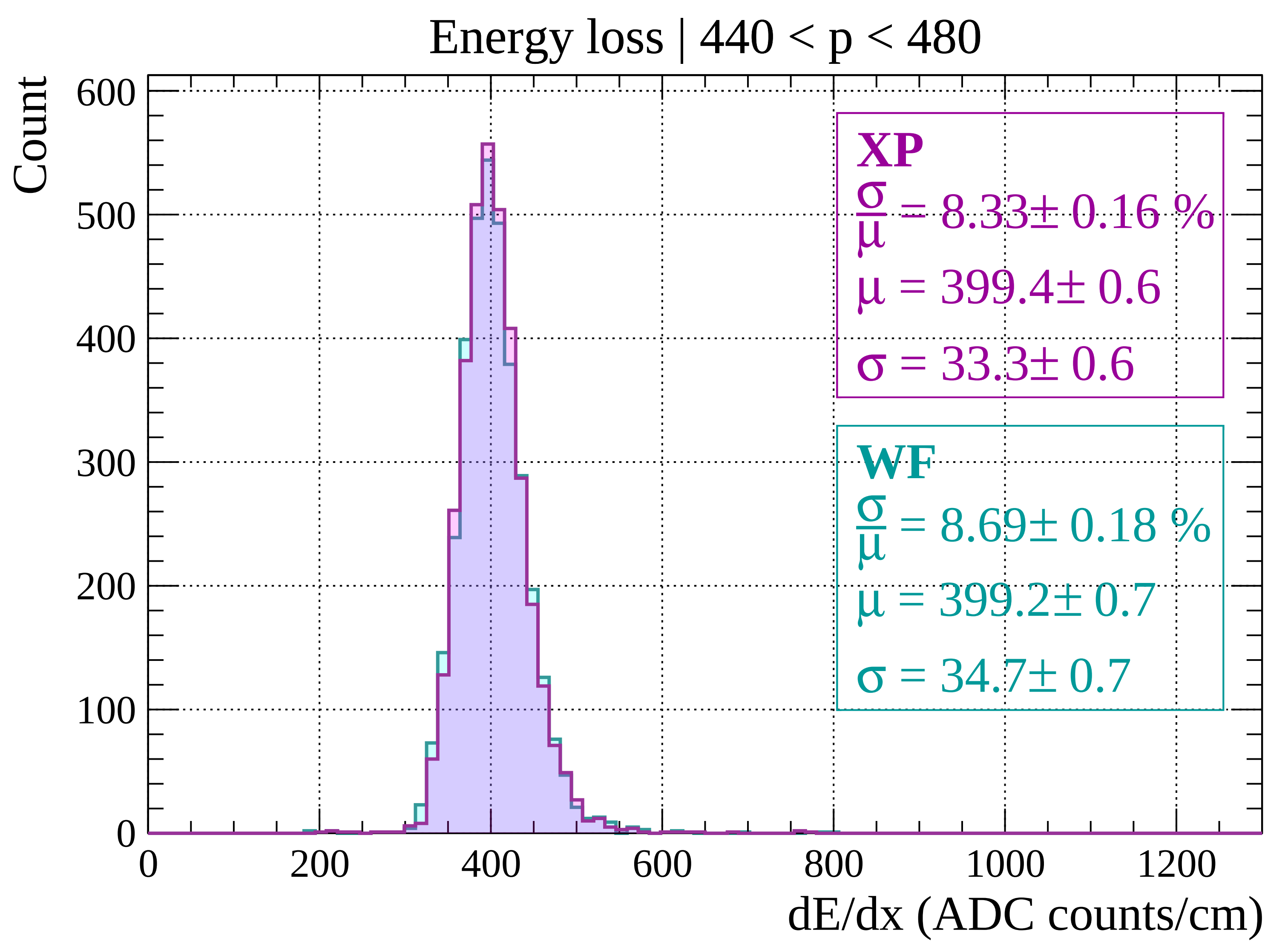
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.66± 0.18 %
μ = 404.4± 0.7
σ = 35.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.87± 0.17 %
μ = 403.7± 0.7
σ = 35.8± 0.6


Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.34± 0.19 %
μ = 399.0± 0.7
σ = 33.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.52± 0.17 %
μ = 398.4± 0.7
σ = 34.0± 0.6

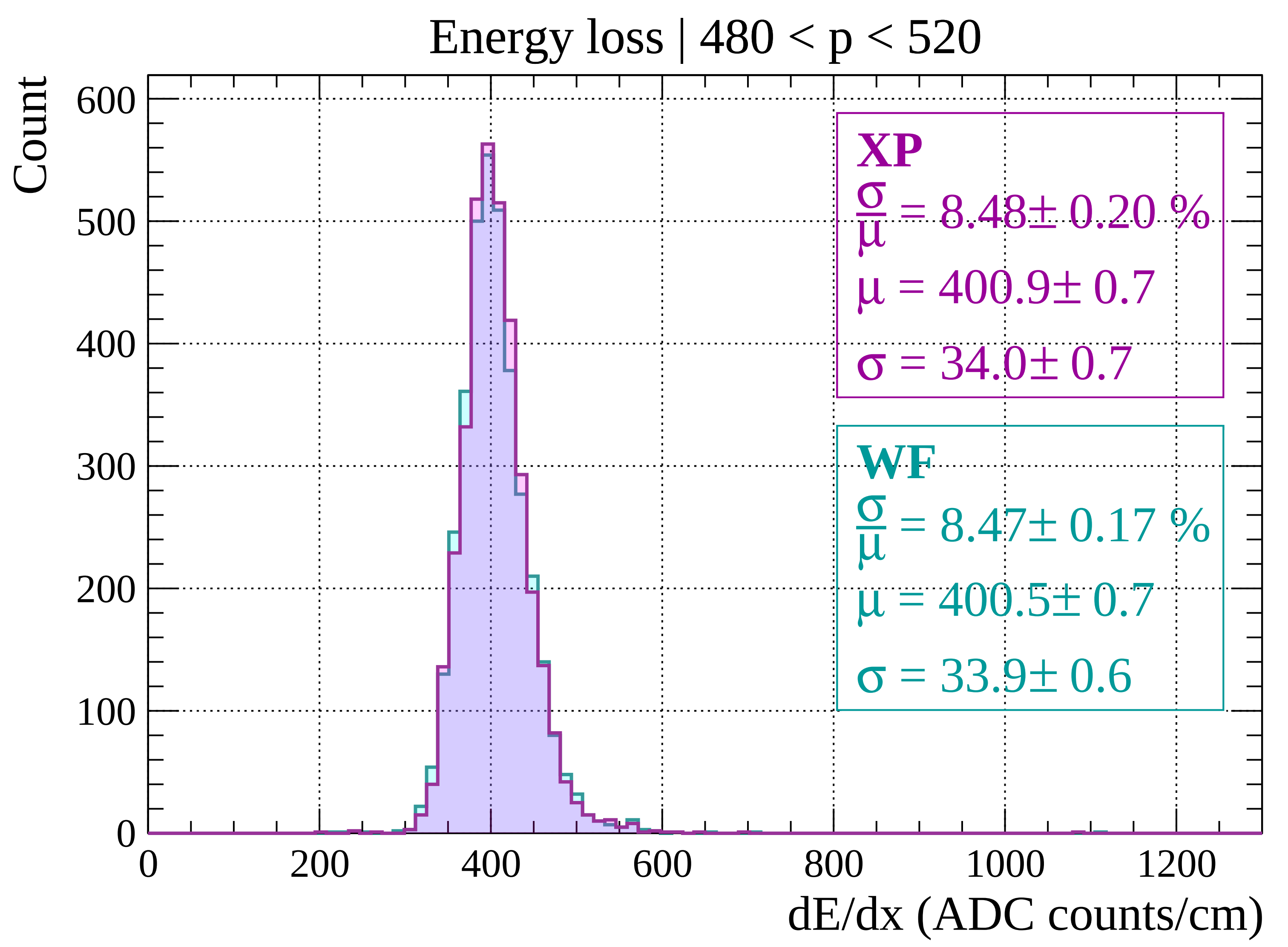

Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.44± 0.16 %
μ = 396.9± 0.7
σ = 33.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.97± 0.19 %
μ = 395.4± 0.7
σ = 35.5± 0.7

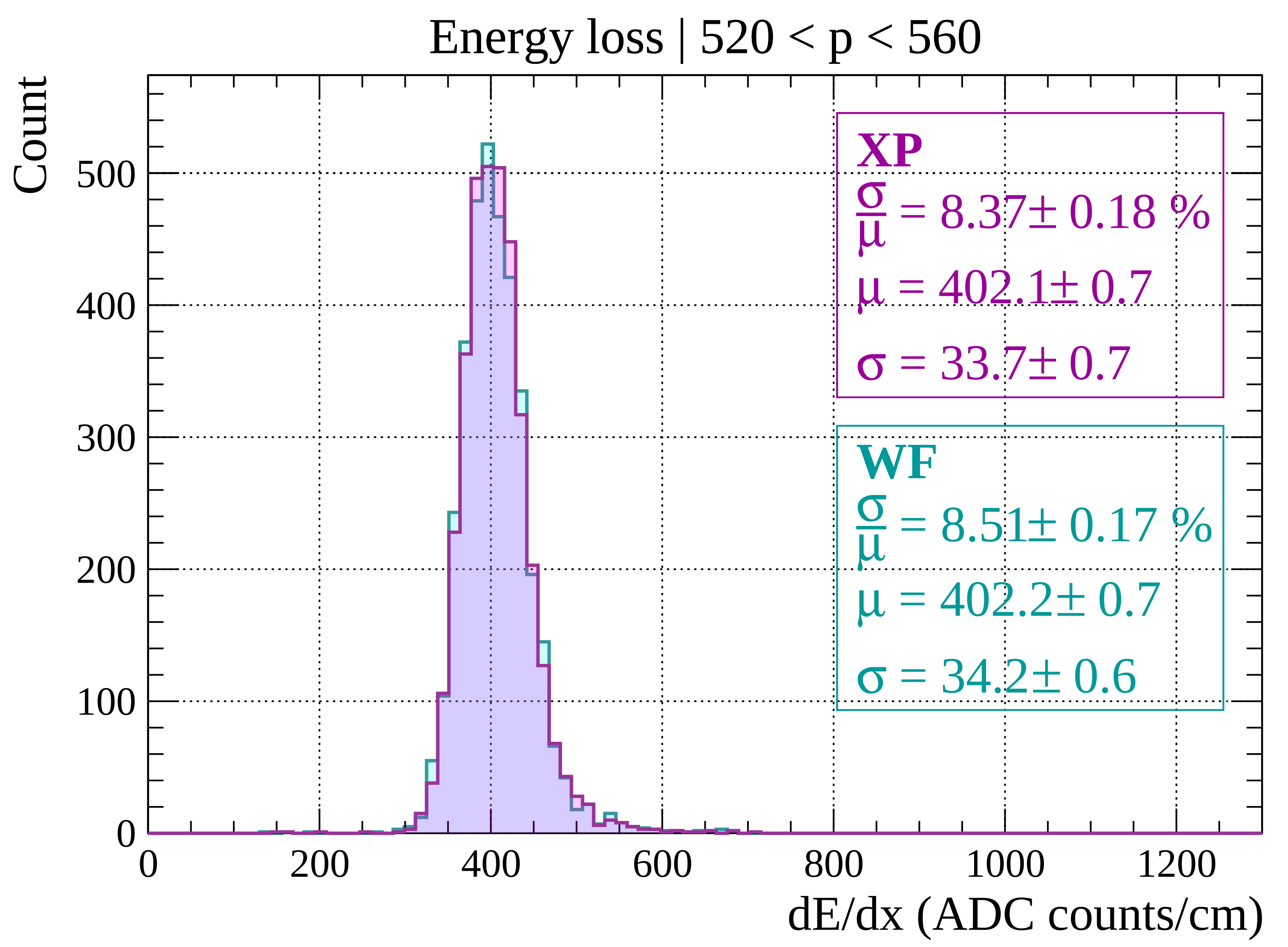


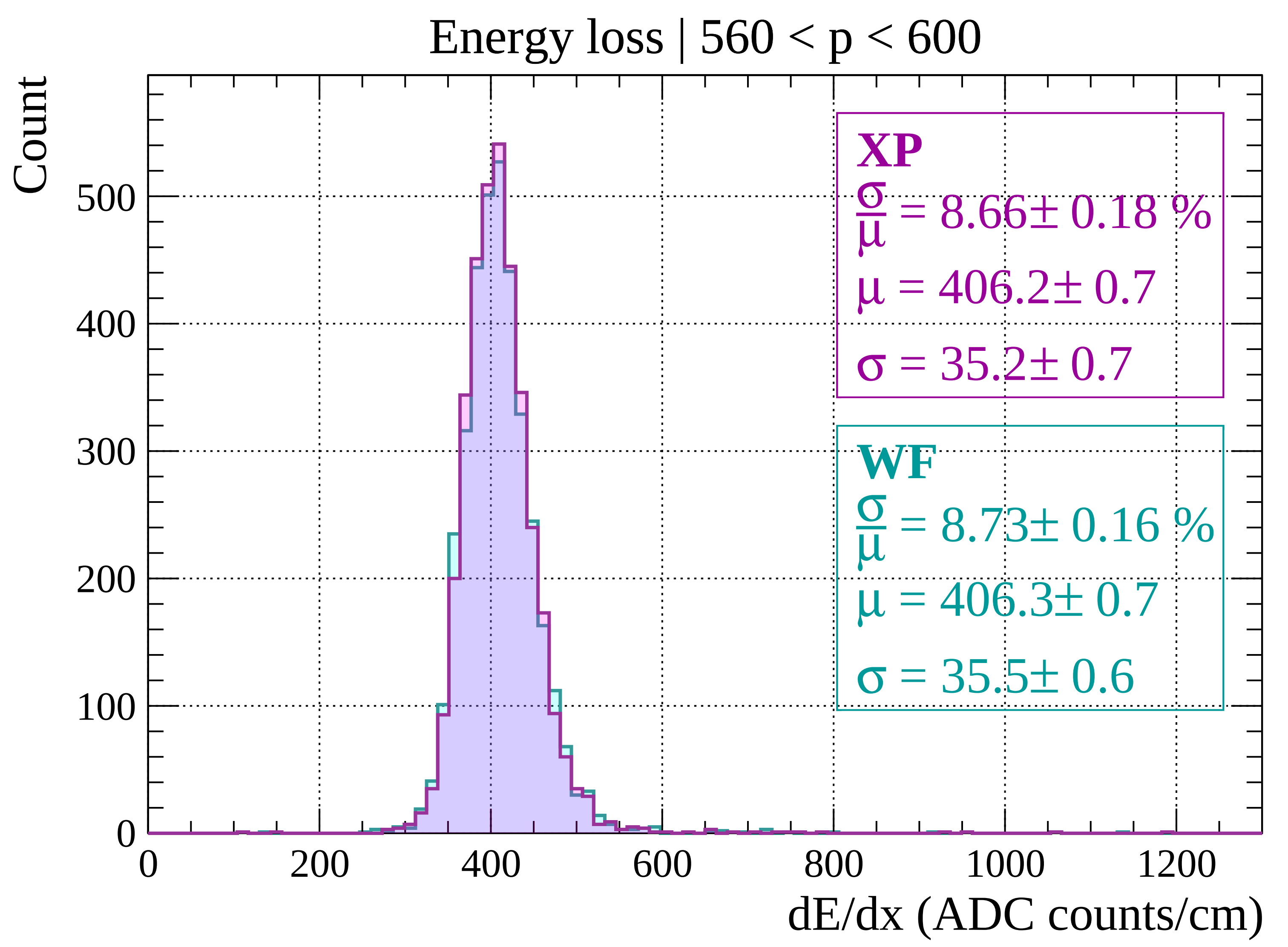
Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.36 ± 0.16 %
μ = 399.1 ± 0.6
σ = 33.4 ± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.69 ± 0.17 %
μ = 398.3 ± 0.7
σ = 34.6 ± 0.6

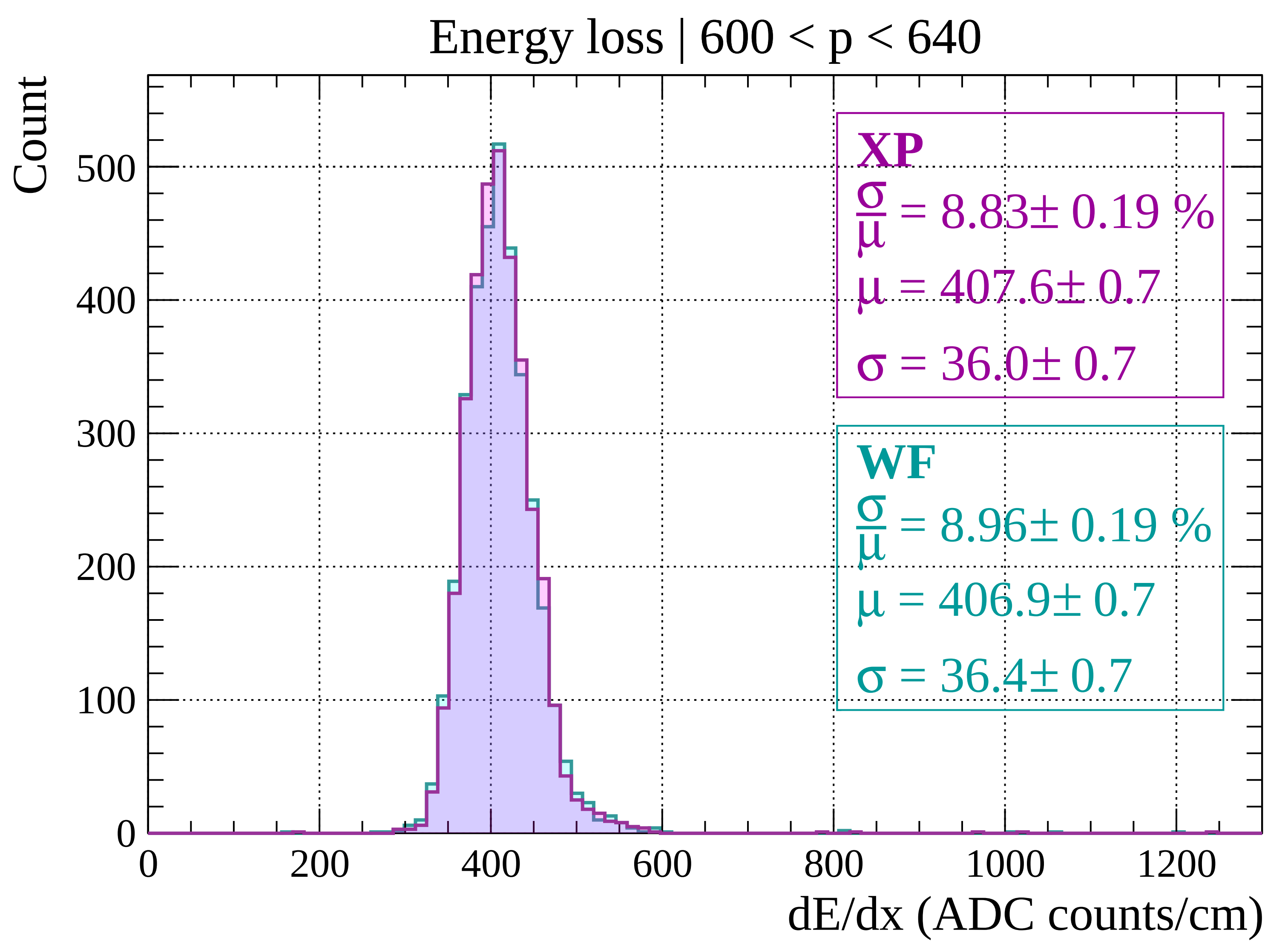


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.33± 0.16 %
μ = 399.4± 0.6
σ = 33.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.69± 0.18 %
μ = 399.2± 0.7
σ = 34.7± 0.7

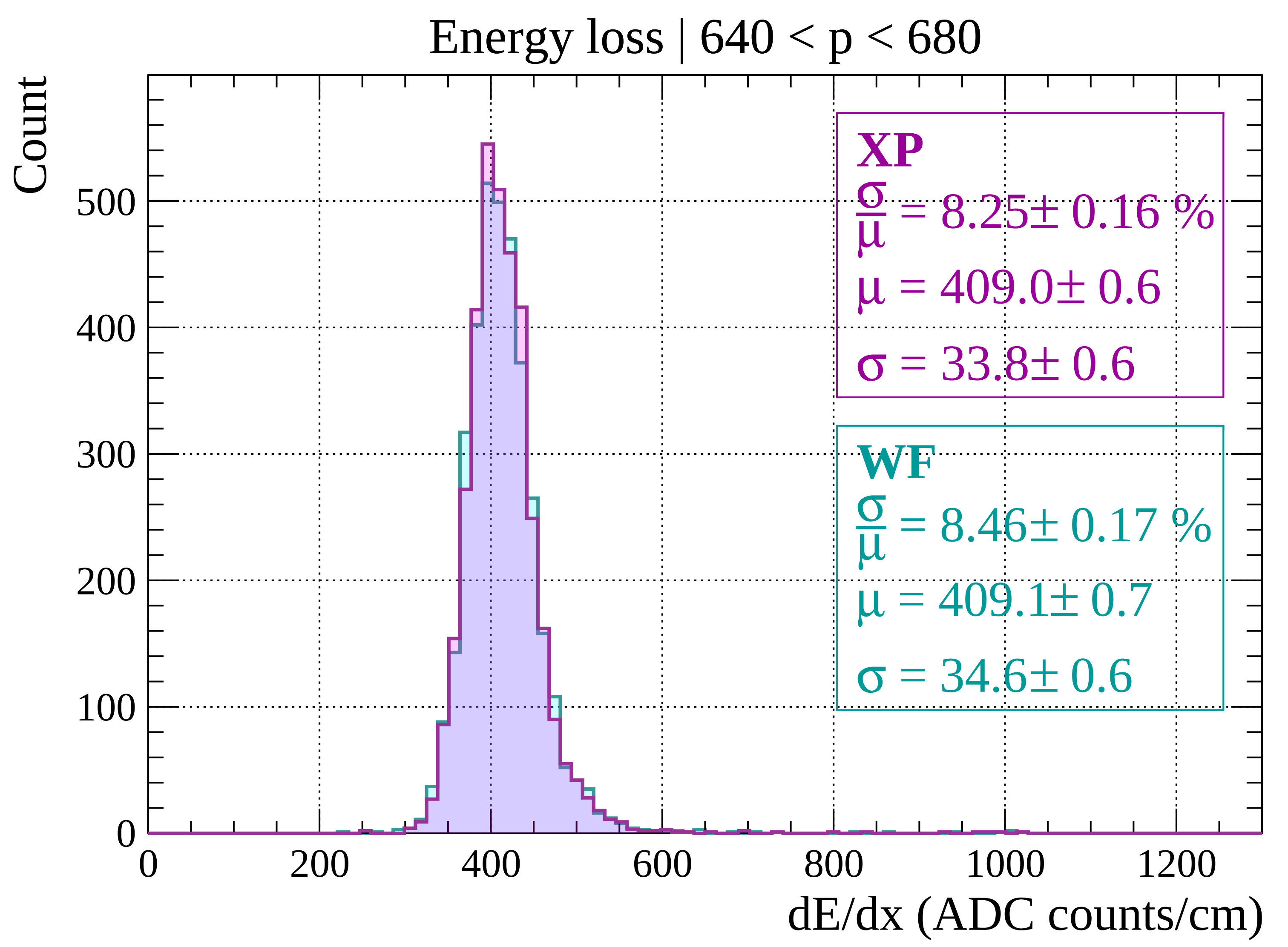

Energy loss | 480 < p < 520
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.48± 0.20 %
μ = 400.9± 0.7
σ = 34.0± 0.7
WF
σ/μ = 8.47± 0.17 %
μ = 400.5± 0.7
σ = 33.9± 0.6


Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.37 \pm 0.18$ %
$\mu = 402.1 \pm 0.7$
$\sigma = 33.7 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 8.51 \pm 0.17$ %
$\mu = 402.2 \pm 0.7$
$\sigma = 34.2 \pm 0.6$

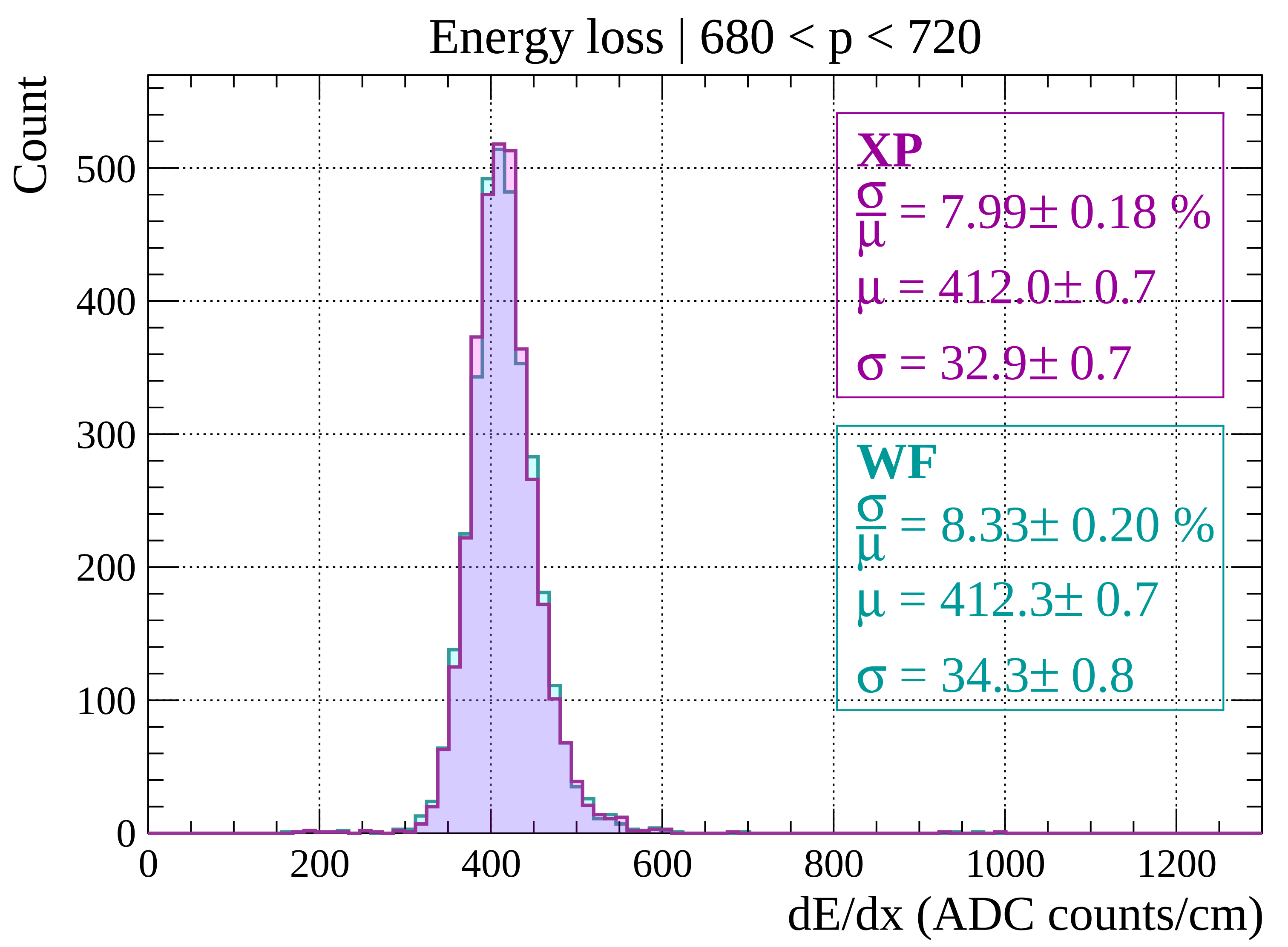

Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.66 \pm 0.18$ %
$\mu = 406.2 \pm 0.7$
$\sigma = 35.2 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 8.73 \pm 0.16$ %
$\mu = 406.3 \pm 0.7$
$\sigma = 35.5 \pm 0.6$



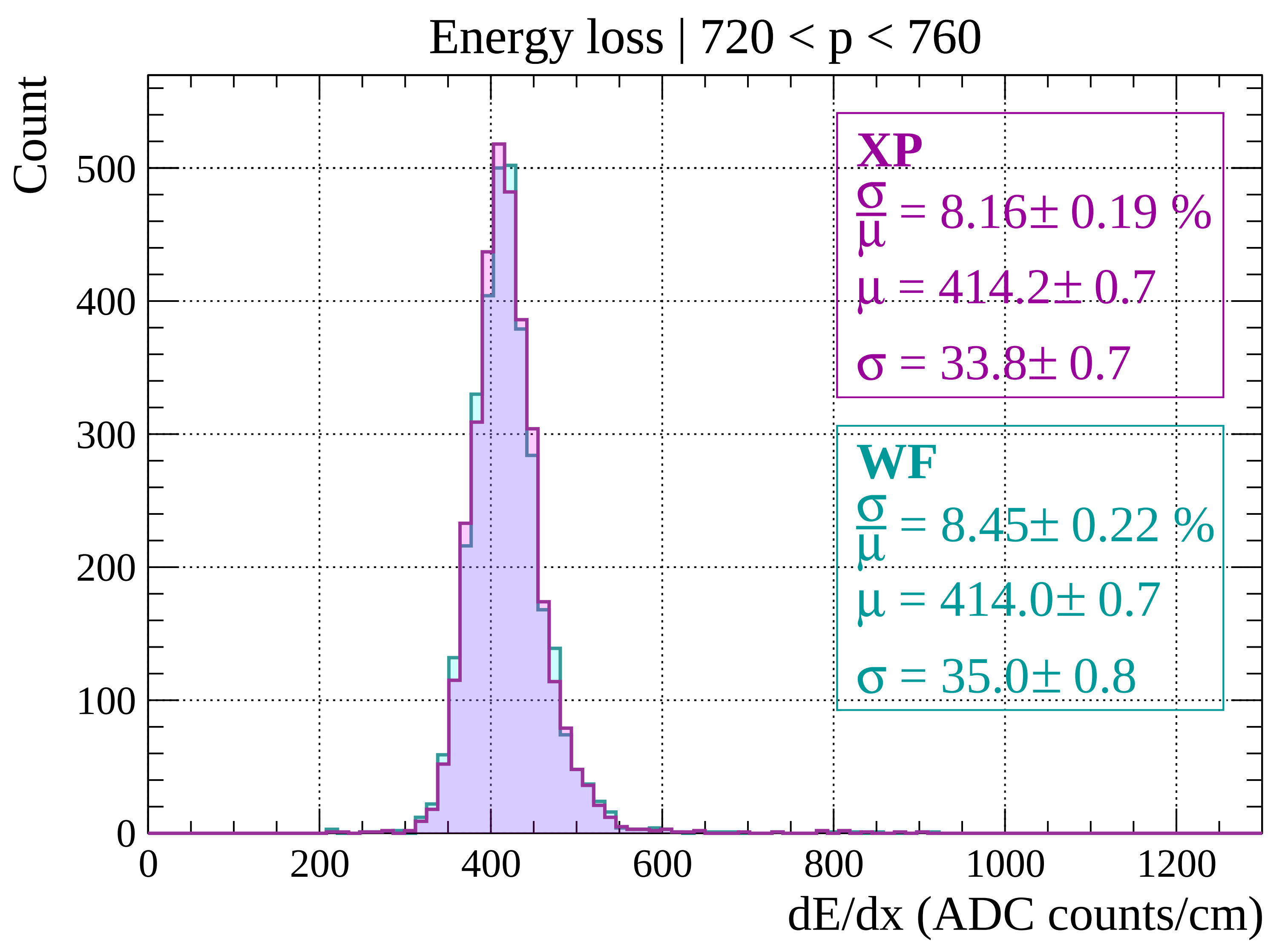
Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.83± 0.19 %
μ = 407.6± 0.7
σ = 36.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.96± 0.19 %
μ = 406.9± 0.7
σ = 36.4± 0.7



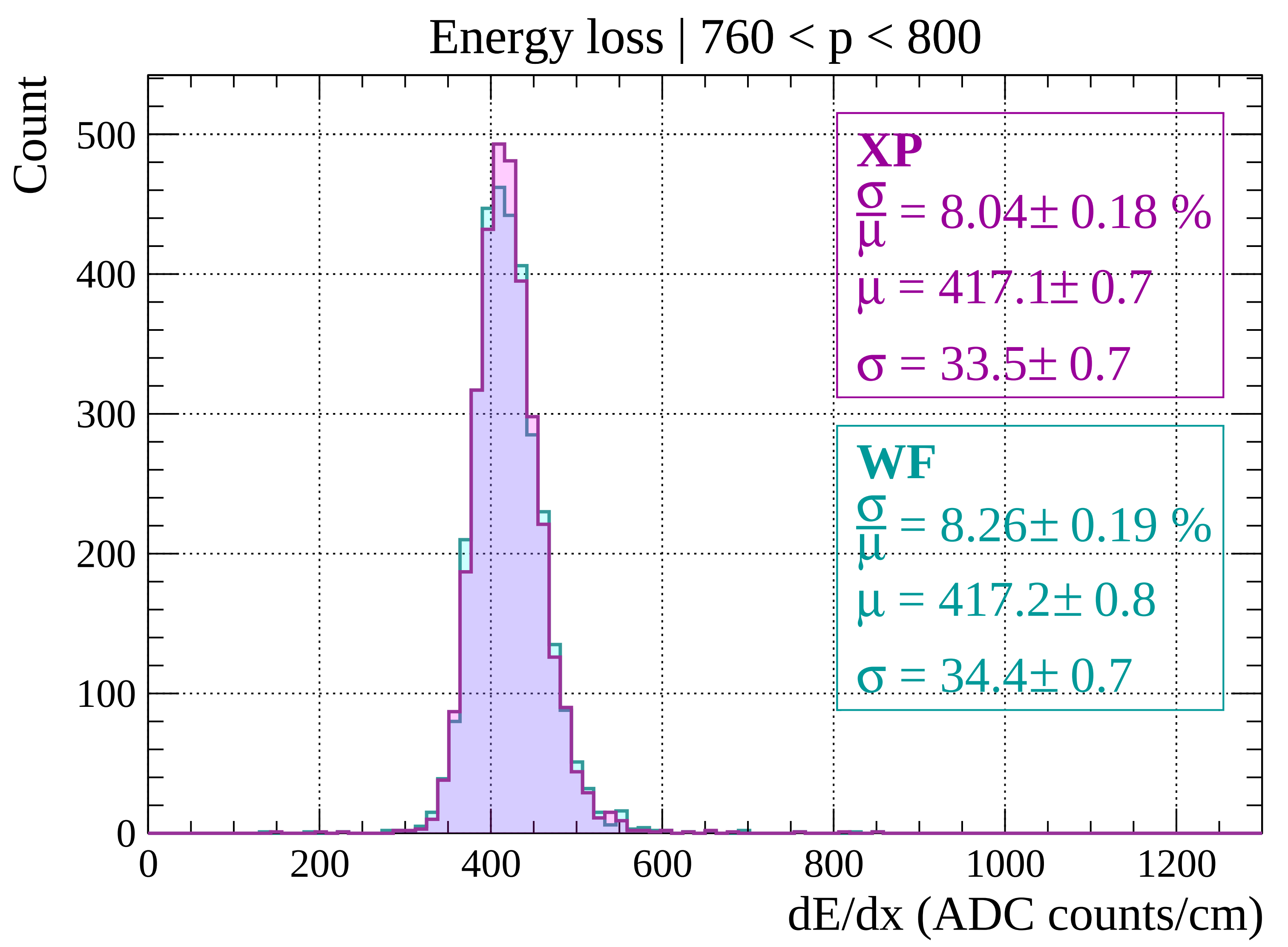
Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.25± 0.16 %
μ = 409.0± 0.6
σ = 33.8± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.17 %
μ = 409.1± 0.7
σ = 34.6± 0.6

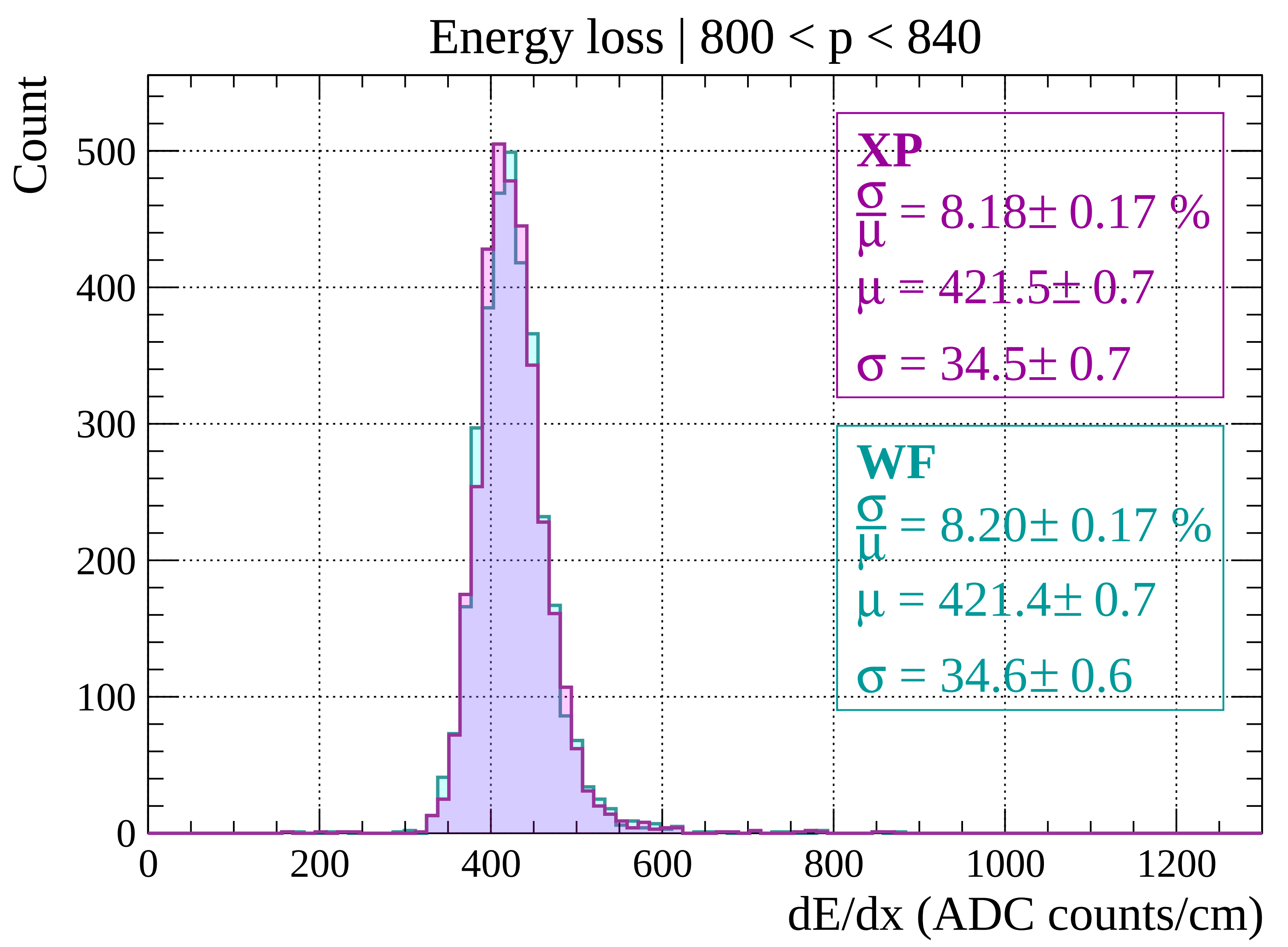


Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 7.99± 0.18 %
μ = 412.0± 0.7
σ = 32.9± 0.7
WF
σ/μ = 8.33± 0.20 %
μ = 412.3± 0.7
σ = 34.3± 0.8

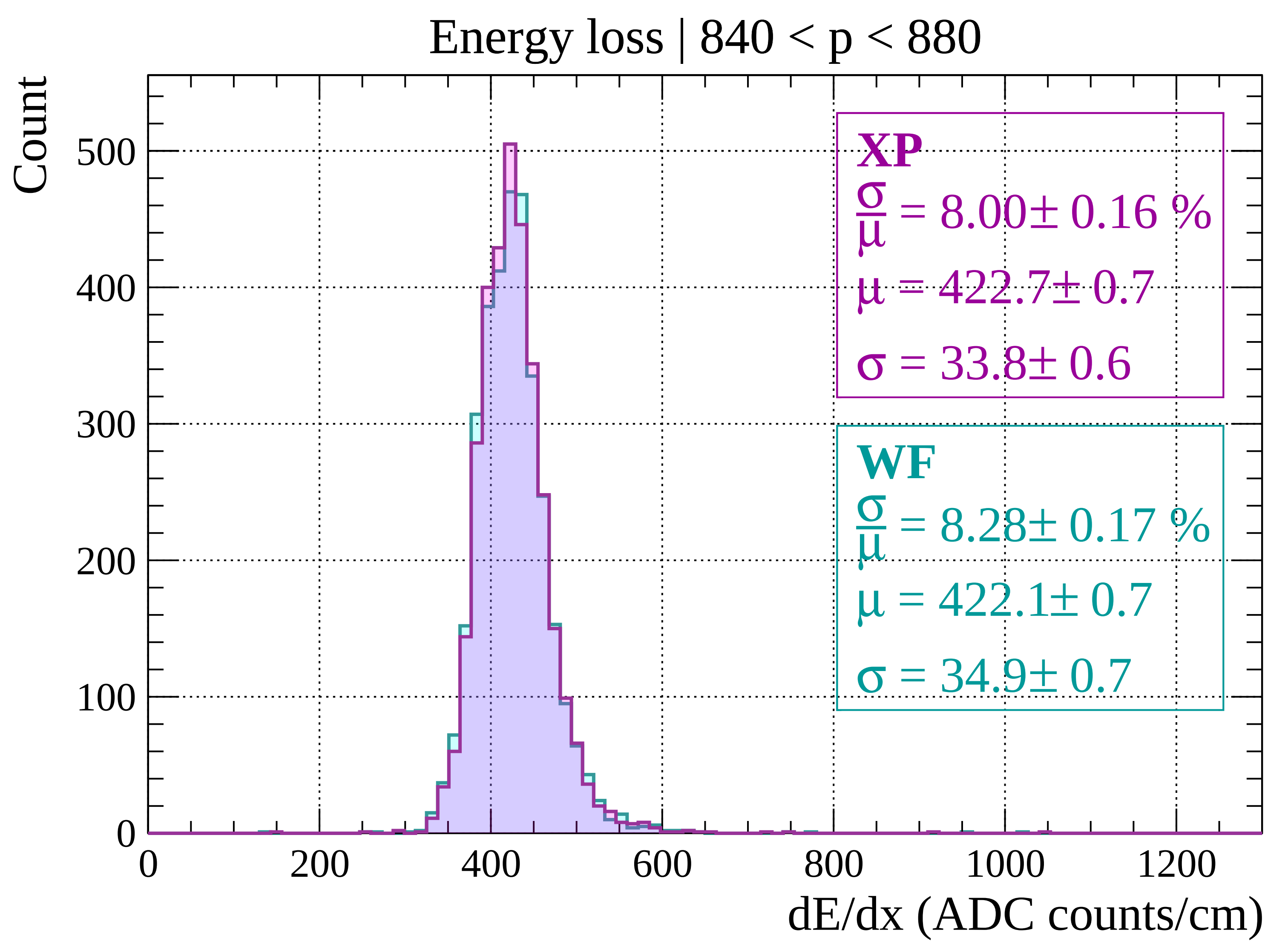


Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 8.16± 0.19 %
μ = 414.2± 0.7
σ = 33.8± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.45± 0.22 %
μ = 414.0± 0.7
σ = 35.0± 0.8

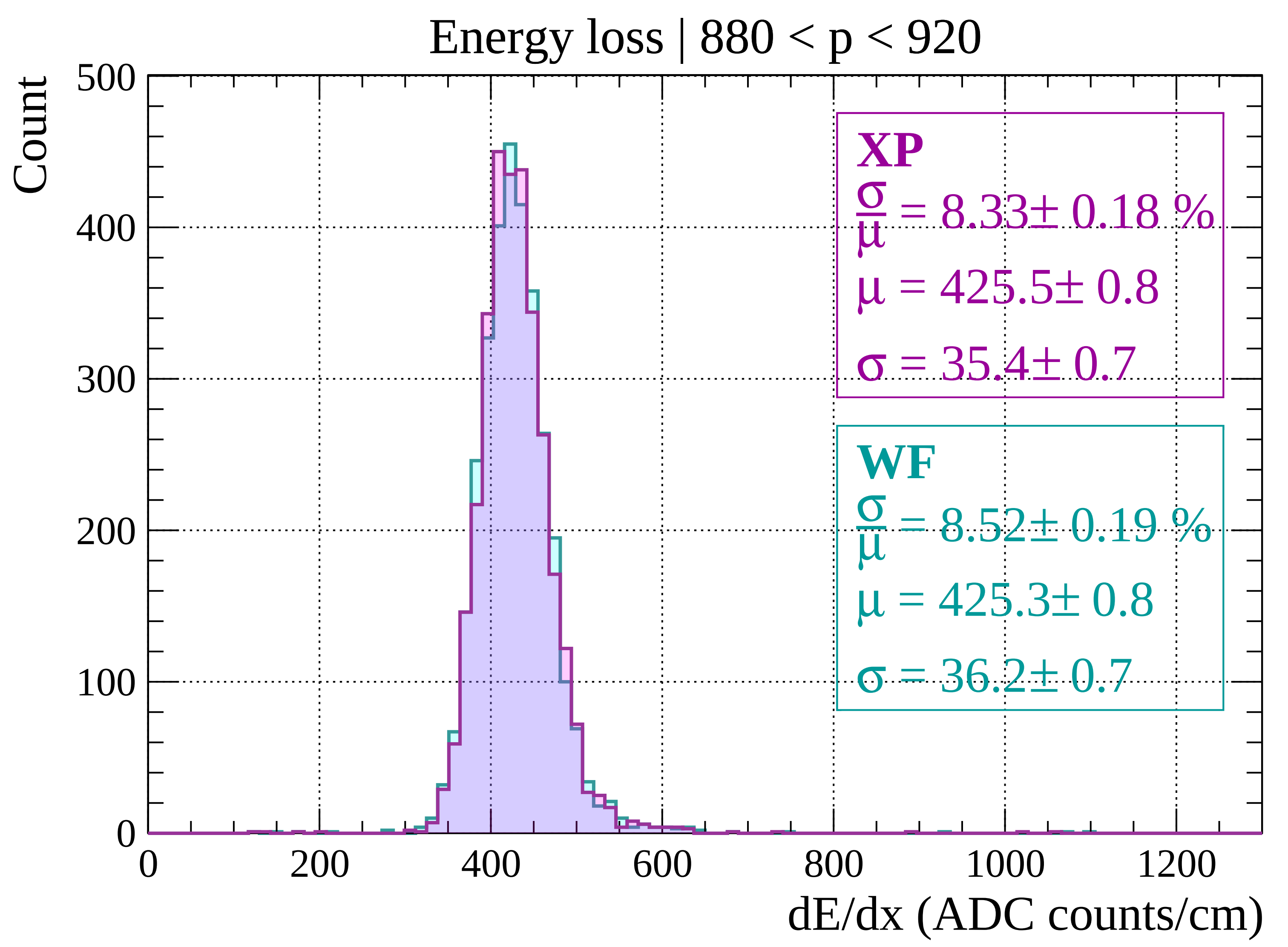

Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.04± 0.18 %
μ = 417.1± 0.7
σ = 33.5± 0.7
WF
σ/μ = 8.26± 0.19 %
μ = 417.2± 0.8
σ = 34.4± 0.7



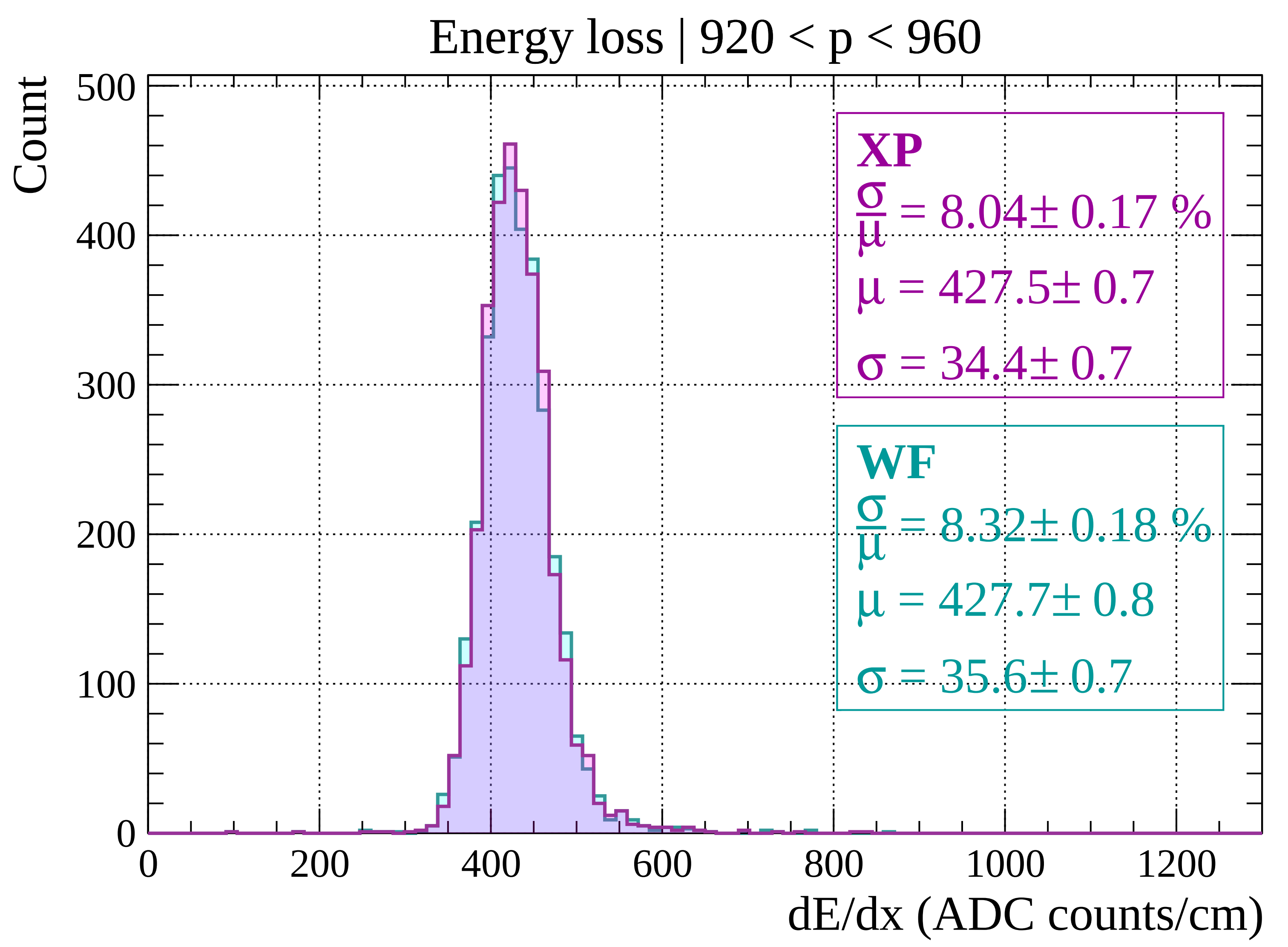
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.18± 0.17 %
μ = 421.5± 0.7
σ = 34.5± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.20± 0.17 %
μ = 421.4± 0.7
σ = 34.6± 0.6



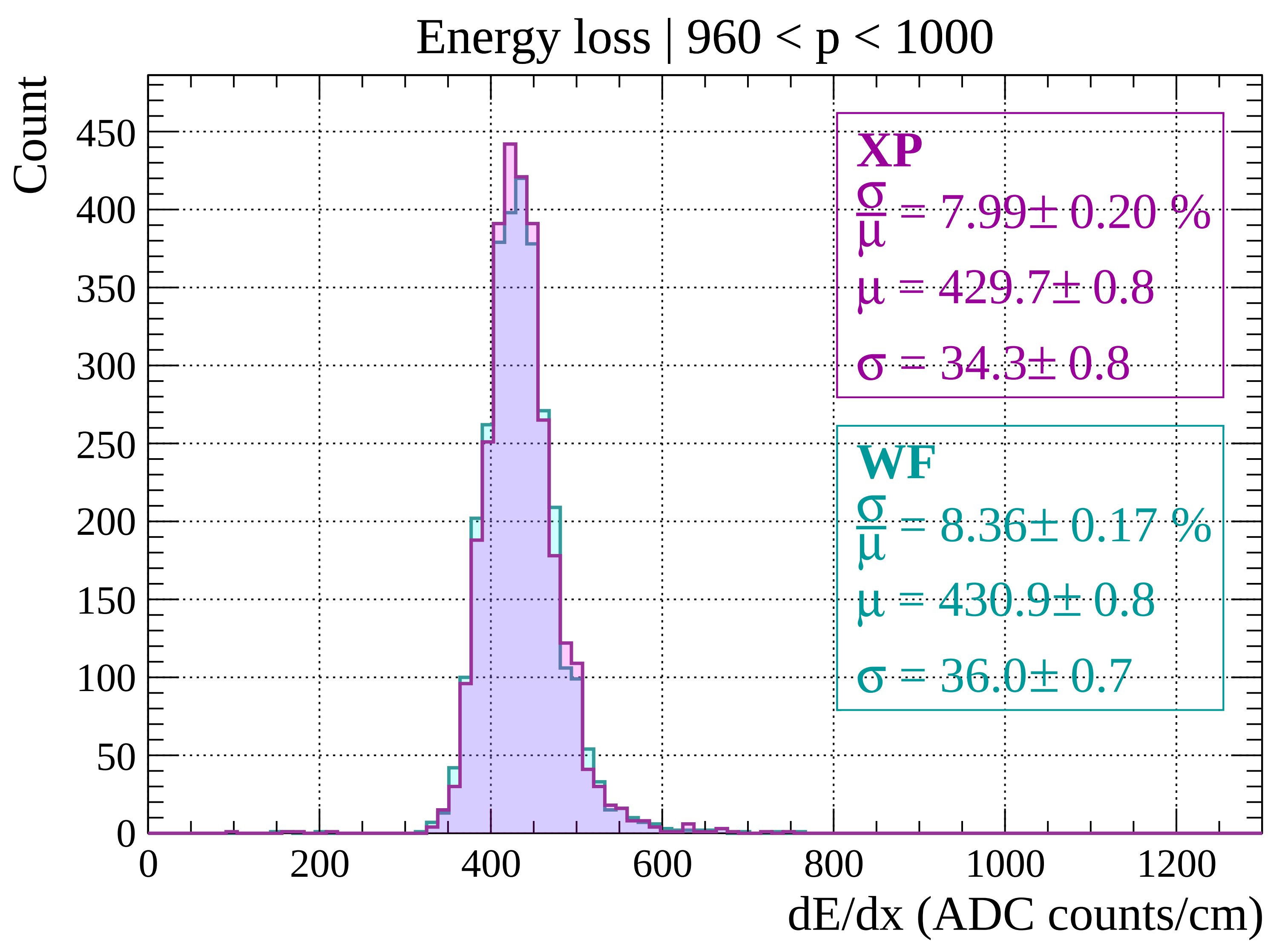
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.00± 0.16 %
μ = 422.7± 0.7
σ = 33.8± 0.6
WF
σ/μ = 8.28± 0.17 %
μ = 422.1± 0.7
σ = 34.9± 0.7

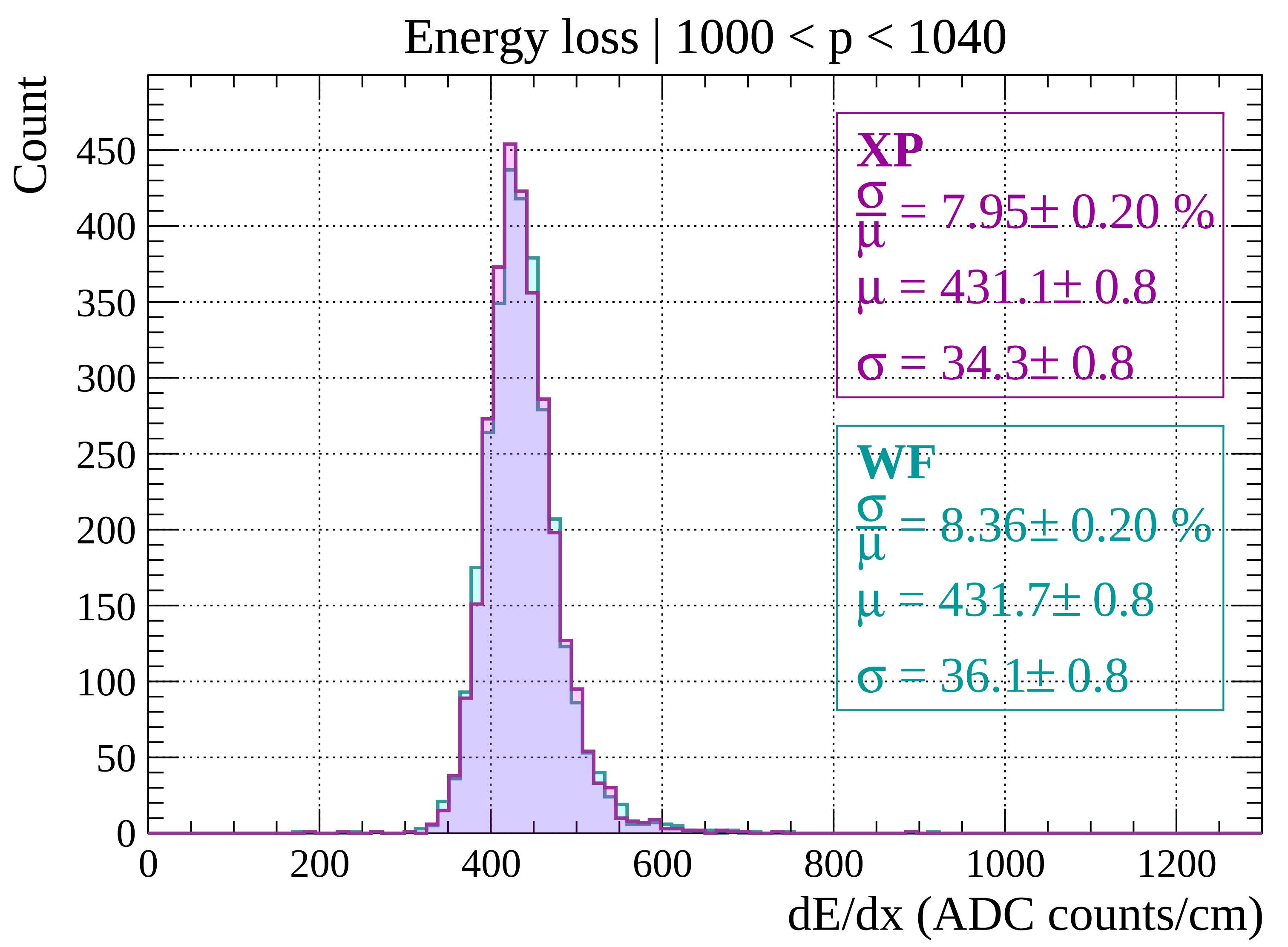


Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu} = 8.33 \pm 0.18\ \%$
$\mu = 425.5 \pm 0.8$
$\sigma = 35.4 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 8.52 \pm 0.19\ \%$
$\mu = 425.3 \pm 0.8$
$\sigma = 36.2 \pm 0.7$

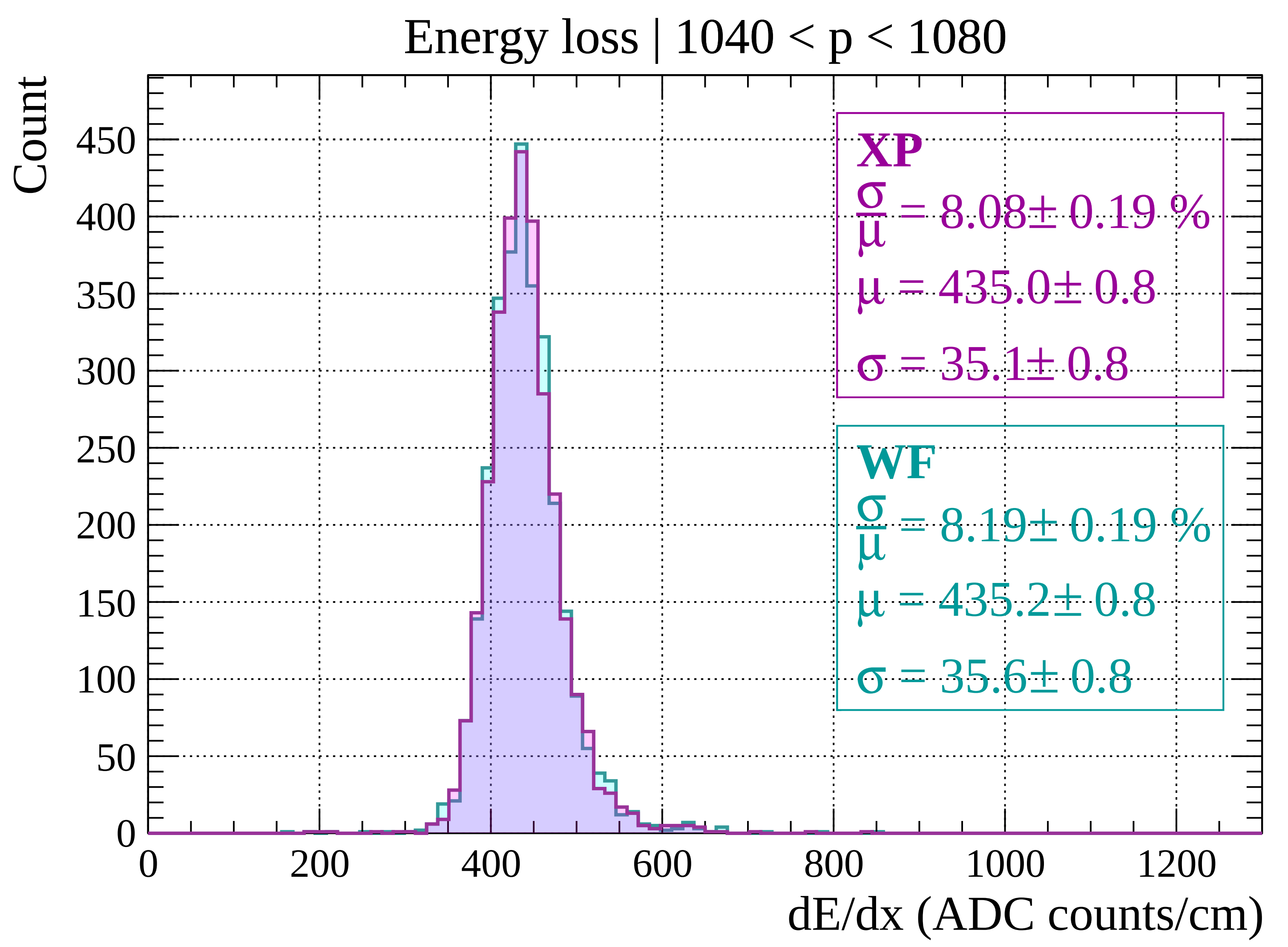


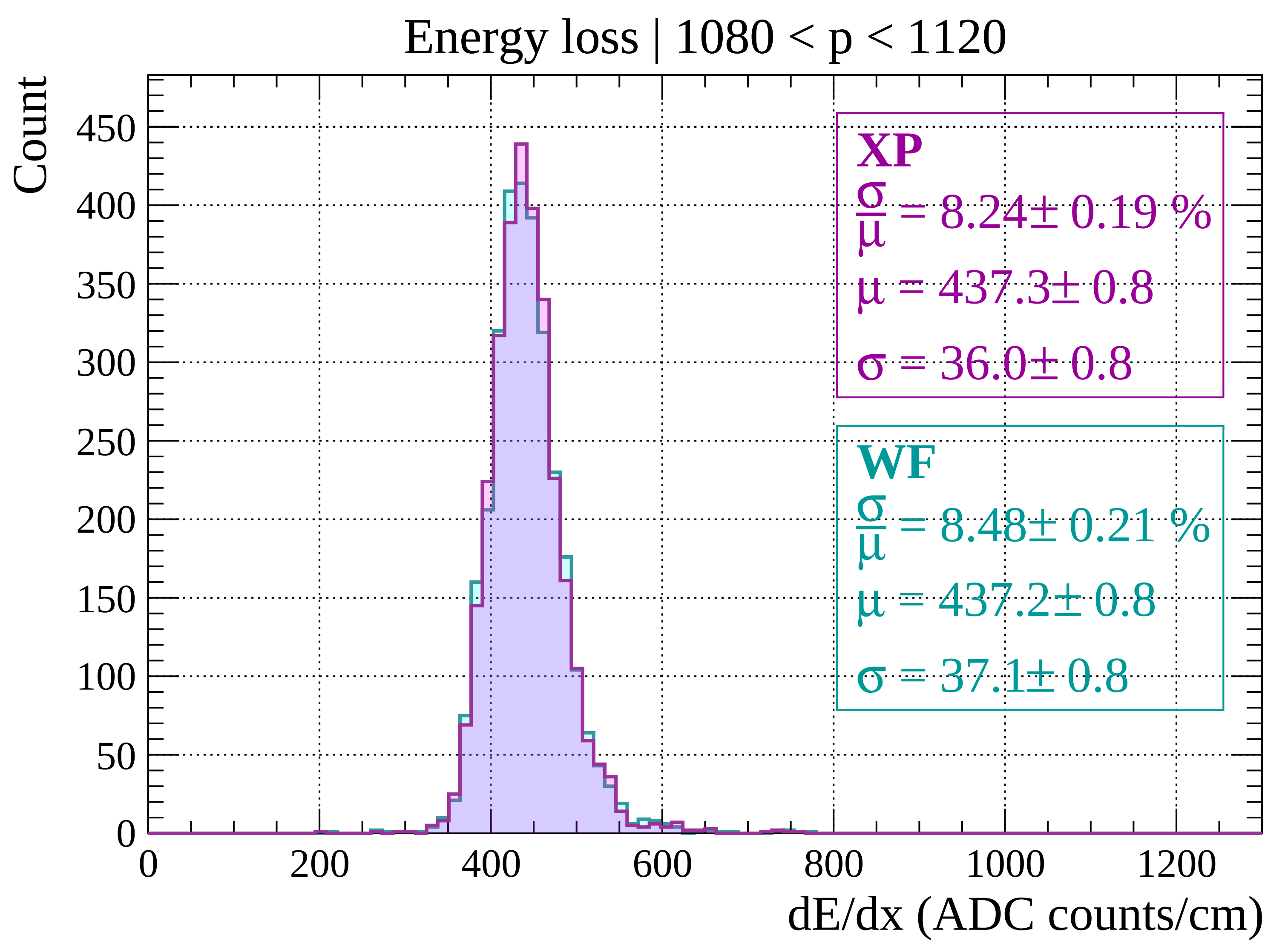
Energy loss | 920 < p < 960
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.04 \pm 0.17$ %
$\mu = 427.5 \pm 0.7$
$\sigma = 34.4 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 8.32 \pm 0.18$ %
$\mu = 427.7 \pm 0.8$
$\sigma = 35.6 \pm 0.7$

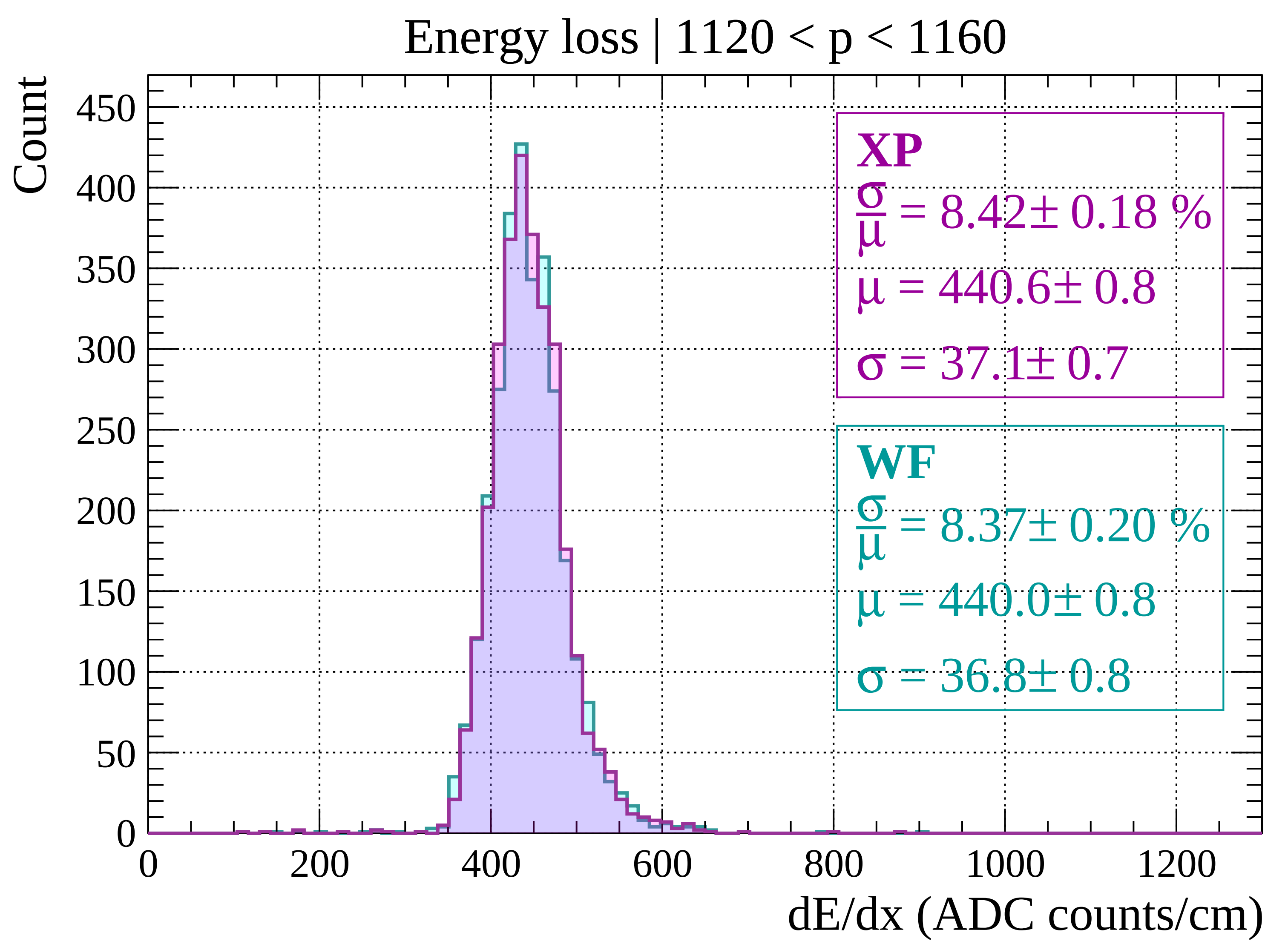

Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.99 \pm 0.20$ %
$\mu = 429.7 \pm 0.8$
$\sigma = 34.3 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 8.36 \pm 0.17$ %
$\mu = 430.9 \pm 0.8$
$\sigma = 36.0 \pm 0.7$



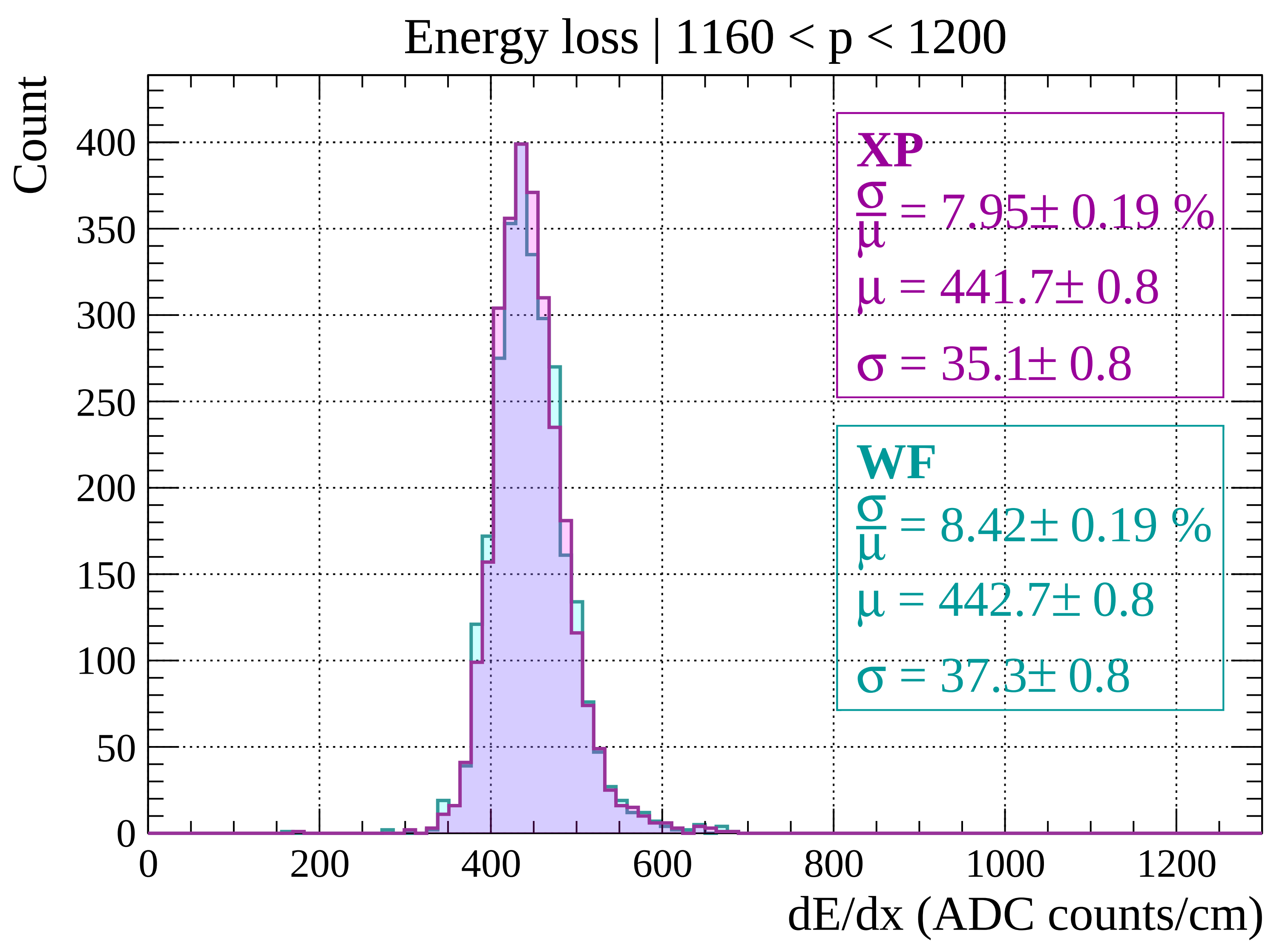
Energy loss | 1000 < p < 1040
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.95± 0.20 %
μ = 431.1± 0.8
σ = 34.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.36± 0.20 %
μ = 431.7± 0.8
σ = 36.1± 0.8


Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.08± 0.19 %
μ = 435.0± 0.8
σ = 35.1± 0.8
WF
σ/μ = 8.19± 0.19 %
μ = 435.2± 0.8
σ = 35.6± 0.8
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

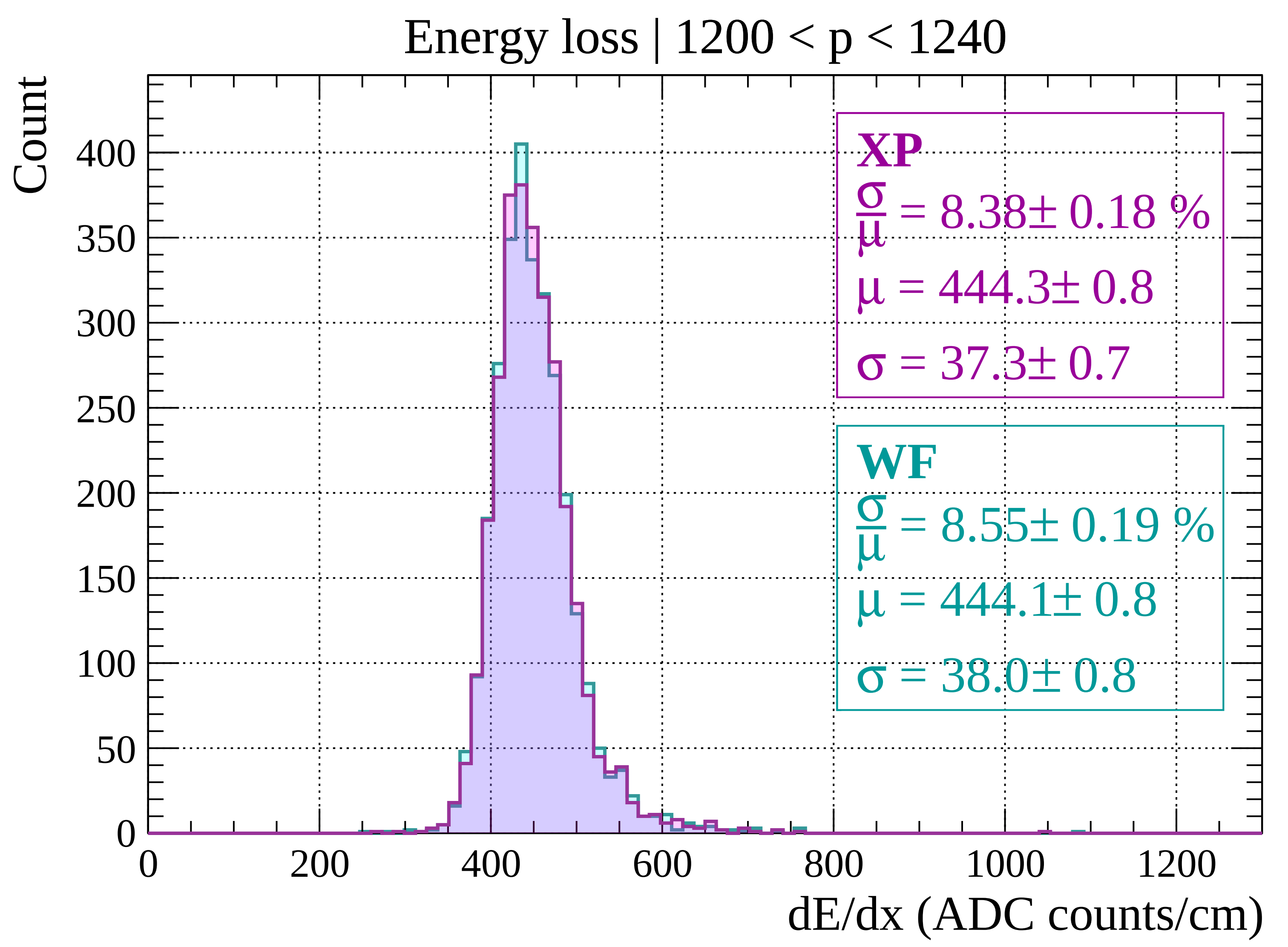

Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 8.24± 0.19 %
μ = 437.3± 0.8
σ = 36.0± 0.8
WF
σ/μ = 8.48± 0.21 %
μ = 437.2± 0.8
σ = 37.1± 0.8



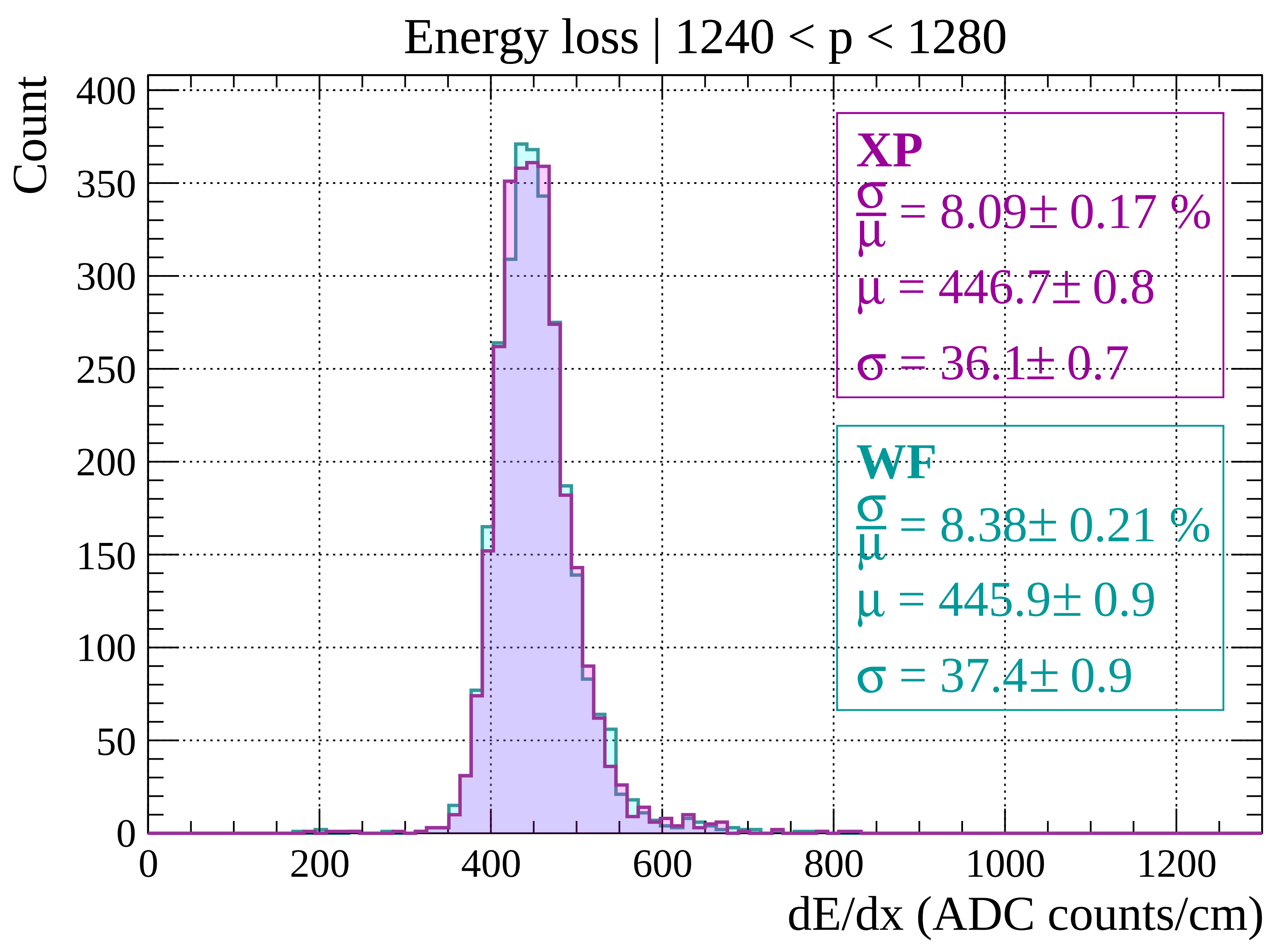
Energy loss | 1120 < p < 1160
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.42± 0.18 %
μ = 440.6± 0.8
σ = 37.1± 0.7
WF
σ/μ = 8.37± 0.20 %
μ = 440.0± 0.8
σ = 36.8± 0.8



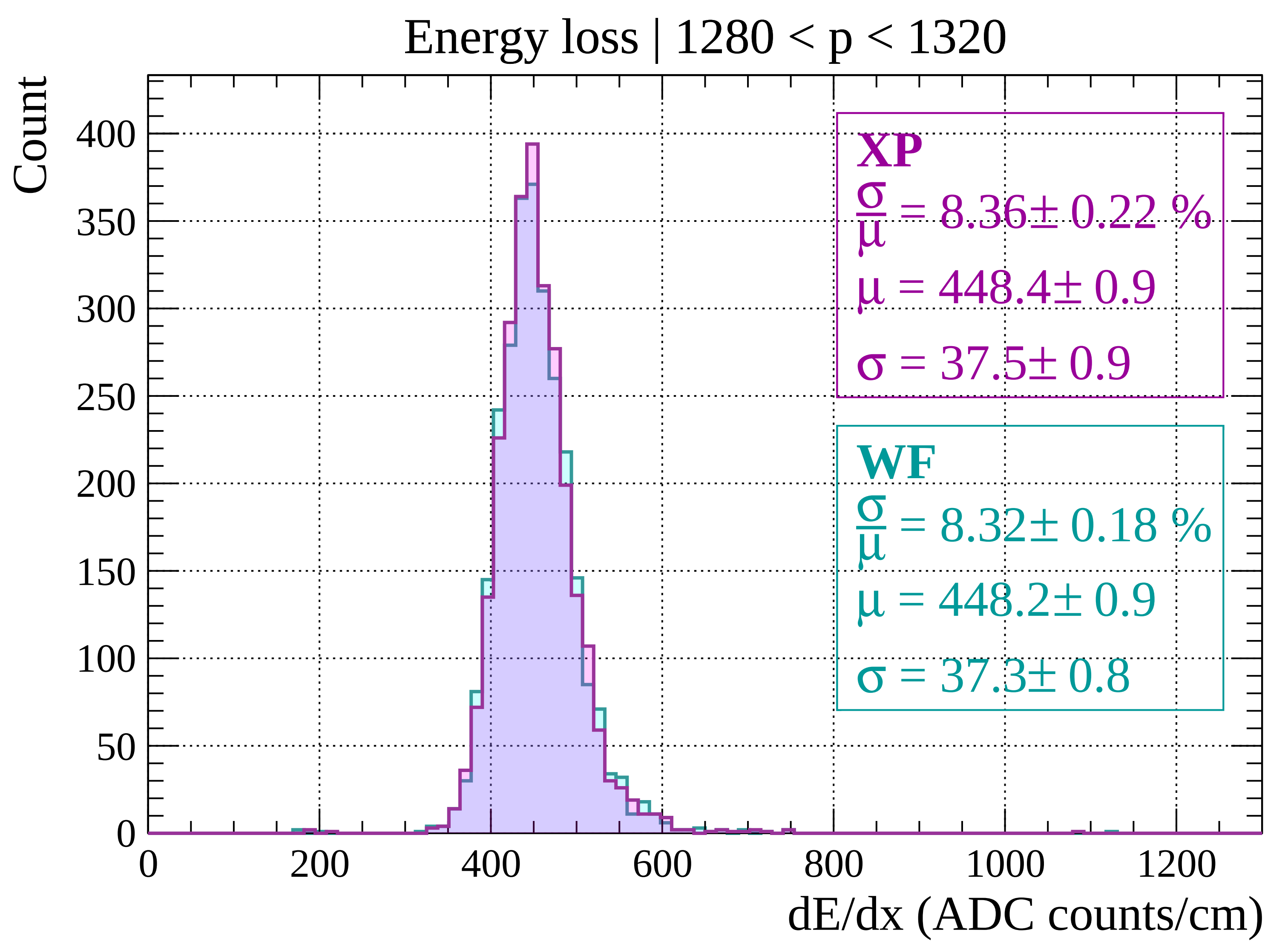
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 7.95± 0.19 %
μ = 441.7± 0.8
σ = 35.1± 0.8
WF
σ/μ = 8.42± 0.19 %
μ = 442.7± 0.8
σ = 37.3± 0.8



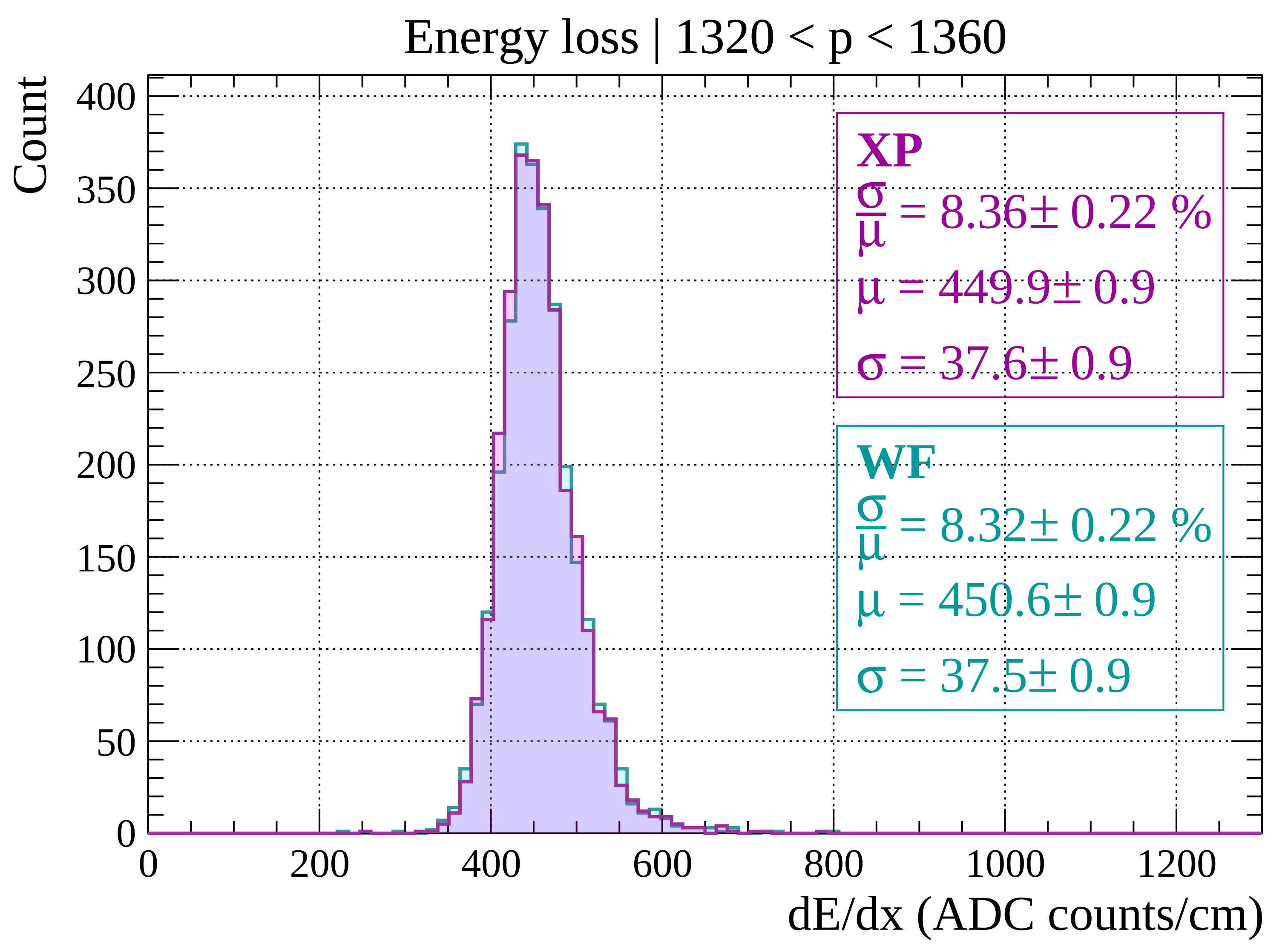
Energy loss | 1200 < p < 1240
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.38± 0.18 %
μ = 444.3± 0.8
σ = 37.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.55± 0.19 %
μ = 444.1± 0.8
σ = 38.0± 0.8



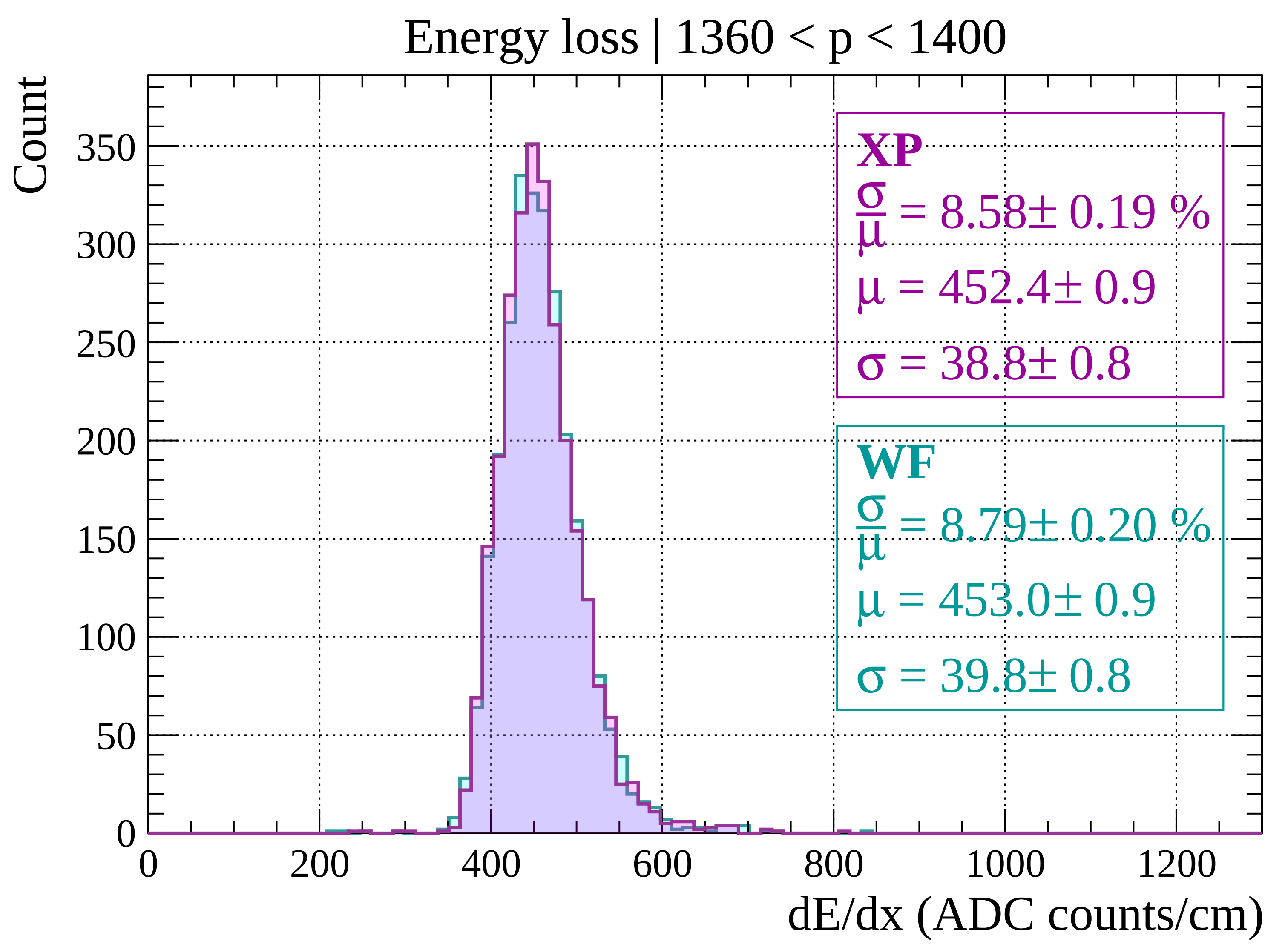
Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.09 ± 0.17 %
μ = 446.7 ± 0.8
σ = 36.1 ± 0.7
WF
σ/μ = 8.38 ± 0.21 %
μ = 445.9 ± 0.9
σ = 37.4 ± 0.9



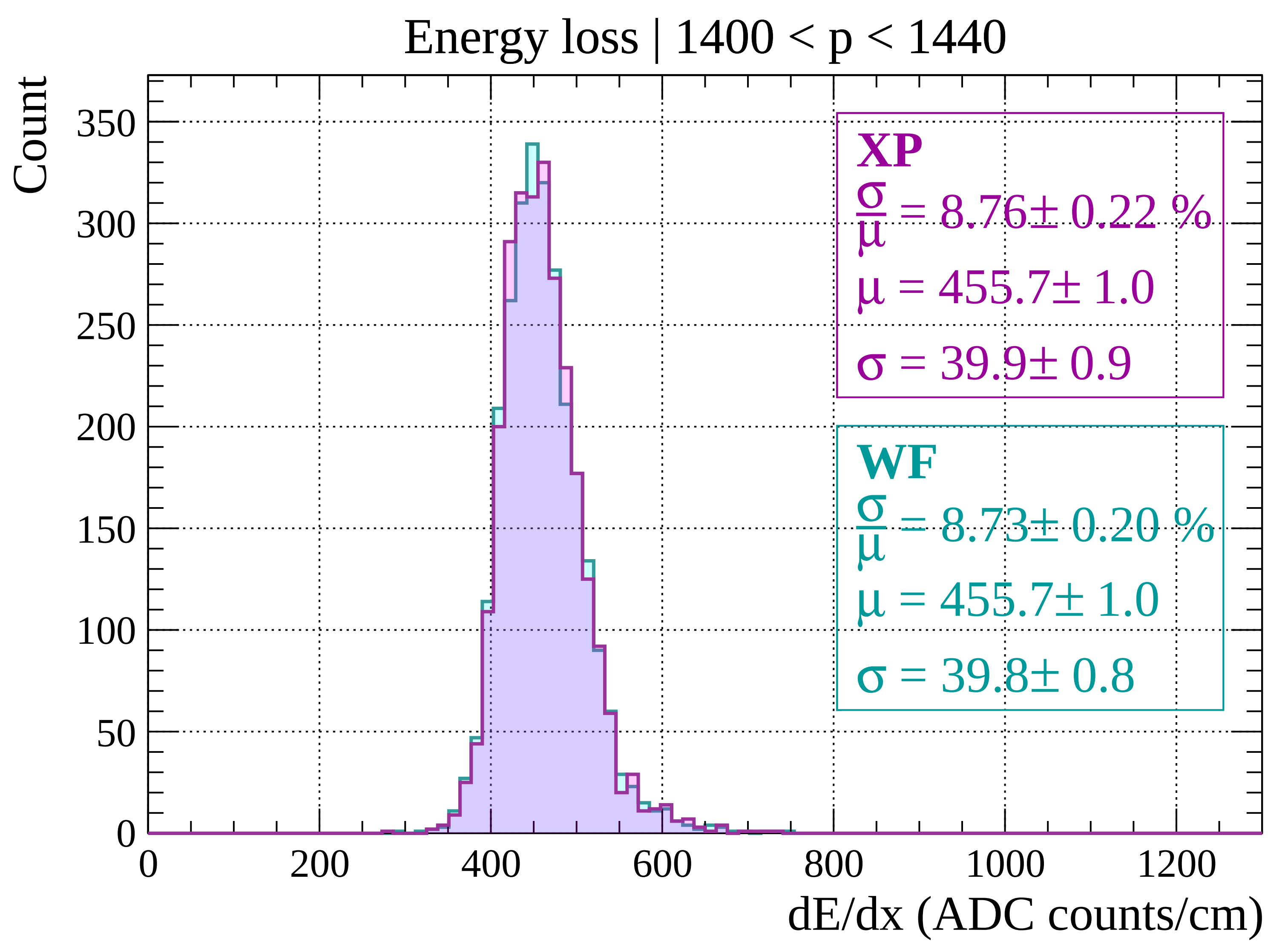
Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.36± 0.22 %
μ = 448.4± 0.9
σ = 37.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.32± 0.18 %
μ = 448.2± 0.9
σ = 37.3± 0.8



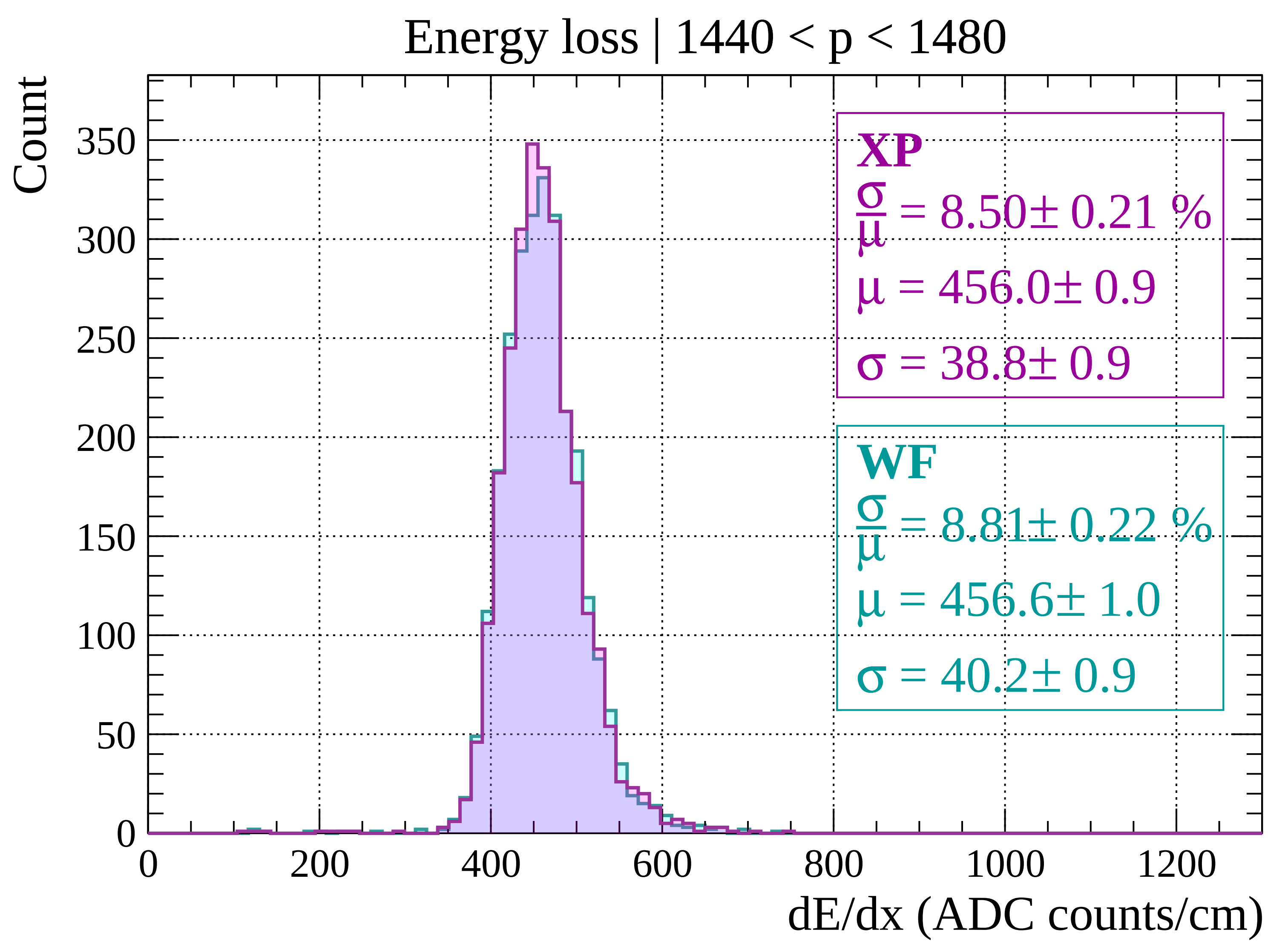
Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.36± 0.22 %
μ = 449.9± 0.9
σ = 37.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.32± 0.22 %
μ = 450.6± 0.9
σ = 37.5± 0.9



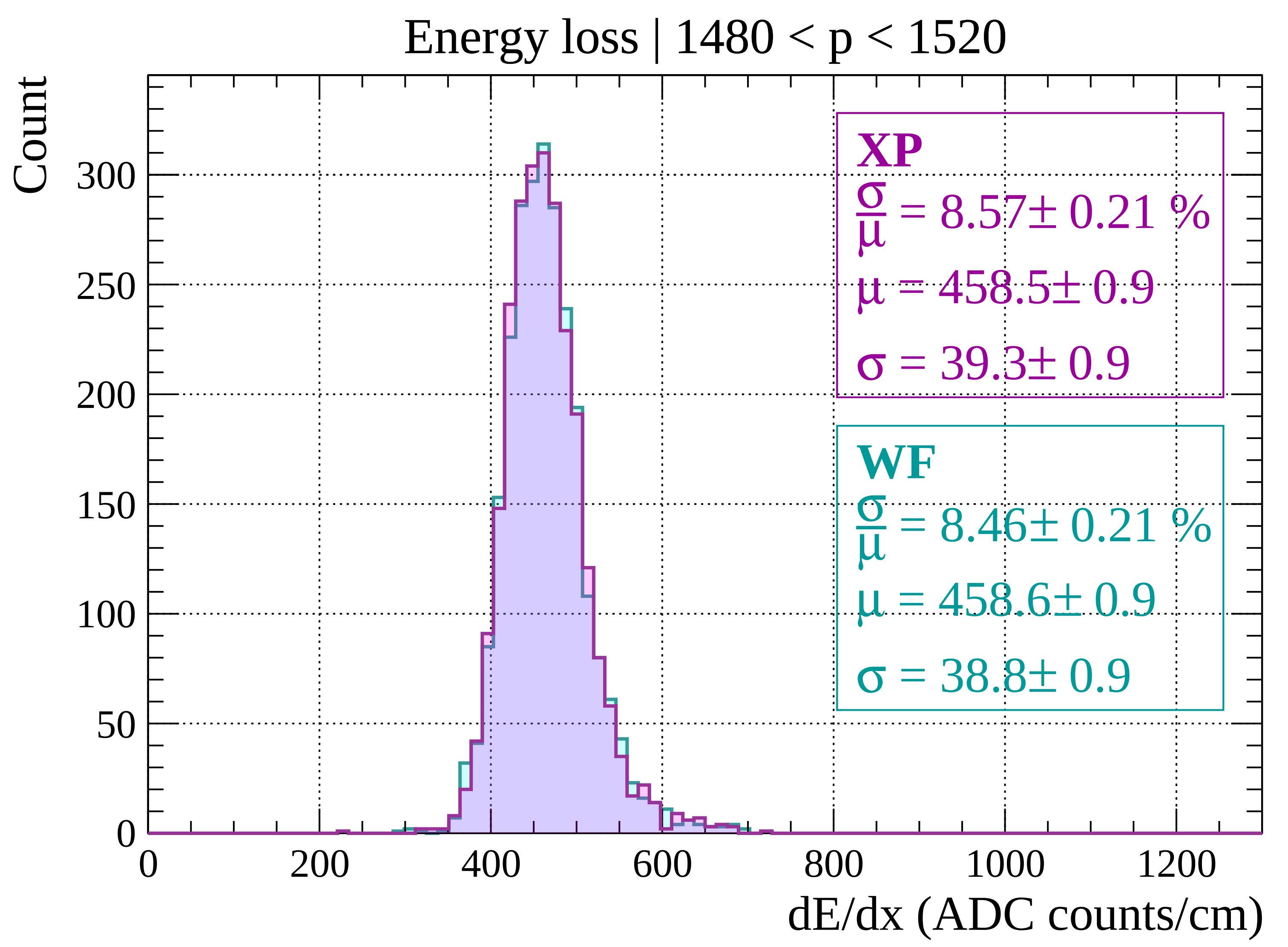
Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.58± 0.19 %
μ = 452.4± 0.9
σ = 38.8± 0.8
WF
σ/μ = 8.79± 0.20 %
μ = 453.0± 0.9
σ = 39.8± 0.8



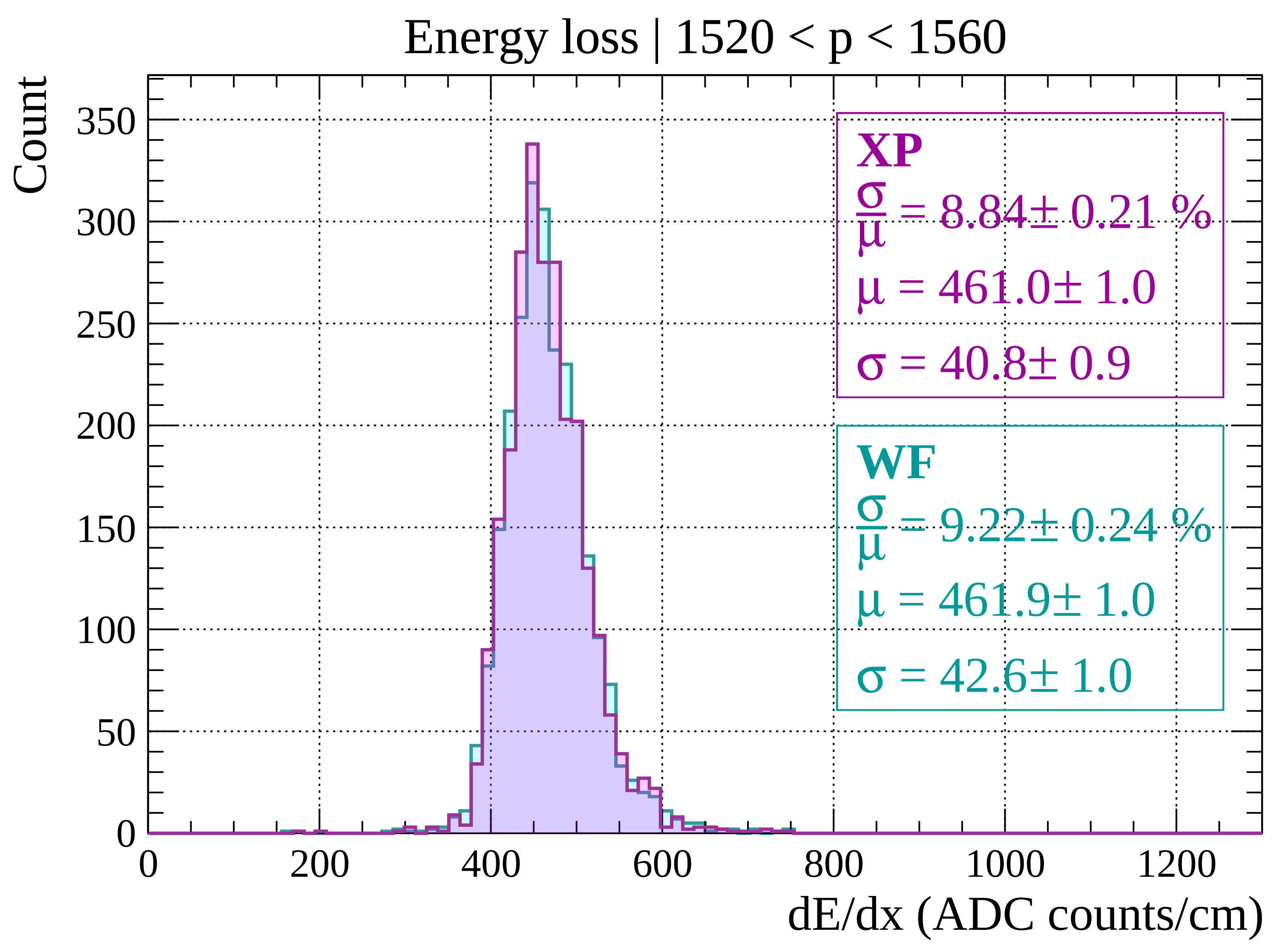
Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.76± 0.22 %
μ = 455.7± 1.0
σ = 39.9± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.73± 0.20 %
μ = 455.7± 1.0
σ = 39.8± 0.8

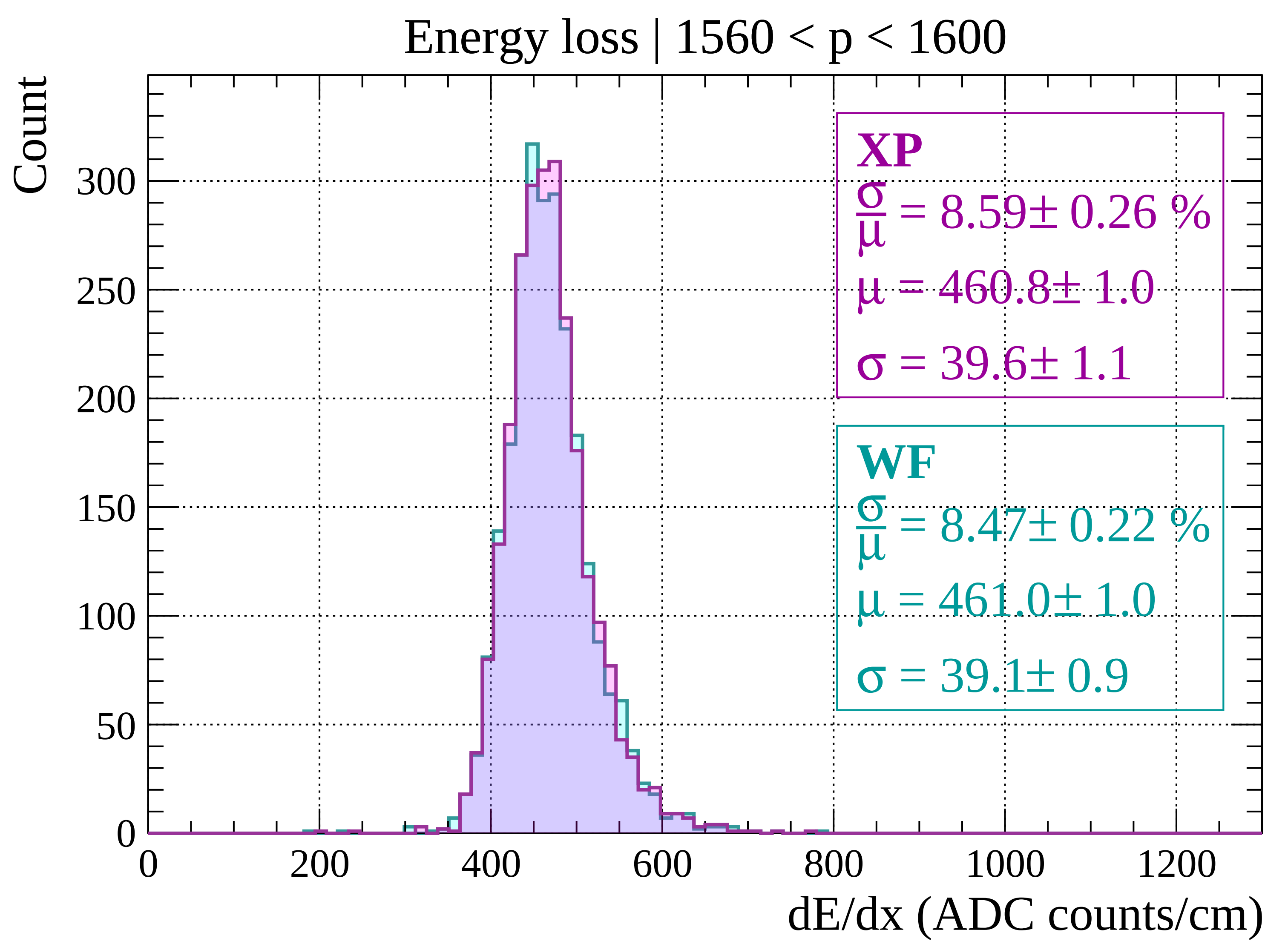


Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu} = 8.50 \pm 0.21$ %
$\mu = 456.0 \pm 0.9$
$\sigma = 38.8 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 8.81 \pm 0.22$ %
$\mu = 456.6 \pm 1.0$
$\sigma = 40.2 \pm 0.9$

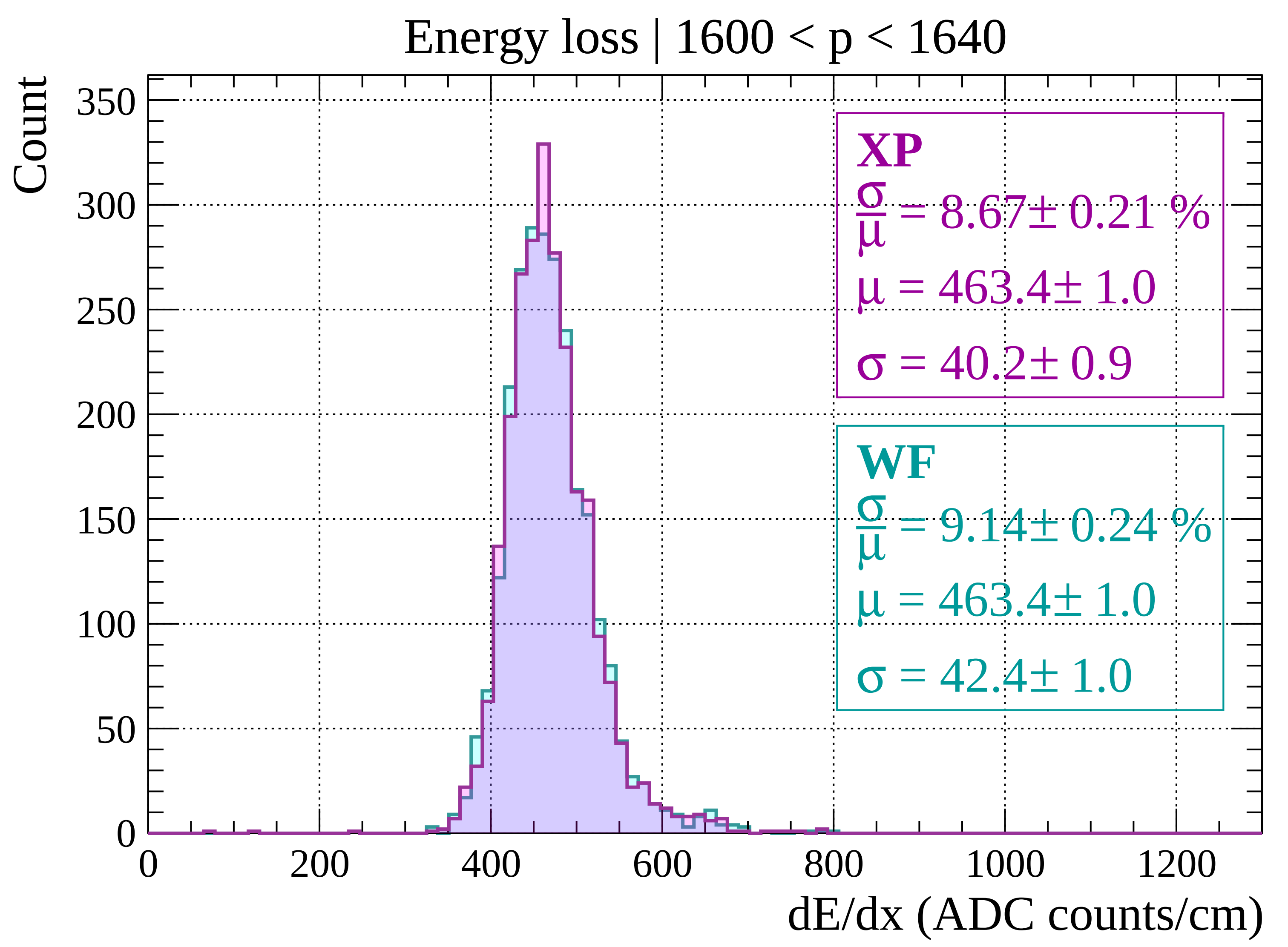


Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.57± 0.21 %
μ = 458.5± 0.9
σ = 39.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.21 %
μ = 458.6± 0.9
σ = 38.8± 0.9

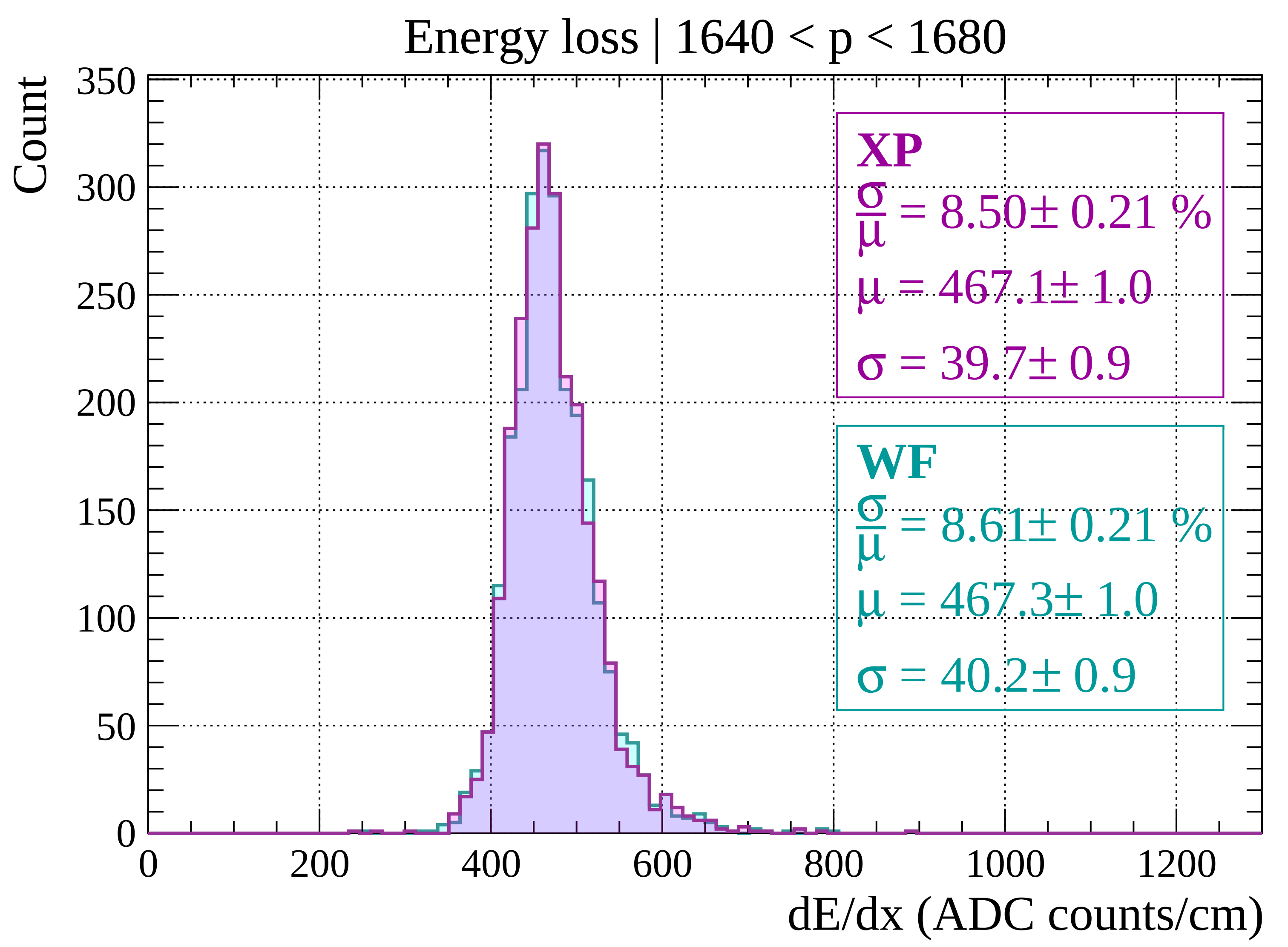

Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.84± 0.21 %
μ = 461.0± 1.0
σ = 40.8± 0.9
WF
σ/μ = 9.22± 0.24 %
μ = 461.9± 1.0
σ = 42.6± 1.0



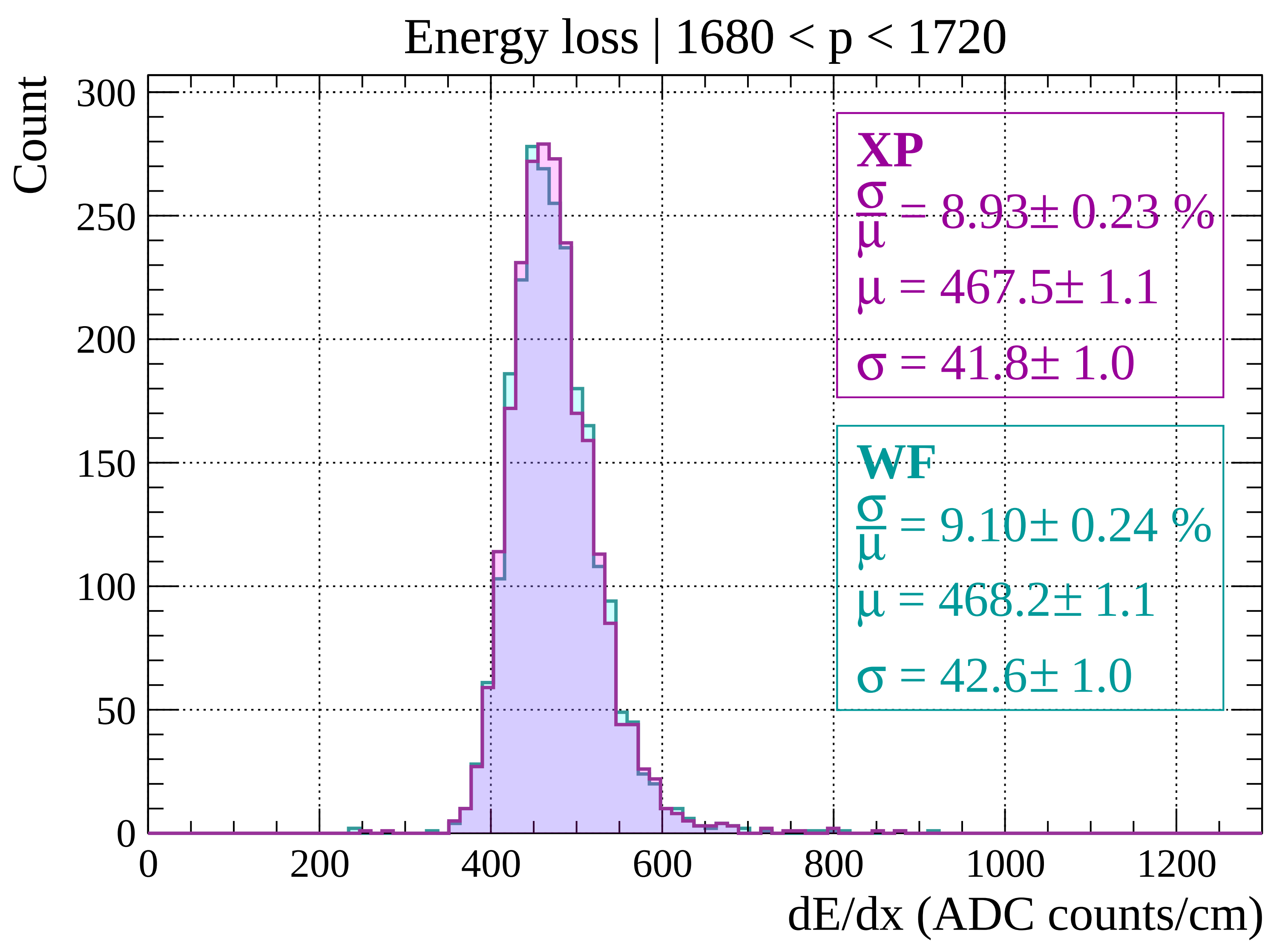
Energy loss | 1560 < p < 1600
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.59± 0.26 %
μ = 460.8± 1.0
σ = 39.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.47± 0.22 %
μ = 461.0± 1.0
σ = 39.1± 0.9



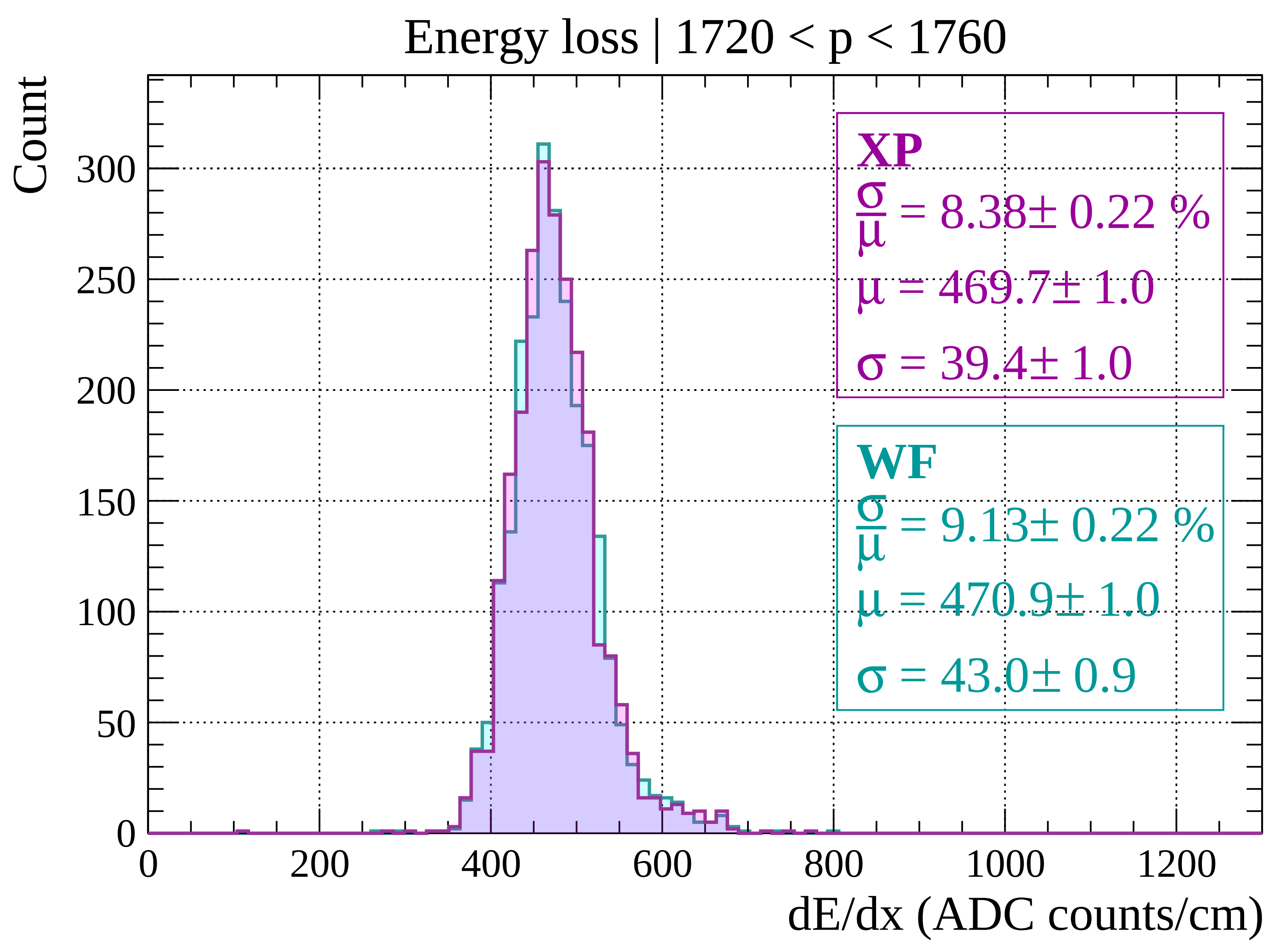
Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.67± 0.21 %
μ = 463.4± 1.0
σ = 40.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.14± 0.24 %
μ = 463.4± 1.0
σ = 42.4± 1.0



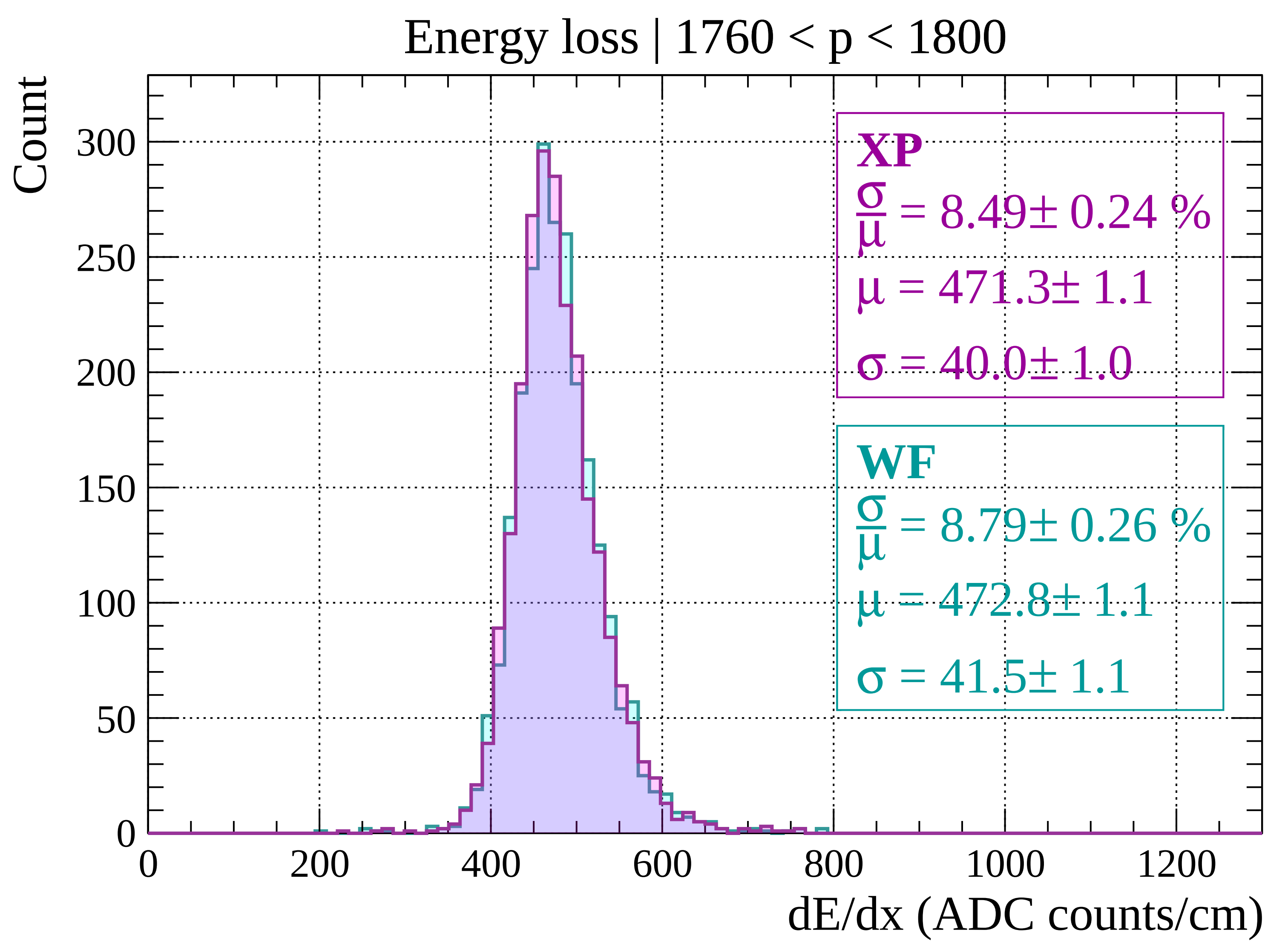
Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.50± 0.21 %
μ = 467.1± 1.0
σ = 39.7± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.61± 0.21 %
μ = 467.3± 1.0
σ = 40.2± 0.9



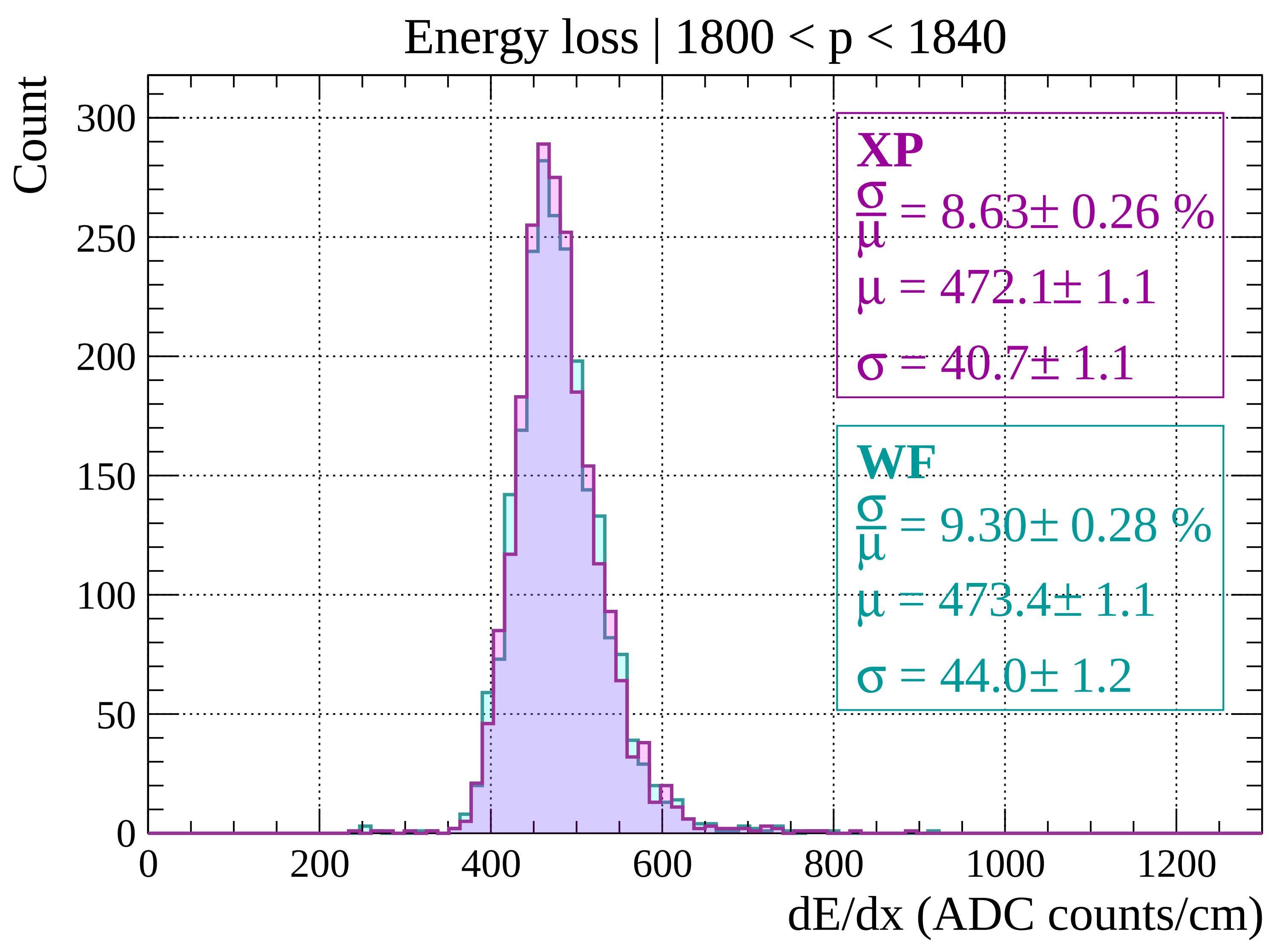
Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.93± 0.23 %
μ = 467.5± 1.1
σ = 41.8± 1.0
WF
σ/μ = 9.10± 0.24 %
μ = 468.2± 1.1
σ = 42.6± 1.0



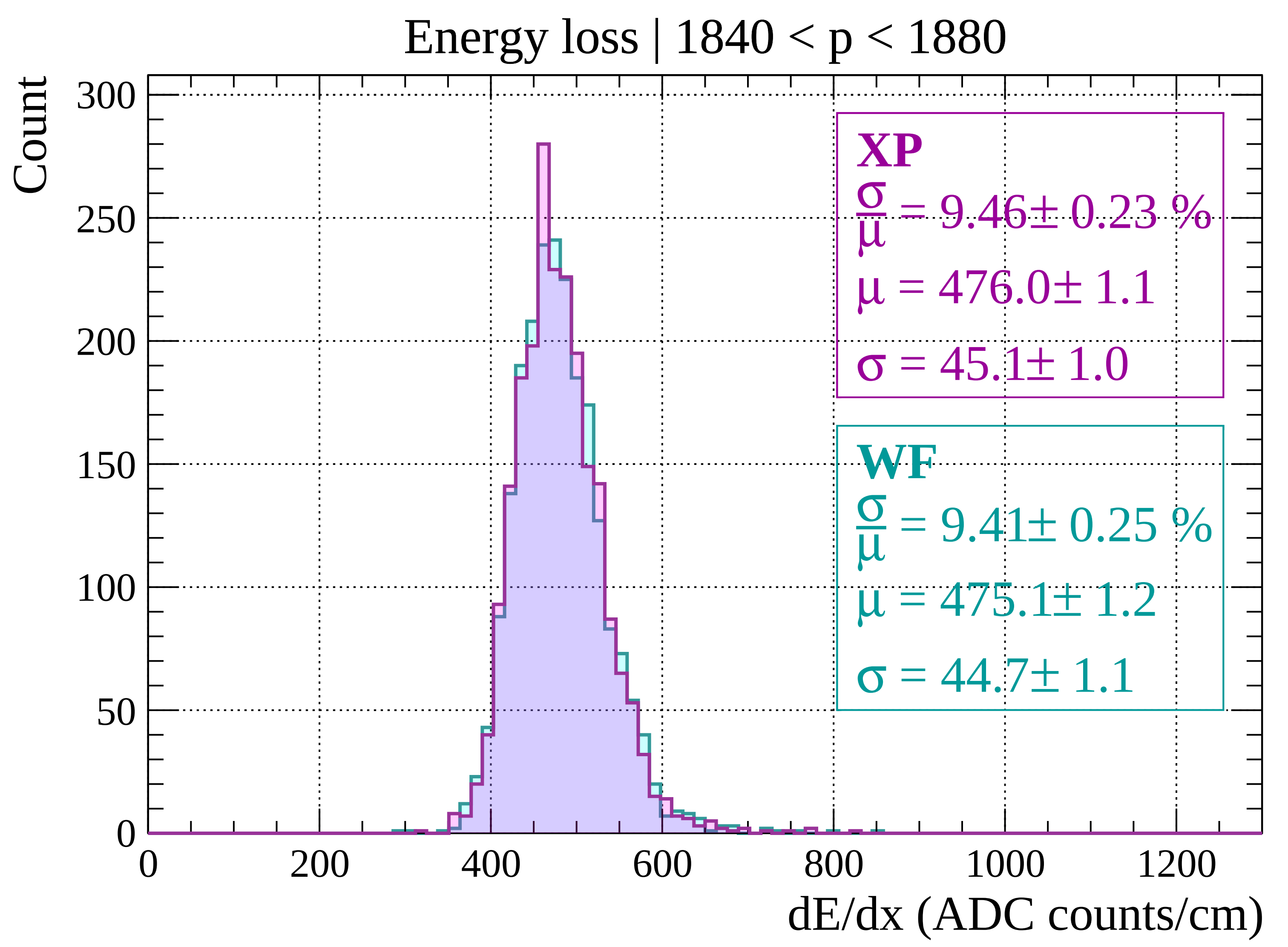
Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.38± 0.22 %
μ = 469.7± 1.0
σ = 39.4± 1.0
WF
σ/μ = 9.13± 0.22 %
μ = 470.9± 1.0
σ = 43.0± 0.9



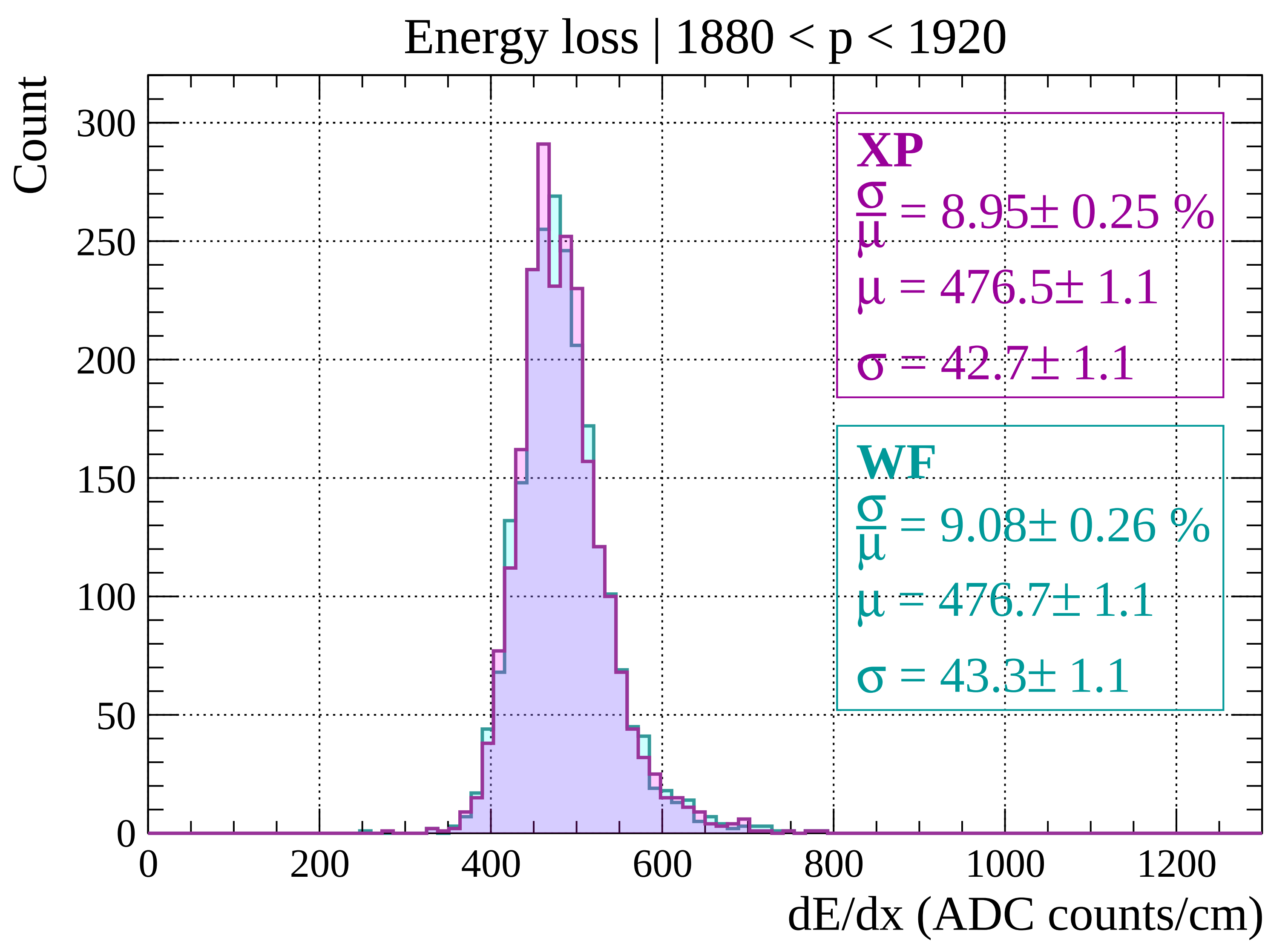
Energy loss | 1760 < p < 1800
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.49± 0.24 %
μ = 471.3± 1.1
σ = 40.0± 1.0
WF
σ/μ = 8.79± 0.26 %
μ = 472.8± 1.1
σ = 41.5± 1.1



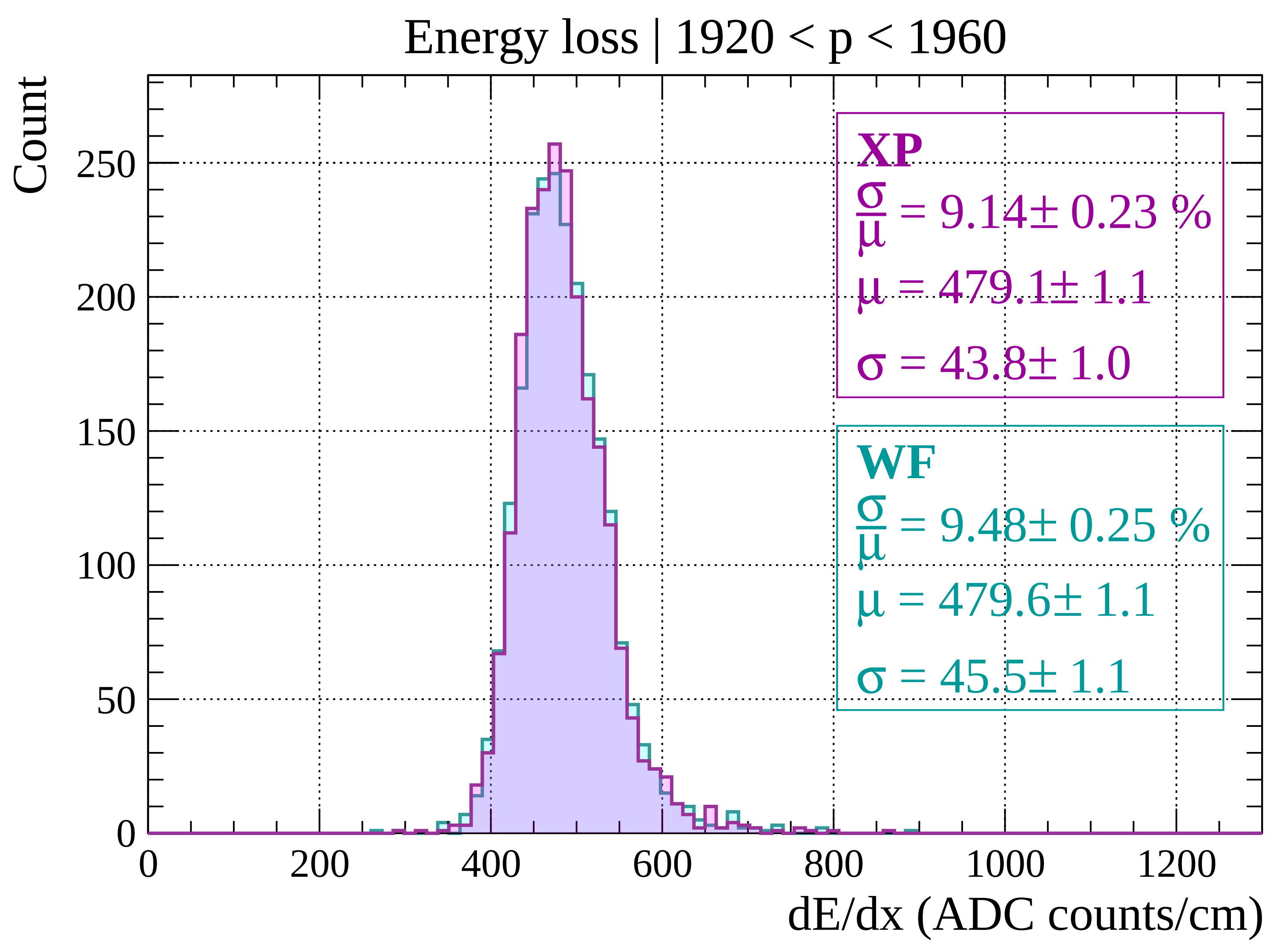
Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 8.63± 0.26 %
μ = 472.1± 1.1
σ = 40.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.30± 0.28 %
μ = 473.4± 1.1
σ = 44.0± 1.2



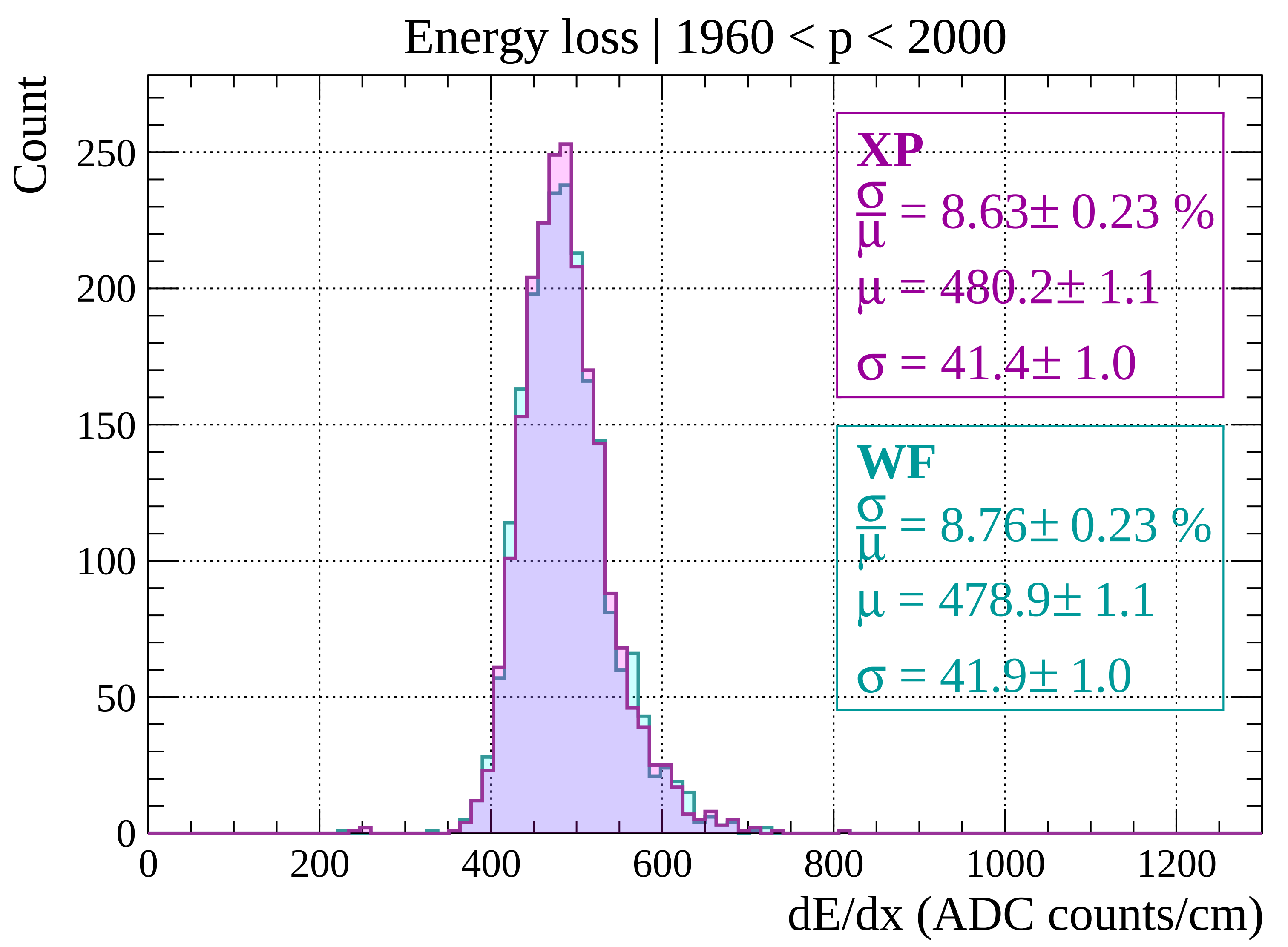
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu} = 9.46 \pm 0.23\ \%$
$\mu = 476.0 \pm 1.1$
$\sigma = 45.1 \pm 1.0$
WF
$\frac{\sigma}{\mu} = 9.41 \pm 0.25\ \%$
$\mu = 475.1 \pm 1.2$
$\sigma = 44.7 \pm 1.1$



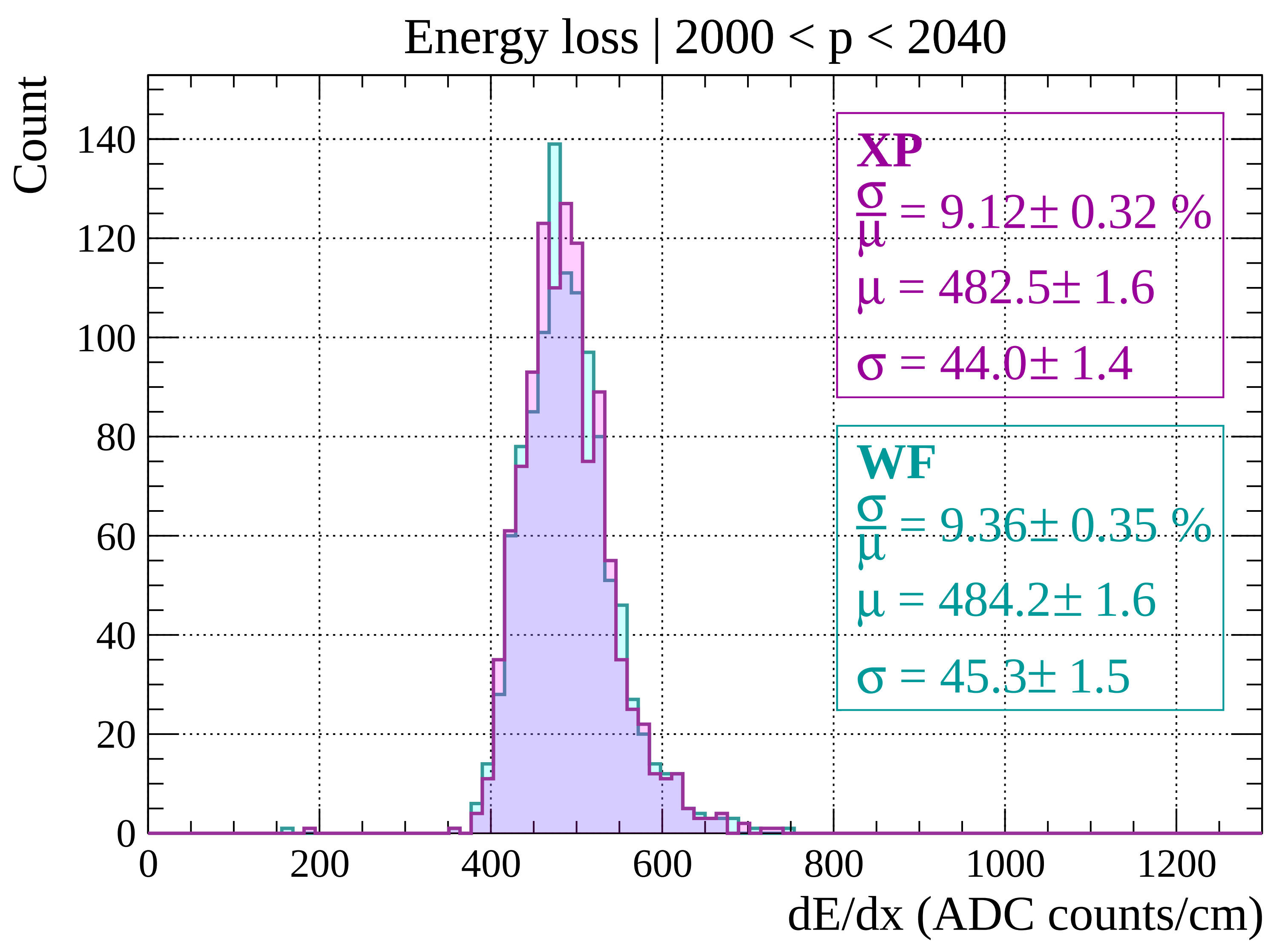
Energy loss | 1880 < p < 1920
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.95 \pm 0.25$ %
$\mu = 476.5 \pm 1.1$
$\sigma = 42.7 \pm 1.1$
WF
$\frac{\sigma}{\mu} = 9.08 \pm 0.26$ %
$\mu = 476.7 \pm 1.1$
$\sigma = 43.3 \pm 1.1$



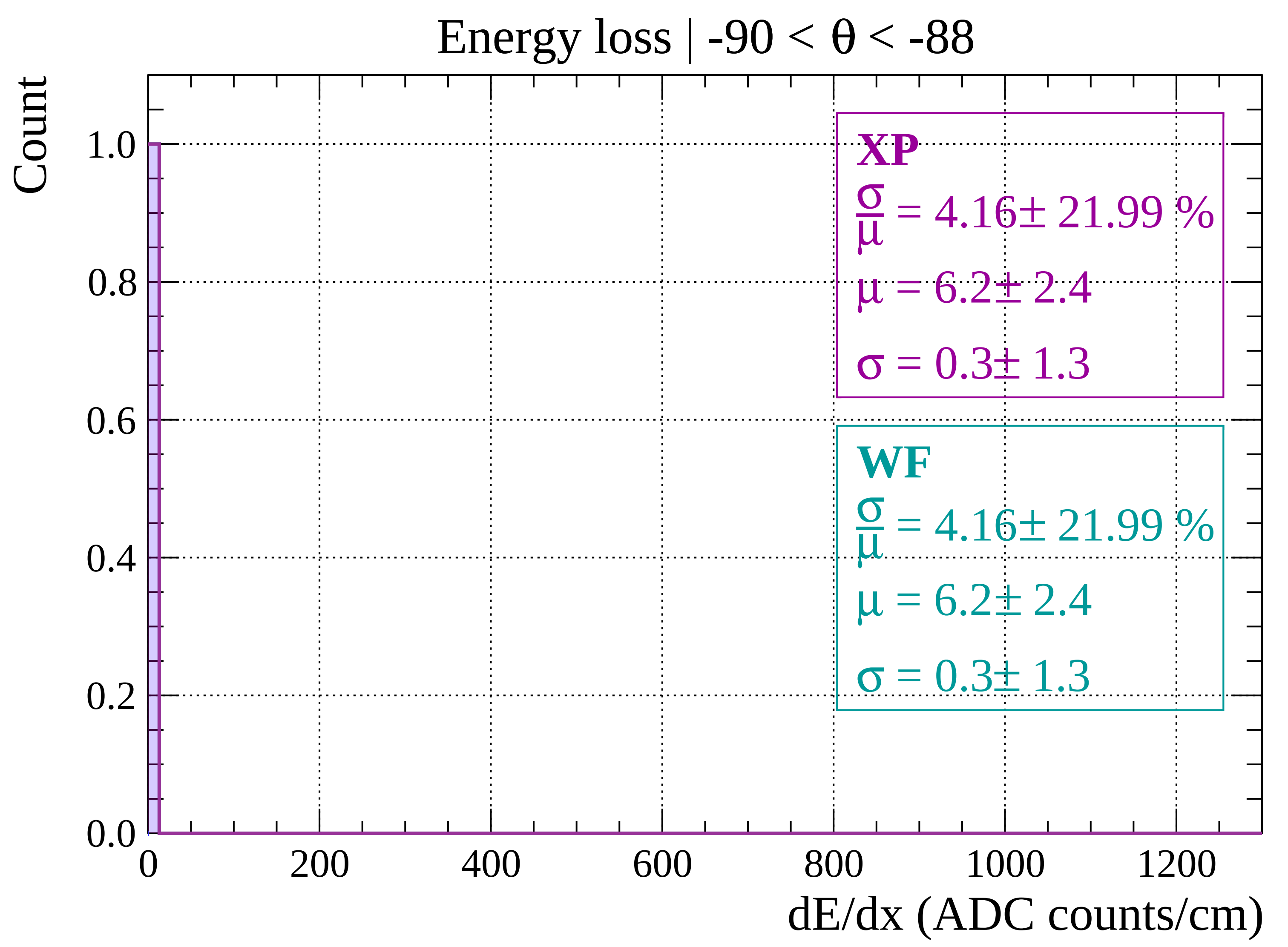
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 9.14± 0.23 %
μ = 479.1± 1.1
σ = 43.8± 1.0
WF
σ/μ = 9.48± 0.25 %
μ = 479.6± 1.1
σ = 45.5± 1.1



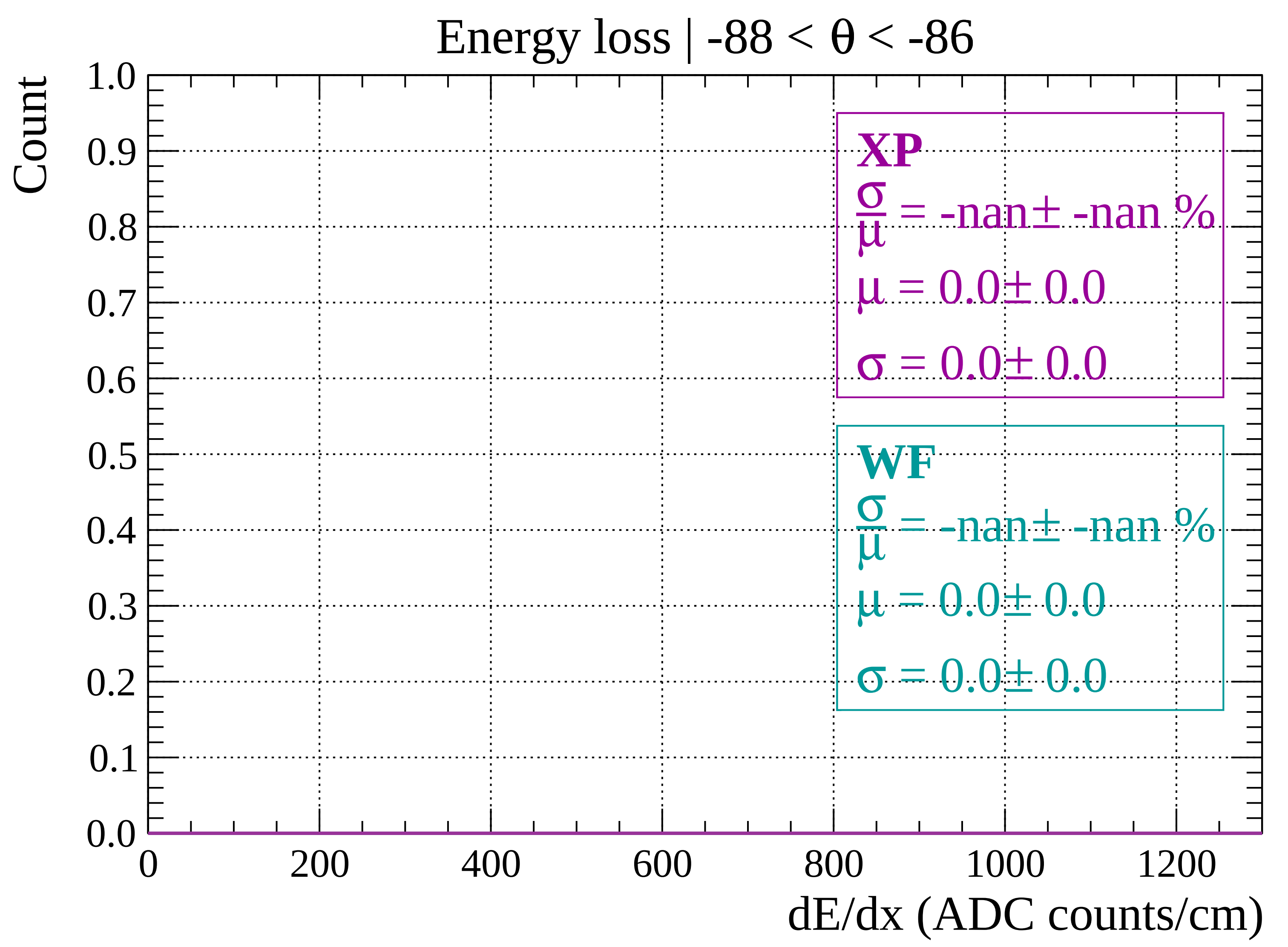
Energy loss | 1960 < p < 2000
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.63± 0.23 %
μ = 480.2± 1.1
σ = 41.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.76± 0.23 %
μ = 478.9± 1.1
σ = 41.9± 1.0



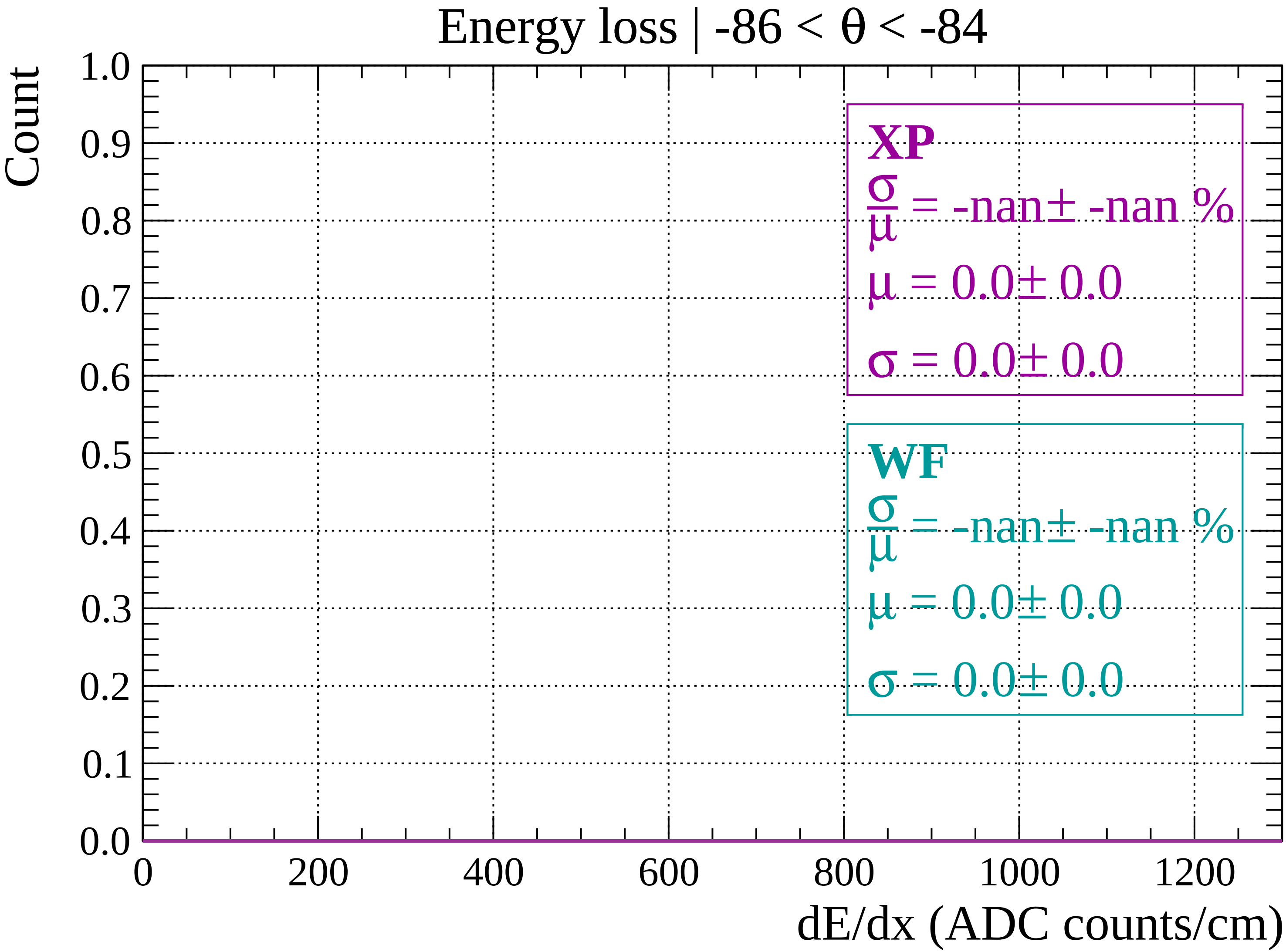
Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.12± 0.32 %
μ = 482.5± 1.6
σ = 44.0± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.36± 0.35 %
μ = 484.2± 1.6
σ = 45.3± 1.5
0
20
40
60
80
100
120
140
200
400
600
800
1000
1200

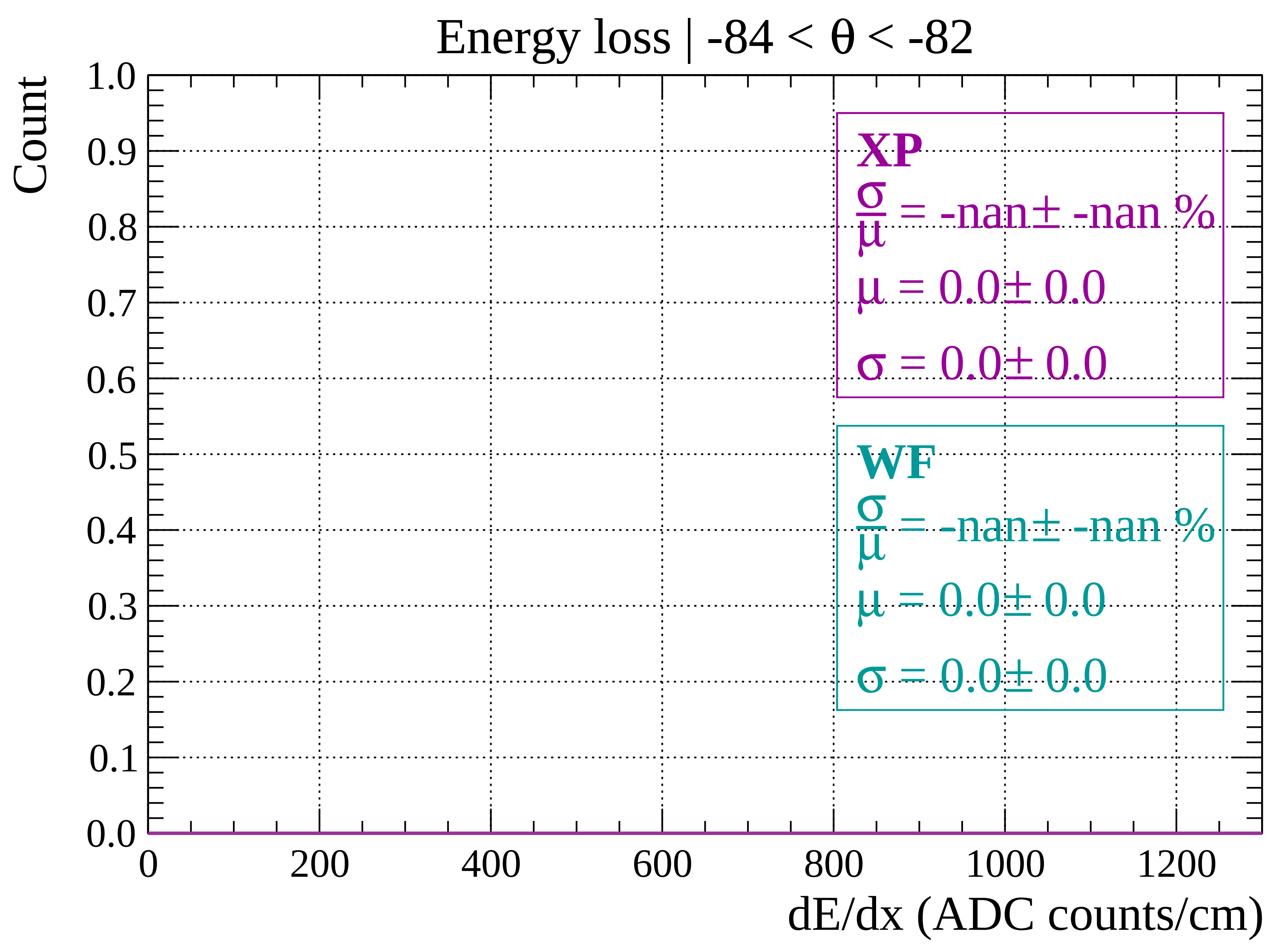


Energy loss | -90 < θ < -88
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3

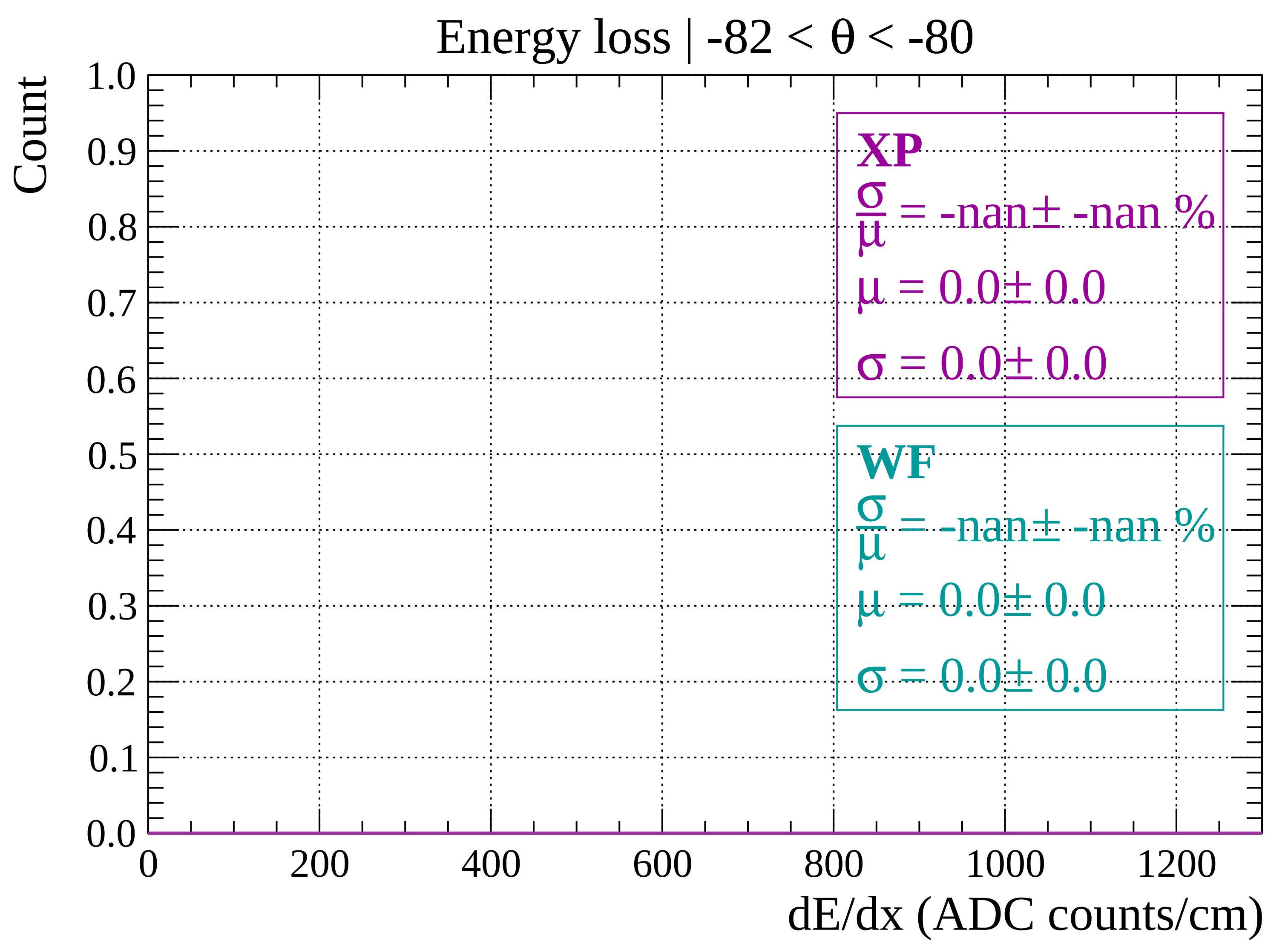


Energy loss | -88 < θ < -86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

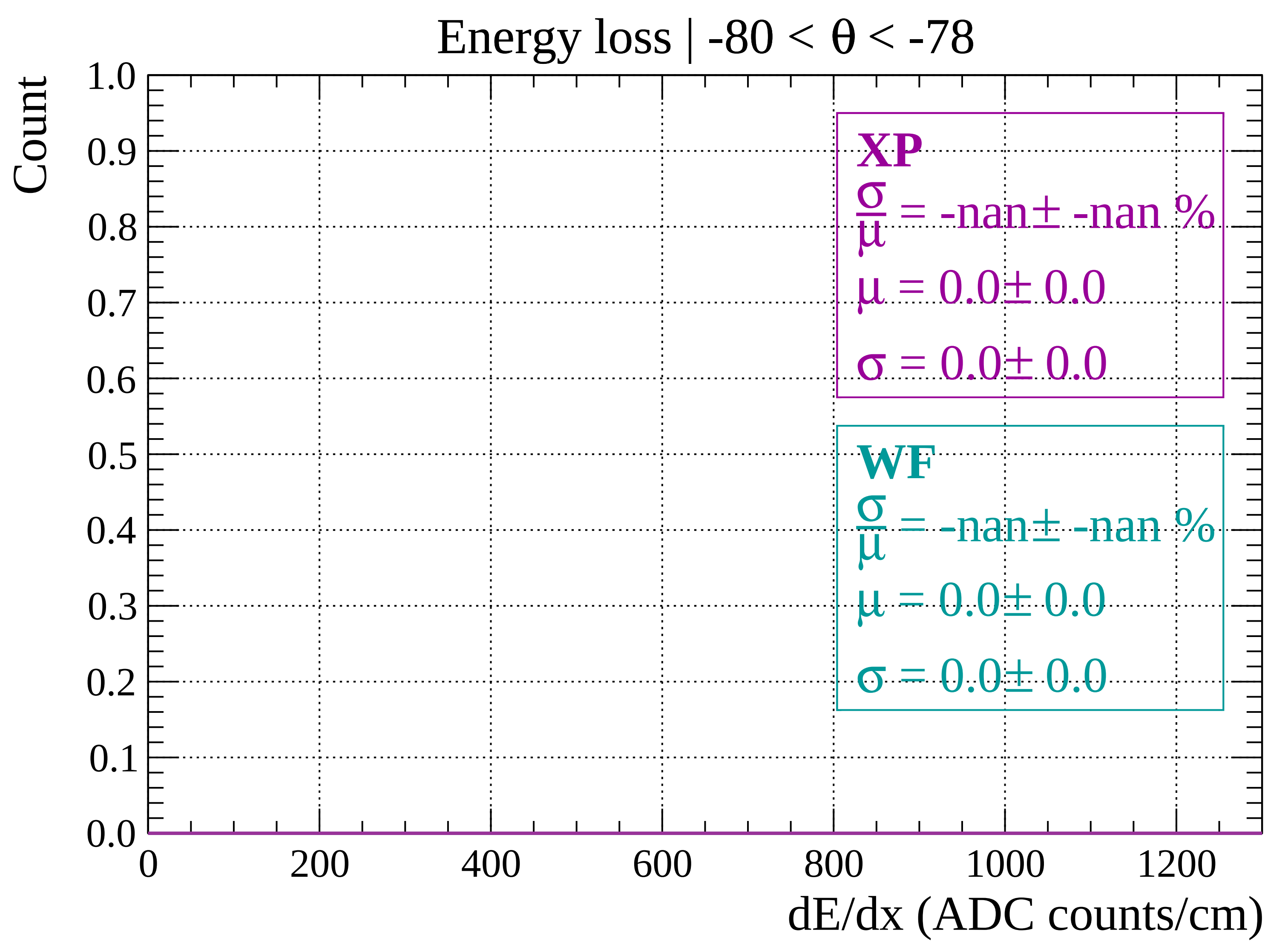

Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

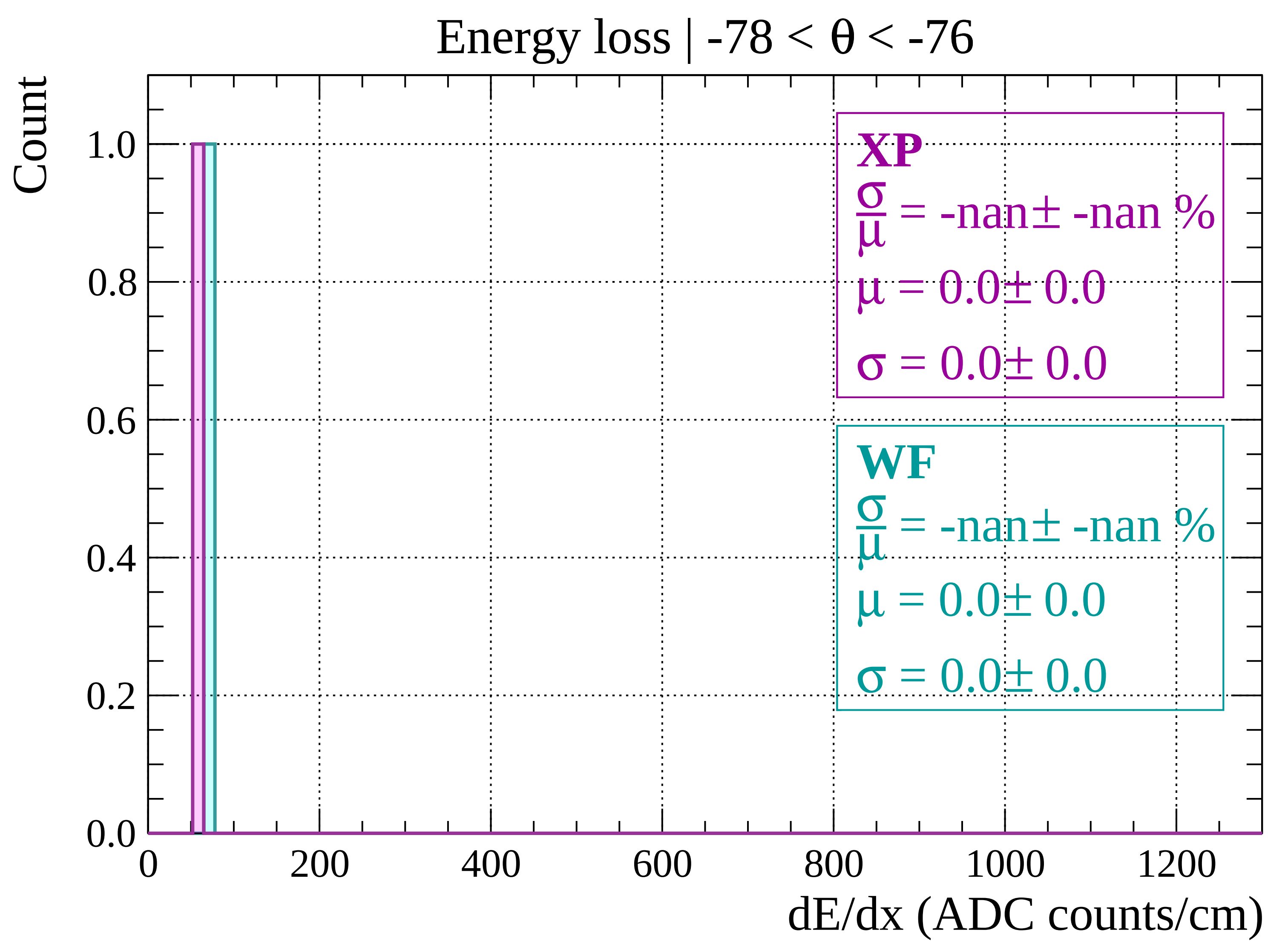


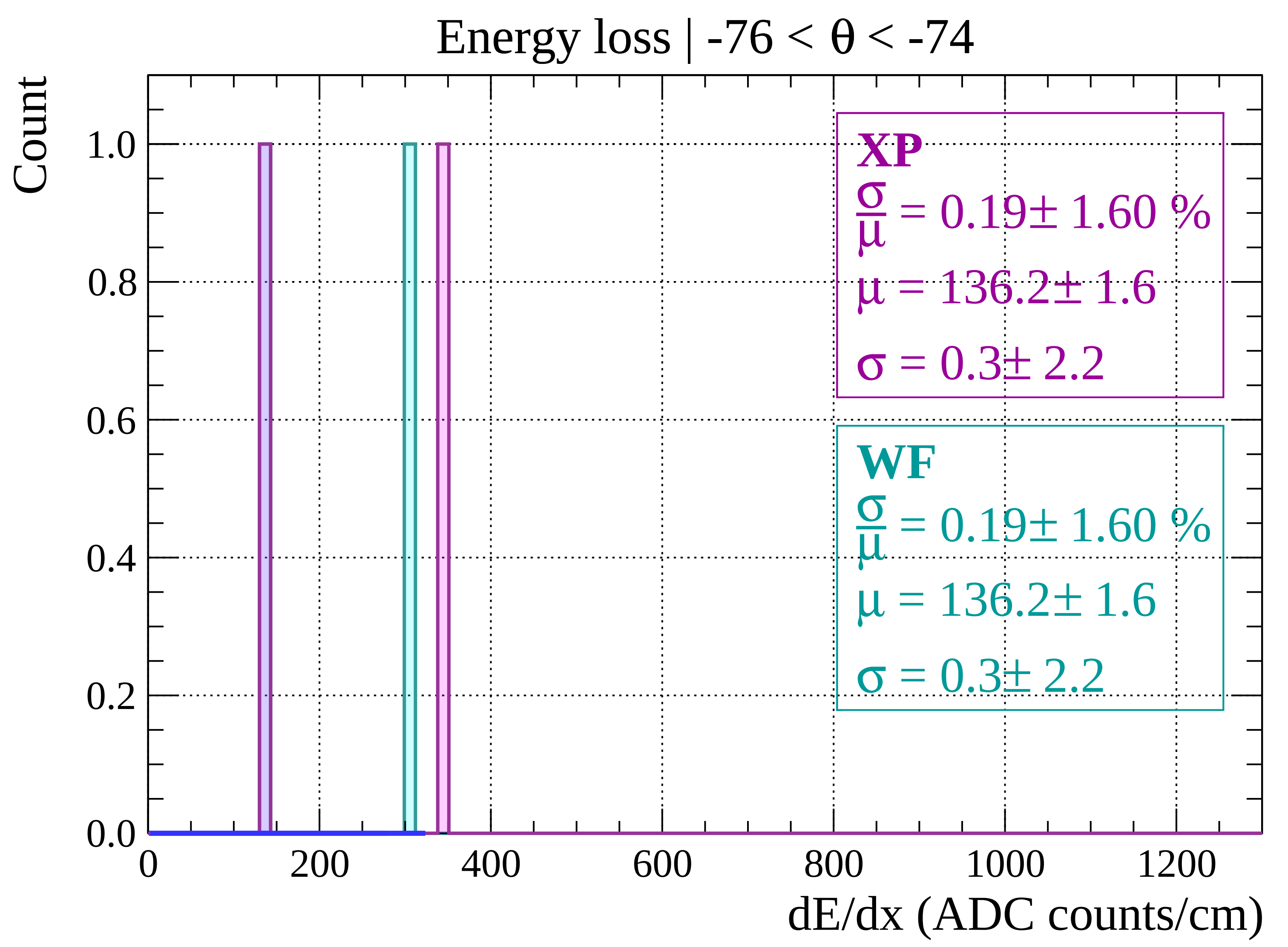
Energy loss | -84 < θ < -82
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



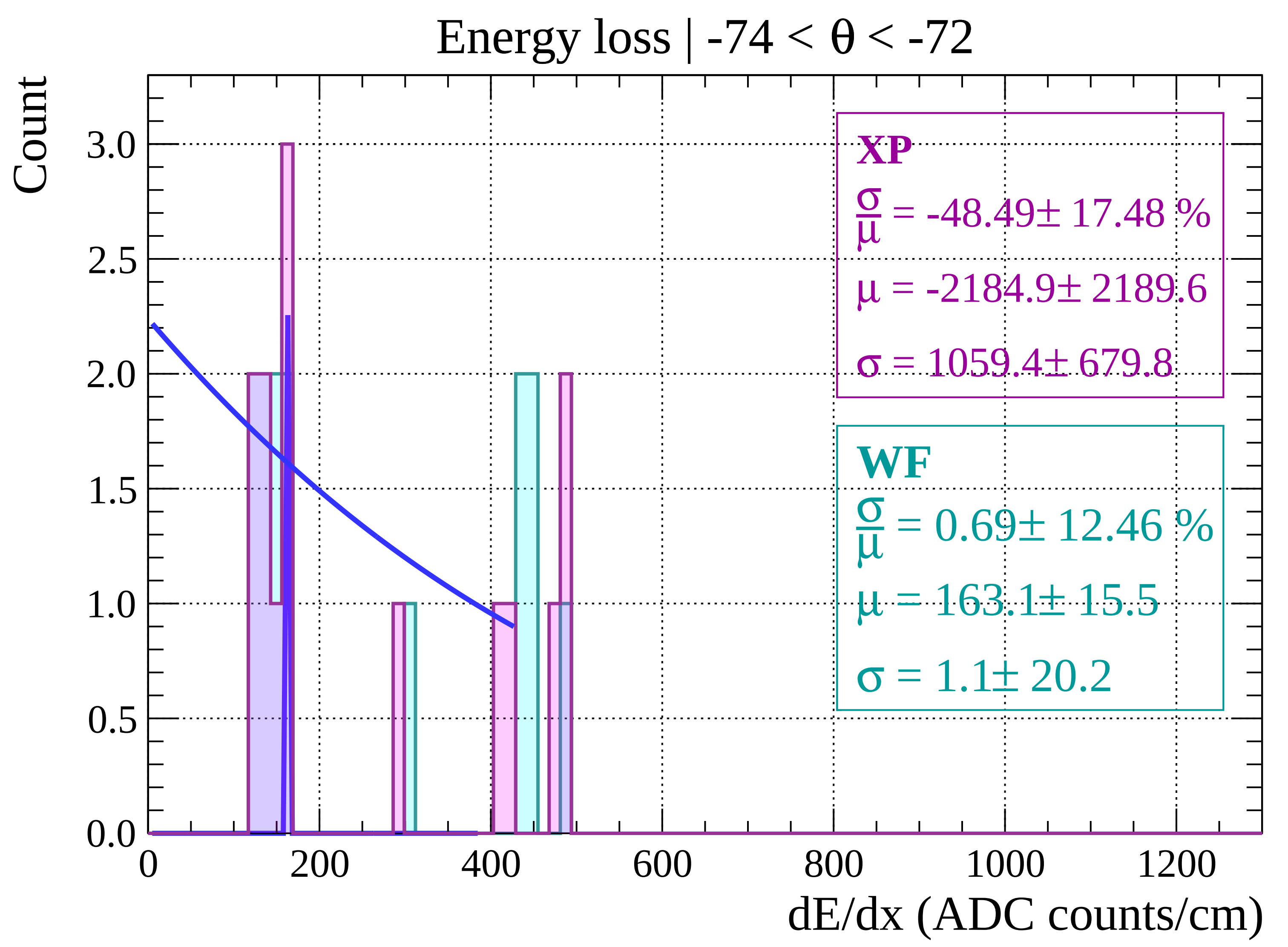
Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

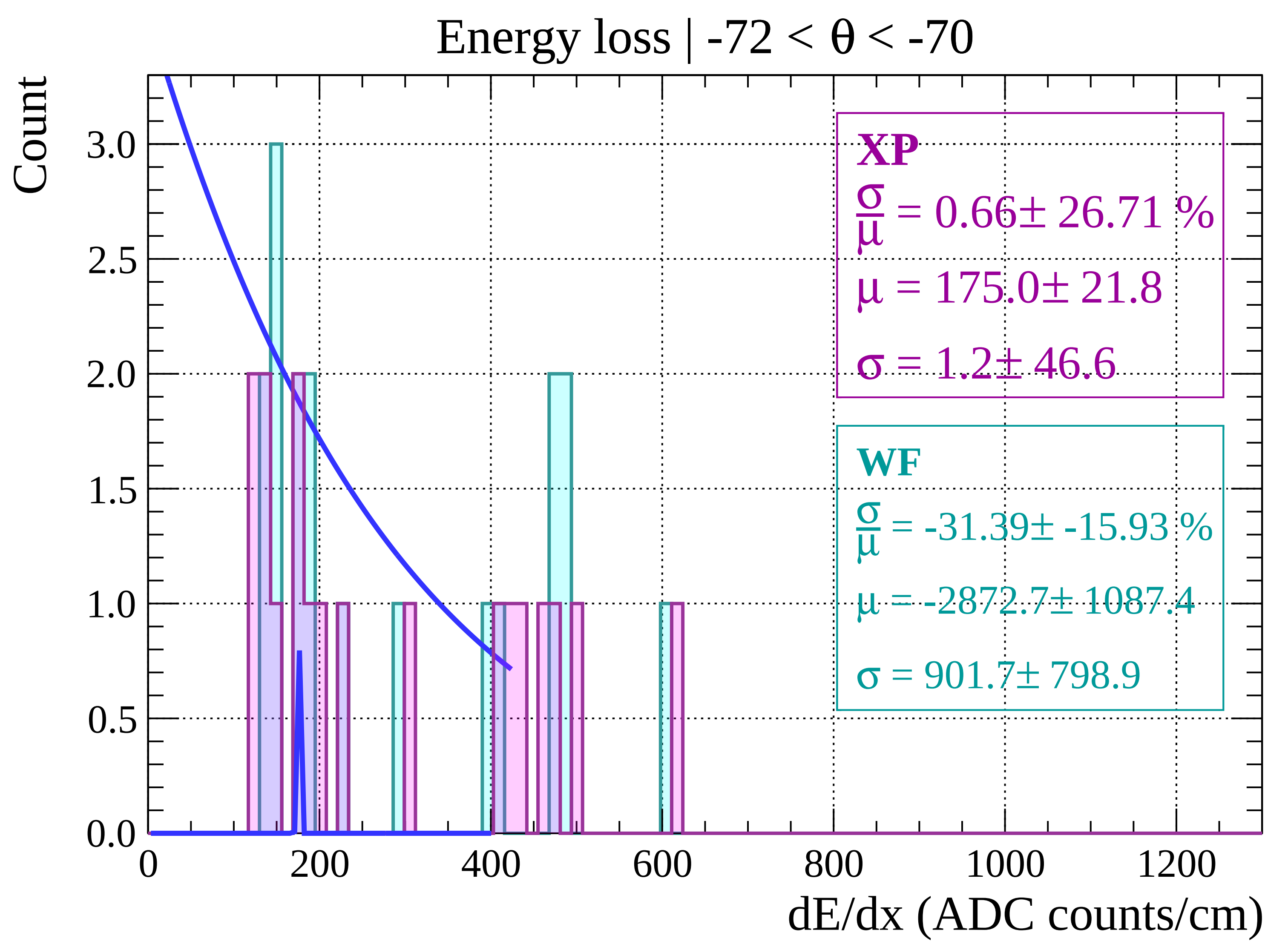

Energy loss | -80 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -78 < θ < -76
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

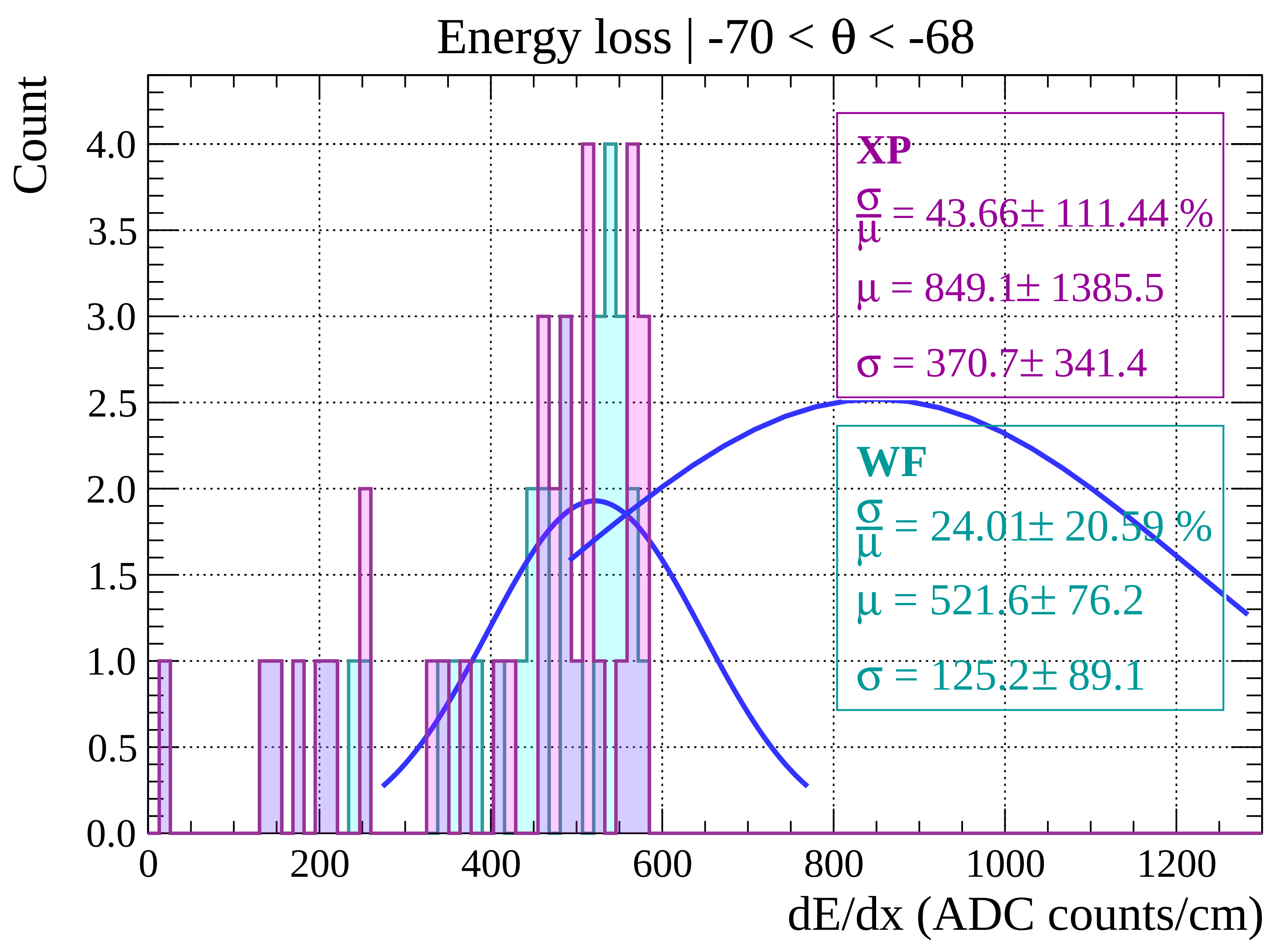


Energy loss | -76 < θ < -74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.19± 1.60 %
μ = 136.2± 1.6
σ = 0.3± 2.2
WF
σ/μ = 0.19± 1.60 %
μ = 136.2± 1.6
σ = 0.3± 2.2

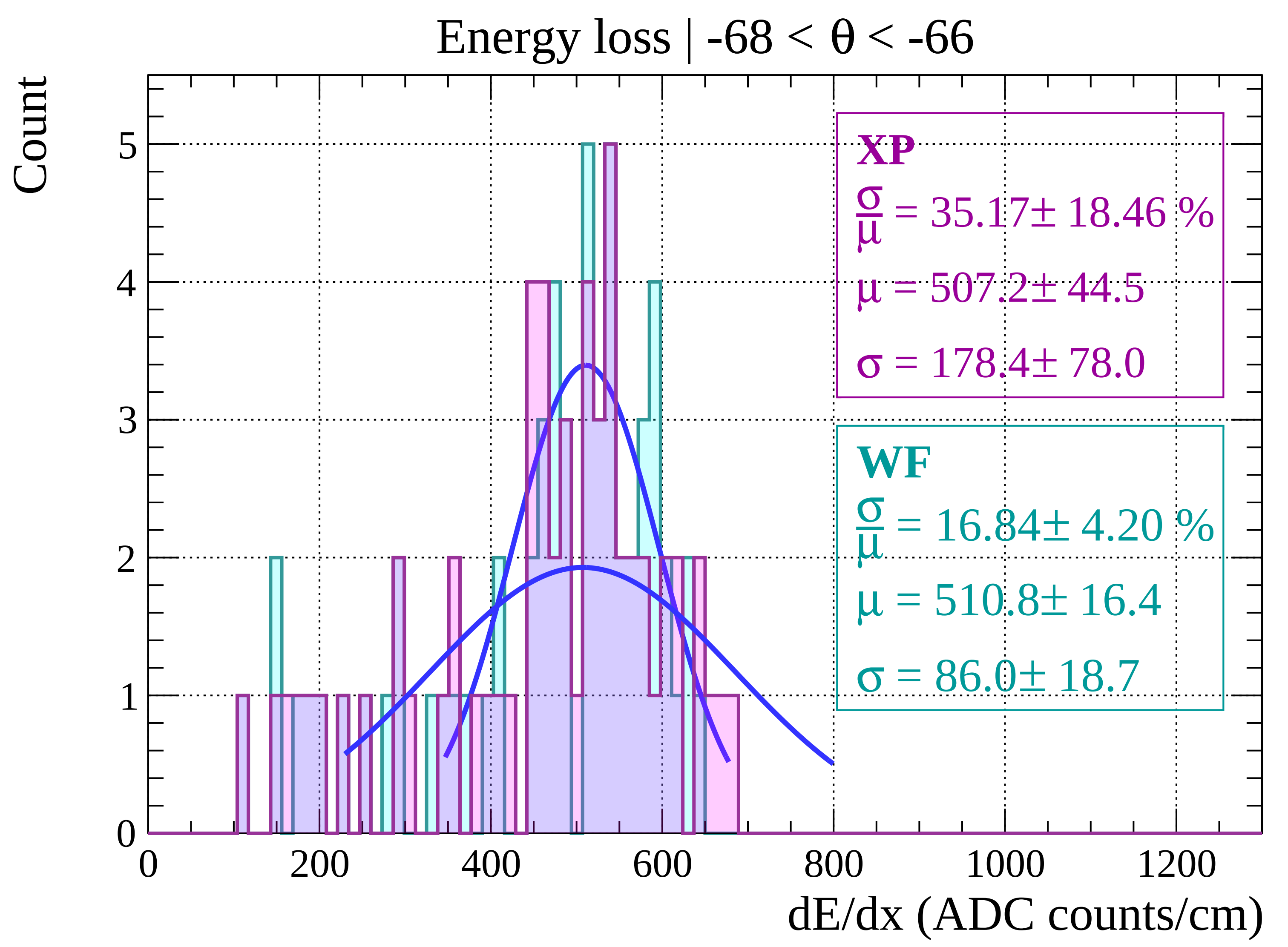

Energy loss | -74 < θ < -72
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = -48.49± 17.48 %
μ = -2184.9± 2189.6
σ = 1059.4± 679.8
WF
σ/μ = 0.69± 12.46 %
μ = 163.1± 15.5
σ = 1.1± 20.2



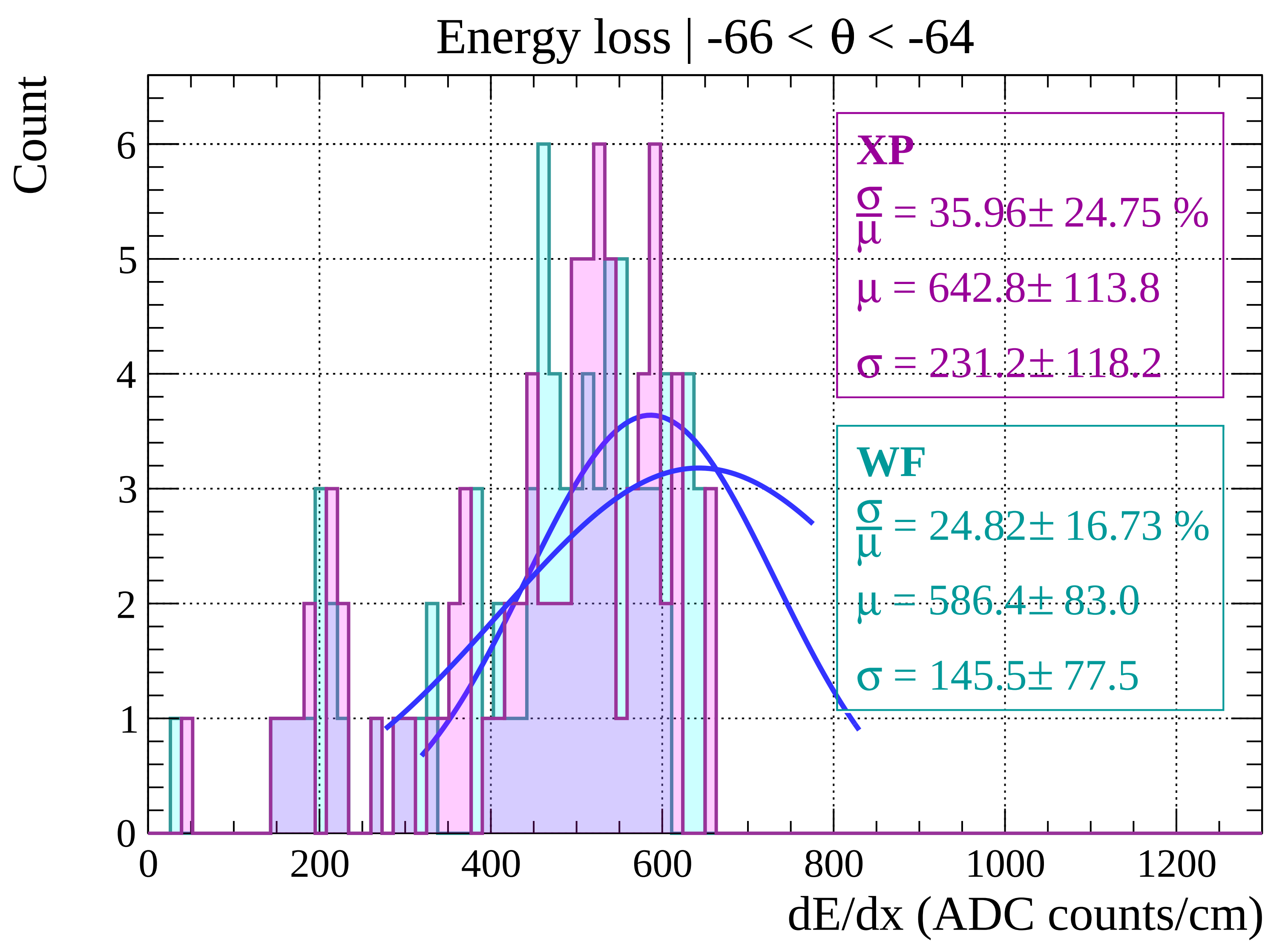
Energy loss | -72 < θ < -70
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.66± 26.71 %
μ = 175.0± 21.8
σ = 1.2± 46.6
WF
σ/μ = -31.39± -15.93 %
μ = -2872.7± 1087.4
σ = 901.7± 798.9

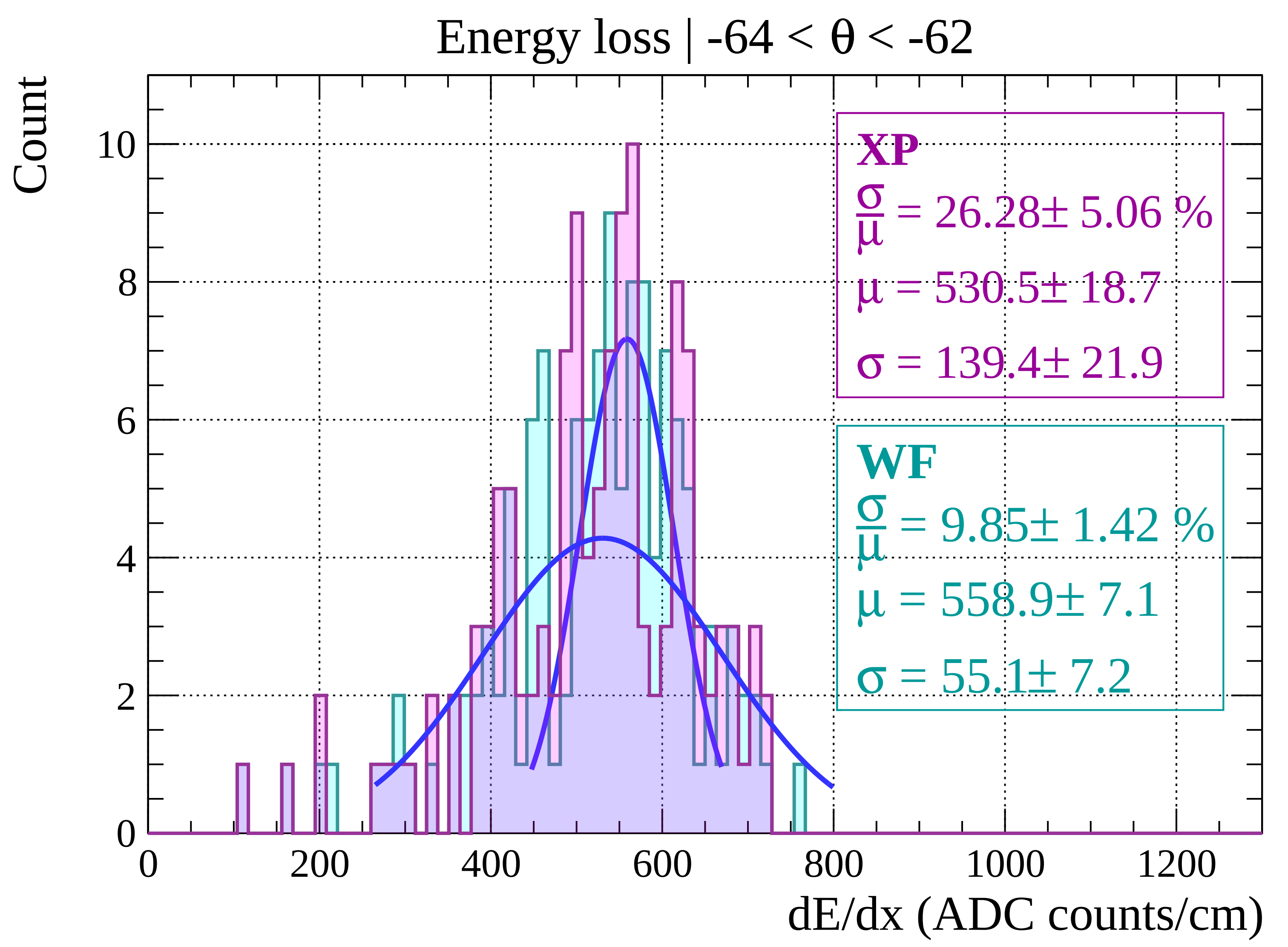


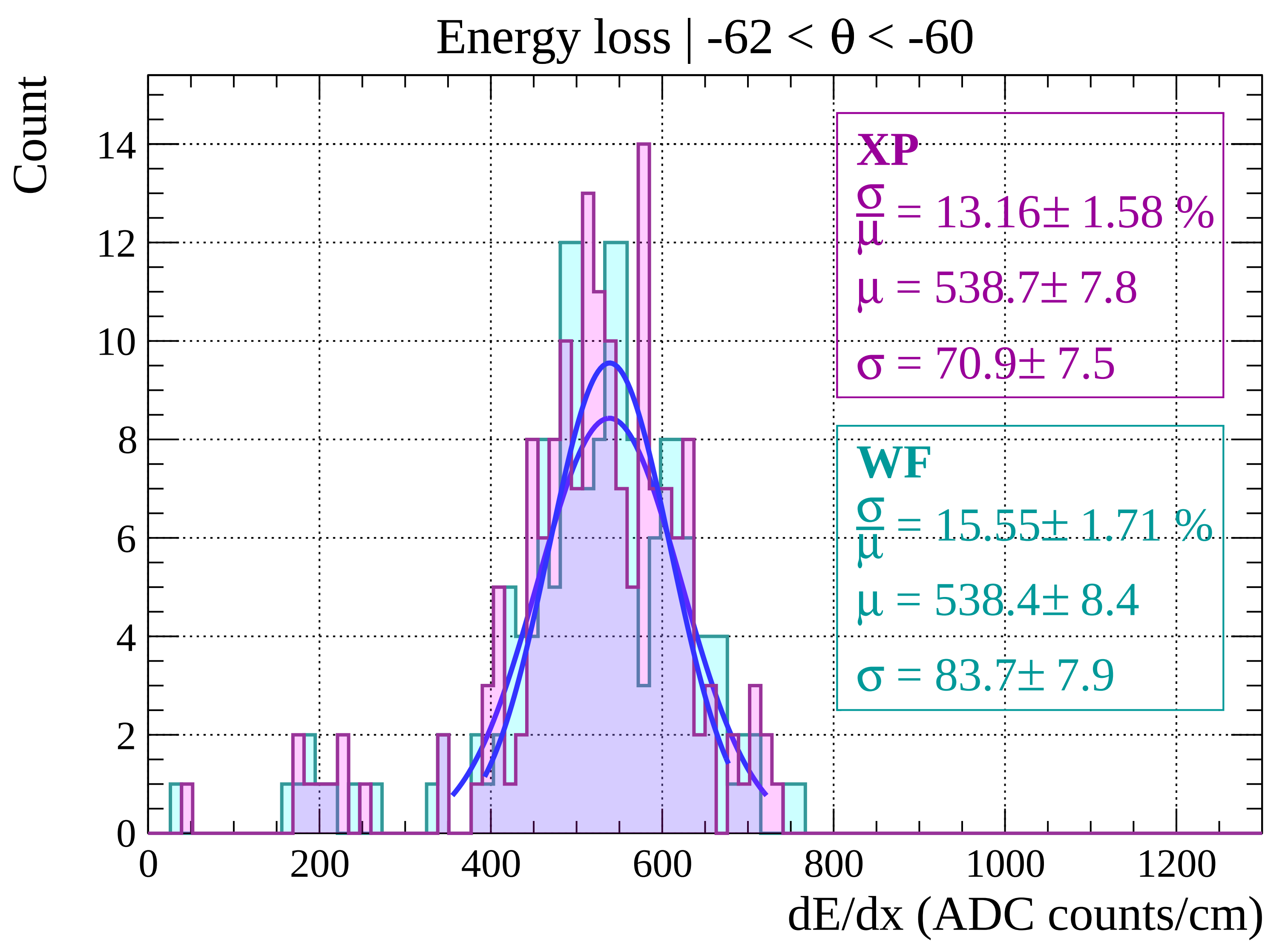
Energy loss | -70 < θ < -68
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 43.66± 111.44 %
μ = 849.1± 1385.5
σ = 370.7± 341.4
WF
σ/μ = 24.01± 20.59 %
μ = 521.6± 76.2
σ = 125.2± 89.1



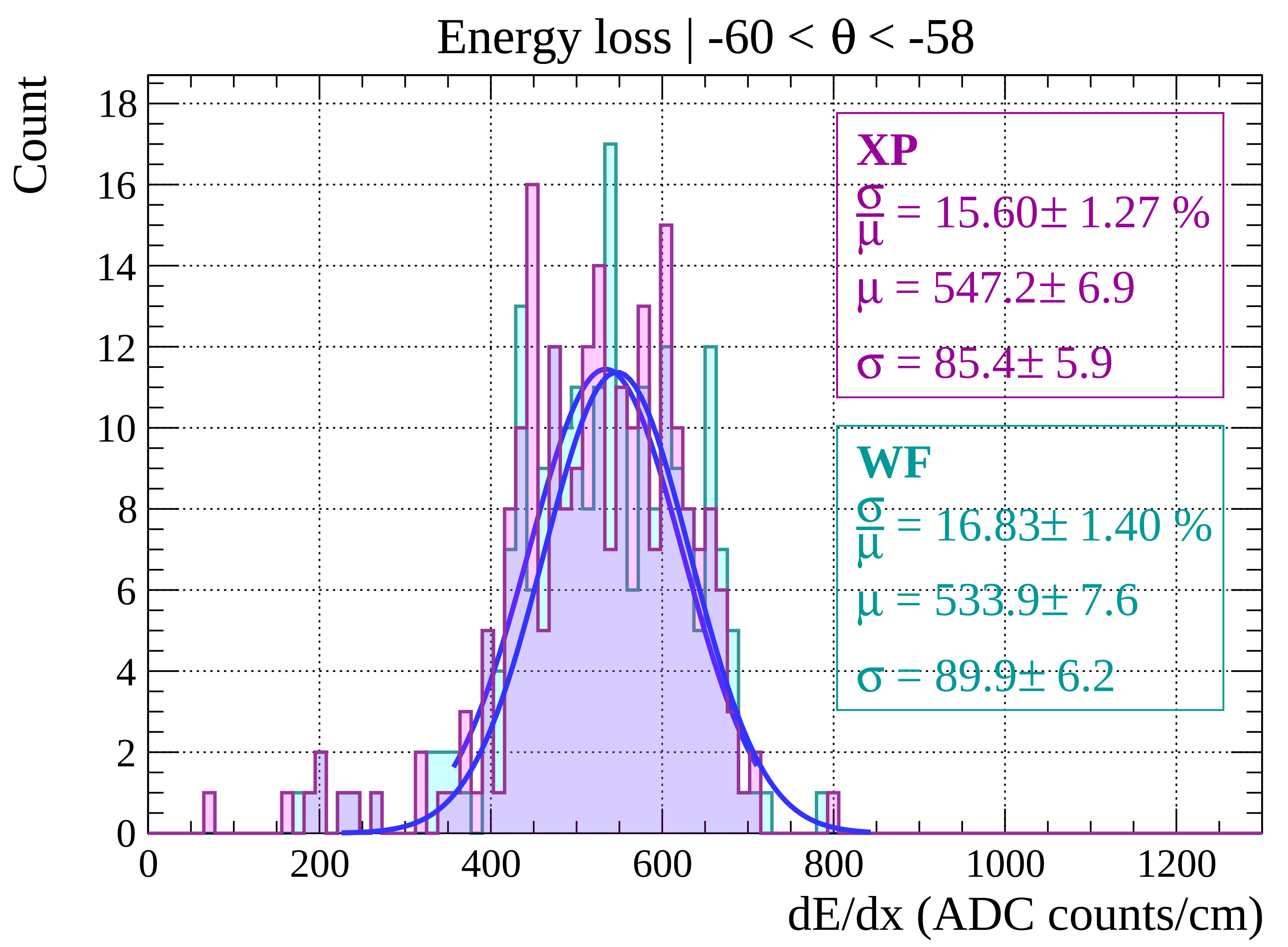
Energy loss | -68 < θ < -66
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 35.17± 18.46 %
μ = 507.2± 44.5
σ = 178.4± 78.0
WF
σ/μ = 16.84± 4.20 %
μ = 510.8± 16.4
σ = 86.0± 18.7


Energy loss | -66 < θ < -64
Count
6
5
4
3
2
1
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 35.96± 24.75 %
μ = 642.8± 113.8
σ = 231.2± 118.2
WF
σ/μ = 24.82± 16.73 %
μ = 586.4± 83.0
σ = 145.5± 77.5

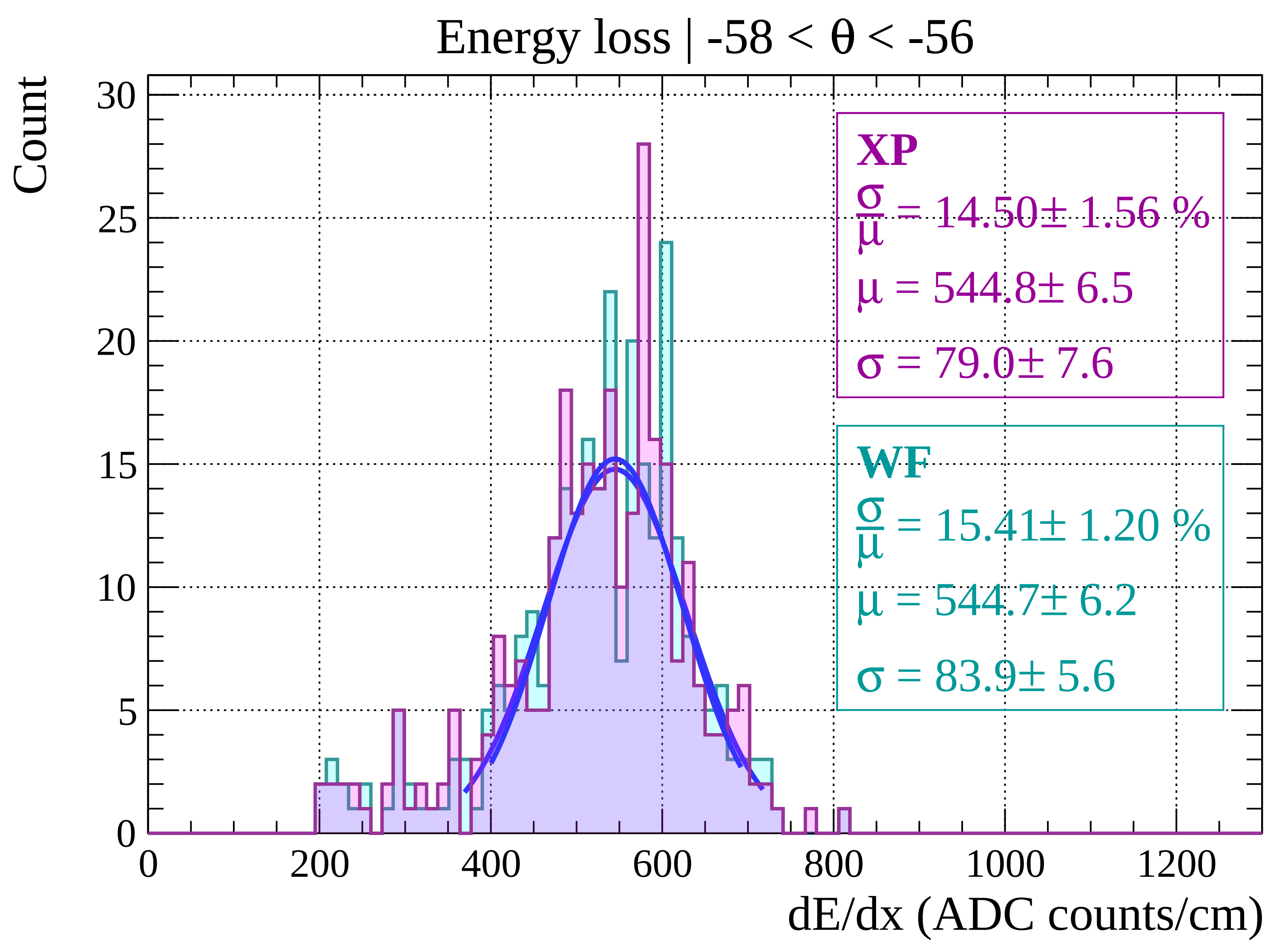

Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
XP
σ/μ = 26.28± 5.06 %
μ = 530.5± 18.7
σ = 139.4± 21.9
WF
σ/μ = 9.85± 1.42 %
μ = 558.9± 7.1
σ = 55.1± 7.2



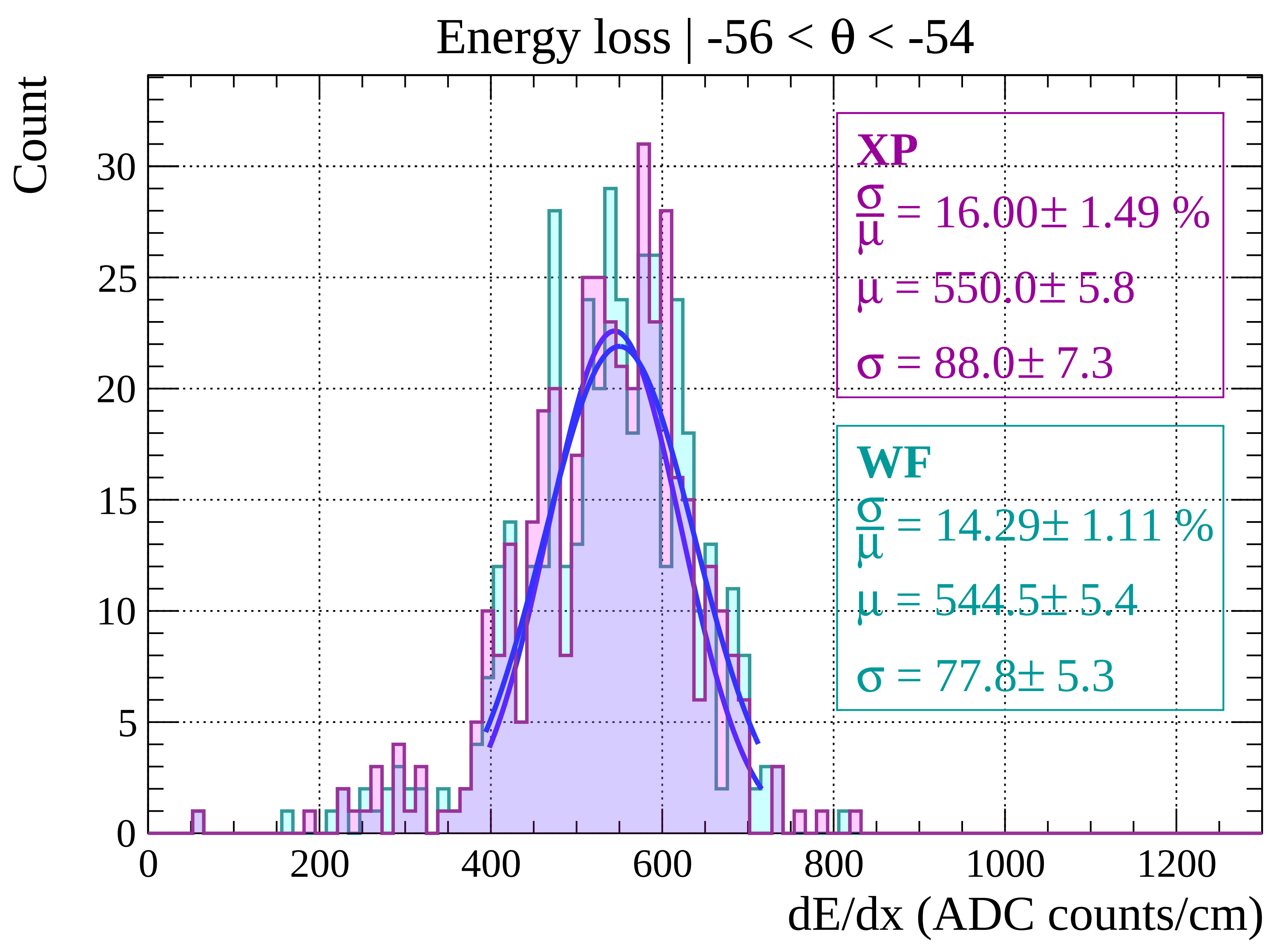
Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 13.16± 1.58 %
μ = 538.7± 7.8
σ = 70.9± 7.5
WF
σ/μ = 15.55± 1.71 %
μ = 538.4± 8.4
σ = 83.7± 7.9

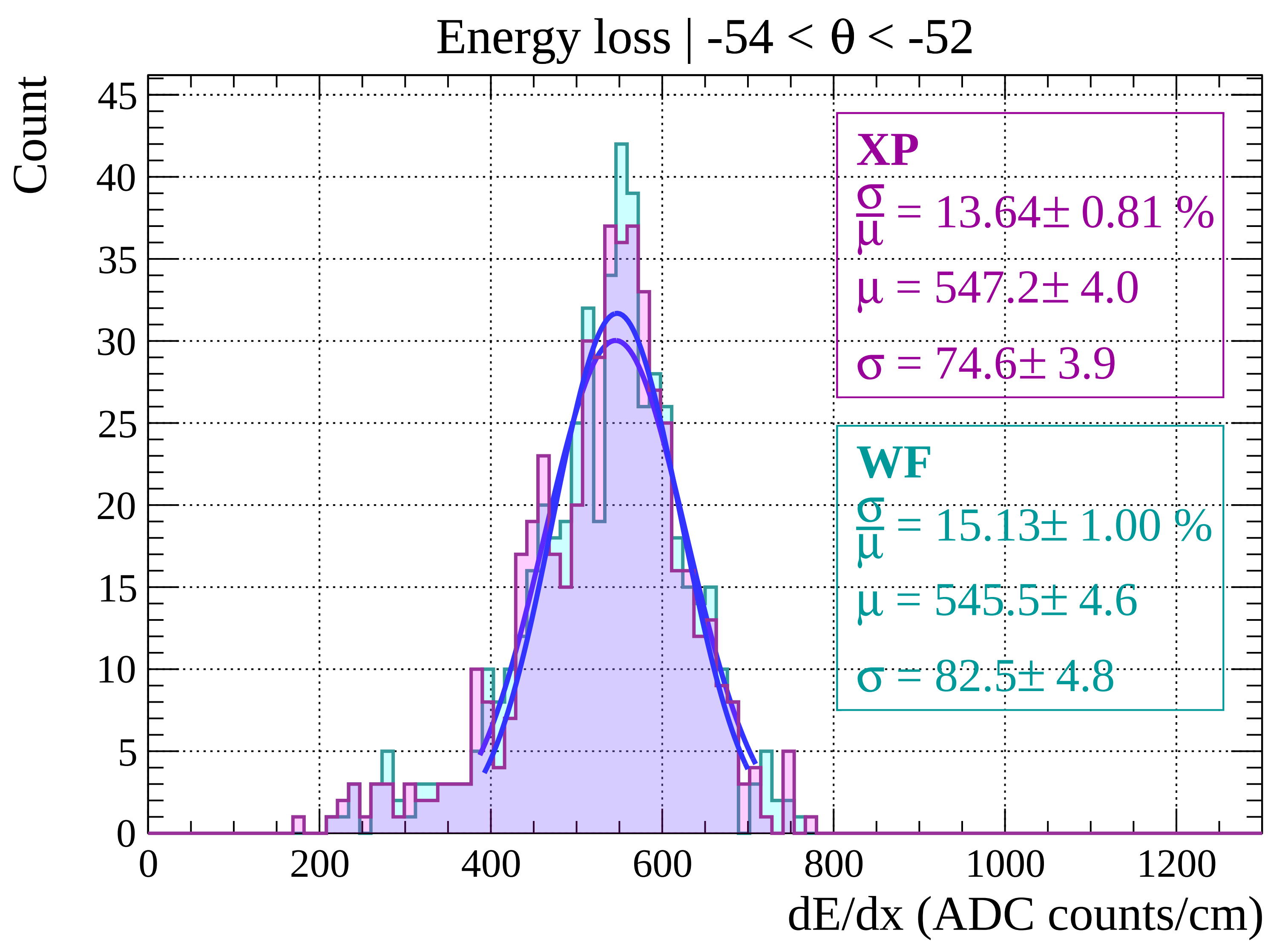


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 15.60± 1.27 %
μ = 547.2± 6.9
σ = 85.4± 5.9
WF
σ/μ = 16.83± 1.40 %
μ = 533.9± 7.6
σ = 89.9± 6.2

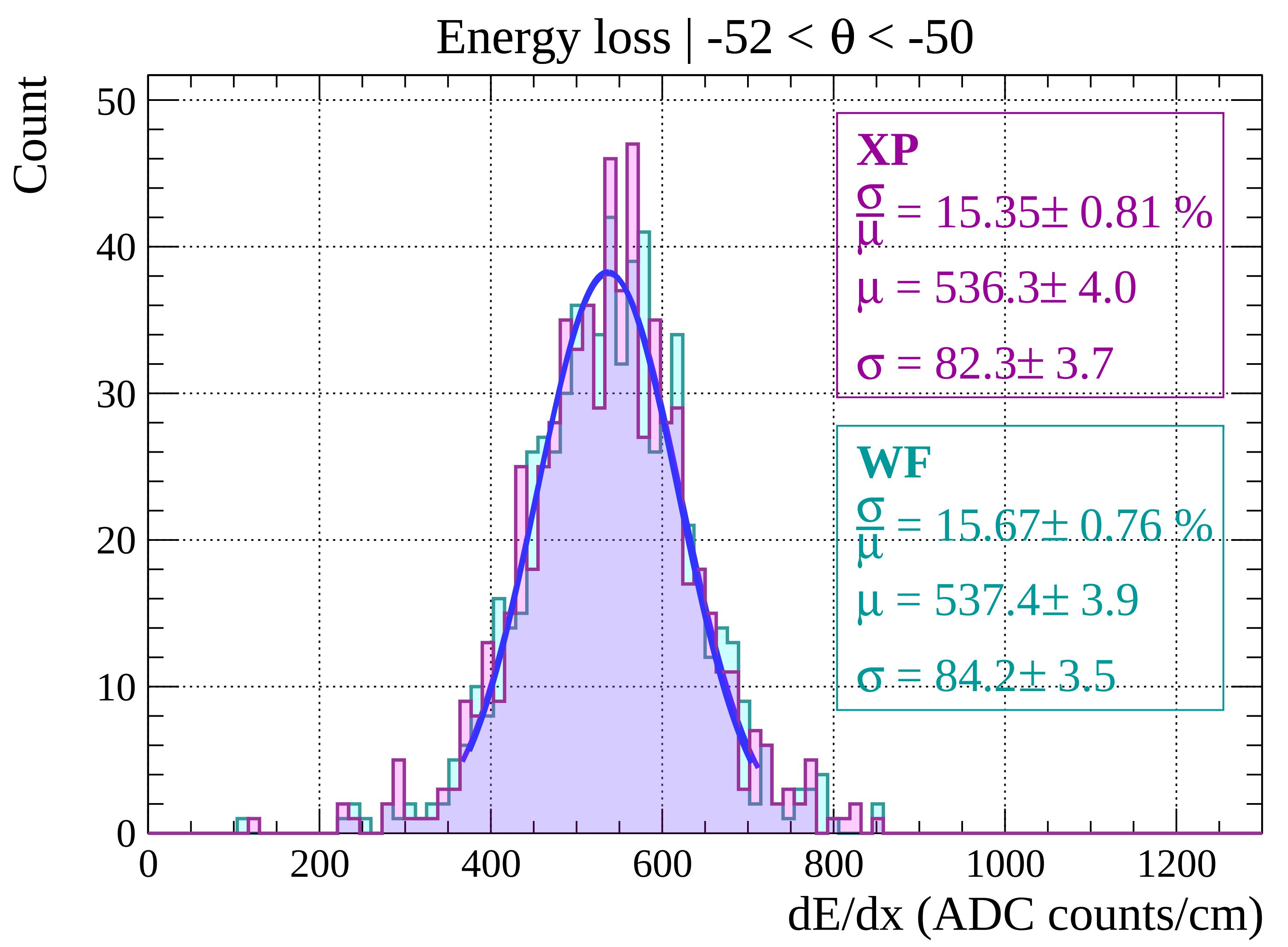


Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 14.50± 1.56 %
μ = 544.8± 6.5
σ = 79.0± 7.6
WF
σ/μ = 15.41± 1.20 %
μ = 544.7± 6.2
σ = 83.9± 5.6

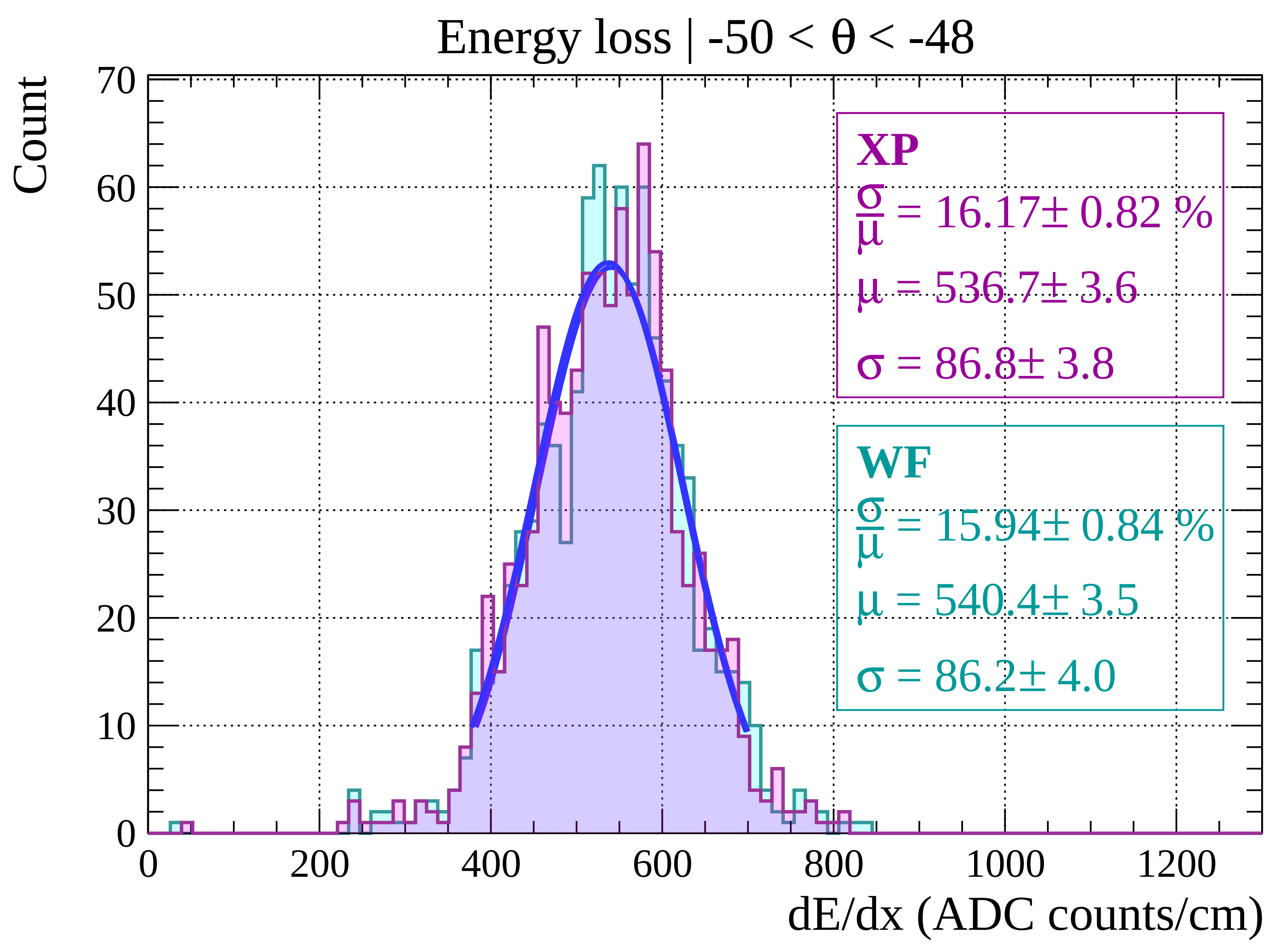

Energy loss | -56 < θ < -54
Count
XP
$\frac{\sigma}{\mu}$ = 16.00± 1.49 %
μ = 550.0± 5.8
σ = 88.0± 7.3
WF
$\frac{\sigma}{\mu}$ = 14.29± 1.11 %
μ = 544.5± 5.4
σ = 77.8± 5.3
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

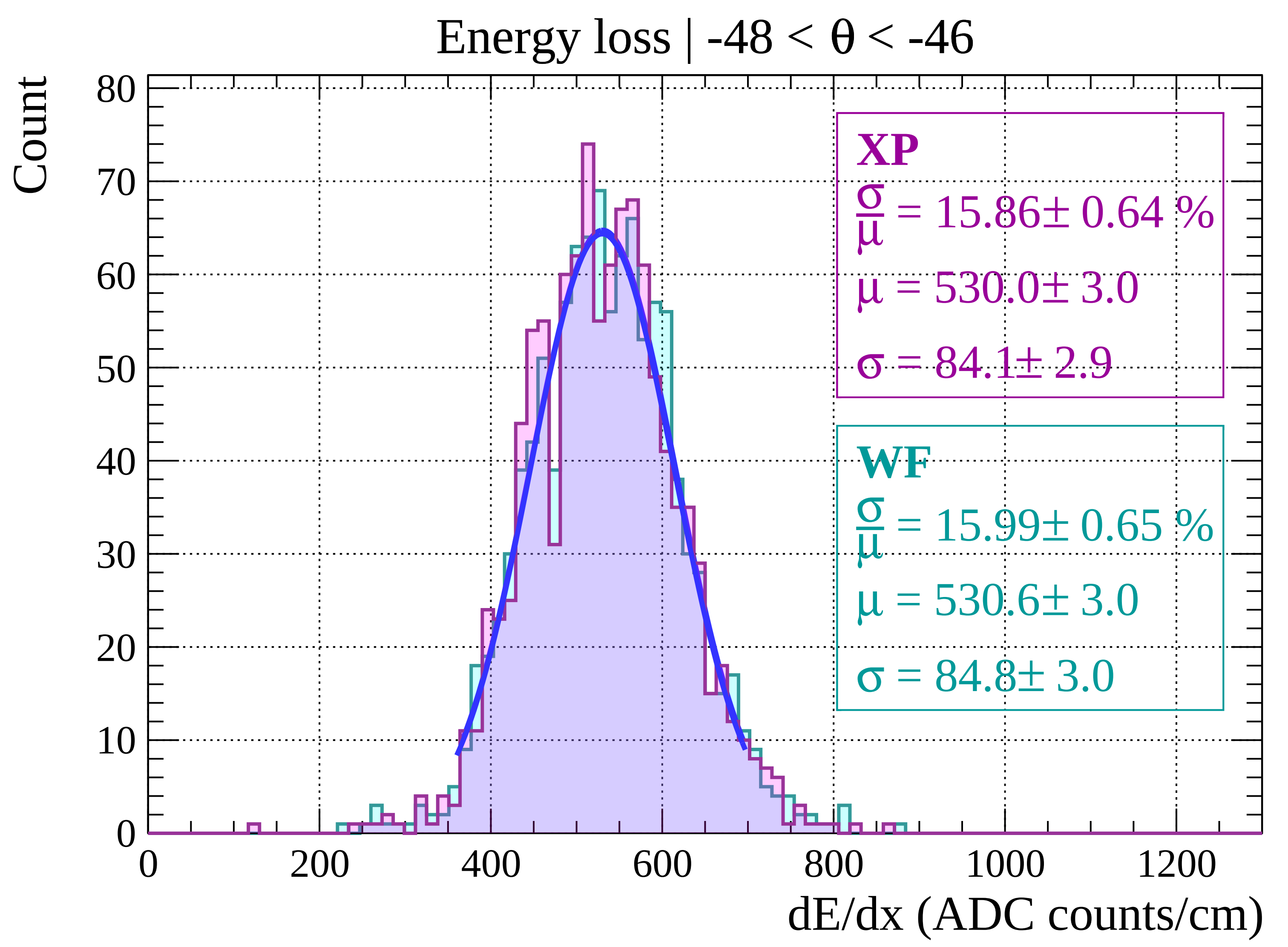


Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40
45
XP
σ/μ = 13.64± 0.81 %
μ = 547.2± 4.0
σ = 74.6± 3.9
WF
σ/μ = 15.13± 1.00 %
μ = 545.5± 4.6
σ = 82.5± 4.8

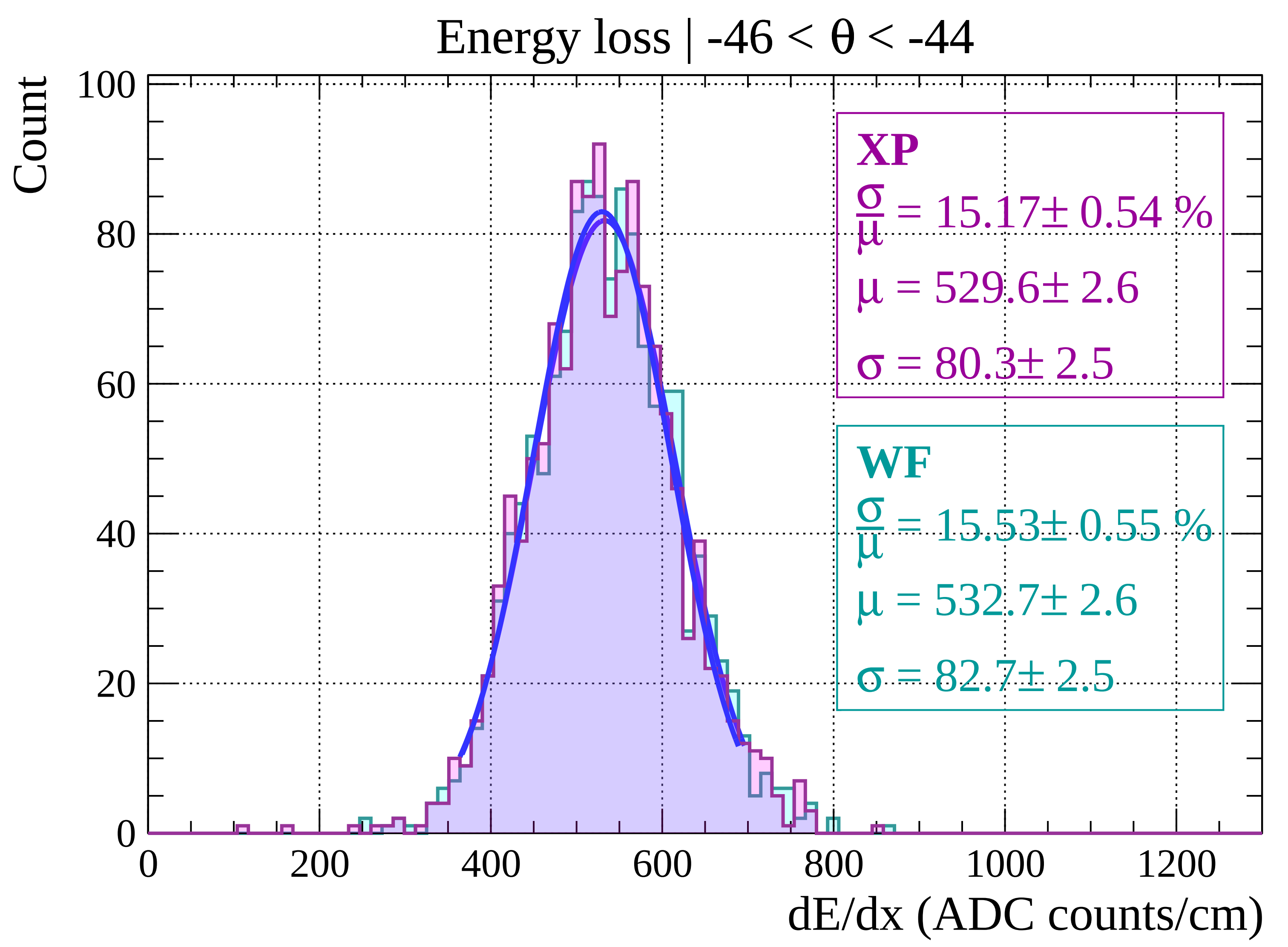


Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 15.35± 0.81 %
μ = 536.3± 4.0
σ = 82.3± 3.7
WF
σ/μ = 15.67± 0.76 %
μ = 537.4± 3.9
σ = 84.2± 3.5

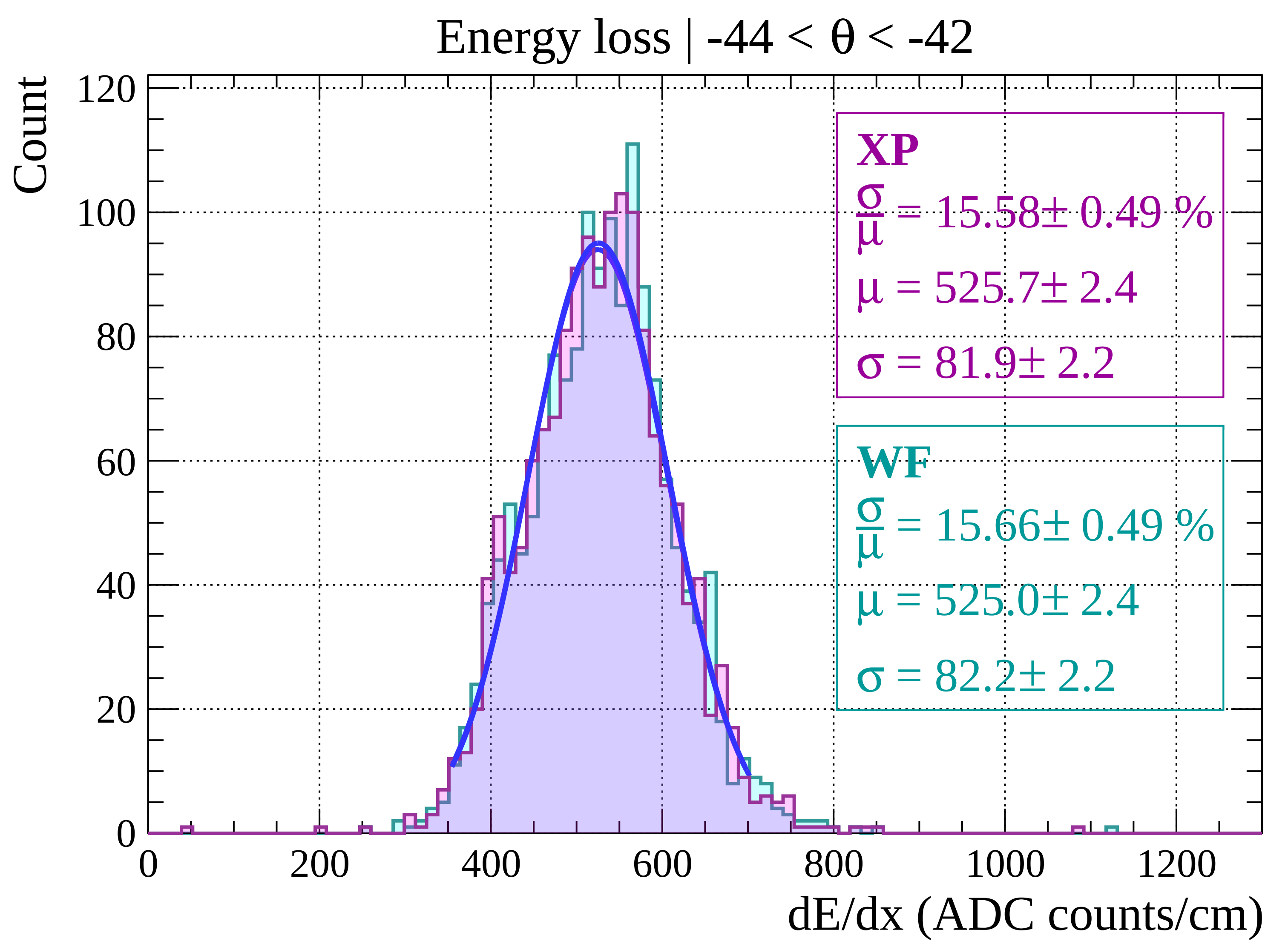

Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.17± 0.82 %
μ = 536.7± 3.6
σ = 86.8± 3.8
WF
$\frac{\sigma}{\mu}$ = 15.94± 0.84 %
μ = 540.4± 3.5
σ = 86.2± 4.0



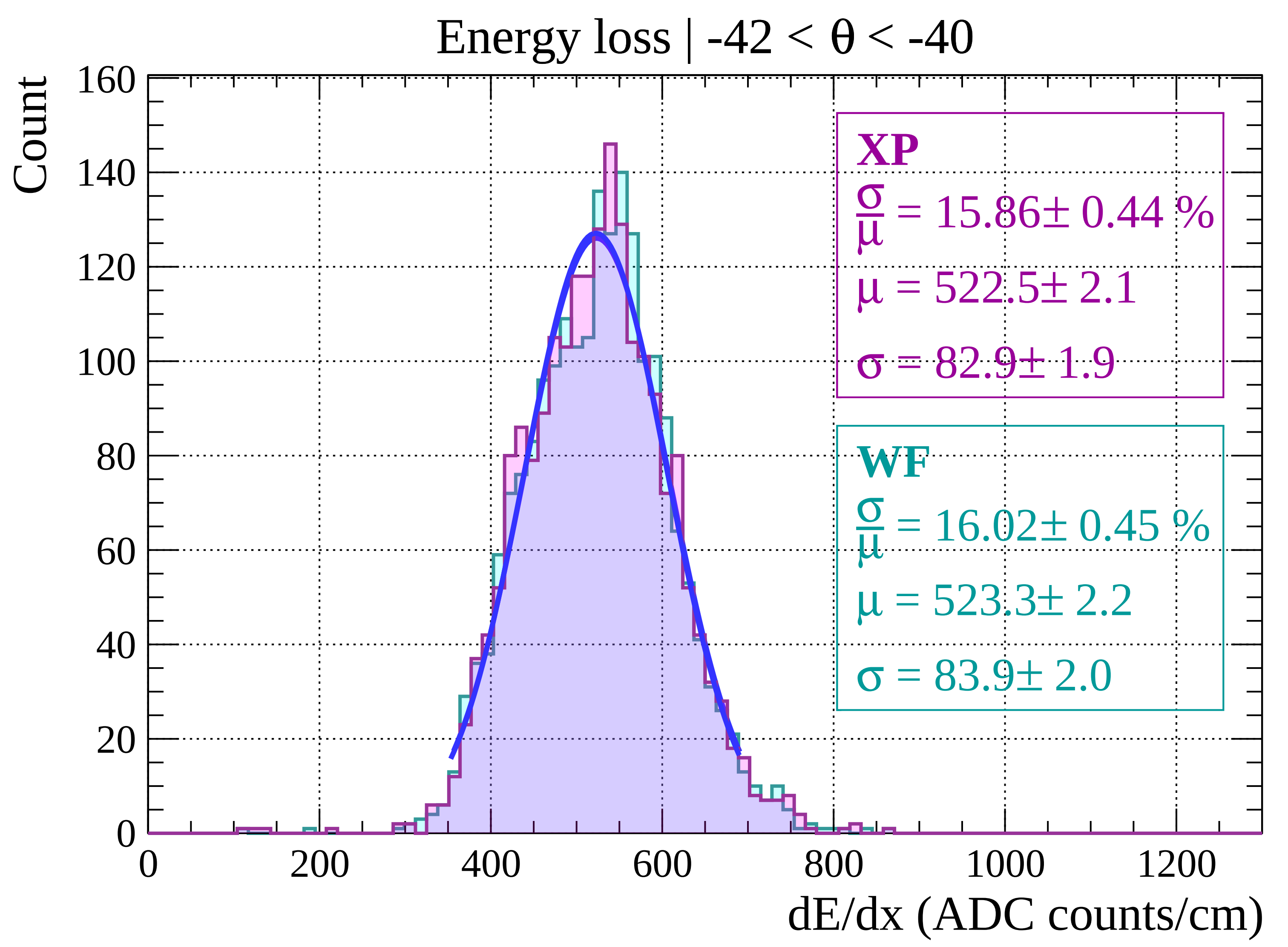
Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
$\frac{\sigma}{\mu}$ = 15.86± 0.64 %
μ = 530.0± 3.0
σ = 84.1± 2.9
WF
$\frac{\sigma}{\mu}$ = 15.99± 0.65 %
μ = 530.6± 3.0
σ = 84.8± 3.0



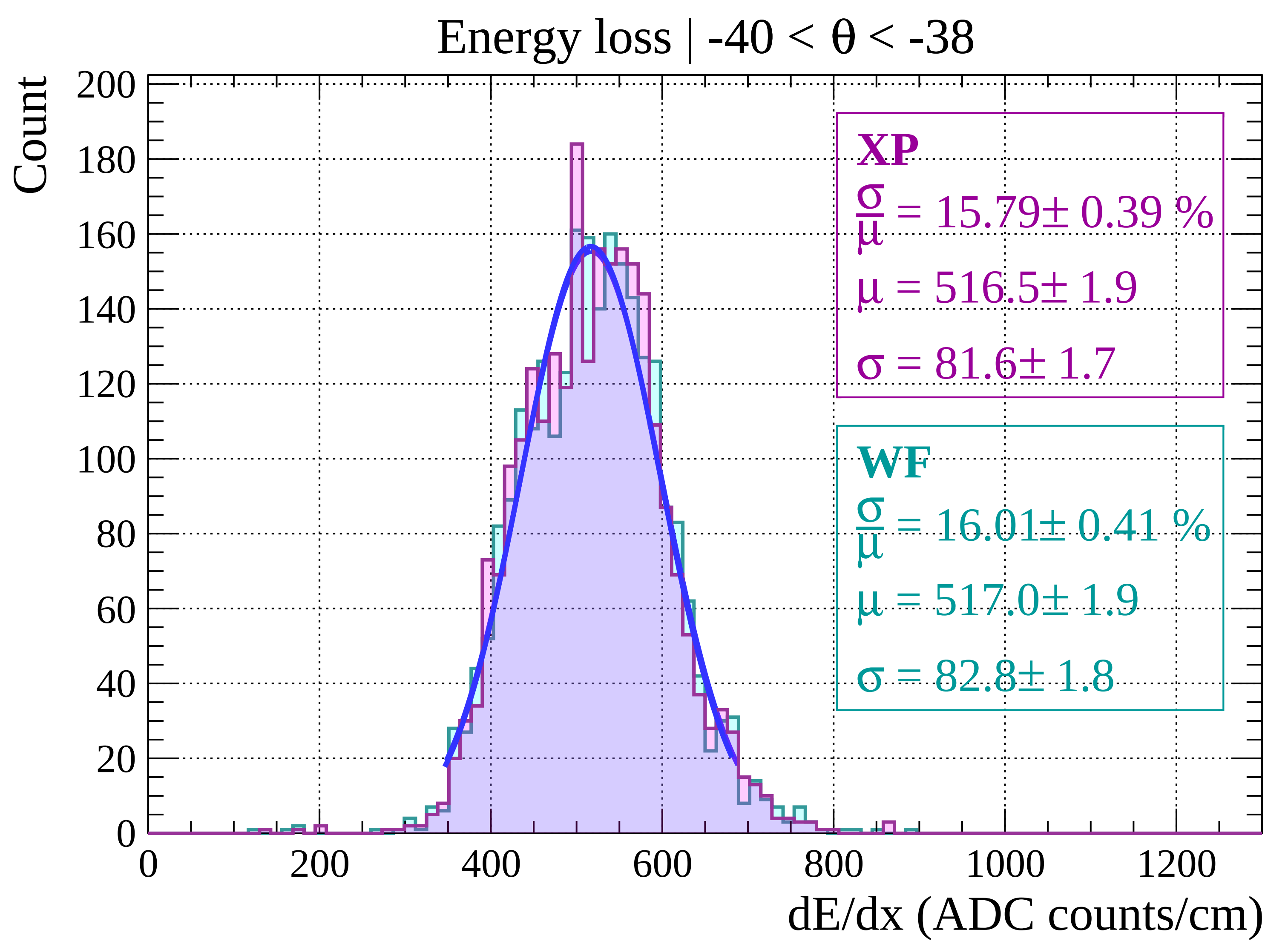
Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 15.17± 0.54 %
μ = 529.6± 2.6
σ = 80.3± 2.5
WF
σ/μ = 15.53± 0.55 %
μ = 532.7± 2.6
σ = 82.7± 2.5



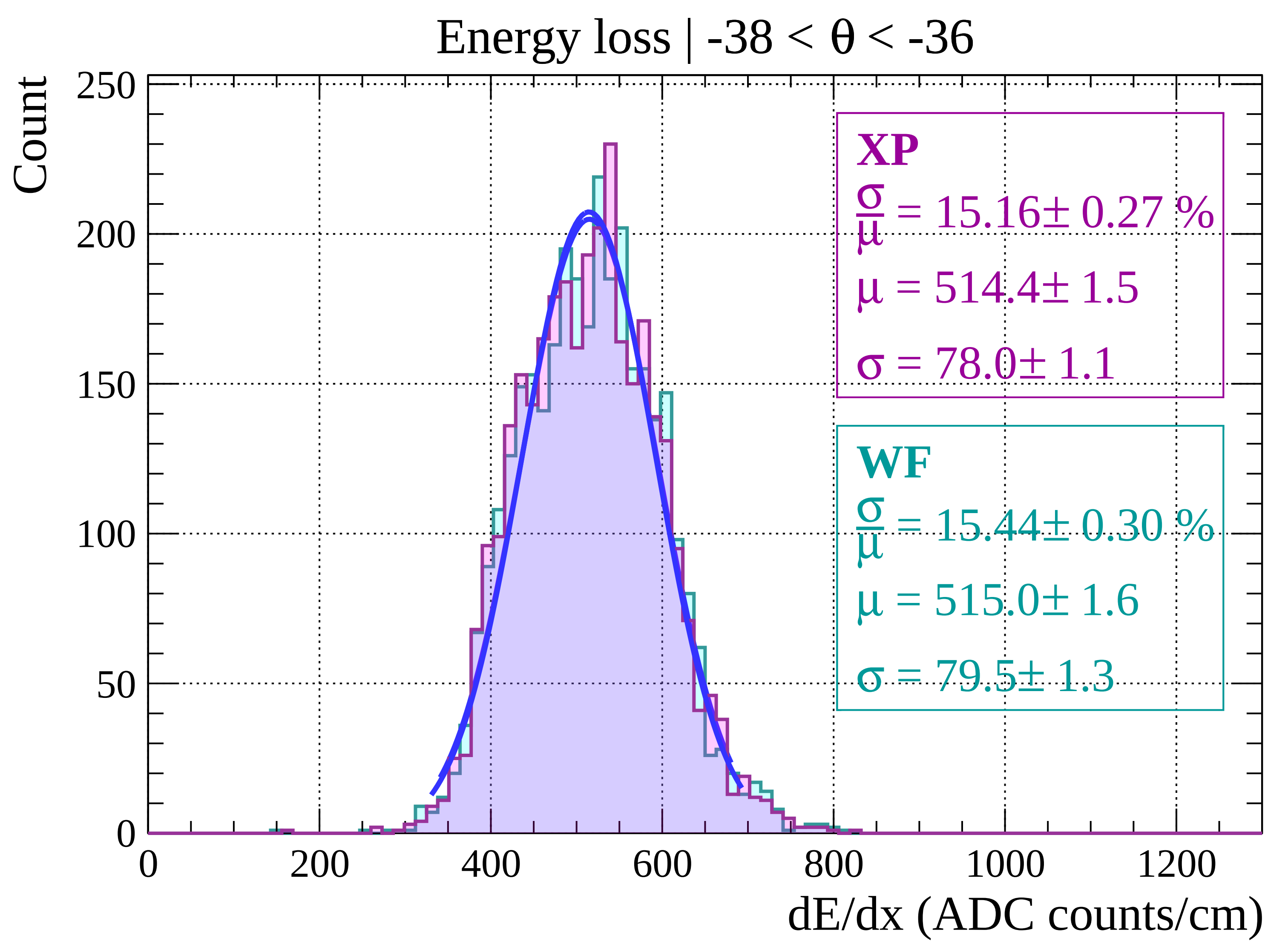
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 15.58± 0.49 %
μ = 525.7± 2.4
σ = 81.9± 2.2
WF
σ/μ = 15.66± 0.49 %
μ = 525.0± 2.4
σ = 82.2± 2.2

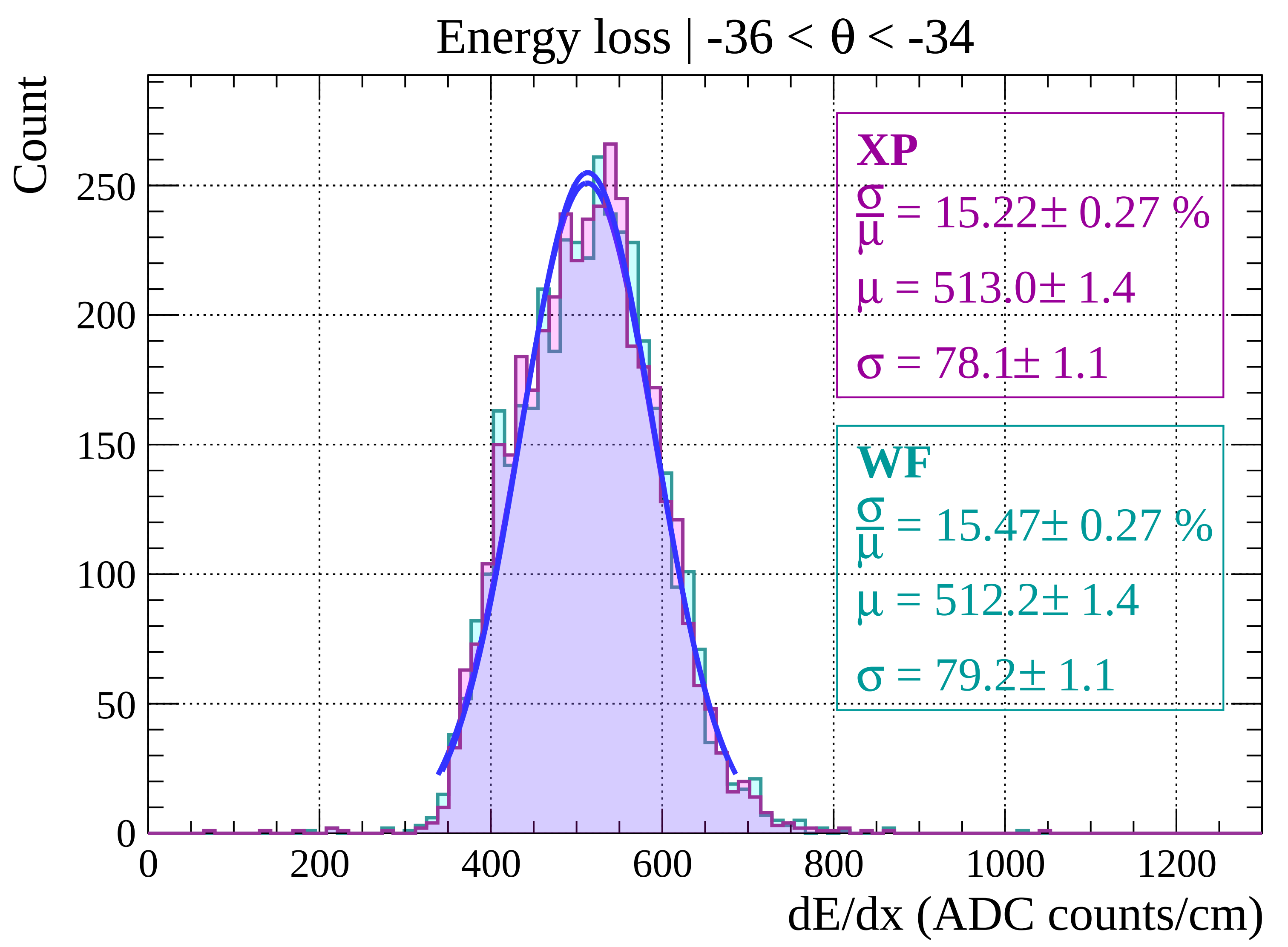


Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 15.86± 0.44 %
μ = 522.5± 2.1
σ = 82.9± 1.9
WF
σ/μ = 16.02± 0.45 %
μ = 523.3± 2.2
σ = 83.9± 2.0

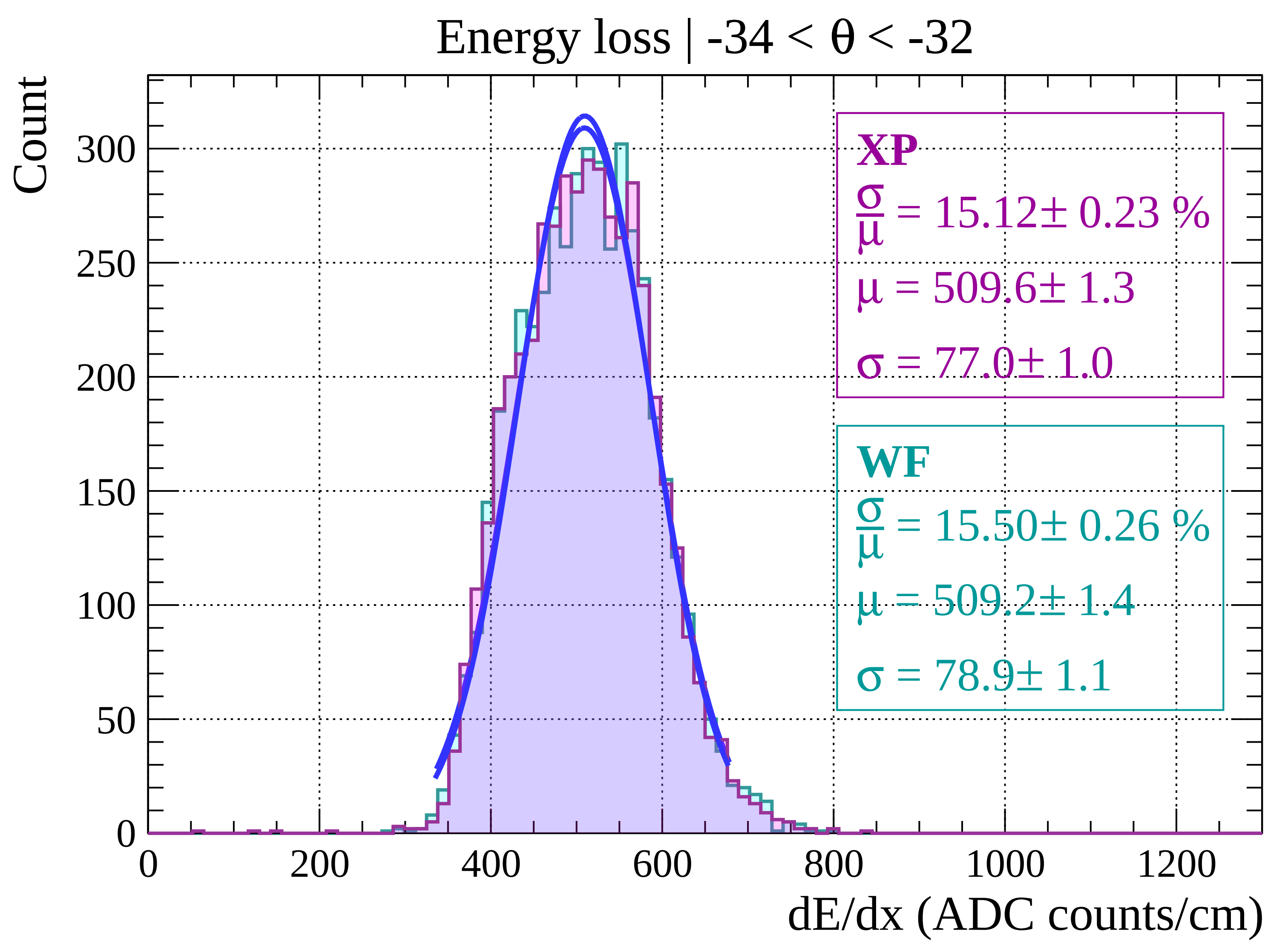


Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 15.79± 0.39 %
μ = 516.5± 1.9
σ = 81.6± 1.7
WF
σ/μ = 16.01± 0.41 %
μ = 517.0± 1.9
σ = 82.8± 1.8

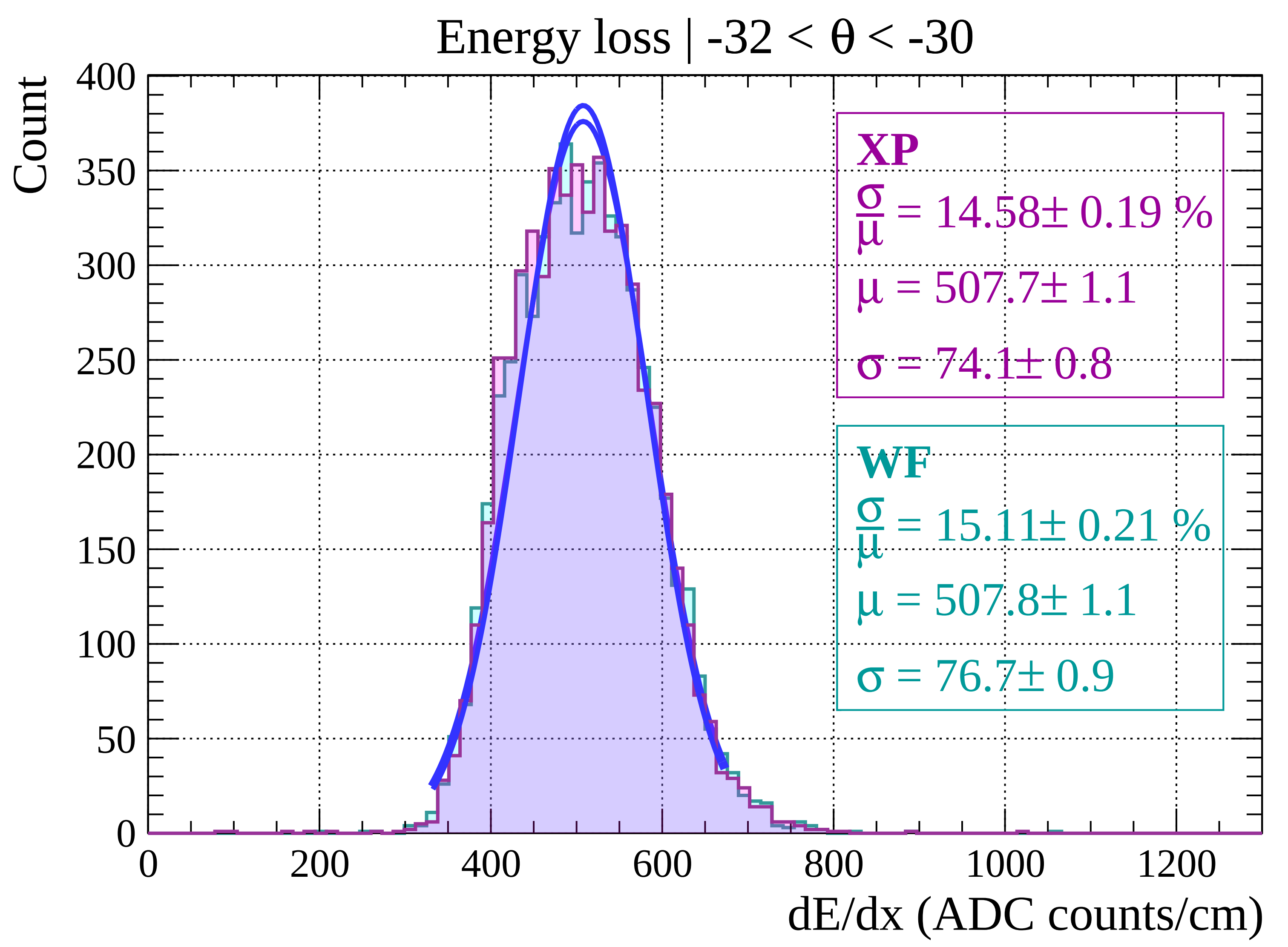

Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.16± 0.27 %
μ = 514.4± 1.5
σ = 78.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 15.44± 0.30 %
μ = 515.0± 1.6
σ = 79.5± 1.3



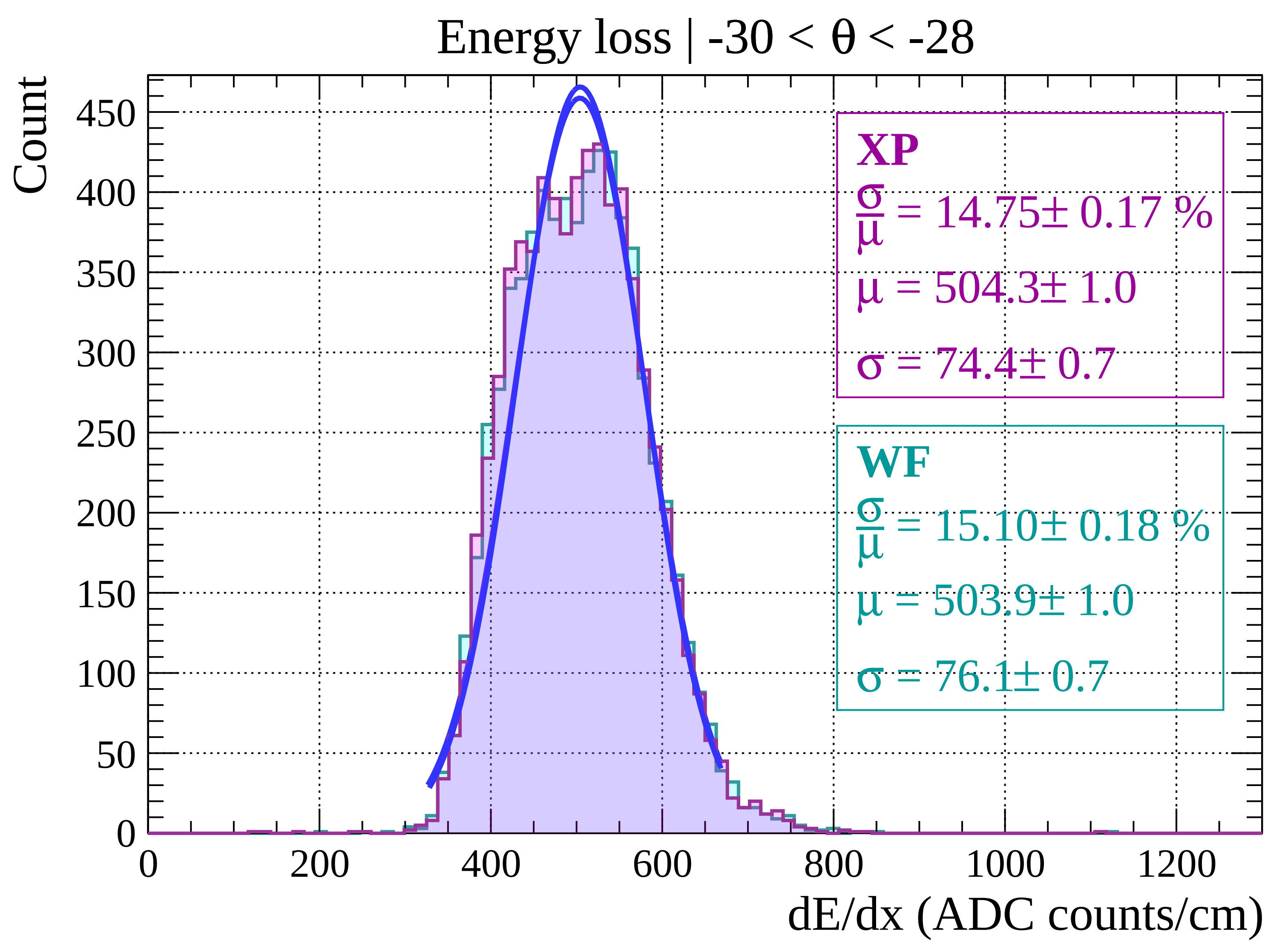
Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 15.22± 0.27 %
μ = 513.0± 1.4
σ = 78.1± 1.1
WF
σ/μ = 15.47± 0.27 %
μ = 512.2± 1.4
σ = 79.2± 1.1



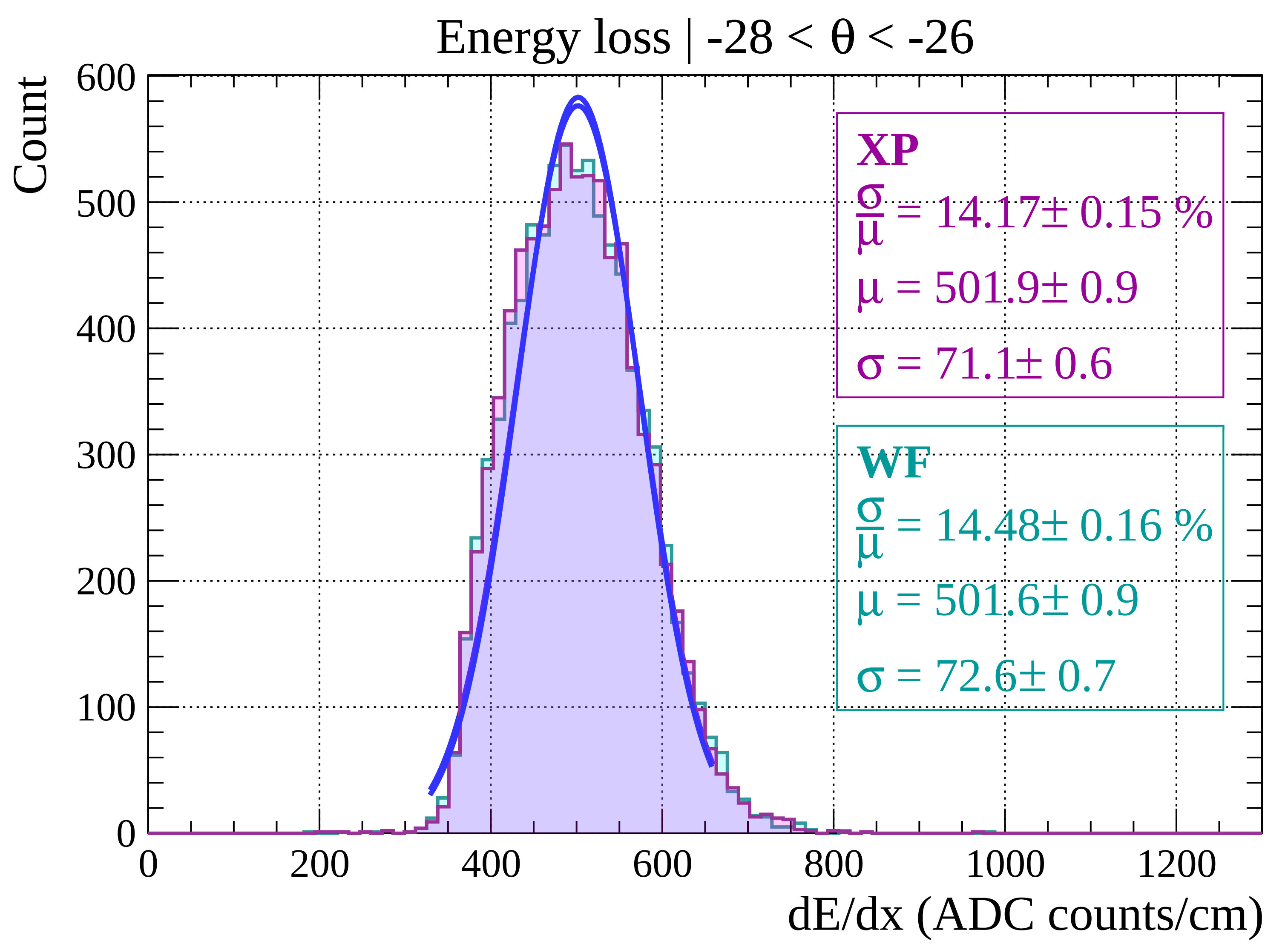
Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 15.12± 0.23 %
μ = 509.6± 1.3
σ = 77.0± 1.0
WF
σ/μ = 15.50± 0.26 %
μ = 509.2± 1.4
σ = 78.9± 1.1

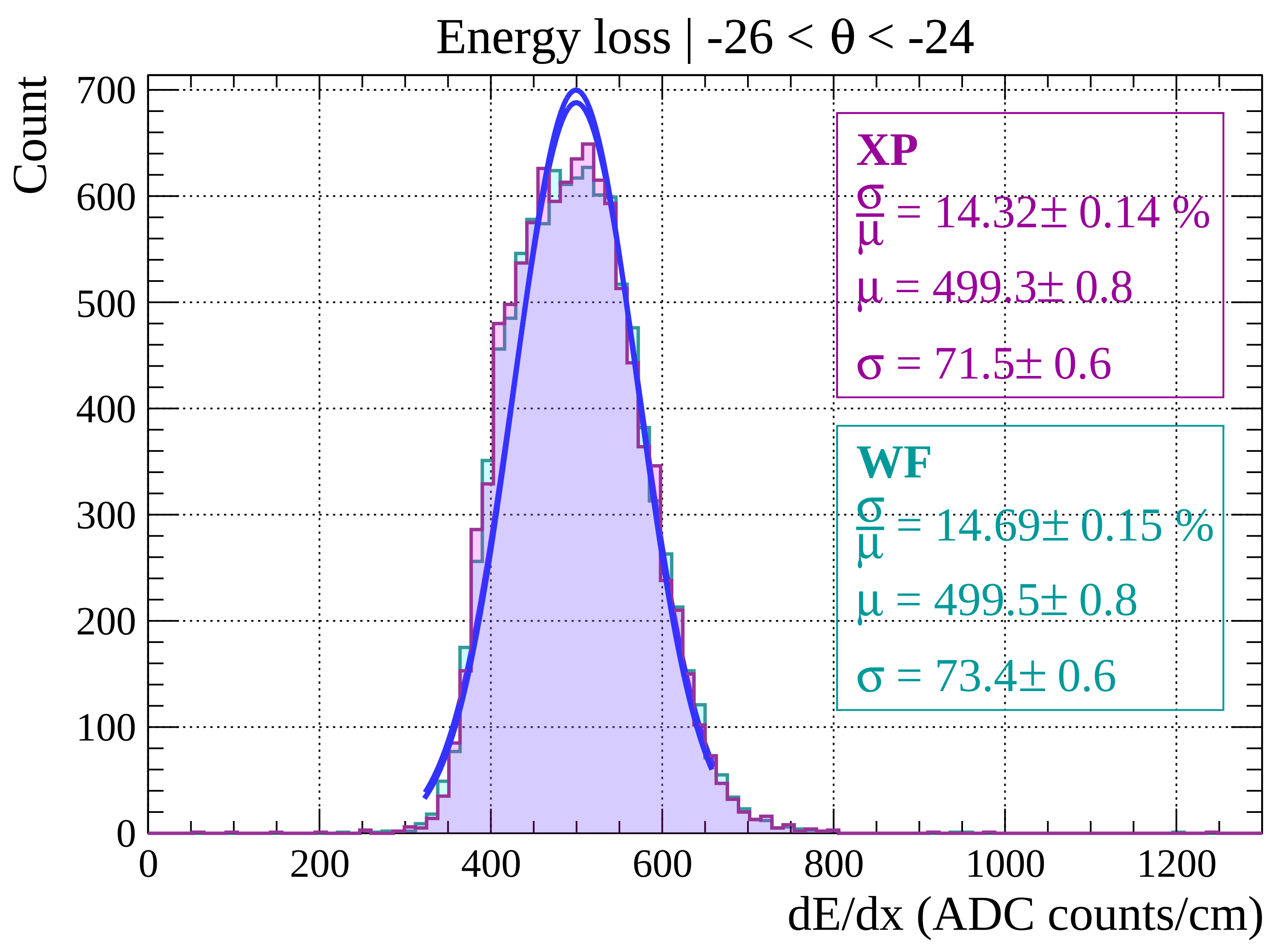


Energy loss | -32 < θ < -30
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.58± 0.19 %
μ = 507.7± 1.1
σ = 74.1± 0.8
WF
σ/μ = 15.11± 0.21 %
μ = 507.8± 1.1
σ = 76.7± 0.9

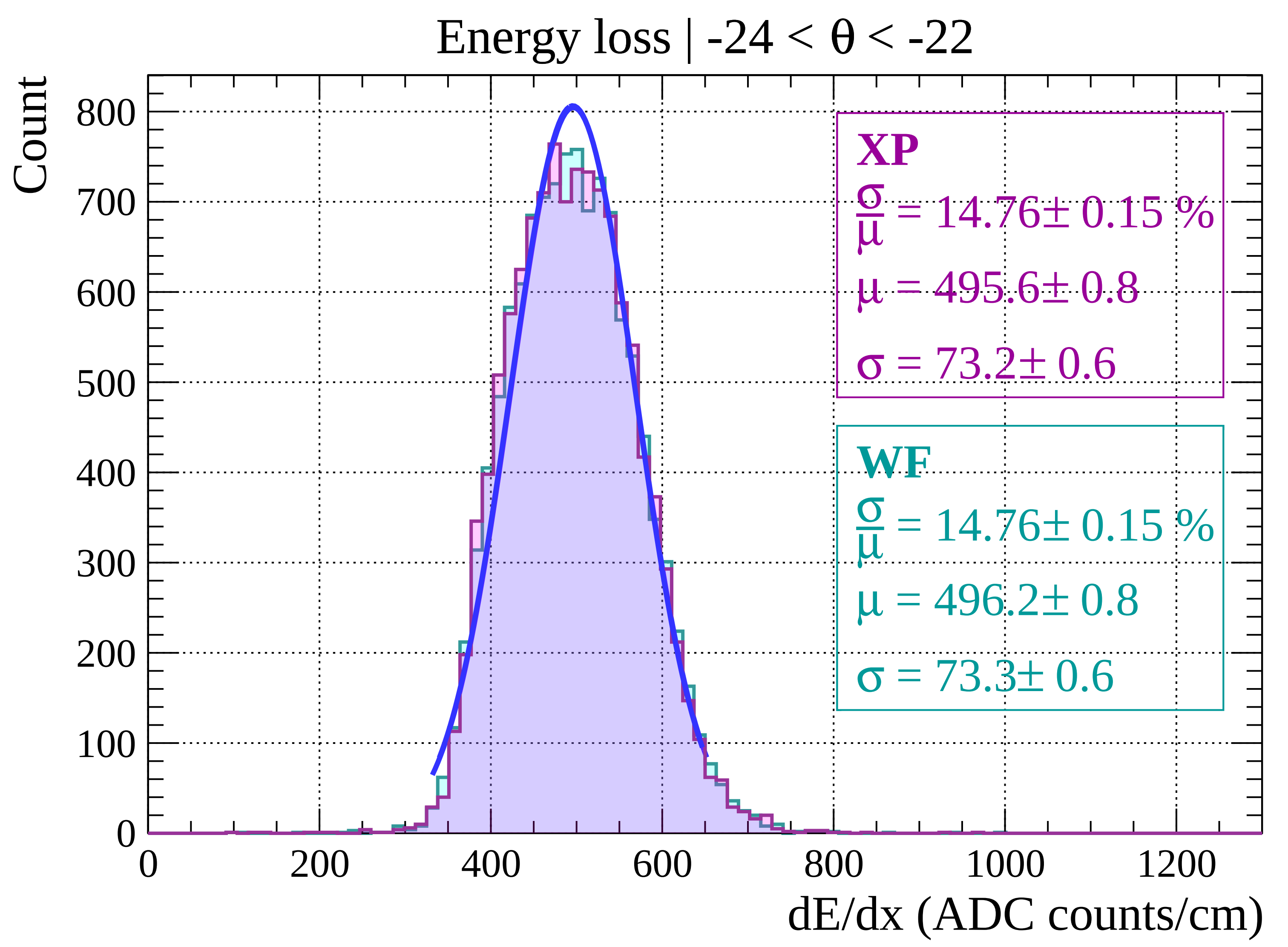


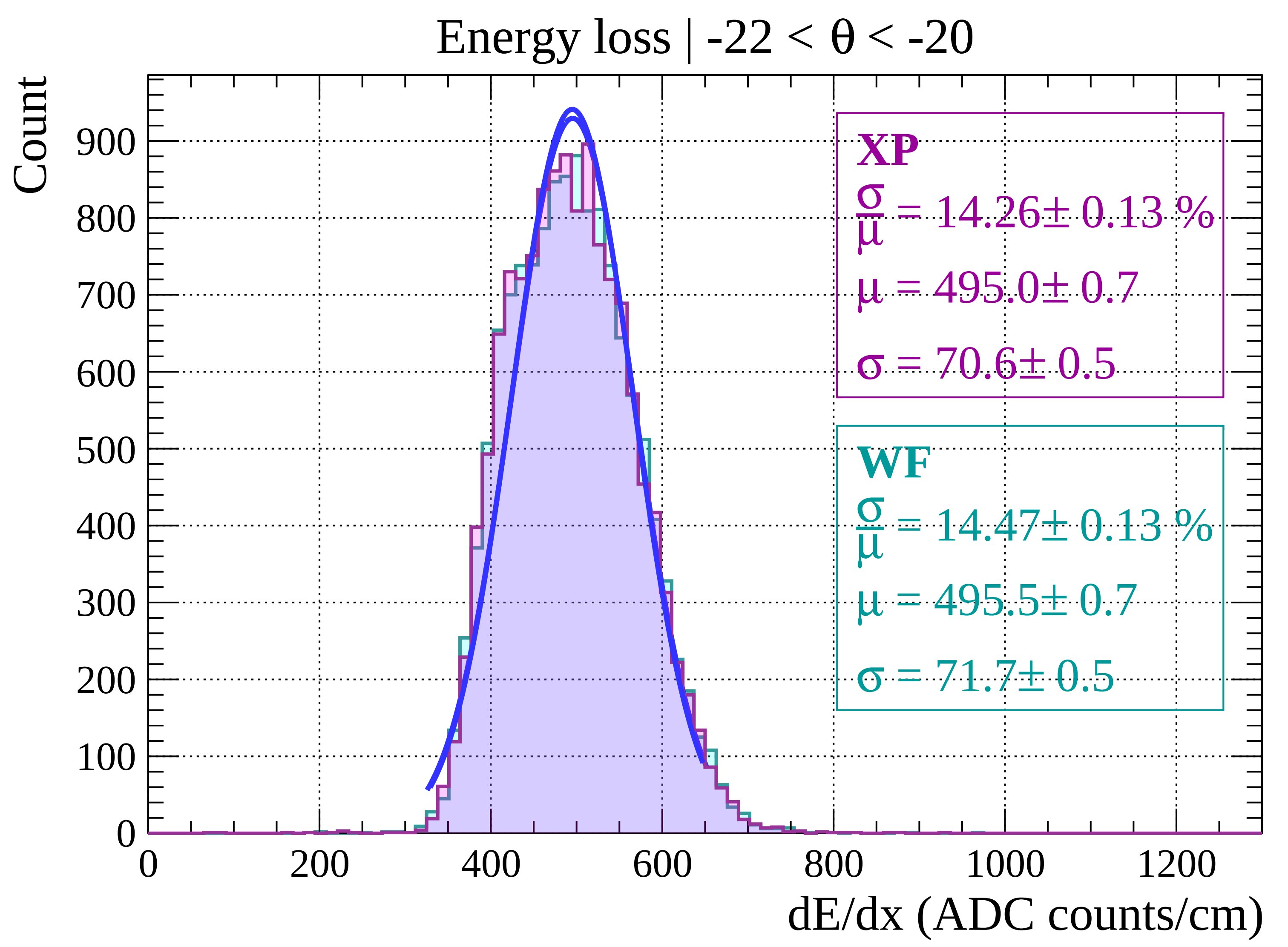
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 14.75± 0.17 %
μ = 504.3± 1.0
σ = 74.4± 0.7
WF
σ/μ = 15.10± 0.18 %
μ = 503.9± 1.0
σ = 76.1± 0.7


Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.17± 0.15 %
μ = 501.9± 0.9
σ = 71.1± 0.6
WF
σ/μ = 14.48± 0.16 %
μ = 501.6± 0.9
σ = 72.6± 0.7
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

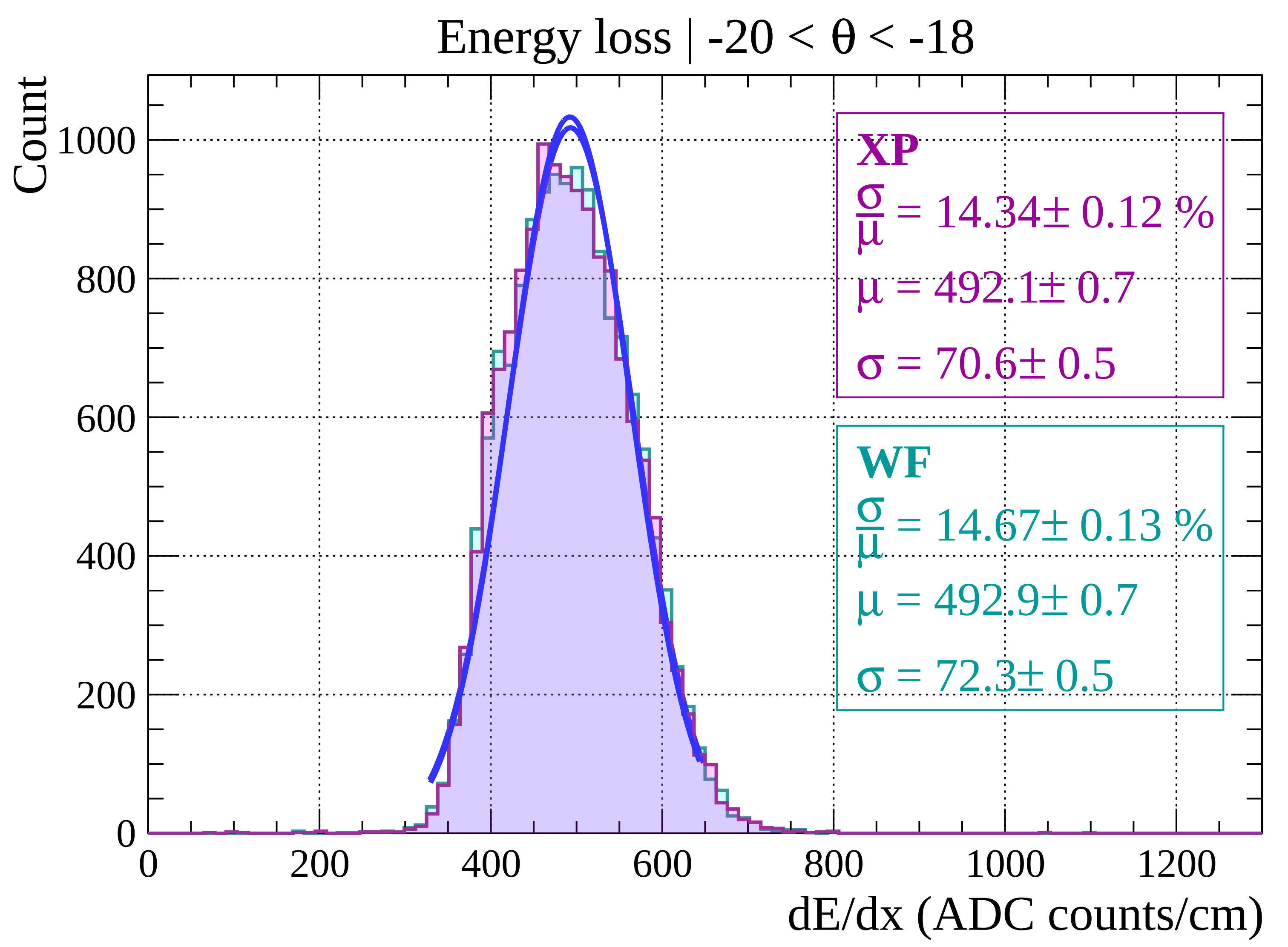


Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.32± 0.14 %
μ = 499.3± 0.8
σ = 71.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 14.69± 0.15 %
μ = 499.5± 0.8
σ = 73.4± 0.6

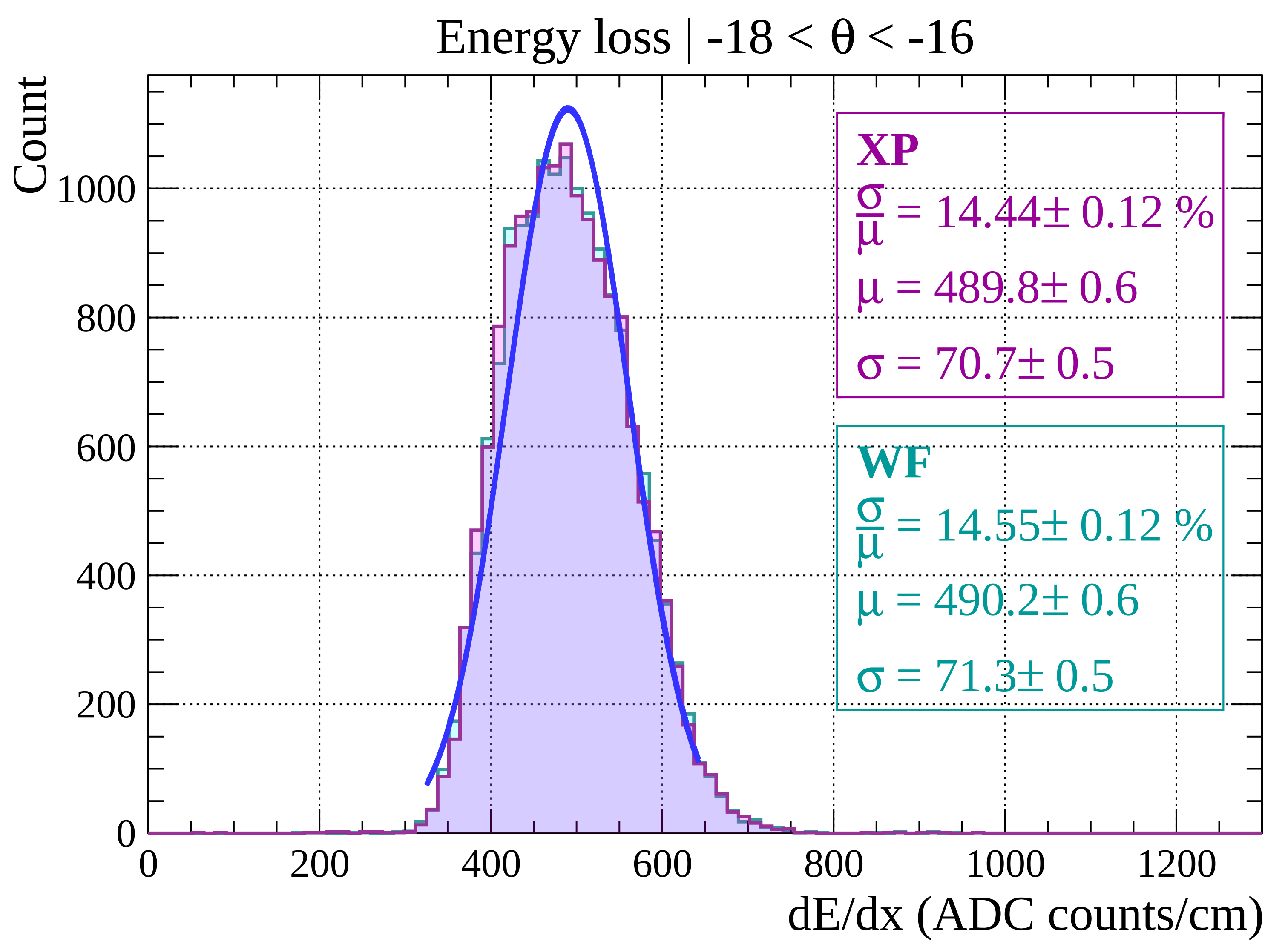

Energy loss | -24 < θ < -22
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.76± 0.15 %
μ = 495.6± 0.8
σ = 73.2± 0.6
WF
σ/μ = 14.76± 0.15 %
μ = 496.2± 0.8
σ = 73.3± 0.6



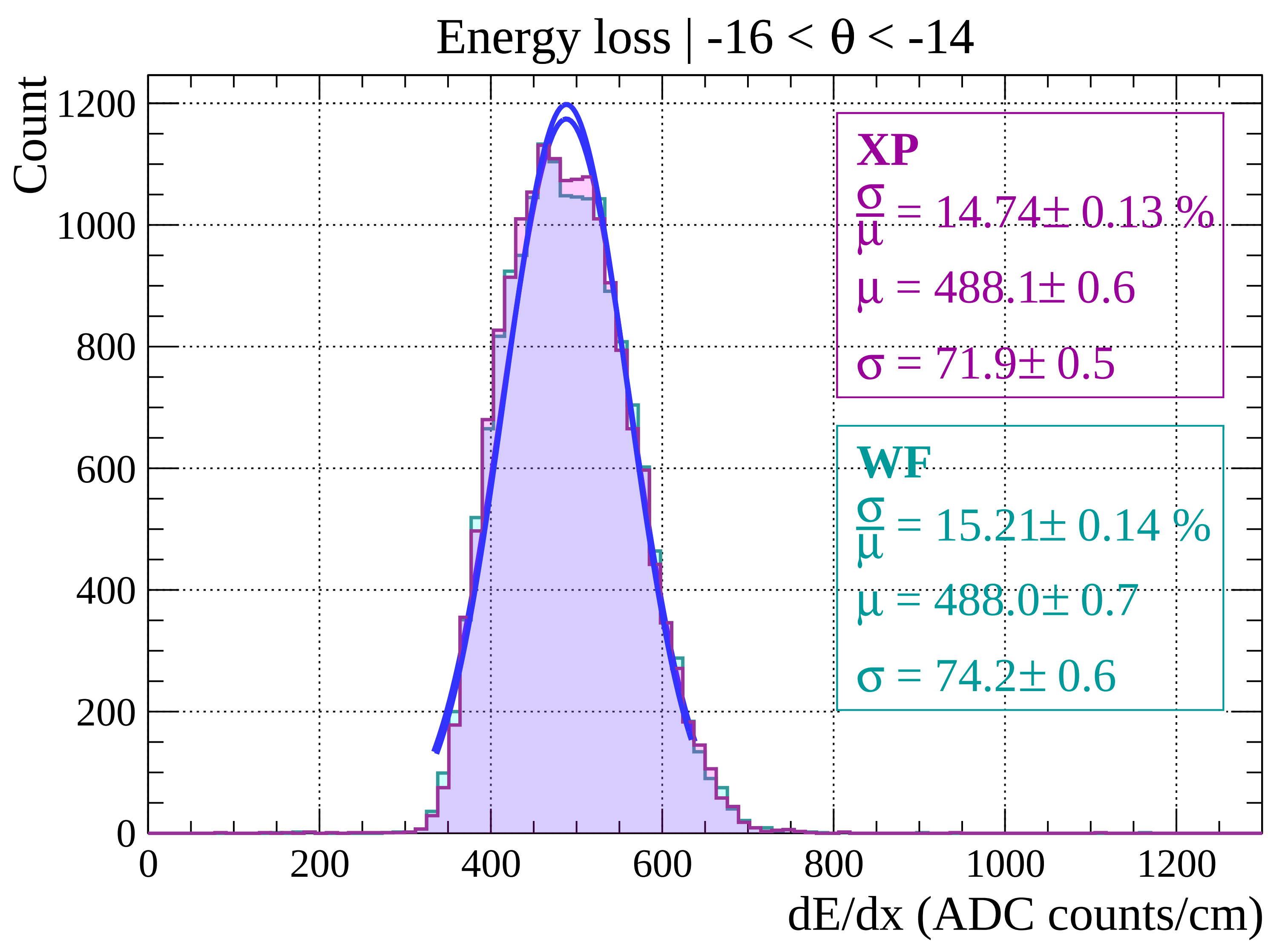
Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 14.26± 0.13 %
μ = 495.0± 0.7
σ = 70.6± 0.5
WF
σ/μ = 14.47± 0.13 %
μ = 495.5± 0.7
σ = 71.7± 0.5



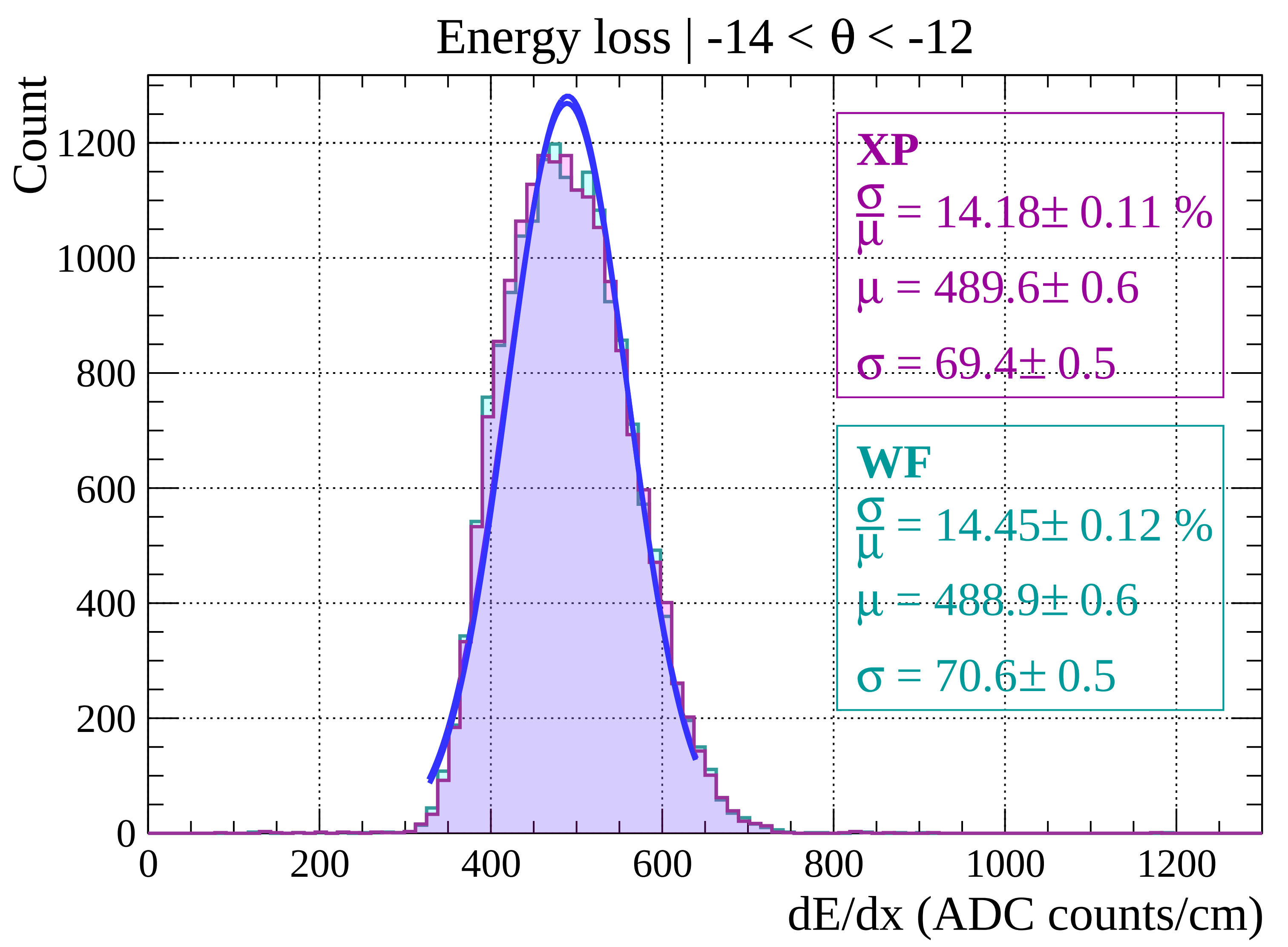
Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.34± 0.12 %
μ = 492.1± 0.7
σ = 70.6± 0.5
WF
σ/μ = 14.67± 0.13 %
μ = 492.9± 0.7
σ = 72.3± 0.5
0
200
400
600
800
1000
1200

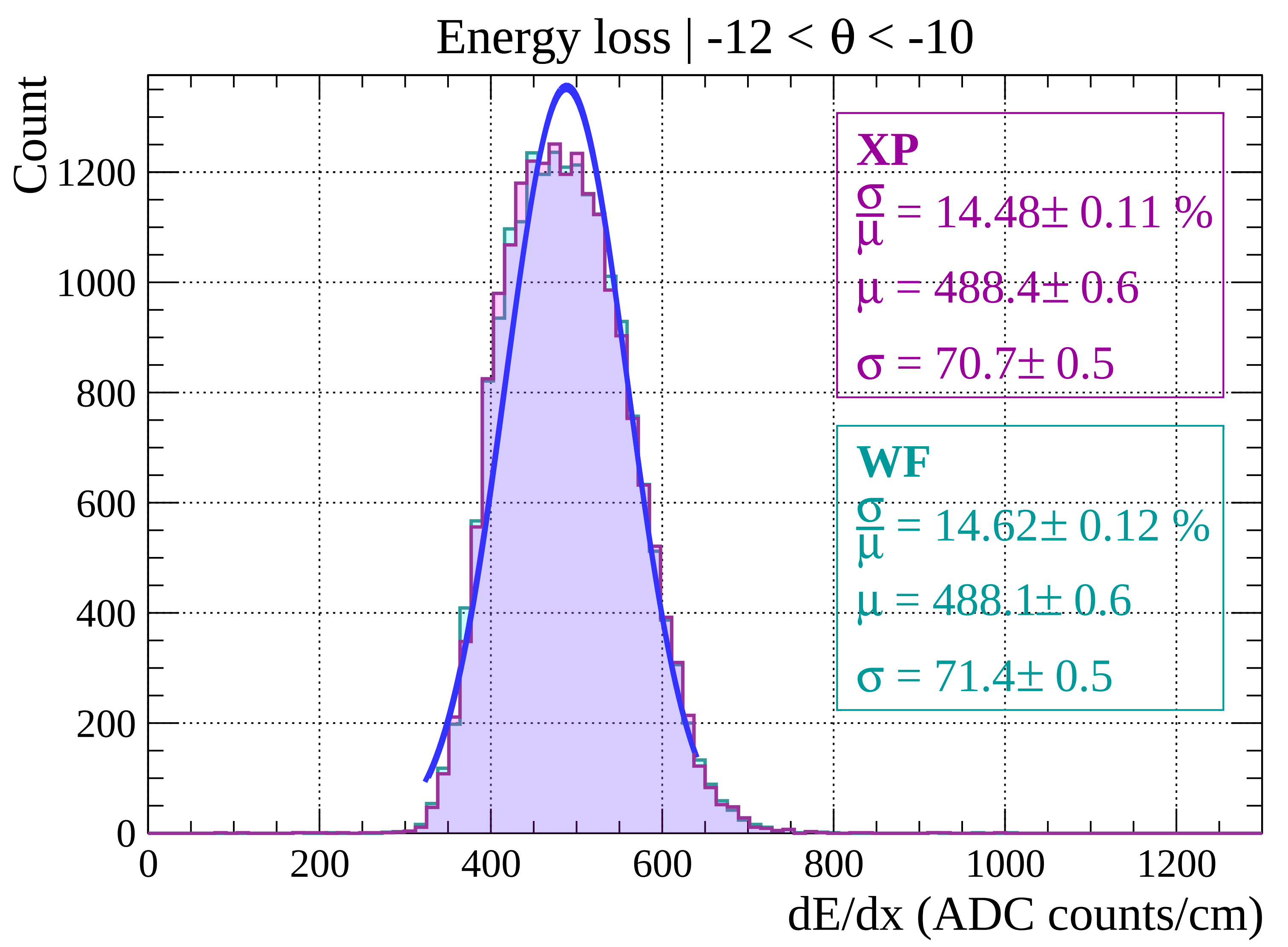


Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.44± 0.12 %
μ = 489.8± 0.6
σ = 70.7± 0.5
WF
σ/μ = 14.55± 0.12 %
μ = 490.2± 0.6
σ = 71.3± 0.5

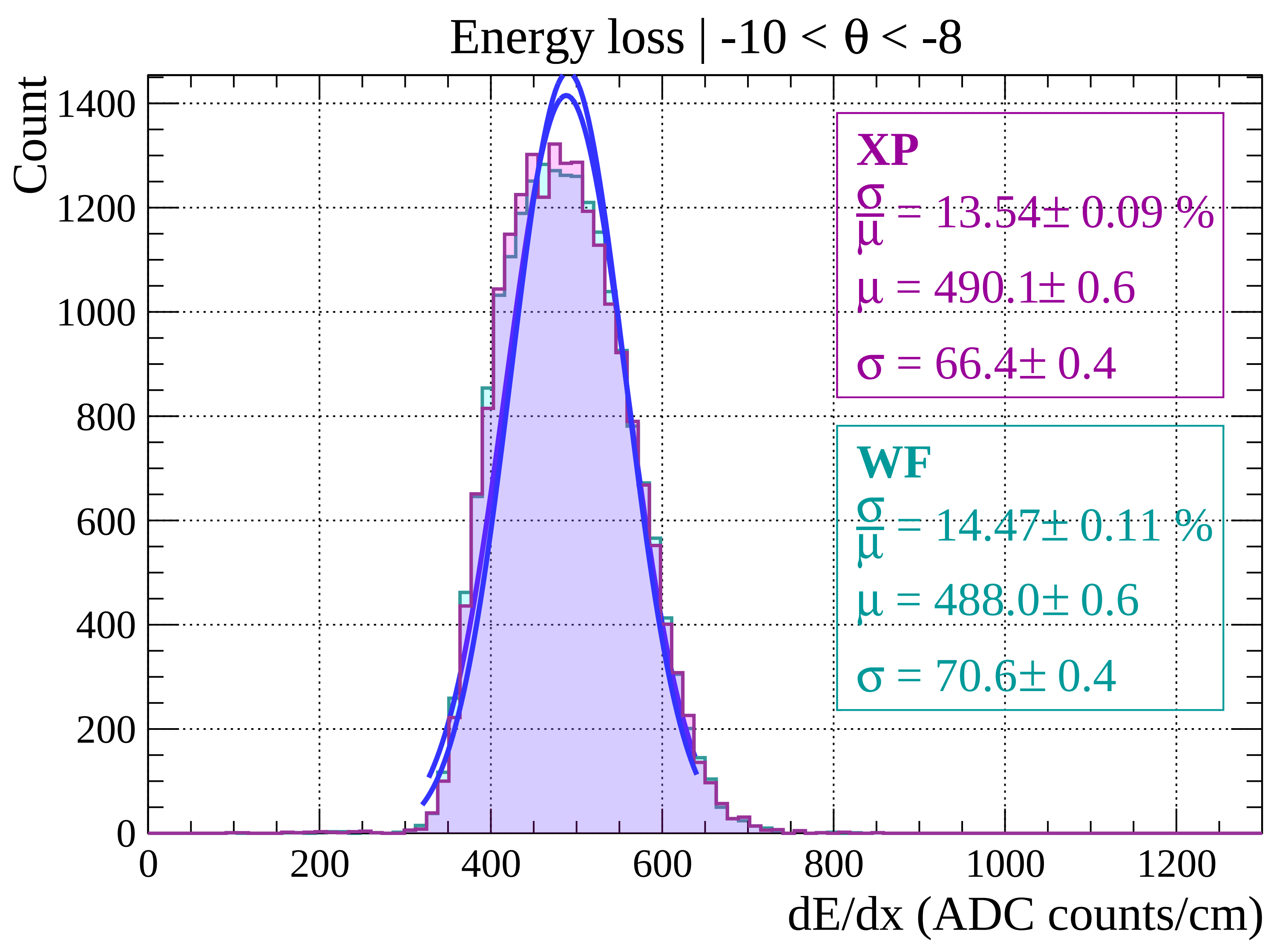


Energy loss | -16 < θ < -14
Count
XP
σ/μ = 14.74 ± 0.13 %
μ = 488.1 ± 0.6
σ = 71.9 ± 0.5
WF
σ/μ = 15.21 ± 0.14 %
μ = 488.0 ± 0.7
σ = 74.2 ± 0.6
dE/dx (ADC counts/cm)

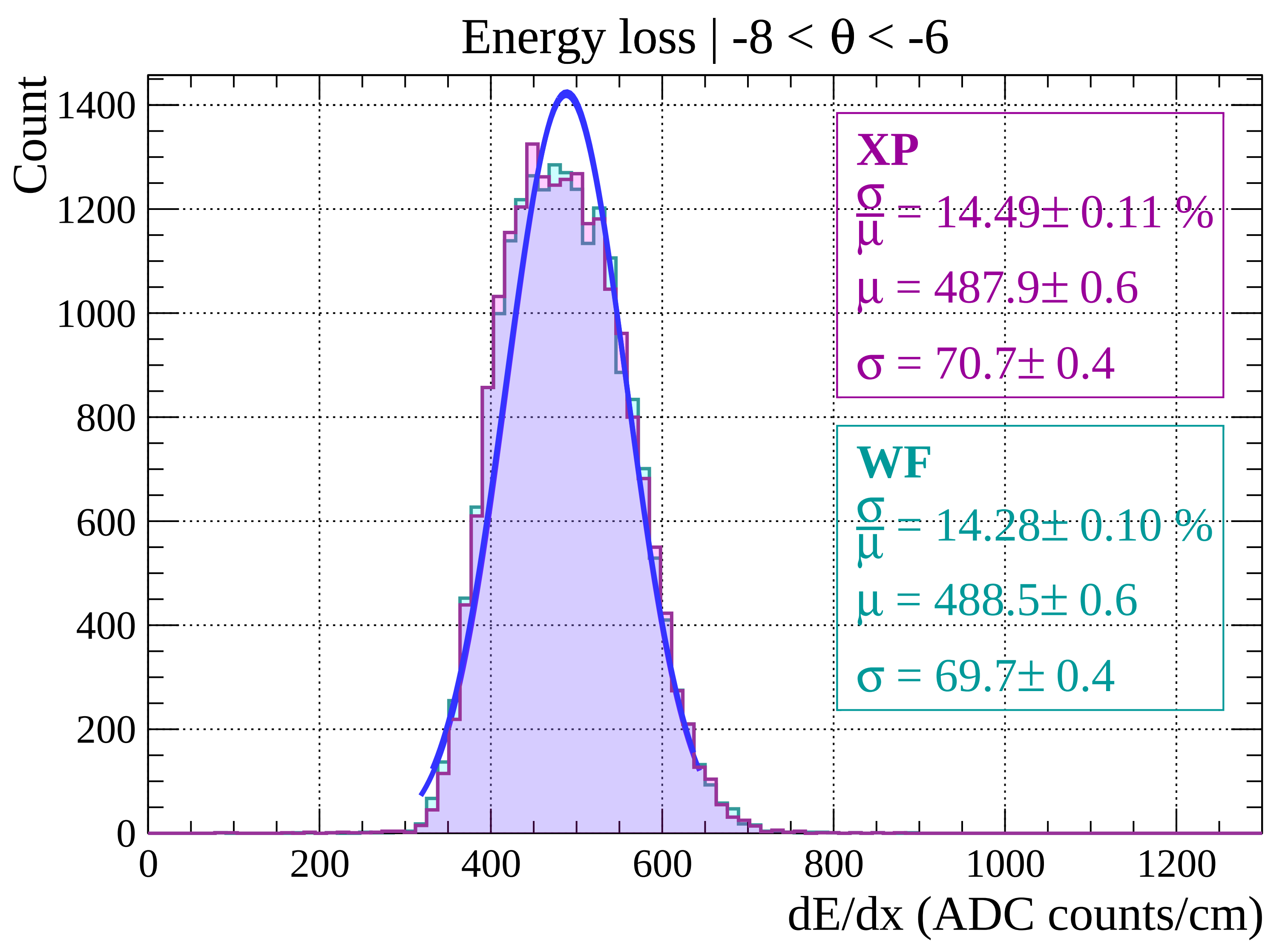

Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.18± 0.11 %
μ = 489.6± 0.6
σ = 69.4± 0.5
WF
σ/μ = 14.45± 0.12 %
μ = 488.9± 0.6
σ = 70.6± 0.5



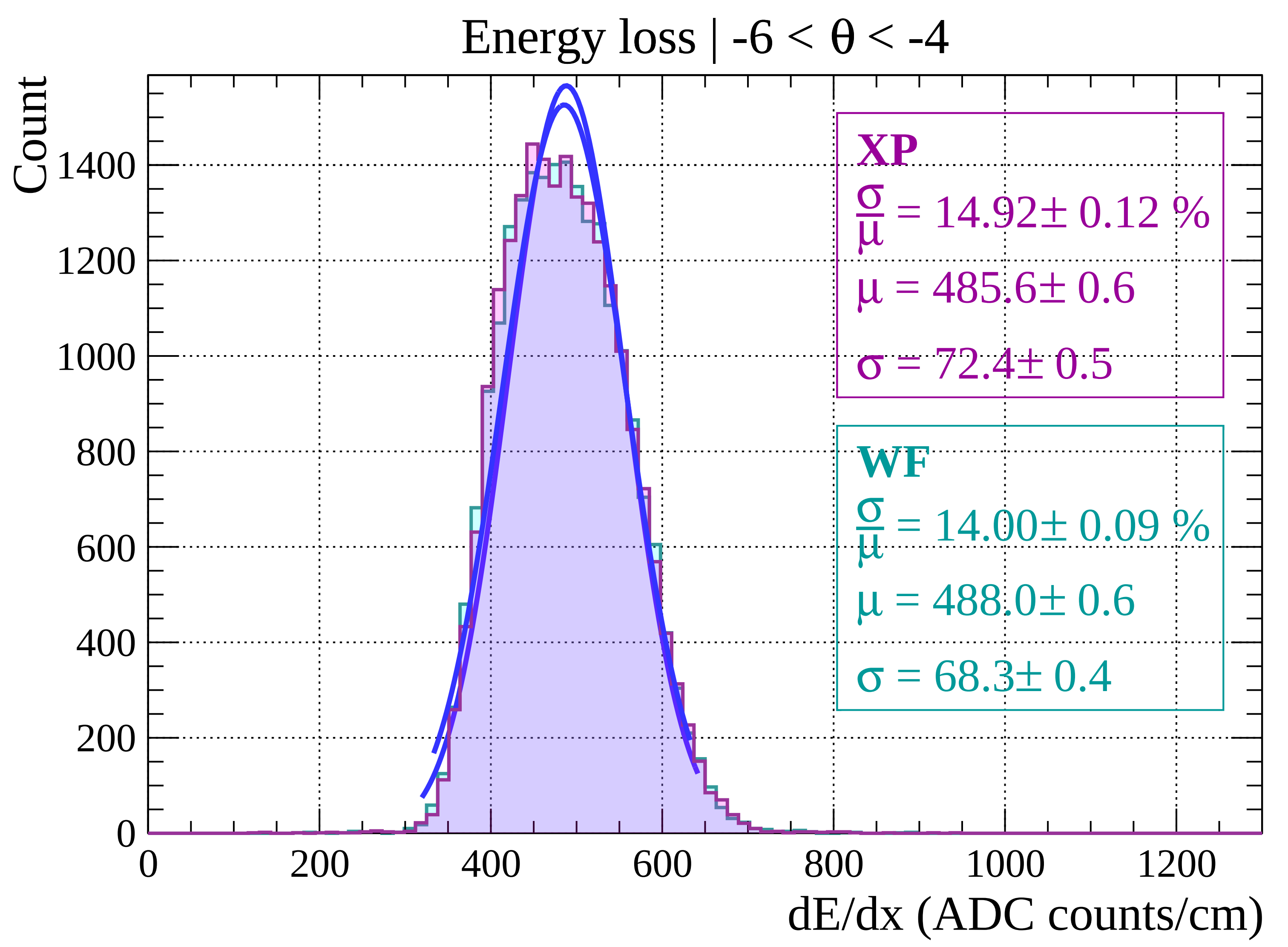
Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.48± 0.11 %
μ = 488.4± 0.6
σ = 70.7± 0.5
WF
σ/μ = 14.62± 0.12 %
μ = 488.1± 0.6
σ = 71.4± 0.5



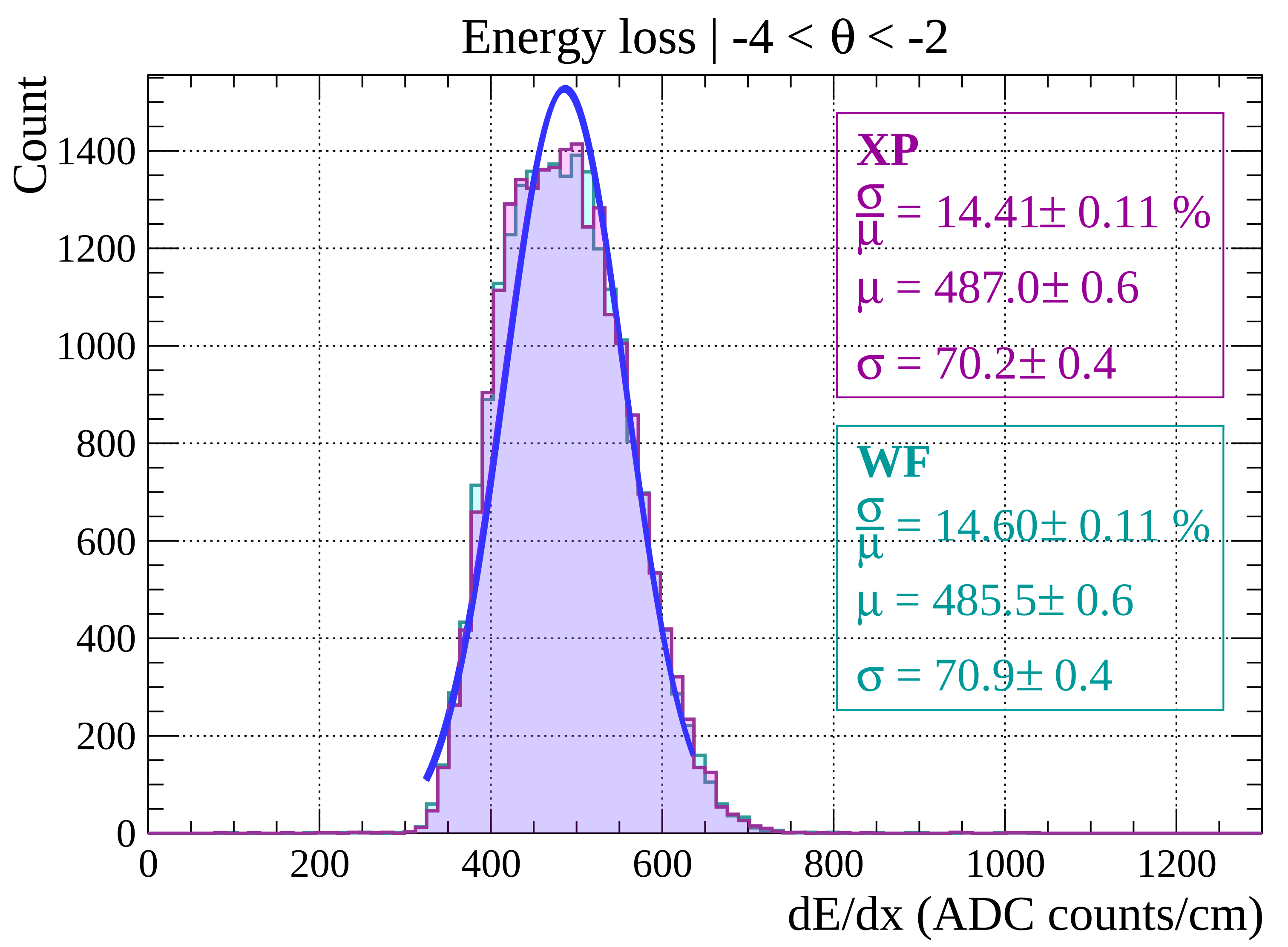
Energy loss | -10 < θ < -8
Count
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.54± 0.09 %
μ = 490.1± 0.6
σ = 66.4± 0.4
WF
σ/μ = 14.47± 0.11 %
μ = 488.0± 0.6
σ = 70.6± 0.4



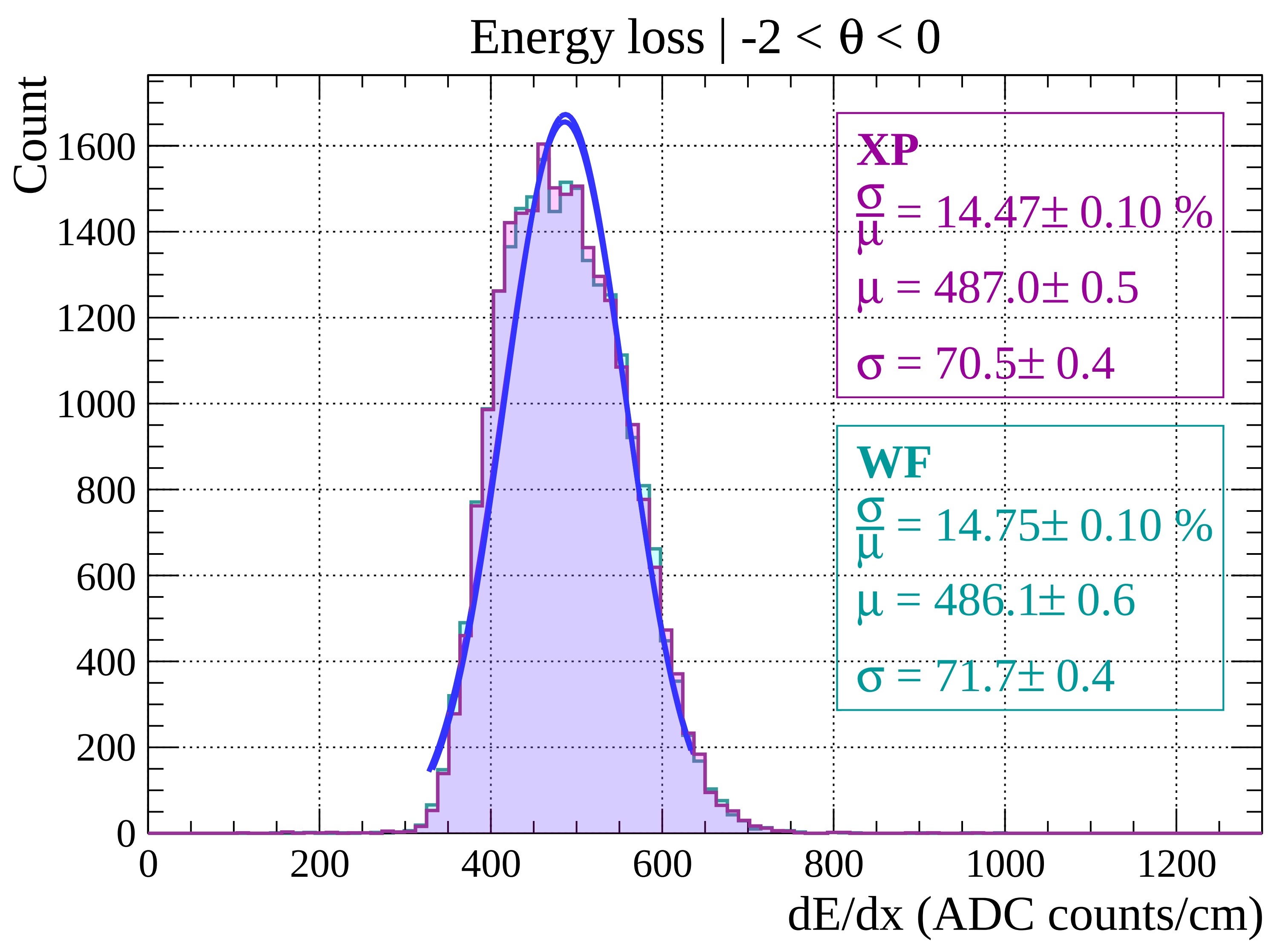
Energy loss | -8 < θ < -6
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.49± 0.11 %
μ = 487.9± 0.6
σ = 70.7± 0.4
WF
σ/μ = 14.28± 0.10 %
μ = 488.5± 0.6
σ = 69.7± 0.4



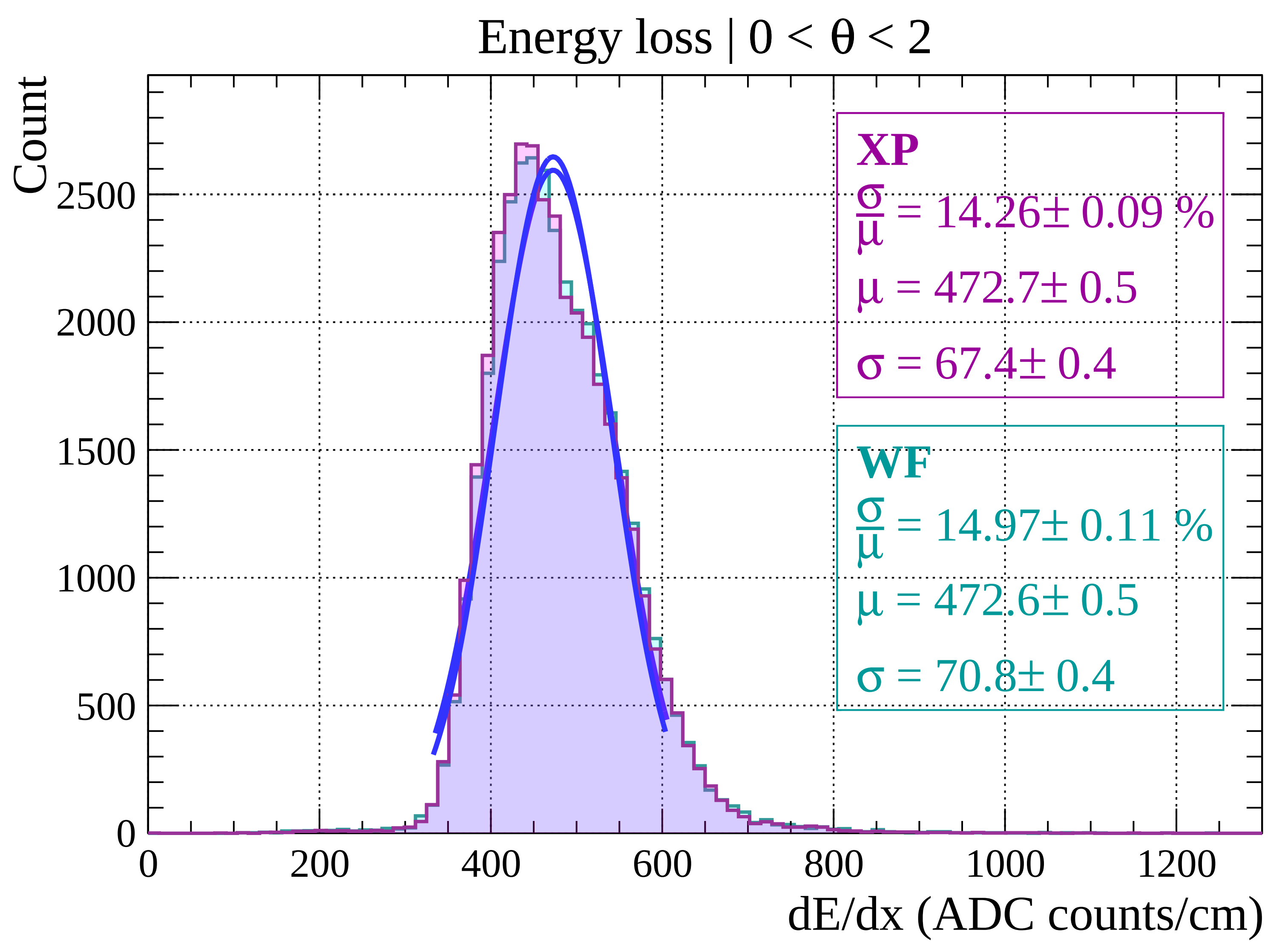
Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.92± 0.12 %
μ = 485.6± 0.6
σ = 72.4± 0.5
WF
σ/μ = 14.00± 0.09 %
μ = 488.0± 0.6
σ = 68.3± 0.4



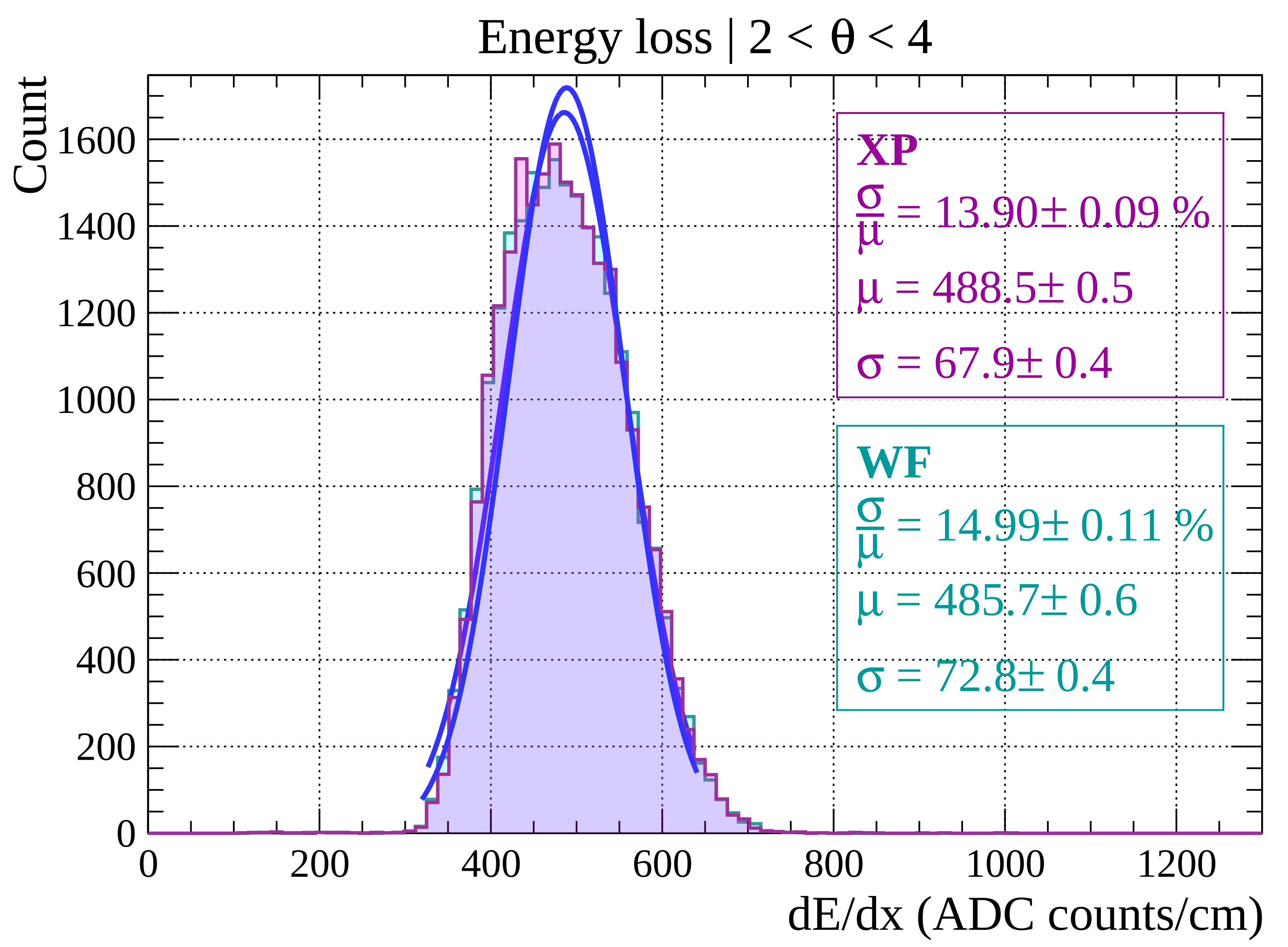
Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
σ/μ = 14.41± 0.11 %
μ = 487.0± 0.6
σ = 70.2± 0.4
WF
σ/μ = 14.60± 0.11 %
μ = 485.5± 0.6
σ = 70.9± 0.4



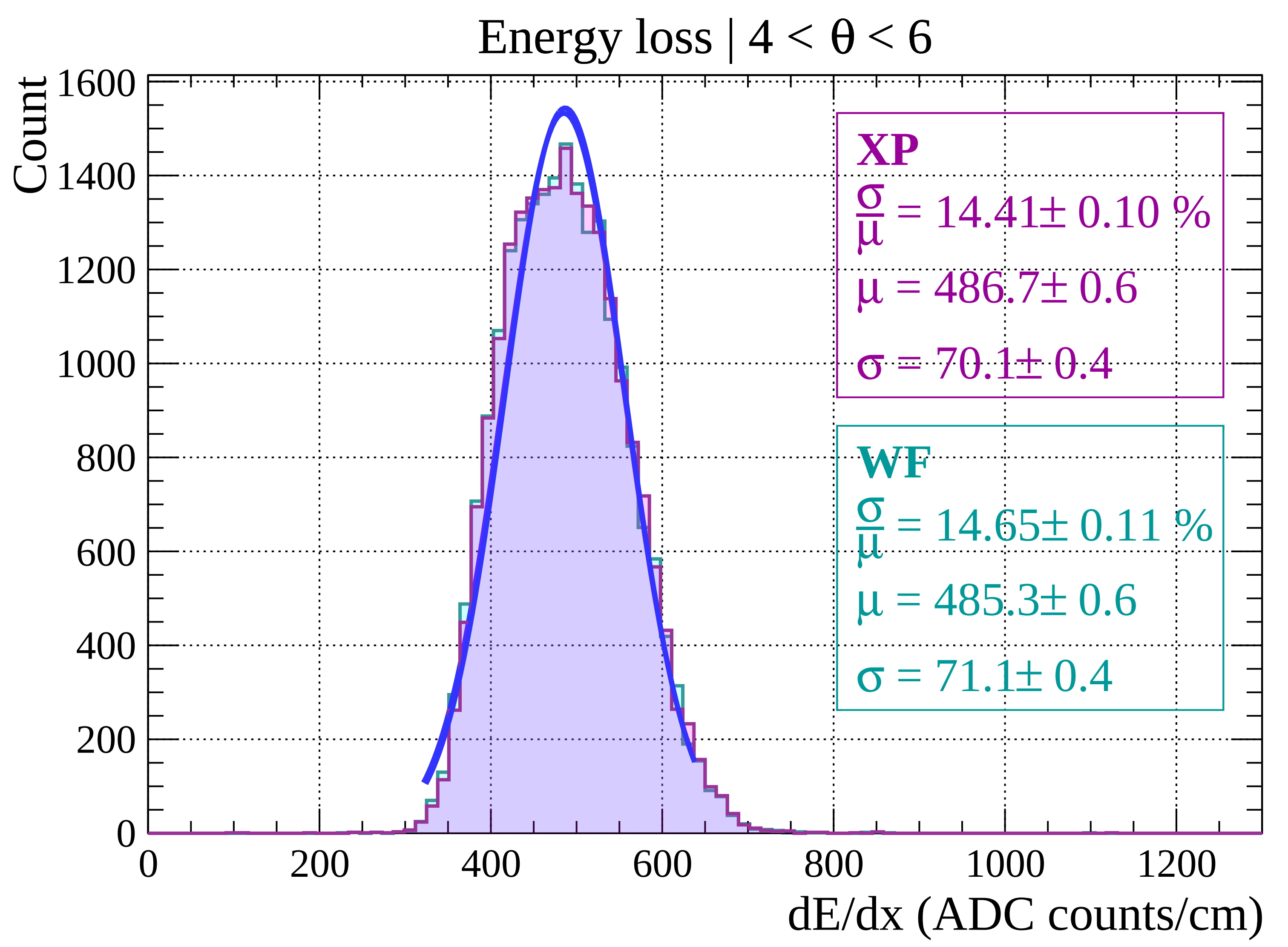
Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP
σ/μ = 14.47± 0.10 %
μ = 487.0± 0.5
σ = 70.5± 0.4
WF
σ/μ = 14.75± 0.10 %
μ = 486.1± 0.6
σ = 71.7± 0.4

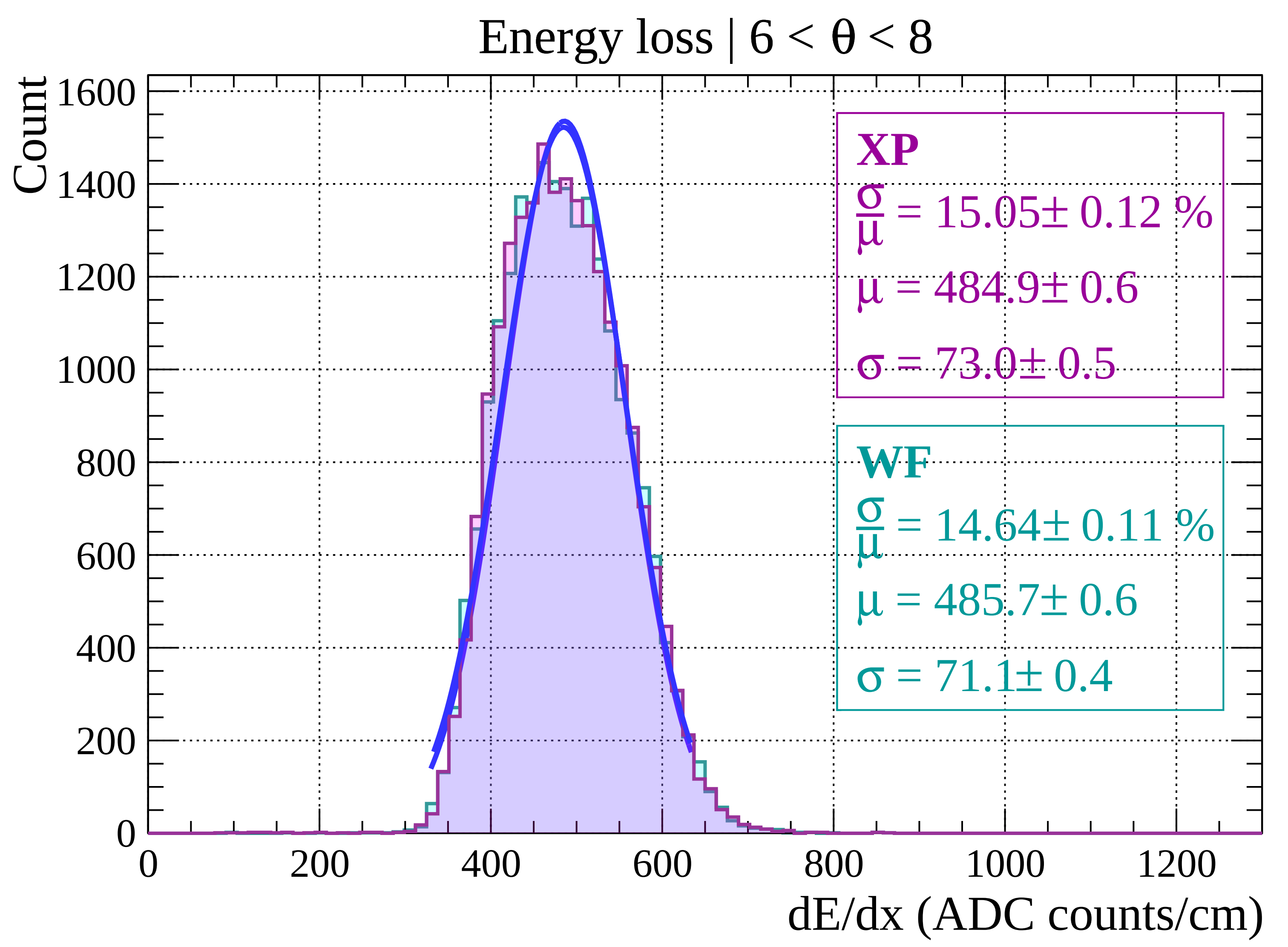


Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
500
1000
1500
2000
2500
XP
σ/μ = 14.26± 0.09 %
μ = 472.7± 0.5
σ = 67.4± 0.4
WF
σ/μ = 14.97± 0.11 %
μ = 472.6± 0.5
σ = 70.8± 0.4

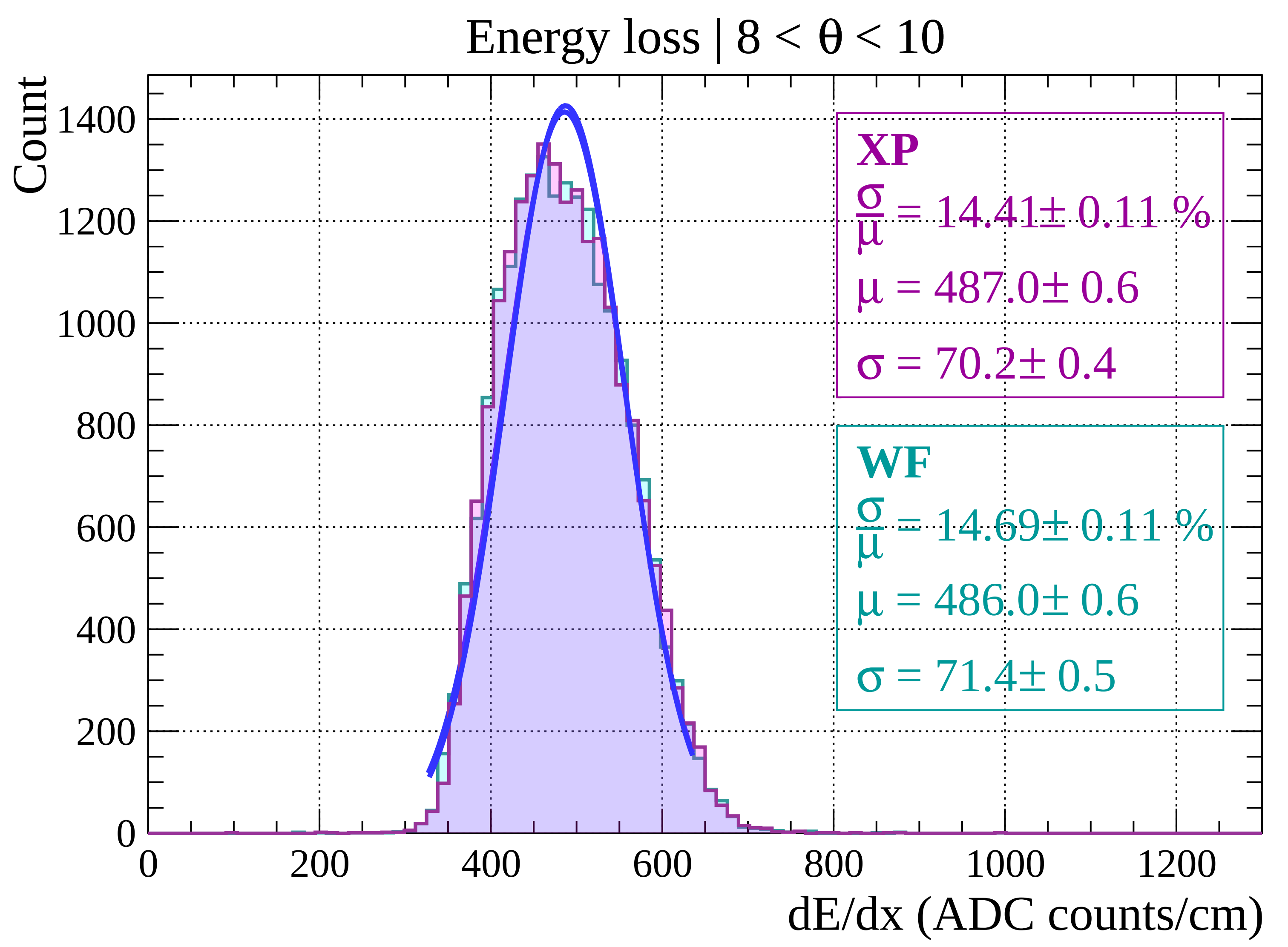


Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.90± 0.09 %
μ = 488.5± 0.5
σ = 67.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 14.99± 0.11 %
μ = 485.7± 0.6
σ = 72.8± 0.4

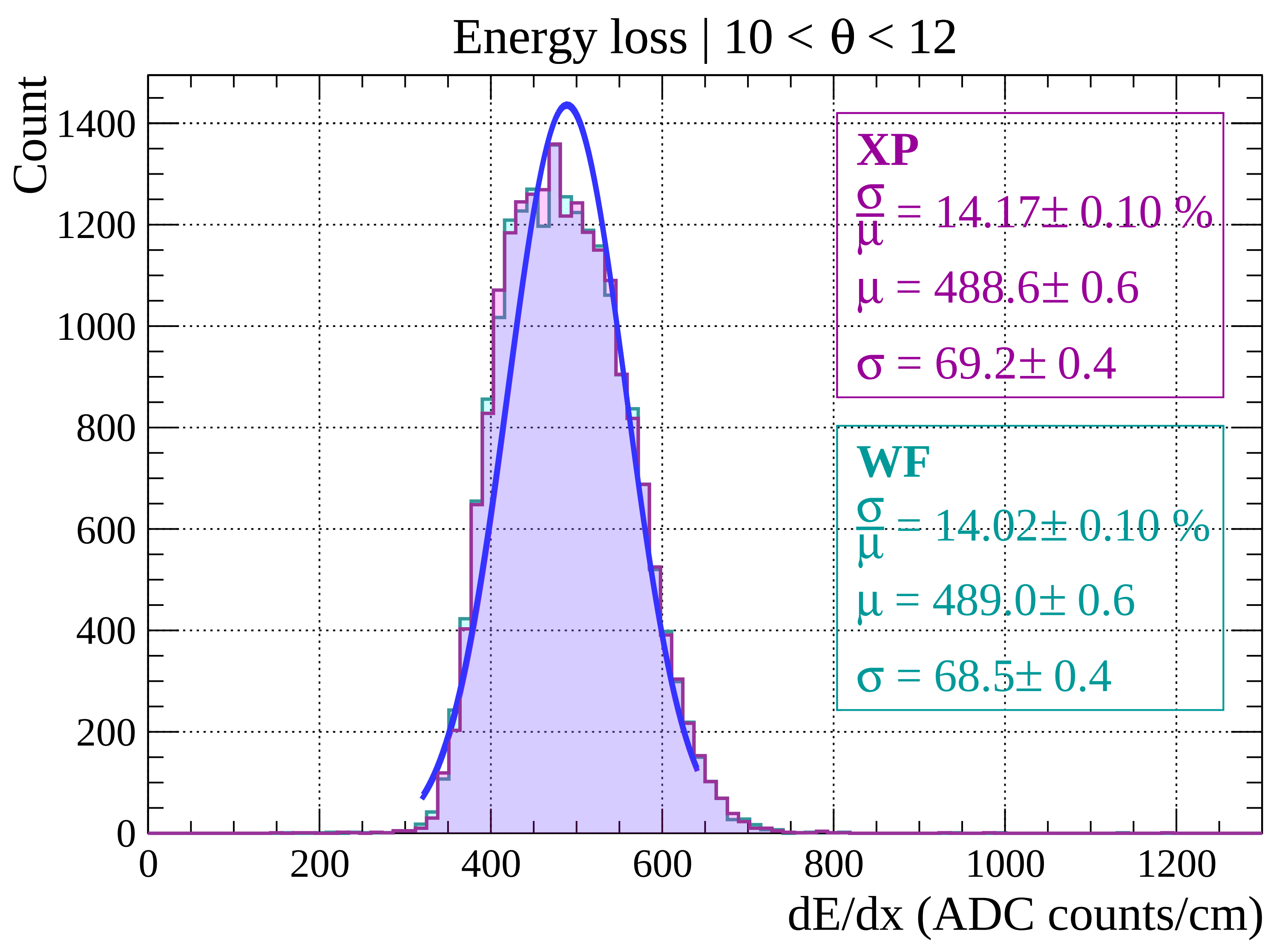

Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
XP
σ/μ = 14.41± 0.10 %
μ = 486.7± 0.6
σ = 70.1± 0.4
WF
σ/μ = 14.65± 0.11 %
μ = 485.3± 0.6
σ = 71.1± 0.4

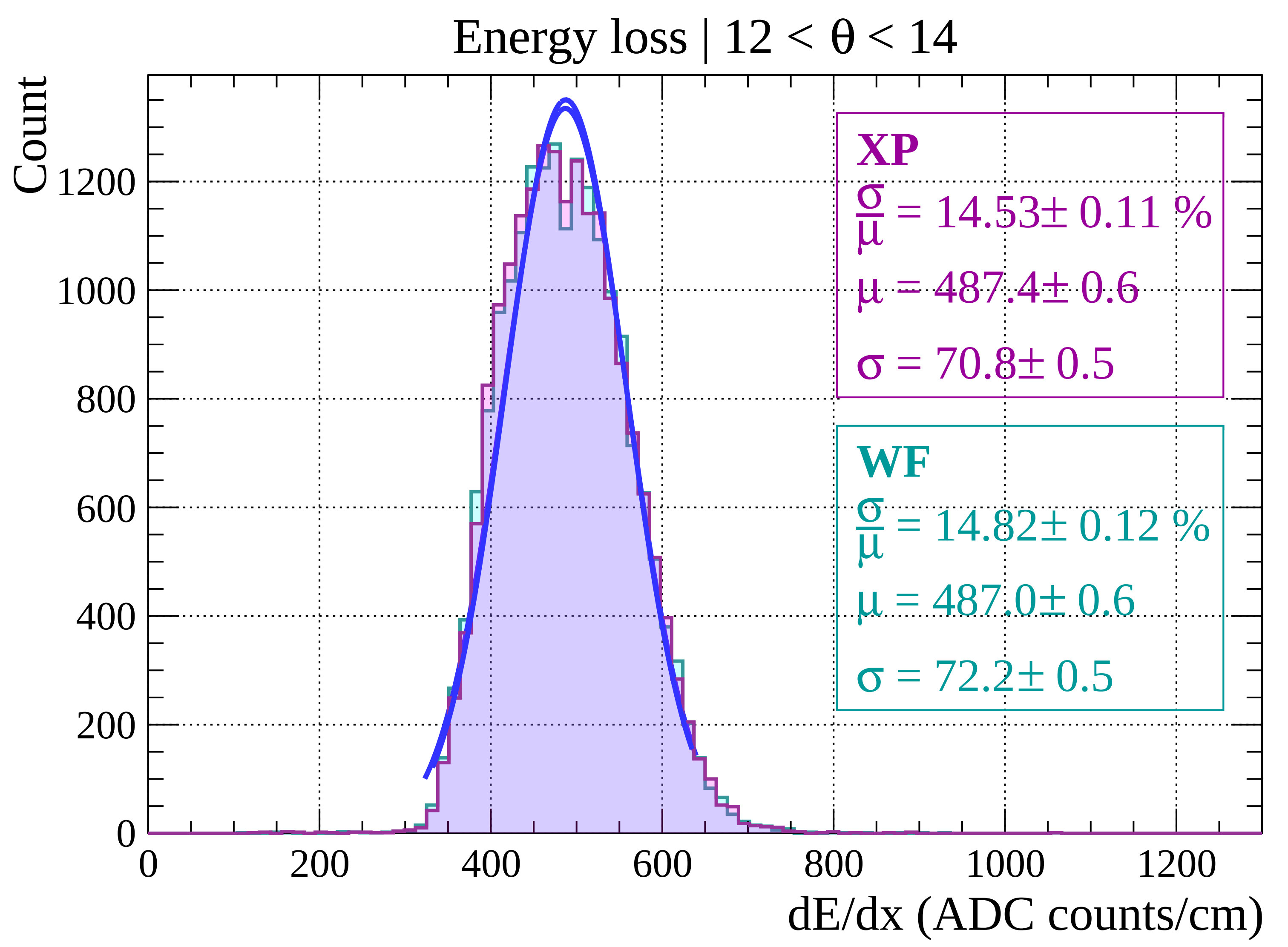


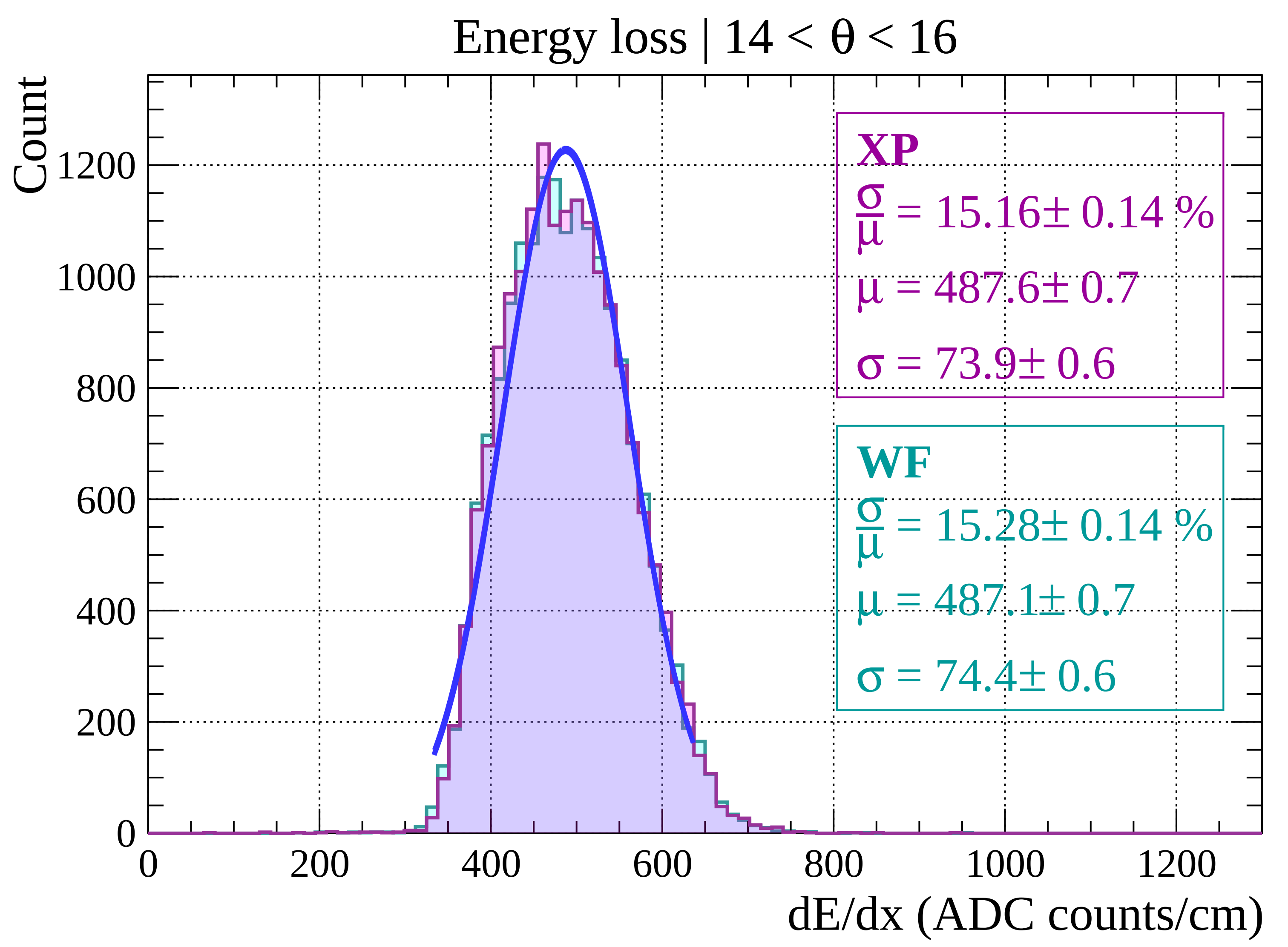
Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP
σ/μ = 15.05± 0.12 %
μ = 484.9± 0.6
σ = 73.0± 0.5
WF
σ/μ = 14.64± 0.11 %
μ = 485.7± 0.6
σ = 71.1± 0.4



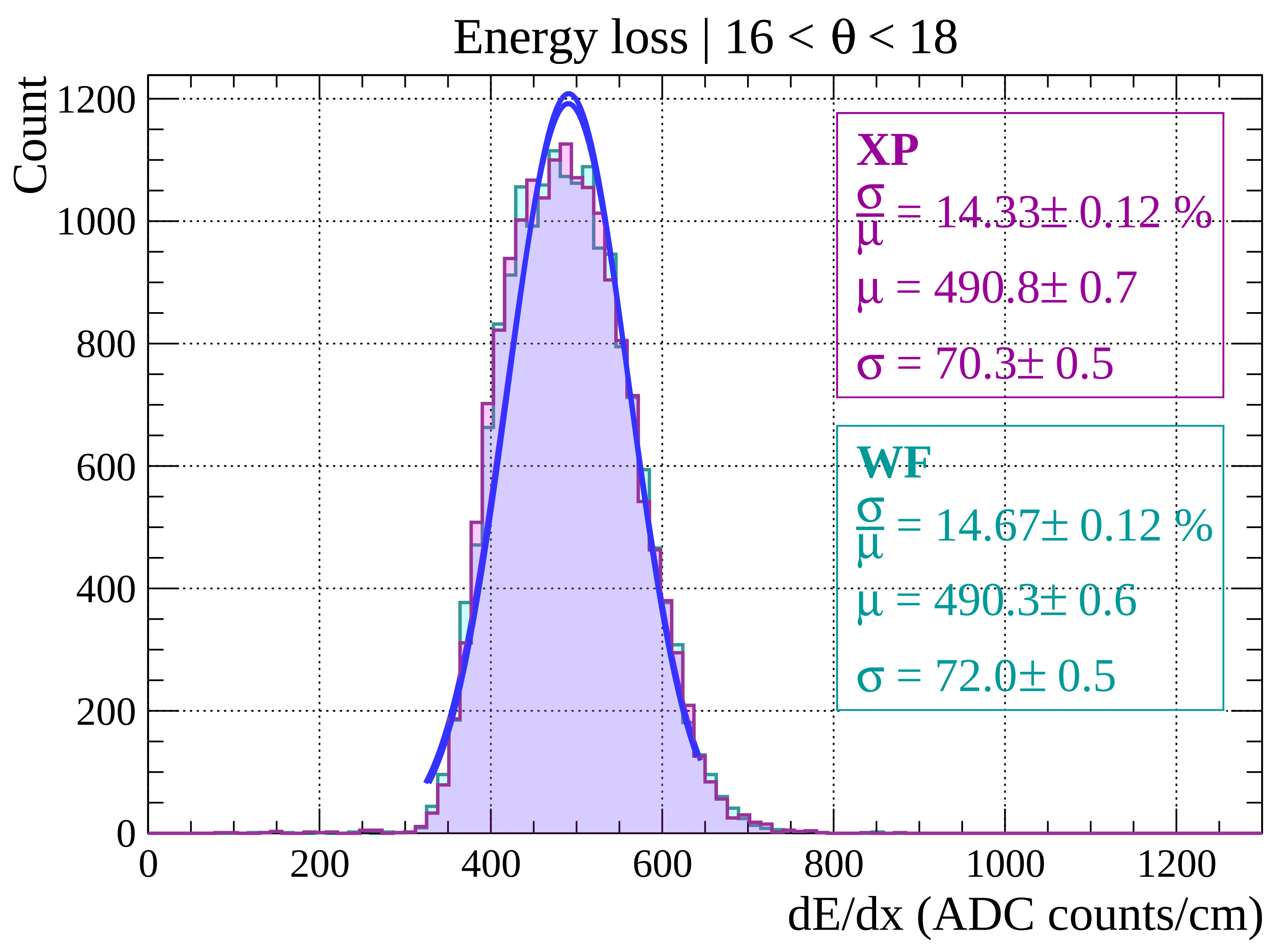
Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
σ/μ = 14.41± 0.11 %
μ = 487.0± 0.6
σ = 70.2± 0.4
WF
σ/μ = 14.69± 0.11 %
μ = 486.0± 0.6
σ = 71.4± 0.5


Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
σ/μ = 14.17 ± 0.10 %
μ = 488.6 ± 0.6
σ = 69.2 ± 0.4
WF
σ/μ = 14.02 ± 0.10 %
μ = 489.0 ± 0.6
σ = 68.5 ± 0.4

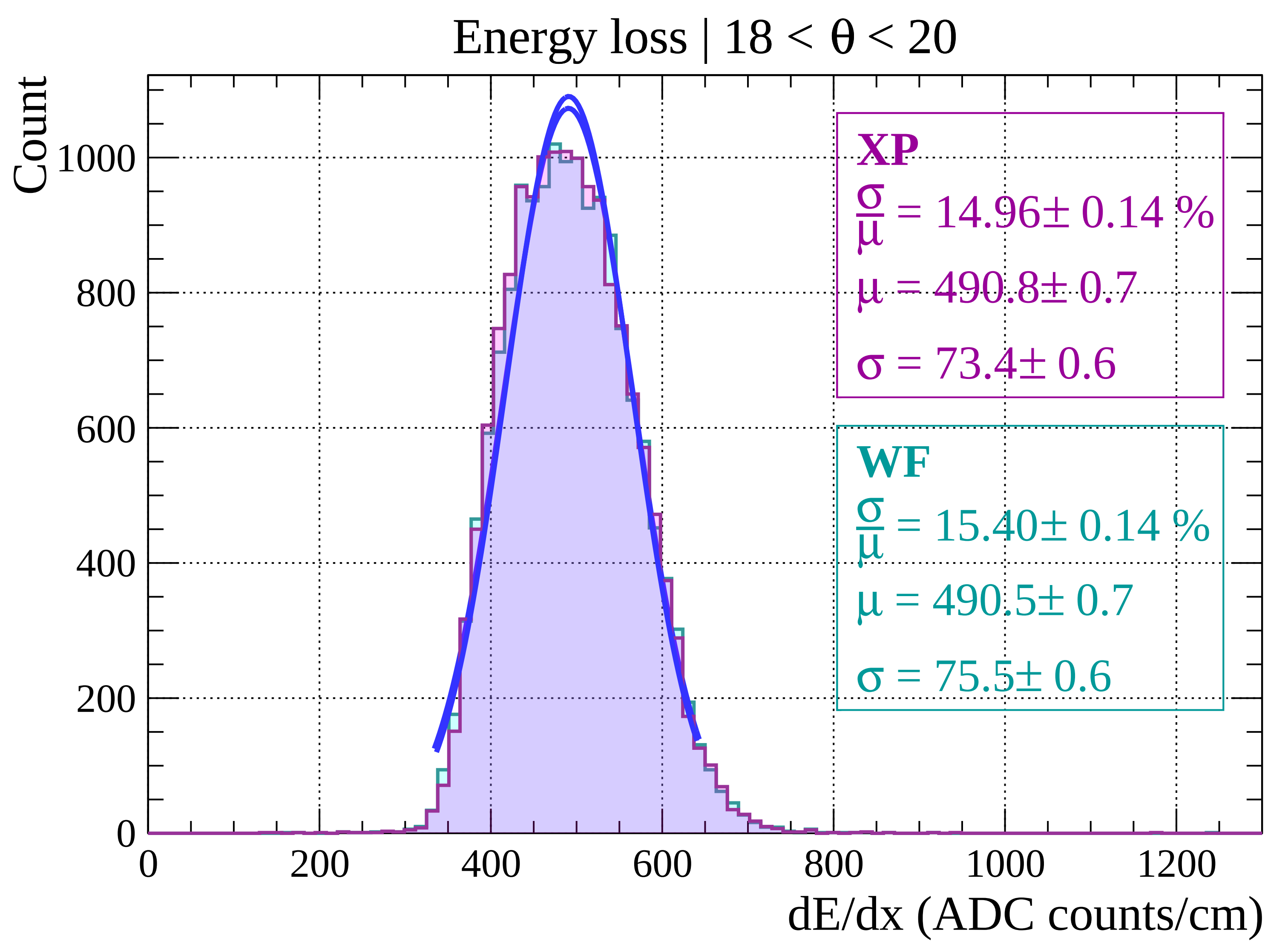

Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.53± 0.11 %
μ = 487.4± 0.6
σ = 70.8± 0.5
WF
σ/μ = 14.82± 0.12 %
μ = 487.0± 0.6
σ = 72.2± 0.5



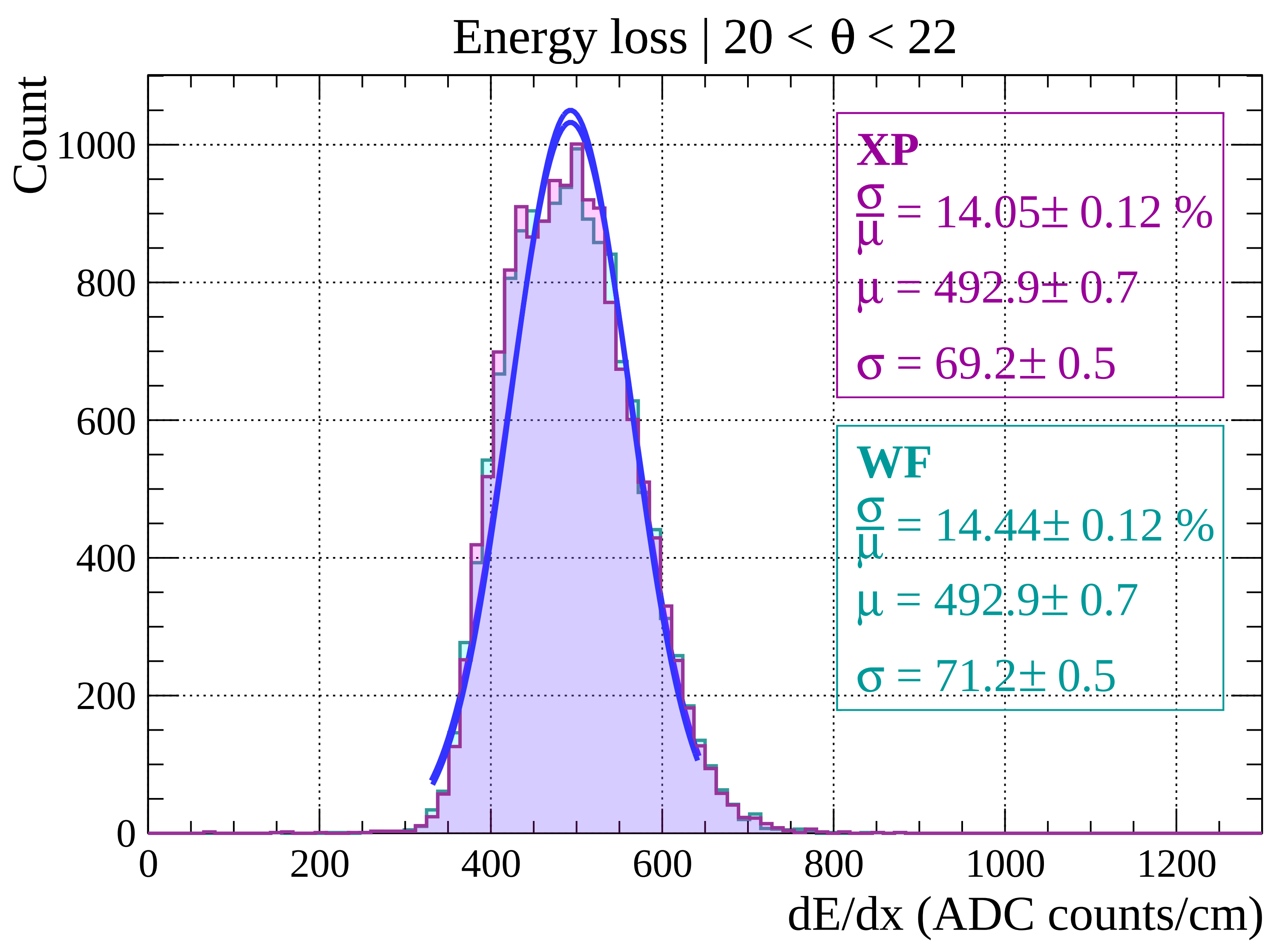
Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.16± 0.14 %
μ = 487.6± 0.7
σ = 73.9± 0.6
WF
σ/μ = 15.28± 0.14 %
μ = 487.1± 0.7
σ = 74.4± 0.6



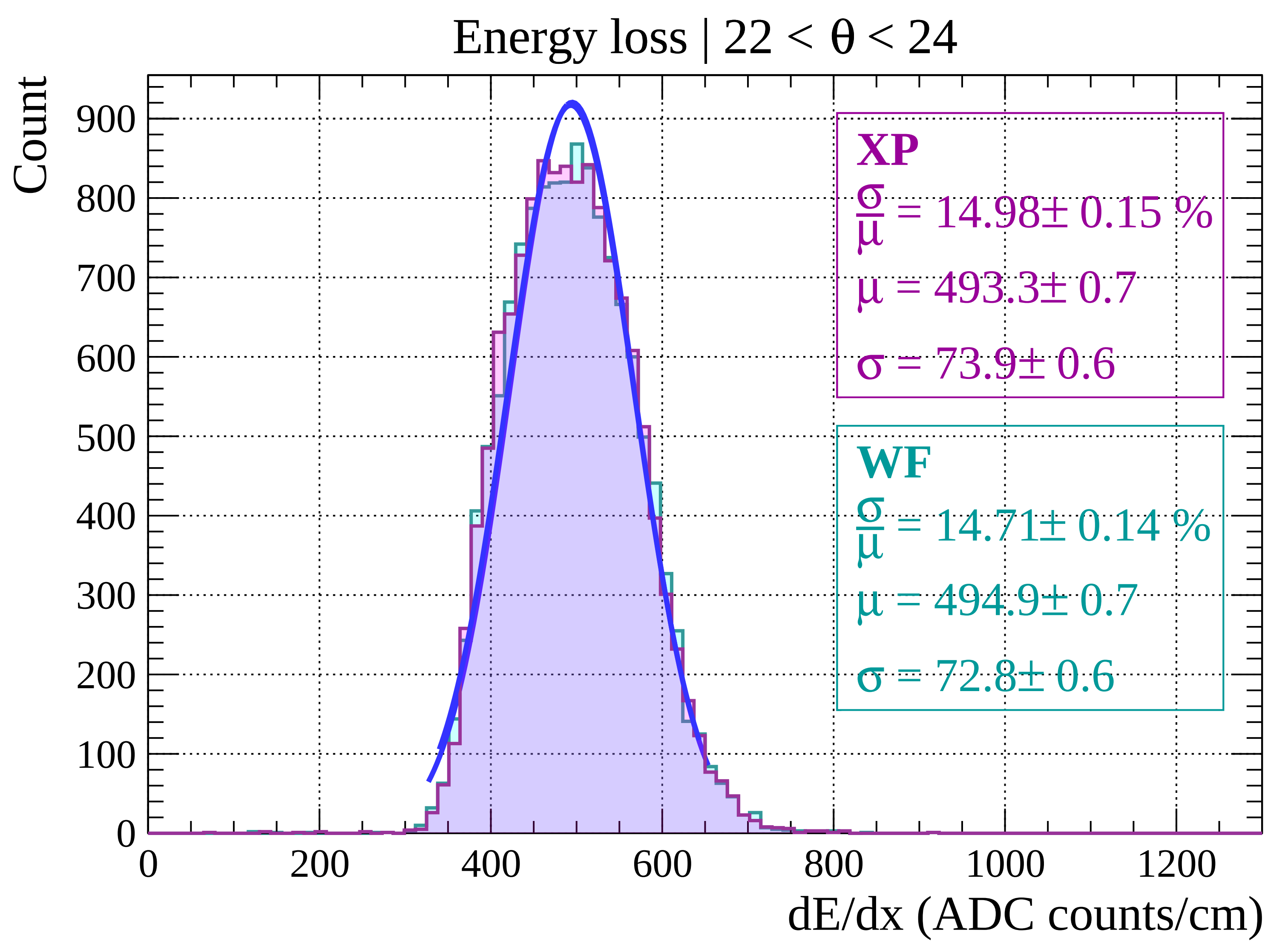
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.33± 0.12 %
μ = 490.8± 0.7
σ = 70.3± 0.5
WF
$\frac{\sigma}{\mu}$ = 14.67± 0.12 %
μ = 490.3± 0.6
σ = 72.0± 0.5



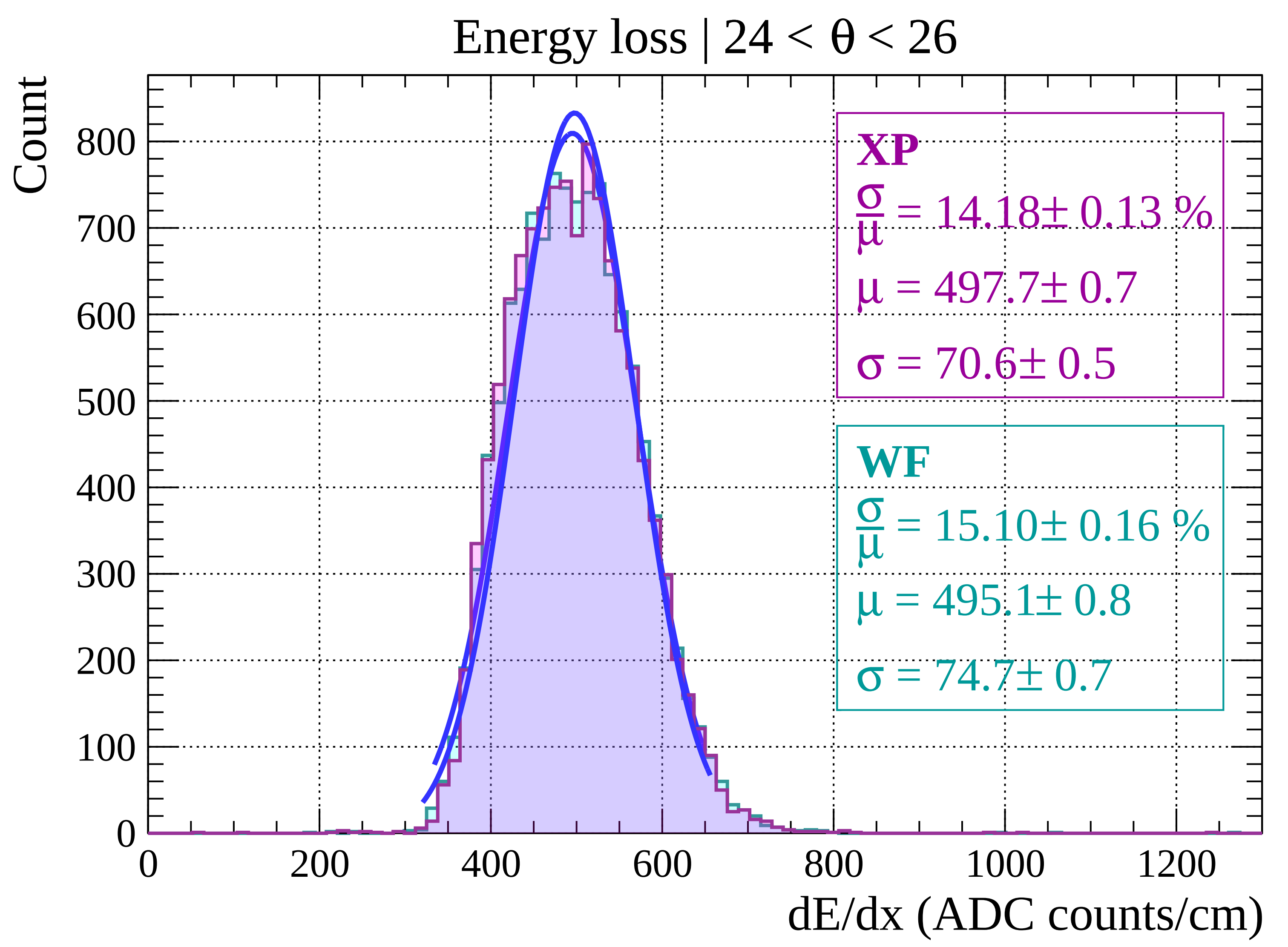
Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.96± 0.14 %
μ = 490.8± 0.7
σ = 73.4± 0.6
WF
σ/μ = 15.40± 0.14 %
μ = 490.5± 0.7
σ = 75.5± 0.6
0
200
400
600
800
1000
1200



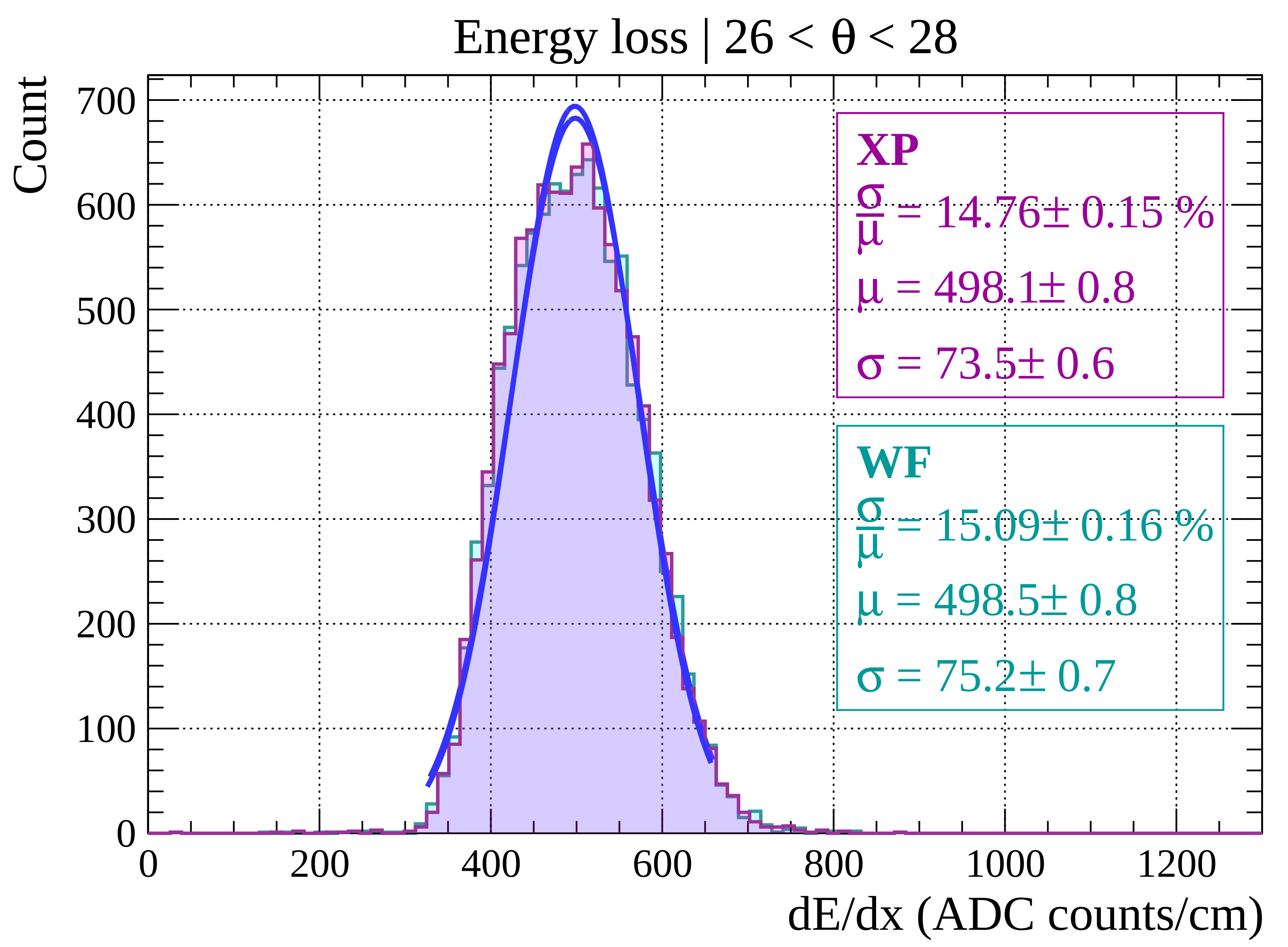
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.05± 0.12 %
μ = 492.9± 0.7
σ = 69.2± 0.5
WF
σ/μ = 14.44± 0.12 %
μ = 492.9± 0.7
σ = 71.2± 0.5

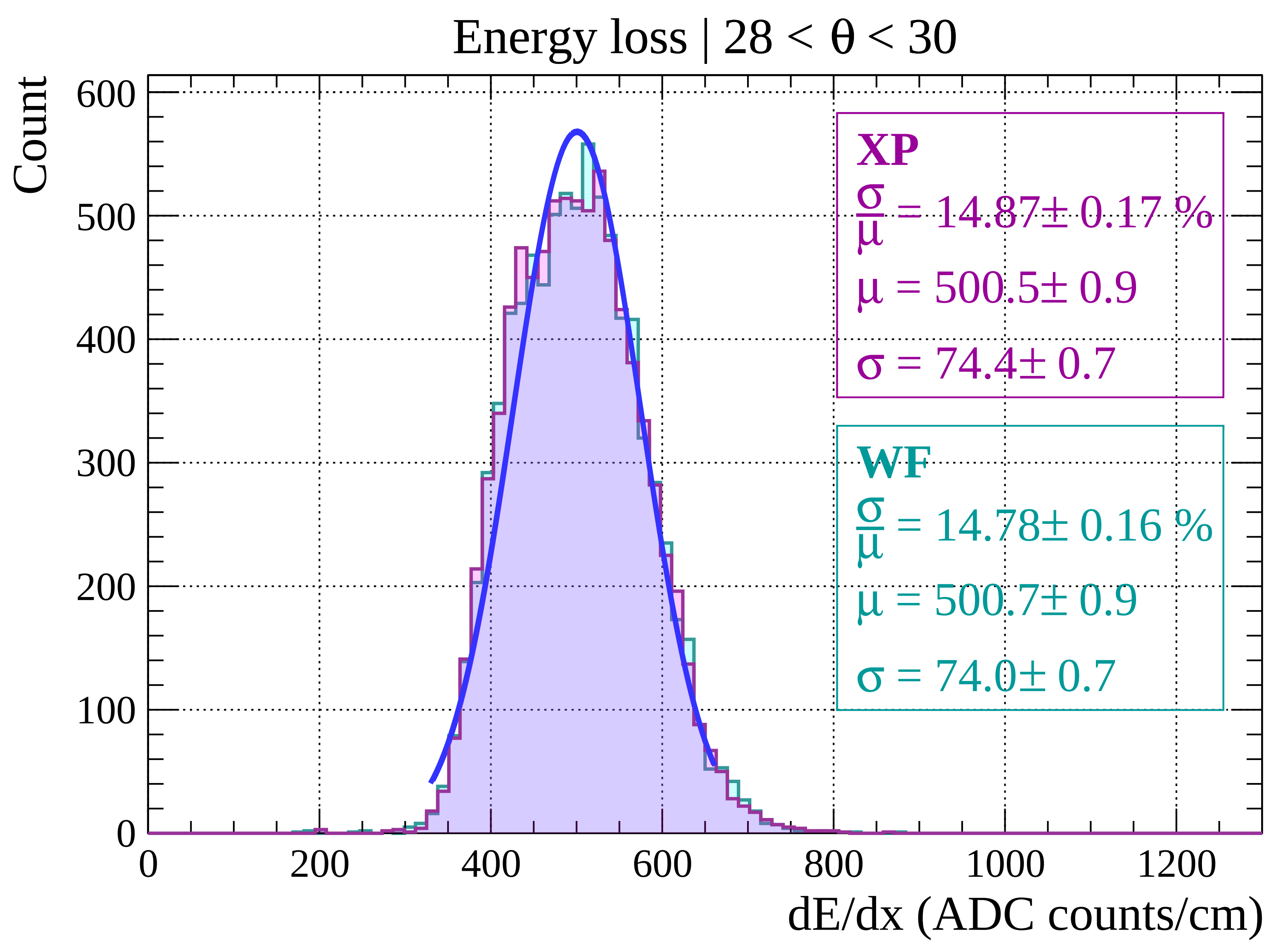


Energy loss | 22 < θ < 24
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.98± 0.15 %
μ = 493.3± 0.7
σ = 73.9± 0.6
WF
σ/μ = 14.71± 0.14 %
μ = 494.9± 0.7
σ = 72.8± 0.6

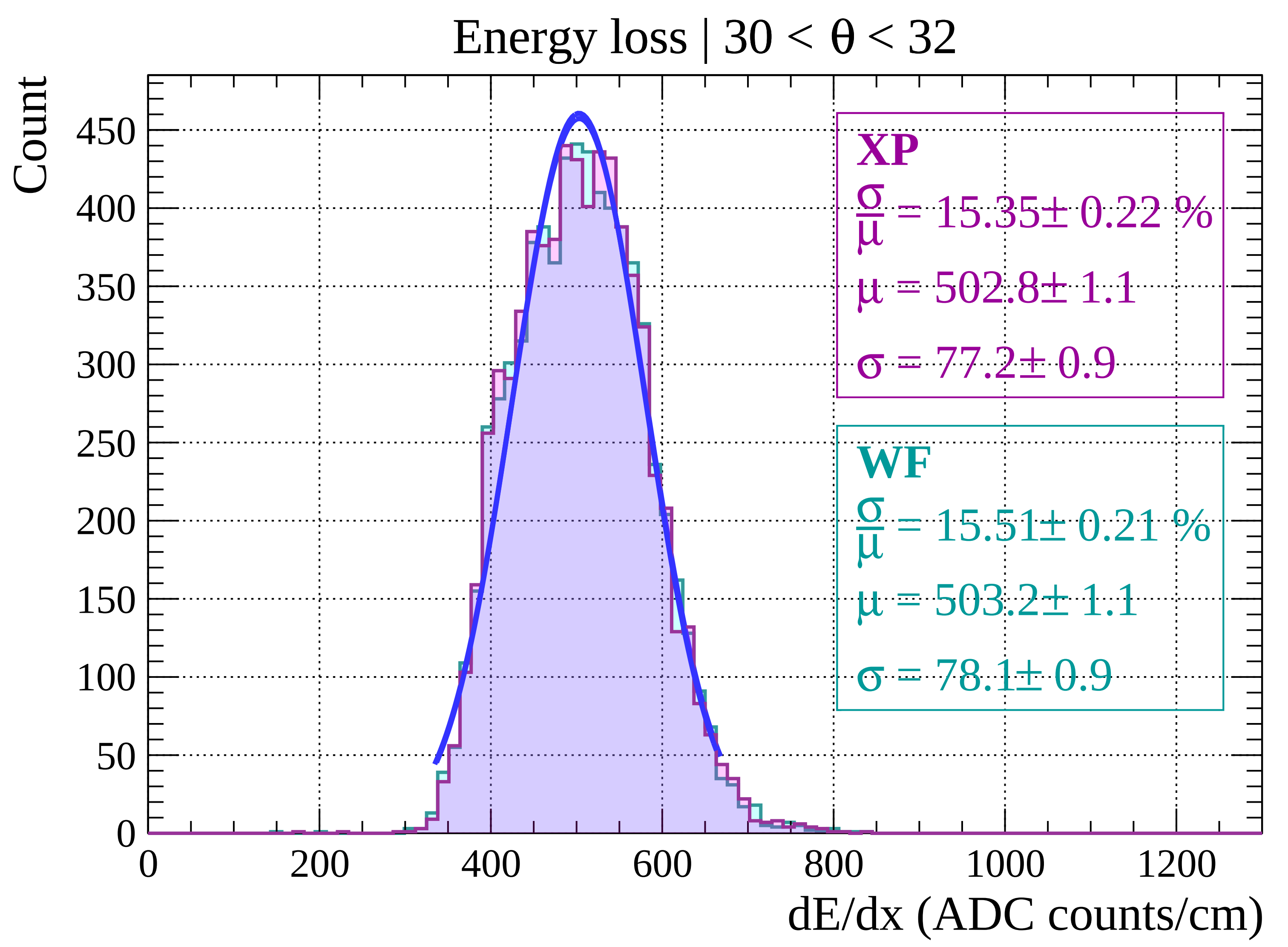


Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 14.18± 0.13 %
μ = 497.7± 0.7
σ = 70.6± 0.5
WF
σ/μ = 15.10± 0.16 %
μ = 495.1± 0.8
σ = 74.7± 0.7

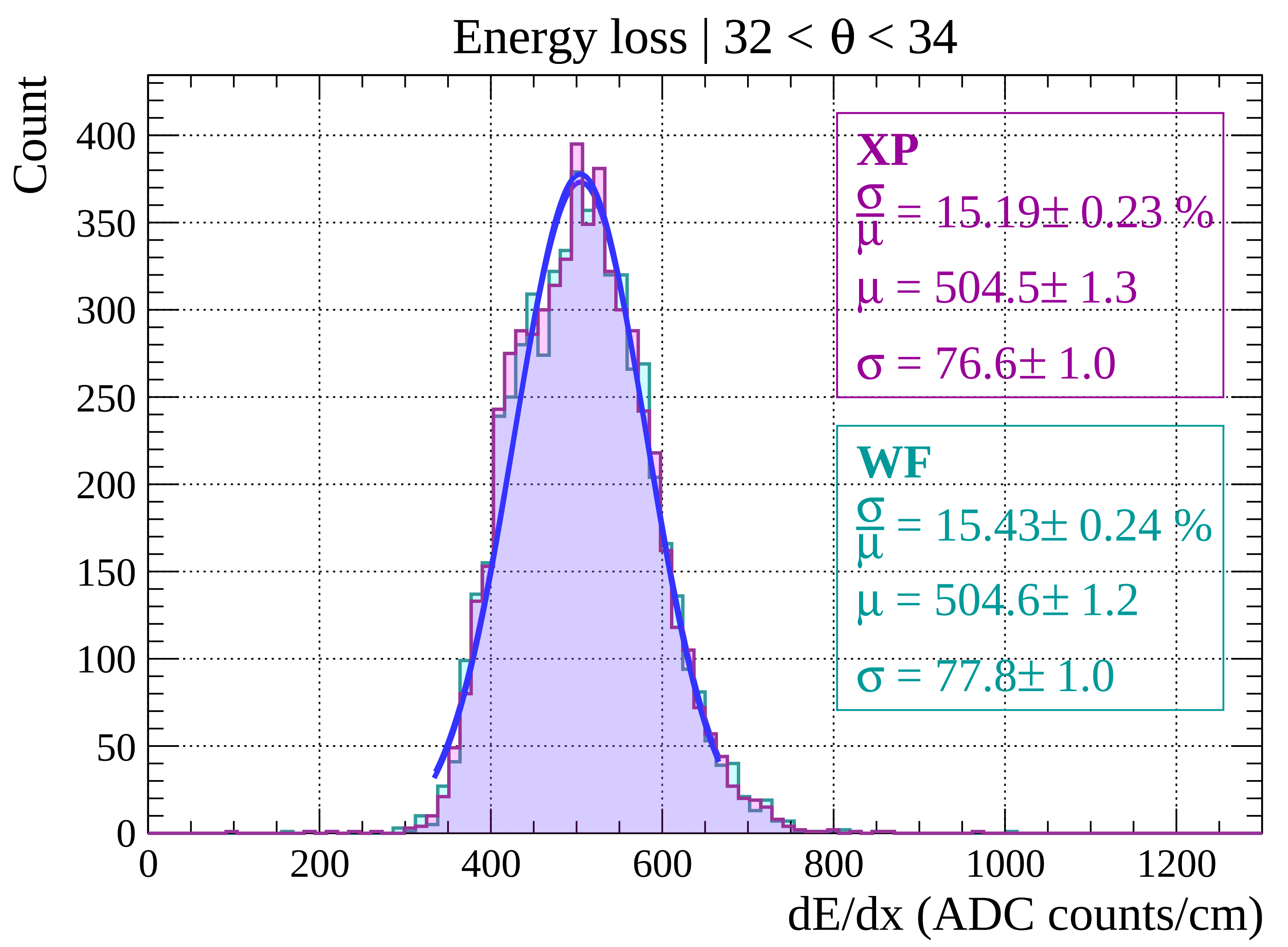

Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 14.76± 0.15 %
μ = 498.1± 0.8
σ = 73.5± 0.6
WF
σ/μ = 15.09± 0.16 %
μ = 498.5± 0.8
σ = 75.2± 0.7



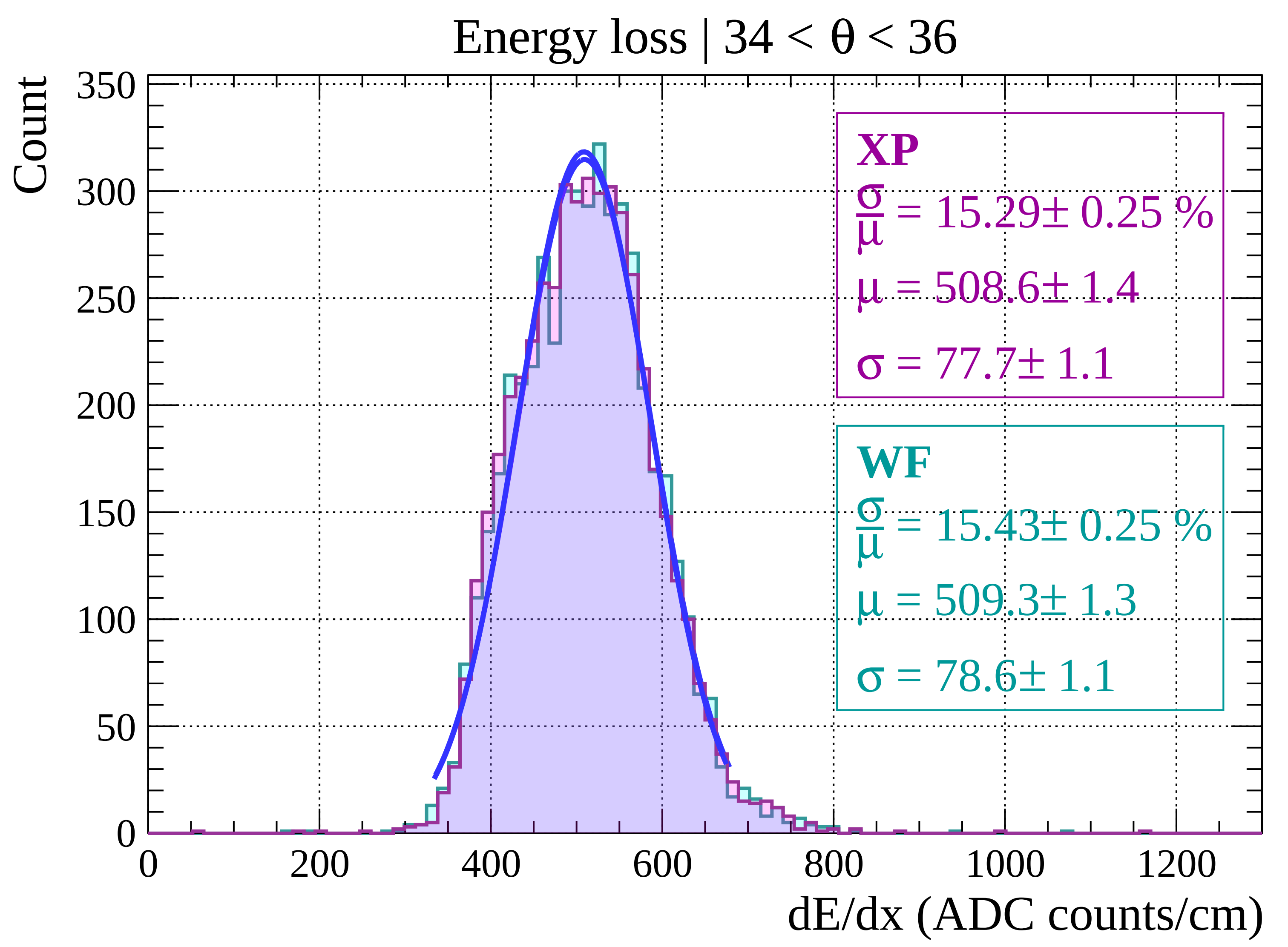
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.87± 0.17 %
μ = 500.5± 0.9
σ = 74.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 14.78± 0.16 %
μ = 500.7± 0.9
σ = 74.0± 0.7



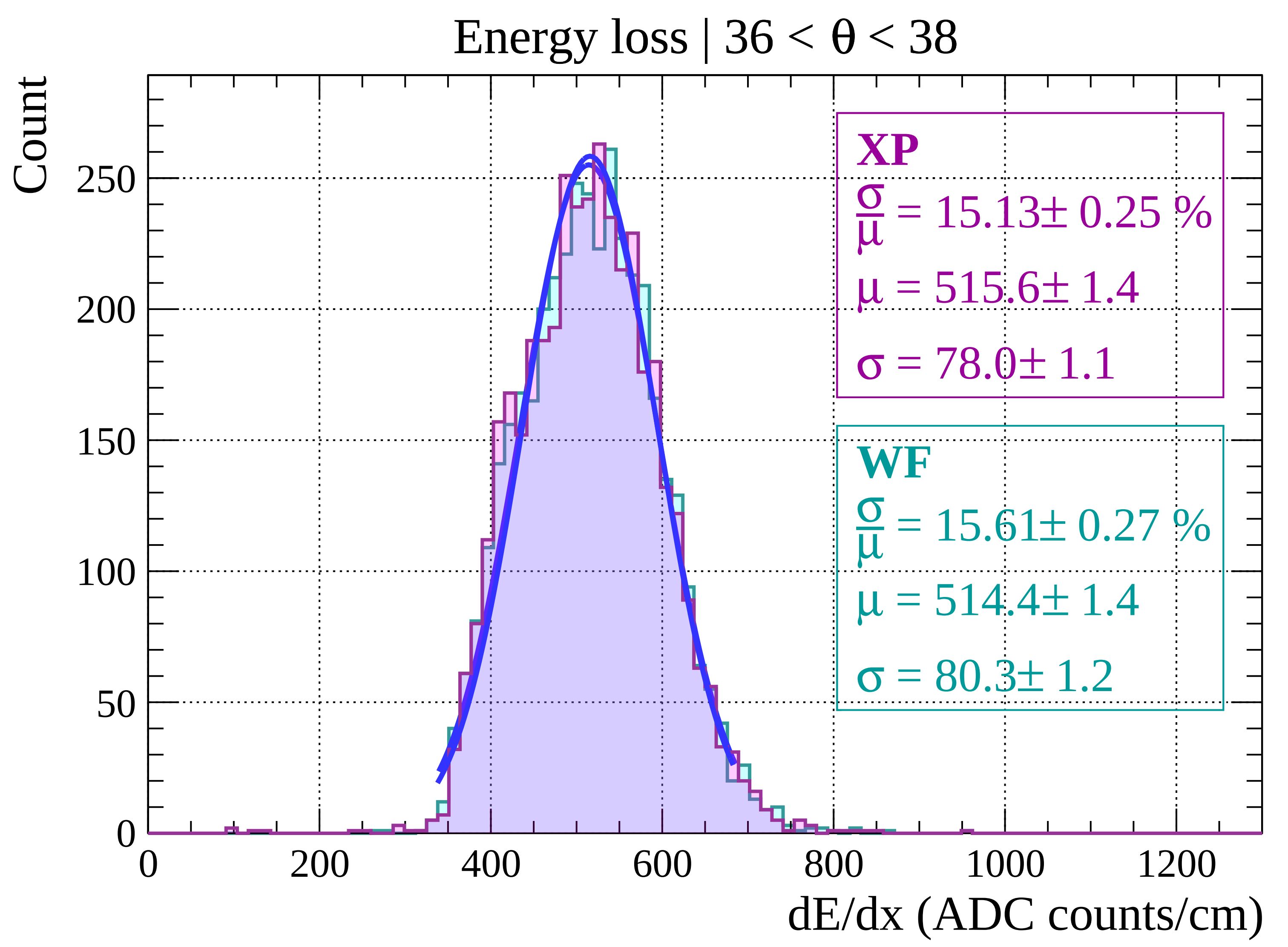
Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP
σ/μ = 15.35± 0.22 %
μ = 502.8± 1.1
σ = 77.2± 0.9
WF
σ/μ = 15.51± 0.21 %
μ = 503.2± 1.1
σ = 78.1± 0.9



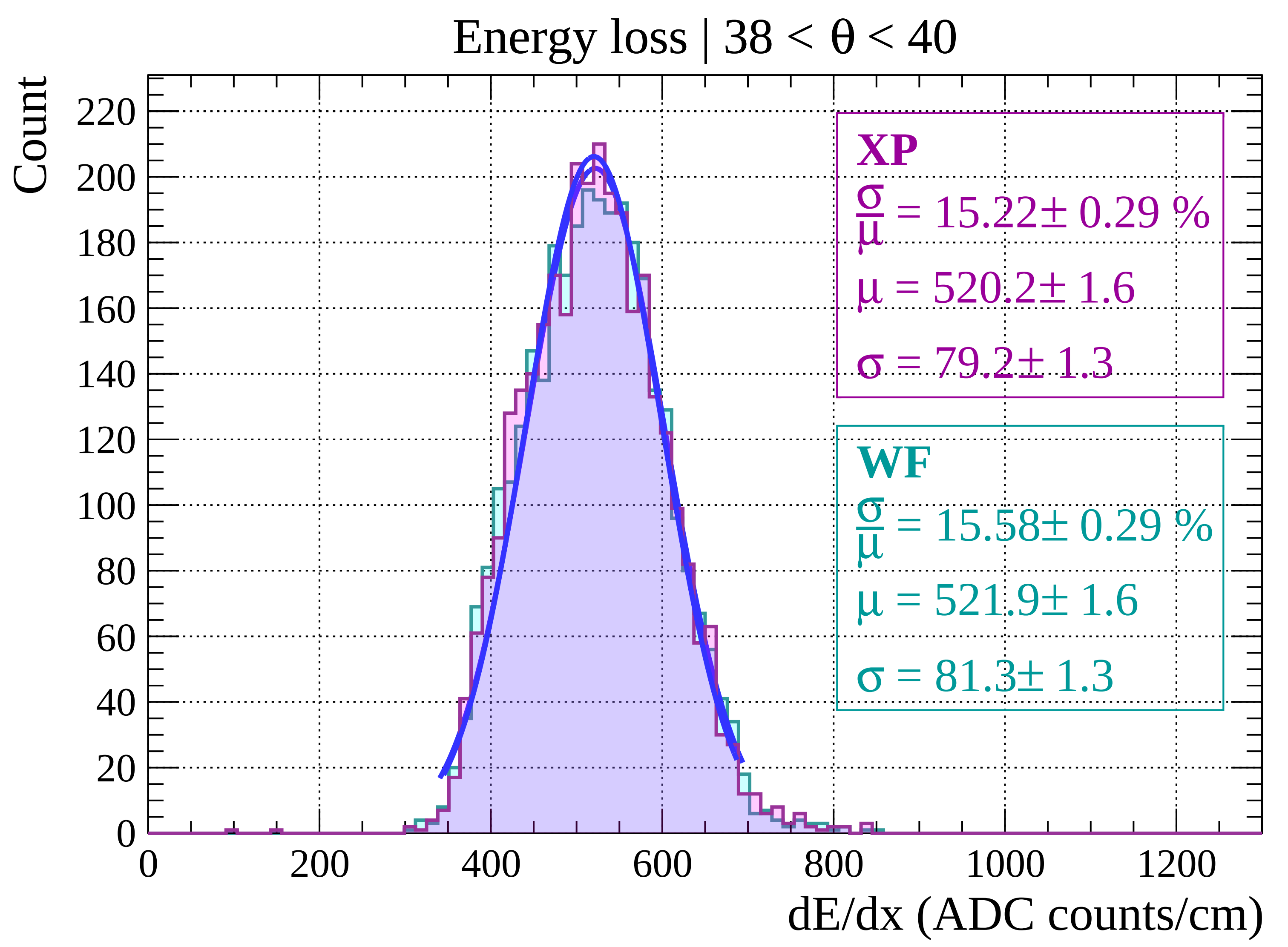
Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.19± 0.23 %
μ = 504.5± 1.3
σ = 76.6± 1.0
WF
$\frac{\sigma}{\mu}$ = 15.43± 0.24 %
μ = 504.6± 1.2
σ = 77.8± 1.0



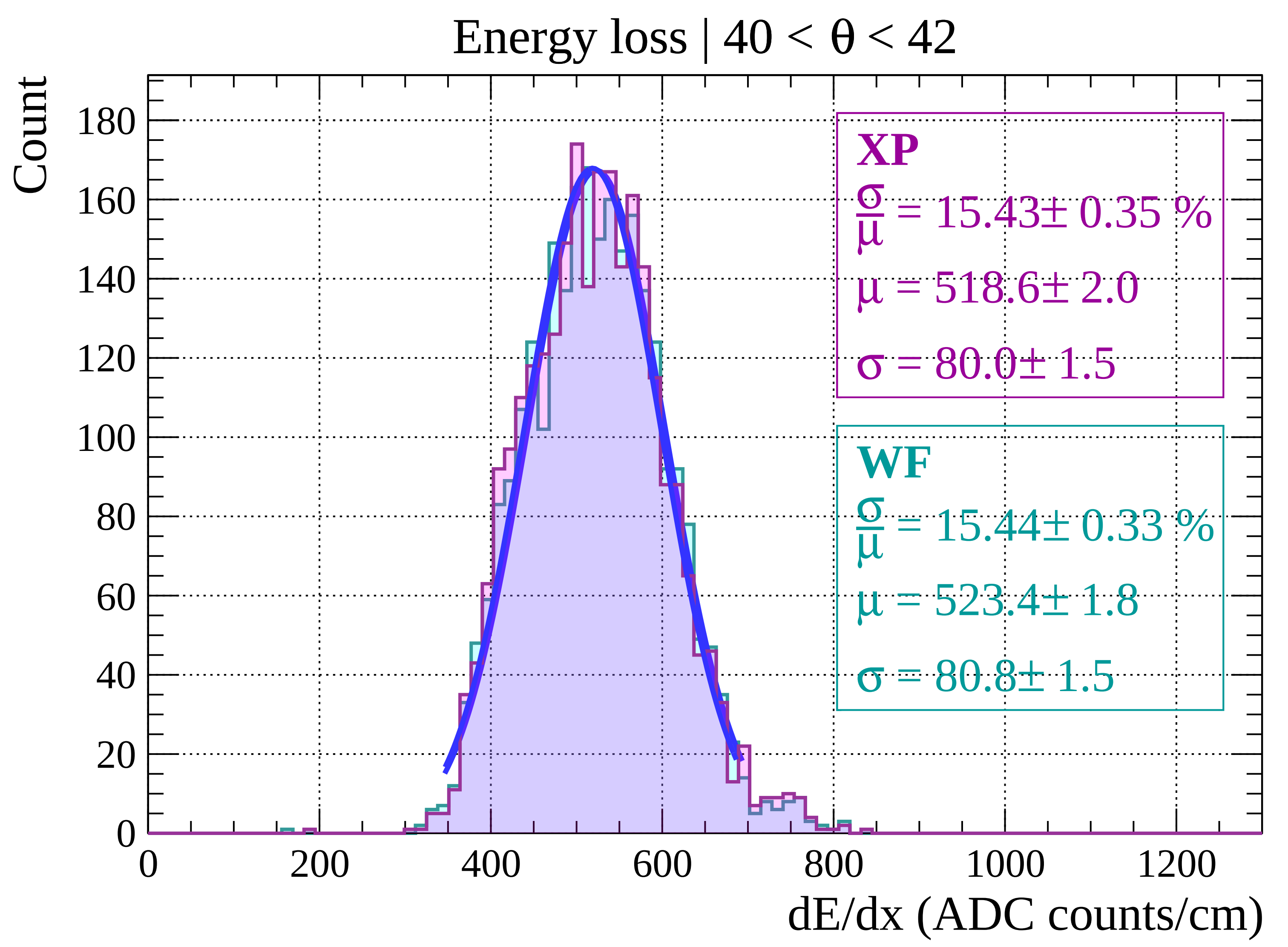
Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.29± 0.25 %
μ = 508.6± 1.4
σ = 77.7± 1.1
WF
σ/μ = 15.43± 0.25 %
μ = 509.3± 1.3
σ = 78.6± 1.1

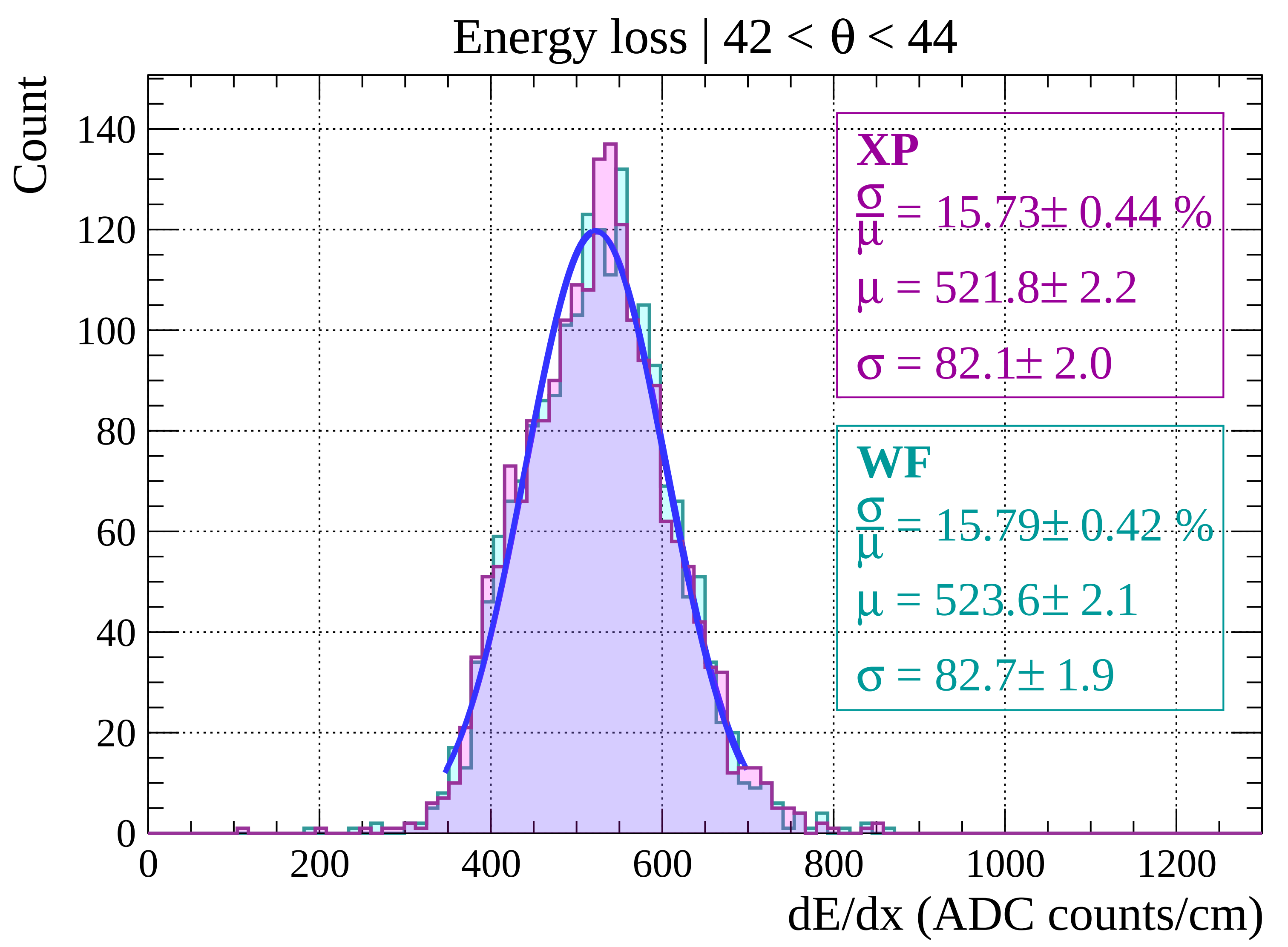


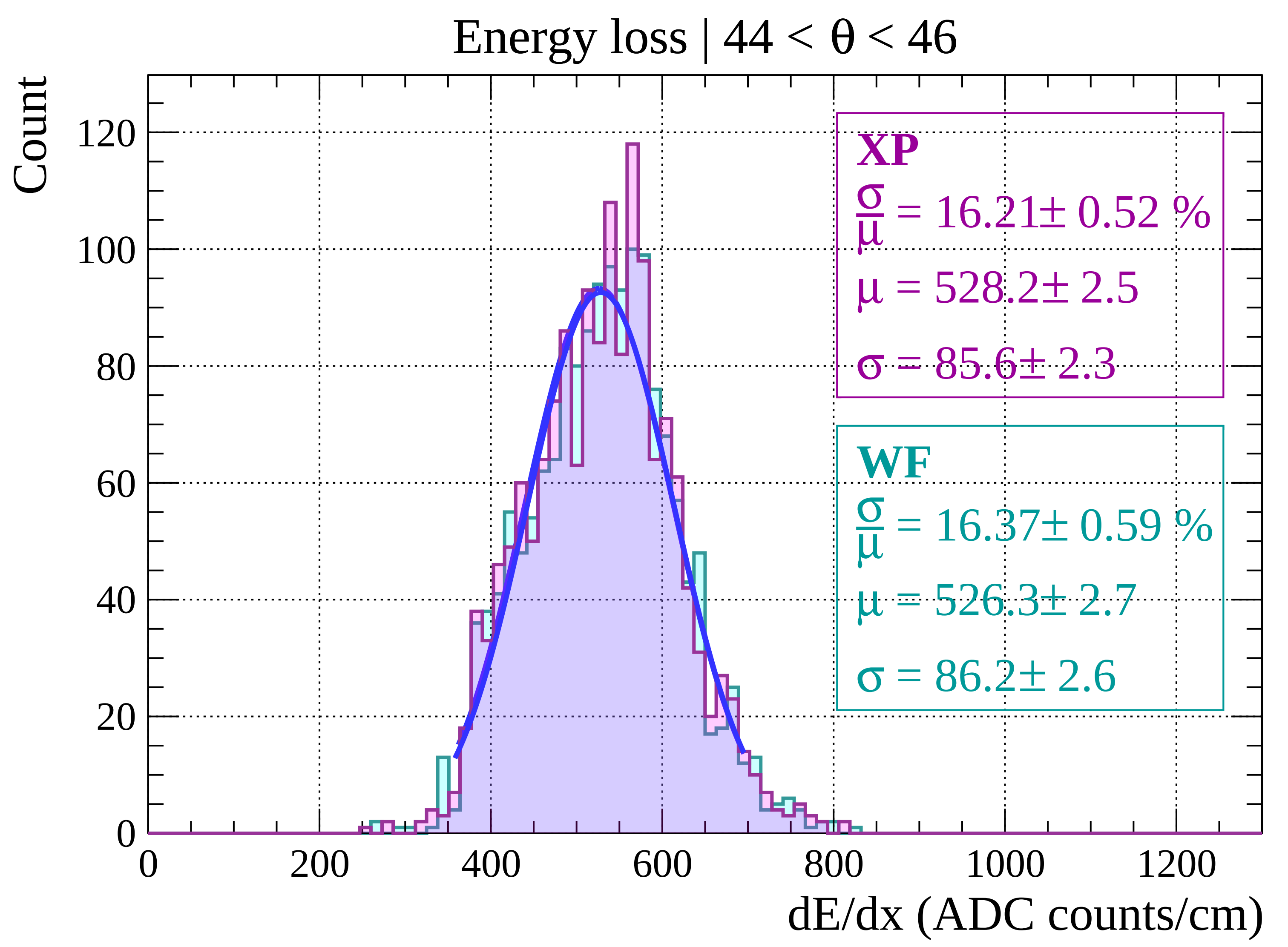
Energy loss | 36 < θ < 38
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.13± 0.25 %
μ = 515.6± 1.4
σ = 78.0± 1.1
WF
σ/μ = 15.61± 0.27 %
μ = 514.4± 1.4
σ = 80.3± 1.2



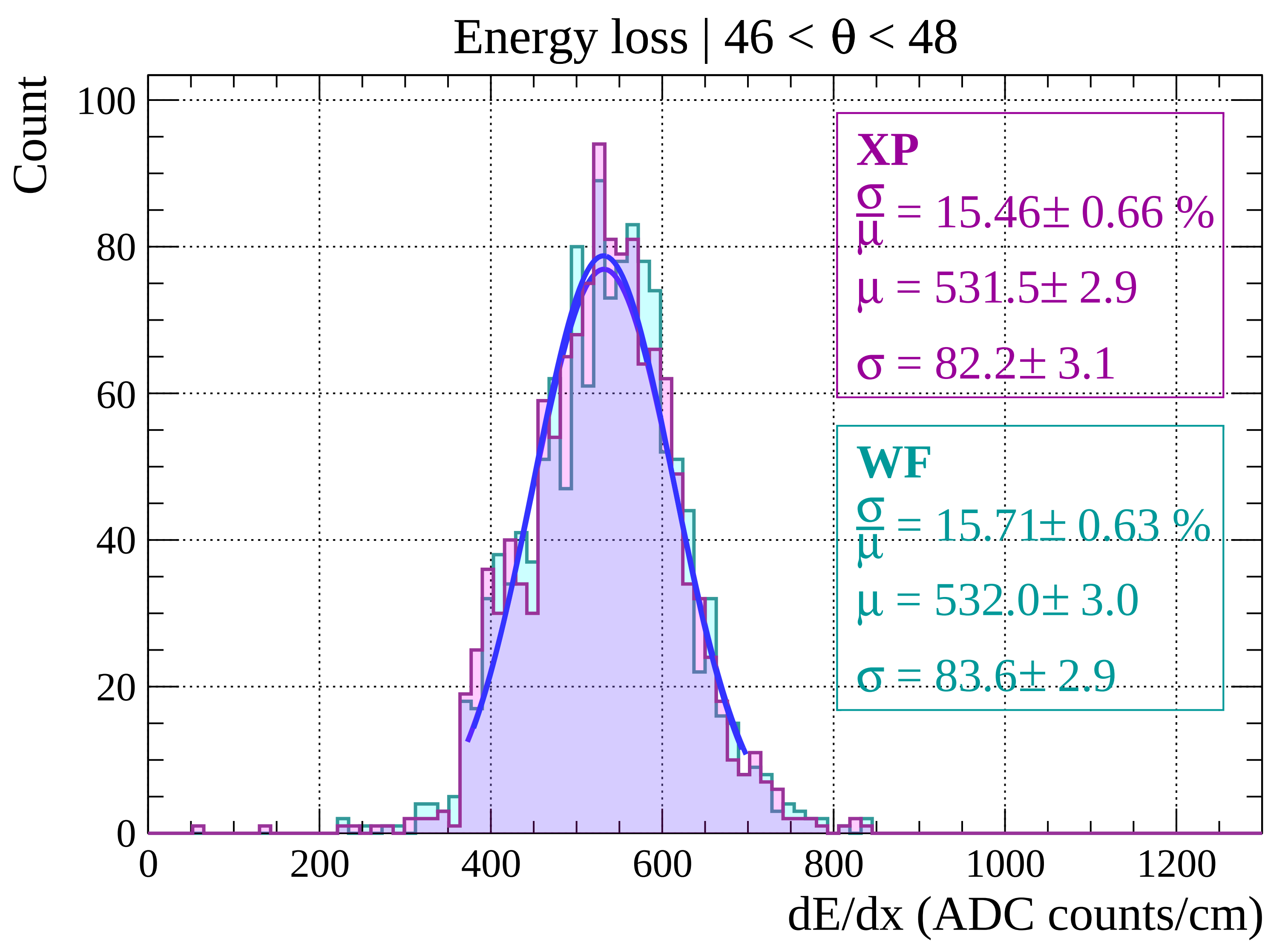
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.22± 0.29 %
μ = 520.2± 1.6
σ = 79.2± 1.3
WF
σ/μ = 15.58± 0.29 %
μ = 521.9± 1.6
σ = 81.3± 1.3


Energy loss | 40 < θ < 42
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.43± 0.35 %
μ = 518.6± 2.0
σ = 80.0± 1.5
WF
σ/μ = 15.44± 0.33 %
μ = 523.4± 1.8
σ = 80.8± 1.5

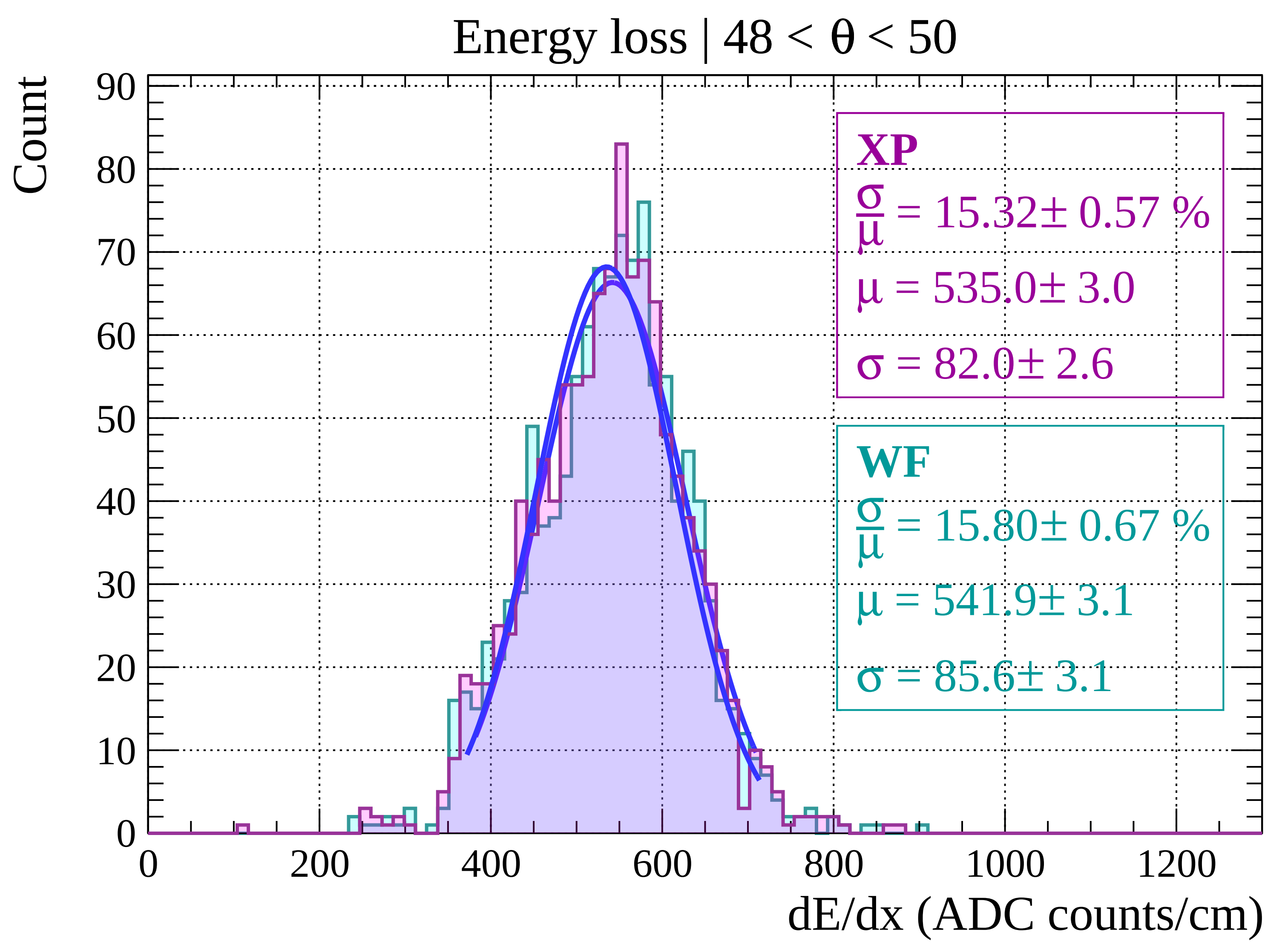

Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 15.73± 0.44 %
μ = 521.8± 2.2
σ = 82.1± 2.0
WF
σ/μ = 15.79± 0.42 %
μ = 523.6± 2.1
σ = 82.7± 1.9



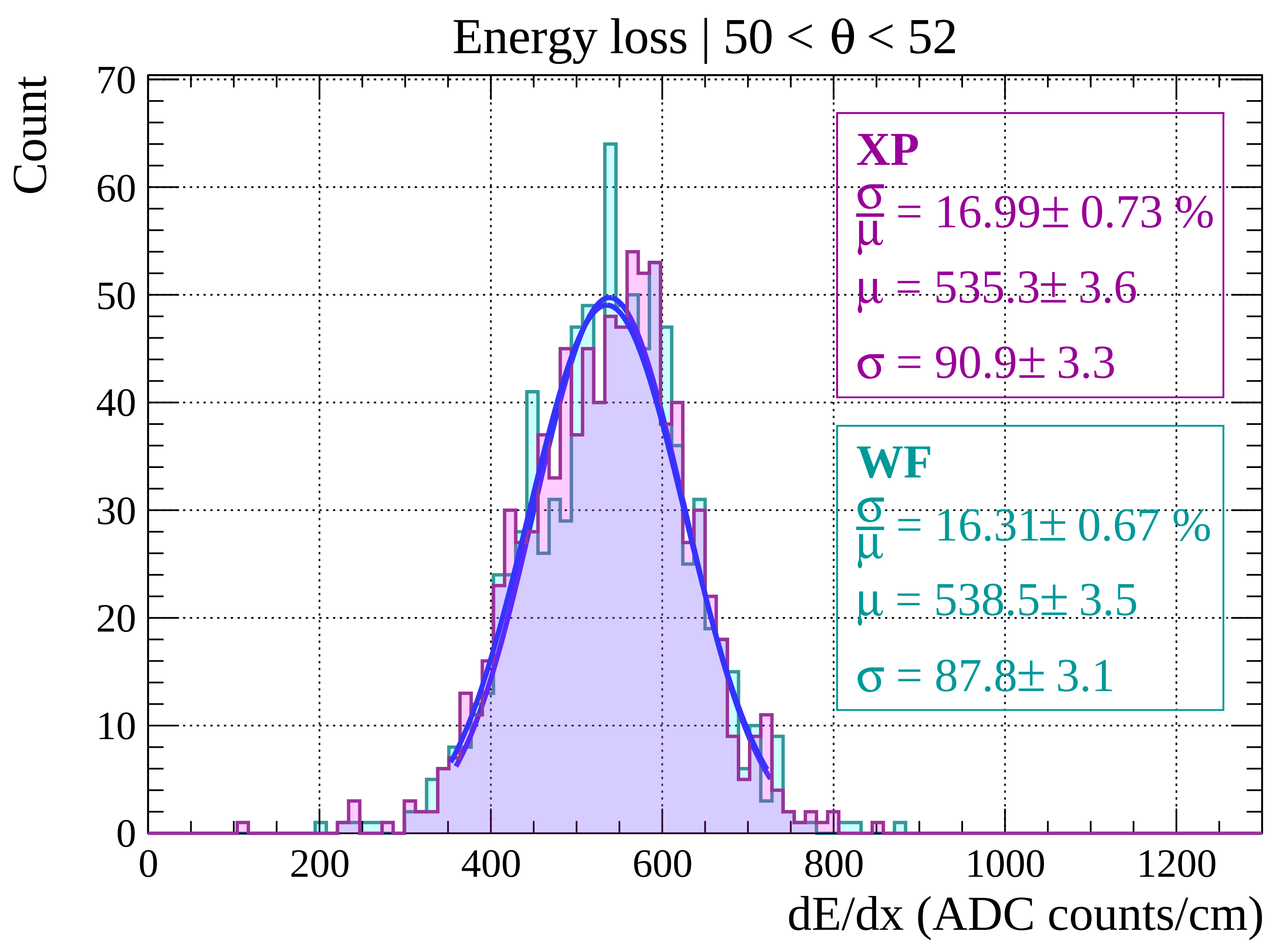
Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 16.21± 0.52 %
μ = 528.2± 2.5
σ = 85.6± 2.3
WF
σ/μ = 16.37± 0.59 %
μ = 526.3± 2.7
σ = 86.2± 2.6

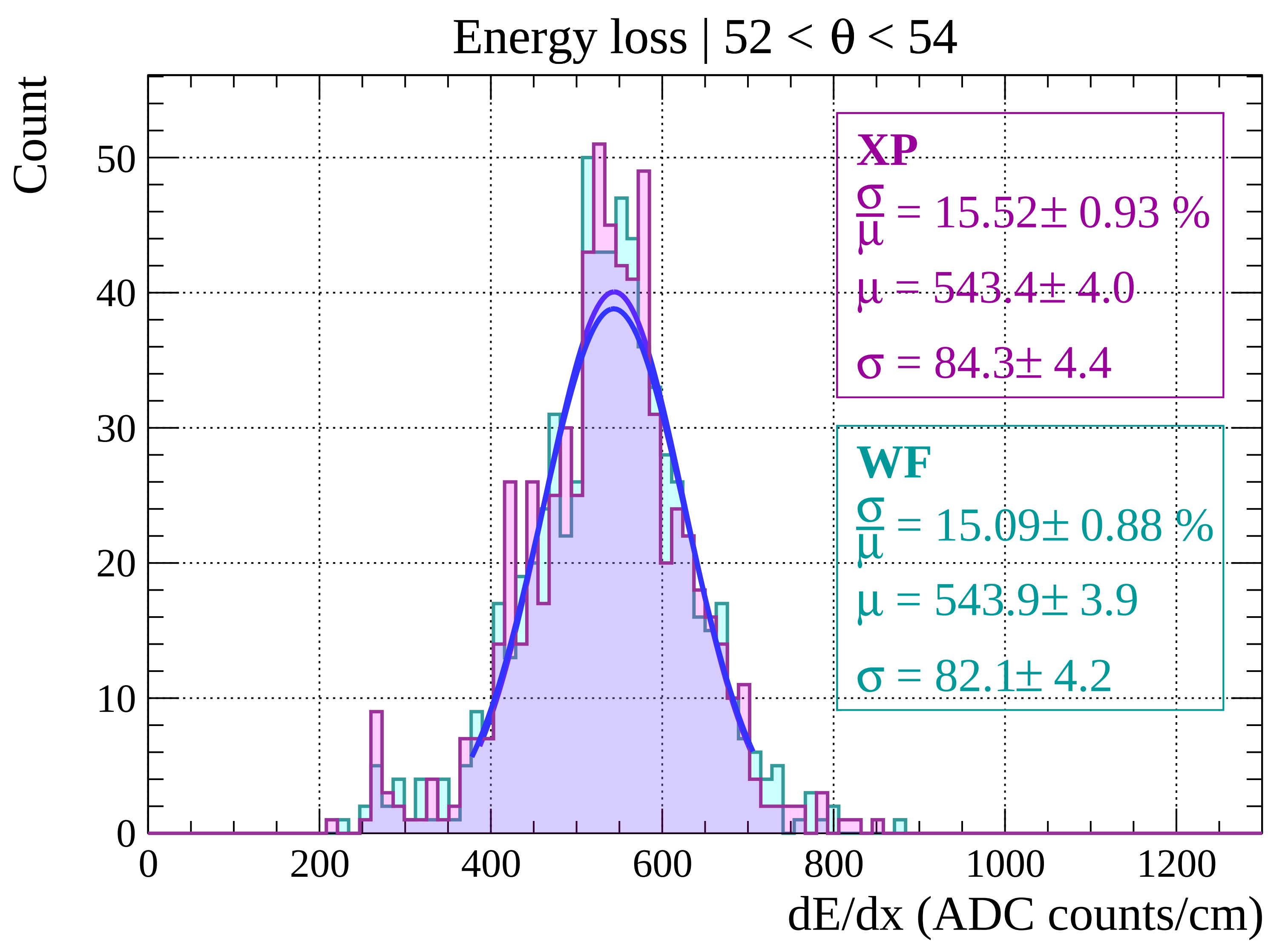


Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 15.46± 0.66 %
μ = 531.5± 2.9
σ = 82.2± 3.1
WF
σ/μ = 15.71± 0.63 %
μ = 532.0± 3.0
σ = 83.6± 2.9

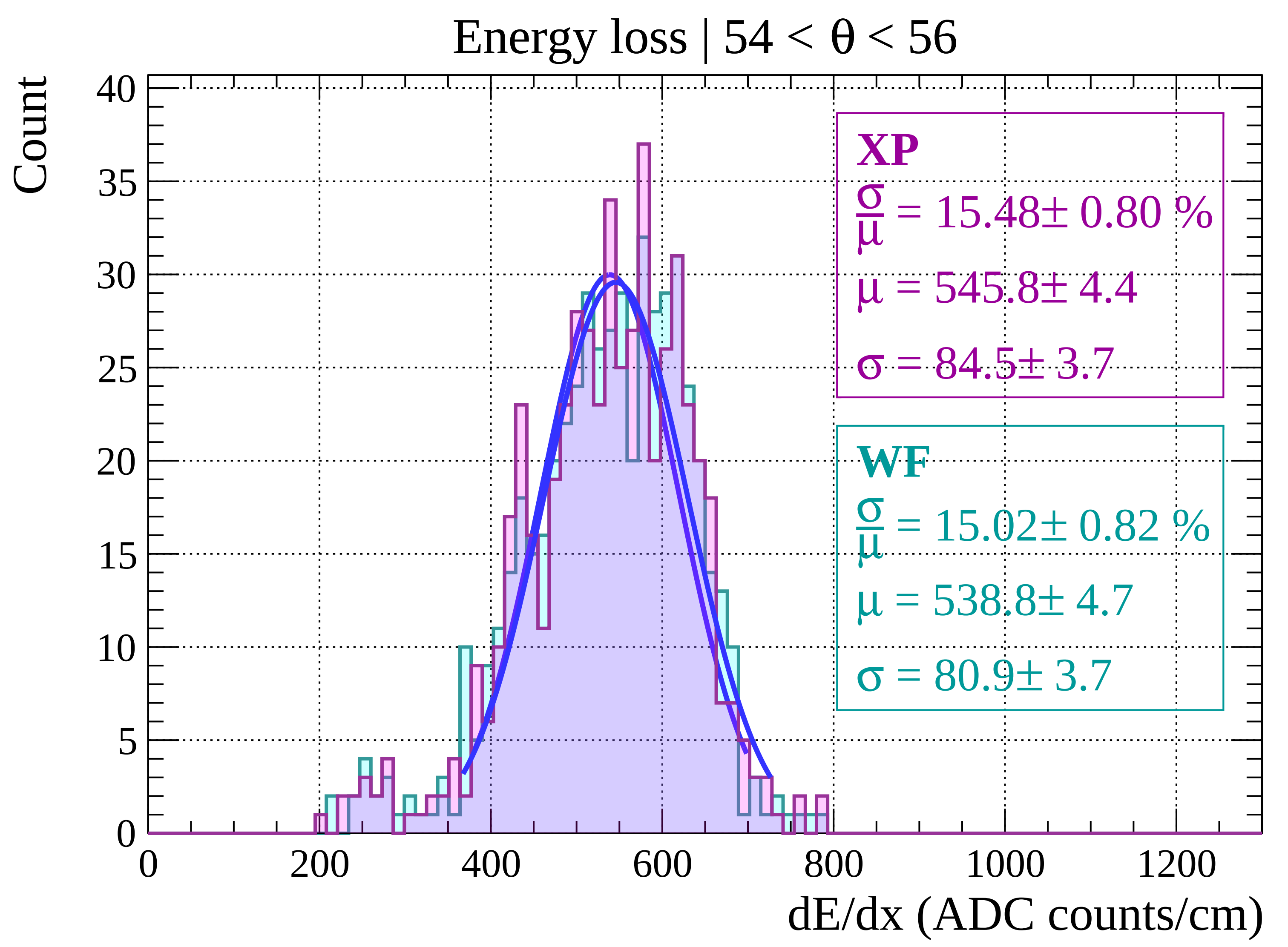


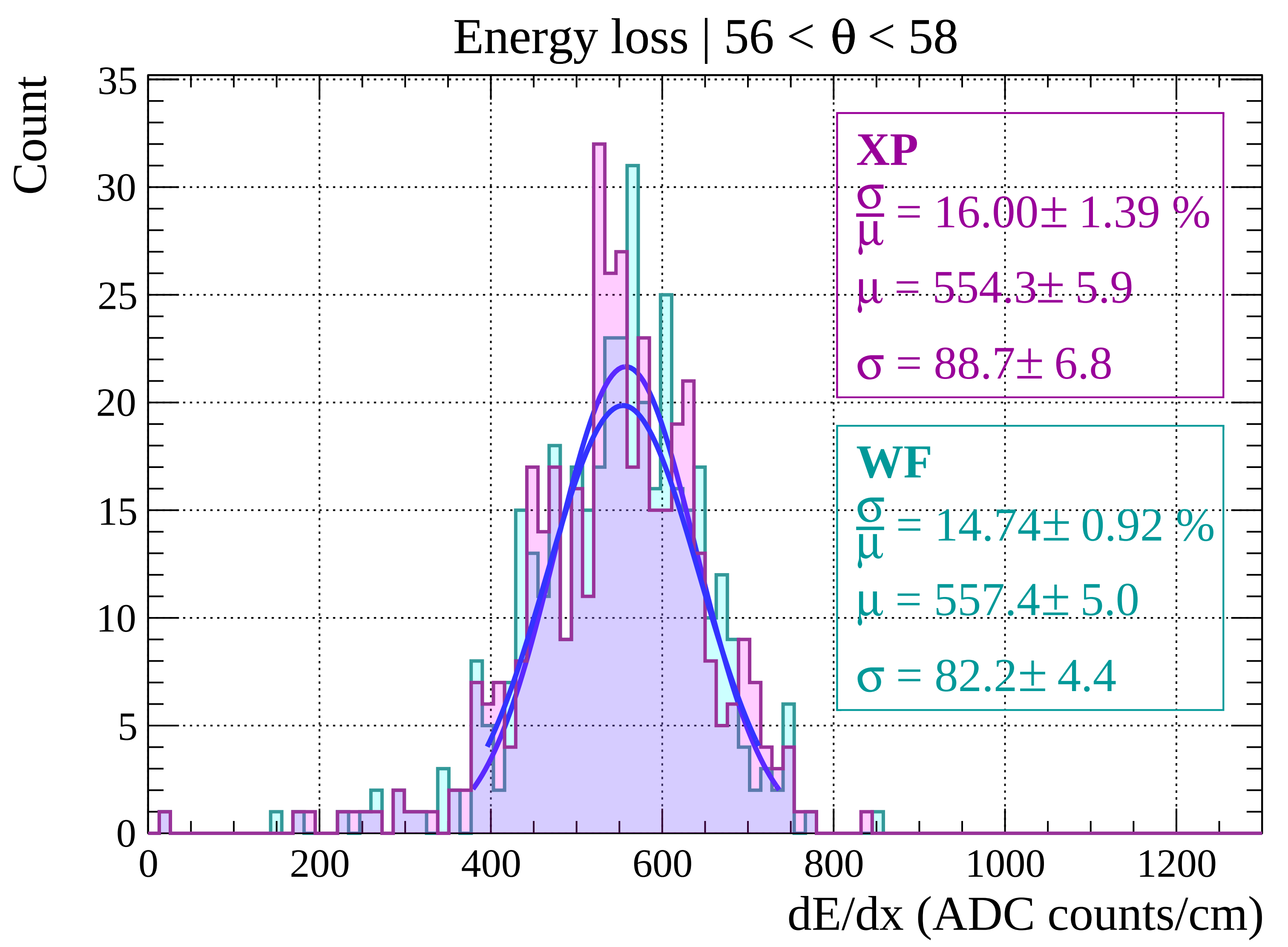
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 15.32± 0.57 %
μ = 535.0± 3.0
σ = 82.0± 2.6
WF
σ/μ = 15.80± 0.67 %
μ = 541.9± 3.1
σ = 85.6± 3.1


Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 16.99± 0.73 %
μ = 535.3± 3.6
σ = 90.9± 3.3
WF
σ/μ = 16.31± 0.67 %
μ = 538.5± 3.5
σ = 87.8± 3.1

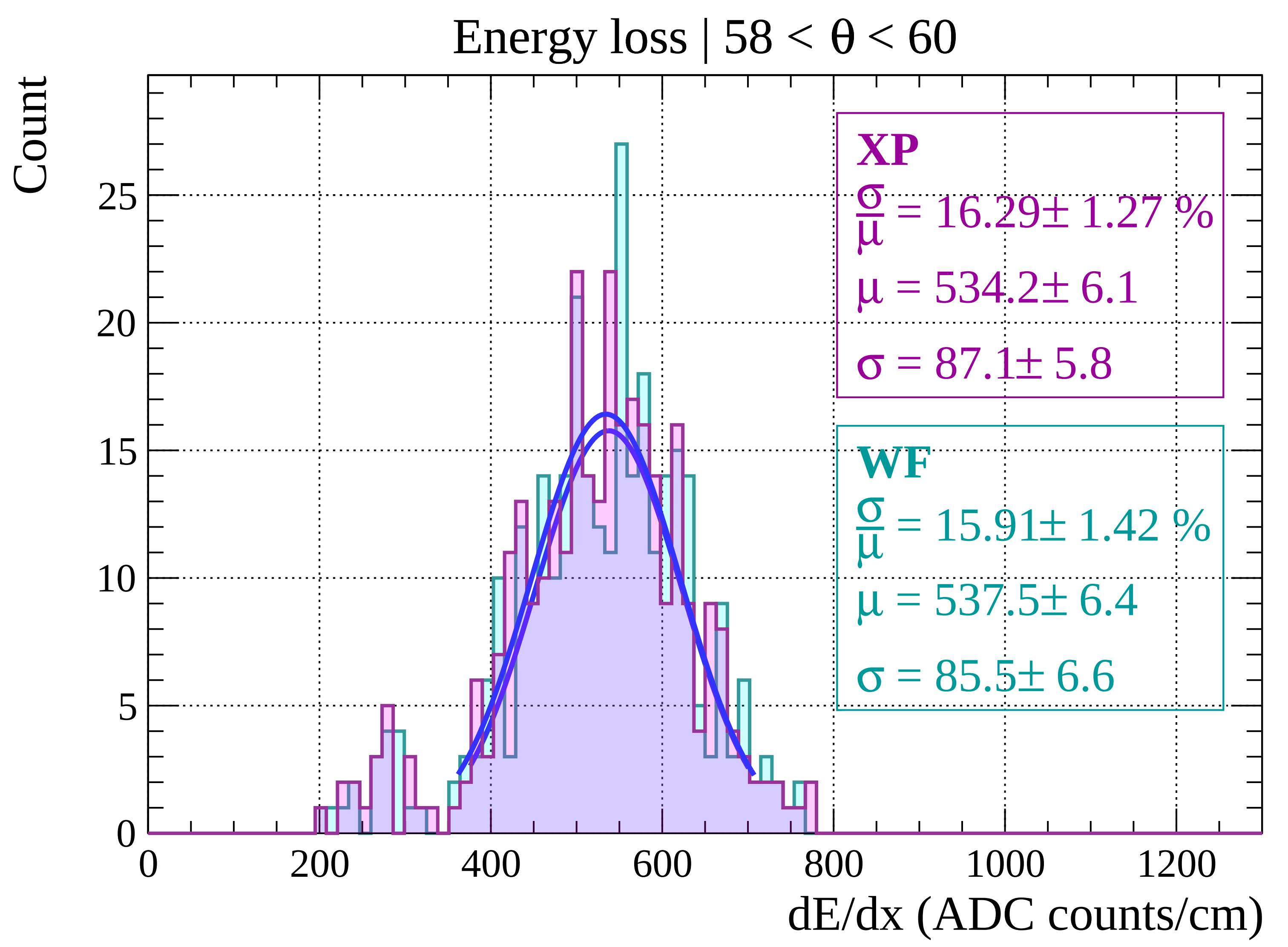


Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 15.52± 0.93 %
μ = 543.4± 4.0
σ = 84.3± 4.4
WF
σ/μ = 15.09± 0.88 %
μ = 543.9± 3.9
σ = 82.1± 4.2

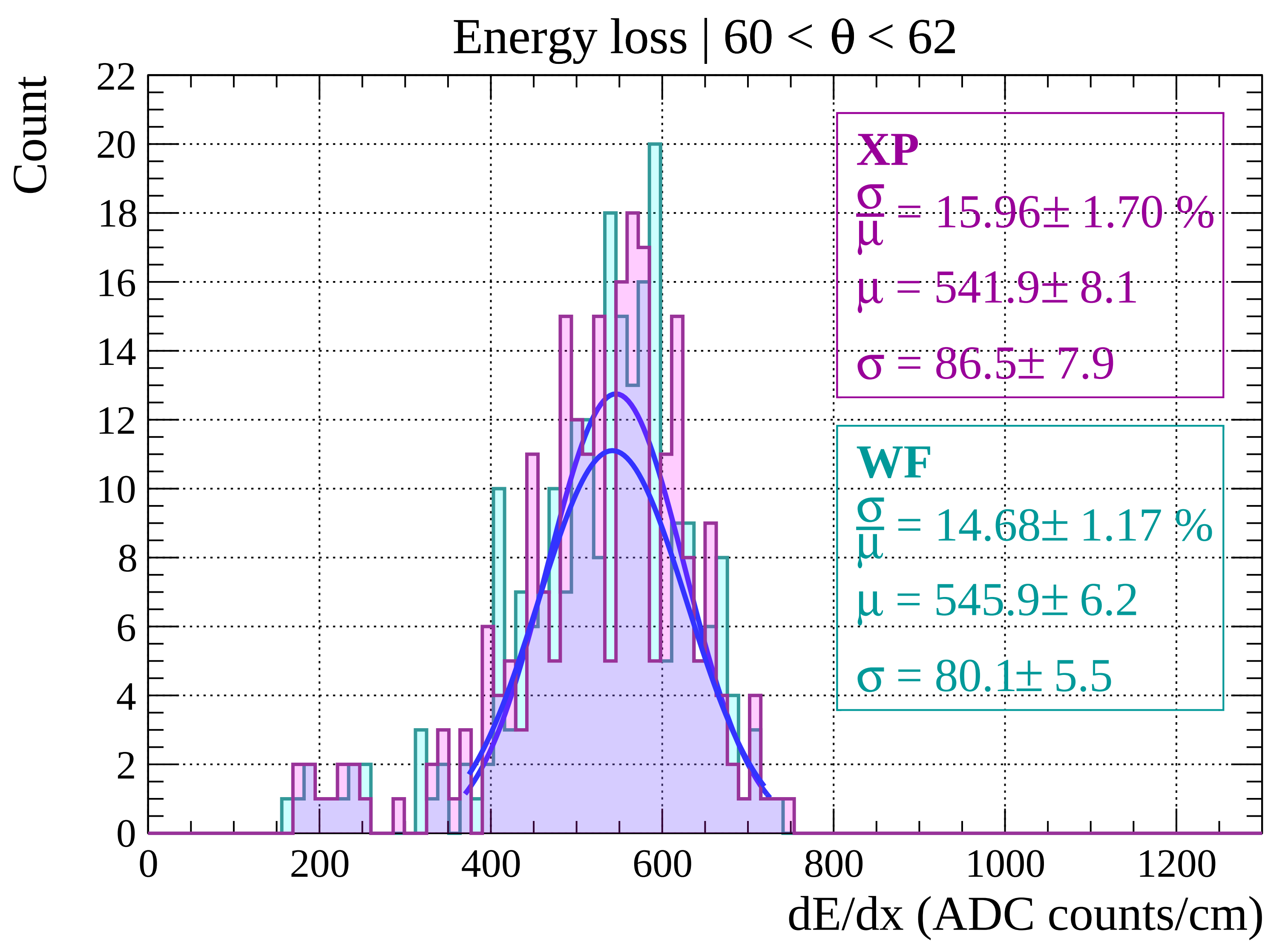

Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP
σ/μ = 15.48± 0.80 %
μ = 545.8± 4.4
σ = 84.5± 3.7
WF
σ/μ = 15.02± 0.82 %
μ = 538.8± 4.7
σ = 80.9± 3.7



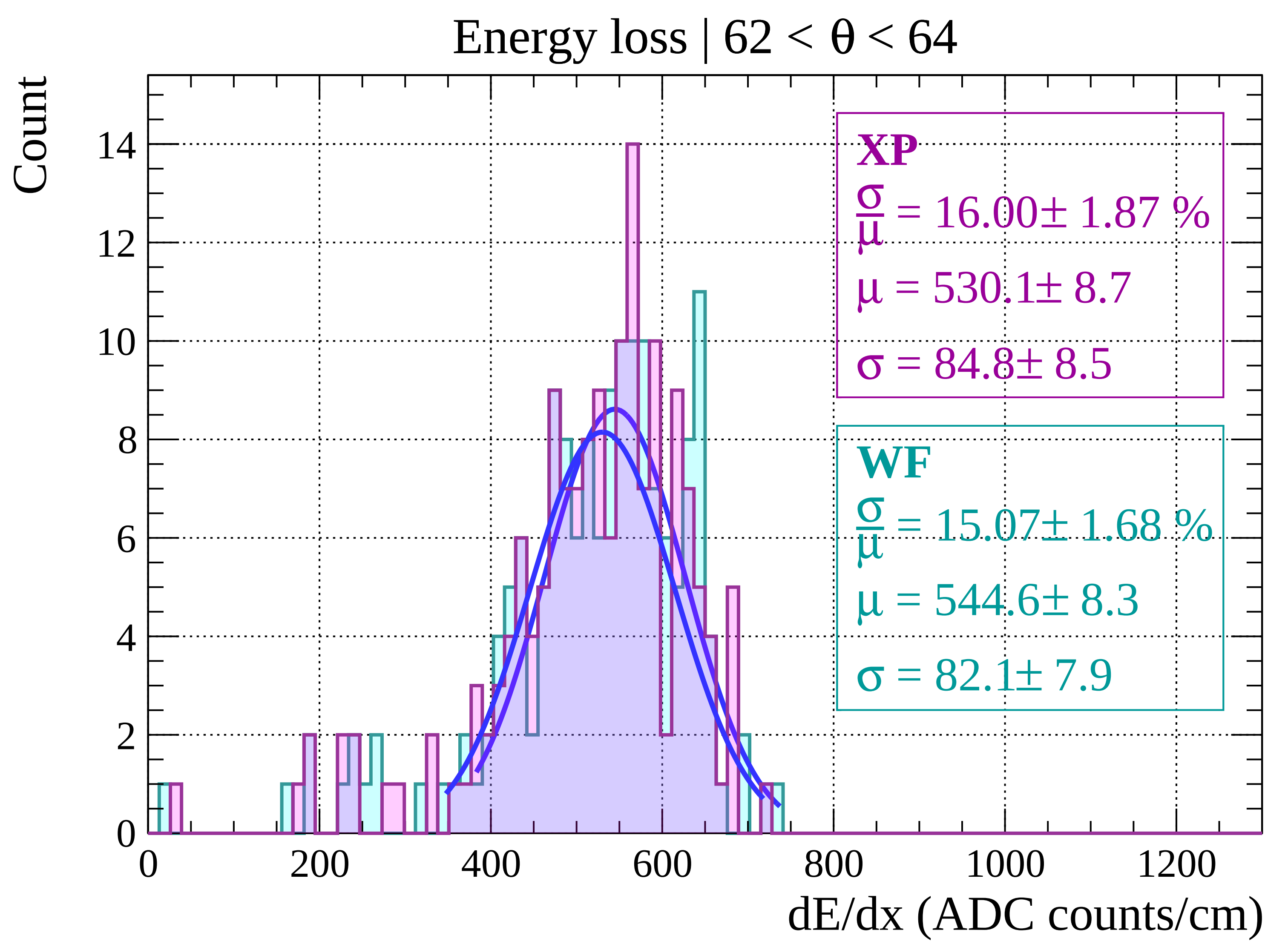
Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 16.00± 1.39 %
μ = 554.3± 5.9
σ = 88.7± 6.8
WF
σ/μ = 14.74± 0.92 %
μ = 557.4± 5.0
σ = 82.2± 4.4

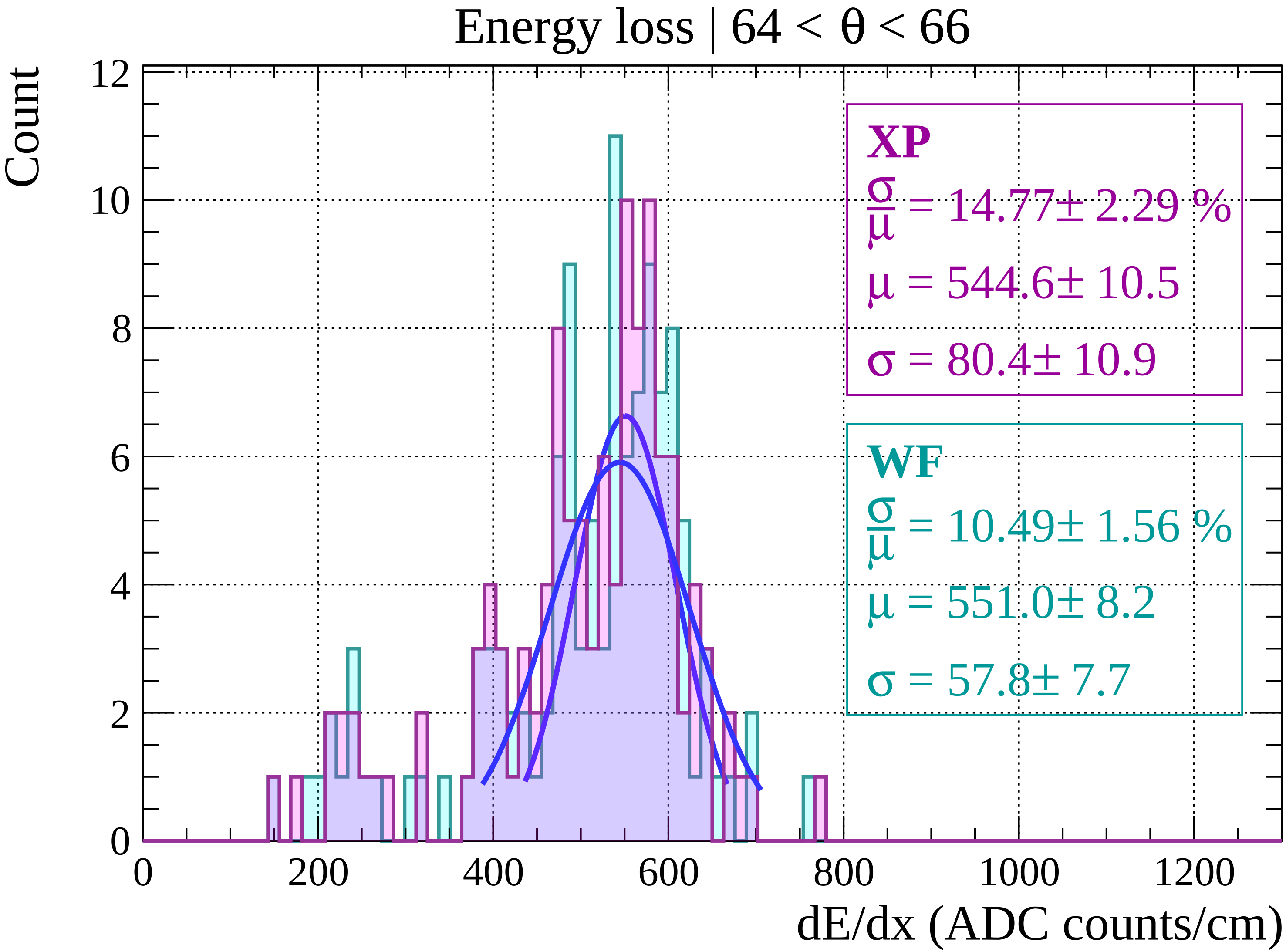


Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 16.29± 1.27 %
μ = 534.2± 6.1
σ = 87.1± 5.8
WF
σ/μ = 15.91± 1.42 %
μ = 537.5± 6.4
σ = 85.5± 6.6

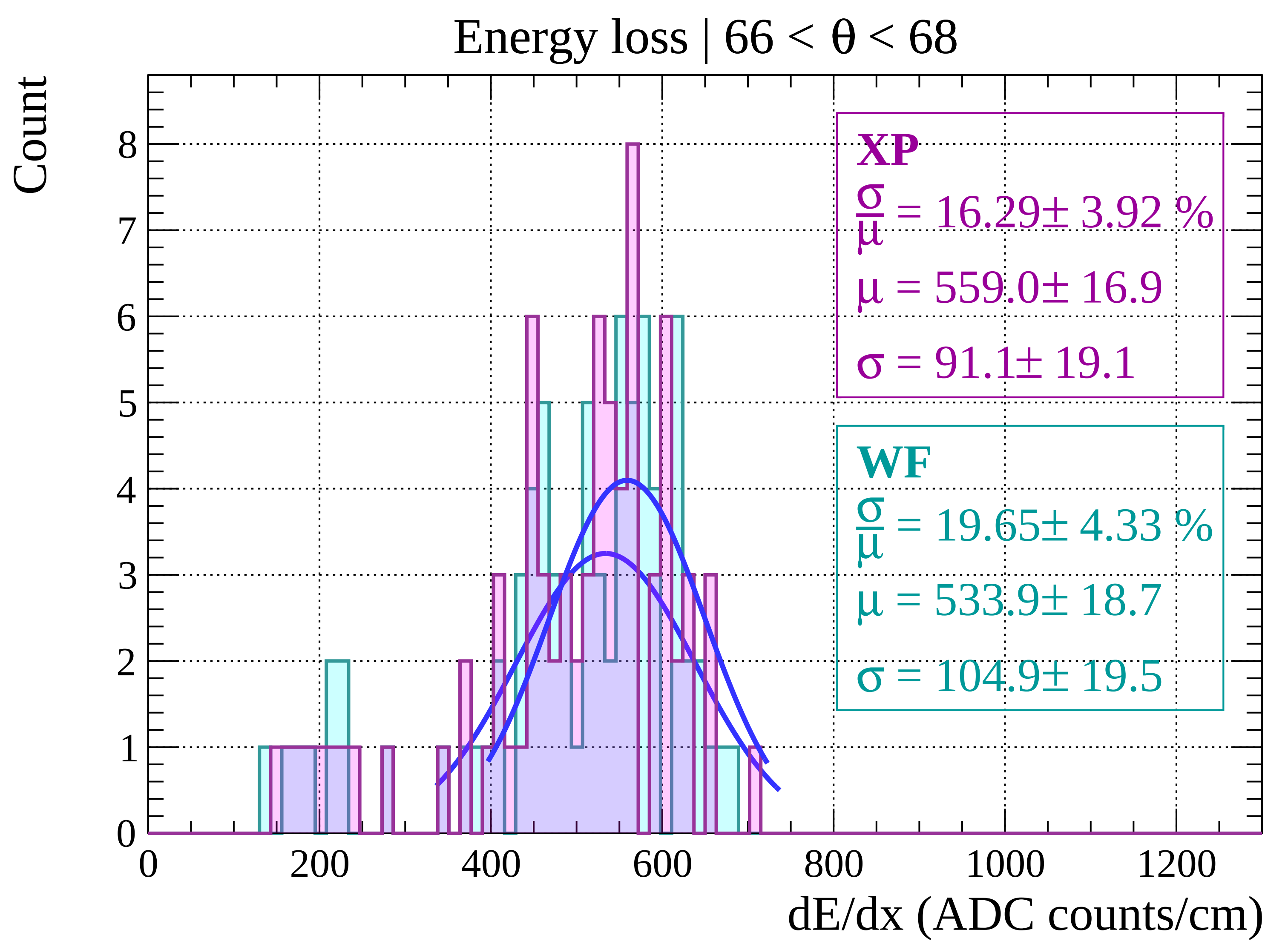


Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.96± 1.70 %
μ = 541.9± 8.1
σ = 86.5± 7.9
WF
σ/μ = 14.68± 1.17 %
μ = 545.9± 6.2
σ = 80.1± 5.5
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22

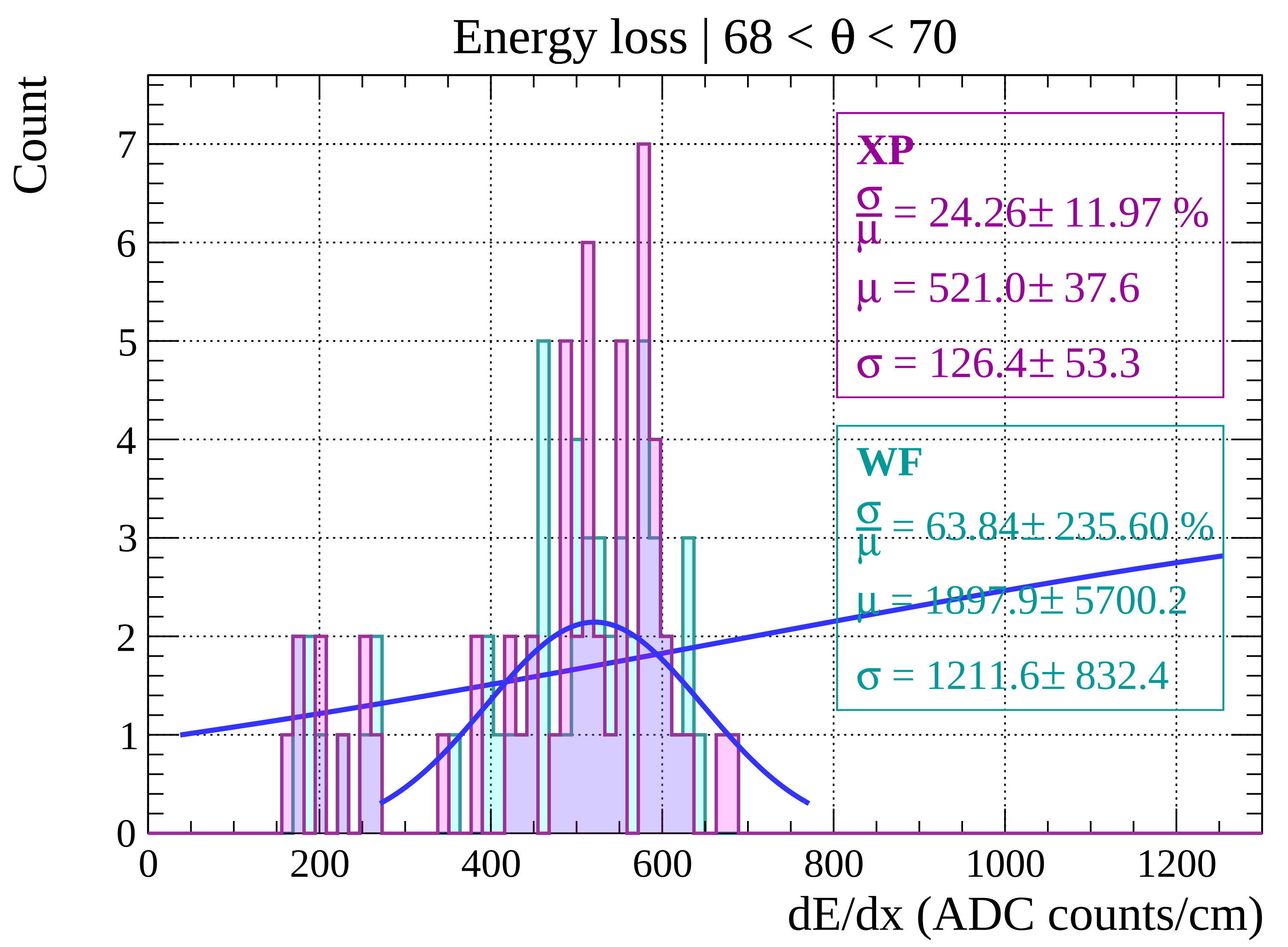

Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 16.00± 1.87 %
μ = 530.1± 8.7
σ = 84.8± 8.5
WF
σ/μ = 15.07± 1.68 %
μ = 544.6± 8.3
σ = 82.1± 7.9



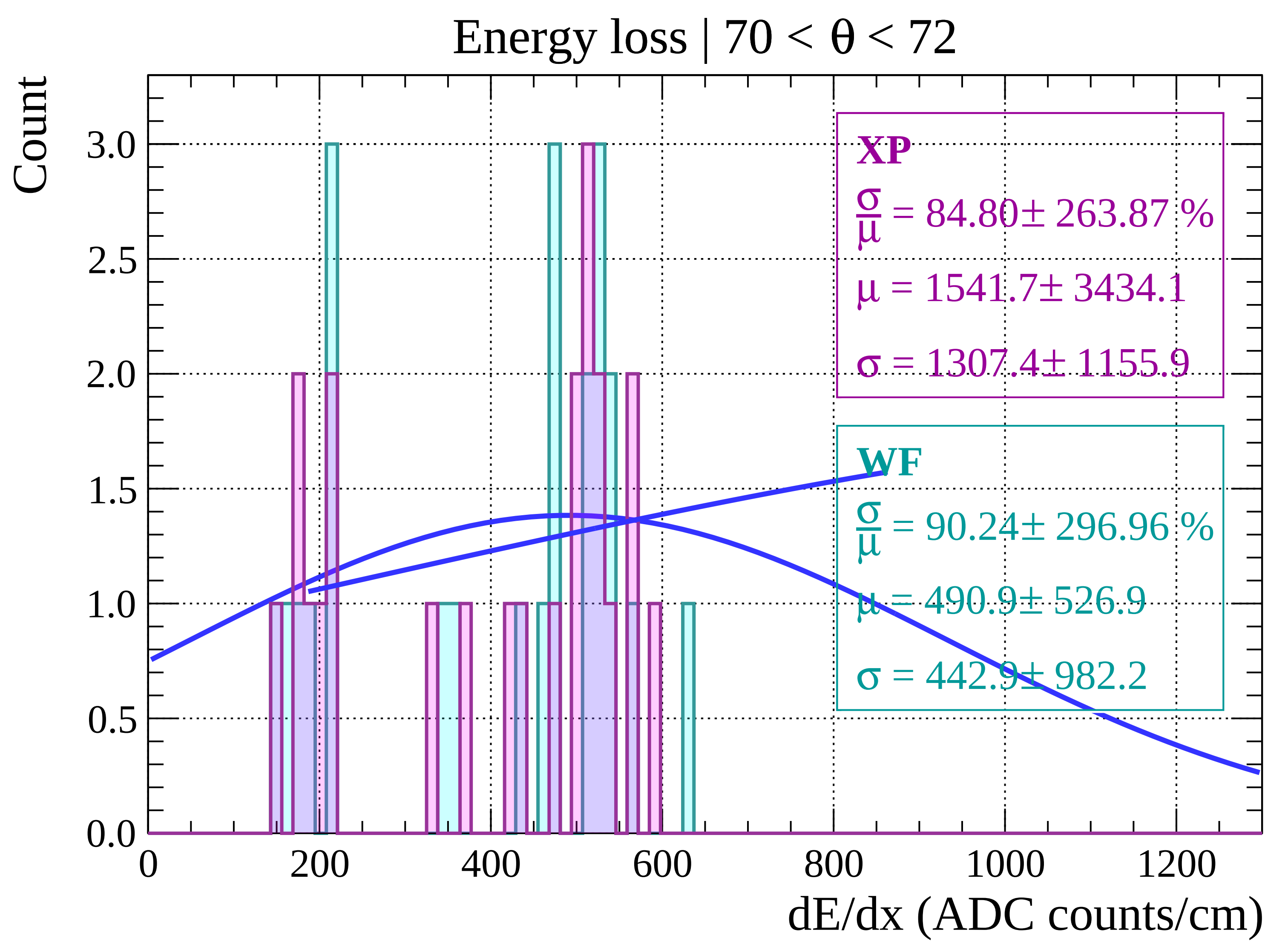
Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 14.77± 2.29 %
μ = 544.6± 10.5
σ = 80.4± 10.9
WF
σ/μ = 10.49± 1.56 %
μ = 551.0± 8.2
σ = 57.8± 7.7



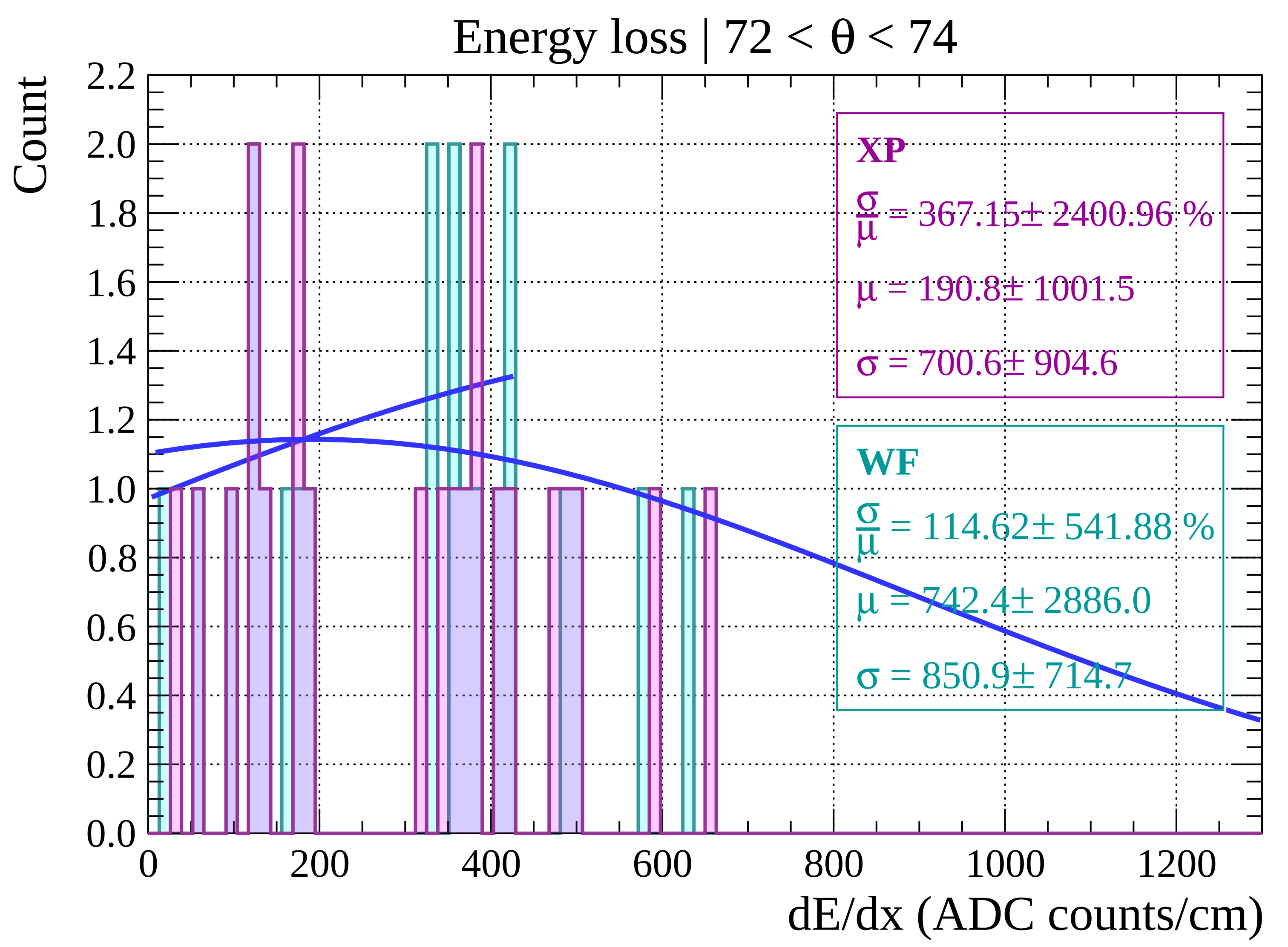
Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.29± 3.92 %
μ = 559.0± 16.9
σ = 91.1± 19.1
WF
σ/μ = 19.65± 4.33 %
μ = 533.9± 18.7
σ = 104.9± 19.5

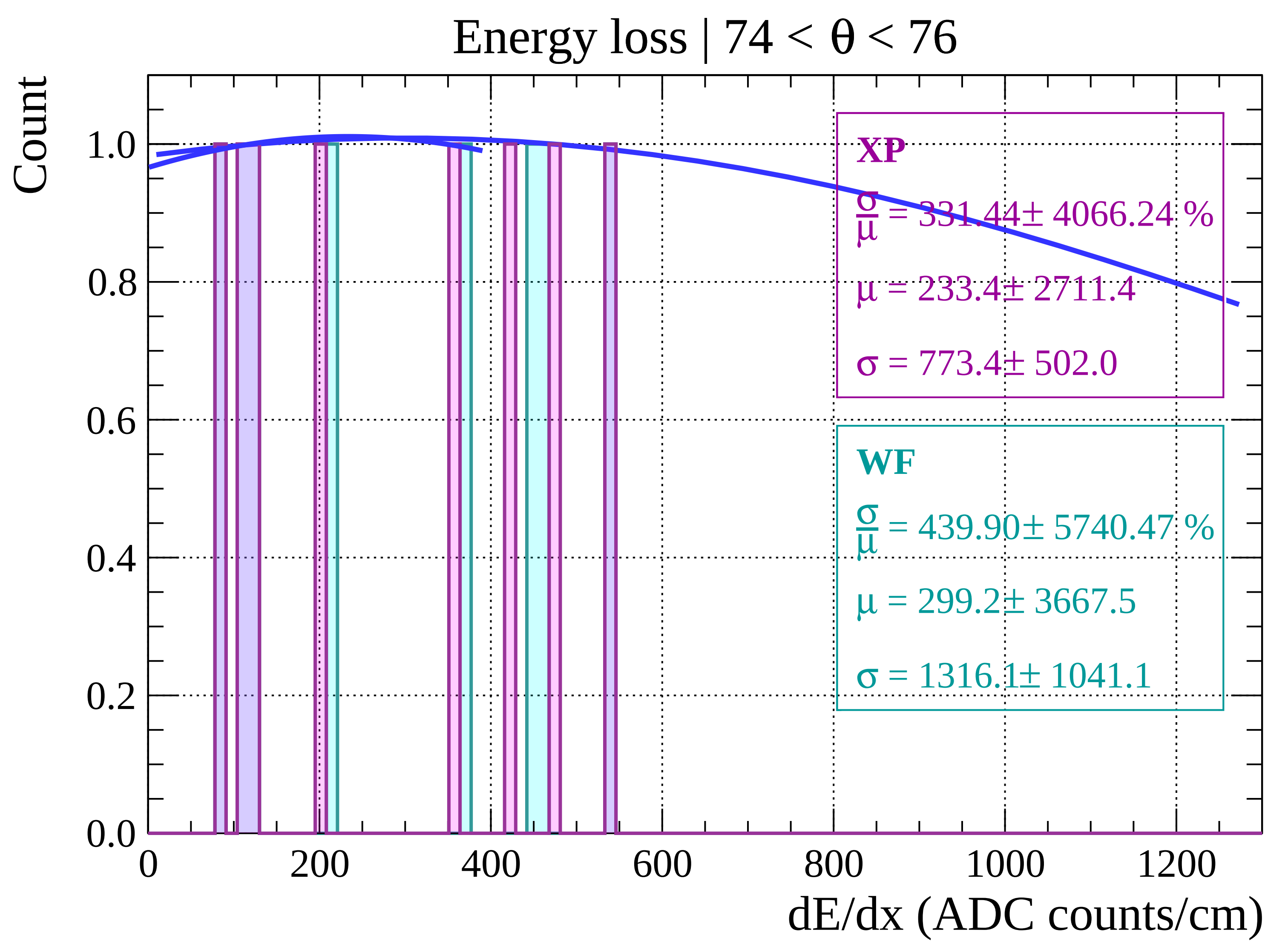


Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 24.26± 11.97 %
μ = 521.0± 37.6
σ = 126.4± 53.3
WF
σ/μ = 63.84± 235.60 %
μ = 1897.9± 5700.2
σ = 1211.6± 832.4
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200

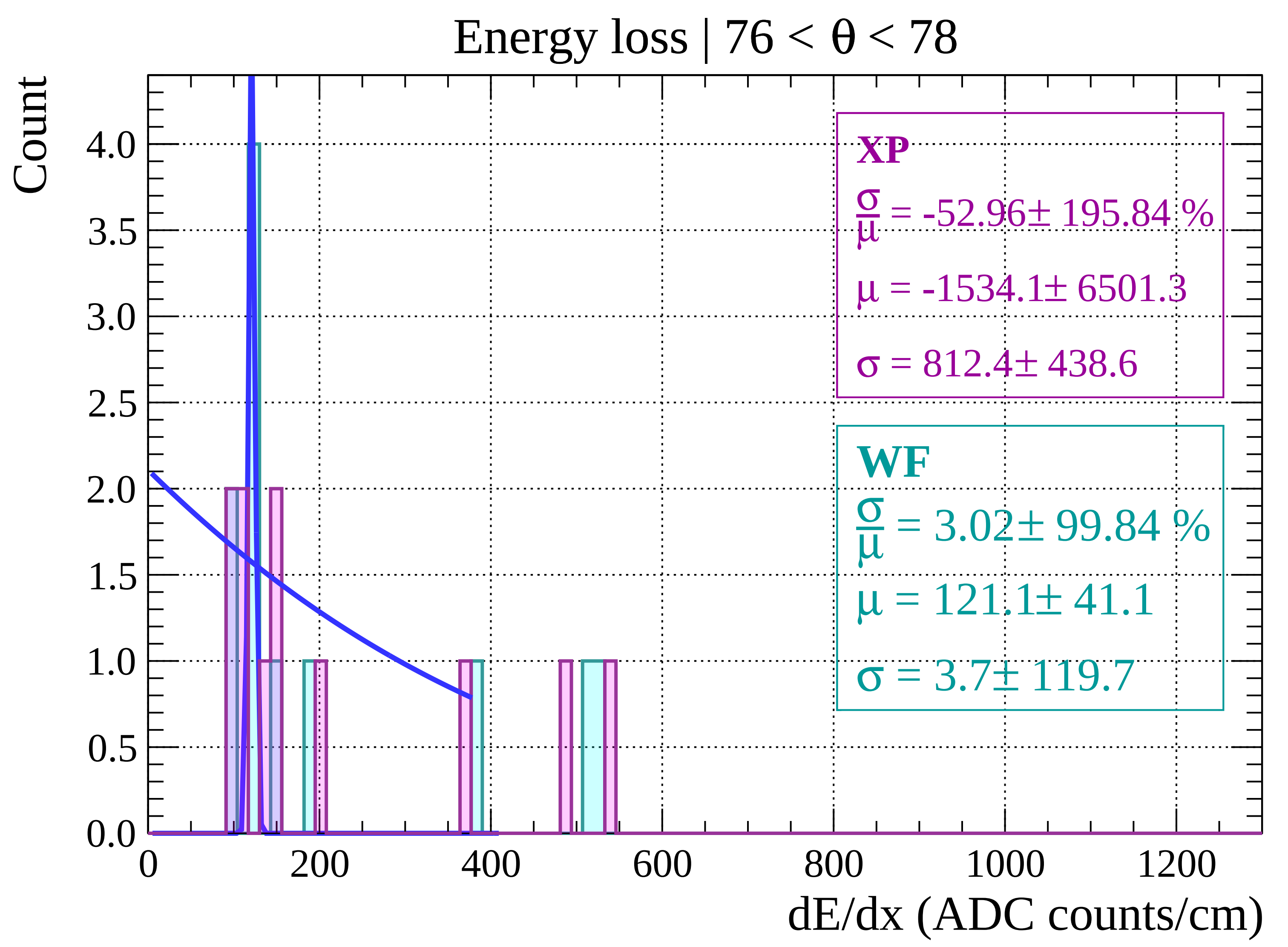


Energy loss | 70 < θ < 72
Count
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 84.80± 263.87 %
μ = 1541.7± 3434.1
σ = 1307.4± 1155.9
WF
σ/μ = 90.24± 296.96 %
μ = 490.9± 526.9
σ = 442.9± 982.2

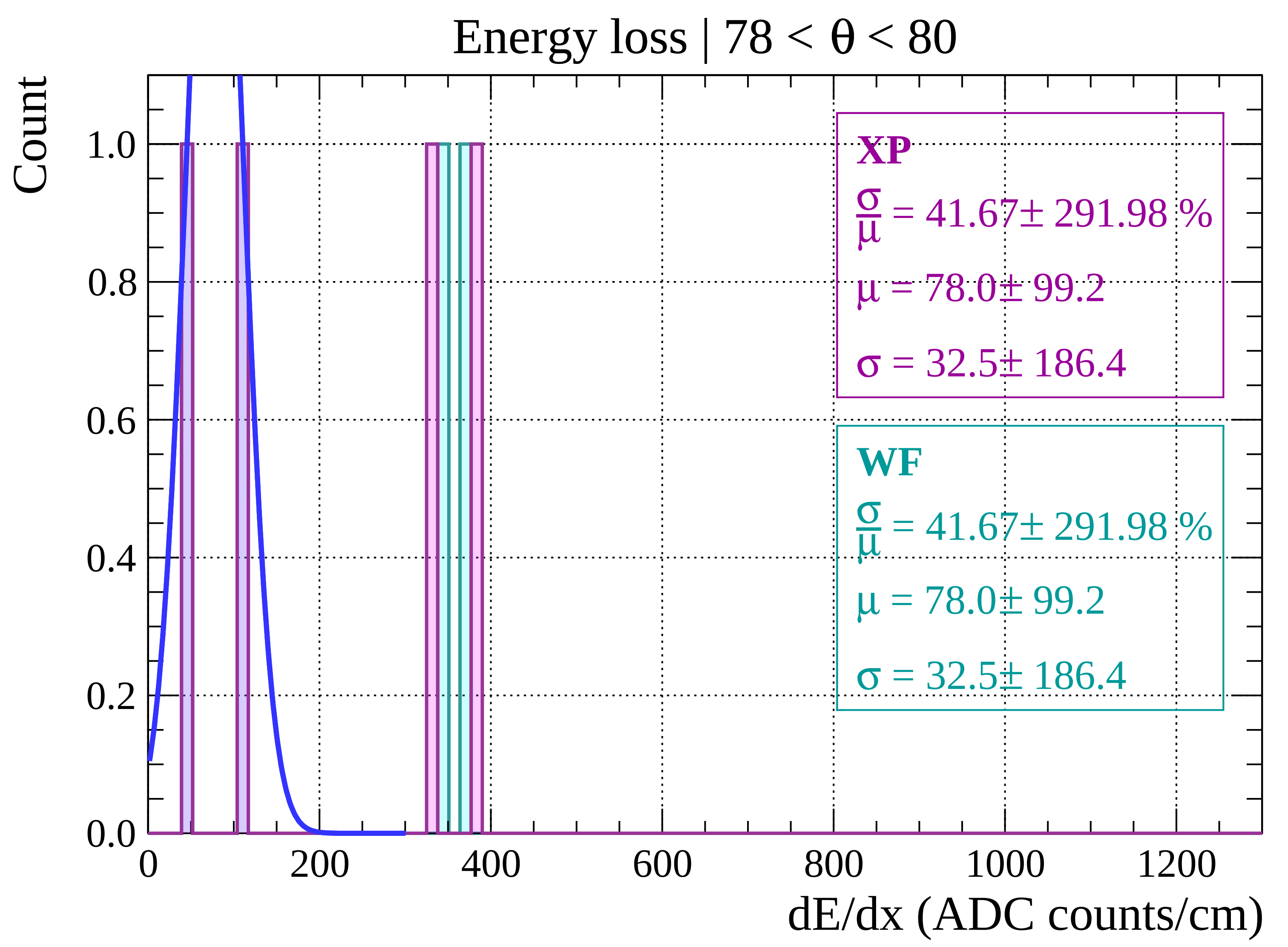

Energy loss | 72 < θ < 74
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 367.15± 2400.96 %
μ = 190.8± 1001.5
σ = 700.6± 904.6
WF
σ/μ = 114.62± 541.88 %
μ = 742.4± 2886.0
σ = 850.9± 714.7



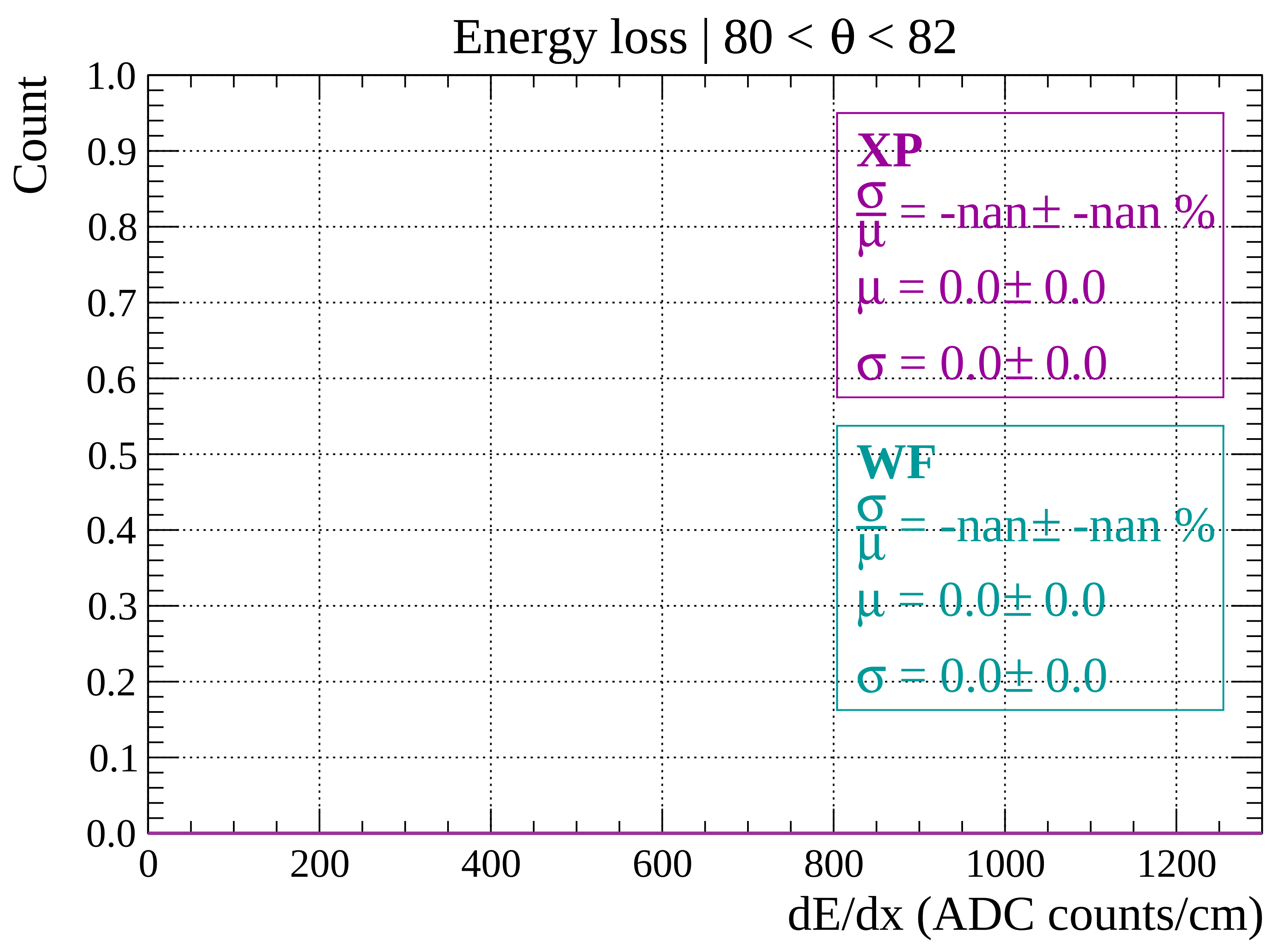
Energy loss | 74 < θ < 76
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = 331.44± 4066.24 %
μ = 233.4± 2711.4
σ = 773.4± 502.0
WF
σ/μ = 439.90± 5740.47 %
μ = 299.2± 3667.5
σ = 1316.1± 1041.1



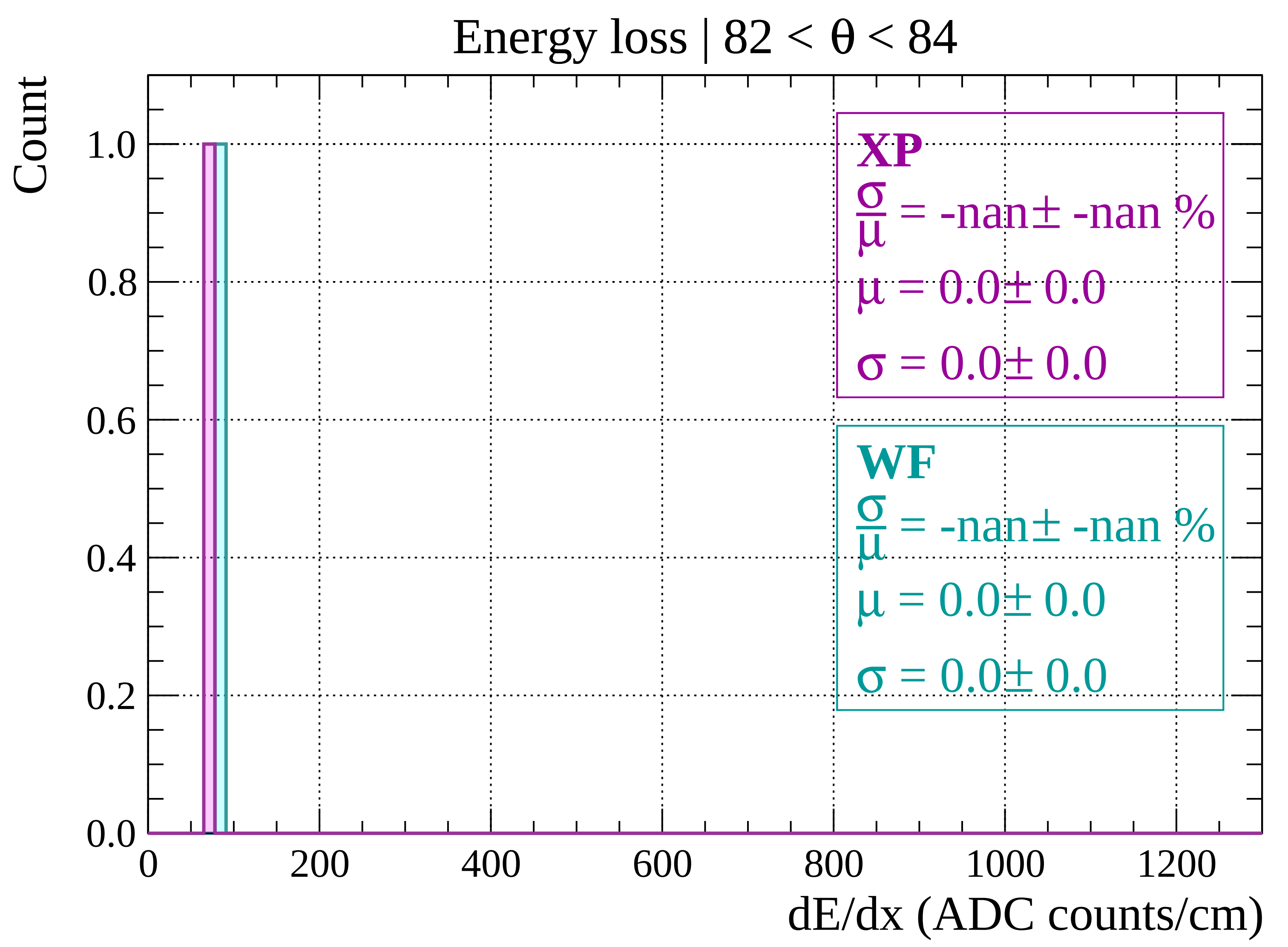
Energy loss | 76 < θ < 78
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -52.96± 195.84 %
μ = -1534.1± 6501.3
σ = 812.4± 438.6
WF
σ/μ = 3.02± 99.84 %
μ = 121.1± 41.1
σ = 3.7± 119.7



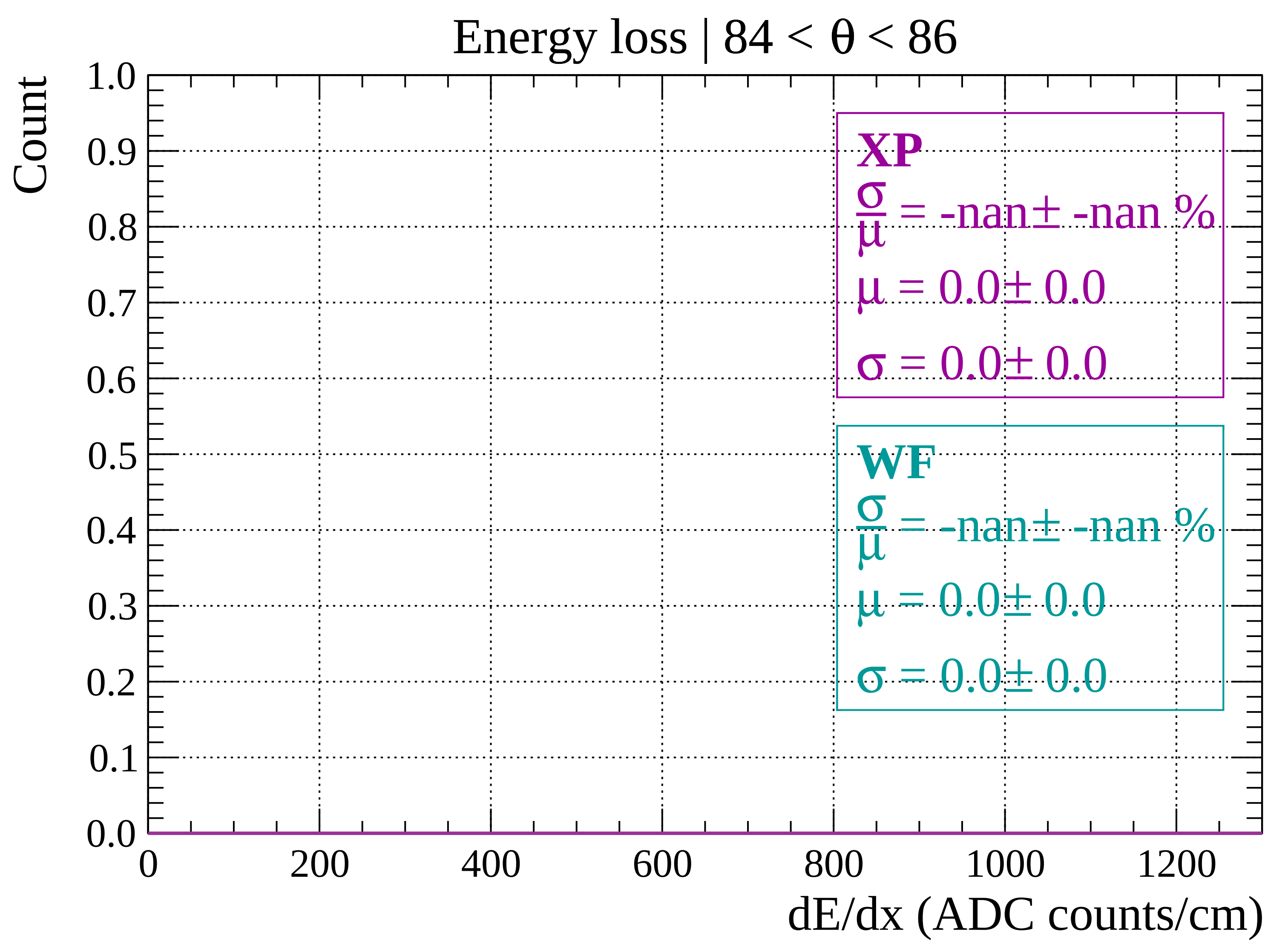
Energy loss | 78 < θ < 80
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 41.67± 291.98 %
μ = 78.0± 99.2
σ = 32.5± 186.4
WF
σ/μ = 41.67± 291.98 %
μ = 78.0± 99.2
σ = 32.5± 186.4



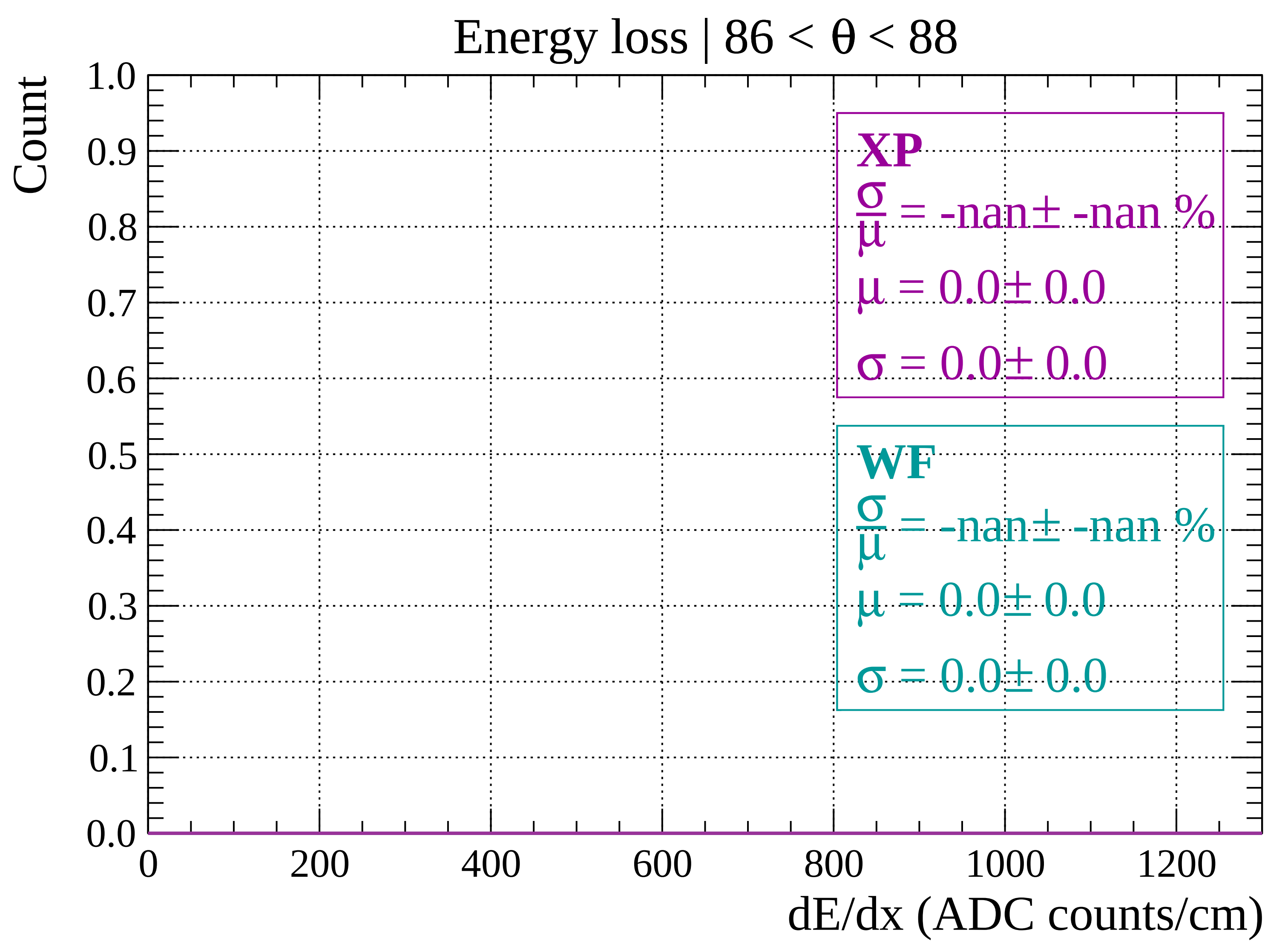
Energy loss | 80 < θ < 82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



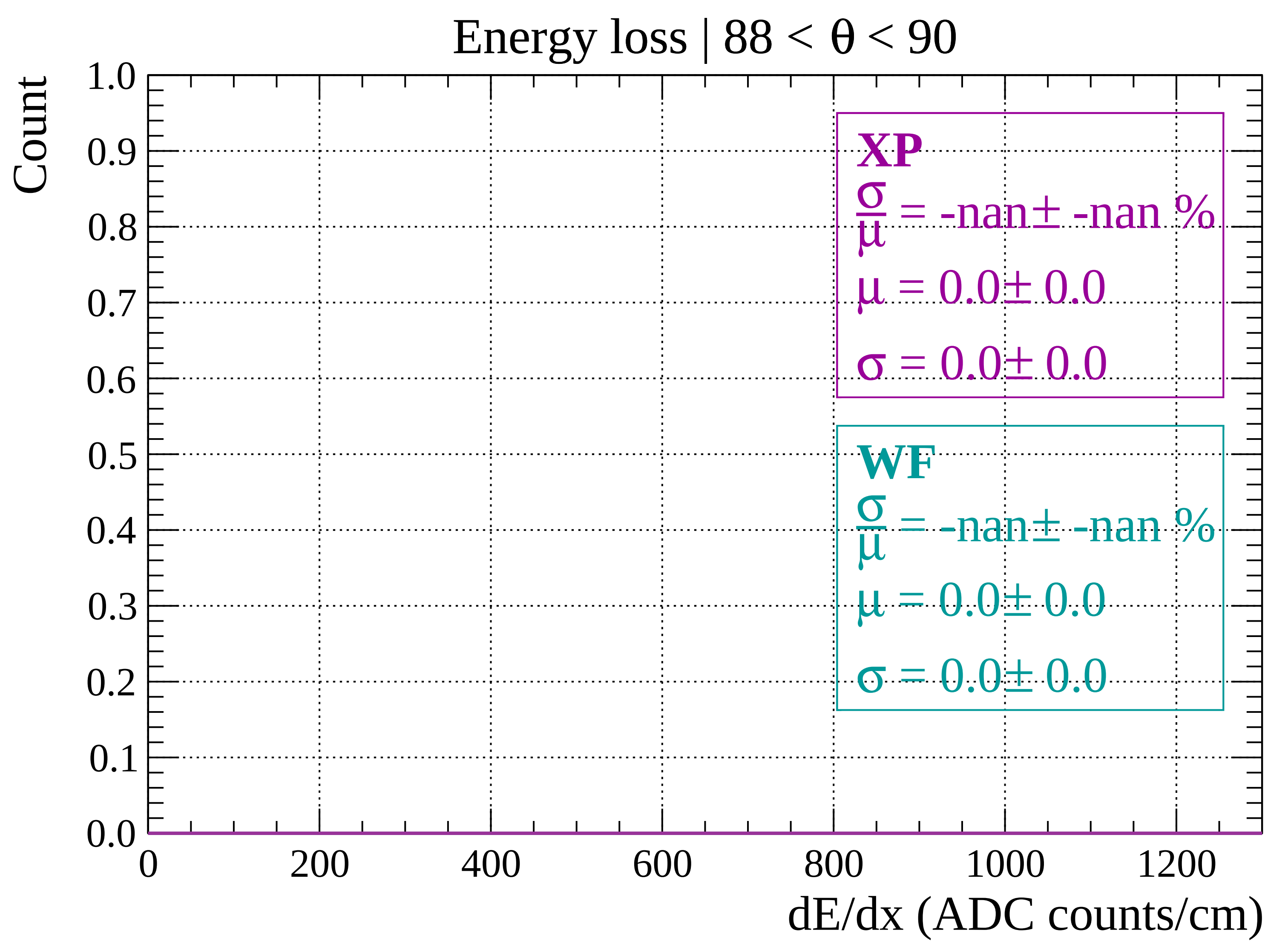
Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



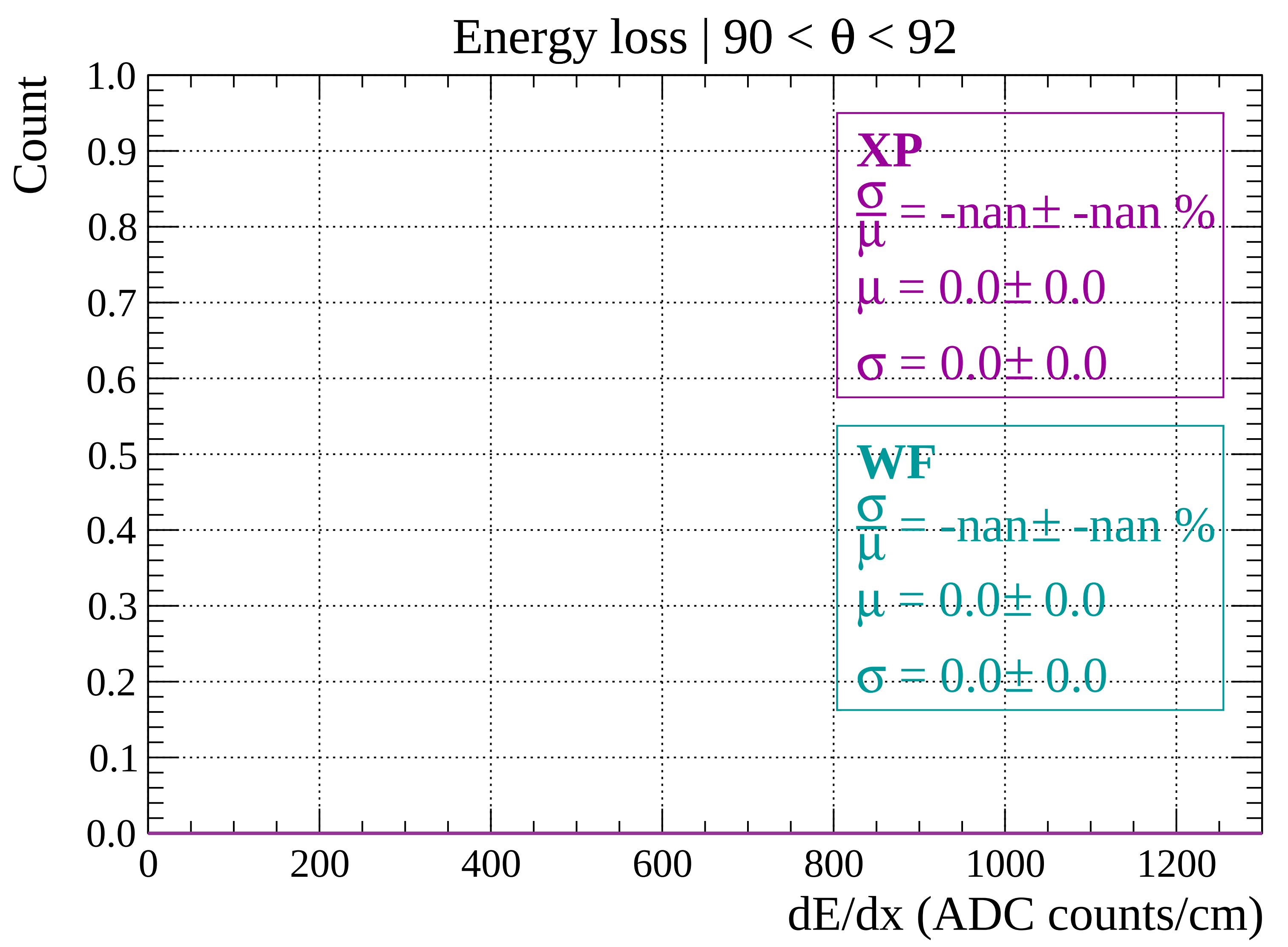
Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

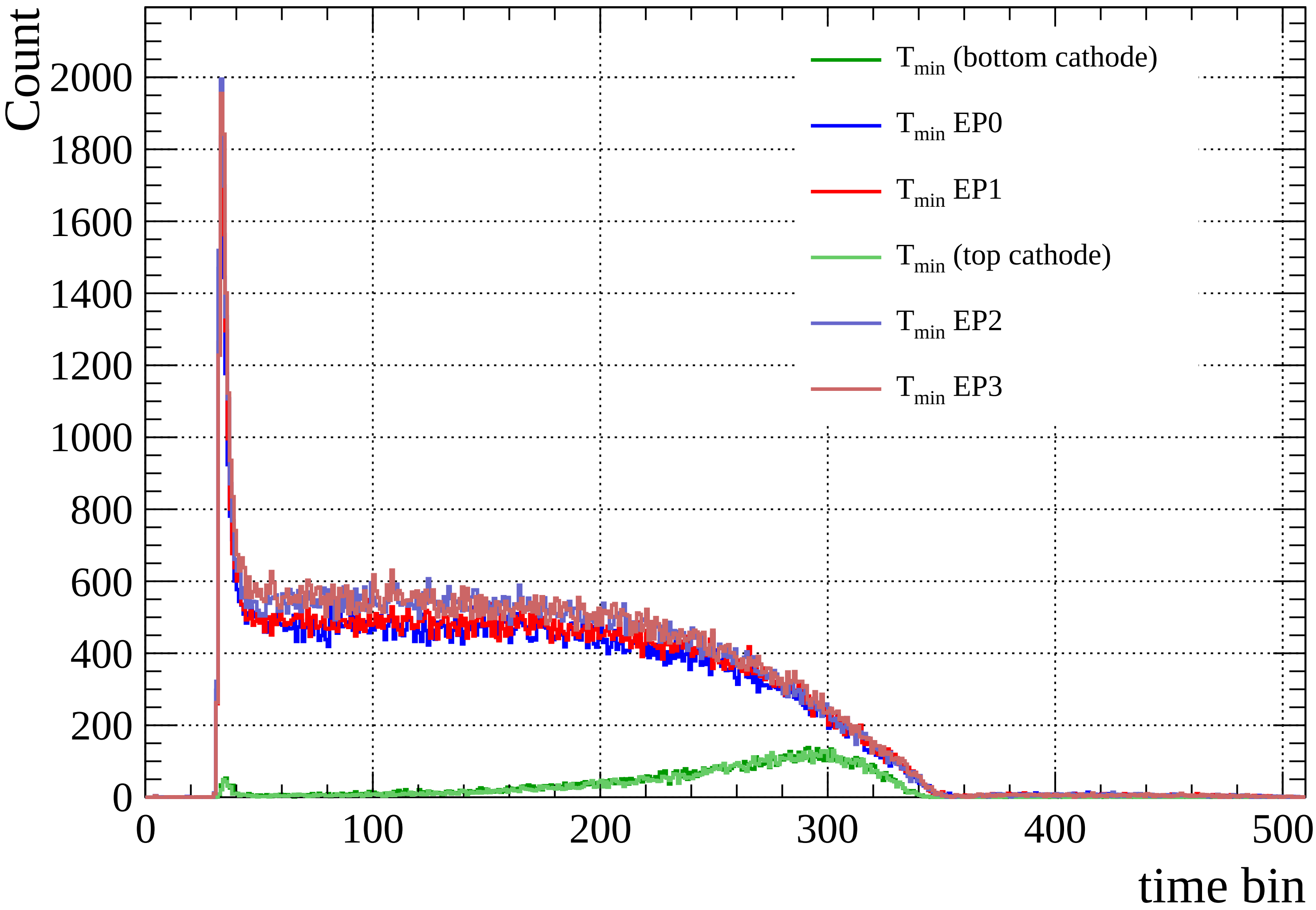


Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

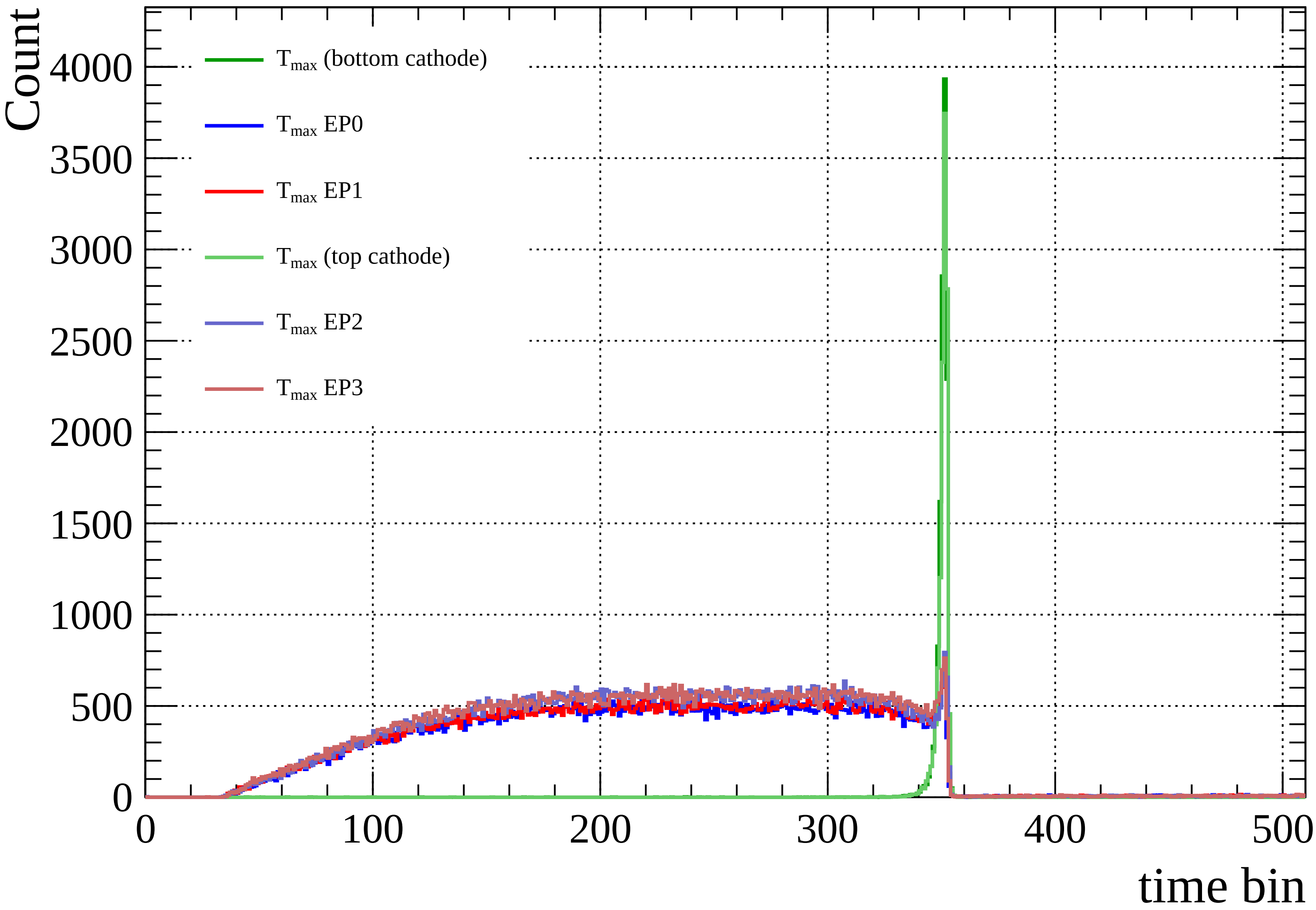


Energy loss | 88 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 92
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
100
200
300
400
500


Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3